Topics in the semantics of Slavic languages

Edited by

Berit Gehrke

Radek Šimík

Open Slavic Linguistics 11

language
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Open Slavic Linguistics

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ISSN (print): 2627-8324
ISSN (electronic): 2627-8332

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Berit Gehrke & Radek Šimík (eds.). 2026. *Topics in the semantics of Slavic languages* (Open Slavic Linguistics 11). Berlin: Language Science Press.

This title can be downloaded at:
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ISBN: 978-3-96110-589-2 (Digital)
978-3-98554-206-2 (Hardcover)

ISSN (print): 2627-8324
ISSN (electronic): 2627-8332
DOI: 10.5281/zenodo.20757287
Source code available from www.github.com/langsci/573
Errata: paperhive.org/documents/remote?type=langsci&id=573

Cover and concept of design: Ulrike Harbort
Typesetting: Berit Gehrke, Radek Šimík, Olga Borik, Mojmír Dočekal, Hana Filip, Atle Grønn, Olav Mueller-Reichau, Adrian Stegovec, and Marcin Wągiel, Kristýna Drobná
Proofreading: Matthew Korte, Tim Morgan
Fonts: Libertinus, Arimo, DejaVu Sans Mono
Typesetting software: XƎLaTeX

Language Science Press
Scharnweberstraße 10
10247 Berlin, Germany
http://langsci-press.org
support@langsci-press.org

Storage and cataloguing done by FU Berlin

FREIE
UNIVERSITÄT
BERLIN

Contents

Preface

Topics in the semantics of Slavic languages contains twelve chapters devoted to a particular topic in formal semantics, as well as an introductory chapter. We are also grateful to Hana Filip, a key figure in the history of Slavic formal semantics, for contributing a personal note, which follows this preface. The topic chapters all consist of two parts: an overview part and a new analytical contribution or case study. The volume fulfills a number of goals: (i) to summarize existing analyses and theories, (ii) to provide a representative bibliographic overview of the relevant literature, (iii) to demonstrate what Slavic data and theories built upon them have to offer to the general linguistic discourse, and (iv) to provide a novel theoretical and/or empirical contribution.

The collection is of interest to a broad readership – Slavic as well as general linguistic scholars, experienced researchers as well as advanced undergraduate or postgraduate students. Individual contributions can serve as background reading to topical seminars. The volume covers areas of semantics in which Slavic languages have traditionally or more recently played an important role in the general formal semantic discourse (e.g. aspect, event structure, genericity, tense, information structure, negation and NPIs, (in)definiteness and specificity, bare NP semantics), as well as topics that have received less attention in their application to Slavic languages, but are pertinent to the theory of semantics, including its interfaces (e.g. focus participles, questions, imperatives, numerals, passives).

All the chapters have solid empirical grounding and, where applicable, the authors emphasize the existing empirical generalizations. The use of state-of-the-art empirical methods, including corpus and experimental research, were encouraged, but not required. Although the focus of the volume lies on semantics, authors were free (and in some cases encouraged) to discuss the pertinent semantic issues in the context of any relevant interface, especially morphosyntax and pragmatics. Authors were encouraged to include cross-Slavic comparisons and to include discussion of Slavic languages that are less well studied. The chapters in this collection discuss data from the following Slavic languages: Bosnian/Croatian/Montenegrin/Serbian (BCMS), and more specifically Croatian and Serbian, Bulgarian, Czech, Macedonian, Polish, Resian, Russian, Slovak, Slovenian, Upper Sorbian, and Ukrainian. The contributions successfully completed a one-round review process in which every article was evaluated and commented on by one internal reviewer, i.e. someone who also contributed a chapter, as well as two external reviewers, one of whom was a PhD student at the time of reviewing.

This book would not have been possible without the help of the following reviewers: Daniel Altshuler, Maša Bešlin, Petr Biskup, Joanna Błaszczak, Olga Borik, Donka Farkas, Hana Filip, Berit Gehrke, Ljudmila Geist, Nina Haslinger, Tania Ionin, Peter Jenks, Dorota Klimek-Jankowska, Natasha Korotkova, Arkadiusz Kwapiszewski, Dimitra Lazaridou-Chatzigoga, Fabienne Martin, Ora Matushansky, Sergey Minor, Olav Mueller-Reichau, Rick Nouwen, Stefan Milosavljević, Maria Onoeva, Mareike Philipp, Agata Renans, Deniz Rudin, Radek Šimík, Frank Staniszewski, Peter R. Sutton, Elena Vaikšnoraitė, Marcin Wągiel, Hedde Zeijlstra, and Karolina Zuchewicz. Many thanks for your reviews! Furthermore, we thank the following people for assisting in LaTeX typesetting and not leaving all that work to us two: Olga Borik, Mojmír Dočekal, Hana Filip, Atle Grønn, Olav Mueller-Reichau, Adrian Stegovec, and Marcin Wągiel. We also wish to acknowledge the technical support of the Language Science Press editorial team, especially Sebastian Nordhoff, Felix Kopecky, as well as Matthew Korte for proofreading the whole book. Finally, Radek Šimík would like to acknowledge the institutional funding Cooperatio LING (Faculty of Arts, Charles University) for enabling the successful completion of this book project.

We initiated this project in 2019 with the aim of having the volume published earlier than it was published now, but unforeseeable circumstances (including the pandemic) led to delays, both on the side of the authors and editors. Nevertheless, we hope that the readers will find it interesting and inspiring.

Berit Gehrke
Radek Šimík
Berlin & Prague, March 2026

Slavic semantics: A personal note

Hana Filip

The start of my life-long preoccupation with semantics of Slavic languages goes back to 1989. It is connected to two momentous events for me, which were surprisingly related, on December 29, 1989: namely, my first meeting with Barbara Partee at my first LSA (Linguistic Society of America) talk on the occasion of its annual meeting in Washington D.C., and Václav Havel's swearing-in ceremony as President of Czechoslovakia in Prague, the pinnacle of the Velvet Revolution that started on November 17 of that year.

I was a third-year PhD student in the Department of Linguistics, at the University of California at Berkeley. My talk, entitled *Nominal reference and aspect: Aspectually determined sort-shifting in Czech*, was scheduled in the session Semantics II, at 3:50 pm. The analysis integrated insights from lexical semantics, in the style of my advisors Charles Fillmore and Paul Kay at UC Berkeley, with compositional semantics building on my previous studies at the Ludwig-Maximilians-University in Munich. The integration of these two semantic traditions also characterized much of Barbara Partee's research, starting in the mid 1990s. Specifically, in her project on the Russian genitive of negation, funded by the National Science Foundation (NSF) and inspired by her extended stays in Moscow, she integrated Russian (Moscow) lexical semantics with "Western" compositional semantics, particularly in the guise of Montague Semantics (she coined this term in her earliest introductions of Montague's work to linguists, see Partee 2005, 2015). Barbara Partee inspired many semanticists working on Slavic languages by showing how these two seemingly disparate semantic traditions can be successfully combined, leading to breakthroughs in our understanding of a variety of topics in Slavic semantics.

Perhaps it was the mention of "Czech" in the title of my talk and the formal semantics term "sort-shifting" that raised Barbara's attention. When I was getting ready to start my talk, she took the seat in the first row, right below the speaker's

Hana Filip. 2026. Slavic semantics: A personal note. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, v–viii. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078567

podium. That this raised the level of my jitters is an understatement. After my talk, she immediately came up to me: "May I introduce myself, I'm Barbara Partee." – "It's so nice to meet you, I know who you are." Barbara then started telling me with great enthusiasm and excitement that she had just returned from Prague, where she had spent the whole fall with Eva Hajičová, Petr Sgall, and their colleagues at the Institute of Formal and Applied Linguistics (Faculty of Mathematics and Physics, Charles University), as well as with philosophers Jaroslav Peregrin, Pavel Materna, and others, and most importantly, she took part in the demonstrations during the Velvet Revolution in Prague. She directly experienced this seismic political and societal change in Czechoslovakia, and tremendously enjoyed every moment of it. After my talk, and having just made acquaintance with Barbara Partee, I went back to my hotel room and switched on TV, the CNN news, showing President Václav Havel walking across one of the courtyards at Prague Castle, shortly after his swearing-in ceremony. It was one of the most memorable moments of my life indeed, and one of the most iconic moments of the Czechoslovak history. For me it was also directly linked to the conversation that I just had with Barbara. This was the beginning of our friendship, Barbara mentored me and helped me along, like she did so many linguists. It has been her tireless, energetic and consistent support of many linguists who focus on Slavic semantics that was fundamental for this field to take off and flourish in the past almost forty years.

The initial decisive impetus came from the foundation of two conferences dedicated to Slavic linguistics: namely, the *Annual Workshops on Formal Approaches to Slavic Linguistics* (*FASL*) with its first meeting in 1992 at the University of Michigan, Ann Arbor, in the United States, and its European counterpart *Formal Description of Slavic Languages* (*FDSL*) followed in 1995 with its first meeting at the University of Leipzig. The first FASL workshop was a rather small gathering of six Slavicists, John F. Bailyn, Steven Franks, Gerald Greenberg, Ljiljana Progovac, Maaike Schoorlemmer, and Jindřich Toman, who focused on *Functional Categories in Slavic Syntax*, which is also the subtitle of the first proceedings, published in 1993 by Michigan Slavic Publications and edited by Jindřich Toman (University of Michigan). The second FASL workshop already had sixteen participants, and it was open to all the fields of Slavic linguistics, not just syntax. It took place at the Massachusetts Institute of Technology (MIT) in 1993. The volume with papers based on the presented talks contained six semantics talks, five syntax talks, two morphology talks (one delivered by Morris Halle) and one prosody talk. My paper in *The MIT Meeting* volume, entitled *Quantificational Morphology: A Case Study in Czech*, was one of my first publications on the topic of the generic *-va-* in Czech; the most recent update on this topic is published in the present volume. The editors of *The MIT Meeting* volume, Sergey Avrutin (MIT), Steven Franks (Indiana University), and Ljiljana Progovac (Wayne State University), stated in their introduction: "We believe that the rapid success of the *Workshop on Formal Approaches to Slavic Linguistics* reflects an increased interest in this general area of linguistics. It is also our hope that the FASL workshop will become a regular annual event, and that it will contribute to creating a sense of community and mutual awareness among researchers from diverse backgrounds with similar professional interests." Their hope was fulfilled, because in 2025 the FASL workshop celebrated its thirty-fourth instantiation at Cornell University, having enjoyed continued success throughout the years.

The inaugural conference of *Formal Description of Slavic Languages* (*FDSL*) was organized in Leipzig in 1995 by Gerhild Zybatow, Uwe Junghanns, and Dorothee Fehrmann. Initially, it took place biennially, either at the Leipzig University or the University of Potsdam, and later at other institutions, including the University of Nova Gorica (Slovenia), the Independent University of Moscow (Russia), Masaryk University in Brno (the Czech Republic), the University of Göttingen, Humboldt University of Berlin, the University of Graz, and the University of Wrocław. From 2006 to 2017, also "halftime" or *x.5* conferences were organized at institutions outside Germany. Starting in 2018, the conference became a regular annual event. In 2025, FDSL18 was hosted by the University of Wrocław (Poland), where more than fifty presenters from across the world delivered papers spanning all the areas of Slavic linguistics.

Around the same time when the FASL workshop was initiated, one of its founders Steven Franks and his colleague George Fowler at the Indiana University (Bloomington) launched a brand new journal, the *Journal of Slavic Linguistics (JSL)*. Their editorial manifesto in the first 1993 volume poses two "unanswerable" questions: "Why are we devoting so much time and energy to *JSL*? We have long felt that there is a real niche waiting to be filled by something like *JSL*, and eventually came to ask ourselves 'If not us, then who? And if not now, then when?' Those questions are unanswerable, and so we concluded that since there is no perfect occasion for launching a time and energy-intensive endeavor, we might as well go ahead on our own." Since 2006 the *JSL* has also been the official journal of the *Slavic Linguistics Society (SLS)*, which was established in December 2004, also thanks to Steven Franks, who served as its first chair, and thrives not the least thanks to its annual conference. For more than thirty years now, *JSL* has been an important venue for disseminating state-of-the-art work on Slavic semantics. One of the *JSL* Special Issues in 2003, guest edited by Wayles Browne and Barbara Partee, is dedicated to Slavic semantics.

This short note conveys some of my memories, and highlights only some key milestones that early on fostered the research on Slavic semantics. This is not to say that many other key events and inspiring researchers did not contribute to it and without whom Slavic linguistics, and Slavic semantics, would not be what it is today. This note is by no means a comprehensive summary. It is my hope that this volume will be one piece in the rich mosaic of the work done in Slavic semantics, creating a sense of community among all researchers who share the excitement for its enigmas.

Hana Filip, Düsseldorf, December 18, 2025.

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Chapter 1

The semantics of Slavic languages: An introduction

Berit Gehrke[a] & Radek Šimík[b]
[a]Humboldt-Universität zu Berlin [b]Charles University

In this volume, we bring together researchers who have had a consistent interest in doing formal semantic research based on the material of Slavic languages. Over the past thirty years or so, the research in Slavic formal semantics has experienced a steady growth in quantity and quality. Nowadays, there are dozens of formal semanticists in the field, producing prominently published research. We believe that it is the right time to bring Slavic semanticists together and produce a strong volume that will not just summarize what has been achieved, but also move the research in the field forward.

1 Introduction

Formal approaches to natural language semantics go back to the early 1970s, but the study of crosslinguistic semantics, as well as the semantics of individual languages other than English gained momentum only in the 1980s and 1990s. Among the exceptionally early birds was Anna Szabolcsi and her work on the semantics of topic-focus articulation in Hungarian (Szabolcsi 1981, 1983). Later examples include Veneeta Dayal's work on Hindi (cor)relativization and other wh-constructions (Srivastav 1991, Dayal 1996), Molly Diesing's work on indefinites in German (Diesing 1992) and Helen de Hoop's in some sense related 1992 dissertation on weak and strong case and its role for scrambling in Dutch (published as de Hoop 1996), Maria Bittner's work on Inuit or Yoruba (Bittner 1994), Roberto Zamparelli's research on DPs at the syntax-semantics interface (Zamparelli 1995), Lisa Matthewson's investigation into Salish determiners and quantifiers (Matthewson 1996), or Veerle Van Geenhoven's work on the semantics of incorporation in

Berit Gehrke & Radek Šimík. 2026. The semantics of Slavic languages: An introduction. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 1–39. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078569

West Greenlandic (Van Geenhoven 1998). Concerning Slavic languages, there is Ljiljana Progovac's work on negative polarity (Progovac 1993), Hana Filip's work on aspect and eventuality types (Filip 1993, 1996), or Roumyana Pancheva's work on tense/aspect (Izvorski 1997) and wh-constructions (e.g., Izvorski 1998).

Since the advent of the 21st century, the field has witnessed a remarkable increase in the interest in crosslinguistic semantics and it is no longer uncommon that semantic investigations of once less-studied languages, including Slavic languages, have a great impact on the whole field of formal semantics and neighboring fields. An example of this is Olga Borik's (2006) *Aspect and reference time*, which deals with the semantics of the Russian (im)perfective and which boasts 484 citations on Google Scholar (as of November 2025). Major semantic conferences (e.g. *Semantics and Linguistic Theory, Sinn und Bedeutung*) routinely host crosslinguistic talks or talks focused on single less-studied languages. New crosslinguistic semantics conferences have been established in both Europe (*The Semantics of African, Asian and Austronesian Languages*) and America (*Semantics of Under-Represented Languages in the Americas*). Finally, semantic papers constitute an integral part at conferences that are devoted specifically to the linguistics of Slavic languages (e.g. *Formal Description of Slavic Languages* in Europe, *Formal Approaches to Slavic Linguistics* in North America). It is this thriving tradition that this volume grows out of and that it contributes to.

The chapters in the volume constitute a combination of a synthetic overview and an analytical contribution or case study, whereby the space dedicated to the two individual parts varies depending on the topic and each author's approach. The goal of the overview part is to summarize the existing research in the pertinent area of Slavic semantics and to explain how it fits into the context of general formal semantic research. Unlike in classical research papers, the overview part can go well beyond what is necessary for understanding the novel analysis and can – in some cases – even be conceived as a self-contained contribution. This way, we provide a venue for contents that would otherwise not find its way into standard research publications but which we believe to be of great value to the linguistics community.

The volume fulfills a number of goals: (i) to summarize existing analyses and theories, (ii) to provide a representative bibliographic overview of the relevant literature, (iii) to demonstrate what Slavic data and theories built upon them have to offer to the general linguistic discourse, and (iv) to provide a novel theoretical and/or empirical contribution in the pertinent area. The volume is of interest to a broad readership – Slavic as well as general linguistic scholars, experienced researchers as well as advanced undergraduate or postgraduate students. Individual contributions can serve as background reading to topical seminars.

There are a number of general properties of the volume which we consider important and worth highlighting. On the one hand, the volume covers areas of semantics in which Slavic languages have traditionally or more recently played an important role in the general formal semantic discourse, including aspect (see the contributions by Gehrke, Mueller-Reichau, Tatevosov), event structure (Tatevosov), genericity (Filip), tense (Grønn), information structure (partially touched upon in the contributions by Borik and Tomaszewicz), negation and NPIs (Dočekal), (in)definiteness and specificity (Borik, Geist), or bare NP semantics (Borik). On the other hand, there are chapters that touch upon traditional semantic topics that have not received much attention in their application to Slavic languages, such as focus particles (Tomaszewicz), questions (Šimík), imperatives (Stegovec), numerals (Wągiel), or passives (Gehrke).

All the chapters have solid empirical grounding and, where applicable, the authors emphasize the existing empirical generalizations. The use of state-of-the-art empirical methods, including corpus and experimental research, were encouraged, but not required. Although the focus of the volume lies on semantics, authors were free (and in some cases encouraged) to discuss the pertinent semantic issues in the context of any relevant interface, especially morphosyntax and pragmatics. Finally, among all the Slavic languages, Russian has received – by a wide margin – the most attention. Although we realize that our power in this respect is limited, we intended to do our best to improve on this situation. Authors were encouraged to include cross-Slavic comparisons and to write – especially in the analytical part – about Slavic languages that are less well studied. In the end, the chapters in this volume discuss data from the following Slavic languages: Bosnian/Croatian/Montenegrin/Serbian (BCMS), and more specifically Croatian and Serbian, Bulgarian, Czech, Macedonian, Polish, Resian, Russian, Slovak, Slovenian, Upper Sorbian, and Ukrainian.

2 Topical areas covered by the chapters in this volume, and beyond

The initial goal of the volume was to cover all traditional areas of research in formal semantics and to give equal room to the different Slavic language groups (South, West, East). In the end we managed to cover many areas, but not all, and due to some authors' withdrawing from the project, the language coverage is unfortunately slightly skewed towards the North Slavic languages (i.e. West and East). Table 1 contains the list of authors that contributed to this volume, in alphabetical order, along with the title of their chapter.

Table 1: Authors and chapter titles

Author	Chapter title
Olga Borik	Bare nominals and definiteness in Russian
Mojmír Dočekal	Negative polarity items in Slavic
Hana Filip	Genericity and habituality in Czech
Berit Gehrke	Cross-Slavic aspect, passives and temporal definiteness
Ljudmila Geist	Indefiniteness and specificity in Slavic languages
Atle Grønn	Tense in Slavic
Olav Mueller-Reichau	The morphosemantics of Russian aspect
Radek Šimík	Polar question semantics and bias: Lessons from Slavic/Czech
Adrian Stegovec	Lessons from Slovenian imperatives
Sergej Tatevosov	Event structure and derivational morphology: A few reflections on the system of Russian
Barbara Tomaszewicz	Family of exclusives in Polish
Marcin Wągiel	Numerals and their kin: The view from Slavic

In what follows we list the core areas of interest, include a selected bibliography, and, where applicable, indicate why Slavic languages are interesting or suitable for investigating the particular area.

2.1 The nominal domain

2.1.1 Bare noun phrases

Most Slavic languages are articleless languages and as such have a wide distribution of bare (determinerless) noun phrases. While there are many syntactic papers concentrating on the question whether or not one should postulate an empty D for the use of bare nominals in various Slavic languages, in particular in argument position (see, e.g., Bošković 2007, Köylü 2021, or Dayal 2026 for recent surveys that go beyond Slavic), there is less work that focuses on the semantics of such nominals, and until only very recently all existing work was on Russian. While Pereltsvaig (2006) is one of the first to also include semantic considerations in her discussion of "small nominals" in Russian, the semantics of Russian bare NPs is also discussed explicitly e.g. in Geist (2010a), Kagan & Pereltsvaig (2011a,b), Mueller-Reichau (2015), Borik (2016), Šimík & Demian (2020), Borik et al. (2020), Seres (2020), Seres & Borik (2021). The semantics of Polish bare noun phrases is investigated experimentally in Šimík & Demian (2021) and there are corpus investigations into bare NPs in Czech (Šimík & Burianová 2020), Russian in contrast with Hindi and Mandarin (Liu et al. 2024, 2023), as well as a comparison between Macedonian, Polish, and Russian (Borik et al. 2025). Further issues related to the lack of (overt) functional elements in the nominal domain concern the semantics of number (Ionin & Matushansky 2006), specificity (Geist 2008b), kind reference (Mueller-Reichau 2011, Kwapiszewski & Fuellenbach 2021), or superlative semantics (see Section 2.1.3).

Borik (2026 [this volume]) provides an overview of the readings that bare nominals can have in Russian (definite, indefinite, generic) and also outlines factors that facilitate one over the other reading, such as word order, topic position, predicate type, sentence type. In the second part of the chapter, she addresses different theories of definiteness, zooming in on the uniqueness approach. She shows that Russian (singular) bare nouns do not presuppose uniqueness, even in contexts in which they are intuitively interpreted as definite, and this leads her to argue that all bare nouns (in argument position) in Russian should receive a quantificational analysis in terms of indefinites. This does not, however, preclude that there cannot be other types of definiteness with bare nouns (e.g. anaphoric definiteness), or that non-bare nouns (e.g. demonstratives, possessives) do not express semantic definiteness.

2.1.2 (In)definiteness, pronouns, determiners, quantifiers

Many Slavic languages boast a rich repertoire of indefinite pronouns and determiners, often wh-based (cf. Haspelmath 1997). These indefinites fall into various categories, including negative concord indefinites, polarity, free choice, or dependent indefinites, (non-)specific indefinites, and epistemic indefinites. The literature in this area is relatively rich and includes studies on Bulgarian (Izvorski 1994, Koev 2017b), Czech (Dočekal 2011, Šimík 2015, Strachoňová 2016), Polish (Błaszczak 2001, 2003), Russian (Pereltsvaig 2000, 2004, 2008, Yanovich 2005, 2008, Geist 2008b, Kagan 2011, Ionin 2013, Martí & Ionin 2019), BCMS (Progovac 1993, Gajić 2016a), and Slovak (Richtarcikova 2015).

The semantics of Slavic demonstratives is explored in Šimík (2016, 2021) (Czech) and Arsenijević (2018) (Serbian). Work on the semantics of personal pronouns and issues of pronominal anaphora and binding is not very well represented, although see Podobryaev & Ivlieva (2008) or Vassilieva & Larson (2005).

Butschety (2025) explores the semantics of the so-called plural pronoun construction, where a plural pronoun is combined with a comitative phrase, whose denotation is "included" in the denotation of the pronoun (e.g., literally, 'We with Pete will go home' with the truth conditions of 'Pete and I will go home.'). Quantification is explored from a theoretical point of view (on the basis of Slovenian, among other languages) by Živanović (2007); subatomic quantification with part-whole structures are explored in part-whole structures in Wągiel (2021); proportional quantification, in particular percentage expressions are addressed in Gehrke & Wągiel (2023); quantifier scope and its interaction with information structure is studied by Ionin (2002), Antonyuk (2015), Ionin & Luchkina (2018) (Russian), Godjevac (2003) (BCMS), Abels & Grabska (2022) (Polish). Dočekal & Šimík (2021) explore the semantics of binominal quantifiers in Czech.

Apart from Bulgarian and Macedonian, which have nominal suffixes assumed to grammatically mark definiteness, primarily addressed in the syntactic literature (for recent discussion, see Rudin 2021), other Slavic languages lack articles.[1] However, there is discussion on 'one' occupying different positions in different Slavic languages when it comes to the grammaticalization path from a numeral to an indefinite article (see, e.g., Hwaszcz & Kędzierska 2018, Seres 2024, Molinari 2025). For example, while Russian 'one' primarily serves the function of marking specificity (see also Ionin 2013), Bulgarian 'one' behaves like an indefinite marker in a wider variety of contexts (see, e.g., Geist 2013, Gorishneva 2013, 2016).

Geist (2026 [this volume]) addresses indefiniteness and specificity in various Slavic languages, distinguishing between two types of specificity: epistemic and scopal. Investigating three different types of indefinite pronouns in Russian, she shows how they are sensitive to these different types of specificity, and she provides an overview of different theoretical approaches to the semantics of such pronouns in Russian. In the second part of the chapter, she compares epistemic indefinites in Bulgarian, Czech, Russian, and Slovak, discusses the relative emergence of 'one' as an indefinite article in Bulgarian, Croatian, Macedonian, Polish, and Resian, and provides a case study on the emergence of the indefinite article in Russian.

2.1.3 Adjectives, degree semantics, gradability

Slavic languages often exhibit a morpho(phono)logical difference between two adjective forms, often called short and long. BCMS has been of particular inter-

[1] Sorbian is arguably near to having a definite article grammaticalized. For Lower Sorbian, see Pankau (2021); for Upper Sorbian, see Scholze (2012). To the best of our knowledge, there is no formal semantic account of Sorbian definite articles or demonstratives.

est as the short–long distinction, found with attributive adjectives only, has been claimed to have impact on the referential properties of the NP it is contained, in that the long form has been argued to correlate with definiteness or with specificity; see Aljović (2002), Trenkic (2004), Stanković (2015), among others. In contrast, in Russian the short–long form distinction is only found with predicative adjectives, and has often been associated with the stage- vs. individual-level distinction, although a direct correlation has also been called into question (see, e.g., discussion in Geist 2010b).

The dependence on a judge in determining the standard for comparison in adjectives is addressed in Bylinina (2013), who also takes into account Russian data (in comparison to other data). Goncharov (2015) investigates adjectival intensification in Russian. The semantics of adjectives and their nominal use is addressed by King (2018) for Slovak. Slavic comparative constructions have been studied by Pancheva (2006), Ionin & Matushansky (2013). The superlative has received attention as well (see e.g. Stateva 2003), also for its interaction with focus and NP/DP-syntax, which turns out to be particularly interesting due to the articleless nature of many Slavic languages; see Živanović (2007, 2008), Bošković & Gajewski (2011), Pancheva & Tomaszewicz (2012), Tomaszewicz (2015). Dočekal (2025) studies the semantics of equative constructions and its relevance to the licensing of negative concord items.

2.1.4 Phi-features

The semantics of phi-features, including person, number, and gender, in pronouns, nouns, or agreement, received relatively little attention. The relevance of number agreement with coordinated subjects for scalar implicatures (in Russian) is explored in Ivlieva (2012a,b, 2013). The semantics of person features in Russian is briefly addressed in Podobryaev (2017) and the semantics of gender in BCMS in Murphy et al. (2018).

2.1.5 Plurality, countability, numerals

Plurality and various related issues have recently received attention in the works of Wągiel (2015a), Grimm & Dočekal (2021), or Butschety (2025). The semantics of various types of numerals, numeral-like determiners, or countability-related affixes is studied by Ionin & Matushansky (2006), Dočekal (2012), Wągiel (2015b, 2020, 2021), Dočekal & Wągiel (2018), Khrizman (2021), Geist et al. (2023).

Wągiel (2026 [this volume]) provides a general cross-linguistic overview of the uses of simple and complex numerals, such as ordinals, multiplicatives, taxonomic and group numerals. In the second part of the chapter, he provides a morpho-semantic analysis of label numerals in Slavic, illustrated with examples from BCMS, Bulgarian, Czech, Polish, Slovak, Slovenian, and Russian. He argues that the arithmetical use of such numerals is basic, while the quantificational and label uses are derived.

2.1.6 Case alternations

Some Slavic languages exhibit systematic case alternations, particularly the alternation between the nominative and the instrumental in the predicative position, and the alternation between the accusative and the genitive in object (and in some cases also subject) position, in negative, intensional, or quantificational contexts. Literature on genitive of negation or quantification includes Pereltsvaig (1999), Partee & Borschev (2002, 2004), Kagan (2007, 2013), Khrizman (2013), Geist (2015) (Russian) and Husić & Renans (2022) (BCMS). Literature on predicative instrumental, which partially overlaps with the literature on bare nominals mentioned above, includes Filip (2001, 2005) on Czech, and Pereltsvaig (2006), Geist (2008a), Bogatyreva (2011), Bogatyreva & Espinal (2014) on Russian. Kagan's (2020) monograph discusses a range of issues including the semantics of datives. An attempt at a more comprehensive account of the syntax and semantics of Russian cases can be found in Zimmermann (2003).

2.1.7 Prepositional phrases

The syntax-semantics interface of Slavic prepositions (and their corresponding prefixes) is explored more recently in Biskup (2019). Romanova (2007) addresses the syntax (and semantics) of Russian prepositions, Gehrke (2008b) the semantics and syntax of Russian and Czech prepositions (and their corresponding prefixes) in the description of motion events, also in comparison with other European languages.

2.1.8 Nominalizations

Nominalizations do not primarily constitute a formal semantic topic, but they are of relevance for the lexicon-syntax interface, and particular in the discussion of Slavic languages, deverbal nominalizations have been investigated to explore the role aspect morphology plays in event structure and for aspect semantics, and how much of the event structure (in addition to the argument structure) and aspect information of the underlying verb is preserved in such nominalizations. Schoorlemmer (1995, 1998) addresses Russian nominalizations and shows that diagnostics to distinguish between simple event nominals, complex event nominals, and result nominals, can also be extended to Russian, whereas Zimmermann (2002) spells out the syntax and semantics of Russian process nominalizations. Tatevosov (2011), in turn, uses the observation that Russian nominalizations contain morphology typically associated with aspect (particular prefixes and suffixes) but lack aspect semantics, as an argument for dissociating the two and in favor of his proposal that grammatical aspect is syntactically high (where it sits cross-linguistically), whereas the morphemes in question are low and contribute to event structure (see also Tatevosov 2026 [this volume] and Section 2.2.1). This line of research is further elaborated on and extended to Czech in Biskup (2023), who points out differences between Czech and Russian in whether nominalization can contain Aspect (AspP) (Czech) or not (Russian; see also Schoorlemmer 1995). The implications of such differences for the general semantics of aspect in these two languages, and the question whether grammatical aspect in Russian (but not in Czech) interacts closely with finiteness, is explored in Gehrke (to appear).

2.2 The verbal and clausal domain

2.2.1 Aktionsart, event structure, aspect

Given that all Slavic languages grammaticalize the distinction between perfective and imperfective and have a rich inventory of verbal affixes that make some (direct or indirect) contribution to event structure and aspect, Slavic languages form a substantial part of the literature on aspect semantics as a whole. There is influential work on the semantics of aspect in terms of partial vs. total events, also in interaction with the semantics of nominal arguments (in particular incremental themes) (Filip 1993, 1999), or in terms of relations between temporal intervals (Klein 1994, 1995, Borik 2002, 2006, Grønn 2004, 2015, Tatevosov 2011, 2015, Altshuler 2014). Related to this, there are a number of works that are devoted to the semantics and syntax of verbal affixes (Filip 2003, 2005, Młynarczyk 2004, Součková 2004, Arsenijević 2006, Tatevosov 2008, Gehrke 2008a,b, Žaucer 2009, Kagan 2015, Zinova 2016, Biskup 2019, 2025, Naumov 2020, Dočekal et al. 2023, Karagjosova 2024), or the use of imperfective forms in (seemingly) semantically perfective contexts (Grønn 2004, 2015, Altshuler 2012, Mueller-Reichau 2013, 2015, Alvestad 2014, Borik & Gehrke 2018, Klimek-Jankowska 2022, Gehrke 2023). There are also important differences in the use of aspectual forms across Slavic languages; see, e.g., Dickey (2000) for an empirical overview of differences between ten Slavic languages in the use of aspect in various contexts. There are only few formal-semantic approaches that address these differences, and they do so only for a limited set of contexts (Alvestad 2014, Mueller-Reichau 2018, 2025, Gehrke 2022, Klimek-Jankowska 2022). The interaction between the aspect of attitude predicates and the factivity of their complements is explored – mainly on the basis of Polish – in Zuchewicz (2018, 2020).

Tatevosov (2026 [this volume]) takes as a point of departure the well-known observation that (im)perfective semantics does not come with a uniform morphological marking on the verb, and he continues his research program that dissociates aspect semantics from aspect morphology (Tatevosov 2011, 2015). In his chapter, he diligently maps Russian aspect morphology in simple imperfectives, perfectives, and secondary imperfectives, to event structure, arguing that morphological complexity coincides with event structural complexity. In the second part of the chapter, he provides a typology of lexical prefixes, which relate to result states, event-structurally. He argues that lexical prefixes can be divided into two groups, depending on whether the verb upon prefixation maintains its lexical semantic core or undergoes a semantic shift. In the former case, the prefixes are further divided based on their argument structure, into monadic and dyadic ones. Monadic prefixes can be argument-sharing or not, and when they are, they can be restrictive with respect to the possible result states they denote, or not.

Also Mueller-Reichau (2026 [this volume]) assumes that aspect semantics is to be dissociated from aspect morphology. After providing a thorough overview of different theoretical approaches to the semantics of aspect in Slavic (primarily, but not exclusively Russian), he advocates for only one null aspectual operator in Russian. This operator is argued to be concerned with state change and to express default aspect, based on the morphological input it gets. He then spells out a detailed account of how the mapping from aspect morphology to aspect semantics is mediated via this aspectual operator, employing the framework of Discourse Representation Theory.

Gehrke (2026 [this volume]) addresses cross-Slavic differences in the use of aspect, primarily focusing on the differences between Czech and Russian, as discussed in the (mostly descriptive) literature. She exemplifies the differences in four contexts: sequences of events, habitual (or what Filip 2026 [this volume] calls generic) contexts, the historical present, and general-factual contexts. A new context of investigation is added, namely the domain of passives (reflexive and participial passives); for example, it is shown that while Russian past passive participles are derived predominantly from perfective verbs, whereas there are morphological and semantic restrictions on such derivations from imperfectives (see also Borik & Gehrke 2018), we regularly find both imperfective and perfective past passive participles in Czech, expressing the aspect semantics we also find in other contexts. A summary of the (few) existing formal-semantic approaches to capturing cross-Slavic differences in aspect use is provided. Since all of them employ some notion of temporal definiteness (or sometimes also specificity), the chapter concludes with outlining a research program to explore parallels between individuals, events, and times regarding definiteness.

2.2.2 Tense

Among the various possible topics in the domain of Tense, some that are of particular relevance to Slavic languages include perfect meanings, also in comparison to perfect meanings in non-Slavic languages, e.g. the semantics of the perfect in Bulgarian (Izvorski 1997, Iatridou et al. 2001) or also perfect meanings associated with past passive participles in Russian (Paslawska & von Stechow 2003). Further topics include sequence of tense in Russian (Babyonyshev & Matushansky 2006, Khomitsevich 2007, Grønn & von Stechow 2010), the concept of (in)definite tense in Russian (and more generally) (Grønn 2015), the semantics of future tense in Polish (Błaszczak et al. 2012), as well as temporal adverbs in embedded contexts (Vostrikova 2018).

Grønn (2026 [this volume]) provides an overview of issues relating to the semantics of tense in Slavic languages, illustrated with examples from Bulgarian, Croatian, and Ukrainian, also providing insights into the history of Slavic tense semantics. He spells out a formal semantics for the Ukrainian tense system that takes into account the role of aspect as well. In the second part of his chapter, he addresses the semantics of temporal auxiliaries used in future and perfect tenses, with a particular focus on Bulgarian, also addressing evidentiality. Primarily based on data from Ukrainian and Polish, he furthermore addresses subordinate tense, e.g. in complements and adjuncts, and he concludes the chapter with a discussion of relative tense and sequence of tense phenomena.

2.2.3 Generic sentences

Generic sentences, such as *Cats meow* or *Mina rides her bicycle to school*, are often discussed as topics pertaining to aspect or tense, or both. Generic statements are aspectually stative, and in some Slavic languages (e.g. Russian), this leads to the predominant use of the imperfective aspect, so that one could get the impression that one function of the imperfective could be to describe generic statements. However, generic statements can also be made with perfective verbs, even more so in other Slavic languages; see, for instance, Filip & Carlson (1997), Mueller-Reichau (2017) for discussion of West Slavic languages.

Filip (2026 [this volume]) argues that genericity, expressed in generic sentences, is a grammatical category in its own right, distinct from both aspect and tense. She investigates the generic marker *-va-* in Czech, which enforces a generic interpretation of the verb it is attached to and is assumed to function the same in Polish and Slovak. She shows that *-va-* shares many commonalities with the null generic GEN-operator but is more restricted than it, since it can only be used in a subset of generic sentences; one restriction, for instance, is that it requires the event in question to have been instantiated (what Filip calls the actualization requirement). She furthermore addresses the question whether a unified semantics for generic sentences, which she views as desirable, is possible, or whether one needs to posit subtypes, with the possibility that *-va-* could represent one such type.

2.2.4 Negation

Negation has received attention especially due to its interaction with negative polarity and negative concord phenomena (Progovac 1993, Błaszczak 2001, Gajić 2016a). An in-depth recent investigation of Czech negation, including its interactions with polarity, aspect, or wh-questions, can be found in Dočekal (2015). Dočekal (2020, 2025) explores the licensing of negative polarity and negative concord items in Czech, based on experimental evidence. Šafratová (2020) presents an experimental investigation of the interaction of negation and comparatives in Czech. The semantics and pragmatics of negation in polar questions is investigated in Staňková & Šimík (2025).

Dočekal (2026 [this volume]) provides an overview of three general approaches to NPIs (syntactic, semantic, pragmatic), illustrating them with examples from Slavic languages (Czech, BCMS, Polish, Slovenian), among others. He then offers a pragmatic account of Czech scalar NPIs with a meaning component that corresponds to the contribution of 'even', building on the theory of Crnič (2011). Cross-linguistically, there are three types of scalar particles (weak, strong, concessive), and he illustrates these types with Slavic examples.

2.2.5 Mood, modality, propositional attitudes, evidentiality, imperatives, and related issues

Modality also plays a role in the interaction with other phenomena already mentioned above, such as the evidential/epistemic inferences of the present perfect in Bulgarian (Izvorski 1997), ability readings of Czech imperfectives (Dočekal &

Kučerová 2009), or modal superlatives (Goncharov 2015). Alvestad (2014) investigates various Slavic languages in the way they differ in aspect choice in imperatives. Stegovec & Kaufmann (2015) analyze Slovenian imperatives, concentrating on their embeddability. Bulgarian tense, mood, epistemic modality, and evidentiality are analyzed in Smirnova (2011), Koev (2017a), Goncharov & Irimia (2020), or Karagjosova (2021). Močnik (2019, 2025) addresses epistemic modality and propositional attitudes in Slovenian. The Russian subjunctive in the context of related mood and tense phenomena is discussed in Zimmermann (2016). Arsenijević (2021) provides a semantic analysis of five subordinate clause types – conditional, counterfactual, concessive, causal, and purposive – which crucially relies on the idea that they are all a type of relative clause, a so-called situation relative.

Stegovec (2026 [this volume]) provides an overview of forms and functions of imperatives across Slavic languages (illustrated mostly with data from BCMS, Russian, Serbian, Slovenian). He then focuses on embedded imperatives in Slovenian and argues that they are also semantically imperative. He shows that other Slavic languages, such as BCMS, Czech, Macedonian, Polish, lack embedded imperatives. He also addresses surrogative imperatives in Slovenian, and concludes the chapter with discussing consequences of the empirical investigation into Slovenian for the semantic analysis of imperatives in general.

2.2.6 Questions and relatives

Slavic questions, relatives, and more generally wh-constructions have received plenty of attention primarily in the syntactic literature, but some progress has been made in formal semantics as well. Izvorski (1996, 1998), Pancheva (2000) deals extensively with Bulgarian/Slavic modal existential wh-constructions, correlatives, free relatives, and related wh-constructions. Zimmermann (2008) analyzes Russian questions with *kakoj* and *čto za* 'which/what kind of'. The semantics of (multiple) wh-questions is explored in Šimík (2010) (Czech) and Simeonova (2025) (South Slavic). Slavic languages have played a central role in Šimík's analyses of modal existential wh-constructions (Šimík 2011, 2013) and ever free relatives (Šimík 2018). The semantics/pragmatics of (alleged) wh-scope marking in Russian wh-questions received attention in Korotkova (2012). The meaning of Slavic polar questions and polar question particles is investigated in Korotkova (2023) (Russian) or Oikonomou & Ilić (2023), Todorović (2024) (Serbian) from a theoretical perspective, in Mitkovska et al. (2024) (Macedonian), Simeonova & Kamali (2024) (Bulgarian), or Onoeva & Staňková (2025) (Czech and Russian) from a corpus perspective, and in Jordanoska & Meertens (2021) (Macedonian), Staňková & Šimík (2025) (Czech) or Repp & Geist (2025) (Russian) from an experimental perspective. The meaning of Slavic response particles (especially 'yes' and 'no') received attention in Gruet-Skrabalova (2015) (Czech), Esipova (2021) (Russian), and more recently also experimentally in Geist & Repp (2023) (Russian) and Hrdinková & Šimík (2025) (Czech).

Šimík (2026 [this volume]) investigates the interpretation of polar questions in Czech, also focusing on different types of bias (epistemic, evidential, preferential) such questions can have. He provides an overview of cross-Slavic differences in encoding polar questions, discussing data from Macedonian, Polish, Russian, Serbian, Slovak, Slovenian, Ukrainian, and Upper Sorbian, and outlines the bias profile for two Slavic languages, Czech and Serbian. In the second part of the chapter, he spells out a semantic account of negative polar questions, also addressing pleonastic negation and particles appearing in such questions, such as Czech *náhodou* 'by (any) chance', or the question particles *(da) li* in Bulgarian, Macedonian, Russian, and Serbian.

2.2.7 Connectives

There is no large body of literature on the semantics of Slavic connectives (conjunctions and disjunctions). The pragmatics of some Russian connectives (e.g. 'but') has been studied by Katja Jasinskaja (Jasinskaja & Zeevat 2008, 2009, Jasinskaja 2010). The phenomenon of "hybrid coordination" (e.g. coordination of subject and object) in Russian is studied by Paperno (2012). A decomposition-based view of conjunction and disjunction, partly based on South Slavic data (in the context of Indo-European and diachronic development), is found in Mitrović & Sauerland (2014). A semantic, strong NPI-view of the Serbian *ni*-disjunction is defended in Gajić (2016b, 2018, 2022).

2.3 Pragmatics, discourse, and information structure

2.3.1 Information structure

Information structure has been an essential topic in Slavic linguistics (see, e.g., Junghanns 2002), and contributions within the field of formal semantics also exist. A recent survey of information structure in Slavic languages (with no specific focus on semantics, however) can be found in Jasinskaja (2016). The semantics of givenness and its relation to word order (scrambling) and prosody (deaccenting) is investigated in Kučerová (2007, 2012), Šimík & Wierzba (2015). Focus has received much attention in the syntactic literature (especially regarding focus-related scrambling; cf. Bošković 2009). Relevant semantic work on focus and focus sensitivity includes the contributions of Šimík (2009) and Kimmelman (2009) on monoclausal cleft-like structures in Czech and Russian, respectively, and recent work of Tomaszewicz (2013a,b, 2015) on focus-association and focus-sensitive particles in Polish (and Bulgarian). The semantics and pragmatics of Slavic (*ė*)*to*-clefts is explored in Kimmelman (2009), Yerbalanov & Philippova (2023), Shipova (2023) (Russian), Šimík (2009) (Czech), and Tajsner (2018) (Polish), and of predicate clefts in Karagjosova & Jasinskaja (2015) (Bulgarian). The interaction of information structure and quantifier scope is studied e.g. by Ionin & Luchkina (2018), and the interaction of information structure and aspect is addressed in Grønn (2004), Alvestad (2014), Borik & Gehrke (2018).

Tomaszewicz-Özakın (2026 [this volume]) addresses the semantics and pragmatics of four exclusives in Polish: *tylko, zaledwie*, and *dopiero* are argued to have direct English counterparts (*only*, *merely*, temporal *only*, respectively), whereas *aż* lacks a direct counterpart and can sometimes be translated by *only*, sometimes by *even*. She identifies the conditions and contexts in which these exclusives appear and spells out a formal account that captures their semantics and pragmatics and highlights the commonalities and differences between one another, as well as with *only*. Some space is also dedicated to word order and focus projection behavior of the Polish exclusives, in comparison to English and German 'only'. The chapter concludes with a brief comparison of the Polish exclusives to similar words in BCMS, Bulgarian, Czech, and Slovak.

2.3.2 Discourse particles

The pragmatics of Russian discourse particles and conjunctions (additive, adversative) is explored by Jasinskaja & Zeevat (2008, 2009), Jasinskaja (2010). Contrastive discourse particles like *to* or *že* in Russian are studied by McCoy (2001), Hagstrom & McCoy (2003), McCoy-Rusanova (2017). The pragmatics and discourse properties of the Czech particle *že* are addressed in Gruet-Skrabalova (2012) and Kaspar (2017). Judging from the description of Serbian *bre* in Mišković-Luković et al. (2015), it also seems to be a contrastive discourse particle, but there is no formal account. Rich inventories of discourse particles are described for Czech (Nekula 1996), Russian (Zybatow 1990), and South Slavic (Dedaić & Mišković-Luković 2010), but formally oriented literature on their semantics and pragmatics is missing.

3 Conclusion

We hope to have demonstrated above that formal semantics of Slavic languages has grown to become a thriving field with plenty of junior and senior researchers contributing regularly to the analysis of a wide variety of empirical domains. Despite our comprehensive approach to covering the relevant bibliography, it turned out to be impossible to be completely exhaustive. The field is undergoing a steady growth and we have clearly reached a point where it becomes very difficult to keep track of everything.[2] We apologize to everybody whose work we failed to include, despite its relevance.

We wish Slavic semantics and Slavic semanticists a lot of new and exciting discoveries and fruitful years ahead.

Acknowledgments

We are grateful to the contributors of the volume who read (parts of) this introductory chapter and took time to comment on it, specifically Mojmír Dočekal, Hana Filip, and Atle Grønn. Any misinterpretation or omission is solely our own responsibility.

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[2] A fun fact: 10 of our citations are from before 1990s, 24 from 2000s, 54 from 2010s, and 43 from 2020s, suggesting a 100 + % growth every decade.

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Chapter 2

Bare nominals and definiteness in Russian

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This article is devoted to the semantics of bare nominals (primarily bare singulars) and their interpretation in Russian, a language with no articles. After an extensive empirical overview of the distribution and interpretation of Russian bare nominals, I turn to a more detailed overview of definiteness as a semantic notion, eventually narrowing down the discussion to the theory of uniqueness. Possible semantic analyses of Russian bare nominals perceived as definite will be reviewed. One of the most important theoretical claims that is defended in this chapter is the absence of uniqueness presupposition with Russian (singular) bare nominals seemingly interpreted as definite, which has major repercussions for an adequate semantic account of Russian bare nominals. I will also briefly show that demonstratives and possessives do not present any threat to the absence of uniqueness claim.

1 Introduction

The discussion of bare nominals (henceforth BNs) and their possible theoretical analyses has been a popular topic in the formal semantic literature over the last decades. The literature, however, has mostly been focusing on those cases and those languages where bare nominals are rather an exception than a rule (cf. Stvan 1998 for English, Munn & Schmitt 2005, 2020 for Brasilian Portuguese, Doron 2003 for Hebrew, Borthen 2003 for Norwegian, Espinal & McNally 2011 for Spanish and Catalan, Lazaridou-Chatzigoga & Alexandropoulou 2013 for Greek, de Swart 2015 for Dutch, English and French), although there have been exceptions, most notably Dayal (2004, 2011) and Munn & Schmitt (2020), who discussed the semantic properties of bare nominals in languages without articles.

Olga Borik. 2026. Bare nominals and definiteness in Russian. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 41–104. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078571

This article will be devoted to bare singular NPs and their interpretation in Russian, a language with no articles. However, the reader should not be led to believe that all languages without articles are predicted to behave the same with respect to the interpretation of BNs. In fact, there is evidence to the contrary, as has recently been argued by Liu et al. (2023, 2024). Slavic languages do exhibit some variation both with respect to the use of BNs and with respect to the distribution of some (in)definiteness markers like *one* (see Weiss 2004, Geist 2013, Hwaszcz & Kędzierska 2018, Czardybon 2021, Šimík 2021), not to mention Bulgarian and Macedonian, which have overt definite articles (Tomić 2006, Rudin 2021). Broader cross-linguistic studies can also give surprising results with respect to languages well outside the Slavic family, e.g. Mandarin, Hindi, or Hebrew, as recently reported in Liu et al. (2023), Borik et al. (2025), Liu et al. (2024). Thus, the behaviour of BNs, and especially bare singulars (BSs) in Russian discussed in this chapter is just one of the patterns, which does not have to be replicated in all languages without articles.

In what follows, I will first give a rather broad empirical overview of the distribution and interpretation of Russian BNs in Section 2 and describe the conditions and contexts where a BN can get a generic, a definite or an indefinite reading. After that, I will move on to a more detailed discussion of one of the interpretations available for BNs: a definite reading. After a brief survey of theoretical approaches to definiteness in Section 3.1, we narrow down the discussion to the theory of uniqueness in Section 3.2 and examine possible semantic analyses of Russian BNs perceived as definite within the uniqueness theory of definiteness in Section 4. One of the most important empirical claims that will be defended in this chapter is the absence of uniqueness presupposition with Russian BNs interpreted as definite (Section 4.3) which will have major repercussions for a possible semantic account of the Russian facts, as discussed in Section 4.4. I will also show that demonstratives and possessives do not pose any threat to the absence of uniqueness claim in Section 5 and summarise and conclude in Section 6.

2 Bare nominals in Russian: an empirical overview

It is not surprising that, in the absence of an article system, BNs in Russian render all types of interpretations generally available for nominals in natural languages. This section will provide an empirical overview of the distribution and interpretation of BNs in Russian, as well as discuss some possible constraints on the interpretation that Russian BNs seem to exhibit.

To begin with, Russian BNs can appear both in predicate and argument position. In predicate position, BNs can render interpretations that appear to be identical, or at least comparable, to bare, indefinite and definite nominals in languages with articles. Those parallels are illustrated in the examples below.[1]

(1) a. Dolphins are *fish*.
 b. My sister is *a manager*.
 c. John is *the author of this book*.

(2) a. Del'fin – ėto *ryba*.
 dolfin this fish
 'Dolphins are fish.'
 b. Moja sestra – *menedžer*.
 my sister manager
 'My sister is a manager.'
 c. Vasja – *avtor' ėtoj knigi*.
 Vasja author this.GEN book.GEN
 'Vasja is the author of this book.'

In this chapter, I will not go into the details of the semantic analyses proposed for NPs in predicate position. It suffices to say that in a classical (by now) theory of type shifting based on Partee's (1987) proposal, the denotation of NP can shift from a predicate type $\langle e,t\rangle$ to an individual of type e or a quantifier type $\langle\langle e,t\rangle,t\rangle$. Combined with one of the approaches to the denotation of copula *be*, the type shifting theory opens up various options to account for the behaviour of nominal predicates and arguments. In this chapter, I will only consider nominal arguments and leave out the discussion of (in)definiteness in predicate position, although to the best of my knowledge there is no articulated proposal for the semantics of predicative nominals in languages without articles, so this niche should eventually be filled.[2]

As for argument position, BNs in Russian do not really have any distributional restrictions per se, which is what we intuitively expect in a language without articles. In particular, BNs appear freely as subject and direct or prepositional objects, with a full range of interpretations available for arguments, although some interpretations appear to be constrained to or preferred for certain argument positions, as we will discuss in more detail below. In the rest of this section,

[1]All examples are in Russian or English, unless indicated otherwise.

[2]There is, however, a considerable body of literature on the interpretation of predicative nominals in relation to their case, Nominative or Instrumental (see, for instance, Pereltsvaig 2001, Geist 2008, to mention just a few). No connections to (in)definiteness have been made or suggested in this literature, with an exception of Bogatyreva (2014), who claims that instrumental predicative nominals are weakly referential.

I will discuss BNs in subject and object position, leaving aside prepositional BN arguments, as they do not exhibit any additional relevant properties that are not present in other types of arguments. The empirical overview below is organised by grouping together various types of interpretations, the main division line being drawn between a generic/kind vs. an individual object interpretation.

2.1 Generic/kind BNs

Both morphologically singular and plural bare nominal phrases in Russian can have a generic reference (see Chierchia 1998, Doron 2003, Dayal 2004), as exemplified in (3) and (4):[3]

(3) a. *Kit* naxoditsja na grani isčeznovenija.
whale.nom.sg is.found on verge extinction.gen
'The whale is on the verge of extinction.'

b. *Poezd* kak sredstvo peredviženija očen' udoben.
train.nom.sg as means transportation.gen very convenient.sg
'The train as a means of transportation is very convenient.'

(4) a. *Kity* naxodjatsja na grani isčeznovenija.
whale.nom.pl are.found on verge extinction.gen
'Whales are on the verge of extinction.'

b. *Poezda* kak sredstvo peredviženija očen' udobny.
train.nom.pl as means transportation.gen very convenient.pl
'Trains as a means of transportation are very convenient.'

In contrast to Russian, in a language with overt determiners like English, the nominals corresponding to the subject in (3) will be expressed by means of a definite generic (cf. Carlson 1977) or the singular generic (cf. Chierchia 1998), that is *the whale* or *the train*, as illustrated in (5a). On the other hand, English also allows bare plurals to refer to kinds. This is shown in (5b). Borik & Espinal (2015) propose a term definite kind in relation to a kind referring singular in (5a) and reserve the term *generic* for plurals exemplified in (5b).

(5) a. *The whale* is on the verge of extinction.

b. *Whales* are on the verge of extinction.

[3]This discussion is limited to a generic/kind interpretation of nominal phrases. There are other well-known means of expressing genericity. For a broader overview, the reader is referred to Krifka et al. (1995). Filip (2026 [this volume]) discusses a verbal marker of genericity in Czech.

A common background assumption that seems to be supported by experimental evidence (see Ionin et al. 2011) is that plural generics are more frequent and less restricted. The observation that singular generics as in (5a) are more restricted in their distribution goes back to Carlson (1977), which is probably the most influential approach to kind reference to date. Carlson assumes that the two generic expressions in (5) refer to the same ontological object, namely, a kind, as opposed to an entity. However, a definite generic, in Carlson's terminology, can only be used to refer to well-established kinds (see also Krifka et al. 1995).

The claim that generic expressions in (5a) and (5b) refer to the same ontological object has been challenged by Borik & Espinal (2015) for Spanish and by Seres (2020) for Russian. Borik & Espinal (2015) claim that at least in Spanish, definite kind expressions of the type *el búho* in (6a) denote the kind or species itself (cf. Jespersen 1927), while plural subjects, which are also definite in Spanish, as illustrated in (6b), "refer to a (maximal) sum of representatives of the kind, but do not name the kind itself" (see Borik & Espinal 2015: 170).

(6) a. El búho {es común / está por todas partes / desaparece
the owl is common is at all parts disappears
rápidamente / a menudo es inteligente}.
rapidly often is intelligent
'The owl is {common / widespread / quickly disappearing / often intelligent}.'

b. Los búhos {son comunes / están por todas partes / desaparecen
the owls are common are at all parts disappear
rápidamente / a menudo son inteligentes}.
rapidly often are intelligent
'Owls are {common / widespread / quickly disappearing / often intelligent}.'

(Spanish; Borik & Espinal 2015: 171)

As has already been mentioned, the term DEFINITE KINDS is proposed by Borik & Espinal (2015) for kind referring nominals exemplified in (6a). The proposed analysis derives definite kinds by applying an iota operator directly to the denotation of a noun, which is assumed to denote properties of kinds (cf. McNally & Boleda 2004, Dobrovie-Sorin & Pires de Oliveira 2008, Espinal 2010, Espinal & McNally 2011). A definite entity interpretation arises if Number intervenes: Number is defined as a semantic instantiation operator that applies to properties of kinds (the denotation of a common noun) and yields properties of objects (cf. Déprez 2005).

Crucially, definite kinds are numberless in Borik & Espinal's (2015) analysis.[4]

As for plural generic expressions as the one exemplified in (6b) above, Borik & Espinal (2015) argue that in Spanish, a generic reading for definite plural subject arguments is treated more like an epiphenomenon: this reading is derived by an intensionalizing operator $^\wedge$ applied to the definite description, i.e., $^\wedge\iota$, an operation that is triggered only when the nominal expression is in argument position of a kind- or an individual-level predicate (henceforth also k-/i-level), largely following the analysis of Chierchia (1998) for Romance.

To sum up, Borik & Espinal (2015) attribute two different semantic analyses to the two types of generic expressions in (6a) and (6b), the former being analysed as referring to a kind entity itself, and the latter being conceptualised as intensionalized definite plural expression referring to a maximal set of individual entities satisfying the descrptive content of a noun. From the ontological point of view, hence, morphologically singular and plural expressions refer to essentially different types of entities. Borik & Espinal (2015) suggest that this analysis reflects two different conceptual strategies that human beings use to refer to kinds: one is a direct kind reference and another is reference to a kind via its instantiations.

The analysis presented above has been applied to Russian in Borik & Espinal (2019), Seres (2020) and Borik & Espinal (2020). While the interested reader is referred to the original sources for the details of the respective analyses, I outline the empirical coverage of these works. Definite kind expressions, apart from subject position of kind-level predicates exemplified in (3a) above, include modified kinds in (7a) (see Borik & Espinal 2019, 2020) and are also found in definitional generic sentences in (7b), as well as in the object position of *invent*-type predicates in (7c) (see Seres 2020), as illustrated below:

(7) a. *Amurskij tigr* zanesen v Krasnuju knigu.
amur tiger registered in Red book
'The Siberian tiger is registered in the IUCN Red list.'

b. *Kaban* – ėto dikaja svinja.
wild.boar.NOM.SG that wild swine.NOM.SG
'The wild boar is the wild swine.'

c. Steve Jobs izobrël *iPad.*
Steve Jobs invented iPad.ACC.SG
'Steve Jobs invented the iPad.'

[4]This is just a brief summary of the analysis proposed by Borik & Espinal (2015), the reader is referred to the original paper for a detailed justification of this particular theoretical proposal.

As for plural generics, according to Seres (2020), they are found in subject position of kind-level and individual level predicates. With individual level predicates, a generic interpretation arises only if a sentence is interpreted as characterising, i.e., ascribing an essential, characterising property to a subject in the absence of any spatiotemporal anchoring, as illustrated below:

(8) a. *Sobaki* široko rasprostraneny. GEN, K-LEVEL
dogs wide spread
'Dogs are wide spread'

b. *Sobaki* lajut. GEN, I-LEVEL
dogs bark
'Dogs bark.'

c. *Sobaki* lajut vo dvore. INDIV, I-LEVEL
dogs bark in yard
'Some/The dogs are barking in the yard.'

Note that the counterpart of (8a) with a singular subject *sobaka* 'dog.SG' would sound strange, as *widespread* is a distributive kind-level predicate that requires plurality of its argument (Chierchia 1998). In (8b) and (8c), the generic interpretation of the whole sentence comes from a predicate and is presumably due to a GEN operator (see Krifka et al. 1995 for sentence-level genericity). The only difference between (8b) and (8c) is the presence of a locative prepositional phrase which immediately renders a generic reading of the subject *sobaki* 'dogs' impossible. Singular subjects are possible in both cases, but the interpretation of (8b) would be then comparable to the English *A dog barks*, wheareas (8c) would just be a statement about an individual definite or indefinite dog.

BNs with a kind interpretation bear an intimate connection to the notion of definiteness, which will be discussed in detail in Section 3. In particular, it is common to derive a kind (a definite kind, in the terminology of Borik & Espinal 2015) reading of *the whale* from example (5a) by means of an iota operator, which is also responsible for a "regular" definite interpretation in formal approaches, as proposed, for instance by Chierchia (1998), Dayal (2004). Borik & Espinal (2015) also employ an iota operator in their analysis of definte kinds, which they apply to Russian in Borik & Espinal (2019, 2020). All these proposals will require adjustments in the light of the claims that will be made in Section 3, so we will come back to this issue in Section 4.4. To proceed with the empirical part, we will now turn to an overview of BNs with an individual/object reference in Russian in relation to their position in a sentence.

2.2 (In)definite BNs and word order

If a bare nominal refers to an individual object, rather than to a class or a kind, it seems that, at least in Russian, it can get both a definite and an indefinite reading, although there are some factors that appear to constrain the apparent freedom of interpretation. One such factor, probably the most discussed and well known one, is word order and hence we will now look at the effects of linear position of a nominal on its interpretation.

In the descriptive literature on Russian, it has often been pointed out that the following correlation holds: preverbal BNs tend to be interpreted as definite, whereas postverbal ones usually get an indefinite reading (cf. Pospelov 1970, Fursenko 1970, Chvany 1973, King 1995). Examples in (9), modelled on Krámský's (1972) examples from Czech, illustrate this effect:

(9) a. Kniga ležit na stole.
book.NOM lies on table.LOC
'The book is on the table.'

b. Na stole ležit kniga.
On table.LOC lies book.NOM
'There is a book on the table.'

A similar correlation between distribution and interpretation has been reported for other articleless languages, such as Mandarin, where preverbal bare nominals are interpreted only as generic or definite, while postverbal bare nominals can be interpreted as either indefinite or definite or generic (Cheng & Sybesma 2014). In fact, the correlation "preverbal–definite" and "postverbal–indefinite" has even been claimed to be universal (Leiss 2007). Thus, at least at first impression, the contrast between definite and indefinite readings of nominals can be perceived, understood and even encoded in a language without articles. This observation is supported by the literature: a common wisdom seems to be that languages without articles can express definiteness distinctions by other means, despite the absence of an article system.

However, perceived definiteness of preverbal nominals may result from an interplay of various factors, one of which is topicality. In Section 2.2.1, I will discuss possible constraints on the interpretation of BNs in sentence initial position, which is usually assumed to be a position for topics in Russian (see, for instance, Geist 2010, Jasinskaja 2016).[5] Several other studies have raised the question of

[5] This, however, does not mean that a "topic position" in a strictly syntactic sense is necessarily postulated for Russian. What I understand by a "topic position" is simply the leftmost position

whether preverbal vs. postverbal linear position of a nominal also plays a role in its interpretation: a corpus study for Czech by Šimík & Burianová (2020) and an experimental study for Russian by Borik et al. (2020), Seres et al. (2019), which will be discussed in 2.2.2.

2.2.1 BNs in topic position

It is well known that information structural principles can impose various constraints and conditions on the interpretation of nominal arguments (as well as other elements of a sentence). Russian belongs to a group of so-called "free word order" languages, which means that changes in the order of constituents, when permitted, do not influence the syntactic function of affected elements but rather the information structural properties of a sentence. It has often been suggested in classical descriptive and functional literature that word order alternations in Slavic languages are determined by information structure (see Mathesius 1938, Sgall 1972, Hajičová 1974, Isačenko 1976, Yokoyama 1987, Comrie 1981: i.a.).[6]

Topicality, one of the main notions of information structure (see, for instance, Krifka 2008), is particularly relevant for a discussion of referential properties of nominal phrases. Although the concept of topicality has been around for many years and its relevance for linguistics is not a subject of any considerable debate, there are still many ways in which topics are understood. For concreteness, only sentential topics will be discussed here. A sentential or aboutness topic is what a sentence is about, following Reinhart (1981) and Endriss (2009), among many others. The following example taken from Endriss (2009: 20) can provide an elementary illustration of a sentential topic in the sense adopted here.

(10) a. Yesterday, Clarissa visited Dena.
     b. Yesterday, Dena was visited by Clarissa.

The sentence in (10a) is naturally understood as saying something about Clarissa, whereas (10b) is rather a statement about Dena. In this sense, *Clarissa* is a topic of (10a) and *Dena* is a topic of (10b).

For Russian, both preverbal subjects and objects are considered topics when they appear in sentence initial position (cf. Jasinskaja 2016). As illustrated in the

(in a linear sense) in a sentence. If there is a topic in the leftmost position of a sentence, it does not entail (at least in my view) the presence of a specific syntactic projection associated with Topic in syntactic structure.

[6]For a broader discussion of factors that can influence the word order in Slavic languages see Jasinskaja & Šimík (In press).

examples below, both subject (see (11a)) and object (see (12a)) can also be left-dislocated, a construction that is considered to be a reasonable, although not a clear-cut diagnostic for topichood (see Reinhart 1981).

(11) a. Tolya včera razgovarival s Anej.
Tolya.NOM yesterday talked.IPFV with Anja.INSTR
'Tolya talked to Anya yesterday.'

b. Čto kasaetsja Toli, to on včera razgovarival s Anej.
what concerns Tolya that he yesterday talked.IPFV with Anya
'As for Tolya, he talked to Anya yesterday.'

(12) a. Varen'e ja včera s"jel.
jam.ACC I yesterday ate.PFV
'I ate the jam yesterday.'

b. Čto kasaetsja varen'a, to ja ego včera s"jel.
what concerns jam that I it yesterday ate.PFV
'As for the jam, I ate it yesterday.'

The connection between definiteness and topicality is based on a descriptive generalization that is accepted in a lot of semantic literature on topics in general: elements that appear in topic position can only be strongly referential, i.e., of type *e*, and hence either definite or specific indefinite (see, for instance Reinhart 1981, Erteschik-Shir 1998, Portner & Yabushita 2001, Endriss 2009), where specificity is understood as having a particular referent and being of type *e*. An appealing intuitive idea behind this generalisation is that if the nominal topic does not refer to any entity, it cannot serve as an aboutness topic because then there is no entity to be talked about.[7]

[7]Some examples from various Romance languages have been brought up in the literature to show that a topicalized left dislocated element can, in fact, be interpreted non-specifically. The following examples from Leonetti (2010) illustrate the point for Spanish (i.a) and Italian (i.b):

(i) a. Buenos vinos, (los) hay en Castilla.
good wines CL.MASC.PL have.PRES.3SG in Castille
'There are good wines in Castille.'

b. Libri in inglese, (li / ne) può trovare al secondo piano.
books in English CL.MASC.PL CL.PART can.PRES.3SG find on.the second floor
'English books can be found on the second floor.'

As suggested by Leonetti (2010), non-specific or weak indefinites topics do exist, although are highly restricted. In general, the examples above illustrate that the correlation between topic and definiteness and/or topic and referentiality is not a strict dependency but rather a strong tendency.

As far as Russian is concerned, it has been suggested that BNs in Russian that appear in preverbal subject position are subject to an indefiniteness constraint. The restriction is proposed and extensively discussed by Geist (2010), who gives the following example to illustrate the phenomenon:

(13) a. *Context:* I saw a boy and a girl.
Devočka vošla v dom.
girl.NOM came into house.
'The girl entered the house.'

b. *Context:* The door opened and…
* Devočka vošla v dom.
girl.NOM came into house.
Intended: 'A girl entered the house.' (Geist 2010: 193, ex. (5))

In these examples, the sentences in brackets set a context, which induces a definite interpertation of *devočka* 'girl' in (13a), and an indefinite one in (13b). In the second case a bare singular nominal is not acceptable in the intended indefinite reading. To convey this meaning, the subject can either be explicitly marked with one of the indefinite determiners (actualizers, in Padučeva's (1985) terminology) available in Russian, as illustrated in (14), or appear postverbally, as in (15):[8]

(14) *Context:* The door opened and…
{Odna / kakaja-to} devočka vošla v dom.
one some girl came into house
'A / Some girl entered the house.'

(15) *Context:* The door opened and…
V dom vošla devočka.
into house came girl
'A girl entered the house.' (Geist 2010: 193, ex. (6))

However, as Geist (2010) also notices, in other cases a preverbal subject can be interpreted indefinitely, as illustrated in (16):

(16) *Context:* Why is it so noisy?
Rebënok plačet.
child is.crying
'A child is crying.' (Geist 2010: 194, ex. (7))

[8]See Ionin (2013) for arguments that unstressed *odin* 'one' is a specificity marker in Russian.

After carefully considering the empirical data, Geist (2010) arrives at a conclusion that it is not pre- or postverbal position per se that restricts the indefinite interpretation of BNs, but the topic-comment structure. Thus, in thetic sentences, which are sentences without a topic/comment division like (16), a bare nominal that appears preverbally can be interpreted as indefinite, whereas the indefinite interpretation is excluded for bare NPs that are aboutness topics, as in (13).

To explain the data, Geist (2010) adopts as a starting point the generalization according to which a topic of a sentence has to be strongly referential, i.e., either definite or specific indefinite. Given that indefinite BNs in Russian seem to be banned from topic position, Geist suggests that this restriction can be explained by the absence of a *specific* indefinite interpretation in BNs in Russian. Non-specific (weakly referential) indefinites cannot be aboutness topics (although see footnote 7 for exceptions in other languages), as witnessed by (13). If BNs, indeed, cannot render a specific indefinite interpretation in Russian, they cannot be topics. Hence, only BN with a definite interpretation are acceptable aboutness topics in Russian.

Geist's (2010) analysis sets a milestone in research on BNs in articleless languages as it is one of the few careful, systematic and thorough studies of possible constraints on their interpretation in Russian. However, despite the undeniable relevance and value of Geist's proposal, I have two objections to it, both of which are based on a more careful and extended examination of the empirical data. The first one is that indefinite BNs do actually appear as aboutness topics in sentence initial position, albeit (possibly) not very frequently.[9] The second objection is based on examining the behaviour of bare singular nominals in object position and showing that they do, in fact, get a specific reading in typical ambiguity contexts.

As for the first point, I provide an example below that is very similar to Geist's example (13) discussed above. My example is taken from from the ruTenTen corpus, available through Sketch Engine (Kilgarriff et al. 2014).[10]

(17)	Odnaždy	v	maje	mesjace,	v	pasxal'nye	dni	my	priexali	v	detskij
	once	in	may	month	in	Easter	days	we	arrived	in	children

[9] Frequency is an intuitive guess here, no specific frequency study has been done for this work. See, however, Šimík & Burianová (2020) for a corpus study in Czech, which reveals that indefinite BNs, although existent, are very infrequent in sentence initial position (which is not necessarily a topic position). I think it would be safe to assume that Russian and Czech would not show significant differences in this respect.

[10] I am grateful to Daria Seres for her collaboration in obtaining these data. The Russian ruTenTen corpus is available here: https://www.sketchengine.eu/rutenten-russian-corpus/.

prijut. ... Ja sprosil detej: "Čem vas obradoval segodnja
orphanage. I asked children what you make.happy today
Gospod'?" *Odin mal'čik* skazal: "Ja utrom prosnulsja i
Lord one boy said I in.morning woke.up and
poblagodaril Gospoda za to, čto on prigotovil nam etot den'." On
thanked Lord for that what he prepared us this day he
obradovalsja, čto opjat' uvidel svet. *Devočka* vspomlina: "Ja uvidela
got.excited that again saw light girl remembered I saw
segodnja aista." *Drugaja devočka* rasskazala: "A ja segodnja
today stork other girl told and I today
babušku uvidela vo sne."
grandmother saw in dream
'Once we went to an orphanage in may around Easter. ... I asked the children: "How did the Lord make you happy today?" One boy said: "I woke up in the morning and gave thanks to the Lord for the fact that he had given this day to us." He was excited to see the light again. A girl remembered: "I saw a stork today." Another girl said: "And I saw my grandmother in my dream." '

This is a rather long example but I would like to draw the reader's attention to the BNs in italics. The first one, *odin mal'čik* 'one boy', appears with an indefinite marker *one*, just as predicted by Geist's account. However, the first occurrence of *devočka* 'girl' can only have an indefinite reading, witnessed by the fact that in the next line *drugaja devočka* 'another girl' is successfully used, hence it is obvious that there is more than one girl in the relevant group of children. This is, hence, a clear instance of an indefinite BN that occurs in a sentence initial position as aboutness topic. The example is not exceptional, as a corpus search yields quite a few sentences with an indefinite bare singular nominal in sentence initial position, like, for instance, (18):

(18) *Služaščij* prišel na rabotu na pjat minut ran'še. Ego posadili za
clerk came on work on five minutes earlier him imprisoned for
špionaž. *Drugoj služaščij* prišel na pjat minut pozže. Ego posadili
espionage other clerk came on five minutes later him imprisoned
za sabotaž.
for sabotage
'A/One clerk came to work five minutes earlier. He was imprisoned for espionage. Another clerk came to work five minutes later. He was imprisoned for sabotage.'

Just like in the case of (17), it should be obvious that the only available interpretation for the BN *služasčij* 'clerk' in (18) is an indefinite one, which contradicts the conclusion that indefinite bare nominals are ruled out from sentence initial position.[11] Our corpus search provides a few examples where a bare, non-modified singular nominal appears in sentence initial position and can only have an indefinite interpretation. This empirical fact, together with the scope argument developed below, suggests that we should reassess the generalisation put forth in Geist (2010) and look for an alternative theoretical explanation.

One question that invites itself with respect to the Russian data discussed above is how can examples like (13) and (17) or (18) coexist in the same language.[12] Even though I will not provide an elaborate and/or detailed answer to this question, I would like to offer some tentative explanation which can hopefully be properly worked out (or properly ruled out) in the future. The key observation, I believe, lies in the fact that although Russian lacks articles, it does use specificity markers for indefinites, as was illustrated in (14), repeated below:

(19) {Odna / kakaja-to} devočka vošla v dom.
 one some girl came into house
 'A / Some girl entered the house.'

Elements like *odna* 'one' or *kakaja-to* 'some' are used to explicitly mark specificity of the referent of a nominal expression. Although, unlike articles in English, they are never obligatory, native speakers might have a strong preference to use them, especially in those contexts where only strongly referential nominals are allowed, like in sentence initial topic position, or in any other context where it might be crucial to disambiguate a BN. Specificity markers are the only elements in the nominal domain, apart from demonstratives, which will be discussed in Section 5.1 below, that explicitly set the referential status of a nominal, so it is natural that speakers prefer to use them in the contexts where they are most useful. The contrast between a bare indefinite, which is ambiguous between a definite and an indefinite interpretation, as opposed to an indefinite with a specificity marker seems to be best described in terms of (strong) preference rather than a

[11] Ljudmila Geist (p.c.) notes that (18) is not necessarily an example with an aboutness topic. I think it is, although it might be less obvious than in the case of (17). In any case, we both agree that a more extensive corpus study is needed where topichood is controlled for.

[12] In my own native judgment, (13) is not really ungrammatical but still rather odd. Nevertheless, the difference in acceptability of (17) and (13) needs to be explained, although if the contrast is weaker, then the explanation does not necessarily have to be stated as a strict rule, but can rather be formulated as a tendency or a (strong) preference.

difference in grammaticality, so a formal account of this phenomenon might call for mechanisms similar to the ones adopted in optimality theory.

Let me now move to the second objection, which concerns the availability of a specific interpretation for indefinite BNs in Russian. I remind the reader that the claim defended in Geist (2010) is that BN objects are necessarily weak, i.e., always have a narrow scope. Geist formulates the following descriptive generalisation about the available interpretations for BN objects in Russian:

(20) *Scope behaviour of bare indefinite NPs in Russian:*
Bare indefinite NPs have narrow scope and cannot scope out of syntactic islands. (Geist 2010: 212, ex. (61))

I will narrow down the discussion below to bare singular (BS) objects and argue that they can, in fact, have both a narrow and a wide scope reading in Russian and hence behave as proper full fledged indefinites. In other words, they exhibit the same properties as the English indefinite nominals with the indefinite article *a* (*a*-indefinites). But let us first have a look at the example that Geist provides to support the claim in (20) (see Geist 2010: 211–212):

(21) Džon xočet ženit'sja na francuženke.
John wants marry on French.woman.PREP
'John wants to marry a French woman.'

According to Geist (2010), the prepositional object *francuženke* 'French woman' in (21) cannot scope out of the opaque context created by the intensional verb *xočet* (wants), so the sentence would be incompatible with a continuation like 'He met her last summer', which would require a specific/referential interpretation of the indefinite object. On the other hand, a continuation like 'But he doesn't know any French women yet' is fine, as in this case the indefinite object has a weak quantificational reading. A specific (or referential) interpretation of bare singluar objects can only be triggered by a specificity marker, as in (22). In that case, as illustrated by the example, the continuation 'He met her last summer' is fully acceptable.

(22) Džon xočet ženit'sja na odnoj francuženke. On poznakomilsja s
John wants marry on one French-woman he met with
nej prošlym letom.
her last summer
'John wants to marry a French woman. He met her last summer.'

A closer look at the data, however, presents some complications for the empirical generalisation of Geist (2010). Consider the following example:

(23) a. Vasja ne polučil grant (xotja zajavlenije bylo sostavleno očen' xorošo).
Vasja not received grant although application was composed very well
'Vasja did not get a grant, although the application was written very well.'

b. Vasja ne polučil grant (ėto den'gi iz drugogo proekta).
Vasja not received grant this money from another project
'Vasja did not get a grant, the money comes from a (different) project.'

The examples in (23) illustrate scope ambiguity of a BS object not in an opaque context, but with respect to another scopal element, namely, negation. The first context favours a specific interpretation, as we are clearly talking about one specific grant that Vasja has applied for. In this case the existential operator associated with the indefinite object scopes out of negation (i.e., ∃ > NEG). In the second case, a grant is understood as a possible source of funding and hence does not have to be specific, which means that it takes a narrow scope with respect to negation (i.e., NEG > ∃). Thus, a BS object *grant* 'grant' can be interpreted either specifically, as in (23a), or as a weak non-specific indefinite, as in (23b), and the BS object in (23) exhibits exactly the same pattern with respect to scope as a canonical *a*-indefinite in English.

An objection that could be raised with respect to (23a) is that the object nominal in this sentence is definite. But the truth is, it does not have to be definite. Although *grant* in (23a) can be interpreted as something contextually identifiable, familiar and possibly unique in the given context, it can also be successfully used in a context that induces an indefinite reading, for instance (24) below:

(24) It is the first time that Vasja applied for funding and only his close friend among his colleagues knew about that. Seeing that Vasja looks gloomy one day, one colleague asks that friend: 'What's the matter with him?' The friend replies: 'Vasja did not get a grant… although the application was written very well.'

This context clearly facilitates an indefinite interpretation of *grant* in (23a), it is a classical case of a new discourse referent, which would be rendered by an indefinite article in English (using a definite *the grant* would probably lead to a presupposition failure in this context). The context promotes a specific reading, as this is a grant that Vasja has applied for, not just any grant.

To sum up, we have seen that bare singular indefinites in object position in Russian behave like full fledged indefinites. In particular, they can scope out of opaque contexts and show scopal ambiguity with respect to other scope taking operators like negation, the type of behaviour typical for *a*-indefinites in English. This argument underlies an important objection to Geist's (2010) proposal based on the claim that indefinite BNs in Russian cannot get a specific reading. The argument will also be of crucial importance later in the discussion of a definiteness hypothesis for the interpretation of BNs in Russian (see Section 4.1). For now, we conclude that at least indefinite BSs are similar to the English *a*-indefinites, as they show the same scopal ambiguities. Indefinite BNs can also appear in sentence initial position as witnessed by the examples from corpus discussed above, although not all examples with BNs as aboutness topics are equally acceptable (see also next section), the fact that still needs an explanation. However, the restriction on BNs in topic position is not absolute and can possibly be explained by the fact that BNs compete with and lose to those indefinites which are overtly marked for specificity, which makes their referential status explicit and hence allow them to appear in topic position more freely.

As was pointed out at the beginning of this section, there is a common belief that the interpretation of BNs in Russian depends on their position in a sentence. We have by now reviewed the claim that BNs that appear in sentence initial position as aboutness topics can only be interpreted as definite. We have seen that there is a strong tendency to associate topicality with definiteness in Russian, just like in the majority of other languages, with or without articles, but this tendency cannot be formulated in strong definitional terms: it is not the case that topicality is a sufficient condition for definiteness. In the next section, we will review another proposal, which associates (in)definiteness with pre- or postverbal position in a sentence, independently of topicality.

2.2.2 BNs in preverbal position

Studies of the effects of word order on the interpretation of nominal phrases in Russian and other Slavic languages usually consider two factors. One is the influence of the information structure notions like topic on the (in)definiteness of BNs, something that we discussed in the previous section. Another factor is the linear position of a nominal constituent with respect to the verb and a possible influence of this position on the range of available readings for a BN in question. The predictions are mostly made for preverbal nominals, as postverbal position, as documented in the literature, imposes very few restrictions on the interpretation of BNs (see Brun 2001, Šimík & Burianová 2020 for further references). As for preverbal position, Czardybon et al. (2014) recently claimed, based on a corpus experiment conducted for Polish data, that preverbal position increases the probabily of definiteness. However, as pointed out by Šimík & Burianová (2020), the study did not differentiate between sentence initial and preverbal position, so the results that they report can just as well be explained by the likelihood of definiteness in sentence initial position, as discussed in the previous section. Šimík & Burianová (2020) themselves insist that a dependency can be established between sentence initial (absolute, in their terms) position and definiteness, but pre- or postverbal position does not have a significant influence on the interpretation of BNs. However, Seres et al. (2019) and Borik et al. (2020) report the results of an experiment conducted for Russian, where the link between definiteness and preverbal position reemerges as a possible factor for definitness, so we will review this experiment in more detail.

In the experimental study reported in Borik et al. (2020), Seres et al. (2019), 120 native speakers of Russian were asked to judge the acceptability of sentences containing pre- and postverbal bare nominals in two types of contexts, definiteness- and indefiniteness-suggesting contexts. To induce a definite reading, anaphoric or bridging contexts have been used. The empirical coverage was limited to subjects of stage-level intransitive verbs. The main conclusion drawn from the experiment is that linear word order in Russian cannot be considered the means to either encode or express definiteness and indefiniteness of BNs. The study, however, confirms a clear correlation between linear position and interpretation of bare nominals, in the sense that preverbal bare subjects are mostly interpreted definitely and postverbal subjects indefinitely. Another result that has been reported is that some (not all) indefinite preverbal subjects are judged as acceptable by native speakers.

These are some examples of the experimental items that have been used:

(25) *Preverbal subject in an indefiniteness-suggesting context*:
Ėto že pustynja, ėto samaja nastojaščaja pustynja. V ėtoj mestnosti do six por ne bylo ni odnogo živogo suščestva. No na prošloj nedele *pticy* prileteli.
'This is a desert, a real desert. In this area there has never been a living creature. But last week birds came (lit. birds came flying).'

(26) *Postverbal subject in an indefiniteness-suggesting context*:
Čto-to strannoe stalo proisxodit? v našej kvartire. V kuxne vsegda bylo očen' čisto, nikogda ne bylo ni odnogo nasekomogo. No nedelju nazad obnaružilis' *tarakany.*
'Strange things started happening in our flat. It has always been very clean in the kitchen, there has never been a single insect. But a week ago cockroaches were found (lit. found themselves cockroaches).'

(27) *Preverbal subject in a definiteness-suggesting context*:
Kogda Katja i Boris vernulis' iz otpuska, oni obnaružili, čto ix dom ograblen. Pervym delom Katja brosilas' v spal'nju i proverila seif. Ona uspokoilas'. *Dragocennosti* ležali na meste.
'When Katja and Boris came back from holidays, they found out that their house had been burgled. First of all, Katja rushed into the bedroom and checked the safe. She calmed down. The jewelries were still there (lit. jewelries lay in place).'

(28) *Postverbal subject in a definiteness-suggesting context*:
Oživlenije spalo, publika potixon'ku potjanulas' domoj. "Počemu vse uxodjat?" – sprosil Miša. "Gonki zakončilis'. V garaži vernulis' *mašiny*."
'The agitation declined, the public slowly started going home. "Why is everybody leaving?" Miša asked. "The race has finished. The cars have returned to their garages." (lit. to garages returned cars).'

The target nominal is always the one of the last sentence, marked here in italics. As can be seen from these contexts, the target BNs do not have to be topics, so topicality was considered as a factor that might influence the interpretation of a BN. The results of the experiment are presented in the Figure 1.

First of all, we see in the graph a rather strong preference for interpreting preverbal NPs definitely and postverbal NPs indefinitely. This is the main result of the study that seems to confirm the relevance of pre-/postverbal position for the interpretation of a BN. However, there is no clear one-to-one correspondence, which is also to be expected and which suggests that linear position per se cannot be considered a means of expressing (in)definiteness. The correlation between a linear position and interpretation is, in fact, a reasonably strong tendency but it remains a tendency. This result is in agreement with the results of previous experimental work on Slavic languages (cf. Czardybon et al. 2014, Šimík & Burianová 2020) and with descriptive literature on Slavic world order. The results show that preverbal nominal subjects are much more likely to be interpreted

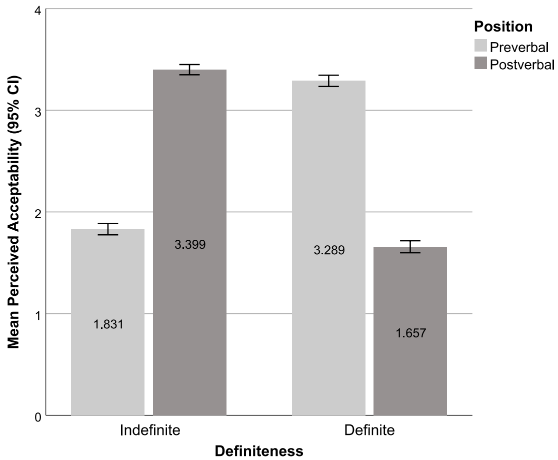

Figure 1: Average perceived acceptability of the experimental sentences

as definites. However, indefinites are not rare in this position either and their acceptability is reasonably high.

One of the important question that this experimental study brings up relates to a long standing debate in the literature of what definiteness actually means in semantic terms (see Section 3 for further discussion). In the experiment presented above, Seres et al. (2019) and Borik et al. (2020) followed Farkas (2002) in assuming that the familiarity and the uniqueness approaches to definiteness could converge. In particular, as has already been pointed out, BNs in definiteness-suggesting contexts are anaphorically (directly or indirectly) linked to a referent in the preceding discourse. This makes all the experimental items fully legitimate candidates for semantically definite nominals because they are familiar to the speaker. However, this also means that all BNs in definiteness-suggesting contexts are *given*, so the question that offers itself is whether the results of the experiment are influenced by the givenness status of the tested items.

Givenness is a category related, although not equivalent to definiteness. An element is given if there is an antecedent for it in a preceding discourse, so givenness is an information-structural category that is also closely related to anaphoricity. Any constituent of a sentence can have a status of given, including, of course, nominal arguments. The relationship between definiteness and givenness is not straightforward: in principle, both definite and indefinite arguments can be either given or new. For instance, any contextually unique definite mentioned for the first time is new (e.g., *The UV is very high today*, or *The head of* *the department just called me*), whereas any anaphoric definite is given. Crucially, however, the given/new status seem to correlate with stress: deaccentuation and word order are common ways to indicate givenness of a certain constituent (cf. Krifka 2008). As for Slavic languages, Šimík & Wierzba (2015) present a thorough study of the interaction between givenness, word order and stress in Czech.

In the Russian experiment discussed here a stress factor was controlled for in the sense that all example sentences were recorded with a neutral intonation, flat pitch, and a phrasal stress at the end of a sentence. Hence, all preverbal items (in definiteness and indefiniteness contexts) were unstressed and postverbal items (in definiteness and indefiniteness contexts) were stressed only when they also appeared in a sentence final position. It might be that givenness is the factor that had a certain influence on the acceptability judgments in the experiment, especially in the case of nominals in postverbal position in definiteness contexts, because the speakers might be less willing to accept a post-verbal stressed given nominal. However, stress considerations should not considerably alter the results of acceptability of BNs in preverbal position, so the conclusion that preverbal non-topic position still increases the likelihood of definiteness still holds.

Needless to say, the interaction of definiteness, anaphoricity, givenness and stress is the area that still needs to be thoroughly and carefully explored and Slavic languages, including Russian, present fruitful ground for further experimental work on the topic. Stress might be a particularly important factor for the availability of an indefinite interpretation in BNs, as it has been noticed in the literature that stressed indefinites become more acceptable in preverbal position. More experimental work in this area should be on the research agenda of Slavicists, and I hope that it will shed light on various puzzles of nominal interpretation in the future.

To sum up, this section was dedicated to the empirical overview of Russian BNs with the focus on word order as a factor that has been claimed to influence the availability/likelihood of (in)definite interpretation of BNs in Russian. I have discussed the interpretation of BNs in topic position and in pre/postverbal position and have hopefully shown that both sentence initial position and preverbal position increase the probability of a definite interpretation of BNs. It is important to emphasise that this tendency is only valid for "real" BNs: once the referential status of a nominal is marked in some way (by a modifier, a specificity marker, etc.), its linear position does not affect its referential status. Importantly, we have also seen that indefinite nominals are not fully excluded from either sentence initial/topic position or from preverbal position, so the reported dependencies between definiteness and the position of a BN should be empirically treated as – possibly strong – tendencies. With this result in mind, we will now move on to an in-depth discussion of one of the interpretation of Russian BNs, namely, a definite one. We will start by a brief theoretical overview of theories of definiteness (Section 3) and the consequences and predictions that one of these theories, namely, a uniqueness theory, makes for a possible semantic analysis of BNs in Russian (Section 4).

3 Definiteness

Definiteness is one of the most debated and most fascinating topics in the semantic literature as it touches upon some very fundamental questions of natural language semantics, such as reference and referential properties of nominal phrases. A typical and perhaps the most familiar way to express definiteness in a languages is by means of articles. English uses a definite (*the*) and an indefinite (*a*) article to mark definiteness and indefiniteness on count nominal phrases. Many languages have asymmetrical article systems, where only a definite (Hebrew or Burgarian) or only an indefinite (Persian or Turkish) article is used. Finally, there are many languages which do not use any articles at all, Russian and the majority of other Slavic languages are among those. Thus, there is a lot of variation in the expression of definiteness and indefiniteness across languages (for typological overviews, see Dryer 2005, 2013, Givón 1978), but from a semantic viewpoint the most relevant question to be asked is whether the grammatical category of definiteness has a uniform semantics across languages, independently of the way definiteness is expressed or marked. The question is far from trivial, despite the fact that the affirmative answer is often taken for granted or at least assumed to be a null hypothesis. A related theoretical question is, of course, what an adequate theoretical representation of definiteness would be and how many definiteness theories we need to adequately represent a wide range of empirical data related to definiteness and indefiniteness. While we are still in the search for convincing answers to both questions, I would say that the recent interest for languages without articles in relation to definiteness can lead to significant advances in the semantics theories of definiteness and related categories. This section will provide a brief overview of the most common types of definite nominal phrases in 3.1, and the semantic theories that have been proposed to account for definiteness, with a special focus on the uniqueness theory of definiteness in 3.2. For a detailed discussion of indefiniteness, the reader is referred to Geist (2026 [this volume]).

3.1 How many types of definiteness?

This section will give a brief overview of semantic theories of definiteness. We will begin with the most familiar uniqueness and a familiarity theories, but will also mention less mainstream approaches, such as von Heusinger (2013) or Löbner (1985). The main goal of this section is to remind the reader about the main points that those approaches try to make, whereas a more detailed discussion can be found in the vast literature on the issue.

To begin with, let us look at some examples of definite nominals in English, where definiteness is expressed by means of articles. Definite NPs have various uses, the most typical of which are the following: situational definites in (29), anaphoric definites in (30), cases of bridging in (31) (Clark 1975), and weak definites in (32):[13]

(29) It's so hot in the room. Could you open the window, please?

(30) I saw a man in the street. The man was tall, slim and elegantly dressed.

(31) I'm reading an interesting book. The author is Russian.

(32) Every morning I listen to the radio.

A uniqueness theory of definiteness that originates in Frege (1892) and Russell (1905) treats a (singular) define description as having a unique referent, i.e. there is only one object that fits the corresponding description. The uniqueness part of the meaning of a definite description is usually assumed to be presuppositional, as was famously argued by Strawson (1950) and more recently by Elbourne (2013), hence the term UNIQUENESS PRESUPPOSITION commonly used in formal semantic literature on definiteness. The uniqueness theory is a well-articulated, clear and precise formal semantic approach that takes situational definites exemplified in (29) to be prototypical definites.

However, this characterisation of definites is clearly too idealistic for natural languages, as can be illustrated even by our example above: in (29) the window referred to by the definite NP does not have to be the only window in the room or in the house. Moreover, it is not even very likely to be a unique window-object in a given situation, but it has to be the one that both the speaker and the hearer are able to uniquely identify. This brings us to the rule of domain restriction, which seems to be an unavoidable amendment to the uniqueness

[13] For much more detailed empirical descriptions see Hawkins (1978) or Lyons (1999).

theory: a uniqueness condition has to be restricted by the relevant situation in which a definite description has to have a uniquely identifiable referent.

Another approach takes the anaphoric uses of definites as in (30) as default and is based on different discourse properties of definite and indefinite descriptions. This is a familiarity approach, developed by Kamp (1981) and Heim (1982). The main idea that lies behind this approach is that each nominal phrase represents a discourse referent, but in the case of indefinite nominals the correspondent discourse referent is new, whereas the referent of a definite description has to be identified with the one previously introduced. The theory is simple and elegant, which makes it intuitively very appealing. Moreover, it makes the semantic description of definite and indefinite nominal phrases fit nicely into a general discourse representation theory and connects it to anaphora resolution mechanisms (Kamp & Reyle 1993). However, situational uses of definites that constitute one of the core cases of definites, can only be accounted for with additional assumptions.

Needless to say, there have been attempts to unify both theories to capture as many uses of definite nominals as possible. Farkas (2002) introduces the notion of determined reference to "capture what is common to anaphoric and unique reference" (Farkas 2002: 221). Determined reference simply means that the value assigned to the variable introduced by a definite NP is fixed: there is no choice of entities that can satisfy the descriptive content of a definite nominal. Definite descriptions, according to Farkas (2002), always have a determined reference, although it can be achieved in different ways: a definite nominal can have a unique referent in a context or it can refer anaphorically.

Another approach that unifies situational and anaphoric definites is Löbner (1985), although Löbner's theory of definiteness is not based on uniqueness, but on a broader notion of "non-ambiguity" of reference. Löbner (1985) distinguishes between semantic and pragmatic definiteness, the unambiguous reference for the latter type being fully dependent on a particular situation. This group hence includes both situational and anaphoric definites, which renders the property of uniqueness as a special case of unambiguous reference. Semantic definites for Löbner are relational nominals like *the son of a famous actor* or *the author* from (31) above. Those definites have unambiguous reference "due to general constraints" (see Löbner 1985: 229).

The uniqueness property and the uniqueness theory of definiteness is also questioned in von Heusinger (1997, 2002, 2013), who suggests that the concept of SALIENCE should be central to definiteness. In von Heusinger's proposal, nominal referents are ordered along the salience hierarchy, and the context crucially contributes to this ordering. The NP with the most salient referent in a given situation is marked as definite. The hierarchy is constantly changing and different referents can become salient with a slight change of a discourse situation or a speaker's intention. Salience is a fairly intuitive notion and can relatively easily capture all the main uses of definites illustrated above in (29) to (32).[14] The concept of salience, however, is very difficult to formalise, and the attempt made in von Heusinger (2013) to use choice functions as a formal tool to represent definiteness, leaves it unclear how to differentiate between definites and specific indefinitenes, which have traditionally been given a choice function analysis (see Reinhart 1997, Winter 1997). Finally, in the case of weak definites in (32), there is no requirement for the definite DP to have a single referent. One of the proposals that can be found in the literature, namely, Aguilar-Guevara & Zwarts (2014), treats weak definites as kind nominals.

This relatively brief overview of the main approaches to definiteness forms a basis for a more detailed discussion of a definiteness theory based on the property of uniqueness in the next section. As has already been pointed out, the uniqueness theory is perhaps the best understood one among formal theories and, despite the fact that there are some very well known exceptions to the uniqueness theory, it is still the one that most formal semanticists adopt while dealing with definiteness. The discussion of the main approaches to BN semantics in Russian, which will be presented in Section 4 will also rest on the main assumptions and postulates of the uniqueness theory, which justifies a more detailed overview of this theory provided in the next subsection.

3.2 Definiteness as uniqueness

Let me begin this section by reminding the reader about the intuition that underlies the uniqueness approach to definiteness. Consider first an indefinite NP in (33a) and a definite one in (33b):

(33) a. I've just heard a mouse squeak.
b. I've just heard the mouse squeak.

The common intuition is that (33b) can only be uttered successfully if the nominal phrase *the mouse* has a contextually unique and identifiable referent. In contrast, (33a) is felicitous if there is at least one referent for the indefinite nominal *a mouse*. The intuition appears to be very robust among all the native speakers of English and many other languages with definite articles. This intuition is fundamental for the theory of uniqueness.

[14] Perhaps with the exception of weak definites, which form a very peculiar class of definites.

In Russell's (1905) theory of definite descriptions, the uniqueness of the referent of a definite description is part of the assertion. This means that the famous example of Russell's in (34) comes out as false under the Russellian account.

(34) The king of France is bald.

Strawson (1950) argued that Russell was attributing wrong truth conditions to (34). In particular, following the ideas of Frege (1892), Strawson suggested that the uniqueness condition should be treated as part of the presupposition associated with the definite description and this is the most accepted view nowadays. This means that a definite description presupposes existence and uniqueness of the referent and when this presupposition fails, the sentence cannot be judged as either true or false. Thus, (34) is interpreted as neither true nor false, as there is no such entity that could (in our world and relative to the present) satisfy the description of being the king of France, but the uniqueness of the king of France is still presupposed in this example.

Formally, semantic definiteness in argument position is standardly associated with the semantic contribution of the definite article itself, represented by the ι (iota) operator.[15] The iota operator shifts the denotation of a common noun from type $\langle e, t\rangle$ to type e, i.e., from a predicate type to an argument type (see Heim 2011: 998), and thus, denotes a function from predicates to individuals (Frege 1879, Löbner 1985, Elbourne 2005, 2013, Heim 2011). Predicative uses of definites can either be derived from argumental ones (Partee 1987, Winter 2001) or taken as basic (Graff Fara 2001, Coppock & Beaver 2015).

The meaning of the definite article can be represented as in (35):

(35) $[\![\text{the}]\!] = \lambda P : \exists x \forall y [P(y) \leftrightarrow x = y].\iota x.P(x)$,
where ιx abbreviates 'the unique x such that'. (Heim 2011: 998)

Plural definite NPs naturally violate the presupposition of uniqueness. In this case uniqueness is reformulated as maximality (Sharvy 1980; Link 1983), i.e. reference to a maximal individual in the domain, which is picked out by the definite article.

The uniqueness approach to definiteness is given a different twist in Coppock & Beaver's (2015) proposal. Crucially assuming the predicative uses of definite descriptions to be the basic ones, they differentiate between definiteness and determinacy, the two notions that are formally dissociated in their model. A nominal phrase is determinate if it has an individual denotation and semantically,

[15]With the exception of Coppock & Beaver (2015), which will be discussed right below.

this is the contribution of the iota operator. Definiteness in their proposal is contributed by the definite article and semantically only conveys a uniqueness presupposition. Contrary to more common proposals, in Coppock & Beaver's (2015) analysis definiteness does not signal or require existence. This property of definites can be witnessed, as Coppock & Beaver (2015) argue, by predicative uses of definites where it becomes clear that existence is not required by the definite article *per se*. Argumental definites do require existence, but this is an effect of the type shifting operator (the iota operator in this case).

Without going into the intricacies of Coppock & Beaver's (2015) non-trivial argumentation about existence presuppositions, the most important and the most relevant contribution of their proposal for our purposes can be summarised as follows. First, this is a proposal that clearly and explicitly differentiates between the semantic contribution of the definite article and the semantic contribution of the iota operator. Second, in argument position the overall semantic effect of definiteness marking will be the same as in more conventional proposals (cf. Partee 1987, Winter 2001, Elbourne 2005, Heim 2011), as the contributions of the iota operator and the definite article coincide in this case.

The above-mentioned concepts related to definiteness (i.e. uniqueness, existence, maximality) have all been postulated in relation to languages with articles and therefore, in one way or another, are associated with the presence of the definite article on the nominal. The relevant question that arises when one faces a language without articles is whether the expressions perceived as definite in such a language would give rise to the same effects as the definite nominals marked with overt articles.

From a theoretical perspective, there are several possible answers to this question. The first one is based on attributing definiteness effects such as uniqueness or maximality to the presence of the article itself. In this case, the uniqueness of a definite description, be it asserted or presupposed, will follow directly from the semantics of the definite article, as in classical uniqueness/type-shifting theories (e.g., Frege 1892, Russell 1905, Partee 1987). We expect that languages without articles do not show the same type of definiteness effects as languages with overt articles, simply because the former do not have any lexical element that would make the same semantic contribution as a definite article.

Another option is to follow Chierchia's (1998) and Dayal's (2004) influential proposals that claim that articleless languages use the same inventory of type shifting operators with the only difference that these operators are not lexicalized.[16] Crucially, the semantic contribution of such operators remains the same,

[16]See Section 4.1 for a more detailed discussion of Dayal (2004).

whether or not an operator is lexicalized in a given language. Thus, if the iota operator is responsible for deriving a definite interpretation of nominal arguments in both English and Russian, the predictions are clear: the uniqueness effects associated with definite descriptions in English should also exist in Russian. However, the empirical facts that will be discussed below, in particular in Section 4.3, seem to indicate that the perceived definiteness in Russian does not give rise to the same semantic effects as in English, which, in principle, argues against the Chierchia/Dayal type of analysis.

As a side note, I would also like to point out that I do not associate the iota operator with any particular syntactic projection or any particular syntactic head. Thus, the question about a possible syntactic structure of referential bare nominals in Russian and, in particular, the presence or absence of the D-layer, is not straightforwardly connected to whether or not a language employs a certain semantic operator to derive its arguments. The iota operator, should it exist, does not straightforwardly entail any syntactic projection, because the function of this operator (namely, to derive expressions of type *e*) is semantically defined and whether or not all expressions of this type should have the same syntactic structure across languages or within a language is an independent question.

4 Possible semantic analyses of BNs in Russian

In this section, I will discuss possible theoretical accounts of the semantics of BNs in Russian. Adopting a uniqueness theory of definiteness as a point of departure, I will argue that, unlike in English or other languages with articles, there is no uniqueness/maximality presupposition carried by Russian BNs that are perceived as definites. Among different implementations of the uniqueness approach to definiteness we will discuss the following proposals. Section 4.1 will be devoted to the influential proposal by Dayal (2004), according to which BNs in languages without articles can get either a definite or a kind interpretation. I will refer to this claim as the DEFINITENESS HYPOTHESIS. In Section 4.2, we proceed to check the predictions and implications of an ambiguity approach based on Chierchia (1998), which claims that BNs are semantically ambiguous between a definite and an indefinite reading, and contrast it with the indefiniteness hypothesis of Heim (2011), which claims that BNs in languages like Russian are semantically indefinite. The empirical argument against uniqueness that I develop in Section 4.3 will, as I argue, favour Heim's (2011) theoretical approach to BNs and the section will conclude with the general discussion of possible implications that the Russian data can have for theoretical accounts of definiteness.

4.1 A definiteness hypothesis

Dayal (2004) presents an influential proposal that determines the universal principles of type shifting in languages with and without articles, and proposes a hierarchy of type shifting operators. Drawing on the selection of empirical data from Hindi, Russian and Mandarin Chinese, she claims that BNs in languages without articles can have either a definite or a kind interpretation, but cannot be *bona fide* indefinites. She builds her theory mostly on Chierchia's (1998) well-known proposal for kind reference, assuming, among other things, that English bare plurals directly refer to kinds and get an indefinite interpretation only in those contexts where it can be derived from their kind interpretation via Derived Kind Predication (DKP) rule (cf. Chierchia 1998: 364).

Applying these principles to languages without articles, Dayal argues that an indefinite interpretation of bare plurals in these languages is also derived from the underlying kind interpretation, whereas bare singluars simply cannot be *bona fide* indefinites, as the DKP rule is explicitly defined only for plural properties.[17] Thus, Dayal concludes that all bare nominals in languages without articles get either a kind or a definite reading derived by applying corresponding semantic type shifting operators, i.e., a $^\cap$ operator that yields a kind interpretation when applied to plural properties, and an ι operator that yields a definite interpretation when applied to singular or plural properties. This motivates the proposed ranking of semantic type shifters where an existential quantifier associated with an indefinite interpretation ranks lower than an iota and a down operator.

In order to support her claims, Dayal (2004) first shows that there is a crucial difference in the interpretation of bare singular vs. plural subjects in languages like Hindi or Russian, illustrated in the following examples:

(36) a. # caroN taraf cuuha hai
four ways mouse is
'The mouse/A particular mouse (the same one) is everywhere.'

b. caroN taraf cuuhe haiN
four ways mice are
'There are mice (different ones) everywhere.'

(Hindi; Dayal 2004: 395)

[17] In Chierchia's (1998) proposal, this restriction reflects the fact that in English, bare singulars virtually do not exist (although see Stvan 1998, de Swart 2015: for exceptions) and the kind interpretation of definite singulars is subject to a different complex semantic mechanism, whereas bare plurals directly refer to kinds. This last claim is challenged in Borik & Espinal (2015).

(37) a. # Sobaka byla vesde
dog was everywhere
'The dog/A particular dog (the same one) was everywhere.'

b. Sobaki byli vesde
dogs were everywhere
'There were dogs (different ones) everywhere.'
(Russian; Dayal 2004: 395)

The contrast illustrated above is attributed to the fact that singular subjects in (36a) and (37a) can only get a definite reading, yielding therefore an implausible interpretation of 'the (same) mouse/dog was everywhere'. In the case of plural subjects, this interpretation does not arise. Note, however, that the examples above are based on Carlson (1977), who used the corresponding English examples to illustrate one of the several differences that can be observed between singular and plural indefinites in English. Carlson's (1977) original examples are given in (38):

(38) a. A dog is everywhere.

b. Dogs are everywhere. (Carlson 1977: 13, ex. (26)–(27))

In Carlson's judgment, (38a) "has only an absurd reading, one which means that there is some omnipresent dog" (ibid.). This means that the infelicity of the corresponding Russian example in (37a) does not provide any evidence for the presence of definiteness in Russian: a singular indefinite in English creates exactly the same effect, as illustrated by (38a). Hence, at least for Russian, the contrast between (37a) vs. (37b) does not constitute any evidence in favour of a definite interpretation of the subject in (37a). Neither does it rule out an indefinite interpretation of the subject *sobaka* 'dog' in (37a). All the above examples, including the English (38a) and (38b), only illustrate the contrast between singular and plural indefinite subjects in distributive sentences.

Dayal (2004) does not discuss the interpretation of bare singulars in object position, as empirical facts for Hindi are complicated by the fact that object bare nominals are/can be pseudo-incorporated (cf. Dayal 2011, 2015). In Russian, however, pseudo-incorporated objects, if they exist at all (see Mueller-Reichau 2015, Seres 2020 for some discussion), are not really the most common type of objects, so object BNs should not *a priori* be excluded from consideration. Another issue with limiting the discussion to subject position is that BNs in preverbal subject position can also be topics, at least in Russian (see Section 2.2.1), and it is well known that topics tend to be interpreted as definite. This tendency can seriously blur the overall empirical picture, as a definite (or, rather, a definite-like) interpretation of preverbal bare nominals might arise due to informational structure considerations and not necessarily due to their semantic definiteness. Given this, it becomes almost necessary to examine the behaviour of object nominals in a language that does not require bare objects to be pseudo-incorporated.

As Geist (2010) shows, bare singular objects *can*, in principle, be indefinite, even if their interpretation, as argued, is limited to weak indefiniteness. This fact is not predicted or expected in Dayal's (2004) approach, due to the fact that an indefinite interpretation in languages without articles, according to Dayal, can only be derived from a kind interpretation. A semantic mechanism for this "derived" indefinite interpretation does exist for bare plurals (the DKP rule discussed above), but crucially not for bare singulars, at least not without any serious modification of the DKP rule. The empirical data discussed by Geist (2010), thus, constitute a first piece of evidence against the definiteness hypothesis.

Moreover, as I have argued above, the objection can be made much stronger. The reader should recall the scope argument from Section 2.2.1, which shows that singular BNs in object position, just like *a*-indefinites in English, display scope ambiguities with respect to other scopal elements and in opaque contexts. Let me repeat the crucial example that I used in Section 2.2.1 as an objection to Geist's (2010) hypothesis that indefinite singular BNs cannot have a specific interpretation. The example is repeated in (39):

(39) Vasja ne polučil grant (xotja zajavlenije bylo sostavleno očen'
Vasja not received grant although application was composed very
xorošo).
well
'Vasja did not get a grant, although the application was written very well.'

In this example, the context favours a specific interpretation of a bare singular object *grant* 'grant', as we are clearly talking about one specific grant that Vasja has applied for. Hence, the existential operator associated with the indefinite object scopes out of negation, yielding the interpretation of 'there is a grant that Vasja did not get', with a wide scope of the indefinite object. The conclusion to be drawn is that singular BNs in object position in Russian behave like proper indefinites with different scope possibilities. This, in turn, means that the definiteness hypothesis as it is set out in Dayal (2004) cannot be maintained as the empirically adequate one for Russian BNs.

Even though the scope properties of bare singular indefinites constitute a crucial argument in favour of their indefinite status, there are some other types of examples that would be difficult to explain should the definiteness hypothesis be adopted. For instance, BNs are perfectly acceptable in distributive contexts, as illustrated in (40):

(40) V každom dome igral rebenok.
in every house played child.NOM
'A child (a different one) was playing in every house.'

In sentence (40), a nominal phrase cannot be interpreted as either a kind, or a definite. As for a kind interpretation, morphologically singular kinds generally cannot facilitate access to individuals (cf. Krifka et al. 1995) and hence cannot distribute, so a kind interpretation is ruled out in this example. A definite interpretation, on the other hand, would lead to an implausible reading that there is a unique child, such that it is playing in every house. This is, however, not the interpretation that this sentence naturally obtains. Hence, *rebenok* (child), has to be indefinite. Dayal's crucial examples for the absence of an indefinite interpretation with bare singulars in Russian are also based on distributive contexts, but, as (40) illustrates, distributive sentences do not rule out singular indefinites on a regular basis.

Finally, let us look at yet another canonical indefinite environment, i.e., existential there sentences. The Russian counterpart of *there*-sentences in English are formed by fronting a locative phrase, as in (41):

(41) V komnate ležal kover.
in room lied carpet
'There was a carpet in the room.'

In this example, singular BN *kover* 'carpet' can only be interpreted as an indefinite, given the context.

Another piece of evidence that shows that BNs in Russian can be interpreted indefinitely is based on the examples where the same BN is used twice with different referents:

(42) Učenyj vsegda pojmet učenogo, daže esli oni
scientist always understand.PFV.PRES scientist even though they
rabotajut v soveršenno raznyx oblastjax nauki.
work in absolutely different areas science.GEN
'A scientist will always understand (another) scientist, even if they work in completely different areas.'

In (42), two occurrences of *učenyj/učenogo* 'scientist' differ in case, but in terms of interpretation, they do not lead to the oddness that would be expected with two definite NPs with the same descriptive content, as below:[18]

(43) # The scientist will always understand the scientist.

The most reasonable conclusion to be drawn about the interpretation of BNs *učenyj* and *učenogo* in (42) is that both BNs are interpreted as (non-specific) indefinites.

Finally, yet another argument against the definiteness hypothesis rests on the following difference between English and Russian:

(44) a. I bought a new car. #The wheel broke down immediately.
(cf. The motor broke down.)

b. Ja včera kupil novuju mašinu. V pervyj že den' spustilo koleso.
I yesterday bought new car in first PTCL day deflated wheel
'I bought a new car yesterday. I got a flat tire on the very first day.'

The contrast between (44a) and (44b) above shows that in those contexts where it would be odd to use a definite NP in English, a BN in Russian is nevertheless perfectly acceptable. Even thought BN *koleso* 'wheel' in (44b) is in the final (postverbal) position, the word order considerations do not affect Dayal's predictions. This example illustrates the absense of full parallelism between the English definite and the Russian BN in bridging contexts and this contrast would be difficult to explain in Dayal's theory.

Thus, we have seen plenty of evidence that singular BNs in Russian freely appear in typical indefinite environments and, moreover, exhibit scopal ambiguities just as bona fide indefinites in English. We can therefore conclude that BNs do function as proper full-fledged indefinites and the definiteness hypothesis advocated by Dayal (2004) will have considerable obstacles in handling the interpretation of BNs in Russian.[19]

[18]Dayal (2004) notes that examples like (i) are unacceptable, and I agree with her judgement, but the status of (i) is probably due to the individual denotation (Löbner 1985), rather than to definiteness.

(i) # V ėtoj kletke tigr spit i tigr est. (Dayal 2004: 411)
in this cage tiger sleeps and tiger eats

[19]Similar conclusions have been reached for Russian in Borik et al. (2025), Liu et al. (2023) and Liu et al. (2024), studies based on parallel translation corpora that include empirical data from Russian, Hindi and Mandarin, among other languages.

4.2 An ambiguity vs. an indefiniteness hypothesis

Since the definiteness hypothesis for Russian bare nominals cannot be maintained, we are left to decide between the two remaining hypotheses: the one based on ambiguity and the one that takes bare nominals to be indefinite. In a sense, the ambiguity approach is the default one for languages without articles, also in descriptive grammars, that often state that the interpretation of a bare nominal can be either definite or indefinite depending on the context (Geist 2010). Formally, the ambiguity hypothesis explicitly or implicitly follows from some type shifting approaches, for instance, Chierchia (1998). Since formal theories have the undeniable advantage of making clear predictions, in what follows I will base my considerations on Chierchia's (1998) approach.

In Chierchia's (1998) theory of nominal reference, Russian and English belong to the same typological group of languages as far as the semantics of nominals is concerned: both languages have a mass/count distinction and their nominals can be of two basic types, argumental [+arg] or predicative [+pred]. Both languages apply type shifting operators like $\exists$, ι and $^\cap$ to derive a required nominal type in a given context. However, the first two operators are lexicalized in English (as *a* and *the*, respectively), hence their application is restricted in such a way that they cannot apply covertly (Blocking principle; Chierchia 1998: 360). The $^\cap$ operator, however, is not overtly realised in English and hence can be applied covertly, but only to pluralities. The latter restriction is inherent to the $^\cap$ operator and holds for all languages.

In Russian, there is the same inventory of available operators, but, unlike in English, none of the type shifting operators are lexicalised. This means that all the operators apply freely, with the only restriction that the $^\cap$ operator only applies to pluralities. Hence, bare nouns are semantically ambiguous: bare singulars can have a definite or an indefinite interpretation and bare plurals are three-way ambiguous between a kind, a definite and an indefinite reading.

While Chierchia (1998) mentions Russian only very briefly, appealing to the necessity of a scrutinous study of possible interpretations of BN in languages without articles, one explicit prediction made by this approach is that Russian bare count nominals with an indefinite interpretation should have the same scopal properties as indefinite singulars in English (Chierchia 1998: 362). This prediction is fully supported by the data, as was argued in 2.2.1. Apart from that, we can infer that in Chierchia's model, both $\exists$ and ι operators are available for BNs in argument position.[20] Since the semantic contribution of the type shifters is presumably universal, we expect definite nominals in English and BNs with a

[20]We will abstract away from the $^\cap$ operator for the current purposes.

definite interpretation in Russian to show the same semantic effects, in particular, to give rise to presupposition of uniqueness associated with the iota operator.

Another proposal relevant for this section is an indefiniteness hypothesis of Heim (2011), which rests on the idea that a definite interpretation in general is the result of a pragmatic strengthening mechanism involved in establishing the referential status of a nominal. The idea is based on the observation that in (45) below, the sentence in (a) entails the one in (b), but not the other way around.

(45) a. The mat belongs to the cat upstairs.
b. The mat belongs to a cat upstairs.

The interpretation of both definite and indefinite nominals is construed with a covert restrictor, whose function is to limit the number of elements that satisfy the descriptive content of the nominal to those that are considered relevant in a given situation. The narrower restrictor is the only one compatible with a definite interpretation. Indefinites, on the other hand, are compatible with any type of restrictor, a narrower, a broader or no restrictor at all. In the case of a narrower restrictor, however, the definite expression has to be used because it is a pragmatically stronger alternative, which is only compatible with this type of restrictor. Therefore, if the speaker uses (45b), the hearer concludes that it is the only statement the speaker can commit to under the circumstances (given that we follow the Grice's maxim of quantity) and hence infers that the stronger statement in (45a) is false or its presuppositions are not satisfied.[21]

The above reasoning applies to languages that have both a definite and an indefinite article, where only the former is in fact used to convey a narrower domain restriction. In languages without articles like Russian, the above mechanism is simplified significantly since one and the same type of nominal phrase, namely, a bare nominal, is compatible with all possible domain restrictions. Semantically, a BN is indefinite and since there is no competing expression for the narrower domain restrictor, indefinite BNs will also be compatible with a (contextually triggered) narrower domain restrictor, resulting in a definite interpretation. Thus, if the contextual factors favour a narrower domain restriction, the nominal phrase will be interpreted as definite, but crucially, this interpretation will be implied rather then derived by hard-core semantic mechanisms like type shifting.

[21]Coppock & Beaver (2015) also suggest a pragmatic mechanism to account for the definite/indefinite contrast in predicative nominals. Their proposal is based on Maximize Presupposition (MP!), a pragmatic principle introduced by Heim (1991) which urges speakers to presuppose as much as possible.

The two approaches discussed here have not really been carefully explored and thoroughly checked against the data of languages without articles. The predictions they make seem to be the following. Given that the ambiguity approach derives definiteness semantically by means of an iota type shift, which is expressed by the definite article in English and not lexicalised in Russian, we expect the same definiteness effects to be present in both English and Russian, since the semantics of the type shifting operator should be language independent. In particular, Russian BNs with a definite interpretation are expected to give rise to uniqueness presupposition, just like their English counterparts.

The indefiniteness approach, on the other hand, would not make any predictions with respect to uniqueness for bare nominals in Russian. The difference between a definite and an indefinite interpretation in this approach is accounted for by different domain restrictors, and hence Russian bare nominals are compatible with both a narrow and a broader domain restrictor, their interpretation is derived by contextual factors that determine the domain restriction. In the case of a narrower restrictor, the uniqueness is implied, but not presupposed.

Thus, the difference between an ambiguity vs. an indefiniteness hypothesis applied to Russian boils down to whether BNs in Russian convey a uniqueness presupposition in their definite use. The question is empirical and the argument developed in the next section affects only those theories that predict uniqueness presuppositions to hold for Russian BNs.

4.3 An empirical argument for the absence of uniqueness in Russian

To check the predictions that the two approaches discussed in the previous section make for Russian, we will compare two sets of examples, one from English and another one from Russian, and compare the relevant interpretive effects in both languages. To narrow down our empirical coverage, we only look at singular bare preverbal subjects, considering them the strongest candidates for definiteness, due to their position and a default definite-like interpretation that they receive in native speakers' judgments.[22] The data presented in this section partially comes from Seres & Borik (2021), where the same argument was first developed and discussed. Let us first have a look at the English examples:

(46) The director of our school appeared in a public show. #The other / #Another director of our school…

[22]The expectation is that for any other type of BN (a plural BN or/and a BN in object position) the relevant examples will be much easier to construct, as the indefinite interpretation will always be available.

(47) A director of our school appeared in a public show. The other / Another director of our school…

In example (46), the subject of the first sentence is definite: it is marked by a definite article, semantically derived by the ι operator and has a strong uniqueness presupposition that cannot be cancelled, as witnessed by unacceptability of suggested continuations. The only possible interpretation of the second sentence in (46) would be 'the other director of the other school', which would not violate the presupposition of uniqueness of the definite description *the director of our school* from the first sentence. However, any continuation with *our school* in the second part of (46) is impossible.

In (47), on the other hand, the first subject is indefinite and does not give rise to any uniqueness effects. In this case, as the example illustrates, it is possible to conceive the interpretation 'another director of the same school', even though it might require some adjustment. The two examples thus clearly illustrate the effects created by the uniqueness presupposition of a definite description.

Now let us have a look at similar data from Russian.

(48) a. Direktor našej školy pojavilsja v tok-šou.
director.NOM our.GEN school.GEN appeared in talkshow
'The director of our school appeared in a talkshow.'

b. Drugoj direktor (našej školy) vystupil na radio.
other director.NOM our.GEN school.GEN spoke on radio.LOC
'The other director (of our school) spoke on the radio.'

The Russian example in (48a) taken in isolation seems to be equivalent to the first part of the English example in (46), in the sense that the nominal phrase '(the) director of our school' in both cases is interpreted as definite and, thus, the default interpretation of the expression is the uniquely identifiable director in a given context in both languages. The question is, however, whether this interpretation is semantically encoded in both languages. If what appears to be a definite nominal in Russian is also associated with the uniqueness presupposition, just like a definite description in English, the effects of violating this presupposition should be comparable to those observed in the English example (46). Should we find the same type of uniqueness effects both in English and in Russian, we can conclude that the same semantic operator, namely, the ι operator, is responsible for deriving definiteness in both languages. In search for an answer, we turn to (48b).

Crucially, we observe a substantial difference in the interpretation of example (46) on the one hand, and example (48), on the other hand. In particular, the subject in (48b) can be interpreted as 'another director of the same school', as opposed to the English example in (46). This means that there seem to be no uniqueness presupposition associated with the subject 'director of our school' in (48a). Examples (49) and (50) show the same effect, i.e., there seems to be no uniqueness presupposition associated with bare nominals that are perceived as definite. In the examples below, the judgments are given for 'another doctor of the same patient' and 'another author of the same essay', respectively.

(49) Vrač prišel tol'ko k večeru. Drugoj vrač prosto pozvonil.
doctor.NOM came only to evening other doctor.NOM simply called
'One doctor came only towards the evening. Another doctor simply called.'

(50) Avtor ėtogo očerka polučil Pulitcerovskuju premiju. Drugoj
author.NOM this essay.GEN received Pulitzer prize.ACC other
avtor daže ne byl upomjanut.
author.NOM even not was mentioned
'One author of this essay got a Pulitzer prize. The other/another author was not even mentioned.'

Taking into consideration the English and Russian data discussed in this section, we can conclude that the mechanism that yields a definite interpretation for bare nominals in Russian is crucially different from the mechanism that derives definiteness in English. If both definite descriptions in English and bare singulars in Russian perceived as definites were derived by the same semantic operation, we would expect the same semantic effects associated with definite expressions in both languages. The data, however, show that uniqueness effects are, indeed, very prominent with definite nominals in English, but seem to be absent in Russian. This means that what we call a definite interpretation in Russian is of a different nature. In particular, there is no violation of a uniqueness presupposition in the Russian examples discussed in this section, unlike in (46) in English.

Rather, the effects that we observe in (46) vs. (48) can be explained as a difference between presupposition and implicature: unembedded presuppositions (the case of (46)) are not cancellable, whereas unembedded implicatures (the case of (48)) are (see, for instance, Beaver et al. 2024). A real presupposition violation in Russian can be illustrated by the following examples with factive predicates ('know', 'be glad', etc.). The continuations in (51)–(52) are clearly unacceptable, whereas in examples (48)–(50) above only some pragmatic adjustment is needed.

(51) Tolja znaet, čto Anja zavalila ėkzamen. #Ona polučila otlično.
Tolja.NOM knows that Anja.NOM failed exam.ACC she got excellent.
'Tolja knows that Anja failed her exam. #She got an 'excellent'.'

(52) Tolja ne znaet, čto Anja zavalila ėkzamen. #Ona polučila otlično.
Tolja.NOM not knows that Anja.NOM failed exam.ACC she got excellent.
'Tolja doesn't know that Anja failed her exam. #She got an 'excellent'.'

The effect of presupposition violation in the examples above is completely different from the effect of cancelling an implicature of uniqueness of bare nominals in Russian. Ideally, the two effects should be compared experimentally, either by an acceptability judgment task or by measuring the response time in each of the cases. For now, the only experimental data available for Russian are reported in Šimík & Demian (2020), who show the absence of uniqueness (with singulars) or maximality (with plurals) effects with bare nominals in Russian, including those nominals that appear in sentence initial position.

Šimík & Demian's (2020) series of experiments are designed to check whether uniqueness/maximality is conveyed by definite descriptions in German and bare nominals in Russian. The experiment was based on the so-called "covered box" design, where the participants are presented with two pictures turned face down, one of which flips open after the participant has listened to a short audio message describing some situation. The "open" picture accurately depicted the described situation apart from one critical condition. In case of definite descriptions, the uniqueness/maximality condition was the critical one and the depicted situation contradicted the uniqueness/maximality conveyed by the corresponding definite description in the audio message. The task that the participants faced was to choose which one of the two pictures (the open one or the closed one) represented the situation correctly. The closed card was never displayed, so the participants basically had to judge whether the open card fully corresponds to a given situation or if there should be a better, more faithful depiction of the same situation. The participants were warned that there is only one "correct" option/card.

The experiment showed that in German, a language with a definite article, definite descriptions do indeed convey uniqueness and maximality, as expected, although the uniqueness condition appears to be weaker than expected.[23] In

[23]The reader is referred to the original paper for detailed results and discussion.

Russian, however, the uniqueness/maximality effects were not found at all, independently of the grammatical number of the nominals (singular vs. plural), their position in a sentence (sentence initial vs. non-initial position) or prosody (stressed vs. unstressed nominals). The results of the experiments conducted by Šimík & Demian (2020) are striking and show very convincingly that none of the approaches that predict uniqueness/maximality for bare nominals in Russian is supported by experimental data.

Interestingly, Šimík & Demian (2020) also conducted a series of experiments testing sensitivity to various presuppositions and inferences (exhaustivity, scalar implicatures, cardinality inferences with *both* or *two* etc.), using the same experimental design as for maximality/uniqueness. The results of those experiments clearly show that Russian speakers are highly sensitive to presupposition violation. In fact, there was barely any difference between German and Russian participants in this respect, as in both groups the participants made the correct and expected choice (i.e., the covered picture) in about 90 % of the cases. Thus, the experiments fully confirm the conclusions reached on the basis of the data discussed above: Russian bare nominals do not give rise to uniqueness effects, as the contextually imposed definite interpretation can be easily overruled in Russian. True presupposition violations are very clearly manifested and judged as virtually unacceptable by a large percentage of speakers.

In addition to the data and the experiments discussed above, it should be pointed out that corpus data also support the conclusions of this section. As was discussed in Section 2.2.1, examples of a BN in initial position followed by the same lexical nominal specified by ‘another’ are also found in corpus data. The reader is referred back to examples (17) and (18).

A different type of a corpus study is reported in Liu et al. (2023), Borik et al. (2025), Liu et al. (2024). These papers present an analysis of the distribution of nominal phrases across definite and indefinite contexts in various languages, including Russian, based on a parallel translation corpus.[24] The translation corpus used for the study was built on the first chapter of J. K. Rowling’s *Harry Potter and the Philosopher’s Stone*, a novel written in English and translated into many languages. It is shown that Russian can be considered a truly articleless language that freely allows BNs to take on definite and indefinite readings across domains. Thus, the study also confirms the main claims made in this section: Russian freely uses BNs in contexts where English uses indefinite NPs. Another conclusion of this corpus study is that not all languages without articles necessarily show the

[24]For a more detailed discussion of the translation corpus methodology see Le Bruyn et al. (2022), Le Bruyn & de Swart (2022).

same distributional patterns of BNs across contexts: the differences between Russian, Hindi and Mandarin are discussed in detail in Liu et al. (2023).

To sum up, the data discussed in this section all firmly point to a conclusion that there is no uniqueness effects associated with BNs in Russian. In the next section I will conclude the discussion of definiteness in Russian by examining a number of theoretical implications that these empirical data seem to suggest.

4.4 Definiteness and uniqueness in Russian: interim conclusions

The most important results of this section so far can be summarised in two points. First, there is ample empirical evidence that BNs in Russian behave like *a*-indefinites in English: they appear in the same contexts and exhibit the same scopal ambiguities. Second, there is no strict correspondence between definite descriptions in English and bare nominals perceived as definites in Russian. In particular, a careful analysis of Russian BNs let us conclude that there is no uniqueness presupposition associated with those nominals, whereas their definite counterparts in English do, indeed, give rise to strong uniqueness effects. Moreover, those uniqueness effects in English, when detected, cannot be cancelled in non-embedded contexts, which is a typical feature of presupposed content.

These results strongly suggest that approaches to BN semantics based on covert type shift operations (like, for instance, Chierchia 1998 and Dayal 2004) will not successfully handle Russian data if not substantially modified. The claim can presumably be extended to other languages without articles that behave similar to Russian with respect to the absence of uniqueness effects in BNs. In other words, we do have a very good reason to believe that the differences between definite descriptions in a classical sense and bare nominals perceived as definites lie deeper than just the surface realisation of the corresponding article and should be properly reflected in their semantic representation.

This, of course, does not imply that all languages without articles should necessarily behave the same (see Liu et al. 2023 and Liu et al. 2024). The differences in definiteness marking between languages with articles are widely attested (for instance, *la literatura* 'the literature' in Spanish vs. *literature* in English), so there is no a priori reason to believe that all bare nominals in languages without articles show a semantically uniform behaviour. Thus, Dayal's (2004) proposal might be a perfect account for Hindi data, but should not be blindly transferred to any other language without articles without an in-depth analysis of the relevant empirical facts, as argued in Liu et al. (2023). Of course, the opposite is true as well: if there are no uniqueness effects associated with bare singular nominals in Russian, bare nominals of some other language might very well yield uniqueness.

The results reported here do not mean that there is no definiteness in Russian at all, either. The argument developed in this section has to do only with the uniqueness theory of definiteness and its possible extension to articleless languages. We have established that bare nominals in Russian do not bear any uniqueness effects, but that does not mean that they do not express anaphoric definiteness or saliency (cf. von Heusinger 1997, 2013), or are not determinate in the sense of Coppock & Beaver (2015). Which of these properties can be associated with BNs whose interpretation is – in some sense – definite is a fascinating research question which deserves an accurate and precise answer. There is much to be done. For now, we can be fairly sure that if there is a category of definiteness in Russian, then it cannot be successfully characterised in familiar uniqueness/maximality terms.

From a theoretical viewpoint, the question that arises is which of the existing theoretical approaches would be better equipped to model the interpretation of bare nominals in Russian. Future research might still have a significant impact on a possible answer, but we may begin by sorting out the options. Let us assume that BNs in Russian denote properties (following Chierchia 1998, among many others), as there is not much reason to believe otherwise. Assuming the type-shifting approach to nominal reference with Partee (1987), this means that nominal arguments need to undergo a type shift to be used in argument position. Which type shifting operators need to be postulated for languages like Russian? In particular, is there an iota shift in Russian?

At this stage, the answer to the last question seems to be negative. To the best of my knowledge, there is no account that does not attribute uniqueness to the iota operator. As has already been pointed out, classical approaches to type shifting, like Partee (1987), do not distinguish between the semantic contribution of the ι operator and the definite article itself, so their prediction seems to be fairly straightforward: if there is no definite article, there is no iota operator and no uniqueness effects. Lexicalization theories based on Chierchia (1998), Dayal (2004) do postulate an iota shift for articleless languages and hence predict uniqueness for bare nominals with a definite-like interpretation, the prediction that has just been argued to be incorrect. The predictions that Coppock & Beaver (2015) make for articleless languages are not entirely clear and seem to hinge on a pending question of whether indefinite nominals have a determinate interpretation or not (cf. Coppock & Beaver 2015: 424). Coppock & Beaver (2015) also define ι as an operator that yields uniqueness, but the only other operator that they assume is the existential type shift, which does not derive determinate (i.e., type *e*) interpretations. Russian bare nominals can definitely be determinate, that is, have an individual denotation, but at the same time, cannot be derived by the ι operator that denotes uniqueness. Hence, what we need is an operator that derives determinate indefinites, exactly the one that is missing in Coppock & Beaver's (2015) analysis.

Perhaps the best candidate to handle the Russian data so far is the indefiniteness hypothesis proposed by Heim (2011). The proposal is that all bare nominals in articleless languages are inherently indefinite, as was explained in Section 4.2, and definiteness effects are achieved by pragmatic strengthening mechanisms. Following Heim (2011), I propose that for any bare nominal phrase in Russian, an indefinite interpretation is the only one derived semantically. Although a formal semantic analysis for bare nominals in Russian remains to be developed, we can make a first step by assuming that there are two semantic mechanisms involved in the derivation of indefinites in Russian, just like in other languages (cf. Reinhart 1997): existential quantification and a choice function, as defined in (53) below:

(53) a. $\exists x[P(x) \wedge Q(x)]$
b. $f_{\mathrm{CH}}[x : P(x)]$

Quantificational indefinites are considered to be non-referential, whereas a choice function analysis could account for those cases where an indefinite refers to a (specific) individual. The representation in (53b) is based on the choice function analysis of specific indefinites (cf. Reinhart 1997, Winter 1997, Kratzer 1998) and states that there is a function f that is a choice function and, applied to a predicate P, yields a member satisfying this property. A full formal analysis of bare nominals in Russian will need to determine how precisely the labour is divided between the two mechanisms (or, perhaps, just one mechanism suffices, as proposed by Winter 1997), whereas we can conclude this section by stating that the perceived definiteness of Russian BNs, under the indefiniteness hypothesis presented here, must be of a pragmatic nature. In practice, this means that there is no need to postulate an iota operator for Russian. Note that Šimík (2021) makes a similar proposal for Czech BNs: in his analysis, argument BNs are treated as referential expressions of type e derived from the basic predicate denotation by a Skolemized choice funcion. Šimík (2021) also argues that BNs have the property of inherent uniqueness, which is derived as a property of a resource situation. Crucially, in Šimík's (2021) analysis BNs do not carry any presupposition of uniqueness.

Seres & Borik (2021) argue that there are at least three factors that are responsible for triggering definiteness effects in Russian: ontological uniqueness, topichood, and anaphoricity. The interpretation of bare nominals in topic position was already discussed in Section 2.2.1 and anaphoric definiteness in Russian is an open field of research, which is impossible to embrace within the limits of this chapter.[25] As for ontological uniqueness, this term is used to refer to those objects which are perceived as unique by default in the current situation/world, such as *the earth, the sun, the moon*, etc. in English.[26] In Russian, those unique objects are usually referred to by bare singular nominals, as illustrated in (54):

(54) Solnce svetit.
sun.NOM is.shining
'The sun / #a sun is shining.'

In this case, uniqueness is conveyed not so much by the definite article, but by the descriptive content of a nominal phrase itself. For instance, when we want to use an expression with the noun *sun*, a usual case is that we want to refer to the sun of our solar system, which is a unique object. In this case the definite article in English does not contribute but rather reflects the uniqueness of the object in the actual world. If we use *sun* with an indefinite article in English, we would overrule the assumption that we are talking about the sun of our solar system. In Russian, both options are compatible with the BN in (54), so that this case is essentially not different from examples like (48) discussed in Section 4.3.

Note that if we cast doubts on the necessity of the iota operator in Russian, we have to specifically mention another type of nominal interpretation, namely, the kind reading. In Section 2.1 it was stated that the analyses of singular kinds (or definite kinds) usually rely on the iota operator for semantic representation. I would suggest that kinds can, in principle, be handled like ontologically unique objects (see Šimík 2021 for a similar idea). Note that kinds in English obligatorily appear with the definite article, just like ontologically unique nouns like *sun*. Any other type of determiner would give rise to a different interpretation, cancelling the uniqueness of the object or kind that we are referring to. The function of the definite article *the* is the same with kinds and with ontologically unique nouns: the article simply marks a unique referent. In the absence of articles, the unique interpretation arises as a result of the descriptive content of the noun.

In sum, I have argued that definiteness effects that can be associated with some uses of Russian bare nominals should not be encoded in their semantic representation. I have also proposed that the indefiniteness hypothesis by Heim (2011) seems to be better equipped to model the behaviour of bare nominals in

[25] Although see Borik et al. (2020), Seres et al. (2019) for some data concerning anaphoricity.

[26] In Löbner's (1985) terms, these nominals are "semantically unique" and in Šimík's (2021) terms they are "inherenly unique".

Russian and makes correct empirical predictions, unlike uniqueness approaches postulating a (covert) ι (iota) operator for a representation of those bare nominals that exhibit a definite-like interpretation in Russian.

In general, definite nominals is a very broad class that includes not only definite descriptions, but also proper names, demonstratives, personal pronouns and possessive constructions, which are all regarded as definite (cf. Lyons 1999). So far, the scope of the discussion of the Russian data has been restricted to BNs in Russian whose interpretation is intuitively similar to the interpretation of definite descriptions in English. The next section will extend the empirical coverage of the chapter to include demonstratives and possessive constructions.

5 Marked definiteness: demonstratives and possessives in Russian

Demonstrative and possessive constructions are often discussed in the definiteness literature (cf. Hawkins 1978, Lyons 1999: etc.) because semantically, these constructions seem to somehow incorporate definiteness. In this section, I will briefly describe some general proposals in the literature regarding a semantic analysis of demonstratives and possessives in terms of definiteness and examine the repercussions of these proposals for the corresponding Russian data. It will be shown that neither demonstratives nor possessives in Russian seem to pose an immediate threat to the hypothesis put forward and defended in the previous section: the absence of definiteness/uniqueness in Russian BNs.

5.1 Demonstratives

There is a strong tendency to associate *canonical* demonstrative nominal phrases like the one in (55) with definite descriptions (cf. Lyons 1999, Elbourne 2008, Šimík 2016):

(55) This dog [pointing at some dog] looks dangerous.

In this example, the demonstrative pronoun *this* is quite similar in meaning and can probably even be substituted by the definite article *the*: the demonstrative pronoun combined with the nominal *dog* restricts its denotation to a (unique) individual dog identifiable in the context. There is, however, a whole class of demonstrative constructions that do not seem to fit this description. One example is given below:

(56) This John Blake might be the smartest guy in the whole universe but boy, is he annoying!

Unlike in the previous example, in this case the demonstrative does not restrict the denotation of the descriptive content to a unique individual or shift the meaning of its complement to an individual denotation, as it combines with a proper name. Rather, it contributes an emphatic pragmatic meaning, or some sort of evaluation, the meaning described by Lakoff (1974) as AFFECTIVE.

Both uses exist in Russian and other Slavic languages (see Šimík 2016 for Czech), as exemplified below:

(57) a. Posmotri na ėtu kartinu, kakaja krasota!
look on this painting what beauty
'Look at this painting [pointing at some painting in a museum], what a beauty!'

b. Ėta Irina, novaja sekretarša šefa, očen' mne ne ponravilas'.
this Irina new secretary boss.GEN very me.DAT not like.REFL
'I really didn't like this Irina, a new secretary of the boss.'

Affective demonstratives will not concern us here, despite their many intriguing properties, as the case that is directly relevant to the discussion of uniqueness in the previous section is really a canonical use of demonstratives.

Going back to canonical demonstratives exemplified in (55), there are formal analyses that capitalize on their definiteness and the uniqueness of the referent of a demonstrative description. For instance, Elbourne (2008) states that demonstrative nominals do not receive any denotation in a context if there is no unique individual that satisfy the corresponding description (Elbourne 2008: 26). Consequently, uniqueness forms an integral part of his analysis of demonstratives which integrate the definiteness semantics by means of employing an iota operator in the semantic representation of demonstratives.

However, certain crucial differences between demonstrative and definite descriptions have not gone unnoticed either. Roberts (2002) points out the following difference between the two classes of NPs:

(58) a. You [nodding to Mary] sit in that chair [pointing at chair *a*], and you [nodding to Jonathan] sit in that chair [pointing at chair *b*].

b. # You (nodding to Mary) sit in the chair [pointing at chair *a*], and you (nodding to Jonathan) sit in the chair [pointing at chair *b*].

The examples illustrate that a deictic demonstrative description is appropriate in a context with more than one potential referent if accompanied by a gesture, as in (58a), whereas a definite description is clearly unacceptable in the same situation. This fundamental difference underlies a range of linguistic analyses which are, to a bigger or lesser extent, built on Kaplan's (1989) original idea that the semantic treatment of demonstratives should rely on an additional referential means, something like DEMONSTRATION.

This additional component finds its way into a syntactic (and semantic) structure associated with demonstrative nominal phrases in recent literature (e.g. Elbourne 2008, Simonenko 2013, Šimík 2016):

(59) [Dem [Det [NP]]]

Semantically, the structure in (59) reflects two crucial components of the complex meaning of a demonstrative: a relational/anaphoric/demonstrative component encoded by Dem and a definite component encoded by Det. In most formal analyses, the Det component is represented by the familiar iota operator, the same that derives a uniquely referring individual expression in the case of definite descriptions.[27]

The scope of this chapter cannot be extended to include a detailed and theoretically comprehensive discussion of the semantics of demonstratives, but I think that one question directly relevant to the discussion in this chapter should nevertheless be addressed. This question is whether demonstratives presuppose uniqueness of their referent.

Without looking into technical details, the point that I would like to make is the following. It seems to me that uniqueness in the case of definite descriptions and uniqueness in the case of demonstratives is not of the same nature. Whether I say *Look at this/that book!* or *Look at the book!*, I am, indeed, referring to one single individual representative of the *book*-class, but the referent of *this/that book* is not semantically unique in the same sense as the referent of the definite description *the book*. For instance, I can use the expression with a demonstrative while looking at the books on the bestsellers stand in a bookstore (and pointing at the relevant book at the same time, perhaps), but a definite description would sound odd in this context (cf. Roberts 2002). In Russian, in the same type of situation, a demonstrative (or any other nominal modifier) is almost required, as a bare nominal can only be successfully used if accompanied by some sort

[27] Even though this structure is semantically motivated, the additional question that it poses for Slavic languages is the presence of a DP layer which is not commonly assumed or adopted in Slavic syntax. For a recent overview, see Geist (2021).

of gesture or other identifying information, otherwise the immediate reaction would be *Which book?*

If the uniqueness of a referent is not satisfied in the case of a definite description, this definite description cannot be successfully used in the corresponding context. However, in the case of demonstratives the individual referred to does not have to be unique, the speaker *makes* it unique by singling it out from the rest of the individuals that satisfy the descriptive content of the NP. As a result, the demonstrative expression will refer uniquely but this uniqueness seems to have a different *source* than the uniqueness of a definite description.

This reasoning is general and applies to demonstrative nominals in both English and Russian. If this line of argumentation is on the right track, no specific argument against uniqueness of Russian demonstratives needs to be developed. This also means that if we want to use the structure in (59) to represent the syntax and the semantics of a demonstrative NP, then perhaps the semantic operator that corresponds to Det should not be the same as the one used in definite descriptions, as the great majority of analyses, including Roberts (2002), Wolter (2006), Elbourne (2008) suggest.

Postulating the ι operator in the semantic representation of demonstratives raises some questions concerning the Russian facts discussed in Section 4. In particular, if the indefiniteness hypothesis turns out to be the correct one for representing bare nominals in Russian, it might be the case that the ι operator is redundant for Russian, unless it is demonstrated that this operator is needed for other types of nominals, not bare nominals.[28] Then, for instance, if Det in (59) is adopted for Russian, it should correspond to some other semantic operator, different from ι. But once this turns out to be an option for Russian, why should we expect Det to be exclusively associated with ι in English and other languages?

To sum up, the point that I have tried to make in this section is of a more general nature and does not draw so much on Russian data: I have questioned that uniqueness properties of a definite description and of a demonstrative nominal are of the same nature and should be attributed to the same semantic operator. I tentatively suggest that the answer to this question is negative. It would be wrong to claim that the differences in referential properties of definite and demonstrative descriptions have not been tackled in the literature. Different solutions that have been proposed include modifying the notion of uniqueness (cf. "informational uniqueness" of Roberts 2002), postulating different contextual domain restrictions for definite vs. demonstratives (e.g., Wolter 2006) or differentiating between inherent (pragmatically presuppositional) vs. accidental

[28] See Šimík (2021) for a proposal based on the indefiniteness hypothesis.

(presuppositional) uniqueness (Šimík 2021). To this, I would add that another option should be explored: the one that does not include semantic/presuppositional uniqueness for the analysis of demonstratives at all.

5.2 Possessives

It is often suggested in the literature that at least some possessive descriptions are definite. For English Saxon genitives, i.e., those expressions that begin with a possessive pronoun or with a genitive nominal phrase expressing a possessor (e.g., *Peter's house*), it is common to postulate (or to implicitly assume) a hidden *the* in the semantic analyses (cf. Partee & Borschev 2003, Vikner & Jensen 2002). This position is countered by Peters & Westerståhl (2013), who argue that possessive are not definite, and Coppock & Beaver (2015), who argue that Saxon genitives are (typically) determinate, but are not marked for definiteness.

In Russian, there are two types of possessive constructions exemplified in (60):

(60) a. portret Peti
portrait.NOM Petja.GEN
'the portret of Petja'

b. Petin portret
Petja.POSS.ADJ portrait.NOM
'Petja's portrait'

The construction in (60b) is restricted to proper names, pronouns, kinship terms, professions/titles, or animal names, which are grammatically and semantically singular (Babyonyshev 1997) and often compared to the English Saxon genitive. Two relevant properties of prenominal possessors in (60b) have been noticed and mentioned in the literature.

The first one is that they exhibit an agreement pattern similar to demonstratives but not to regular adjectives, as illustrated in (61):

(61) ėtot-∅ Vanin-∅ krasiv-yj dom-∅
this-M.SG.NOM Vanya.POSS-M.SG.NOM beautiful-M.SG.NOM house-M.SG.NOM
'this beautiful house of Vanya's'

As can be seen in the example, a "proper" adjective in Russian marks agreement with the head noun by means of a special ending (*-yj* in (61)), whereas both demonstratives and possessives lack this ending. Supposedly, the presence of the ending signals a proper adjectival status of the agreeing element, and the absence of it indicates that the properties of the agreeing expression combine adjectival and nominal characteristics.

Another characteristic of prenominal possessors is that they introduce a referent that can serve as an antecedent for a reflexive pronoun. Possessive adjectives do not exhibit this property:

(62) a. sosedkin$_i$ rasskaz o svoix$_i$ problemax
neighbour.poss.m.sg.nom story.m.sg.nom about self.pl.prep problem.pl.prep
'the neigbour's story about their own problems'

b. * sosedskij$_i$ rasskaz o svoix$_i$ problemax
neighbour.m.sg.nom story.m.sg.nom about self.pl.prep problems.pl.prep
Intended: 'the neighbour's story about their own problems'

On the basis of these two properties, Babyonyshev (1997) proposed that prenominal possessors in Russian should be treated as D-elements, since they are reminiscent of demonstratives and have referential properties of their own.

However, it is likely that the agreement pattern of possessors has an independent explanation and the referential properties of possessors are not that robust. As for agreement, both demonstratives and possessive adjectives with the suffix *-in* in Russian, like *Vanin* and *sosedkin* in the examples above, belong to a so-called mixed type and fully coincide with proper (long) adjective forms in all cases except for Nominative and Accusative (Vojejkova & Xolodilova 2022), as illustrated below:

(63) a. ėtot Vanin krasiv*yj* dom
this.m.sg.nom Vanya.poss.m.sg.nom beautiful.m.sg.nom house.m.sg.nom
'this beautiful house of Vanya's'

b. ėt*omu* Vanin*omu* krasiv*omu* dom*u*
thism.sg.dat Vanya.poss.m.sg.dat beautiful.m.sg.dat house.m.sg.dat
'(to) this beautiful house of Vanya's'

As for the second property, i.e., the referential status of prenominal possessors, Trugman (2007) points out that they do not always introduce a referent. In fact, she makes a distinction between referential possessives in (64a) and modificational possessives that do not introduce any referent in (64b):

(64) a. babuškiny tufli
grandma.POSS.PL.NOM shoes.PL.NOM
'grandma's shoes'

b. babuškiny sredstva
grandma.POSS.PL.NOM remedy.PL.NOM
'folk remedies'

To sum up, neither of the two characteristic properties of prenominal possessives can form basis for any solid generalisation concerning their syntactic status and their structural position within the nominal phrase.[29]

As for the definiteness status of possessives, Coppock & Beaver (2015) argue for English that possessive descriptions are not marked for definiteness, although they typically have individual reference and hence are determinate. In particular, Coppock & Beaver (2015) show that possessives do not encode (weak) uniqueness presupposition, whereas definite descriptions do. They base their main arguments on predicative uses of possessives and show that they lack the uniqueness requirements on the basis of the following examples:[30]

(65) a. Is that your bicycle?
b. Is that the bicycle you own?

(66) Yes, and that one there is also mine.

The sentence in (66) cannot be a felicitous answer to (65b), which contains a definite description that imposes the uniqueness requirement. However, it can be successfully used as an answer to (65a), which contains a possessive instead of a definite description. If possessives required uniqueness, just as definite descriptions do, we would expect that (66) would not be acceptable as an answer to either (65a) or to (65b).

In the case of Russian, it is fairly easy to show that possessive descriptions with prenominal genitive in Russian do not show any uniqueness effects. Consider the following example:

[29] See Gepner (2021) for similar conclusions.

[30] The argument is more complex for the possessives in argument position.

(67) a. V každoj komnate našego doma visit mamin
in every room our.GEN house.GEN hangs mom.POSS.M.SG.NOM
portret.
portrait
'There is a portrait of mom in every room of our house.'

b. Mamin portret visit v každoj komnate našego
mommy.POSS.M.SG.NOM portrait hangs in every room our.GEN
doma.
house.GEN
'A portrait of mom is hanging in every room of our house.'

Both examples in (67) would get an implausible interpretation should the possessive nominals yield uniqueness effects: the sentences would have to state that the same portrait was hanging in every room of the house. However, no such oddness is observed in either of the examples, the natural interpretation that arises is that every room of the house contains a different portrait of the same person. Due to the differences in word order, which cannot be easily reflected in the translation, the interpretation of the possessive phrase in (67b) is intuitively "more definite" than the one in (67a) due to the initial position of *mamin portret* in the second sentence. Despite the position, none of the possessive nominals encode uniqueness and hence we can conclude that possessive phrases in Russian do not pose any threat to the absence of uniqueness hypothesis developed in the previous section.

6 Conclusions

Despite the fact that Russian does not have articles, the contrast between definite, indefinite, and generic interpretations of BNs are still perceptible to the native speakers. This is true for other articleless languages as well, so the question of how to model the interpretations of BNs in languages without articles is an important linguistic question.

This chapter includes two parts: a broader empirical overview of the interpretive possibilities of BNs in Russian, and a comprehensive theoretical discussion of one of the possible interpretations of BNs, namely, the definite reading. In the first (empirical) part, I have shown that, pretty much in accordance with reasonable expectations, BNs in Russian can get a generic, definite or indefinite reading in argument position and discussed some factors that give rise to or facilitate a particular interpretation, in particular, word order and topichood. We have seen that for a generic reading, the type of predicate, the position in a sentence (pre- vs. post-verbal for subjects), the type of a sentence (characterising vs. episodic) are among the crucial factors that facilitate this interpretation. I have also discussed a linear position as a factor for obtaining an (in)definite interpretation, and one of the most important generalizations concerning a constraint on an indefinite interpretation for Russian BNs in topic position proposed in Geist (2010). I have shown that BNs in Russian do show scope ambiguity, just like English indefinite nominals, hence a specific indefinite interpretation of BNs in Russian cannot be excluded *a priori*.

The second part of the chapter discussed the relation between BNs and definiteness, both from a theoretical and from an empirical perspective. I have argued that BNs in Russian cannot be treated as definite descriptions and provided various arguments against the semantic definiteness of those nominals. In particular, I have shown that Russian BNs which appear to be interpreted definitely do not give rise to uniqueness presupposition, unlike their definite counterparts in English, and hence should not be modelled in the same way as definite nominals in languages with articles. The absence of uniqueness presupposition with those BNs whose interpretation at first sight appear to be definite raises a fundamental question about a possible semantic analysis of BNs in articleless languages. In this chapter, I proposed that an indefiniteness hypothesis as sketched by Heim (2011) appears to be the best fit for the Russian data discussed here.

This, however, does not necessarily mean that there is no definiteness in Russian. The only point that this chapter defends is that a uniqueness based theory of definiteness, depending on its specific implementation, makes incorrect predictions for Russian. There might be other types of definite nominals in Russian, those whose semantics do not have to be modelled based on uniqueness effects. Anaphoric definite BNs, for instance, exist in both Russian and other languages without articles. There might be other types of definite nominals whose semantics do not have to be modelled by means of the iota operator that conveys a uniqueness presupposition.

A lot remains to be done. A detailed, motivated and empirically adequate semantic analysis of BNs in Russian and other articleless languages is still to be developed. My hope is that this chapter lays the groundwork for developing such an analysis by providing empirical generalizations and arguments that have so far received little (in relative terms) attention in the literature.

Abbreviations

3	third person	PRS	present tense
ACC	accusative	PTCP	participle
INF	infinitive	REFL	reflexive
M	masculine	SG	singular
NOM	nominative		

Acknowledgments

I am grateful to the reviewers of this chapter for their valuable comments and suggestions, and to the editors of this volume for their endless patience, as well as their dedication and investment in this volume. All errors are mine. At the time of writing this chapter I was profiting from the financial support of the Spanish MINECO grant PID2020−112801GB−100.

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Chapter 3

Negative polarity items in Slavic

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This chapter describes Slavic expressions dependent on negation. It summarizes three approaches to such expressions which appeared in the history of formal linguistic Slavistics. First, the syntactic analysis of negative dependent expressions (see Progovac 1994 a.o.) is discussed; next the semantic approach of Błaszczak (2001); and last, the pragmatic stance of Crnič (2011) is introduced. The chapter shows how Slavic data of this kind were important from the point of view of formal linguistics, and their description mirrored the changes in the general linguistic theories.

1 Introduction

Negation and negative dependent or sensitive expressions belong to the core grammatical constructions of natural languages. There are no known languages that would lack expressive means to communicate negation (see Bernini & Ramat 2012 a.o., where the fact of negation being present in all natural languages is called a "pragmatic universal"). Slavic languages are no different, of course. So there is a plethora of phenomena (many of them already described by formal linguists) concerning negation, which are perfect candidates for inclusion in this chapter. However, space and other limitations force me to narrow down the domain described in the present chapter (but see Horn 1989 for the book aiming at the description of all linguistic things connected to negation). Therefore, I will limit myself to discussing various types of Slavic negative dependent expressions. And I will only mention many valuable contributions which would surely deserve appropriate space but are beyond the horizon of negative dependent expressions.

Mojmír Dočekal. 2026. Negative polarity items in Slavic. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 105–148. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078573

Criteria for delimiting this chapter to Slavic negative dependent expressions were as follows. I wanted to summarize such areas of Slavic negation where (i) Slavic linguistics contributed to general linguistics theories (even if now the particular approach seems to be more of a historical value) and (ii) where Slavic data, and more specifically expressions interacting with negation or which in some respect are negative themselves offer valuable insights from a typological perspective. And it seems more than reasonable that Slavic negative dependent expressions (especially negative polarity items) pass both criteria.

First, the obvious thing to note is that Slavic languages do offer much more expressivity than the (still) most studied language, English. Consider the all-time favorite English polarity expression *any* (other types of English polarity expressions like *even one* will be discussed in Section 2.3). In the following English example, *any* is licensed by clause-mate negation (1a), adversative predicate (1b) (which I, for expository reasons, treat as containing negation but see Section 2.2.1 for more details of adversative predicates) and, finally, by a possibility modal (1c). In (1a) and (1b) *any* is usually considered as a NEGATIVE POLARITY ITEM (NPI), and the example like (1c) as FREE CHOICE ITEM (FCI). Although it seems typologically more frequent (than the opposite pattern) that one item (like English *any*) acts both as the NPI and the FCI, it is not a rule.[1] For the sake of exposition, let us assume that at least in English, *any* is an item with roughly existential semantics, and such a polarity item can be described by a unified theory.[2] Be that as it may, NPI and FCI are united by a licensing requirement. In all three cases, the scope of the existential NPI/FCI must be narrow relative to the scope of negation or existential modal. And, of course, the use of NPIs/FCIs leads to ungrammaticality without embedding in an appropriate environment: (2). Negative polarity items are the main topic of this chapter, and as an avid reader most probably knows, they are the class of negative dependent expressions which are usually discussed the most (at least by semanticists), since their distribution can be very well-expressed in terms of entailment and other related properties of the sentences where they occur. A more formal characterization will be introduced in Section 2.1.

[1] See Horn (1972) and Haspelmath (1997) for an early observation and a typologically broader picture, respectively. As one of the anonymous reviewers correctly points out, languages like Latin, Romanian, and Russian are clear examples of distinguishing FCIs from NPIs series.

[2] Nowadays, this seems like a standard approach, following Carlson (1980) and then Kadmon & Landman (1993), but at least for FCIs a universal quantifier analysis was offered by Dayal (1998); a.o. Thanks to all three reviewers for reminding me of the importance of *any* in research history.

(1) a. Peter didn't visit anyone.
  i. $\neg > \exists$
  ii. # $\exists > \neg$
 b. John refused any help.
  i. $\neg$ACCEPT $> \exists$
  ii. # $\exists >$ $\neg$ACCEPT
 c. You can take any card from the deck.
  i. $\Diamond > \exists$
  ii. # $\exists > \Diamond$

(2) #Peter visited any student.

Compare this with the Czech translation of the three environments showing the grammatical but also ungrammatical possibilities of three types of negative dependent expressions (which is still a subset of possible expressions interacting with negation in various ways, see Section 2.3 for more details). The translations, in some cases intended interpretations, are approximate.[3]

(3) a. Petr nenavštívil {ani jednoho / #byť jediného / #jakéhokoliv}
 Petr neg.visited even one just one any
 studenta.
 student.
 Approximately: 'Petr didn't visit {even one / a single / any} student.'
 b. Petr odmítl {jakoukoliv pomoc / #ani jednoho studenta / pohnout
 Petr refused any help even one student move
 se byť o jediný centimetr}.
 REFL just by one centimeter
 Approximately: 'Petr refused {any help / any student / to budge an inch}.'
 c. Můžeš si vzít {jakoukoliv / #byť jedinou / #ani jednu} kartu.
 can.2SG REFL take any just one even one card
 Approximately: 'You can take {any / even a single / not even one} card.'

[3] All non-English examples in this chapter are in Czech, unless explicitly indicated otherwise. Also, unless a source is provided, the Czech examples and their acceptability stem from the author's intuition.

The negative dependent expressions in Slavic come in many forms. Some seem to be vanilla NPIs but are limited to clause-mate negation (like *ani* in (3)). Others bear presuppositions of unlikelihood (as *byť jediný* in (3), which resembles English NPIs like *even one*), and yet others seem to prefer free choice contexts (as *jakýkoliv* from (3)). The variation of Czech (and generally Slavic NPIs) is reflected in the patterns of their distribution. Or, to put it the other way round: the flavors of NPIs determine which contexts we can find them in, and the varieties of Slavic polarity items carve up the landscape in a different way from the more usually discussed Romance or Greek polarity expression (see Laka 1990, Giannakidou 1997, Dalmi et al. 2024; a.o.).

When looking at the history of approaches to Slavic negative dependent expressions, we can clearly distinguish three types of theories: (i) syntactic theories of NPIs (discussed in Section 2.1), (ii) semantic theories (Section 2.2), (iii) pragmatic theories (Section 2.3). The particular details of each approach will be discussed in the respective sections. But what unites them all is the observation about the richness of Slavic expressions, which offers a much clearer picture than the sparse English *any*. Moreover, the various subsystems of negative dependent expressions create beautifully intricate territories where individual subtypes compete for insertion. Consequently, their accurate description brings new data not only for the polarity agenda but also for our understanding of concurrence in grammar. In summary, various approaches to Slavic NPIs will be the main topics of the current chapter, simply because they were at the center of attention in the last 30 years of formal Slavic linguistics.

But before we begin, let us quickly address a closely related phenomenon which I will describe just briefly in the current article (see Dočekal 2020 for full description). Next, to semantically dependent expressions, NPIs exemplified above, there are (in all Slavic languages) negative dependent expressions, so-called NEG-WORDS. Neg-words are described in current standard semantic theories as syntactically, not semantically, dependent. Neg-words are syntactically indefinites but clearly distinct from specific, nonspecific, and other kinds of indefinites (see Geist 2026 [this volume] for an overview). But focusing on the distinction between neg-words and NPIs, the theoretical distinction (neg-words: syntax, NPIs: semantics) is operationalized using the following criterion from Giannakidou & Zeijlstra (2017):

(4) X qualifies as an neg-word iff:
  a. X can be used with structures with sentential negation or other X with meaning equivalent to ¬;
  b. X provides a negative fragment answer.

Especially the second part of the criterion is the most reliable diagnostic to distinguish NPIs from neg-words because neither of the 3 types of NPIs in (3) can be used as a fragmentary answer to a question. See ($5B_1$), where all three NPIs are unacceptable, contrasting with the acceptability of the neg-words in the same context, see ($5B_2$).

(5) A: Kdo byl dneska večer na náměstí?
who was today evening on square?
'Who was on the square today in the evening?'

B_1: # {Ani jeden člověk. / Byť jediný člověk. / Kdokoliv.} NPIs
even one human just one human anyone
Approximately: 'Not even one man.' / 'Not a single man.' / 'Anyone.'

B_2: Ni-kdo. neg-word
NEG-person
'Nobody.'

It is important to note that both the syntactic approach of Progovac (1994) and the semantic approach of Błaszczak (2001) discussed in Section 2.1 and Section 2.2 respectively do not accept the distinction between neg-words and NPIs as delineated above. And it is a problem which has caused both empirical and theoretical flaws in both approaches. But to be fair, both approaches were published before there was a solid framework for the treatment of neg-words in negative concord languages (Zeijlstra 2004). Moreover, either syntactic treatment of all NPIs as in Progovac (1993) or fully semantic treatment of neg-words (and their sub-kinds as neg-pronouns) as in Błaszczak (2001) were compatible with the linguistic state of art valid at the time of their publication.

In this chapter, I will use the following terminology, which is based partially on Błaszczak (2001) and exemplified with Czech examples in (6).

(6) a. NEG-WORDS
nikdo 'nobody' (glossed as NEG.person), *nic* 'nothing' (NEG.thing), *nikde* 'nowhere' (NEG.place)

b. A-NPIs
ani jeden 'not even one' (NPI-one), *ani jednou* 'not even once' (NPI-once)

c. K-PRONOUNS
kdokoliv 'anyone', *kdekoliv* 'anywhere', *cokoliv* 'anything'

d. B-NPIs
byť jediný (glossed/translated depending on context as 'even/just/at least one'), *byť jednou* ('even/just/at least once')

e. REGULAR (WEAK) NPIs
as Czech *vůbec* 'at all'

2 Three types of approaches to NPIs (from the Slavic perspective)

2.1 Syntactic approach of Progovac

Let us first start with the syntactic approach of Progovac (1994) (for a compatible approach to PPIs see Szabolcsi 2004). Progovac's (1993) approach is partially inspired by the influential work of Linebarger (1987) where some serious problems of early semantic approaches (like Fauconnier 1978, 1980, Ladusaw 1979) to NPIs were recognized. The syntactic theory of NPI licensing claims that the domains and licensors of NPIs should be described by syntactic rules: either via transformations, as in the earlier approaches of Klima (1964) or via binding, as in Progovac's work.

Progovac's work should be understood with the 1980's semantic theory of NPIs as a background. In the 1980s, the semantic theory of NPIs, in a nutshell, claimed that NPIs occur in DOWNWARD ENTAILING (DE) contexts. Downward entailing contexts are such that speakers can reason from sets to subsets (see Section 2.2 for more details). Let us see this more clearly in an example: consider an example of a downward entailing context in (7b) as opposed to an UPWARD ENTAILING context in (7c) (and note the subset–superset relationship in (7a)). In (7a), the quantifier *every* allows one to draw an inference about the truth of the predicate to all subsets of dogs. The inference is valid only for the first argument of *every* (natural language quantifiers have two arguments like their formalization in predicate logic with λ-abstraction, $\lambda P \lambda Q \forall x[P(x) \rightarrow Q(x)]$), since from the truth of *Every dog barks* it doesn't follow that *Every dog barks loudly*. On the other hand, a quantifier like *some* in (7c) allows inferences about the truth of the predicate from subsets (e.g., *dachshund*) to supersets (*dogs*), in both its arguments. The quantifier *every* is then called downward entailing (in its first argument) and *some* upward entailing (in both arguments). The quantifier *every* is downward entailing, and because of that (in semantic theories of NPIs), it licenses NPIs: (7d) while upward entailing quantifier *some* does not: (7e). Quantifiers like *every* and *some* are then called downward and upward monotonic, respectively, since they are monotonic in the same sense as monotonically increasing or decreasing functions in mathematics.

(7) a. dachshund $\subseteq$ dog
b. Every dog barks. $\vDash$ Every dachshund barks.
c. Some dogs barks. $\nvDash$ Some dachshunds bark.
d. Every dog which won any medal barked.
e. *Some dogs which won any medal barked.

The semantic approach works in simple cases like (7d), but in its basic form, it has many drawbacks. One of the first influential critiques is Linebarger (1987), who noticed some serious problems with the semantic approach. First, Linebarger correctly pointed out that many NPI licensing contexts are not, in fact, downward entailing. Her examples involved conditionals and adversative predicates; see Section 2.2.1 for more details and refinement of the semantic approach, which can deal with the majority of such problems. And her second line of attack concerned locality restrictions between the licensors and their NPIs. One such example is in (8a). The example cannot have the reading (8b): it is not true that Mary wears earrings to every party (to some yes but not to all) but only has the reading in (8c): there are no such earrings that Mary wears to every party. According to Linebarger, this shows that some scopal (maybe syntactic) locality constraint (requiring the NPI to be in the immediate scope of its licensor) is needed. This seems to be a valid objection that cannot be explained by a theory of NPI licensing depending only on the monotonicity of its context. But see Chierchia (2013) for a modern pragmatic/semantic framework that covers even these supposedly syntactic locality problems.

(8) a. Mary didn't wear any earrings to every party.
b. # $\neg\forall y[\textsc{party}(y) \rightarrow \exists x[\textsc{earring}(x) \wedge \textsc{wear at}(\textsc{Mary}, x, y)]]$
c. $\neg\exists x[\textsc{earrings}(x) \wedge \forall y[\textsc{party}(y) \rightarrow \textsc{wear at}(\textsc{Mary}, x, y)]]$

It is appropriate to understand Progovac (1993) on the background of (in its time very important and influential) Linebarger's (1987) theory. Progovac bases her analysis in syntax, trying to explain NPI properties from the principles of binding theory but applied to various types of NPIs instead of the standard pronouns and anaphors. Next, Progovac (1993) was very influential in its time and still is one of few systematic analyses of Slavic NPI data. Moreover, the whole analysis does not only cover Slavic languages but also provides a general framework designed for a typologically diverse sample of languages. But let us begin with empirical claims. Progovac distinguishes three classes of polarity-sensitive expressions and claims that there are two NPI licensors: negation and an operator in Comp.[4] The polarity operator is licensed either by syntactic movement or by DE predicate/operator from a c-commanding clause.[5] Progovac brings data from Bosnian/Croatian/Serbian (BCS). Nevertheless, her analysis is meant for all Slavic languages. And it is true that in the majority of Slavic languages, similar data can be found (see Section 2.2 for analogous data from Polish, described in a different framework, though). According to Progovac (from the NPI point of view), we should distinguish the following three classes of expressions:

1. BCS pronouns which begin with the negative prefix *ni*, e.g., *ni-(t)-ko* 'no one', *ništa* 'nothing', *nikud* 'nowhere' are NPIs, in our terminology neg-pronouns. Progovac, following Laka (1990), claimed that neg-words in natural languages with negative concord should be analyzed as NPIs, which is one of the possible positions but definitely not the only one (see Section 1). Neg-pronouns are claimed to be anaphoric, subject to Principle A: they have to be bound by negation in their governing category. Data supporting the anaphoric nature of NPIs are in (9)and (10). The Grammaticality/ungrammaticality of examples like (9)/(10) is then simply accounted for as syntactic locality licensing or, more specifically, as a failure to find an antecedent that would be local enough.[6]

   (9) Milan *(ne) vidi *ni-šta*.
   Milan not sees NEG-thing
   'Milan cannot see anything.' (BCS; Progovac 1994: 41)

   (10) * Milan ne tvrdi [da Marija poznaje *ni(t)-koga*].
   Milan not claims that Mary knows no-one.ACC
   Intended: 'Milan doesn't claim that Mary knows anyone.'
   (BCS; Progovac 1994: 41)

2. Progovac further claims that another class of BCS pronouns, beginning with the prefix *i*, e.g., *i-(t)-ko* 'anyone', *išta* 'anything', *ikud* 'anywhere',

[4] With Comp I refer to complementizers, the functional category exemplified with English *that* or *whether*. Complementizers are heads of embedded clauses (CPs).

[5] What falls under syntactic movement is the licencing of NPIs in questions, conditionals, and some other constructions where inversion or movement can license the existence of the polarity operator in Comp.

[6] In Slavic languages, the local domain for the majority of anaphors (governed by Principle A) is a tensed clause, TP. See Büring (2005) for a good textbook on Binding Theory, which also discusses cross-linguistic data.

and the like are polarity items as well, in our terminology k-pronouns. K-pronouns are, according to Progovac, anaphoric pronominals. They are subject to Principle B: they need to be free in the local clause but bound in the sentence (licensed from a higher position) – after Progovac (1994: 64). Environments allowing only k-pronouns, not neg-words, are (according to Progovac): questions, conditionals, adversative predicates, restrictive clauses of universal quantifiers, and superordinate negation. For all these environments, she offers a syntactic explanation of the possible occurrence of k-pronouns (and the impossibility of neg-words): an operator in Comp (in all five environments) is supposed to license k-pronouns but is too non-local for neg-words. Some examples (after Progovac 1994) are shown in (11); the explanation is the same for them all (but see Progovac 1994: 64 for a full range of examples).

(11) a. Da li Milan voli *i(t)koga* / **ni(t)koga*?
that Q Milan loves anyone NEG.person.ACC
(Intended:) 'Does Milan love anyone?'

b. Akor Milan povredi *i(t)koga* / **ni(t)koga*, bi-će kažnjen.
if Milan hurts anyone.ACC NEG.person.ACC be.FUT punished
(Intended:) 'If Milan hurts anyone, he will be punished.'

c. Sumnjam da Milan voli *i(t)koga* / **ni(t)koga*.
doubt.1.SG that Milan loves anyone.ACC NEG.person.ACC
'I doubt that Milan loves anyone.'

d. Svako (t)ko povredi *i(t)koga* / **ni(t)koga*, mora bit kažnjen.
everyone who injures anyone.ACC NEG.person.ACC must be punished
'Everyone who injures anyone must be punished.'
(BCS; Progovac 1994)

3. Finally, Progovac analyzes BCS pronouns beginning with the prefix *ne*, e.g., *ne-(t)-ko* 'someone,' *nešto* 'something' as POSITIVE POLARITY ITEMS (PPIs). PPIs (*ne(t)ko, nešto*, etc.) are according to her pronominals (in the sense of binding theory). They are subject to Principle B and have to be free in their governing category. Data supporting PPI analysis of *ne* pronouns come from the grammaticality of these pronouns in positive sentences. Their

occurrence within the scope of superordinate negation allows for possible narrow scope (added formalization, MD) with respect to a superordinate negation, yet they obligatorily take wider scope than their clausemate negation, as argued by Progovac (1994).

(12) Milan je uvredio ne(t)koga.
Milan is insulted someone.ACC
'Milan has insulted someone.' (BCS; Progovac 1994)

(13) a. Marija ne smatra da je Milan uvrdio ne(t)koga.
Mary not thinks that has Milan insulted someone
'Mary does not think that Milan insulted someone.'
(BCS; Progovac 1994)

b. $\neg$THINK(MARY, p) $\wedge$ $p = \exists x$[PERSON(x) $\wedge$ INSULTED(MILAN, x)]

(14) a. Ne(t)ko nije došao.
someone NEG.is come
'Someone has not come.' (BCS; Progovac 1994)

b. i. Available: $\exists x$[PERSON(x) $\wedge$ $\neg$COME(x)]

ii. Unavailable: $\neg\exists x$[PERSON(x) $\wedge$ COME(x)]

To take stock: Progovac analyzes three classes of BCS expressions: neg-words, k-pronouns, and indefinite pronouns. She explains their various polarity sensitivity as a syntactic phenomenon: basically as the need of an expression to be bound either locally (neg-words) or non-locally (k-pronouns). In the third case (indefinite pronouns) as the obligation to be free locally (indefinite pronouns must be locally free but might be bound outside of their governing category). The negation acts as a binder for the first (and to some extent second class). The operator in Comp is the long-distance binder for the second class; she offers a similar analysis for English, though with parametric changes.

Progovac distinguishes only two classes of English expressions: (i) PPIs and (ii) NPIs. The items she considers uncontroversial PPIs for English are *some* or the temporal adverbial *already*. English PPIs (analogically to Slavic data discussed above) have to be locally free (Principle B). NPIs, like *any*, have to be bound locally (Principle A). There is an obvious difference between *any* and *ni-* BCS pronouns – *any* can be licensed across the clausal boundary (either by negation or by another DE operator). Such possibility to be licensed non-locally (against

Principle A) is, according to Progovac, the result of a language variation: the English type of NPIs can raise at LF, BCS NPIs cannot.

For English then, Progovac predicts that *any* can be bound either by a clausemate negation: (15a) or by an OP licensed from, e.g., a root clause with predicates that are not upward-entailing: (15b) – *be sorry* is a downward entailing predicate: if you are sorry that people kill animals, you are sorry that people kill pigs. This is not possible for Slavic neg-words since they cannot raise at LF; their clausemate negation must license them. English-type NPIs cannot be licensed only in a clause without negation and with a verb which is upward entailing like (15c); if you forgot keys, you forgot an object but not the other way round.[7] This all is correct and seems like a beautiful cross-linguistically sensitive syntactic theory of NPIs.[8]

(15) a. Peter didn't write any letter.
b. Peter is sorry that he said anything.
c. * Peter forgot anything.

But there are some problems. Progovac's approach predicts (as correctly pointed out by Krifka 1995: 212–213) that the English-type NPI can never be licensed in the non-clausal argument of a non-negated root clause (which works as an explanation of the ungrammaticality of (15c)). There is no Comp for an operator in the root clause, and there is no negation to license it either. But this prediction is wrong both for English and for Slavic languages. Consider the English example from Krifka (1995: 213), which was later modified by Błaszczak (2001) to show that Slavic-type NPIs (neg-words) can be licensed without any proper syntactic licensor in Progovac's sense, (16a). The same point can be made for k-pronouns as well: they are licensed by adversative predicates, usually licensing NPIs; see

[7]I follow here Krifka (1995) and his slightly Janus-face approach to *forget*: for Krifka, the clausal argument of *forget* is (at least) not upward entailing since "If Mary forgot that a woman came yesterday, she might not have forgotten that a person came yesterday", but the NP argument of *forget* is upward entailing since "if Mary forgot a poem by Goethe, then she forgot a poem, but not necessarily vice versa" (Krifka 1995: 212). That explains correctly why NPIs in the clausal arguments of *forget* are licensed (*I forgot to bring anything*), unlike in (15c). But this simply seems like a restatement of NPI licensing behavior in terms *forget*'s entailment profile, as one of the anonymous reviewers correctly remarks. It would be best to thoroughly test NPI-licensing patterns of *forget* across languages and then hopefully come up with the right theory of *forget*, but that has to remain a project for future work.

[8]I simplify here a bit. Progovac, in fact, observes that there are some cases of acceptable *any* in the non-clausal argument of non-negated root sentences, but she classifies such examples as FCI usage of *any*. Her explanation was then questioned by Horn & Lee (1995), Hoeksema (1996). Thanks to one of the anonymous reviewers for pointing out the debate.

the naturally-sounding Czech example in (17). (17) is a positive root sentence, so according to Progovac's approach, the k-pronoun should be free in this domain. Nevertheless, it is precisely the main predicate of the sentence which allows the grammaticality of the k-pronoun. But if the predicate is the licensor here, we have a Principle B violation pace Progovac (1993). At least in Czech, the set of predicates that license k-pronouns by themselves seems to overlap with adversative predicates. The most frequent collocations of positive verbs occurring with k-pronouns – from Czech National Corpus – are verbs as *zakázat* 'forbid,' *odmítnout* 'refuse' and *vyhýbat* 'avoid.' But there are also many examples of licensing by positive existential modals or imperatives. Such cases are not explainable as "inherently negative" in the style of adversative predicates; see Strachoňová (2016) for many valuable data points.

(16) a. John lacks any sense of humor.
b. John came without any present.

(17) Petr odmítal jakoukoliv pomoc.
Petr refused any help
'Petr refused any help.'

In this section, a historically very influential syntactic theory of NPIs licensing was discussed. There are obvious shortcomings of the theory, discussed partially above and further in Section 2.2. Maybe the most serious problem is the attempt to unify three Slavic classes of expressions under the roof of the Binding Theory. Today it seems more reasonable to treat neg-words in syntax. One of today's standard theories of neg-words is Zeijlstra (2004) – different in formalization but similar to Progovac (1994) in taking locality constraints seriously as syntactic in nature. And as for k-pronouns and indefinite pronouns, they ought to be explained via some appropriate semantic or pragmatic theory (already the problems of adversative predicates pointed out the difficulty of finding a reasonable syntactic licensor for k-pronouns or English NPIs in a purely syntactic framework like Progovac 1994). But despite these problems, Progovac (1994) is the first serious attempt to deliver a theory of various NPI classes, starting with Slavic data but aiming at a general framework of cross-linguistically conceived polarity licensing.

2.2 Semantic theories

The semantic approach to NPI licensing is still the standard theory of the semantic negative dependent expressions. There are various ways of approaching the idea that NPIs are licensed purely semantically. The most widely accepted reasoning is based on the downward entailing approach of Ladusaw (1992), but the idea can be found in influential works of Heim (1984), Ladusaw (1992), Kadmon & Landman (1993), Krifka (1995), Giannakidou (1997), Lahiri (1998) too. It usually starts with the observation (introduced quickly already in Section 1) that next to negation, there are many other prototypical NPI licensing contexts. Such contexts include downward entailing quantifiers, antecedents of conditionals, the scope of the exclusive particle *only*, adversative predicates, comparatives, and superlatives. All the contexts are claimed to share the property of reversing the direction of entailment. The monotonicity reasoning is demonstrated in the predicate logic implications in (18). In the non-negated formula (18a) (corresponding to a natural language positive sentence under (18b)), the entailment goes from a subset (intersection of P and Q) to a superset (union of P and Q). In a negated formula (18c) (and corresponding natural language sentence under (18d)), the entailment is reversed and proceeds from a superset (union) to its subset (intersection). The same entailment reversal can be observed in the difference between upward-entailing quantifiers as *some* versus downward entailing quantifiers like *few* demonstrated in (19).

(18) a. $\exists x[P(x) \land Q(x)] \rightarrow \exists x[P(x) \lor Q(x)]$
b. $\exists x[\textsc{red}(x) \land \textsc{wine}(x)] \rightarrow \exists x[\textsc{red}(x) \lor \textsc{wine}(x)]$
John likes red wine. → John likes wine.
John likes wine. ↛ John likes red wine.
c. $\neg\exists x[P(x) \lor Q(x)] \rightarrow \neg\exists x[P(x) \land Q(x)]$
d. $\neg\exists x[\textsc{red}(x) \lor \textsc{wine}(x)] \rightarrow \neg\exists x[\textsc{red}(x) \land \textsc{wine}(x)]$
John doesn't like wine. → John doesn't like red wine.
John doesn't like red wine. ↛ John doesn't like wine.

(19) a. Some people drank red wine. → Some people drank wine.
b. Some people drank wine. ↛ Some people drank red wine.
c. Few people drank red wine. ↛ Few people drank wine.
d. Few people drank wine. → Few people drank red wine.

The semantic approach is very successful: it explains the distribution of so-called weak NPIs (like English *any*; more on different types of NPIs in Section 2.2.2) via a unified semantic property. The property of NPI licensing is that they occur in downward entailing (entailment reversing) environments (but see Section 2.2.1 for many qualifications). Despite some problems discussed below, the DE-based explanation of NPI licensing is still considered a benchmark of semantic reasoning and appears in standard textbooks on formal semantics (Portner 2005, Coppock & Champollion 2024; among many others). The proper formalization of DE licensing can be stated as (20), and DE is defined in (21), both after von Fintel (1999). The explanation of weak NPI acceptability in a sentence like *John didn't drink any wine yesterday* (contrasted with ungrammaticality of **John drank any wine yesterday*) is as follows: (22) states the entailment pattern for *red wine* and *wine* (x and y from (21)), subset relation on sets corresponds to the entailment in (21) as $\forall x[[\textsc{red}(x) \wedge \textsc{wine}(x)] \Rightarrow \textsc{wine}(x)]$. But negation ($F$ from (21)) reverses the sub-set superset relationship: $\forall x[\neg\textsc{wine}(x) \Rightarrow \neg[\textsc{red}(x) \wedge \textsc{wine}(x)]]$. And because of this DE entailment reversal, *any* is licensed as stated in (20).

(20) Fauconnier-Ladusaw's Licensing condition (von Fintel 1999: 100)
An NPI is only grammatical if it is in the scope of an α such that $[\![\alpha]\!]$ is DE.

(21) Downward Entailment (von Fintel 1999: 100)
A function f of type $\langle\delta,\tau\rangle$ is downward entailing iff for all x, y of type δ such that $x \Rightarrow y$: $f(y) \Rightarrow f(x)$ ['$\Rightarrow$' stands for cross-categorial entailment].

(22) $\{x : x \text{ is a red wine}\} \subseteq \{y : y \text{ is a wine}\}$

2.2.1 Some problems of purely semantic theories

Even if the criterion in (20) is very successful, many problems appeared during years of NPI exploration. The problems have led to some ramifications (and in some cases abandonment or replacement of the criterion; see Giannakidou 1997, Linebarger 1987). Very famous problems arise when we scrutinize monotonic properties of *only*, conditionals, superlatives, and adversative predicates. All these expressions seem to license NPIs but do not fit the defining property of downward entailment, at least at first sight. A proper exposition of such problems goes beyond the scope of this chapter; see von Fintel (1999) for the classic reference and Gajewski (2011) among many others for a recent refinement.[9] But let me demonstrate the nature of the problems of simple DE theory with

[9] As one of the anonymous reviewers correctly notes, von Fintel (1999) is just one of the attempts (even if very influential and empirically successful) to overcome the limitations and problems of a purely semantic approach to NPIs. Historically sorted, at least the following researchers – Heim (1984), Kadmon & Landman (1993), Israel (1996) – proposed various pragmatic relativizations of the entailing conditions necessary for NPI licensing, be it contextual entailment, fixation of contexts, or integrating background knowledge of communication participants into the reasoning important for NPI licensing.

an example. Despite the validity of the sentential logic tautology in (23a), conditionals in natural language seem not to allow the strengthening of antecedents. Consider counter-intuitive reasoning from (23b) to (23c), but even if such reasoning can sound paradoxical when viewed intuitively, it would be formalized in classical logic as the correct strengthening in (23a). As a consequence, it seems that natural language conditionals do not exhibit downward entailing property, but despite that, they are one of the prototypical NPI licensors. Similar problems were observed with the other contexts mentioned above (adversative predicates, superlatives, …) too. Even if the environments are not intuitively downward entailing, they license NPIs like *any*: see (24).

(23) a. $(p \rightarrow q) \rightarrow ((p \wedge r) \rightarrow q)$
b. If [you pet a dog] it will be happy. $\nrightarrow$
c. If [you pet a dog and kick it] it will be happy.

(24) a. If you pet any dog, it will be happy.
b. I am sorry to give you any trouble.
c. Only Peter drank any tea.

The analysis of the problems concerning conditionals (and other problematic environments) proceeds in the following way (after von Fintel 1999). First, the basic notion of downward entailment is replaced by Strawson downward entailment (SDE), and SDE adds to the standard argument a presupposition of the sentence. Second, the added presupposition plus the at-issue meaning of the sentence then qualify for the downward entailing property as described in (21).[10] For conditionals like (23b), the added presupposition would concern the current modal horizon. Modal horizons are sets of worlds that are compatible with everyday reasoning about conditionals. For (23b), possible worlds where an agens both pets and kicks a dog are beyond the current modal horizon. Consequently, adding a presupposition to the effect that the current modal horizon is compatible only with worlds where only petting a dog would cause happiness of a dog would save the downward monotonicity of conditionals and explain the grammaticality of an NPI in (24).

This gives an impressionistic introduction to SDE reasoning, but let us demonstrate the SDE on a simpler pattern since a proper treatment of reasoning in conditionals is surely beyond the scope of this article. Gajewski (2016) offers exactly such a type of example; he discusses the cases of NPI licensing by plural

[10] The formal definition of SDE is in (27).

definite NPs like in (25). But the entailment in (26) is invalid since knowing that the supremum of (in the context) salient students drank beer does not tell us anything about their nationality. Exactly a similar problem as discussed above with respect to the pattern in (24) is encountered – expressions that are not DE do license NPIs.

(25) The students who have any beer are sharing it. (after Gajewski 2016: ex. 2)

(26) The students drank beer. $\nrightarrow$ The German students drank beer.

The exact definition of Strawson downward entailment is in (27). In (26), the conclusion wasn't entailed by the premise. But according to (27), the presupposition of the conclusion (the existence and uniqueness presupposition of the definite description in conclusion) should be granted as a premise in the argument as demonstrated in (28). Then the argument becomes valid, although the price is obvious: the entailment must take into account pragmatic notions like presupposition. Despite that, SDE has been successfully applied to many cases of NPI licensing in environments that simply are not DE, so it is a useful linguistic tool.

(27) Strawson Downward Entailment (after von Fintel 1999: 104)
A function f of type $\langle\sigma,\tau\rangle$ is Strawson-DE iff for all x, y of type σ such that $x \Rightarrow y$ and $f(x)$ *is defined*: $f(y) \Rightarrow f(x)$.

(28) The students drank beer.
There are salient German students.
$\vDash$ The German students drank beer.

But even if the solution via Strawson downward entailment is widely accepted today, there still remain problems that are not well-understood. One of them (important later – see Section 2.4.2) is the following: von Fintel (1999) and his SDE predicts that adversative predicates as *sorry, regret, amazed, difficult,* or *refuse* can license NPIs because adding a (usually factive) presupposition un-masks their hidden DE properties. But predicates like *want, glad,* or *like* cannot license NPIs because they are not even SDE (they are usually analyzed as upward monotonic but see von Fintel 1999 for qualifications). The prediction seems to be mostly right: compare the contrast between (29a) and (29b). But there is a special type of context (discussed by Kadmon & Landman 1993 already) where *glad* can license NPIs. Such contexts permitting NPIs under *glad* have a special interpretation called "settle for less" by Kadmon & Landman (1993). The core description of such contexts demonstrated in the example (30) is the following: the speaker wanted an alternative (higher) number of tickets, but because he considered the situation so dire that even lousy tickets were the only possibility, he settled for less than he would normally want. For some reasons (not totally clear, but see Alonso-Ovalle 2016 for some recent insights), it seems that other adversative predicates like *want, hope,* or *wish* at least in English do not allow the settle-for-less reading and consequently cannot license NPIs. Nevertheless, to sum up, the current sub-section: the semantic approach (if enriched with a pragmatic notion like SDE) can deal with the majority of more complicated cases where allegedly non-downward entailing expressions license NPIs (and which were correctly pointed out already by Linebarger (1987) as not simply downward entailing).

(29) a. I regret there is any idea like that.
b. # I am glad there is any idea like that.

(30) I am glad that we got any tickets.

2.2.2 Types of NPIs

Another step in broadening the empirical coverage of semantic NPI theories was a seminal work of Zwarts (1998) where various types of NPIs are, for the first time systematized, and a purely semantic explanation of their differing distribution is offered. This classification of NPIs was especially important from the perspective of Slavic languages as it inspired the most thorough semantic approach to Slavic NPIs: Błaszczak (2001) and some following works of her like Błaszczak (2003) and Błaszczak (2008); Błaszczak's contributions are discussed in Section 2.2.3. Consider the first examples of the two most prominent classes of NPIs: weak in (31) and strong in (32). As evident, only negation and negative quantifiers license STRONG NPIs (such as *until midnight* in (32)), while for the licensing of WEAK NPIs as *any*, the downward entailing environment is enough; see (31).

(31) a. John didn't leave any message.
b. No teacher left any message.
c. Few teachers left any message.
d. At most ten teachers left any message.
e. # Some teachers left any message.

(32) a. John didn't leave until midnight.
b. No teacher left until midnight.

c. # Few teachers left until midnight.
d. # At most ten teachers left until midnight.
e. # Some teachers left until midnight.

The logical property which licenses strong NPIs share is a strengthened form of entailment reversal and usually is named ANTI-ADDITIVITY.[11] The definition of anti-additivity is in (34): it represents a half of the well-known De Morgan's laws of conjunction/disjunction transformation via negation (Tarski 1994: 46). Standard logical negation obeys De Morgan's two laws in (33).

(33) a. $\neg(p \vee q) \leftrightarrow \neg p \wedge \neg q$
b. $\neg(p \wedge q) \leftrightarrow \neg p \vee \neg q$

Anti-additivity picks one of them, so we can view anti-additivity as a subset of negation properties. (35) illustrates the anti-additivity (the quantifier *no* is anti-additive since negation is always anti-additive, as is clear from De Morgan's laws). But DE quantifiers like *few* in (36) are not anti-additive – imagine a scenario with 10 students, three of them drinking and three of them smoking, then $\vee$ part of (36) is false while $\wedge$ part of (36) is true. That explains why *no* is licensed by the strong NPI *until midnight* in (32) and not licensed by just downward entailing quantifier *few*.[12]

(34) Anti-additive function: $F(x \vee y) \leftrightarrow F(x) \wedge F(y)$

(35) No student smokes or drinks.
$\leftrightarrow$ No student smokes and no student drinks.

(36) Few students smoke or drink.
$\nleftrightarrow$ Few students smoke and few students drink.

[11] A popular alternative explanation of strong NPIs, and their behavior can be found in Gajewski (2011). Gajewski describes their stricter distribution via downward entailing properties but checked both in at-issue meaning and in the presupposition/implicature part of the meaning. More pragmatic approaches to NPI licensing will be presented in Section 2.3.

[12] As one of the reviewers correctly remarks, anti-additivity as presented in this section and its contrastive logical behavior with respect to DE is just a part of the whole picture, presented carefully and systematically in Zwarts (1998). Zwart's main idea then is that there is a hierarchy of negative strength: downward entailing < anti-additive < anti-morphic. Anti-morphicity is defined as: F is anti-morphic iff $F(\neg p) \leftrightarrow \neg F(p)$. Classical verbal negation is, of course, anti-morphic, since $\neg(\neg p) \leftrightarrow \neg\neg p$ but universal quantifiers, which are anti-additive, since $\forall x((P(x) \vee Q(x)) \rightarrow R(x)) \leftrightarrow \forall x(P(x) \rightarrow R(x)) \wedge \forall x(Q(x) \rightarrow R(x))$, are not anti-morphic because $\neg\forall x(P(x) \rightarrow Q(x)) \nleftrightarrow \forall x(P(x) \rightarrow \neg Q(x))$. The relative strength of the licensor is, for Zwarts (1998), picked up by weak, strong, and superstrong classes of NPIs, respectively.

2.2.3 The semantic theory applied to Slavic data

Błaszczak (2001) and other works by her apply this classification of NPIs to Slavic data. First, Błaszczak observes that, generally, k-pronouns (Polish pronouns with the suffix *-kolwiek*) are licensed in prototypical NPI contexts. Such contexts include higher clause negation, questions, conditionals, the scope of downward entailing quantifiers, comparatives, adversative predicates, *before*-clauses, etc. Neg-pronouns (Polish pronouns usually beginning with *n-*) are generally licensed only by clause-mate negation: see (37) and (38).

(37) a. * Ewa nie chciała, żeby Jan nikogo zapraszał.
Eve NEG wanted that.SBJV John NEG.person invited
Intended: 'Eve didn't want John to invite anyone.'
(Polish; Błaszczak 2003: 2)

b. Ewa chciała, żeby Jan nie nikogo zapraszał.
Eve wanted that.SBJV John NEG NEG.person invited
'Eve wanted John not to invite anybody.'
(Polish; Błaszczak 2003: 2)

(38) a. Czy widziałeś tam kogokolwiek?
whether saw.2SG there anybody
'Have you seen anybody there?'

b. Jeśli ktokolwiek przyjdzie, daj mi znać.
if anybody comes let me know
'If anyone comes, let me know.'

c. Ewa nie chciała, żeby Jan kogokolwiek zapraszał.
Eve NEG wanted that.SBJV John anybody invited
'Eve didn't want John to invite anyone.'
(Polish; Błaszczak 2003: 2)

This is a familiar observation concerning the partial complementary distribution of two classes of Slavic pronouns, which was discussed already in Section 2.1 dedicated to the syntactic approach of Progovac (1994). But Błaszczak's (2001) solution is different, and her critique of Progovac (1993) is correct. Błaszczak (2001) criticizes syntactic approach of Progovac (1993) mainly on empirical grounds. One of her main arguments concerns the licensing of neg-pronouns by a preposition *bez* 'without.' The grammaticality of (39) is unaccounted by the syntactic approach of Progovac since there is no negation or polarity operator in Comp. But since (at least in Polish) neg-pronouns are fully licensed in the complement of *bez* 'without,' Błaszczak concludes that the semantic approach is more in accordance with the Slavic data. For more general problems of Progovac's syntactic theory of NPI licensing, see Section 2.1.

(39) Został bez nikogo.
was-left.3SG without NEG.person
'He was left without anyone.' (Polish; Błaszczak 2003)

The core of Błaszczak's (2001) proposal concerning Polish NPIs is the following:

1. K-pronouns are licensed in downward entailing or anti-additive environments. So in contrast to the syntactic binding-theoretic analysis of Progovac (1993), we see the standard semantic treatment of NPIs, which is more in accordance with the variability of NPI licensors: quantifiers, polarity operators, various types of embedding, etc., which cast doubt on any purely syntactic approaches to NPIs.

2. According to Błaszczak (2001), neg-pronouns can occur only in anti-morphic environments.[13] The definition of the anti-morphic functor (adding one condition on top of anti-additivity) is in (40). An anti-morphic function is a classical negation because it obeys both requirements of (40) (De Morgan's laws). And it seems that only clausal negation fits such a requirement in a natural language. And because of its nature, it also satisfies another condition in (41): first negation before the equation sign in (41b) corresponds to f in (41a). But notice that anti-morphic predicatees are also negated truth-value ascribing predicates and neg-raising predicates, as in *it is not true that…* or *x does not believe that…* Both contexts will be discussed at the end of the current section.

Applied to examples in (37): a negated modal verb does not license neg-pronouns in its embedded clause, since from $\neg\forall[P]$ it does not follow $\forall[\neg P]$. Nevertheless, the ungrammaticality of (37) is not caused by the interfering modal since modals generally do not interrupt NPI licensing. Verbal negation, of course, licenses the clausemate neg-words. For k-pronouns in (38), the semantic analysis predicts that they should be grammatical in anti-additive conditionals (they are, in fact, Strawson anti-additive, see Gajewski 2011) or in a simple downward entailing context like a negated embedding modal verb.

[13] As one of the reviewers correctly points out, it is not clear whether this claim is valid even for sentences with ≥ 2 occurrences of neg-words, since the scopally lowest neg-word is in the anti-additive environment, but the sentence is still grammatical.

(40) A functor F is anti-morphic iff:

a. $F(x \wedge y) \leftrightarrow F(x) \vee F(y)$

b. $F(x \vee y) \leftrightarrow F(x) \wedge F(y)$

c. $\neg(x \wedge y) \leftrightarrow \neg(x) \vee \neg(y)$

d. $\neg(x \vee y) \leftrightarrow \neg(x) \wedge \neg(y)$

(41) a. $f(\neg x) = \neg f(x)$

b. $\neg(\neg x) = \neg\neg(x)$

The semantic approach is much more appropriate than the syntactic approach, especially in its treatment of k-pronouns. The set of environments where k-pronouns appear forms a natural semantic class. But there are some problems with the semantic approach too. Some of them were observed in recent general linguistic work, which usually argues for a semantic and pragmatic approach to NPIs instead of a purely semantic one (see Krifka 1995, Gajewski 2011, Chierchia 2013; among many others). Nevertheless, the standard approach today is to treat neg-words as a separate class of negative dependent expressions licensed in syntax (see already Krifka 1995 for some insightful remarks and Zeijlstra 2004 for the current *de facto* standard theory of neg-words). But let us discuss some empirical problems which are particularly interesting and concern Slavic data:

1. As (42a) shows, only NPIs are licensed in neg-raising contexts; real neg-words are not acceptable there (see Dočekal & Dotlačil 2016a for the experimental data confirming this claim). But neg-raising predicates are anti-morphic even in their embedded clauses, so a purely semantic approach predicts that both NPIs and neg-words should be licensed there (contrary to facts). This asymmetry raises serious problems for any semantic treatment of neg-words.[14]

[14]One of the anonymous reviewers correctly remarks that the status of neg-raising predicates as either anti-morphic or as anti-additive is, to some extent, theory-dependent. In approaches where the negation would really be present syntactically and semantically in the embedded clause, they would be anti-morphic, but for Gajewski (2005, 2007), the embedded clauses of neg-raisers are just anti-additive. Gajewski (2007: 305) empirically supports his classification with the following entailment pattern where (for him) the first entailment is valid, but the second one is not. And since, for him, the crucial test for anti-morphicity is the entailment $f(x \wedge y) \rightarrow f(x) \vee f(y)$, he argues for the mere anti-additive status of neg-raising embedded clauses.

(i) John doesn't think Mary left and John doesn't think Bill left.
$\rightarrow$ John doesn't think Mary left or Bill left.

2. The same point holds for truth-value ascribing predicates: (42b) and (42c) are equivalent, but only (42e) allows negative concord items. Again a purely semantic theory of neg-words is the problem here, while syntactic locality constraints fare better.[15]

(42) a. Petr nechce, aby {ani jeden student / #nikdo} propadl.
Petr NEG.wants COMP even one student NEG.person failed
(Intended:) 'Petr doesn't want even one student/anyone to fail.'

b. It's not true that Petr was sleeping. ↔

c. It's true that Petr wasn't sleeping.

d. * Není pravda, že nikdo přišel.
NEG.is true that NEG.person came
Intended: 'It's not true that nobody came.'

e. Je pravda, že nikdo nepřišel.
is true that NEG.person NEG.came
'It's true that nobody came.'

Let us summarize the current section. The relationship between verbal negation and the morphological negation on neg-words is a syntactic phenomenon; trying to explain it in semantics leads to empirical problems. There are NPIs in Slavic languages, though, which seem to behave in the way described by Błaszczak (2001), namely as anti-morphic NPIs (Czech *ani jeden* is an example of such an NPI). But the Slavic neg-words are (it seems from today's perspective) better delegated to purely syntactic rules.

(ii) John isn't certain that Mary left and John isn't certain that Bill left.
↛ John isn't certain that Mary left or Bill left.

I'm not sure whether his argument is very strong since the predicate logic implication $\neg\forall x(Px \wedge Qx) \rightarrow (\neg\forall x Px \vee \neg\forall x Qx)$ is valid. Recent work on necessity modals (Agha & Jeretič 2022) opens up a possibility to formalize certain necessity modals not as universal quantifiers. Going in this direction would weaken Gajewski's argument even more. Nevertheless, following this route of argumentation would be intriguing but would lead us too far away from the merit of the current chapter.

[15] Gajewski (2007: Appendix to Sect. 2) offers a semantic treatment of the fact that strict NPI is not licensed in the clauses embedded under *it's not true that* predicates. The proposal works with the idea of presupposition-canceling properties of *true*; see Gajewski (2007: 308–310) for details. Thanks to one of the anonymous reviewers for reminding me of this passage.

2.3 Pragmatic theories

After summarizing two types of NPI theories applied to Slavic data in the previous sections (a syntactic theory in Section 2.1 and a semantic one in Section 2.2), I will focus on the last type of approach to NPIs – a pragmatic one. The pragmatic approach has not been systematically applied to Slavic languages (at least not to the same extent as semantic and syntactic theories). But there is some work in this direction (as Tomaszewicz 2013, Gajić 2018, 2022, Dočekal & Dotlačil 2016b or Dočekal & Šafratová 2019). But let us start with the general-linguistic properties of this approach. Currently, it seems that the pragmatic theories of polarity constraints are *de facto* standard, one of the recent formalizations, namely Chierchia (2013), is probably the most widely used framework in the area. One of the applications of Chierchia (2013) to Slavic data can be found in Gajić (2018). For related work discussing exclusive particles in Polish, see Tomaszewicz-Özakın (2026 [this volume]). The shared idea of all pragmatic theories of NPI is that in describing polarity effects, we have to consider both the semantic properties of the expressions under consideration and general pragmatic mechanisms as well. The formalizations usually agree on the idea that NPIs introduce alternatives, and polarity effects emerge when something goes wrong during the computation of alternatives and their inclusion into the truth conditions. The execution of these ideas can then proceed either via the postulation of exhaustification operators present in syntax (like Chierchia 2013) or in the neo-Gricean spirit where the pragmatic mechanisms are more related to general rules of rational communication.

In particular, I will follow the NEO-GRICEAN PRAGMATIC APPROACH to NPIs as introduced by Krifka (1995) and followed by many others (especially the work of Crnič 2011 and Alonso-Ovalle 2016 will be of most relevance). But for the present purposes, nothing hinges too much on the adopted framework, and it seems to me that it would be possible to reformulate the present section in the framework of Chierchia (2013). Nevertheless, let us first illustrate the typical problems pragmatic (unlike syntactic or semantic) theories can deal with. Consider the Czech SCALAR PARTICLES (SP) *i* and *ani*, which have a complementary distribution to some extent and resemble the complementary patterns described by Progovac (1993) and Błaszczak (2001). Until now, I have described the polarity properties of NPIs, but from now on, I will focus on scalar particles, which represent an extension of the polarity landscape we discussed so far. Czech scalar particle *ani* behaves to some extent like a strong NPI requiring negation, but simple anti-additivity is not enough to license its appropriate occurrence. In the context of the familiar three books of Tolkien's *Lord of the Rings,* consider its acceptability (for experimental data supporting the following judgments, see Dočekal & Šafratová 2019). As (43a) shows, *ani* cannot be licensed in downward entailing contexts only, but even the anti-additive operator is not able to license all possible occurrences of *ani*: (43b). As the example shows, *ani* associates only with the bottom of the scale conveyed by the scalar expression (the contextual scale being ⟨first volume, second volume, third volume⟩). Association with the top of the scale leads to *ani*'s unacceptability. The need for a pragmatic explanation, intuitively put, is the following: reading the first volume is the most likely case, the second less likely, and reading the third one is the least likely. *Ani* then requires not only a particular type of monotonicity but also ranking over the probability of alternatives where alternatives (required by *ani*) come from focus computation. Similarly, as can be seen from the contrast in (43c), *i* shows (at first sight) just the opposite pattern to *ani*: it occurs in positive sentences, and pragmatically it can associate only with the top element of a contextual scale. When it occurs in a negated sentence, it cannot associate with the top or bottom elements of a scale; see (43d).[16] Again, monotonicity alone is not enough to license its grammaticality. In summary: there are both semantic and pragmatic requirements of both *i* and *ani*. Technical explanation of these kinds of patterns follows in Section 2.3.1.

(43) a. # Málo lidí přečetlo ani {první / třetí} díl Pána prstenů
few people read not.even first third volume Lord Rings
Intended: 'Few people have even read the first/third volume of the Lord of the Rings.'

b. Petr nepřečetl ani {první / #třetí} díl Pána prstenů.
Petr NEG.read not.even first third volume Lord Rings
'Petr didn't read even first/third volume of the Lord of the Rings.'

c. Petr přečetl i {#první / třetí} díl Pána prstenů.
Petr read even first / even third Lord Rings
'Petr read # even the first / third volume of Lord of the Rings.'

d. # Petr nepřečetl i {první / třetí} díl Pána prstenů.
Petr NEG.read even first third volume Lord Rings
Intended: 'Petr have even read the first/the third volume of Lord of the Rings.'

[16]Example (43d) is unacceptable under the scalar interpretation. But since *i* can be used also as an additive particle, there can be non-scalar contexts where (43d) would be acceptable. Thanks to Radek Šimík for this point.

2.3.1 Crnič's decompositional theory of scalar particles

I will introduce the pragmatic theory of Crnič (2011) where the constraints on scalar particle distribution and their semantic properties are explained via a pragmatic theory of presuppositions. The presuppositions are contributed by NPIs or scalar particles as *i*/*ani*. The description of scalar particles is related to Crnič's (2011) pragmatic approach to English *even* as a polarity item. But especially in chapter 4 and chapter 5, where he describes scalar particles, he pays close attention to cross-linguistic variation and discusses many intriguing properties of Slovenian scalar particles. So unlike in Section 2.1 and Section 2.2, where I summarized two contributions to polarity research explicitly focused to (mostly) Slavic data, in this section, I will introduce a more general framework. The framework was only partially applied to Slavic data more as a form of demonstration. This, of course, has some consequences; the nice one is that Crnič's framework is an explicit formal theory of presupposition and polarity effects of NPIs/scalar particles. The worse one is that the approach still lacks proper empirical meat, and naturally concerning Slavic data (even if it offers many insights) there are many points where it has to be fine-tuned or slightly changed to be in accordance with the data. Be it as it may, another strong aspect of Crnič's theory is its further component which describes the distribution of scalar particles as determined partially by the spectrum of the relevant scalar particles in the particular language. In other words, there is a competition between various types of scalar particles, and their behavior is a result of two factors. The first factor is the calculation of presuppositions. The second factor is a competition between various expressions available for the lexical insertion.

2.3.2 Three basic types of scalar particles

I will start by introducing fundamental dichotomies in the territory of scalar particles. I will deal mostly with so-called weak and strong scalar particles (e.g., German expressions *sogar* and *auch nur*). And I will only scratch the surface of concessive scalar particles like Slovenian *magari* or Spanish *siquiera*.

2.3.2.1 Nondiscriminating scalar particles The first type of scalar particles is the one associating both with strong and weak elements in their immediate surface scope. The most famous linguistic example is English *even*: its strong association is free, (44a), and it can associate with weak expressions in non-upward monotonic embedding too: (44b). There are no known Slavic examples of such nondiscriminating type of *even*, Crnič (2011: 129) cites French *même* as a second example, next to English *even* (the subscript *F* indicates focus).[17,18]

(44) a. John drank even ten$_F$ beers.
  b. Paul didn't drink even one$_F$ beer.

2.3.2.2 Strong scalar particles The second class of particles, STRONG SCALAR PARTICLES, comprises particles which associate with strong scalar elements. This class is well documented for many natural languages, Crnič (2011: 130) discusses German *sogar* and Slovenian *celo*. We can add the Czech scalar particle *i* introduced already above: see a grammatical association of *i* with a strong element in (45) where an unlikelihood presupposition of *i* requires the prejacent to be less likely than the alternatives (drinking $n < 10$ beers) which intuitively is the case (formalizations below).

(45) (V těch letech) Petr vypil i deset$_F$ piv.
  in those years Petr drank even ten beers
  '(In those years) Petr drank even ten beers.'

In Crnič's theory, strong scalar particles seem to be superficially stuck in their base position when they would appear in a downward entailing environment. Otherwise, they would be grammatical contrary to natural language data, see ungrammatical (46a) with the un-acceptability explaining formalization in (46b). If the scope of *i* would be above negation ([i [not [Petr drank one$_F$ beer]]]), the unlikelihood presupposition of *i* would be satisfied.

(46) a. * (V těch letech) Petr nevypil i jedno$_F$ pivo.
  in those years Petr NEG.drank even one beer
  Intended: '(In those years) Petr didn't drink even one beer.'
  b. [not [i [Petr drank one$_F$ beer]]]

[17] It remains to be worked out whether there is no example of Slavic nondiscriminating scalar particles or whether it so far remained hidden. One of the anonymous reviewers suggests that Polish *nawet* 'even' can be used both with strong and weak associated expressions. Similarly, in Czech the scalar particle *dokonce* 'even' seem to associate with both ends of the scale. Nevertheless, as natural occurences suggest, *nawet* is out of the blue more acceptable with weak elements, for strong elements it usually requires some intervening modal or other operator. Czech *dokonce* behaves just the other way round, it occurs most naturally with strong elements. In summary, the proper classification of promissing candidates for Slavic nondiscriminating scalar particles remains a future project.

[18] Out of context, (44a) can sound a bit strange in English but as the following sentence from Sketchengine (Kilgarriff et al. 2014) English Web 2021 shows, English *even* associates with the strong scalar elements: *Some workers even drank 3–4 liters of the contaminated water after mine managers failed to warn them.*

2.3.2.3 Scalar particles that may associate only with weak elements The last type of particles I will write about, namely WEAK SCALAR PARTICLES, are particles which associate with weak elements. Unambiguous examples are German *auch nur*, *einmal*, English *so much as* and Slovenian *niti*. English *even* can be used as a weak particle too, but in that case, it must be embedded under non-upward-entailing operator (other weak scalar particles can appear only in non-upward entailing contexts too but they – unlike English *even* – are limited to such environments). Linguists approaching scalar particles from the typological perspective (like Crnič 2011: chapters 4 & 5 and Gast & Van der Auwera 2011) sub-classify weak scalar particles according to their possible scope under negation or other non-upward-entailing operators. Most constrained are scalar particles like German *einmal* and Slovenian *niti*: they appear only in the immediate scope of negation. The second subtype comprises scalar particles which occur in the scope of the non-upward-entailing operator but may not occur under negation, German *auch nur* or Slovenian *tudi* are scalar particles of this sort. Last, there are nondiscriminating weak scalar particles, like English *so much as* which may occur both in the immediate scope of negation and in other non-upward-entailing environments as well.

2.4 Application of Crnič's theory to Czech NPIs

In the previous section, Section 2.3.2, three kinds of scalar particles were introduced. The first type of scalar particles, nondiscriminating scalar particles, was not found in Slavic languages. The second type of scalar particles, strong scalar particles, is represented by Czech *i* 'even,' and Slovene *tudi* represents the third type of scalar particle. The current section demonstrates the application of Crnič's theory to Czech scalar particles. Both strong and weak scalar particles are represented in Czech, and the theory can be applied to both types of scalar particles. Crnič's theory is taken as a starting point but is not applied in its original form. The main goal of the current section is to show that Crnič's theory can be applied to Czech scalar particles, and the primary aspiration of this section is to open research possibilities for the future. I try to resolve some tensions between the theory and the data, but I do not claim that the theory fully applies to Czech scalar particles. The same is valid for applying Crnič's theory to Slavic languages in general.

Now, let us illustrate the classification with three types of Czech scalar particles corresponding to strong, weak, and negation-requiring weak particles respectively. Following the introductory discussion of *i*, let us consider its type: *i* like German *sogar* requires association with strong elements: if the contextual scale is 1 to 10 beers, then only numerals contextually vaguely around 10 would make an appropriate focus associate of *i*, see (47) with acceptable strong associate and un-acceptable weak associate.

(47) Petr vypil i {#jedno$_F$ pivo / deset$_F$ piv}.
Petr drank even one beer ten beers
'Petr drank even #one/ten beers.'

The second type of Czech scalar particles, are expressions of type *byť jediný* 'at least/even one,' they generally appear in Strawson downward-entailing contexts like an antecedent of conditionals, under super-ordinate negation or generally in the scope of non-upward-monotonic operators like imperatives and some adversative predicates. Depending on their embedding, their correct glossing to English is either 'at least one' or 'even one.' The expression seems to be frozen in cardinality, so is predestined for the weak association: consider (48), unlike *i* which (depending on the type of scale) can associate with different kinds of expressions. Nevertheless, the expression *byť jediný* 'even one' is not an idiom as witnessed by various sortal version of the numeral included in it: it can count cardinality of objects like in (48) but also cardinality of events (*byť jednou* 'even once') or cardinality of a sequence of occurrences like *byť poprvé* 'even for the first time'.

(48) Jestli Petr vypije byť {jediné pivo / #deset piv}, tak bude tančit.
if Petr drink just one beer ten beers then will dance
'If Petr drinks just one beer/#just ten beers he will dance.'

The distinction between *i* and *byť jediný* follows the typology of strong and weak scalar particles assumed in Crnič (2011), Gast & Van der Auwera (2011) and can be formalized as lexical items competing for lexical insertion using a pattern in (49). The scale in (49) uses SEMANTICALLY INTERPRETED FEATURES formalized below but intuitively understandable as follows: EVEN is a feature responsible for the unlikelihood presupposition in examples like (45). SOLO is a feature formalizing the presupposition of likelihood in sentences like (48), where the weak scalar item is embedded in a Strawson downward entailing environment. The presuppositions of EVEN and SOLO are contradictory: the first requires unlikelihood against alternatives, the second one dictates the likelihood of the prejacent. Their conflict can be resolved only in environments reversing entailment.

(49) a. ⟨*i*, *byť jediný*⟩
b. ⟨[EVEN], [EVEN][SOLO]⟩

The features EVEN and SOLO are semantically interpreted in (50) and (51) following Crnič (2011: 134). The part of both formulas, $p \lhd_C q$, means that in the context C (focus induced alternatives) the proposition p is at most as likely as the proposition q. The mechanism works under the assumption introduced to linguistics by Lahiri (1998): that there is a one-way relationship between logical strength and entailment. The relationship is: logically stronger propositions cannot be more likely than logically weaker propositions. Intuitively from the conjunction of two propositions *It rained, and it hailed* any of the propositions follows ($(p \land q) \rightarrow p / (p \land q) \rightarrow q$). The conjunction of the two propositions is logically stronger and less likely than any of its atomic propositions. More formally is the relationship between likelihood and entailment stated in (52).

(50) $[\![\text{EVEN}]\!]^{g,c}(C, p, w)$ is defined only if $\exists q \in C[p \lhd_c q]$.
If defined, $[\![\text{EVEN}]\!]^{g,c}(C, p, w) = 1$ iff $p(w) = 1$

(51) $[\![\text{SOLO}]\!]^{g,c}(C, p, w)$ is defined only if $\forall q \in C[q \neq p \rightarrow q \lhd_c p]$.
If defined, $[\![\text{SOLO}]\!]^{g,c}(C, p, w) = 1$ iff $p(w) = 1$

(52) If $p \rightarrow q$ then $p \lhd_c q$.

I will now apply Crnič's (2011) pragmatic framework to Czech with a simple illustration of possible derivations. First, strong scalar particles as Czech *i* in an appropriate context like (47) would obtain a correct LF like (53) (in pseudo-Czech). The sentence is appropriate in all contexts where drinking ten beers is less likely then drinking alternative $n < 10$-number of beers. If the context is set up for a scale $1, \dots, 10$, the presupposition of EVEN is satisfied and *i* can be spelled out in the structure as (53).

(53) [EVEN C_1] Peter drank ten$_F$ beers.

Now, let us consider weak scalar particles, for a weak scalar particle in the Strawson downward entailing context like (48) a possible LF in (54) is obtained. The presupposition of SOLO is satisfied: to drink one beer is more likely than to drink $n > 1$ beers. The presupposition of EVEN (with a wider scope than the antecedent of implication) is satisfied too. Drinking one beer as a cause of dancing is less likely than drinking $n > 1$ beers as a reason for dancing intuitively. More formally: as usually, the antecedent of a conditional is Strawson downward entailing, so the weak prejacent becomes logically strong. If drinking one beer is a sufficient condition for dancing, then drinking $n > 1$ beers is a sufficient condition for dancing as well. All the contextual alternatives are entailed by the prejacent and due to (52) are less likely.

(54) [EVEN C_1][[if [SOLO C_0] Peter drinks one$_F$ beer] he will dance]

The scale in (49) can be interpreted as the following morphological rules in (55a)/(55b) formalizing both the idea that *i* bears unlikelihood presupposition, while *byť jediný* bears two contradictory presuppositions, and the idea that the items compete for insertion. But before I tackle the competition part, let us again consider the details of weak scalar particles formalization. Weak scalar particles like Czech *byť jediný* bear both the unlikelihood presupposition of strong *i* and the likelihood presupposition formalized as the feature SOLO. The contradictory presuppositions of *byť jediný* explain its limited distribution. While EVEN in (54) scopes over an entailment reversing operator, SOLO scopes under it and consequently both presuppositions can be satisfied. Note as well, that this explains the degraded acceptability of (56). There is no entailment reversing operator over which EVEN would scope and the conflicting presuppositions of the two features (SOLO and EVEN) clash with each other necessarily. The prejacent cannot be both less likely and more likely than its alternatives (in any context).

(55) a. [EVEN] ↔ *i*
b. [EVEN][SOLO] ↔ *byť jediný*

(56) # Petr včera přečetl byť jedinou knížku.
Petr yesterday read even one book
Intended: 'Petr read just one book yesterday.'

2.4.1 Scalar particles and competition

The morphological rules in (55a)/(55b) do not explain the restricted distribution of *i* on their own (recall that *i* avoids weak contexts like (47)). But in the current state of formalization, we predict that an ungrammatical sentence like (57a) would get assigned a correct LF like (57b). Such LF could in principle be spelled out as a sentence containing *i*. The LF (57b) would produce a sensible presupposition of EVEN because it scopes over the entailment reversing operator (the antecedent of the conditional). In such a situation, the weakest scalar expression entails all the other contextual alternatives and becomes less likely than the alternatives.

(57) a. # Jestli Petr vypije i jedno$_F$ pivo, tak bude tančit.
if Petr drink even one beer then will dance
Intended: 'If Petr drinks even one beer he will dance.'
b. [EVEN C_3] [[if ~~[EVEN C_3]~~ Petr drinks one$_F$ beer] he will dance]

But such a spell-out ignores the scale ⟨*i*, *byť jediný*⟩ where the elements COMPETE with each other for insertion. This is quite a frequent situation in linguistics where competition between elements dictates their distribution. There are many possible formalizations of the old intuition that only the most specific item can be inserted and eventually blocks insertion of less specific competitors (called Panini's Principle by many, see Booij 2012 among others). Crnič uses Heim's (1991) Maximize Presupposition reformulation of the Panini's Principle. But I will rely on a neo-Gricean approach essentially following Krifka (1995) and his approach to NPIs (both approaches are well suited for working with competing items differing only in terms of non-at-issue meaning). In the case at hand: there is a competing LF like (58) which is more feature-specific (and would be spelled out as a sentence containing *byť jediný*):

(58) [EVEN C_3][[if ~~[EVEN C_3]~~ [SOLO] Peter drinks one$_F$ beer] he will dance]

There is only one more presupposition of (58) than in the LF (57b): it is triggered by SOLO in its base position and requires its prejacent (drinking one beer) to be more likely than contextual alternatives (drinking $n > 1$ beers). Both presuppositions (EVEN and SOLO) of (58) and the EVEN presupposition of (57b) are satisfied in natural contexts. Moreover, the assertion impact of both LFs is the same, so the structures denote contextually equivalent propositions. Via the usual neo-Gricean reasoning: a speaker communicating (57b) would indicate that she is not in a position to assert (58) but as (58) and (57b) are contextually equivalent, we arrive to a contradiction caused by the competition of elements for insertion. But a speaker communicating (58) does not indicate anything about the non-assertability of (57b), since *byť jediný* (the spell-out of the features in (58)) is logically stronger and its assertion entails the logically weaker alternative (*i*). Put simply: if contextually more feature-specific *byť jediný* can be inserted, it blocks the insertion of the feature-poorer *i*. This doesn't happen in the case of strong contexts like (45) though. The SOLO component of *byť jediný* would produce a contradictory presupposition there. Consequently, *byť jediný* and *i* do not compete for insertion in contexts like that.

2.4.2 Scalar particles and negation

Recall that we have distinguished three classes of weak *even* SP: (i) nondiscriminating (English *even*), (ii) weak particles requiring non-upward-monotonic environments (Czech *byť jediný* or Polish *chociaż/choć/choć* 'just'), and finally (iii) weak scalar particles limited to the immediate scope of negation. Crnič formalizes the distinction between the negation requiring particles (like Czech *ani* or

Polish *ani*) and weak particles like *byť jediný* via a FORMAL UNINTERPRETABLE FEATURE (following essentially Zeijlstra 2004 and Penka & Zeijlstra 2005).[19] For Czech the morphological rules and corresponding scale would be as in (59) (similar to Crnič's (2011) analysis of Slovenian *tudi* and *niti*).

(59) a. [EVEN][SOLO] ↔ *byť jediný*
b. [EVEN][SOLO]$_{[\text{uNEG}]}$ ↔ *ani*
c. ⟨*byť jediný, ani*⟩

This neatly explains the limited occurrence of *ani* type of scalar particles to (in the majority of cases) sentences with negated predicates (but see Dočekal & Dotlačil 2016a,b for qualifications, especially concerning neg-raising contexts). Moreover, it explains some cases of *byť jediný* incompatibility with negation; Crnič shows this "avoid negation" constraint in the following Slovenian sentence.

(60) # Janez ni prebral tudi ene$_F$ knjige.
Janez not read even one book
'John did not read even one book.' (Slovenian; Crnič 2011: 139)

The direct translation of (60) into Czech sounds distinctly odd, and there are other cases of *byť jediný* type of scalar particles which are incompatible with negation (see Crnič 2011: chap. 5 for German, Spanish and Greek data). The blocking is also familiar in Slavic competition of indefinites (or free choice items) and neg-words (see Pereltsvaig 2004). Moreover, the "avoid negation" rule follows the same logic of competition induced above to explain the blocking of strong scalar particles by weak ones. This led Crnič to postulate the following typological implicational relation, which is supposed to be operative in all negative concord languages.

(61) *Implicational relation for weak scalar particles*
There is a scalar particle that may only be weak and that only occurs in the immediate scope of negation in the language. → No other weak scalar particle that may only be weak occurs in the immediate scope of negation in the language. (after Crnič 2011: 131)

But even if the generalization is linguistically reasonable and supported by the data above, I have serious doubts about its validity. First, let us look at the empirical facts. It is easy to find many natural occurrences of both Czech *byť jediný*

[19]Thanks to one of the reviewers for suggesting that Czech and Polish *ani* and Czech *byť jediný*/Polish *chociaż/choć/choć* do not behave differently in this respect.

and Slovenian *tudi* weak scalar particles in negated sentences. Consider (62) and (63).

(62) a. Nemají být' jediný věšák na kabáty.
NEG.have even one stand for coats
'They lack even one coat-stand.'

b. Nikomu ze soutěžících nepoložil být' jedinou otázku.
NEG.person from competitors NEG.ask even one question
'He didn't ask any competitor even one question.'

c. Nikdy by se neodvážily být' jediným slovem projevit svou nespokojenost.
NEG.time SBJV REFL NEG.dare even one word manifest their discontent
'They didn't dare by even one word to manifest their discontent.'

(Czech; examples from the Czech National Corpus; Křen et al. 2020)

(63) Ni izpustil tudi ene same dirke.
NEG pass even one single race
'He didn't pass even one race.' (Slovenian; Lanko Marušič, p.c.)

Even if it is true that in some configurations negating *být' jediný* type of weak scalar particles leads to unacceptability, there is definitely no general ban on negating weak scalar particles even in negative concord languages like Czech or Slovenian (against Crnič's 2011 typological rules as (61)). There are, of course, many possible explanations why the putative blocking between the items on scale (59) does not happen in cases like (62) or (63). A proper investigation of this issue lies beyond the scope of this chapter, though, but I will at least hint at one probable solution. As observed already by Kadmon & Landman (1993) for certain usages of *any* and further scrutinized by Crnič (2011), Alonso-Ovalle (2016) for weak *even* carries in many contexts peculiar semantics which is usually termed settle for less (see also Section 2.2.1).

Let me demonstrate the semantics on example in (64a). The example has the following meaning ingredients: (i) the speaker of the sentence would prefer to buy more than one ticket; (ii) but he settled for one ticket. Slavic b-NPIs like Czech *být' jediný* (or Slovenian *tudi*) can have a settle-for-less reading, at least in some contexts. But this is not a general feature of weak scalar particles; compare (64b) with a-NPI *ani* (64b), which is compatible with a speaker who did not care about how many tickets he bought at the end, *ani* would be strange in a settle-for-less context. Consequently, at least some Slavic languages with b-NPIs seem to carry the settle-for-less semantics, unlike weak a-NPIs.[20]

(64) a. Nakonec jsem nekoupil byť jediný lístek.
finally AUX NEG.bought even one ticket
'At the end, I didn't buy even one ticket.'

b. Nakonec jsem nekoupil ani jeden lístek.
finally AUX NEG.bought even one ticket
'At the end, I bought not even one ticket.'

2.4.3 Short note on concessive scalar particles

In the previous section, we observed that Slavic b-NPIs (unlike a-NPIs) could carry the settle-for-less meaning. The main idea of explaining the lack of blocking between *ani* and *byť jediný* under negation would be the following: it seems reasonable that b-NPIs (*byť jediný*) can spell out a semantics unavailable for a-NPIs (*ani*); if they do, like in (64a), they are not less specific than a-NPIs, and the usual blocking is gone. In cases like (60) used by Crnič, there is no plausible settle-for-less semantic component, which would be spelled out by b-NPIs and blocking kicks in. In other words, I claim that weak b-NPIs sometimes behave like concessive scalar particles. But to get this idea working, we have to make a short detour to proper concessive particle territory.

Concessive scalar particles are expressions like Greek *esto*, Spanish *aunque sea*, *siquiera*, Slovenian *magari* and Czech *alespoň*. They occur in downward entailing contexts, in questions, and in some modal contexts. Moreover, there is a partial overlap of concessive scalar particles with weak scalar particles.[21] Con-

[20] One of the anonymous reviewers notices that if *byť jediný* in examples like (62) can be substituted with the a-NPI *ani* without any meaning difference, the competition story offered in Section 2.4.3 would be weakened. Intuitions of Czech native speakers I consulted support the subtle difference discussed in this section, although (for obvious reasons) it is not easy to demonstrate them with (62). Nevertheless, the difference can be made more visible in the following scenario: imagine a man who always wanted to stay a bachelor without children; in such a context, he can say *Vyšlo to, nemám ani jedno dítě* 'It worked. I don't have even one child' (a-NPI) but to say *Vyšlo to, nemám byť jediné dítě* 'It worked. I don't have even one child' (b-NPI) is conceived by native speakers as incoherent.

[21] As one of the anonymous reviewers notes, there are approaches to sub-types of NPIs that resemble the reasoning in the current section. One such example is Rullmann (1996), where two different types of Dutch NPIs are described via two different theoretical approaches. The first via the scalar theory of *any* from Lee & Horn (1994) and the second via the non-scalar, domain-widening approach to *any* from Kadmon & Landman (1993). Despite some common properties, the approach in this section is different from Rullmann (1996) since both concessive and weak scalar particles are described as scalar.

sider desire predicates like *want* in Czech (65a). Note that in (65) I translate both items (*byť jedenkrát* and *alespoň jedenkrát*) as 'at least once', while in (48) *byť jediný* is translated as 'just one', which corresponds to the 'even one' semantics discussed below.[22]

(65) a. Petr chce, aby Karel navštívil Grónsko {byť jedenkrát / alespoň jedenkrát}.
Petr wants that Karel visited Greenland at.least once at.least once
'Petr wants Karel to visit Greenland at least once.'

b. * Petr ví, že Karel navštívil Grónsko {byť jedenkrát / alespoň jednou}.
Petr knows that Karel visited Greenland at.least once at.least once
Intended: 'Petr knows that he visited Greenland at least once.'

So far, we have discussed only the properties which are shared both by weak and concessive scalar particles. But there is an obvious difference between them: concessive scalar particles are additive, so they contribute a meaning of multiplicity, unlike the weak scalar particles (even if the additive presupposition can be masked in some contexts). The difference between them can be clearly observed in some contexts where concessive scalar particles are licensed but weak particles are not, see (66). The core meaning of (66) is the necessary condition: obtaining $n \geq 1$ scout badges is *sine qua non* for participating in a scout camp. This means that earning 2, 3, or more scout badges is even more positive; the additive component stems from the concessive scalar particle *alespoň*.

[22] Both weak scalar particles and concessive particles are grammatical as well in the scope of desire predicates and positive emotive factives like in (i). And as one anonymous reviewer correctly remarks, this is a non-trivial problem since *glad* is not SDE and desire predicates like *want* are Strawson upward entailing (see von Fintel 1999). Crnič (2011: section 4.2.5) offers a technical solution to this problem which (in bare essentials) rests on the free choice interpretation of concessive scalar particles and its exhaustification interpretation leading to a non-entailing set of alternatives. Discussing Crnič's solution would lead us too far away from the goals of the current chapter, but all the details of Crnič's analysis should carry on to the cases of Czech concessive particles in contexts like (i).

(i) Petr je rád, že {byť jedenkrát / alespoň jednou} navštívil Grónsko.
Petr is glad that even once at.least once visited Greenland
'Petr is glad that he visited Greenland at least once.'

(66) (Abys mohl jet na skautský tábor), musíš získat {#byť
in.order.to be.able participate in scout camp have.to obtain even
jediného / alespoň jednoho} bobříka.
one at.least one beaver
'(For you to be able to participate in a scout camp) you have to obtain even one / at least one scout badge.'

Based on the observations above, I postulate that Slavic b-NPIs (Slavic *byť jediný*, Slovenian *tudi*) can realize some of the concessive scalar particles' meaning. More specifically, I will use the feature formalization of concessive particles from Crnič (2011: 109), below as (67). There are two components of [AT-LEAST]: (i) the familiar presupposition part: the prejacent has to be more likely than all its alternatives (this part is shared with the presupposition of SOLO, see (51)); (ii) the assertive meaning (different both from EVEN and SOLO since both EVEN and SOLO are vacuous in at-issue meaning; see (50) and (51)): there is at least one alternative to the prejacent which is at most likely and is true (can be the prejacent itself). Let us see the working of the formalization demonstrated for the adversative predicate *regret* in (68). For Czech *alespoň*, I postulate a morphological rule in (69). There are two presuppositions in (68): (i) the scalar presupposition triggered by AT-LEAST – it is more likely to visit Greenland once than $n > 1$ times; (ii) the scalar presupposition of EVEN – because *regret* is SDE, the weak associate becomes strongest (if somebody regrets that he visited Greenland once, he regrets that he visited it $n > 1$ times). Because both presuppositions in (68) are satisfied in natural contexts, *alespoň* can be inserted. The additive multiplicity part of the at-issue contribution is in the scope of the adversative predicate, so correctly not projecting, unlike the presupposition part of AT-LEAST. But recall that (68) is the LF of (65a) where both *alespoň* and *byť jedenkrát* were acceptable with the same interpretation. This then shows that, at least in some cases, the b-NPI can spell out features like (69).

(67) $[\![\text{AT-LEAST}]\!]^{g,c}$
$= \lambda C.\lambda p : \forall q \in C[p \neq q \rightarrow q \lhd_c p].\lambda w.\exists q \in C[q \lhd_c p \wedge q(w) = 1]$
Crnič (2011: 109)

(68) [EVEN C_2][Petr regrets [that [AT-LEAST C_0] he visited Greenland once$_F$]]

(69) [EVEN][AT-LEAST] ↔ *alespoň*

And if b-NPIs can spell out AT-LEAST features, we get the formal implementation of the idea from the beginning of the current section. Consider modified morphological rule for *byť jediný* in (70a) and repeated morphological rule for *ani* in (70b). In terms of features, the competition between the two items is then in equilibrium. Both items realize the same number of features; neither of them is more specific, stronger, and blocking the other. That explains the occurrences of b-NPIs under negation in examples like (62) – when b-NPIs realize the same number of features as a-NPIs, they can be inserted even in the immediate scope of negation. In cases like (60), the blocking works as proposed by Crnič (2011: chap. 5). Of course, proper scrutiny of the proposed leak in the generalization (61) is beyond the scope of this chapter, but I hope to find more supporting data in future work.

(70) a. [EVEN][SOLO][AT-LEAST] ↔ *byť jediný*

b. [EVEN][SOLO]$_{[\text{uNEG}]}$ ↔ *ani*

The empirical claims concerning Czech scalar particles and the features they realize are summarized in Table 1. In Section 2.4, I tried to show how the pragmatic theory in the style of Crnič can be applied to Czech weak and strong scalar particles. The complications we observed in the case of Czech scalar particles are due to the interaction with negation and the competition of the items for insertion.

Table 1: Table of features for Czech scalar particles

Czech expression	features
i	[EVEN]
byť jediný	[EVEN][SOLO]
byť jediný	[EVEN][SOLO][AT-LEAST]
alespoň	[EVEN][AT-LEAST]
ani	[EVEN][SOLO]$_{[\text{uNEG}]}$

Generally, the point of Section 2.3 was to introduce the pragmatic mechanism of NPI licensing. It followed the seminal work of Krifka (1995) where the licensing of NPIs was rethought from the standard semantic downward entailing explanation to a pragmatic theory. In particular, the description of Czech scalar particles was formalized in terms of an *even*-based theory of NPI licensing, following Crnič (2011) (where more details and also references to other literature can be found). As was stated at the beginning of this section, the pragmatic turn, where the environments licensing NPIs are basically characterized via the logical requirements on the focus alternatives for the prejacent, can be cashed out in different frameworks, one widely adopted being Chierchia (2013). See also Gajić (2018) for its application to Slavic data. The pragmatic explanation should be viewed as a continuation of the semantic theories, with which it shares the core idea that NPI licensing is a matter of semantic strength (or entailment patterns; see (52)). Moreover, the pragmatic theory offers more flexibility and also a handle on non-monotonic licensing of NPIs (see again Crnič 2011 for details).

3 Conclusion

In this chapter, I summarized syntactic (Section 2.1), semantic (Section 2.2), and pragmatic theories (Section 2.3) of (not only) Slavic negative polarity items and scalar particles. I summarized the history of ideas in formal Slavistics pertaining to the description and explanation of Negative Polarity Items (and scalar particles). The chapter has shown how a better understanding of Slavic polarity-dependent expressions improves our understanding of the division of labor between syntax, semantics, and pragmatics. In Section 2.4, I applied the pragmatic theory of NPIs to Czech data.

There seems to be a general tendency in the research on NPIs to move from syntactic and semantic theories to pragmatic ones; the move started with Krifka (1995). The pragmatic theories are more flexible and can account for much more data than the syntactic and semantic ones. On the other hand, there is new research on the syntax/semantics interface of NPIs; see influential work of Barker (2018) a.o. Barker's theory seems to be supported experimentally (Alexandropoulou et al. 2020) and is one of the few theories of NPIs that can naturally handle NPIs licensing in non-monotonic environments. So, while the general trend is treating NPIs pragmatically, ignoring the syntactic and semantic approaches would be futile.

However, on the empirical side of the debates, following the pragmatic theories can bring many insights into the nature of NPIs and their licensing. Hopefully, the chapter will help to ignite more research on Slavic NPIs and scalar particles, especially in the light of pragmatic theories and experimental research.

Acknowledgements

The work on this chapter benefited from discussions and correspondence with many people. First, I would like to thank the three anonymous reviewers. Their comments and hints helped the chapter to be more thorough and insightful. Next, I would like to thank people who read my chapter and gave me comments and ideas, namely Hana Strachoňová and Iveta Šafratová. Last, I am very grateful to Berit and Radek, the editors, for their patience and help.

Abbreviations

ACC	accusative	NPI	negative polarity item
AUX	auxiliary	Q	question particle
COMP	complementizer	REFL	reflexive
DE	downward entailing	SBJV	subjunctive
FUT	future	SDE	Strawson downward entailing
NEG	negation	SG	singular

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Chapter 4

Genericity and habituality in Czech

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In Slavic languages, the domain of genericity (and habituality) is traditionally viewed as associated with contextually determined meanings of imperfective forms, and to a lesser extent of perfective forms. What has so far remained largely unexplored, and misunderstood, is the verb marker *-va-*, which is dedicated to deriving verbs that only have a generic (and habitual) interpretation. Based mainly on Czech data, it will be shown that the verb marker *-va-* has distributional and semantic properties that preempt its analysis as a marker of tense and/or aspect (*pace* Dahl 1995, i.a.). Neither can its meaning be reduced to standard (modal) operators that are used in the analysis of progressive or imperfective aspect (*pace* suggestions in Cable 2022, i.a.). Its quantificational and modal (intensional) components strongly suggest that it is a type of generic marker delimiting a subclass of generic sentence. The insights reached here support the view that generic markers in natural languages not only exist, but are not as rare as is generally thought, and by the same token also bolster independent arguments made elsewhere for the independence of genericity (and habituality) from the categories of the TMA (tense, modality, aspect) system.

1 Main data and questions

Generic sentences are sentences like *Dogs bark, Sugar dissolves in water, Sam walks to school on Mondays.* This chapter focuses on their encoding in Slavic languages that have a dedicated marker on the verb enforcing only a generic (or habitual) reading of a sentence, besides using imperfective, and also perfective, verbs for this purpose in an appropriate context that triggers their generic (or habitual) interpretation. Its instantiation in Czech (West Slavic language) is treated as a suffix in standard reference grammars (see Petr 1986: 184, Dahl 1995, Kosek

Hana Filip. 2026. Genericity and habituality in Czech. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 149–229. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078575

2014, Nübler 2017, for instance); some Bohemists treat it as an infix (e.g., Kučera 1981, Bílý 1986). In Genis et al. (2021: 23), it is treated as an infix across different Slavic languages.[1] Henceforth, I will use *-va-* as its citation form (as is also common in Bohemistics), and refer to it as the MORPHEME *-va-* or MARKER *-va-*, given that trying to resolve the suffix-infix controversy would go well beyond the scope of this contribution.

This marker is attested in all Slavic languages (see Genis et al. 2021 for examples from different Slavic languages). However, it is productive only in West Slavic languages, specifically in Czech, Slovak and to a lesser extent in Polish, and commonly used in both spoken and written language (see e.g., Širokova 1963: 62, Comrie 1976: 27, Kučera 1981, Bílý 1985, Bílý 1986, Petr 1986, Genis et al. 2021, among others).[2] In East and South Slavic languages it is also attested, but it is reported to be largely unproductive, at least in their standard varieties.[3]

Here, the main focus is on the instantiation of this Slavic marker in Czech. The constraints on the length of this chapter prohibit me from adducing relevant examples from other Slavic languages. To the extent that the relevant generic verb forms exist in other Slavic languages, e.g., *pisyvat'* in Russian, meaning 'to write repeatedly, sporadically, as a rule' (albeit judged as archaic), it is reasonable to expect that the arguments made here for the corresponding Czech forms, e.g., *psávat*, should apply to them, as well.

Let me start with a few examples, given below:

(1) a. Psi *štěkají.* IPFV
dogs bark
'Dogs bark.' [Intended interpretation: 'Dogs as a kind have the characterizing property of barking.']

[1]Genis et al. (2021: 23): "[T]he most common infix in this [habitual, frequentative/iterative, HF] function across languages seems to be one that might be rendered -/va/-."

[2]The productivity of *-va-* in Czech is also evident in the observation that children use verbs on which it occurs early in their language acquisition, even before they master the grammatical perfective/imperfective distinction (Chmelíčková 2005).

[3]In standard spoken and written Russian, the generic marker only occurs on some verbs that are taken to be marginal in use (Comrie 1976: fn. 1 on p. 27, who cites Isačenko 1962: 405–407). They are attested in fixed collocations (e.g., *V žizni tak byvaet* [gloss: in life so be.GEN.3.SG.PRS] 'That's how life goes. / That's life.'). Dickey (2000: 87) claims that in standard Serbian explicitly "habitual" (in his terms, generic in my terms) verbs can be formed with the morpheme *-va-* (which is an infix, according to him) from biaspectual verbs, e.g.: *kršćavati* (from *krstiti*$^{\text{PFV/IPFV}}$ 'baptize'); *noćivati* (from *noćiti*$^{\text{PFV/IPFV}}$ 'spend the night'); *ručavati* (from *ručati*$^{\text{PFV/IPFV}}$ 'eat lunch'); *večeravati* (from *večerati*$^{\text{PFV/IPFV}}$ 'eat dinner'). However, Dickey's claim is contested in Genis et al. (2021: 26): "[A]ccording to our observations, these verbs do not always have a habitual interpretation but can be used in a durative meaning, as well; other than their base verbs they are explicitly imperfective."

b. Psi se *poštěkají*, že ani nevíte, jak k tomu došlo. PFV
dogs REFL PREF.bark that even NEG.know how to it came
'Dogs (will) bark (out) at each other, and one hardly knows why.'

c. Psi *poštěkávají* ve spánku, vrčí a cukají packami. IPFV
dogs PREF.bark.IPFV in sleep, growl and twitch paws
'Dogs bark now and then while sleeping, growl and twitch their paws.' [i.e., bark a little/on and off]

d. Psi *štěkávají* na ty, které neznají. GEN
dogs bark.GEN at those whom NEG.know
'Dogs bark at those whom they don't know.'

As the English translations of the four Czech generic sentences above show, one and the same English verb form *bark* (3rd person plural, present tense) corresponds to four different verb forms in Czech. All four share the same root *-štěk-* 'bark'. (1a) is headed by a primary (root) imperfective verb, formally unmarked for imperfectivity, while (1c) by a secondary imperfective verb, which is derived by means of the imperfectivizing suffix (glossed with IPFV) from a perfective base, realized in the perfective verb in (1b). In languages with the perfective/imperfective distinction like Slavic or Romance languages, the semantics of imperfectivity is taken to have close affinities to genericity (and habituality), based on the observation that imperfective verbs have common contextually determined generic readings, as we see in (1a) and (1c).

However, there is no incompatibility between the semantics of perfectivity and genericity (and habituality). Perfective verbs are also used in generic sentences, and this use is not as unusual or rare as some assume (*pace* Comrie 1976, Dahl 1995, i.a.). For instance, in Romance languages, and in West Slavic languages (see e.g., Filip & Carlson 1997), perfective verbs have common contextually determined generic readings, as is illustrated by (1b). Here, we have a perfective verb derived by means of the prefix *po-* from the stem of the primary (root) imperfective verb *štěkat* 'to (be) bark(ing)' (used in (1a)).[4]

Most importantly, (1d) has an obligatory generic interpretation in all contexts, and unlike the (im)perfective verb forms in (1a) and (1c), (1d) can never be used in episodic contexts. This is due to the morpheme *-va-* on its main verb. It constitutes a sufficient, but not a necessary, condition for the encoding of generic statements, given that, as we have just seen, such statements can also be expressed by verbs without this morpheme, in an appropriate context.

[4]For a recent study on the use of present tense perfective verbs in Polish generic sentences see Wiemer et al. (2025).

One of the main goals of this chapter is to argue that the morpheme *-va-* (as in (1d)) is best viewed as a generic morpheme sui generis (also building on some previous arguments for Czech in Filip (1993b, 1994), Filip & Carlson (1997), Mendia & Filip (2018), Filip (2021), i.a.). This argument implies a rejection of the prevalent traditional view on which this morpheme is a marker of a (possibly unbounded) plurality of situations of the same type (see Genis et al. 2021, and specifically for Czech see e.g., Petr 1986, Dahl 1995, Nübler 2017, and references therein), and as such, it is treated as a marker that falls under imperfective aspect and delimits an (iterative) *Aktionsart* (Poldauf 1964, for instance), or as a kind of pluralization operator deriving a predicate of plural situations. While the notion of a plurality of situations is relevant to the meaning of the morpheme *-va-*, it is neither sufficient nor necessary to ground its meaning, as I will show. Given the erroneous idea that this marker encodes a mere plurality of situations, it is traditionally referred to as the "iterative" marker, and also the "multiplicative", "frequentative" or "repetitive" marker. That its most common label "iterative" is a misnomer at best, and a category error at worst, is evident in the simple fact that this morpheme is incompatible with iterative adverbials like 'three times' or 'several times' (see section Section 4). It will also be shown that the marker *-va-* does not have a meaning akin to the adverb of quantification 'usually' or the determiner quantifier 'most' (*pace* Dahl 1995, i.a.), as in 'in most situations'.

Related to the main argument to be made here, let me make two clarifications. First, terminology for morphemes and grammatical constructions that are used in sentences that express regularities, patterns, or habits – in short generalizations of various kinds – is not standardized and, as suggested in the previous paragraph, often misleading. Iterativity, multiplicativity, frequentativity, repetitivity, usuality, pluractionality, and the like are often used interchangeably, and also confounded with habituality and genericity.

When it comes to habituality, it is either viewed as separate from genericity, equated with it (e.g., Comrie 1976), or subsumed under genericity as its special case (e.g., Krifka et al. 1995).[5]

In its widest sense, the term "habitual" is used for all verbs, VPs or sentences that are not episodic. Episodic sentences are true relative to a given world and

[5]For instance, Comrie (1976) uses "habituality" for all the uses of imperfective forms that are not episodic. He divides the semantic domain of imperfectivity into a "habitual" subdomain and an episodic subdomain. However, he confusingly refers to the episodic subdomain as "continuous", and divides it into two further subdomains: namely, "nonprogressive" and "progressive". The category "continuous" in (Comrie 1976) is notionally hardly coherent, and formally speaking, there seems to be no language with formal markers for it. For further criticism of Comrie's classification see Filip & Carlson (1997).

reference time, they do not describe regularities. In this widest sense, habituality has been popularized by Comrie (1976), and is commonly used for one subdomain of imperfectivity, as opposed to its other, namely, episodic subdomain in aspect studies and traditional/descriptive linguistics. Habituality in this sense is, however, problematic, because it subsumes uses of sentences describing what is not generally viewed as a matter of habit at all (Lyons 1977: 716, i.a.), i.e., habit in the sense of regularities of action of typically human, or at least animate, individuals, as in *Fred smokes after dinner*. Some examples are: *Glass breaks easily* (a disposition), *Soap is used to remove dirt* (a function), *Bishops move diagonally* (a rule of a game), *Kim works for the government* (an occupation). Therefore, a better term to cover all types of sentences that are not episodic is *generic*, which is prevalent in formal semantics studies that are informed by philosophy, computer science and psychology.

The Czech marker *-va-*, as in (1d), despite being occasionally labeled "habitual" (e.g., Dahl 1995, Danaher 2003, Karlík 2017), can formally mark not only sentences that describe habits proper (as in *Fred smokes after dinner*), but also generic sentences with kind reference, as in our initial example (1d), and also in other types of generic sentences that describe what is not a matter of habit (see e.g., (50) below).

Therefore, here, for the sake of simplicity, I will use the term *generic* as a cover term for all statements that are *not* episodic, while habituality will be treated as a special case of genericity (as in e.g., Krifka et al. 1995). The term *habitual sentence* will be reserved for a subclass of generic sentences that concern habits proper, namely regularities of action of typically human, or at least animate, individuals (e.g., *Fred smokes after dinner*).

The second point of clarification concerns the confounding of the Slavic generic marker *-va-* (as in (1d)) with the imperfectivizing suffix (as in (1c)). Their confounding has to do with the fact that some of their allomorphic variants seem virtually indistinguishable, and both are used in generic (and habitual) contexts.[6] While the generic marker *-va-* enforces a generic reading in all its occurrences, the imperfectivizing suffix on secondary imperfectives is compatible with their

[6]Occasionally this confounding may be also motivated and reinforced by speculations about the diachronic development of one of the morphemes from the other, which, however, seem unfounded. For instance, Kuznetsova & Sokolova (2016) assume, without any evidence and only with reference to some assumptions by Kuznecov (1953), that habituality or iterativity is diachronically the original meaning of the Russian secondary imperfectivizing suffix, standardly cited as *-yva/iva-*. According to Kuznecov (1953: 259), it developed as the result of a reanalysis of the the habitual or "iterative" morpheme *-va-* of the verb *byvati* 'to be usually, typically, normally' or 'to tend to be': *by-va-ti* > *b-yva-ti*. Hence, according to him, the imperfectivizing suffix *-yva/iva-* originally only had the habitual or "iterative" meaning component.

episodic or generic reading. For instance, only the imperfectivizing suffix, but not the generic *-va-*, can be used in sentences with a contextually determined progressive (episodic) reading. I will address the differences between these two morphemes in detail in Section 3, and touch on their differences also elsewhere in this chapter.

With these basic clarifications and caveats out of the way, the main question to be pursued in this chapter is as follows:

Question 1: What properties of the marker *-va-* justify its status as a marker of genericity?

The marker *-va-* constitutes a sufficient, but not a necessary, condition for the expression of generic (and habitual) statements. Such markers are attested in a number of typologically distinct languages. A short list, which also includes Czech, is given in Dahl (1995: 421, fn. 8).[7] In fact, Dahl uses the Czech marker *-va-* as a paradigm example of this class of markers, which, according to him, fail to qualify as generic markers. Instead, they are to be treated as what he calls "habitual tense-aspect markers" that function as a kind of quantifier over situations with the semantics of 'most' or 'usually'. For Dahl (1995), this claim bolsters his overarching argument concerning the reflexes of the episodic/generic distinction in the grammar of natural languages: namely, if all such verb markers like the Czech *-va-* are not generic markers, but rather habitual markers ('habitual' in the sense of Dahl 1995) that fall either under tense or imperfective aspect, then "however sad this may be for those of us to whom the episodic/generic distinction is dear, this distinction more often than not is only indirectly reflected in speakers' choices between grammatical markers" of aspect or tense (Dahl 1995: 425).

These are the claims that will be contested here. Contrary to Dahl (1995), I will show that the Czech marker *-va-* has properties that preempt its subsumption under tense or imperfective aspect (also *pace* Cable 2022). Its meaning cannot be reduced to a quantifier over situations with the meaning amounting to 'most', or 'usually', nor any other single expression of quantity or quantification, also contrary to Dahl (1995).

Having established what the marker *-va-* is not, I will show that it derives verbs denoting predicates that exhibit three hallmark properties of generic predicates (see e.g., Carlson 1977, Krifka et al. 1995): namely, (i) they are aspectually

[7]Dahl (1995: 421, fn. 8) gives the following list: Arabic (Classical), Akan, Catalan, Czech, Didinga, German, Guarani, Hungarian, Kammu, Limouzi, Montagnais, Sotho, Spanish, Swedish, Swedish Sign Language, Yucatec Maya, Zulu.

stative, (ii) have a modal (intensional) import which is tied to the quintessentially predictive force of generics, and as such their evaluation requires access to other possible worlds than just the actual one, and (iii) they express generalizations that crucially involve reasoning with exceptions. This then leads me to the next two questions:

Question 2: What kind of generic sentence does the generic marker *-va-* delimit?

Question 3: What is the relation of the generic marker *-va-* to the phonologically null generic GEN operator?

Given that there is no comprehensive semantic analysis of genericity that is commonly accepted, when it comes to the foundational assumptions and terminology in the domain of genericity (and habituality), I will rely on the theoretical background of the quantificational theory of genericity (see e.g., Krifka et al. 1995). Its key hypothesis is that the semantics of generic sentences can be represented by means of a logical (tripartite) structure, where the generic import is captured by means of the phonologically null, modal generic GEN operator, which is akin to overt adverbs of quantification (Q-adverbs) (see Section 2). The quantificational theory of genericity will here serve as a useful point of reference for examining theoretical and empirical issues raised by the generic morpheme *-va-*. It has the advantage that it has a wide interdisciplinary scope, drawing on insights not only from semantics and pragmatics, but also on insights regarding the nature of generalizations, law-like statements and reasoning with exceptions in philosophy, psychology, computer science and other areas of cognitive science.

As I will show, just as GEN, the generic morpheme *-va-* functions as a quantifier over individuals and/or situations, and exhibits variable binding properties that are akin to those of GEN. Moreover, the meaning of both GEN and *-va-* cannot be reduced to any single quantifier, e.g., 'usually', 'most', or any other quantifier or quantity expression. At the same time, the Slavic generic morpheme differs from GEN in so far as it has two uncancellable inferences: namely, that (i) there are verifying instances in the actual world that count as evidence for the truth of the generalization, and (ii) there are actual or at least potential exceptions or counterexamples to the generalization. The former inference is shared with what are taken to be so-called 'habitual' markers in other languages (see e.g., Cable 2022 on the realization inference of Tlingit habitual morphology), and it is also known as the actualization entailment in the genericity literature (see e.g., Green 2000, Hacquard 2006).

The latter inference is clearly manifested in the observation that *-va-* is semantically anomalous or false in generic sentences that have no exceptions in all the possible worlds (such as universal laws of nature), and it must be used in generic sentences that have known counterexamples and would be false without it, all else being equal (see Section 4.1.2). Generally, as I will show, the truth conditions and felicity of generic sentences that are formally marked with *-va-* crucially depend on the speaker's reasoning with exceptions, which reflect the speaker's commitment to the strength of the generalization calibrated in terms of exceptions that (the speaker believes) it has or might have. In effect, generic sentences formally marked with *-va-* are endowed with two layers of modality (intensionality): one that is tied to their generic, law-like, (g)nomic force, similarly to GEN, and the other to reasoning with exceptions, which may be characterized in terms of the speaker-oriented epistemic and doxastic modality (exploiting general pragmatic principles of interpretation).

The presence of the generic marker *-va-*, and its contrastive absence, in generic sentences differentiates two main types of generic sentences in Slavic languages. Some have argued that different linguistic forms for the encoding of generic statements differentiate different subtypes of generic sentences, each distinguished by different clusterings of formal properties, and possibly requiring different semantic/ontological commitments, and separate semantic/ontological models for their interpretation (see e.g., Greenberg 2003, Pelletier 2009a, Krifka 2012, Boneh & Doron 2013, 2008). Hence, the properties of the Slavic generic marker directly bear on the fundamental question in the domain of genericity (Carlson 1977, 1995):

QUESTION 4: Can we provide a unified semantic analysis for all generic sentences?

A conclusive answer to the above question is still outstanding. A unified analysis for all generic sentences is desirable, and hypothesizing that generic sentences might constitute a single class, markers that are dedicated to the encoding of generic statements, such as the generic marker *-va-*, provide some of the best empirical data for evaluating the claims that address the above question.

The proposal advanced here that the morpheme *-va-* is a generic marker sui generis, as well as the empirical and theoretical arguments in its support, have potentially significant consequences for the theory of genericity. First, if the morpheme *-va-* in Czech, in other Slavic languages where it is attested, is a generic marker, as I argue, it would shed at least some doubts on the widespread belief in linguistics, philosophy and psychology that natural languages have no dedicated markers of genericity (see e.g., Leslie 2008, Johnston & Leslie 2012, Leslie & Gelmann 2012, Rhodes et al. 2018, Cohen 2022, Schiller 2023, to give just a few more recent examples), or only a negligible number (Dahl 1995).[8] Second, it would support the claims independently made elsewhere (see e.g., Carlson 1977, Schubert & Pelletier 1989, Filip & Carlson 1997, Pelletier & Asher 1997, i.a.), and mostly on semantic grounds, that the episodic/generic distinction is relevant for the grammar of natural language (*pace* e.g., Dahl 1995), and bolster the view on which genericity is a category in its own right, and specifically independent from the tense, modality and aspect (TMA) system.

This chapter is structured as follows. Section 2 introduces the theoretical background of the quantificational theory of genericity. In Section 3, the formation of generic verbs by means of the generic morpheme *-va-*, and its place in the Czech verb system will be outlined in a theory neutral way. Its properties will be also shown not to be aligned with markers of tense or imperfective aspect. Section 4 focuses on its main semantic and pragmatic properties, which together with Section 3 invalidate the grounds for the claim made by Dahl (1995) that the Czech morpheme *-va-* fails to qualify as a generic marker. Section 5 addresses the semantic similarities and differences between the phonologically null generic operator GEN and the Czech generic marker *-va-*. The marker *-va-* introduces a modal (intensional) generic operator VA into the logical representation that differs from GEN in so far as it contributes the uncancellable inference regarding the existence/possibility of exceptions or counterexamples to the generalization which, moreover, must have been realized. The use of the generic *-va-* form, when the corresponding non-generic form might have also been used, gives rise to a rank order (see Horn & Abbott 2012, Horn 2017, and references therein), where the non-generic form is the strong alternative (as it allows no exceptions when used in generic sentences) and the generic form the weak one (given that it requires that there be actual or possible exceptions). The rank order generates either the speaker-oriented certainty or ignorance inference concerning the likelihood of exceptions (a kind of quantity-based implicature), which may be characterized in terms of epistemic and doxastic modality, exploiting general pragmatic principles of interpretation.

[8]For instance, Schiller (2023: 541) observes: "Generics are present in every known language, but in no language are generic sentences 'marked' with an explicit generic operator. Generics are studied (by philosophers, linguists, and developmental psychologists) because of their abstruse formal properties and the important role they play in cognitive development."

2 Genericity: Kind reference and generic sentences

The domain of genericity is divided into two independent, but related, phenomena: namely, KIND REFERENCE and CHARACTERIZING GENERIC SENTENCES (Krifka et al. 1995, Carlson & Pelletier 1995). Some examples are given below:

(2) a. *Ravens* are widespread. KIND REFERENCE
 b. *Dodos* became extinct in the 17th century.
 c. *Bronze* was invented as early as 3000 B.C.
 d. *The dodo* / **A dodo* became extinct in the 17th century.

(3) a. *Ravens* are black. CHARACTERIZING GENERIC SENTENCES
 b. *Dogs* bark / *A dog* barks.
 c. Paul plays chess on Saturdays.

There are both formal and semantic reasons for this division. The category of KIND REFERENCE is motivated by the existence of kind-level predicates (KLPs) like *extinct, widespread, invent* that directly select KIND-DENOTING terms for one of their arguments (in italics in the above examples).[9] KLPs attribute a property to a whole kind and one which does not hold of its individual members. For instance, in (2b), the property of being extinct holds of the whole kind DODO, and it makes no sense to attribute it to any individual dodo or a closed set of dodos.

The main topic of this chapter concerns CHARACTERIZING GENERIC SENTENCES, or GENERIC SENTENCES for short. Their generic import often arises from the combined meanings of the subject NP/DP and VP, and possibly also in interaction with pragmatic, prosodic (focus), and discourse factors. Generic sentences with kind-denoting terms, as in (3a) and (3b), express a generalization that holds of a kind as well as of its individual members. Generic sentences like (3c) describe a regularity of action ascribed to an ordinary individual, i.e., a habit in the proper sense of this term. While in Krifka et al. (1995) or Carlson (2011), for instance, they constitute a subclass of generic sentences, some take them to be less typical or marginal members of generic sentences (see e.g. Nickel 2017).

All agree that GENERIC SENTENCES are fundamentally opposed to EPISODIC SENTENCES. The latter refer to particular singular situations, as in (4), or to a closed set of plural of situations, as described in the iterative sentence (5):

[9] A given language may impose specific requirements on the form of a kind-denoting term. In English, for instance, a singular indefinite is often unacceptable.

(4) Fido is barking at the mailman. EPISODIC sentences

(5) Paul played three games of chess last Saturday.

However, there is no general agreement on the positive criteria that delimit all and only generic sentences (see e.g., Dahl 1985, 1995, Nickel 2008, 2016, Pelletier 2009a,b, Carlson 2012b). There is some agreement that prototypical examples are "bare" generics like *Birds fly* that involve kind reference in a generic sentence and have no restrictors like Q-adverbs (e.g., *always, generally, usually*), overt determiner quantifiers (e.g., *some, many, most, all*), modal expressions, or qualifiers that restrict the relevant cases based on which the generalization is made.

Moreover, at least since Carlson (1977), there has been some agreement that generic sentences, in contrast to episodic sentences, have three main properties:

(6) PROPERTIES OF GENERIC SENTENCES
   a. All are aspectually stative.
   b. All have an intensional import, which is tied to their predictive force, with implications for other accessible worlds than just the (actual) world of evaluation, and perhaps all possible worlds.
   c. Most admit exceptions, but may be compatible with no exceptions.

All generic sentences are uniformly stative, be they headed by (i) generic predicates that are derived from stage-level predicates like *bark*, as in *Dogs bark* (on its generic interpretation), or by (ii) individual-level predicates like *(be) intelligent, know (French)*, which are lexically stative predicates (e.g., Chierchia 1995). One standard test for stativity is the lack of reference to particular situations which are locatable at specific spatio-temporal locations. For instance, this is evident in the incompatibility or oddity with adverbials referring to specific time-points and locations:

(7) a. ? Dogs bark right now.
   b. ? John was intelligent on his porch at 4 p.m.

The reason for the oddity of (7a) and (7b) is that they involve characterizing properties of individuals that hold of them during their whole life-time or at relatively long intervals as well as at any moments of such intervals. Therefore, it is pragmatically odd to assert that they hold only at one specific moment of time or location.

Second, all generic sentences have an intensional (modal) meaning component (first observed by Lawler 1973, Dahl 1975), which follows from the nature of generalizations. They transcend our immediate experiences of the world, particular situations, states of affairs, or facts, in so far as they specify what is possible, likely, i.e., they have a predictive, law-like force. This may be based on what actually obtains at given worlds and times, and we might have observed. However, some generics only concern what is *merely* possible and may even never ever be realized, as in (8a) and (8b). They provide the most compelling evidence for intensionality of generics:

(8) a. The Speaker of the House succeeds the Vice President. (Carlson 1995)
(Context: The rule of the United States Constitution that has never been enacted.)

b. This knife cuts cheese.
(Context: The knife has just been delivered in the mail, but its blade got irreparably damaged in shipping.)

The intensional component of generic sentences may be grounded in some underlying causes, principles or factors, or some kind of law that we may surmise behind the pattern, which we may know, but also about which we may be agnostic or ignorant. For instance, *Dogs bark* is true based not just on inductive evidence (e.g., having witnessed barking dogs on a number of occasions), but also on evidence which may not be directly accessible to us, like some underlying causes in the world (such as regularities governing the biology of living organisms) which ultimately ground observable behavior like dog barking.

The third key property of generic sentences is their exception-tolerance. There are generics that admit of no exceptions in all the known conditions, e.g., universal laws of nature like *The Earth revolves around the Sun* or *Water consists of two hydrogen atoms and one oxygen atom*. However, most tolerate exceptions, but may also be compatible with no exceptions whatsoever. For instance, we judge *Dogs bark* to be true, because most dogs bark, even if, as is generally known, there are non-barking dog species like Basenjis and Greyhounds, or individual dogs that do not bark due to injury or illness. These are exceptions that we tolerate. As an extensional consequence, it follows that we cannot validly infer for any particular, arbitrary individual dog that it will have the property of barking. Put differently, that dogs bark is a defeasible fact about dogs. This makes generic sentences particularly amenable to analyses in terms of non-monotonic logics.

What makes any analysis of exception-tolerance so intriguing and challenging is that different types of generic sentences admit a different number and kind of exceptions or non-confirming cases, while still remaining true. Some we judge to be true based on majority satisfaction, such as *Dogs bark*. However, majority satisfaction is neither sufficient nor necessary for the truth of all exception-tolerating generics. It is not sufficient, because, for instance, (9a) is false, despite the majority of printed books being paperbacks. Majority satisfaction is not necessary, because what is characteristic of a kind need not be prevalent among its members. (9b) holds of only a minority subkind, male adult lions. What is characteristic of a kind may also be characteristic of only a minute minority, as (9c) shows. It is judged true, even if a vast majority of mosquitoes (99 %) fail to carry the virus.

(9) a. Printed books are paperbacks. FALSE

b. Lions have a mane. TRUE

c. Mosquitoes carry the West Nile virus (though 99 % do not). TRUE

How many and what kind of exceptions a given generic sentence will admit depends on a variety of factors which may include not only objective facts of the world, but also our kind-specific expectations, which may be influenced by our subjective judgments about what we evaluate as significant, salient or striking in some way for that kind, like a lion's mane or a virus that is highly dangerous to us. Compatibility of generics with a different number and kind of exceptions raise the fundamental question about what we view as "characteristic", "typical" or "normal", on the one hand, and what we view as failing to conform to a given generic statement and why, on the other hand.

The above examples and observations should suffice to illustrate that intensionality and exception-tolerance foil any attempt to analyze all generic sentences in terms of a *single* extensional quantifier (over episodic formulas containing free variables) (e.g., *all, most* or *some*), a *single* quantity expression (e.g., *in a majority of cases* or *in a significant number of cases*), quantity-based criterion, or statistical criterion/correlation, no matter how contextually constrained and/or modalized they might be, or possibly also defined in vague or probabilistic terms (Pelletier & Asher 1997, Carlson 2012b, Nickel 2013, 2017, i.a.).

A number of promising analyses have been proposed for exception-tolerance of generics and related issues, while intensionality has garnered much less attention (see summaries in Krifka et al. 1995, Pelletier & Asher 1997, Mari et al. 2013, Nickel 2017, i.a.). None of the proposed analyses has offered a satisfactory comprehensive semantic framework for all generic sentences, and none so far has emerged as widely accepted.

Therefore, when it comes to representational assumptions, as a useful point of reference and an organizing principle for examining meanings of generic (and habitual) sentences here, I will at various points rely on a logical representation in terms of a tripartite structure with a phonologically null, modalized dyadic operator GEN, akin to overt Q-adverbs, used in quantificational theory of genericity.

In its original formulation, it has the following general form (adapted from Krifka et al. 1995: 32, ex. (57), which in turn relies on suggestions in Krifka 1987):[10]

(10) $\text{GEN}[x_1 \ldots x_i; y_1 \ldots y_j](\text{RESTRICTOR}[x_1 \ldots x_i]; \text{MATRIX}[\{x_1\} \ldots \{x_i\}, y_1 \ldots y_j])$
$x_1 \ldots x_i$: variables bound by GEN
$y_1 \ldots y_j$: variables bound existentially, with scope just in matrix
$\{x_1\} \ldots \{x_i\}$: means $x_1 \ldots x_i$ may or may not occur in matrix

In the tripartite structure (10), just like an overt Q-adverb, GEN relates two arguments, a Restrictor and a Matrix which are separated by ';'. GEN stands for a certain characterizing relation between the conditions stated in the Restrictor (which states the restricting cases) and Matrix (which makes the main assertion of the sentence): intuitively, it sets conditions on the number or proportion of individuals that may satisfy the formula, and if the conditions stated in the Restrictor hold, other conditions stated in the Matrix will also hold.

We may distinguish three main types of instances or particulars that can serve as the base for the generalization (Carlson 2011): (i) instances of particular INDIVIDUALS (as in *Ravens are black*), (ii) instances of particular SITUATIONS involving (stages of) a single individual of an ordinary sort (as in *Paul plays chess*), or (iii) instances of particular SITUATIONS involving instances of different INDIVIDUALS that are members of a given kind (as in *Dogs bark*).[11,12]

The cases (ii) and (iii) involve quantification over situations, which in Krifka et al. (1995) is a defining feature of what they refer to as HABITUAL SENTENCES, a subcase of GENERIC SENTENCES:

(11) A sentence is HABITUAL iff its semantic representation is of the form
$\text{GEN}[\ldots s \ldots; \ldots](\text{RESTRICTOR}[\ldots s \ldots]; \text{MATRIX}[\ldots s \ldots])$,
where s is a situation variable (Krifka et al. 1995: 32, ex. (56))

[10] GEN and overt Q-adverbs are not unselective binders, in contrast to their early treatments, as in Carlson & Pelletier (1995), for instance. Which variables can be bound in a given logical structure depends on a number of factors, including context and topic-focus structure.

[11] The notion of SITUATIONS corresponds to occasions or cases, as Lawler (1972) originally proposed, and are akin to "cases" of Lewis (1975) (Krifka et al. 1995: 30). It plays a role similar to that of STAGES in Carlson (1977, and elsewhere) which is tied to Carlson's distinction between stage-level (episodic) predicates (SLPs) and individual-level (stative) predicates (ILPs).

[12] Generics that fall under (iii) involve a "double generalization" (in Carlson 2011; see also Pelletier & Asher 1997: 1131): *Dogs bark* is true by virtue of (a) *individual dogs* that realize the kind DOG having the characterizing property of barking, and also by virtue of (b) particular *situations* of barking by a stage of an individual dog.

Here, I reserve the term *habitual sentence* for only generic sentences that express habits in the proper sense of the word, as also observed above (Section 1). Most importantly, and as we will see below in Section 4, the generic morpheme *-va-* is not constrained to conveying habits proper, i.e., regularities of action of animate, mostly, human agents (the case (ii) above). Neither does it require that its domain contains situations (the cases (ii) and (iii) above), but it may quantify over individuals only (the case (i) above). In short, it functions as a quantifier over all the three main types of particulars mentioned above, and in this respect it is akin to GEN.

For any theory of genericity the main burden of the analysis is on specifying the relation between the generalization and what it is based on, its basis for generalization, which in many cases (but not all) are some particulars, instances, and particular facts. How this relation comes about is at the core of what it means for a sentence to express a generalization, rule, pattern and the like. We have a rich variety of competing proposals regarding this relation, and by the same token regarding the truth conditions of generic sentences that underpin GEN. One useful way of sorting them out was proposed by Carlson (1995), who suggests that they can be divided into two main perspectives, and mixed proposals combining some elements from both can also be found. One perspective is characterized in terms of an INDUCTIVE MODEL, on which generic sentences express inductive generalizations that are based on some observed (or unobserved) set of instances or particulars in the actual world from which we infer a general rule. Paradigm cases are sentences like *Paul plays chess* that express generalizations over situations of the kind described by their corresponding episodic sentences like *Paul was playing chess when I called him yesterday, Today during the lunch break, Paul played chess with Erwin, Last week, Paul played chess with grandpa and won each and every time.* On this perspective, generics would (mainly) express descriptive (weak) generalizations.

The second main perspective corresponds to what Carlson (1995) dubs a RULES-AND-REGULATIONS MODEL. On this perspective, generics express generalizations that do not (necessarily) depend on particular episodic conditions in the actual world, but rather on some causal structure or underlying forces in the world, which give rise to episodic instances (if there are any), and motivate the patterns that we recognize in the actual world. Paradigm cases are generic sentences expressing various rules, such as game rules that we may learn directly, rather than inferring them from episodic conditions in the actual world, for instance, by reading the relevant rules, manuals or guidelines, such as the rules of chess like *Bishops move diagonally*, or following instructions, customs and the like.

In contrast to the inductive model, the rules-and-regulations model can also handle generic sentences that are independent of particular instances or situations, because they have no actualized verifying instances, such as (8a). On the rules-and-regulations model of genericity, we judge generic sentences to be true or false with respect to a set of rules (or a finite list of propositions), which are considered to be irreducible entities, supporting the truth of possibly an unlimited number of propositions.

Carlson (1995) proposes that the rules-and-regulations model is a better candidate than the inductive one for a unified semantic analysis of generic sentences, and a model which might ground the semantics of GEN. However, it is not a good fit for generics like *Paul plays chess*, "which lend the most prima facie plausibility to the inductive model" (Carlson 1995: 237). A rules-and-regulations model requires us to identify some rule, causal factor or underlying condition that gives rise to or "regulates" Paul's chess playing behavior, which seems rather far-fetched.

These two main perspectives raise an intriguing question which of them might best fit generics marked with the generic morpheme *-va-* in Czech, and in generic sentences that are marked with the cognate marker in other Slavic languages. Given that the the marker *-va-* requires that there be realized instantiations that count as evidence for the truth of generic sentences marked by it, as I will show, generic sentences marked with *-va-* at first blush fit the inductive model of genericity. At the same time, as it will be shown, *-va-* can also be used in generics that express rules, regulations, often with the goal of ensuring plausible deniability, or safeguarding a rule, a regulation against refutation by states of affairs of which its author is ignorant. The exploration of the properties of the generic morpheme *-va-* is highly relevant to the answer to the questions posed at the outset: What kind of generic sentence does the generic morpheme *-va-* delimit? Can we provide a uniform analysis for all generic sentences?

3 Morphological background

3.1 The perfective/imperfective and the generic/non-generic distinctions

The generic marker *-va-*, here glossed as GEN, is productive in West Slavic languages: namely, in Czech, Slovak (e.g., Kučera 1981, Bílý 1986, i.a.), and to a lesser degree also in Polish (Bílý 1986, i.a.). In South and East Slavic languages, its oc-

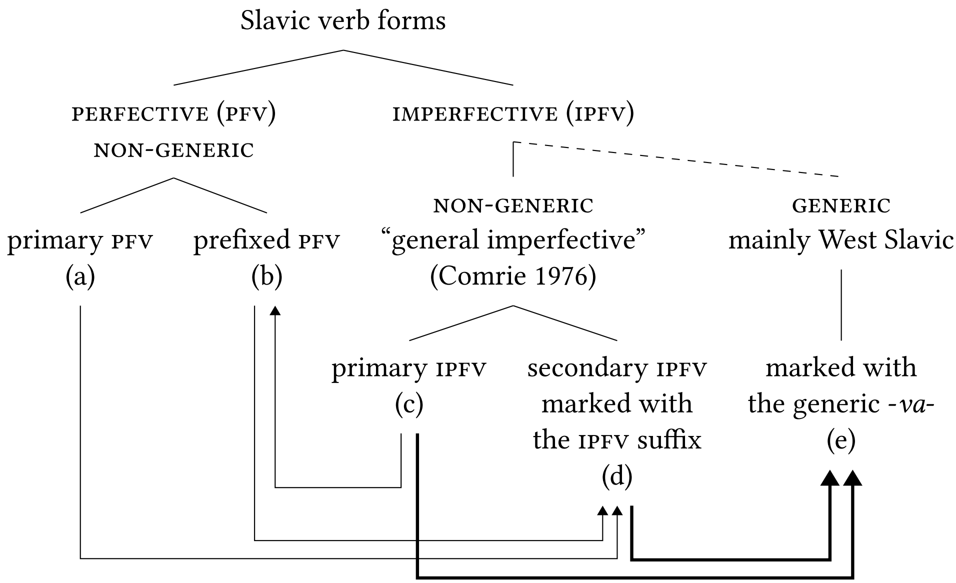

Figure 1: Common formation patterns of non-generic verb forms (perfective and imperfective) and generic verb forms (following e.g. Petr 1986: 180). The dashed line: The traditional subsumption of generic (also known as "iterative") verbs under imperfective aspect is not uncontroversial (see e.g. Kopečný 1948, i.a.) and may best be refuted.

currences are also attested, but they may be treated as a part of a lexicalized combination with a verb base.[13]

Taking Czech as our case in point, in traditional reference grammars, the verb system is structured by two main distinctions: (i) the perfective/imperfective distinction, and (ii) the generic/non-generic distinction. This is sketched in Figure 1, which builds on the rudimentary schema given in Petr (1986: 180). Table 1 contains examples illustrating the different types of verb forms and their morphological relations.

In Bohemistics, the morpheme *-va-* (here glossed as gen) is traditionally labeled "iterative" (*iterativní* in Czech) or "multiplicative" (*násobená* in Czech). It

[13]In Russian, verbs with this marker are taken to be a feature of a colloquial ("substandard") speech (Isačenko 1962: 405–407, Forsyth 1970: 28 and 168–171, Comrie 1976: 27, Kučera 1981: 177, Bílý 1985, Vinogradov 1986: 413–414, Petr 1986, Bílý 1986), while some of them are in fact quite frequent: e.g., *byvat'* (< *byt'* 'to be'), *znavat'* (< *znat'* 'to know'), *govarivat'* (< *govorit'* 'to speak'), *pivat'* (< *pit'* 'to drink'), *siživat'* (< *sidet'* 'to sit') and *xaživat'* (< *xodit'* 'to go'). Interestingly, the generic marker similar to that in West Slavic languages is productive in some Northern Russian dialects (Barnetová 1956).

is clearly separated from the imperfectivizing suffix (see e.g., Petr 1986, Lamprecht et al. 1986, Kosek 2014, Biskup 2024; Genis et al. 2021 also separate the cognate generic morpheme from the imperfectivizing suffix in other Slavic languages). However, at the same time, this morpheme is subsumed under imperfective aspect, treated as a kind of imperfective marker, with one notable exception, namely Kopečný (1948, 1966).

Notice that Figure 1 has a dashed line connecting the IMPERFECTIVE (IPFV) with the GENERIC node. It indicates that the traditional subsumption of verbs with generic morpheme *-va-* under imperfective aspect is not uncontroversial. It is one of the main goals of this chapter to show that verbs marked with the generic morpheme *-va-* do not neatly fit the imperfective paradigm in Slavic languages, and imperfectivity in natural language generally. They have a number of semantic and pragmatic properties that differ from both primary and secondary imperfectives in Slavic languages. This in turn conforms with independent arguments made elsewhere that genericity is is best viewed as independent of imperfectivity, and other categories of the TMA system (see e.g., Filip & Carlson 1997).

In Slavic languages, all the forms without the generic morpheme *-va-* that are inherently episodic alternate between episodic and generic readings depending on context. Hence, they are referred to as non-generic in Figure 1. Slavic non-generic forms are either imperfective or perfective, and members of both classes can be either simple (root, primary) or derived, mostly by means of derivational affixation. Primary and secondary imperfectives correspond to "general imperfectives" (Comrie 1976), meaning that they cover the whole semantic domain of imperfectivity, both its episodic and generic subdomains.

Perfective verbs in Slavic languages, all of which are non-generic (i.e., have either episodic or generic uses depending on context), are for the most part derived. Prefixation operating on primary imperfective bases is the most common way of forming perfective verbs. This corresponds to the derivational relation (c) → (b) in Figure 1. Slavic verb prefixes have hallmark properties of lexical-derivational, rather than inflectional, morphology, and hence are not grammatical markers of perfectivity akin to bona fide inflectional morphology like tense markers (*pace* Borer 2005, i.a., for arguments see Filip 1992, 2000, 2026, and elsewhere).[14] Slavic prefixation derives new verbs, typically with a different lexical meaning and argument structure than their derivational base. In the relevant examples in Table 1,

[14]The argument that Slavic verb prefixes are not grammatical markers of perfective aspect in Slavic languages provides a key support for the view that the Slavic perfective/imperfective distinction is predominantly of lexical-derivational, rather than inflectional, nature. This is also reflected in the common Slavic lexicographic practice of marking lexical entries of verbs as either perfective or imperfective, and some as bi-aspectual.

Table 1: Examples illustrating the morphological relations included in the schema in Figure 1

(a) dát$^{\text{PFV primary}}$ give.INF ‘to give’	→	(d) dávat$^{\text{IPFV secondary}}$ give.IPFV.INF ‘to give’/‘to be giving’
(d) dávat$^{\text{IPFV secondary}}$ give.IPFV.INF ‘to give’/‘to be giving’	→	(e) dávávat$^{\text{generic}}$ give.IPFV.GEN.INF ‘to give regularly, seldom, often, usually, …’
(c) štěkat$^{\text{IPFV primary}}$ bark.INF ‘to (be) bark(ing)’	→	(b) poštěkat$^{\text{PFV prefixed}}$ PREF.bark.INF ‘to bark (a little)’; metaphorically: ‘to bicker’, ‘to squabble’
(b) poštěkat$^{\text{PFV prefixed}}$ PREF.bark.INF ‘to bark (a little)’; metaphorically: ‘to bicker’, …	→	(d) poštěkávat$^{\text{IPFV secondary}}$ PREF.bark.IPFV.INF ‘to (be) bark(ing) (a little)’; metaph.:‘to (be) bicker(ing)’, …
(d) poštěkávat$^{\text{IPFV secondary}}$ PREF.bark.IPFV.INF ‘to (be) bark(ing) (a little)’; ‘to (be) bicker(ing)’, …	→	(e) poštěkávávat$^{\text{generic}}$ bark.IPFV.GEN.INF ‘to (tend to) bark/bicker a little, regularly, seldom, often, usually, …
(c) štěkat$^{\text{IPFV primary}}$ bark.INF ‘to (be) bark(ing)’	→	(e) štěkávat$^{\text{generic}}$ bark.GEN.INF ‘to (tend to) bark regularly, seldom, often, usually, …

the prefix *po-* not only derives a perfective verb from the primary imperfective base, under (c) in Figure 1, but also contributes an attenuative meaning to the resulting prefixed verb which roughly amounts to 'bark (a little)', and this prefixed verb also has a metaphoric sense of 'to bicker', 'to squabble'. Prefixes are also applied to already prefixed perfective bases, so that more than one prefix can occur on one and the same verb.

Secondary imperfectives (under (d) in Figure 1) are derived by means of the imperfectivizing suffix, glossed as IPFV. In all Slavic languages, the imperfectivizing suffix applies to perfective bases and derives their imperfective counterparts, typically with the same lexical meaning and argument structure. The perfective bases to which the imperfectivizing suffix is applied is either primary perfective, as in (a) → (d), or prefixed, as in (b) → (d).[15]

Most importantly, in contrast to the Slavic imperfectivizing suffix, the generic morpheme applies to an imperfective base realized in verb forms that can have either an episodic or a generic reading and derives a form that is unambiguously generic, i.e., it excludes any episodic interpretation.[16] The imperfective base to which the generic morpheme *-va-* applies is either primary, as in (c) → (e), or secondary, as in (d) → (e). When the generic morpheme is applied to a secondary imperfective base, as in (d) → (e), the resultant generic verb contains both the imperfectivizing suffix (IPFV) and the generic morpheme (GEN).

In Czech, the derivation of specifically generic forms from imperfective non-generic ones is transparent, regular and productive, and verbs with the generic morpheme *-va-* are common in all styles of speech (Kučera 1981: 177, Petr 1986).[17] The Czech generic morpheme *-va-* can be reduplicated, which also confirms its productivity: e.g., *mít*$^{\text{IPFV}}$ 'to have' > *mí-vá~vá~va-t* [have-GEN~GEN~GEN-INF] 'to tend to have, very sporadically/very often', 'to have once in a great while'. Attested reduplication examples are fairly easy to find, and two are given below:

(12) a. *Říkávává* se, že po bitvě je každý generál.
say.GEN.GEN.3SG.PRS REFL that after battle is everybody general
'As the saying goes, after the battle, everybody is a general.'

[15]The imperfectivizing suffix may generate so-called "aspectual pairs", i.e., two verbs with the same lexical meaning that only differ in grammatical perfective/imperfective aspect. Therefore, some classify it as an inflectional suffix (Isačenko 1960, Filip 1993a, 1999). It is also often taken to be the only marker in Slavic languages that is solely dedicated to the encoding of aspect (Isačenko 1960, Filip 1993a, 1999).

[16]In Czech, the generic morpheme *-va-* applies to past tense imperfective stems (Petr 1986: 184).

[17]There are also a few irregular generic verbs: e.g., in Czech *číst* > *čítat* 'to read', *vidět* > *vídat* 'to see', *slyšet* > *slýchat* 'to hear', *ležet* > *léhat* 'to lie (down)'.

b. Bude to z jiného konce, než *bývává* zvykem.
will.be it from different end than be.GEN.GEN.3SG.PRS customary
'It will be approached from a different angle than is customary.'

Reduplication is often associated with the speaker's subjective judgment concerning the regularity of instances that form the base for the generalization. It may affectively connote their strikingly high or low frequency.[18]

The formation of generic verbs by means of the morpheme *-va-*, as just illustrated with some Czech examples, follows a cross-linguistic pattern that is attested in a number of typologically unrelated languages that have specifically generic forms: namely, generic forms are derived from non-generic ones; they are the marked case relative to non-generic forms, which are morphologically or syntactically simpler and alternate between episodic and generic interpretations. We do not find generic forms that are basic and episodic forms derived from them (Carlson 1995, 2005, and elsewhere).

Examples of languages that have verb markers enforcing only a generic reading of sentences, but are not a necessary condition for the expression of generic statements, can be found in Dahl (1985, 1995). The Czech marker *-va-* serves as his example par excellence for this class of markers. However, Dahl (1995) also argues that they are not generic markers, but rather habitual markers falling either under tense and/or aspect; they are habitual markers, because, according to him, a bona fide generic marker must constitute not only a sufficient but also a necessary condition on the expression of all and only generic statements. Only 3 languages in Dahl's data base of 76 languages have such a generic marker: namely, Wolof (Niger-Kordofanian), Iṣẹkiri (Niger-Kordofanian), and Maori (Austronesian) (Dahl 1995: 420, 424). Such observations constitute one of Dahl's empirical arguments in support of his main conclusion that the episodic/generic distinction has no direct relevance for the grammar of natural language; rather, it is generally only indirectly reflected in speakers' choices between grammatical markers (Dahl 1995: 425).

To this it might be immediately objected that Dahl's claim here relies on an overly restrictive, unwarranted requirement of what it means for a given marker to count as a grammatical marker of a given category. As is well-known, bona fide grammatical markers (e.g., the English past tense morpheme *-ed*) do not neatly delimit a single semantic domain, and absolute obligatoriness is not a necessary criterion for grammaticalization, as "[s]omething is obligatory relative to the context; i.e., it may be obligatory in one context, optional in another and

[18](12a) retrieved at https://www.vidlakovykydy.cz/clanky/maginotova-linie-ceskoslovenske-opevneni accessed 13/12/2021; (12b) from a native Czech speaker (p.c.).

impossible in a third context" (Lehmann 1995: 12; see also Heine & Kuteva 2002, 2007, and others, for similar observations).

Dahl (1995: 421) also observes that "[a] relatively safe generalization is that habituals in general exhibit a low degree of grammaticalization. They are [...] possible sources of a grammaticalization path yielding what would be the primary candidates for a gram that systematically marks genericity." In West Slavic languages at least, and as we have just seen in Czech, the generic marker is productive, and systematically in all its occurrences only contributes a generic interpretation to the meaning of a verb, in a completely transparent and compositional way. This would seem to indicate that it exhibits a high degree of grammaticalization.

Having sketched some basic morphological facts, the next section will address the independence of the Slavic generic morpheme from tense and imperfective aspect, and specifically its independence from the imperfectivizing suffix. We will also see that it does not neatly fit into the Slavic imperfective paradigm, contrary to standard assumptions in Bohemistics, and also contrary to Dahl (1995).

3.2 Independence of the generic morpheme from tense and imperfective aspect

The generic morpheme *-va-* is independent of tense. It occurs on non-finite forms (including the infinitive and imperative, the relevant examples are given in Filip & Carlson 1997), and with all the overt tense markers (past, present, and future) on the same verb or in the same verb complex, as we see in the following Czech examples:[19]

(13) a. Karel hrával hokej. PAST TENSE
Charles play.GEN.PST hockey
'Charles used to play hockey.' [remote past]

b. Karel hrává hokej. PRESENT TENSE
Charles play.GEN.PRS hockey
'Charles tends to play hockey.' [i.e., regularly, often, seldom, ...]

[19]In the past tense, generic sentences formally marked with *-va-* have a remote past time reference, whereby its reduplication entails a high degree in the past temporal remoteness (Kučera 1981, 1983, Bílý 1986): e.g., *Vídávával* [see.GEN.GEN.1SG.PST] *jsem ho tam ve čtvrtky* 'I used to see him there on Thursdays once in a while / regularly ... a very long time ago.' (Chmelíčková 2005). Generally, *-va-* (possibly also reduplicated) combined with the past tense triggers a conversational implicature that the described situation no longer holds, and can also refer to single extended situations in the past.

c. Karel bude hrávat hokej. FUTURE TENSE
Charles be.AUX.FUT play.GEN.INF hockey
'Charles will tend to play hockey.' [i.e., regularly, often, seldom, …]

The independence of the generic morpheme *-va-* from tense is not surprising, given that generally markers of genericity and tense are orthogonal to one another in languages that have overt markers for genericity and tense. There are tensed languages with generic markers (e.g., Czech, Polish, Slovak), without generic markers (e.g., English), and tenseless languages with arguably generic markers (e.g., American Sign Language) and without them (e.g., Chinese), in which case they convey generic statements by other means (e.g., in Dyirbal and Burmese by means of a modal distinction between realis and irrealis; see Comrie 1985: 51).

Also notionally, genericity (and habituality) and tense are independent of each other. Tense is a deictic category, while genericity (and habituality) are non-deictic. In tensed languages like English, generic sentences can be in any tense form, and refer to present, future or past time: e.g., *Dinosaurs ate kelp; Mary walks to school along the river; When he retires, he will spend more time with his family.* In English, the simple (non-progressive) present tense, as opposed to the present progressive, preferably has a generic interpretation.

Generic sentences are compatible with any sufficiently large temporal interval, contextually located in the present, past, or future. They may also express generalizations that contain overt or covert spatial/temporal restrictors related to the present, past, or future, as shown in (14).

(14) a. *Současný* prezident jídává hamburgery.
current president eat.GEN.PRS hamburgers
'The current president eats hamburgers.'

b. Dinosauři jídávali kelp.
dinosaurs eat.GEN.PST kelp
'Dinosaurs (usually) ate kelp.' [The generalization is restricted to the distant past by virtue of the general world knowledge about the dinosaur kind.]

c. *Od pondělka* se bude tato kancelář otvírávat až ve 2 hodiny.
from Monday REFL be.AUX.FUT this office open.GEN.IPFV only at 2 o'clock
'Starting next Monday this office will open only at 2 p.m.'

Let me now turn to the independence of the generic marker from the imperfectivizing suffix. It is not uncommon for the generic marker *-va-* and the imperfectivizing suffix to be confounded, both notionally and formally, given that their surface allomorphic variants may often seem alike, and both occur in generic sentences. One compelling piece of evidence for their formal independence stems from the fact that they may co-occur on the same verb (see Filip & Carlson 1997). Consider the following attested example:[20]

(15) Bývalý mi taky *dávával* nádherně a originálně
ex me.DAT also give.IPFV.GEN.PST wonderfully and originally
vázané kytice, přímo umělecká díla, …
bound bouquets directly art works
'My ex also used to give me wonderful bouquets in original arrangements, real works of art, …'

In (15), the morpheme sequence which is here glossed with IPFV.GEN is the result of the derivation pattern (a) > (d) > (e), given in Figure 1. In the first step (a) > (d) the imperfectivizing suffix (IPFV) applies to a (primary, or root) perfective base and derives an imperfective form, which alternates between an episodic and a generic reading depending on context. Hence, it is unmarked for genericity, i.e., non-generic or a "general imperfective" form (in the sense of Comrie 1976). In the second step (d) > (e), the generic *-va-* 'GEN' is added "on top of" the imperfectivizing suffix (IPFV), and yields a verb that only has a generic reading.

Following the standard assumption about what it means to be a "formal member of a category system", given that the Slavic imperfectivizing suffix (IPFV) is a marker of imperfectivity, as all agree, the morpheme *-va-* 'GEN' cannot be a different marker of the same imperfective category. A given formal member of a category system stands in a complementary distribution to other formal members of the same category, morphologically and syntactically. For instance, the present and past tense morphemes in English are formal members of the same tense category, hence are in complementary distribution: e.g., **she laugh-s-ed* or **she laugh-ed-s*. The markers of tense and progressive aspect in English can co-occur, because they are *not* formal members of the same grammatical category: e.g., *she was/is/will be laughing*. If we accept this line of reasoning for the present and past tense morphemes in English, we should be ready to accept that the fact that the generic marker *-va-* and the imperfectivizing suffix do not stand in a

[20] The example is from https://abecedazdravi.cz/diskuse/psychicke-problemy-a-psychologie/deprese-i-nova-diskuze/89; accessed on July 9, 2023.

paradigmatic opposition to each other indicates that they are not only two separate morphemes, but also that they cannot be two different markers of the same grammatical category system, and specifically both marking imperfectivity.

To the above argument it could be objected that the morpheme sequence which is here glossed with IPFV.GEN (as in (15)) instantiates two occurrences of one and the same imperfectivizing suffix: 'IPFV + IPFV'. Hence, what we have in (15) is something like a doubly aspectualized imperfective verb associated with a generic (or habitual) reading. Implicit behind this idea would seem to be some encoding economy principle, namely that it is superfluous to apply twice one and the same imperfectivizing suffix with one and the same imperfective function. Therefore, the second application of IPFV must have a different function, i.e., affect a truth-conditional change from a non-generic, or general imperfective, to a generic imperfective meaning.

One immediate prima facie counterargument against the 'IPFV + IPFV' double aspectualization idea has to do with verbs that only have a generic interpretation, but contain a *single* occurrence of the overt alleged IPFV suffix. This is a subclass of generic verbs derived following the pattern (c) > (e) in Figure 1. In order to rescue the double aspectualization idea it might be proposed that simplex (primary) imperfective verbs like *štěkat* 'to (be) bark(ing)', in Figure 1 under (c), contain a phonologically null imperfectivizing suffix $\varnothing^{\text{IPFV}}$, and so generic verbs that are formed following the pattern (c) > (e) contain one covert and one overt instance of the same IPFV suffix: $\varnothing^{\text{IPFV}}$ + IPFV.[21] However, such a move is hardly empirically motivated at all, theoretically costly, and generally supports a proliferation of empty categories for purely theory-internal reasons.

Reduplication without a change in the truth conditions is common in natural languages, it may serve to add affective connotations, intensification, as we have seen above in (12a) and (12b). We would have to ask: Just why should the reduplication of two overt IPFV morphemes have a truth-conditional consequence, and enforce the generic interpretation, when a truth-conditional change is not a necessary consequence of all reduplication processes? Similarly, we may ask: Just why should a double aspectualization consisting of one overt IPFV morpheme and one covert $\varnothing^{\text{IPFV}}$ morpheme have a truth-conditional consequence such that the second overt occurrence of this morpheme restricts the verb to only a generic interpretation?

Moreover, the IPFV morpheme occurs on verbs that are general imperfective verbs, because they can alternate between an episodic (including progressive)

[21]That Slavic primary imperfectives introduce a null imperfective suffix is proposed in syntactic accounts of Slavic aspect, see e.g. Grønn (2026 [this volume]).

and a generic interpretation depending on context (as imperfective morphemes in natural language do, see e.g., Comrie 1976). Presumably the second overt occurrence of the alleged IPFV should be also inherently amenable to either the episodic or generic interpretation, but then it is unclear just why the second overt occurrence of IPFV should enforce the generic interpretation, and not the episodic interpretation of its output. For instance, it is unclear why it does not constrain its output to only the episodic progressive interpretation. The specifically generic properties of verbs containing the sequence IPFV+IPFV or the sequence $\varnothing^{\text{IPFV}}$+IPFV would have to be postulated, because they do not follow from the semantics of either IPFV or $\varnothing^{\text{IPFV}}$.

A more sophisticated version of the reduplication or double aspectualization idea is proposed by Cable (2022): namely, the Czech "habitual" (in his terms, generic in my terms) morpheme, as explored in Filip (2018), is, according to Cable "etymologically an instance of imperfective-aspect combining with another, lower imperfective-form of the verb, a kind of doubly aspectualized verb (Filip 2018). One might wonder, then, whether that higher imperfective-morphology might at all synchronically be a realization of a (bound) T-head, while the lower aspectual morphology is a realization of (non-modal) [IMPRV_{OG}]" (Cable 2022: 44–45) [where OG stands for "ongoing", HF]. This suggestion builds on Cable's analysis of habitual morphology in Tlingit (Na-Dene; Alaska, British Columbia, Yukon), where a verb may be realized in the "habitual perfective-mode" formed with the habitual suffix *-ch*, or in the "habitual imperfective-mode" consisting of the imperfective form followed by the habitual particle *nooch*. The main thesis is that there are two different morphosyntactic structures that underpin habituality both within and across languages. One where the habitual semantics is directly contributed by imperfective aspect, which has the meaning of a modal operator quantifying over alternative worlds/situations (see e.g., Ferreira 2016). In Tlingit, it is only verbs bearing imperfective-mode that can be used to describe pure dispositions, non-actualized regularities. The other structure underpinning habituality corresponds to the bi-partite morphosyntactic structure, where the habitual morphology is the realization of a quantificationally bound T(emporal)-head, licensed when it is bound by a (possibly covert) Q-adverb, which combines with the lower perfective or imperfective Asp-head to yield a complex form with a habitual meaning.[22] The Tlingit habitual morphology is a kind of quantificationally dependent tense, where it is the (potentially covert) Q-adverb that is

[22]This licensing strategy resembles that of Chierchia (1995), who assumes that genericity is subsumed under aspect, and "all languages have a distinctive habitual morpheme (say, Hab) which can take diverse overt realizations" (Chierchia 1995: 197). Chierchia's Hab is a functional head in an (imperfective) aspectual projection, and either realized by explicit aspectual morphemes,

the source of the habitual meaning. It is analyzed as a temporal quantifier, e.g., 'always', 'sometimes', which quantifies strictly over times in the actual world, it does not introduce modal quantification over other possible worlds, because, according to Cable, it comes with the actualization entailment.

Cable's (2022) double aspectualization proposal for Czech does not work for three main reasons. First, it raises the questions posed above about just why the verb "doubly aspectualized" with the same imperfectivizing suffix should have its interpretation constrained to the generic (Cable's "habitual") reading, when the imperfectivizing suffix covers the whole imperfective domain, both its episodic and generic subdomains. Why does not the "double imperfective aspectualization" exclude the generic interpretation, and enforces only an episodic (e.g., progressive) interpretation?

Second, Cable (2022) proposes that habitual morphology, within and across languages, is a kind of quantificationally dependent tense. However, generally, the semantics of generic sentences and habitual sentences as their special case cannot be analyzed in terms of temporal notions, or reduced to the semantics of Q-Adverbs (see Section 2, and references therein). As has been shown above, genericity, and habituality as its special case, is independent from tense, both notionally and formally. Specifically, the Czech generic morpheme *-va-* (a habitual morpheme in terms of Cable 2022) is fundamentally independent of tense.

Third, as many others, Cable (2022) correlates the so-called actualization entailment of habitual sentences with the lack of intensionality (modality). However, all generic sentences, including habitual sentences (i.e., concerning habits in

or, as in English, or it is covert. Hab carries the agreement feature [+Q] which in its specifier requires some overt Q-adverb (Chierchia 1995: 197–198) or the **Gen** operator (Chierchia 1995: 210). The idea that both stand in a local licensing relation to $\mathrm{Hab}_{[+Q]}$ is inspired by the analysis of NPIs (Ladusaw 1979), which effectively makes genericity a polarity phenomenon (Chierchia 1995: 212ff.). The main innovation is the analysis of ILPs as inherent generics, capitalizing on the assumption that polarity phenomena are rooted in the lexicon (with effects on sentences). The lexical entries of ILPs have an abstract habitual morpheme $\mathrm{Hab}_{[+Q]}$, whereby its agreement feature [+Q] can be locally licensed *only* by **Gen**, and never by overt Q-adverbs. ILPs are also taken to involve generic quantification over situations. Their Davidsonian situation variable *s* is bound by **Gen** making it unavailable for binding by overt Q-adverbs. Aside from the problems with Chierchia's (1995) proposal, some of which are detailed in Fernald (2000), the Czech morpheme *-va-* cannot be the spell-out of the covert functional head $\mathrm{Hab}_{[+Q]}$. If it were, it should have the same semantic properties as its covert counterpart in the logical representation of primary and secondary imperfectives on their generic interpretation. However, it does not. Moreover, if *-va-* were the spell out of $\mathrm{Hab}_{[+Q]}$, it should be uniformly incompatible with all ILPs, because their $\mathrm{Hab}_{[+Q]}$ can be locally licensed *only* by **Gen**, and never by overt Q-adverbs. However, *-va-* is compatible with ILPs provided they allow for an "interruption" construal, which also sanctions their co-occurrence with overt Q-adverbs (see Section 4.1.1).

the proper sense), be they formally unmarked or marked by means of dedicated markers like the Czech *-va-*, have modal import and predictive force, precisely because they express some habit or regularity. As such they have implications for other accessible worlds, than just the actual world, and perhaps all possible worlds. The modality of habitual sentences is not only compatible with their actualization entailment, or better realization inference, but is their *sine qua non* semantic property. Hence, a purely temporal analysis in terms of quantifiers that strictly quantify over times in the actual world will not do to capture the meaning of habitual sentences.

Finally, there is another problem faced by Cable's proposal. Cable (2022) assumes that any modality of habituals must be contributed by imperfective morphology, or some other material in the sentence, which has the meaning of a modal operator quantifying over alternative worlds other than the actual one, but never by perfective morphology, which is assumed to be non-modal. However, not only is there no notional and formal incompatibility between perfectivity and genericity as well as habituality (specifically for Slavic languages see Filip 1993b, 1994, Filip & Carlson 1997, and elsewhere), which Cable does not deny, but also perfective forms when used to express generic and habitual statements possess a modal semantics, and so necessitate quantification over alternative worlds other than the actual one. For instance, the English generic sentence (8a), which has a purely intensional interpretation, and is true despite having no actualized instances that count as evidence for its truth, and may never have any, is naturally rendered with a perfective verb in Czech:

(16) Po viceprezidentovi *nastoupí*$^{\text{PFV}}$ na místo prezidenta předseda
after vice.president ascend.PRS.3SG on place president chairman
sněmovny.
assembly
'The Speaker of the House succeeds the Vice President.'

This leads me to make some observations on the use of perfective forms in generic (and habitual) sentences, as a conclusion to this section. Some examples of generic sentences with naturally occurring perfective forms from Slavic languages, Russian, Serbo-Croatian and Czech, as well as Romance (Italian) are given below:[23]

[23]The Czech example (17c) is taken from https://www.kosmas.cz/oko/ukazky/430121/o-panovnici-elite-prezidentu-povysenci-a-polarnikovi-v-nervoznim-pohybu/, accessed July 9, 2023.

(17) a. Nastojaščij drug vsegda pomožet$^{\text{PFV}}$.
real friend always helps
'A real friend will always help you.' (a Russian proverb)

b. Svako jutro popijem$^{\text{PFV}}$ čašu rakije.
every morning PREF.drink glass brandy
'Every morning I drink a glass of brandy.'
(Serbo-Croatian; Mønnesland 1984: 62)

c. To se jeden nalítá$^{\text{PFV}}$ v úřadovnách prezidentského paláce.
it REFL one PREF.fly in offices president palace
'What a runaround one must take in the offices of the presidential palace.' (Czech)

d. Sempre, quando mi vide$^{\text{PFV}}$, il custode aprì$^{\text{PFV}}$ la porta.
always when me saw the janitor opened the door
'Always when he saw me, the janitor opened the door.'
(Italian; Bonomi 1997: 508)

Dahl (1995) claims that the use of perfective forms in generic (and habitual) sentences in Slavic languages is another "case of deviant behavior" (Dahl 1995: 420) of their aspect systems. One must wonder about the justification for this claim, given that perfective forms are commonly used in generic sentences in Romance languages, for instance, as the above example from Italian shows. What is noteworthy though is that perfective forms are commonly used in generics and habituals in West Slavic languages.[24] In East and South Slavic languages they are less common (Forsyth 1970, Fortuin & Kamphuis 2015, Wiemer & Seržant 2017), but perhaps not as rarely used as is generally thought (Wiemer 2023).[25]

Not only is perfectivity notionally compatible with genericity (and habituality), but also their respective overt markers are compatible, because they can co-occur on the same verb form: e.g., in Lithuanian "frequentative past" (Dambriunas et al. 1966), or in Awa (New Guinea) (Loving & McKaughan 1964).

[24]For a recent study on the uses of the perfective present in Polish generic (and habitual) sentences see Wiemer et al. (2025).

[25]In West Slavic language the use of perfective forms for the expression of generic and habitual statements is common, though their frequency as well as the functional range depends on the type of generalization to be expressed, and lexical idiosyncracies of the relevant (im)perfective forms. For instance, in Czech, perfective verbs are common or even preferred in cooking recipes, assembly instructions, game rules like rules of chess, proverbs, to name just a few types of generic statements.

In sum, genericity (and habituality) is orthogonal to the imperfective and perfective distinction (see also Filip & Carlson 1997). Acknowledging this separation of grammatical aspect from genericity (and habituality) is not to deny that there are close semantic affinities between imperfectivity and genericity, as well as habituality, as its special case.[26] Moreover, acknowledging that genericity (and habituality) is orthogonal to the imperfective and perfective distinction means that markers like the Slavic morpheme *-va-* that systematically enforce only a generic (or a habitual) interpretation of sentences need not be subsumed under imperfective aspect.

In support of this point, in the next section, I show that the Czech generic morpheme *-va-* contributes semantic and pragmatic properties to the meaning of generic sentences that their corresponding sentences headed by non-generic imperfective verbs, both primary and secondary, do not have. This also supports my conclusion that generic *-va-*marked verbs do not neatly fit the Slavic imperfective paradigm, and "general imperfective" paradigm in natural language (in the sense of e.g., Comrie 1976).

4 Semantic properties of the Slavic generic morpheme

4.1 When the generic morpheme must, must not and may be used

This section identifies cases in which verbs marked with the generic morpheme *may*, *must* and *must not* be used in contrast to imperfective verbs, which are not formally marked for genericity with this morpheme. The examples come from Czech. However, if a given Slavic language has a verb with the generic marker and could be used in the same context as that of a corresponding Czech verb, it is predicted that the same observations should hold for it that are made for Czech.

4.1.1 When the generic morpheme MUST NOT be used

The generic morpheme *-va-* must not be used in episodic contexts, where the reference is (i) to a particular on-going situation (progressive) or (ii) to a closed set of plural situations at a given world and time; neither can it be used (iii) for the expression of generalizations that permit no exceptions, nor (iv) for the

[26] For instance, aspectually stative predicates in Slavic languages are expressed by imperfective verbs, and aspectual stativity is a key semantic property of generic predicates. Moreover, from the grammaticalization point of view, the expression of genericity (habituality) is tied to markers that become grammaticalized on the path from (present) progressive markers to imperfective ones (Bybee & Dahl 1989).

expression of generalizations with unrealized particulars in the actual world, i.e., in only dispositional, purely hypothetical generalizations.

4.1.1.1 Lack of reference to a particular (on-going) situation

One of the well-known tests tapping into the semantics of the Slavic perfective/imperfective distinction is the possibility of reference to a particular episodic situation that is on-going at a specific moment of time, i.e., the possibility of the progressive interpretation. All imperfective verbs that denote stage-level predicates pass this test, but those denoting individual-level (stative) predicates do not, and neither do perfective verbs.

Most importantly, all verbs marked with the generic morpheme *-va-* also fail this test. In traditional Bohemistics, this property is well-known as their hallmark property, which is referred to as the "non-actuality" or "atemporality" property at least since Kopečný (1948). (See also Kopečný 1962, 1965, Kopečný 1966: 259.) The example below, which is adapted from Kopečný (1948), shows that Czech verbs marked with *-va-* cannot be felicitously used in answers to questions about what is *actually* going on 'now' at the speech time. In contrast, their imperfective counterparts without the morpheme *-va-* can. It is important to emphasize that the evaluation with respect to "extended-now" that covers extended intervals of time is irrelevant for this test:

(18) *Context*: 'What are you doing right now?'

a. # Psávám jeden naléhavý dopis.
write.GEN.1SG.PRS one well-known letter
'I write one urgent letter.' (as a rule/as a habit/often/rarely/usually)

b. Píšu^{IPFV} jeden naléhavý dopis.
write.1SG.PRS one urgent letter
'I am writing one urgent letter.'

The generic morpheme *-va-* if it is attested on a verb in other Slavic languages will also give rise to the oddity that we see in (18a).

The semantic property of "non-actuality" or "atemporality" contributed by the Czech morpheme *-va-* along with its productivity and compositional transparency (see also Section 3) lead Kopečný (1948) to propose that it is a grammatical marker of a "third aspect", namely an "atemporal aspect" that is independent of the Slavic perfective/imperfective distinction (see also Kosek 2014). As a wide-reaching consequence, Kopečný (1948) rejects the claim made by Koschmieder

(1929) that Indo-European languages lack a grammatical category of “atemporality”.[27] On Kopečný’s third “atemporal aspect” hypothesis, the Czech verb system might be schematically represented as in Figure 2, albeit recast in terms of the non-generic/generic distinction, as it is used in the contemporary literature and in this chapter.

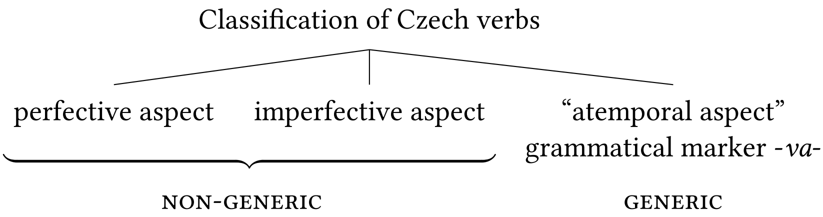

Figure 2: Classification of Czech verbs (following Kopečný 1948 and elsewhere)

Aligned with Kopečný’s “third atemporal aspect” hypothesis, Lamprecht et al. (1986: 325) states that Czech verb forms that are built with the generic morpheme -*va*- (‘generic’ in my terms) are clearly not imperfective forms, because imperfectives are formed by other morphologically means. Also Kosek (2014: 152) observes that in Czech the “third aspect” with its dedicated formal expression by means of the morpheme -*va*- has been established since the 16th century, when it became independent of the perfective/imperfective distinction. However, Kosek, unlike Kopečný, erroneously views it as a means for the expression of iterativity, or frequentativity.

Kopečný’s (1948) “third atemporal aspect” hypothesis is of interest here in so far as it highlights the hallmark “atemporal” or “non-actuality” property of the Czech morpheme -*va*-, i.e., that it formally marks sentences that lack reference to a specific situation in all its occurrences. As observed above (see Section 2), it is one of the key properties of generic predicates in natural languages generally, which makes them aspectually stative. Kopečný’s account of the Czech morpheme -*va*- is a notable improvement on its traditional treatment as an imperfective marker carrying the iterative or multiplicative feature, which it quite clearly lacks (as will be shown below).

[27] Kopečný (1948): “Není tedy zcela správné tvrzení Koschmiedrovo v jeho Nauce o aspektech (33), že indoevropské jazyky nemají [153] gramatickou kategorii atemporálnosti (mimočasovosti, “pozaczasowości”).” [Koschmieder’s claim that Indo-European languages have no grammatical category of atemporality (extra-temporality, ‘pozaczasowość’ [Polish]) is not entirely correct. (my own translation)]

In sum, based on the above observations, the first key property of Czech verbs that are formed with the generic morpheme can be stated as follows:

(19) Semantic Property 1 (to be revised): The generic morpheme *-va-* derives verbs that systematically lack reference to particular situations. It is a property at the core of stativity, and one of the properties of generic predicates in natural languages.

Semantic Property 1 in (19) clearly separates the generic morpheme from the imperfectivizing suffix in Slavic languages, which formally marks secondary imperfectives that, as a whole class, are not aspectually stative. Just like primary imperfectives, such as *píšu* 'I am writing' in (18a), secondary imperfectives freely alternate between episodic and generic readings, and can be used with reference to a specific situation that is on-going at some given reference time, as the following attested example shows:[28]

(20) V křoví právě *přepočítával*$^{\text{IPFV}}$ bankovky v peněžence,
in shrubs just PREF.count.IPFV.PST.3.SG banknotes in wallet
strážníky na sebe upozornil světlem z mobilu.
policemanPL.ACC on himself make.awarePST.3.SG light from mobile
'At that moment he was in the shrubs counting money in his wallet, and the light of his mobile phone alerted the police to him.'

4.1.1.2 Incompatibility with iterativity, or a mere plurality of situations

In general, the lack of reference to a specific situation, which is a hallmark property of generic predicates (related to their stativity), is closely tied to their incompatibility with iterative adverbials (or cardinal count adverbials) like 'three times' or 'several times'. Consider the following example:

(21) # Včera Pavel hrával třikrát šachy s dědou.
yesterday Paul play.GEN.PST three.times chess with grandpa
#'Paul used to play chess with grandpa three times yesterday.'

The oddity of the above sentence must be due to the clash between the generic marker *-va-* and the iterative adverbial 'three times' which counts three episodes that occurred within a single interval of time 'yesterday'. That is, 'yesterday'

[28]The example is from https://www.praha11.cz/redakce/tisk.php?lanG=cs&clanek=9892&slozka=7155&as4uOriginalDomain=www.praha11.cz&as4u_protocol=https&; accessed on August 15, 2025.

enforces an episodic reading of a sentence, which must be evaluated with respect to a particular world/time, but the generic morpheme *-va-* requires evaluation at alternative possible worlds than just one (actual) world.

The generic morpheme *-va-* can scope over the iterative adverbial, if the context does not indicate otherwise, as we see (22). Here the closed sequence of iterated three eventualities forms the basis of the generalization which is a part of a larger pattern that holds every day, and across different possible worlds.

(22) Psával mi aspoň třikrát každý den.
write.GEN.PST me at.least three.times every day
'He used to write to me at least three times every day.'

In (21), however, 'yesterday' restricts the reference to a single interval/world which excludes a generic interpretation of (21), and consequently the use of the generic morpheme is odd. The corresponding non-generic sentence (23) without *-va-* is perfectly acceptable with the iterative adverbial 'three times'. The main verb here is a non-generic primary imperfective verb:

(23) Včera Pavel hrál$^{\text{IPFV}}$ třikrát šachy s dědou.
yesterday Paul play.PST three.times chess with grandpa
'Paul played chess with grandpa three times yesterday.'

Secondary imperfective verbs formed with the imperfectivizing suffix (IPFV), which are also non-generic like primary imperfectives, are compatible with iterative adverbials like 'three times':

(24) Sice jsem již třikrát vyhrávala$^{\text{IPFV}}$, ale nikdy jsem
while AUX already three.times PREF.play.IPFV.PST but never AUX
nevyhrála$^{\text{PFV}}$.
NEG.PREF.play.PST
'Although I was already winning / was about to win three times, I never won.'

The incompatibility with iterative adverbials is a test that generally separates verbs with the generic morpheme *-va-*, on the one hand, from primary and secondary imperfective verbs, on the other hand; only the latter can co-occur with iterative adverbials, modulo their lexical semantics and context in sentences that describe episodic situations that are not a part of a larger pattern. This also means that verbs with the generic morpheme *-va-*, on the one hand, and secondary imperfective verbs with the imperfectivizing suffix (IPFV) along with primary imperfectives, on the other hand, make different contributions to the meaning of a sentence. This provides one piece of evidence in support of separating verbs with the generic morpheme *-va-* from primary and secondary imperfectives.

Generally, the incompatibility of a morpheme with iterative adverbials means that it cannot be assimilated to markers in natural languages that are variously labeled as iterative, multiplicative, frequentative, repetitive, pluractional, and the like, of the kind that are used to encode a mere plurality of situations (i.e., a closed series of particular episodes or situations of the same type at a given world and time). The incompatibility of the generic morpheme *-va-* with iterative adverbials clearly shows that its common labels "iterative" or "multiplicative" and the like are misnomers at best (see also Bílý 1986), or a category error at worst (*pace* standard Czech reference grammars, e.g., Petr 1986, Kosek 2014, Lamprecht et al. 1986, Nübler 2017 or Dahl 1995).

Based on the above data and observations, we may update Semantic Property 1 in (19) as follows:

(25) Semantic Property 1 (final version): The generic morpheme *-va-* derives verbs that lack reference to particular situations. It is a property at the core of stativity, and one of the properties of generic predicates in natural languages.

Corrolary: The generic morpheme *-va-* derives verbs that are incompatible with iterative adverbials like 'three times', 'several times' used to denote a cardinality of a closed set of situations at a given time interval and world (i.e., a set that is not a part of a larger pattern, or generalization).

Semantic Property 1 is related to independent arguments made elsewhere that genericity (and habituality) is clearly separate from iterativity, and the two must not be conflated and confounded, both on formal and notional grounds (see e.g., Carlson 2012a).[29] Iterativity involves a closed set of a plurality of events, eventualities, situations, or cases of the same type (or a plurality of time intervals associated with them), but this has virtually nothing to do with habituality and genericity, i.e., with what gives rise to a generalization, a regularity, or a pattern (see e.g. Krifka et al. 1995, Filip 1993b, Carlson 2005, 2008, 2012a, i.a). It is neither sufficient nor necessary for the truth of habitual and generic statements. For instance, a mere plurality of situations is not sufficient, because it can be a part of the meaning of what clearly are non-generic statements, be it a sequence of temporally close situations that constitute a single complex situation (e.g., *Last night, he knocked three times on the door*), or multiple separate situations (e.g.,

[29] See also e.g. Bertinetto & Lenci (2012) for an overview of notions related to iterativity and their delimitation from genericity and habituality.

During our train ride to Rochester, the conductor came to our car three times to check our tickets), which may also be spread over a large interval (e.g., *During his lifetime Franz met Milena only twice*). A closed set of a plurality of situations of the same type is not necessary for the truth of generic statements, because some have a purely intensional interpretation with no actualized verifying particulars, as in (8a). Moreover, there are generic statements with ILPs whose truth is not inferrable from any array of a particular situations at all, such as *Bob owns a yacht* that presupposes one situation of acquisition like buying (Krifka et al. 1995: 38), or *Bob is a bachelor* for which it makes no sense to try to identify particular situations that count as evidence for its truth (Carlson 1995). Such observations also indicate that genericity and habituality should not be confounded in analyses of phenomena referred to as iterative aspect, event-internal or event-external iterativity, frequentativity, pluractionality, multiplicativity, iterated semelfactivity and the like (*pace* e.g., Nef 1986, Van Geenhoven 2001, 2004, Scheiner 2003).

The conflation of iterativity (i.e., a mere plurality of situations) with habituality and/or genericity is also common in the analyses of habitual and generic readings of imperfective sentences. A common strategy is to add to the imperfective operator a parameter capturing the number requirement on its application. Examples of this strategy can be found in Ferreira (2016) (also Ferreira 2004, 2005, and elsewhere), and Hacquard (2006), for instance. Imperfective forms *either* denote singular eventualities under their episodic progressive interpretation, *or* a plurality thereof under their habitual/generic interpretation.[30] For the latter, Ferreira (2005: 29) assumes a phonologically null "habitual/generic operator *hab*" when they occur without an overt Q-adverb or restrictor. Both the episodic progressive and habitual/generic interpretations share the same temporal component, namely existential quantification over time intervals, and also the same modal component with modal quantification over possible worlds, adapted from the analyses of the progressive aspect (as developed in Dowty 1979, Landman 1992, Portner 1998). In short, the habitual/generic operator *hab* is the plural counterpart of the progressive operator Prog, where Prog is defined on singular (atomic) eventualities. The Prog truth conditions are stated with respect to a pair of an interval and a world $\langle i, w\rangle$, iff there is a singular eventuality e in w and modal continuations in possible worlds in which it terminates or is completed. A habitual (generic) sentence like *John plays soccer (regularly)* is true in all the worlds in which there were plural eventualities of John playing soccer before the utterance time, and there is an ongoing plural eventuality of John playing soccer at some

[30] Assuming that VPs denote sets of time intervals, and VPs like NPs can be singular or plural, as e.g. Ferreira (2016: 358) proposes, the progressive (episodic) interpretation of a VP amounts to singular time intervals at which it holds and the "habitual" (generic) interpretation to plural time intervals.

reference time, whereby all potential external obstacles to the continuation of this plural eventuality are removed.

Setting aside that a proper characterization of the modal continuation component of Prog is still outstanding, and that the meaning of habitual/generic sentences cannot be analyzed in terms of a plurality of events (or plurality of time intervals), one obvious problem with this proposal is that it does not cover sentences with a purely dispositional reading like *Mary handles the mail from Antarctica*, which may be true even if Mary never handled any mail from Antarctica before the utterance time, and such mail may never ever arrive in the future. Neither can it handle all the relevant aspects of the meaning of bare habitual sentences like *John plays soccer*. This sentence may mean that John is not averse to playing soccer, he has an ability or disposition for playing soccer, and I can truthfully assert it, based on my experience of having seen him play soccer once in the past. This allows me to conclude that if John were in a situation to play soccer, he just might likely play soccer. *John plays soccer* is true, even if there are no multiple occasions of John playing soccer at some reference interval including the reference time, and John may never play soccer in the future. Moreover, suppose you asserted that John does not engage in any sports activity. I can contradict you by uttering *John plays soccer*, even if I have only seen John play soccer once in the past. But this cannot be captured if we assume that "habituals are about plural events" (Ferreira 2016: 353) combined with the modality that has been associated with or is akin to Prog.

4.1.1.3 Generalizations permitting no exceptions

One of the key properties of generic sentences is that most are exception-tolerating, but are also compatible with no exceptions (see Section 2). The generic morpheme, as I argue, has a meaning that combines a generalization assertion with an exception or a counterexample inference that there are or may be exceptions to the generalization. *Exceptions* to the generalization are compatible with its truth, while *counterexamples* are incompatible with its truth (see e.g. Nickel 2010). This property was already discussed by Filip (1993b, 1994, 2019, 2021), and here, it is captured as follows:

(26) Semantic Property 2: The generic morpheme *-va-* contributes the uncancellable inference (as part of its truth-conditional content) that there are actual or possible exceptions or counterexamples to the generalization.

Corollary: The generic morpheme *-va-* is incompatible with universal generalizations (e.g., universal laws of nature), or any generalization that is commonly known to permit NO EXCEPTIONS, possibly in ALL THE KNOWN CONDITIONS.

The clearest examples of Semantic Property 2 (in (26)) are sentences in which the generic morpheme *-va-* clashes with universal laws of nature (as in (27a)) or definitional statements that necessarily hold for all the members of a given kind (as in (27b)). Such exceptionless regularities call for verb forms without the generic morpheme, which, in the Czech examples below are primary imperfective verbs.

(27) a. Země se {točí / #točívá} kolem slunce.
earth REFL revolve.3SG.PRS revolve.GEN.3SG.PRS around sun
(Intended:) 'The Earth revolves around the Sun.'

b. Valčík {je / #bývá} ve tříčtvrtečním taktu.
waltz be.3SG.PRS be.GEN.3SG.PRS in three.quarter time
(Intended:) 'A waltz is in three quarter time.'

Semantic Property 2 is also clearly manifested in the incompatibility of the generic morpheme *-va-* with an overt universal quantifier in the same clause:

(28) a. {#Každou sobotu / #vždycky} sedává Honza v hospodě.
each Saturday always sit.GEN.3SG.PRS Honza in pub
'{#Every Saturday / #always} Honza usually tends to sit in a pub.'

b. {Každou sobotu / vždycky} sedí Honza v hospodě.
each Saturday always sit.3SG.PRS Honza in pub
'{Every Saturday / always} Honza sits in a pub.'

(28a) is at least odd, if not involving a contradiction (but see below analogous examples (29a) and (29b), where there is no oddity), assuming that the universal Q-adverbs, *každou sobotu* 'every Saturday' and *vždycky* 'always', are used with their universal quantificational force. If so, they require that the sentence be true just in case Honza's sitting in a pub occurs on each and every Saturday with no exception, or on each appropriate contextually determined occasion. However, the generic morpheme *-va-* contributes the uncancellable inference that this precisely is *not* necessarily the case. That the oddity of (28a) must be due to the generic morpheme *-va-* is evidenced by the fact that (28b) is perfectly acceptable and true, because it minimally differs from (28a) by lacking *-va-*.

However, the generic morpheme *-va-* may co-occur with universal Q-adverbs, if they are used for "intensification of the strength" of the expressed regularity (Danaher 2003: 45), rather than with their inherent universal quantificational force and so allowing no exceptions. The examples below are taken from Danaher (2003: 45):

(29) a. Mládež ve Vídni se zabývala Hebblem – já jsem *vždycky*
youth in Vienna REFL concerned Hebbel I AUX always
býval skeptický k takovým módním proudům.
was.GEN skeptical to such fashion streams
'Viennese youth were all reading Hebbel – I was usually always skeptical about these fashionable influences.'
(from K. Čapek: *Hovory s TGM*)

b. „Je to divný," pokračovala pak rychlým a věcným šepotem, „
is it strange continued then quick and matter.of.fact whisper
jeden šuplík má zamčenej, a *nikdy* ho *nemívá* zamčenej.
one drawer has locked and never.NCI it NEG.has.GEN locked
A nepasuje mi do něj žádnej klíč."
and NEG.fit me into it no.NCI key
'"It's strange," she continued in a quick and matter-of-fact whisper, "one of his desk drawers is locked and he never has it locked. And none of my keys fit the lock."' (from H. Bělohradská: *Poslední večeře*)

The felicitous co-occurrence of the generic morpheme *-va-* with universal Q-adverbs is similar to the felicitous use of non-universal Q-adverbs together with universal Q-adverbs in combinations like *usually always* or *usually never* in English, where the universal Q-adverbs also arguably lack their inherent universal quantificational force, and function as intensifiers of a regularity that is known to have exceptions:

(30) a. I am usually always happy, but today I feel really depressed.

b. I am usually never neurotic about being messy and keeping things tidy, but I can't seem to go to sleep if clothes are hanging up to dry in my room. (Barbara Partee, p.c.)

Semantic Property 2 in (26) also predicts that the generic morpheme *-va-* will be unacceptable, or at least strongly dispreferred, for the encoding of properties that are denoted by inherent individual-level predicates like *know (French), believe, have long arms, be intelligent* that are stable for an individual, in so far as they spread over a relatively large interval of that individual's life-time, and possibly over most/all of its life-time, and including all of its subintervals and moments. One way of thinking about this property is that individual-level predicates involve a kind of implicit universal quantification over intervals and moments at which they are true of individuals. This motivates why the generic morpheme *-va-* cannot (freely and straightforwardly) apply to imperfectives that denote individual-level predicates: precisely because it introduces the uncancellable inference that there are or may be exceptions or counterexamples, i.e., that the characterizing property does not stably hold of an individual at all the moments of its life-time. This is illustrated by the following Czech example:

(31) Je rodilá Francouzska, a proto #znává francouzsky.
is born French.woman and therefore know.GEN.3SG French
(Intended:) 'She is a French native, and so knows French.'

Similarly, *know (French)* in English is odd with a Q-adverb like *sometimes*, as we see in (32a). In contrast, *sometimes* is compatible with the individual-level predicate *be a California resident*, as it can be construed as describing a property holding with interruptions, on and off at different subintervals of evaluation, sometimes it holds, and sometimes it does not. This fluctuation in truth values generates "gaps" or "interruptions" in their application and hence also a plurality of separate cases, which is a prerequisite for the application of the Q-adverb (see e.g. Fernald 2000 and related suggestions by de Swart 1991, i.a.):

(32) a. # Mary sometimes knows French.
b. Max is sometimes a California resident. (Fernald 2000)

The possibility of an "interruption" construal also makes the combination of the generic morpheme *-va-* with individual-level predicates acceptable, and so Semantic Property 2 in (26) is satisfied. Some Czech examples are given below, where (34b) is taken from *Seven poems written on a mirror* by Karel Kryl (1976):

(33) a. myslit (si) 'to think' → myslívat (si) '(to tend) to think' (see (34a))
b. patřit 'to belong (to)' → patřívat '(to tend) to belong to' (see (34b))
c. věřit 'to believe' → věřívat '(to tend) to believe'
d. mít 'to have' → mívat '(to tend) to have'

(34) a. Každopádně nejste sám, kdo si to *někdy* *myslívá*.
in.any.case NEG.be.2PL alone who REFL it sometimes think.GEN.3SG
'In any case, you are not the only one who occasionaly thinks like this.'
b. … *občas* i k slovesu milovat *patřívá* zájmeno zvratné.
at.times even to verb love belong.GEN.3SG pronoun reflexive
'… the reflexive pronoun occasionally belongs even to the verb *love*'

The "interruption" construal of individual-level predicates, which facilitates their use in the scope of non-universal Q-adverbs in generic sentences, might look like a kind of coercion of individual-level predicates into a stage-level predicate interpretation. However, if this were a matter of coerced stage-level predicate interpretations, then they should be also available for individual-level predicates in episodic contexts, for instance, triggered by adverbials denoting particular spatio-temporal locations. However, such coerced stage-level predicate interpretations of individual-level predicates are not available:

(35) #Max is a California resident right now in his garden.

Such considerations lead Carlson (1977), Kratzer (1989), and Fernald (2000) to propose that individual-level predicates in generic sentences like (32b) are not to be treated as having first undergone a shift into a stage-level predicate interpretation, but rather still have arguments that are of the individual sort, and denote individual-level stative properties, albeit having truth values that vary across evaluation times, are true on and off. *Mutatis mutandis* the same holds for individual-level predicates in the scope of the generic morpheme *-va-*.

Another manifestation of Semantic Property 2 in (26) is the oddity, or even unacceptability, of the use of the generic morpheme *-va-* in generic contexts in which what is at stake are characterizing generic properties, such as inherent dispositions, abilities, and acquired professions, that are understood as holding without interruptions for a given individual over an extended period of that individual's life-time, and all its subintervals and moments. Consider the following contrast:

(36) *Question*: 'What is Mary's profession?'

a. Učí na střední škole.
   teach.3SG.PRS on middle school
   'She teaches at high school.'
   Inference: 'Mary is a high-school teacher.'

b. # Učívá na střední škole.
   teach.GEN.3SG.PRS on middle school
   Intended: 'She teaches at high school on and off / sometimes / often…'
   Inference: 'Being a high-school teacher is not Mary's profession.'

The question about Mary's profession is best answered with (36a), which is headed by the imperfective non-generic verb *učí* 'she teaches', here under its generic interpretation, and is paraphrasable with 'Mary is a high-school teacher.' Teaching as one's profession typically presupposes having acquired the requisite training and accreditation, which then holds for an extended interval of one's life, and all its subintervals, without interruptions, including times off work (weekends, holidays, or absence due to illness).

In contrast, (36b) which is formally marked with the generic *-va-*, would be odd or infelicitous in this context, because it conveys the inference that teaching only holds with interruptions of Mary. This invites the inference that Mary only teaches on and off, because she may be a temporary or a substitute teacher, and may also have other alternative jobs or occupations. In short, it invites the inference that Mary is not a teacher by profession.

4.1.1.4 Generalizations with unrealized particulars in the actual world

The generic morpheme *-va-* cannot be used in generic sentences that concern unrealized rules, dispositions, functions, instructions and the like.[31] Consider the following example from Czech:

(37) *Scenario*: We have just bought a new machine for crushing oranges, but we have not yet used it, and it is still packed in its original box.

Tento	stroj	drtí	/ #drtívá	pomeranče.
this	machine	crush.3SG.PRS	crush.GEN.3SG.PRS	oranges

(Intended:) 'This machine crushes oranges.'

What is at stake under this scenario is the intended function of the specific machine token in the domain of discourse, which is as yet not realized. Here, only a verb that sanctions a purely intensional (dispositional) interpretation can be used. In Slavic languages, it is a non-generic verb, which, in the example above, is the primary imperfective verb *drtí* 'it crushes'. The generic verb marked with the generic morpheme makes the whole sentence false under the given scenario. The reason for it is that it requires that this particular machine have already been put to use, and so there have been (observed) instances of it crushing of oranges in the actual world from which the expressed generalization is inductively

[31] In the studies on genericity and habituality this meaning component is commonly (though not uncontroversially) referred to as the "actuality entailment", which was originally coined by Bhatt (1999) to describe the implicative inference that arises with ability modals in perfective aspect. However, in the context of habitual/generic sentences it is perhaps not appropriate, because the requirement that there be actual instantiations of the generalization tends to survive in negated habitual/generic sentences. Therefore, henceforth, I will use the term *realization inference*.

projected. We may formulate this realization constraint on the use of the Slavic generic morpheme as follows:

(38) Semantic Property 3: The generic morpheme *-va-* carries the uncancellable inference (as part of its truth-conditional content) that there are verifying instances in the actual world that count as evidence for the truth of generic sentences it marks.

The realization inference that is often taken to be a signature property of so-called "habitual" sentences is also mistakenly correlated with their lack of modality (intensionality), and used as an argument for not treating "habitual" sentences as a special case of generic sentences which are inherently modal, as all agree. However, as many semanticists also agree, any sentence that describes a habit, regularity, pattern, generalization of any kind has a modal (intensional) import (see also Section 2 above, and references therein). This is because its meaning transcends what we observe in the actual world; they are not only about what was realized, but also predict what is possible or likely. By virtue of having this predictive, law-like force, habitual sentences also have implications for other accessible alternate worlds, rather than just the actual world, and perhaps all possible worlds. Consequently, so-called "habitual" markers also introduce quantification over alternate possible worlds.[32] If the alleged lack of this modal component is the reason for not treating so-called "habitual" markers as falling under genericity, or as not qualifying as generic markers, then this exclusion is in error.

We can illustrate this point with the following example from Czech:

(39) Děti v této škole mívají dobré známky.
children in this school have.gen.3pl.prs good grades
'Children in this school tend to have good grades.'

The truth of (39), which contains the generic marker *-va-*, is not just based on a mere plurality of situations of various children having already actually received good grades, but (39) also gives rise to counterfactual inferences that transcend observable real world conditions we have access to: namely, for instance, if a child were to attend this school, that child too would most likely or probably get good grades. The reason for this is that the truth of (39) also depends on there being some underlying reason(s) or cause(s) for this, such as the quality of instruction, parents' attention, their innate predispositions, etc. While we may not know the

[32] Similar observations are also made by Boneh & Doron (2010) in connection with the Modern Hebrew periphrastic "habitual" construction with *haya* ('was')+participle.

particular reason(s) or cause(s), whatever they might be they also ground the truth of (39), besides our observations of children in this school getting good grades.

The same considerations that motivate the intensionality of (39), which has an overt marker of genericity, also motivate the intensionality of *Children in this school have good grades* in English, which lack any overt generic marker (see e.g., Pelletier & Asher 1997 for similar examples), and which are associated with a logical representation with the intensional generic operator GEN (in Pelletier & Asher 1997 or Krifka et al. 1995, i.a.). If one accepts that the English sentence *Children in this school have good grades* has an intensional (modal) import, as most do, then one should acknowledge that (39) does, as well.

That the realization inference of generic and habitual sentences, including those that have overt so-called "habitual" markers (such as we find in Tlingit or Modern Hebrew), is compatible with the intensionality (modality) which is characteristic of sentences expressing regularities, should not be surprising given that there are other modal operators that introduce the realization inference (see Bhatt 1999 and Hacquard 2006).

4.1.2 When *-va-* MUST be used: Positive counterinstances

Due to its exceptive component, here captured by Semantic Property 2 in (26), the addition of the generic marker *-va-* to a generic sentence may reverse its truth conditions. This also supports its treatment as part of the lexical meaning of *-va-*. A case in point is its use in generics that have counterexamples that count as positive counterinstances (in the sense of Leslie 2008) to the generalization. Generalizations that allow for exceptions partition the domain of their applicability into confirming cases and non-confirming cases. However, not all non-confirming cases are alike, not all render such generalizations false. As Leslie (2008) shows, for a generalization to be true (or false) it matters how exactly non-confirming cases to it are judged to fail it. Consider the following contrast:

(40) a. Birds fly. TRUE
b. Books are paperback. FALSE

(40a) is an example of an exception-tolerating, non-universal generalization that is true by virtue of the majority of birds having the BIRD-kind characterizing property of flying, and it is not falsified by exceptional non-flying bird subkinds like penguins, ostriches or emus, or individual birds that do not fly because of some accidental defect. In contrast, (40b) is false, even if the majority of books are paperbacks.

As Leslie (2008) argues, we are likely to judge a generic sentence to be true, if exceptions to it constitute "negative counterinstances". Non-flying birds constitute negative counterinstances to the positive property of flying characterizing the kind BIRD. They simply lack this positive property without providing any alternative positive property that would be equally (subjectively) salient or striking as flying and that we could pit against it. This makes such negative counterinstances easy for us to ignore, or set aside, when judging a generalization like (40a) as true.

In contrast, if exceptions to a generalization are viewed as positive counterinstances, we are likely to judge that generalization as false. (40b) is judged false, because exceptions to it, even if they are fewer than the confirming paperback members of the kind BOOK, do not simply fail to be paperbacks, but possess a positive distinctive feature, such as being hardback books. I.e., they have an alternative positive property that is subjectively for us just as salient, memorable, significant, or striking as having the positive property of paperback. Positive counterinstances are cognitively salient in our kind-related expectations, which makes them particularly hard for us to ignore when judging the truth of generics like (40b). Other examples of generics that are judged false due to positive counterinstances to them are: *Nurses are female* (there are also male nurses), *People have brown eyes* (there are people with blue, green, grey eye-color, for instance).

In order for a generalization with positive counterinstances to be true, it must contain some overt expression qualifying the generalization that signals that not all the members of the kind share the property that can be truthfully predicated of that kind. For instance, *typically* and *usually* do this qualifying job, as Leslie (2008) observes:

(41) Typically/Usually, books are paperback. TRUE

Other overt qualifiers of generic statements include generic non-universal Q-adverbs like *usually, rarely, seldom*, quantity expressions like *in the majority of cases, in some cases*, spatio-temporal modifiers like *(books) on sale in this bookstore*, and also exceptives like *but*, prepositions *except (for)*, and ceteris paribus clauses.

In Czech, Slovak, and also to a lesser extent in Polish, in addition to the above restrictors, we can also use the generic morpheme *-va-* to qualify a given generalization and so guarantee its truth, which otherwise would be false. Taking examples from Czech, this can be illustrated by the contrast between (42a), on the one hand, and (42b) and (42c), on other hand:

(42) a. Knihy jsou$^{\text{IPFV}}$ brožované. FALSE
books be.3PL.PRS paperback
'Books are paperback.'

b. Knihy jsou^IPFV {obyčejně / nejčastěji} brožované. TRUE
books be.3PL.PRS usually most.often paperback
'Books are {usually / most often} paperback.'

c. Knihy bývají brožované. TRUE
books be.GEN.3PL.PRS paperback
'Books tend to be paperback.'

d. Knihy bývají {obyčejně / nejčastěji} brožované. TRUE
books be.GEN.3PL.PRS usually most.often paperback
'Books are {usually / most often} paperback.'

(42a) contains the primary imperfective verb *jsou* meaning 'they are', and it is judged false based on the same reasoning about exceptions qua positive counterinstances proposed by Leslie (2008) for the English generic sentence (40b). Adding the Q-adverbs 'usually' or 'most often' in (42b) or the generic morpheme *-va-* in (42c), reverses the truth value to true. Q-adverbs and the generic morpheme can co-occur in the same clause, as (42d) shows (and I will return to this point further below).

The contrast between (42a) and (42c) shows that Slavic imperfective verbs have no exceptive meaning component that could function as a safeguard for the truth of the generalization, and the presence of *-va-* is sufficient to make a generic sentence true. This is yet another difference between verbs with the generic *-va-* and imperfective verbs. In (42a), we have a primary imperfective verb. Just as a primary imperfective so a secondary imperfective with the imperfectivizing suffix is insufficient to guarantee the truth of "bare" generics with positive counterexamples. Generally, an exceptive meaning component is not a part of the semantics of imperfective forms in natural languages. In this respect verbs with the generic morpheme *-va-* do not neatly fit the imperfective paradigm.

4.1.3 When *-va-* MAY be used: Pragmatic inferences based on quantity

The exceptive component of the generic marker *-va-*, as given in (26), also has pragmatic consequences in contexts when it *may* be used. Consider the following pair of sentences which only differ in their main verbs, the generic one marked with *-va-* and the related morphologically simpler non-generic verb:

(43) a. Výuka začíná^IPFV od 8.00 a končí v poledne.
teaching start.3SG.PRS from 8:00 and ends in noon
'Classes start at 8 a.m. and end at noon.'

b. Výuka začínává od 8.00 a končí v poledne.
teaching starts.GEN.3SG.PRS from 8:00 and ends in noon
'Classes tend to start at 8 a.m. and end at noon.' [as a rule, usually]

Suppose I registered for a course at school where classes are scheduled every day, and I ask the school administrator when classes start. Under this scenario, either (43a) with the non-generic verb *začíná* here meaning 'it starts' or (43b) with the generic verb *začínává* meaning 'it tends to start', 'it usually / often / as a rule / … starts' can be used as an answer. Suppose the administrator responds with (43a). The next day I arrive at school shortly before 8 a.m., but my classes on this day in fact start only at 10 a.m., and so I waste two hours waiting. And yet, I could not blame the administrator for saying something blatantly false, because generic sentences like (43a) with a verb formally unmarked for genericity, on their generic interpretation, describe regularities that allow for exceptions, i.e., are exception-tolerating like most generic/habitual sentences (see Section 2). Indeed, (43a) can be continued without contradiction with … *ale ne vždycky* '… but not always' or … *s výjimkou pátku* '… except for Friday'.

While I could not blame the administrator for saying something blatantly false by using (43a), I could blame them for being misleading at least. Here is why. (43a) is also compatible with no exceptions, as many generic sentences are (see Section 2). Given that (43a) can be used to express an exceptionless generalization, i.e., the strongest generalization, from the point of view of the generalization strength, (43a) is the stronger alternative than (43b), which requires that there be actual or possible exceptions (Semantic Property 2). From the administrator's choice of (43a), the stronger alternative, when the weaker alternative (43b) with the generic verb could have also been used, I may infer the denial/rejection of the weaker alternative that requires that there be actual or possible exceptions. That is, (43a) can be understood as involving pragmatic strengthening to an exceptionless regularity, namely that classes always start at 8 a.m., without exception.

In contrast, in the described scenario, if the administrator utters (43b), I cannot not blame them for being misleading, or even giving me wrong information about when classes start. The reason for this is that the generic marker, besides its generic import, requires that there be actual or possible exceptions to the generalization, which excludes pragmatic strengthening of (43b) to an exceptionless generalization. Moreover, by using (43b) instead of (43a), the administrator implicates that they do not have enough information to be in the position to commit to the stronger alternative (43a), which allows a pragmatic strengthening to an exceptionless generalization.

Generally, the choice between the two lexical lexical alternatives, a generic verb with *-va-* and a corresponding morphologically related non-generic verb without *-va-*, gives rise to one of the two (defeasible) pragmatic quantity-based inferences which concern either the speaker's *certainty* or *ignorance* about the existence or possibility of exceptions to it. In our example, the administrator may use the alternative with the generic morpheme, because they are *certain* that classes do not always start at 8 a.m., or that they are *uncertain* whether exceptions to their starting at 8 a.m. can be completely excluded. In the latter case, *-va-* may be used as a hedge to safeguard a generalization against refutation by states of affairs of which the speaker is ignorant, and so to ensure its plausible deniability.

This mechanism of quantity-based inferences is also exploited in legal discourse, and in the formulation of various rules, instructions, job descriptions, or customs, which includes generics that have a normative character. Below are some attested examples taken from Czech, a game rule (44a) and an institutional regulation (44b):[33]

(44) a. Zápas *se hrává* na více vítězných her.
match REFL play.GEN.3SG.PRS on more games
'The match tends to be played in several winning sets.'

b. […] z komentáře k Iuliovu zákonu de maiestate z Digest
from commentary to Iulius law de maiestate from disgest
císaře Justiniána: […] 4) kdo *svolávává* k
emperor Justinian: 4) who convene.GEN.3SG.PRS to
povstání, […]
insurrection
'[…] from the commentary to Iulius' law de maiestate from the Digest of the Emperor Justinian: […] who tends to call for an insurrection […]'

For such rules and regulations, using a verb with the generic morpheme is useful as a hedging device. Due to its incompatibility with exceptionless generalizations, the generic morpheme explicitly assures a certain leeway in the application of a rule of law, and hedges against unprecedented or unexpected exceptions to it. In contrast, rules and regulations, such as those above, if they were formulated with morphologically related non-generic verbs (without the generic morpheme *-va-*), would be merely exception tolerating; unless the context explicitly indicated otherwise, they would likely be pragmatically strengthened to strict, exceptionless rules.

[33]Example (44a) is from https://cs.wikipedia.org/wiki/Kuličky, accessed on April 9, 2022. Example (44b) is from https://is.muni.cz/th/nmkho/Frydek_-_Crimen_maiestatis_-_final.pdf; accessed on August 16, 2025.

The use of the generic marker *-va-* in generics describing various rules and norms fits Carlson's (1995) rules-and-regulations model for the truth conditions of generic sentences (see Section 2 above). Recall that according to this model, generic sentences are true by virtue of some underlying rule, principle or cause, which we may directly learn, rather than (just) by virtue of real world particulars from which we induce the generalization. The distribution of the generic marker *-va-* cuts across the divide between the inductive and rules-and-regulations model, and also across the descriptive/normative distinction. It clearly is not a marker of only inductive generalizations, including habits proper, which are paradigm examples for the inductive model (Carlson 1995). However, the rules and regulations that are expressed by generic sentences formally marked with *-va-* must have been already instantiated, which follows from its realization requirement (see Semantic Property 1 in (19)).

4.2 The quantificational properties of the generic morpheme *-va-*

4.2.1 Quantification over situations and/or individuals

One common way of representing the logical structure of generic sentences was introduced in Section 2: namely, the tripartite structure with GEN, which is a phonologically null, modalized dyadic operator. The tripartite structure was originally proposed for the logical structure of sentences with overt Q-adverbs (Lewis 1975). As Chierchia (1995) observes, GEN patterns with overt Q-adverbs when it comes to its variable-binding properties:

(45) Variable-binding properties of the generic operator GEN (see e.g., Chierchia 1995):
  a. a situation variable (in generic sentences without kind terms)
  b. variables provided by singular indefinites and bare plurals
  c. variables provided by kind-denoting definites
  d. more than one variable (in generic sentences with kind terms)
  e. relatively freely select the arguments for its binding, modulo context

The generic marker *-va-* appears to share its variable binding properties with the generic operator GEN (see also Filip 2009, 2021), which serves as one among other arguments given here in support of its status as a generic marker. To illustrate this point, I will adduce some examples from Czech. Comparable uses of this morpheme in other Slavic languages that have analogous verbs with the generic marker will exhibit the same binding properties, as can be easily verified.

The representational style of the simplified tripartite structures that are given below follows Krifka et al. (1995) and Pelletier & Asher (1997) (see also above in

Section 2). The contribution of the generic morpheme *-va-* is represented in the following tripartite structures by means of the quantifier VA, a kind of generic quantifier.[34,35]

(46) BINDING OF SITUATION VARIABLES in generics without kind terms

a. Po práci si Tom dávává pivo.
after work REFL Tom give.GEN.3SG.PRS beer
'After work, Tom has (a) beer.' [i.e., now and then / rarely / often …]

b. VA$[s;]$(TOM in $s \wedge s$ AFTER WORK; TOM HAS (A) BEER in s)
'For all appropriate after-work situations s such that Tom is in s, Tom has (a) beer in s.'

(47) BINDING OF VARIABLES PROVIDED BY INDEFINITES

a. Židle {mívá / mívají} čtyři nohy.
chair(s) have.GEN.3SG.PRS have.GEN.3PL.PRS four legs
'A typical chair/Chairs tend(s) to have four legs.'

b. VA$[x;]$(CHAIR(x); HAVE FOUR LEGS(x))
'When a thing has the property of being a chair, it has four legs.'
'For a given thing x such that x is a chair, x has four legs.'

(48) BINDING OF VARIABLES PROVIDED BY KIND-DENOTING TERMS

a. Člověk se k stáru měnívá.
man REFL toward old.age change.GEN.3SG.PRS
'Man changes as he grows old.' (from K. Čapek: *Obyčejný život*)

b. VA$[x, s;]$(MAN$(x) \wedge x$ in $s \wedge s$ OLD AGE; x CHANGE in s)

[34]The treatment of the proper name *Tom* in the formula (46b) follows the treatment of the proper name *Fred* in *Fred always smokes* in Chierchia (1995: 189, ex. (37)). It is treated as a constant in the logical representation of *Fred always smokes*; the universal Q-adverb *always* only binds the situation variable; its variable binding properties, according to Chierchia and others, pattern with GEN. However, strictly speaking, the generic operator GEN in this type of generic sentence quantifies over different stages of the name bearer participating in different corresponding particular situations in which the characterizing property is manifested. This is implied in an alternative tripartite structure associated with this type of sentence in Krifka et al. (1995), where *Tom* introduces the individual variable x, which is satisfied by different stages of Tom at particular situations: *Tom drinks beer after work* → GEN$[x, s](x =$ TOM $\wedge\, x$ in $s \wedge s$ AFTER WORK; x DRINK_BEER in s).

[35]Slavic languages lack an article system; only South Slavic languages like Bulgarian and Macedonian have what is taken to be a definite article. The bare subject argument in the following Czech examples can be viewed as corresponding to the English singular definite subject, bare plural subject, and where the predicate does not denote a contingent property also to the English indefinite subject argument.

(49) Binding of more than one individual variable

a. Kočka honívá myš.
cat chase.GEN.3SG.PRS mouse
'Cats tend to chase mice.'

b. $\text{VA}[s, x, y;](\text{CAT}(x) \wedge \text{MOUSE}(y) \wedge x \text{ in } s \wedge y \text{ in } s; x \text{ CHASE } y \text{ in } s)$

As we see above, the Czech marker *-va-*, which introduces the quantifier VA into the tripartite structure, semantically functions as a quantifier over (i) situations, as in (46a), (ii) individuals, as in (47a), and (iii) both situations and individuals in sentences with kind denoting terms, as in (48a) and (49a), which is aligned with the quantificational properties of the null generic operator GEN.

In (47a), the Czech marker *-va-* functions as a quantifier whose range only contains individual members of a kind. This invalidates its treatment as an iterative marker denoting a function that returns a set of plural dynamic situations of the same type, at a given world and time. Examples like (47a) clearly show that the standard designations of *-va-* as an iterative, multiplicative, or frequentative marker (e.g., Petr 1986, Kosek 2014, Lamprecht et al. 1986, Dahl 1995) are misleading, if not wrong.

Moreover, examples like (47a) invalidate the claim made by Dahl (1995: 421) that the Czech marker *-va-* serves "as a kind of quantifier over situations with, roughly, the semantics of 'most'. Following the terminology of Dahl 1985, let us refer to this kind of tense-aspect markers as habituals." Given that this claim is false, and given that it is one of the two key claims that Dahl uses to deny that the Czech morpheme *-va-* has a status of a generic marker, there really remains no valid reason, which Dahl adduces, why *-va-* cannot be analyzed as a generic marker sui generis, given that the other argument, namely the obligatory occurrence in all and only generic sentences is also spurious. Given that Dahl uses the Czech marker *-va-* as a paradigm example of comparable markers in other typologically distinct languages (see Dahl 1995: 421, fn. 8) that enforce a generic (and habitual) interpretation, this also raises at least some doubts about their purported status as "tense-aspect habitual markers".

The use of the generic morpheme *-va-* in examples like (47a) also shows that it is clearly not a "habitual" morpheme in the proper sense of "habit", i.e., dedicated to formally marking only sentences that quantify over situations with animate individuals, typically human agents, as in (46a). Another example of a generic sentence marked with *-va-* that has nothing to do with habits in the narrow sense is given below:

(50) Moře tady bývá v tuto dobu vyhřáté na příjemných 28° C.
sea here be.GEN.PRS in this time warmed-up on pleasant 28° C
'At this time the sea here tends to be warmed up to a pleasant 28° C.'

To illustrate the breadth of use of the Czech generic morpheme, we may also mention that it commonly occurs in generic sentences with dedicated kind-denoting terms like 'man', as in (48a). It also occurs in kind reference sentences (see Section 2) that contain kind-level predicates like *be widespread* directly selecting kind-denoting terms for one of their arguments and which assign characterizing properties to kinds that can never hold of its individual members, as we see in (51) below:

(51) Rohozub nachový bývá rozšířený u lidských sídlišť.
ceratodon purpureus be.GEN.3SG.PRS widespread at human dwellings
'The ceratodon purpureus tends to be widespread close to human dwellings.'

There is a growing body of studies on markers that, just like the Slavic generic marker, constitute a sufficient, but not a necessary, condition for the expression of various regularities, and often, similarly as in Dahl (1995), they are treated as "habitual", rather than generic, markers. Some examples are: the Tlingit (Na-Dene; Alaska, British Columbia, Yukon) particle *nooch* (Cable 2022), the *be*-construction in African American English (Green 2000), or the construction with *haya*+participle in Modern Hebrew (Boneh & Doron 2008). It is striking that such studies focus on sentences that describe habits in the proper sense, such as (46a). It then raises the question whether this empirical focus of studies on such "habitual" markers and on habits proper is due to a data elicitation bias, and whether the relevant markers in these languages actually cover a broader swath of data, similar to those examples that we find in Czech. One wonders whether their quantificational properties come close to or perhaps even match those of the null generic operator GEN, or the Czech generic marker *-va-*, and by the same token, whether their treatment as merely "habitual" markers (possibly subsumed under tense and/or aspect) which are claimed to be separate from genericity might not be reconsidered.

4.2.2 The Slavic generic morpheme *-va-* is not an overt Q-adverb: Irreducibility to 'usually' or to any other single quantifier or quantity expression

In standard reference grammars of Czech, the meaning of the Czech morpheme *-va-* is often characterized in terms of "usuality" (*uzuálnost* in Czech, see e.g., Petr

1986: 184, i.a.). Hence, Dahl's (1995) claim that the Czech morpheme *-va-* functions as a quantifier over situations with the meaning of approximately 'most' or 'usually' might seem plausible. In Dahl (1995), this semantic claim is based on two rather flimsy data points: namely, "the glosses given in Kučera (1981)" (Dahl 1995: 421), and a cursory observation (Dahl 1995: 420–421) regarding the distribution of the Czech *-va-* and similar markers from other languages over his two generic contexts: namely, (i) it tends to be optional in the context of 'usually', and (ii) it is typically absent in the context that concerns a job description.[36]

While Dahl (1995) does not motivate the choice of these two generic contexts, they seem well-chosen. The first would seem to elicit generics that are roughly aligned with Carlson's (1995) inductive model of genericity, and concern weak descriptive generalizations, while the second with Carlson's (1995) rules-and-regulations model of genericity. However, as shown above (Section 4.1.3), the Czech generic *-va-* can be used not only in generic sentences that describe weak descriptive generalizations, but also in generics stating a variety of rules (e.g., game rules), including the rules of the law.

Now, if the Czech *-va-* had a meaning of approximately 'most' or 'usually', as Dahl (1995) claims, it would not qualify as a generic morpheme. However, the Czech *-va-* contributes no requirement on a specific quantity of situations or instances that count as evidence for the truth of the generalization. This is a hallmark property of markers of genericity (and habituality as its special case) in natural languages. In this respect, the generic morpheme *-va-* patterns with the phonologically null GEN operator. Consequently, just as GEN, it cannot be reduced to 'most' or 'usually', or the meaning of any other single quantifier or expression of quantity, no matter how modalized (see 2 above). Let me adduce a few concrete examples to illustrate this point.

First, we observe that the Czech marker *-va-* freely co-occurs with virtually any expression of quantification or quantity (see, for instance, corpus studies of Širokova 1963: 62, 81; Širokova 1965, Danaher 2003). Danaher's (2003) corpus study shows that *-va-* occurs less often with *obvykle* 'usually' than with Q-adverb(ial)s denoting a low frequency like *občas* 'from time to time', *někdy* 'sometimes', *málokdy* 'rarely', *tu a tam* 'here and there', *vzácně* 'rarely'.[37] Any of these Q-adverb(ial)s can fill the [Q-ADVERB] slot of (52):

[36]In Dahl (1995: 420–421), the relevant test context is given as follows: [A: My brother works at an office. B: What kind of work he DO? A:] He WRITE letters.

[37]Danaher's (2003) corpus comprises 376 attested examples of verbs marked with the marker *–va-*, collected from a wide variety of text genres (e.g., contemporary literary essays, fiction, memoirs, journalistic prose and scholarly writing).

(52) Po snídani Tomáš Q-ADVERB psává dopisy.
after breakfast Thomas Q-ADVERB write.VA.PRS letters
'Thomas Q-ADVERB writes letters after breakfast.'

Second, *-va-* can be used to formally mark generic sentences that are true even if *most* relevant instances do *not* satisfy the generically-predicated property, as we see in (53), which makes its reductive analysis to 'most' or 'usually' entirely implausible.[38]

(53) Za Putina, ruští oligarchové umírávali pádem z okna.
during Putin Russian oligarchs die.GEN.3PL.PST fall from window
'In Putin's times, Russian oligarchs tended to die by falling out of windows.'

(53) is true even if as a matter of fact most Russian oligarchs who died in Putin's times did not die by falling out of windows, only a very small number did. The reason why we judge this sentence as true does not hinge on the number of Russian oligarchs dying by falling out of windows, but rather on viewing the nature of their death as so highly appalling, striking, unexpected, or memorable that we treat it as their characterizing property. This kind of subjective evaluation is based on the same psychological mechanism that motivates why we judge generic sentences like (9c) *Mosquitoes carry the West Nile virus* as true, even if only about 1 % of mosquitoes carry the virus.

Moreover, adding 'most' or 'usually' to true generic sentences like (53) that are formally marked with *-va-* results in false sentences, as we see in (54a) and (54b). Given that a generic sentence marked with *-va-* does not entail the corresponding sentence with the added quantifier 'most' or 'usually', *-va-* does not have as part of its meaning the quantificational import of 'most' or 'usually'.

(54) a. Za Putina, *většina* oligarchů umírávala pádem z okna.
during Putin most oligarchs die.GEN.PST fall from window
'In Putin's times, most oligarchs died by falling out of windows.'
(53): TRUE ⇏ (54a): FALSE

[38]This example is tailored on Kučera's (1981) example:

(i) Za Stalina ruští generálové umírávali v mladém věku.
during Stalin Russian generals died.GEN in young age
'In Stalin's times, Russian generals tended to die young.'

Kučera translates this sentence as 'Most generals died young in Stalin's times', but this does not seem to be correct, given that what it describes is factually false; most Russian generals did not in fact die young in Stalin's times, as we know.

b. Za Putina, oligarchové *obyčejně* umírávali pádem z okna.
during Putin oligarchs usually die.GEN.PST fall from window
'In Putin's times, oligarchs usually died by falling out of windows.'
(53): TRUE ⇏ (54b): FALSE

Neither can (53) with the generic marker *-va-* be paraphrased by a corresponding sentence with an overt Q-adverb like 'usually' and a non-generic imperfective verb without *-va-*. While (53) is true, (55) is false:

(55) Za Putina, oligarchové *obyčejně* umírali$^{\mathrm{IPFV}}$ pádem z okna.
during Putin oligarchs usually die.PST fall from window
'In Putin's times, oligarchs usually died by falling out of windows.'
(53): TRUE ⇏ (55): FALSE

In sum, the generic morpheme *-va-* in Czech does not function as a "habitual" marker that quantifies over situations only, and its meaning cannot be reduced to that of an overt Q-adverb like 'usually', or any other quantifier or expression of quantity, contrary to Dahl (1995). In Slavic languages that have analogous verbs with the generic marker *-va-*, we can observe the same general interpretation pattern concerning its quantificational import.

4.3 Summary

At the outset, one of the main questions posed concerned the relation of the meaning of the generic morpheme *-va-* to the phonologically null generic GEN operator, which in quantificational theories of genericity is hypothesized to stand for the key meaning component of generic and also habitual sentences (see (11)). Based on the data and observations so far, and assuming that the generic morpheme *-va-* introduces a kind of generic operator VA into the tripartite logical representation, the similarities and differences between VA and GEN can be summarized as follows.

(56) Shared properties of the generic operator VA (in Czech) and the null generic operator GEN:

a. They model the meanings of sentences that only have a generic interpretation (and disallow any episodic interpretation).
b. They derive complex generic predicates that are aspectually stative, one of the signature properties of generic predicates (see above Section 2, and Semantic Property 1 in (25)).

c. Their analysis cannot be reduced to the meaning of any single quantity expression like *usually*, *generally*, *typically*, *most*, or *in a significant number of cases*, of any extensional quantifier (e.g., universal, existential), or any single, purely quantitative criterion or quantity-based measure, regardless whether they are characterized in modal or probabilistic terms (see Section 4.2.2).

d. Both are modal generic operators; their meaning does not (just) depend on a mere plurality of particulars, situations or "cases" (in the sense of Lewis 1975), but rather they add predictive force to the meaning of sentences, which has implications for other accessible worlds (perhaps all), than just one possible (actual) world, and supports counterfactual inferences.

e. They seem to exhibit the same variable binding properties (akin to those of overt Q-adverbs) (see Section 4.2.1).

The generic operator VA differs from GEN in so far as it adds two uncancellable inferences (as part of its truth-conditional meaning) to the meaning of generic sentences. For convenience, they are repeated here as follows:

(57) Two uncancellable inferences

a. Exceptive inference: There are actual (known) or possible exceptions or counterexamples to the generalization (Semantic Property 2 in (26)).

b. Realization inference: There are verifying instances in the actual world that count as evidence for the truth of the generalization (Semantic Property 3 in (38)).

The co-occurrence of these two uncancellable inferences may not be a matter of mere accident. The generic *-va-*, as I argue, semantically functions as a modal generic quantifier sui generis whose use involves reasoning with exceptions and counterexamples; it involves the speaker's (implicit) background understanding of what counts as salient reasons for/against the correctness of applying the generalization, in the light of exceptions or counterexamples which the speaker thinks cannot be safely ignored or tolerated, out of abundance of caution, to ensure plausible deniability and the like. This kind of reasoning with exceptions and counterexamples is motivated by the speaker's more or less direct experience of the facts, particulars in the actual world, of how various states of affairs are realized in the *actual* world. If so, then it should not be surprising that *-va-* also carries the uncancellable inference that there are verifying instances in the actual world that count as evidence for the truth of generic sentences (realization inference).

Due to these two uncancellable inferences, the Slavic generic morpheme *-va-* delimits only a subset of generic sentences that the null generic operator GEN underpins; GEN is introduced into the logical representation of generics that not only sanction exceptions, but also of generics that (necessarily) hold with no exceptions like the universal laws of nature, and also purely hypothetical generics that are true even if there have been no instances that count as evidence for their truth, and may never be.

As in English, the null generic operator GEN may stand for a representation of the generic (and habitual) meaning in the logical structure of Slavic generic (and habitual) sentences that are headed by verbs that are *not* marked for genericity, i.e., non-generic verbs (see 1): these are either primary imperfective verbs (formally unmarked for imperfectivity) or secondary imperfectives (which are marked for imperfectivity with the imperfectivizing suffix), and also perfective verbs, in particular in West Slavic languages where they are commonly used in generic (and habitual) sentences.

5 Proposal

5.1 Reasoning with exceptions

One key difference between generic sentences formally marked with *-va-* and those that lack this marker has to do with the STRENGTH of the generalization they express which is calibrated in terms of the speaker's (public) commitment to the actual or possible exceptions or counterexamples that the generalization *may, must* or *must not* have.[39] (These three salient cases were discussed in Section 4.1.)

Intuitively, the strongest generalizations have no exceptions or counterexamples whatsoever in every possible world/in all the known circumstances (e.g., universal laws of nature *The Earth revolves around the Sun* or definitional statements like *A triangle has three sides*). Lower on this scale are non-universal generalizations (including laws and rules that fall under special sciences such as psychology, biology, and economics, e.g., *As the price of a good or service increases, the quantity demanded decreases, and vice versa.*) which in principle allow for ex-

[39]Assuming that making a statement publicly commits the speaker to its content: A "speaker is committed to a statement when he makes it himself, or agrees to it as made by someone else, or if he makes or agrees to other statements from which it clearly follows" (Hamblin 1971, Hamblin 1987: 136).

ceptions, but may be compatible with no exceptions, followed by generalizations that have (known) exceptions that we tolerate (e.g. *Dogs bark*).

For all these types of generalizations, in Czech there are freely available non-generic (im)perfective verb forms (i.e., forms without the generic *-va-*), unless, they have exceptions that cannot be tolerated (positive counterinstances), in which case, just as in English, some overt expression is needed to qualify the generalization and ensure its truth. By using a non-generic verb in a non-universal generic sentence, which in principle allows for exceptions, the speaker may remain publicly non-committal as to whether there are exceptions to the expressed generalization, but also, in an appropriate context and the nature of the generalization, allow for pragmatic strengthening to an exceptionless generalization.

In contrast, the use of a specifically generic verb with the generic marker *-va-* commits the speaker to there being actual exceptions or counterexamples, or possible exceptions to the generalization that the speaker cannot categorically exclude (due to the lack of sufficient information) in every possible world (see Semantic Property 2 in (26)). This means that the generic morpheme *-va-* can only be used to mark generic sentences that express non-universal generalizations, and which exclude pragmatic strengthening to an exceptionless generalization. It may be used as a hedge to safeguard a generalization against refutation by states of affairs of which the speaker is ignorant, and so to ensure its plausible deniability. Such generalizations are akin to generalizations with exceptives like *but*, prepositions *except (for)*, or with a ceteris paribus clause.

The difference in the strength of the generalization expressed by generic sentences marked with *-va-*, on the one hand, and of the generalization expressed by formally unmarked generics, on the other hand, can be represented, as I propose, in terms of a *rank order* (Lehrer 1974; Hirschberg 1985; Geurts 2010; Horn 1989, 2000, 2010, 2017, i.a.). Rank orders, which have the general format of ⟨⟨strong | weak⟩⟩, are related to, but distinct from, scales for entailment-based scalar implicature. Both rank-order based and entailment based scalar implicatures are a kind of quantity implicature. However, in rank order based implicatures, stronger elements do not entail weaker ones, but rather we can infer their negations:

(58) SCALE ORDER versus RANK ORDER (Horn 2017)

SCALE: In a scale of the form $\langle Y, X \rangle$, … Y … unilaterally entails … X …
Example: $\langle \textit{all}, \textit{most}, \textit{many}, \textit{some} \rangle$

RANK ORDER: In a rank order of the form $\langle\langle Y \mid X \rangle\rangle$, … Y … unilaterally entails … $\neg X$ …

Examples:
⟨⟨*married* | *engaged*⟩⟩: 'if a person is married, it is false that that person is engaged (to be married)'
⟨⟨*dead* | *sick*⟩⟩: 'if a person is dead, it is false that that person is sick'

Unlike values within a scale, ranked elements are mutually incompatible. For instance, being married does not entail being engaged. Applied to generic sentences with and without the generic morpheme *-va-* in Slavic languages, we get the following rank order:

(59) Rank order (preliminary version): ⟨⟨[gen]P | [va]P ⟩⟩
where,
[gen]P: compatible with no exceptions
[va]P: incompatible with no exceptions
In a rank order ⟨⟨[gen]P | [va]P ⟩⟩, [gen]P unilaterally entails ¬([va]P), all else being equal.

The rank order in (59) is based on two main lexical alternatives that Slavic languages have available for the expression of generic statements: namely, (i) verbs that are formally marked for genericity with the generic morpheme *-va-*, which introduces the generic operator va into the logical representation and here represented with va as part of the weak element of the rank order, and (ii) non-generic verbs without the generic morpheme *-va-*, which have contextually determined generic uses, and occur in generic sentences that introduce the null generic operator gen into the logical representation, and here represented with gen as part of the strong element of the rank order. The two ranked alternatives, strong and weak, are equally informative, and do not stand in an entailment relation to each other: namely, the strong element does not a fortiori imply the truth of the weak one, but rather from the strong element we can infer the negation of the weak element. Let me illustrate this with the following contrast:

(60) a. Petra u nás řídí$^{\mathrm{IPFV}}$ autobus.
Petra at us drive.3sg.prs bus
'Petra drives a bus here [i.e., is a bus driver].'

b. Petra u nás řídívá autobus, když je nedostatek řidičů.
Petra at us drive.gen.3sg.pst bus when be.3sg.pst shortage driver.masc.gen.pl
'Petra tends to drive a bus here [i.e., drives a bus now and then, with some regularity], when there's a shortage of bus drivers.'

Unless the context indicates otherwise, from (60a) we may naturally infer that Petra is employed as a bus driver, being a bus driver is her profession. As such it is her characterizing property that holds of her stably over a large chunk of her life-time, i.e., at all its subintervals and moments of time without interruptions, including times when she happens to be off work, e.g., on vacation or sick. In contrast, (60b) is naturally understood as meaning that Petra drives a bus from time to time, often, occasionally, just in case the appropriate occasion arises, and the like. Hence, (60a) does not entail (60b), but rather from (60a) we can infer the negation of (60b). And vice versa, (60b) does not entail (60a). This can be also confirmed by the observation that in (61), B's cancellation with 'yes' is odd or at least dispreferred:

(61) A: Řídívá Petra u Vás autobus?
drive.GEN.3SG.PRS Petra at you bus
'Does Petra {occasionally / from time to time / often…} drive a bus in your city?'

B: {Ne / #Ano}, Petra u nás (dokonce) řídí$^{\text{IPFV}}$ autobus.
no yes Petra at us even drive.3SG.PRS bus
'No, (well) Petra (in fact) is a bus driver here/in our city.'
'#Yes, Petra (in fact) is a bus driver here/in our city.'

Rank orders, similarly to scale orders, give rise to quantity implicatures, which are inferences triggered by an utterance and the alternative utterance(s) a speaker could have made instead. The rank order (59) may trigger one of the two types of inference, depending on context: certainty or ignorance regarding the speaker's commitment to exceptions. The choice of the non-generic alternative, i.e., the strong element in the rank order in (59), may in an appropriate context convey a certainty inference (by pragmatic strengthening) that there are *no exceptions* to the generalization in all the known conditions, in all the speaker's doxastic alternatives. This constitutes the strongest, upper bound of the rank order.

The use of the generic morpheme *-va-*, which corresponds to the weak alternative in the rank order (59), signals that there are exceptions or counterexamples to the generalization that are of the kind that the speaker thinks cannot be safely ignored, given the speaker's (implicit) understanding of the correct application of the generalization, weighing some salient reasons for or against it. The use of *-va-* triggers one of the two inferences, depending on context. The speaker wishes to convey the inference that they know that there are exceptions or counterexamples (certainty inference), *or* the speaker does not have enough information to be in the position to exclude them (ignorance inference).

Let us examine these two inferences in more detail. By using the generic -*va*-verb (the weak alternative) when the verb without the generic morpheme (the strong alternative) could have also been used (salva veritate), the speaker may convey the ignorance inference that they are not in the position to use the strong alternative, because it allows pragmatic strengthening to an exceptionless generalization, unless it is explicitly indicated otherwise, and the hearer might be misled. This case has been illustrated with the minimal pair in (43) above, repeated below.

(62) a. Výuka začíná$^{\text{IPFV}}$ od 8.00 a končí v poledne.
teaching start.3SG.PRS from 8:00 and ends in noon
'Classes start at 8 a.m. and end at noon.'

b. Výuka začínává od 8.00 a končí v poledne.
teaching starts.GEN.3SG.PRS from 8:00 and ends in noon
'Classes tend to start at 8 a.m. and end at noon.' [as a rule, usually]

Let us suppose there is a subset of (62a)'s speaker's doxastic alternatives where the school opens at 8 a.m. and a subset where it does not; in other words, the speaker of (62a) does not know which of these alternatives holds, and considers both as live possibilities (ignorance inference). Uttering (62b) with the generic morpheme, rather than (62a) without it, readily allows for plausible deniability in the face of refutation by information of which the speaker of (62a) is ignorant, and specifically should it turn out that the school does not always open at 8 a.m. In contrast, by uttering (62a) with the non-generic verb, rather than (62b) with the generic one, entails the negation of the latter, and, provided the context does not indicate otherwise, naturally leads to a pragmatic strengthening of (62a) to an exceptionless generalization. I.e., (62a) is compatible with the certainty inference that in all the speaker's doxastic alternatives associated with (62a) the school opens at 8 a.m.

The exceptive inference of the generic verb marked with -*va*- (see Semantic Property 2 in (26)) can also be exploited to convey the certainty inference that there are counterexamples to the generalization that would have made the corresponding sentence without -*va*- false, all else being equal. The best examples are generalizations with counterexamples that count as positive counterinstances (in the sense of Leslie 2008), such as hardbacks that falsify *Books are paperbacks*. These are inadmissible non-confirming cases which *cannot* be (easily) ignored when judging the truth of the generalization. Above, this was illustrated by (42) and the relevant contrast is repeated below. By using the generic morpheme in (63b), as opposed to the alternative without -*va*- in (63a), the speaker (publicly) commits to the existence of counterexamples to the generalization, which, in turn, prohibits pragmatic strengthening to an exceptionless generalization, and guarantees the truth of (63b).

(63) a. Knihy jsou[IPFV] brožované. FALSE
books be.3PL.PRS paperback
'Books are paperback.'

b. Knihy bývají brožované. TRUE
books be.GEN.3PL.PRS paperback
'Books tend to be paperback.'

In representing such speaker-oriented certainty and ignorance inferences, we may rely on the covert doxastic operator $\Box_{\text{SP}(c)}$ (inspired by Meyer 2013), which is understood as universally quantifying over the speaker's doxastic alternatives (Degano et al. 2025):

(64) A covert doxastic operator: '$\Box_{\text{SP}(c)}(P)$' where,
'SP(c)': a doxastic source, the speaker in context c
'$\Box_{\text{SP}(c)}$': a covert doxastic operator
'$\Box_{\text{SP}(c)}(P)$': in all the speaker's doxastic alternatives, the speaker knows that P

Specifically for the representation of the ignorance inference, we may employ the classic definition of ignorance of Hintikka (1962), characterized as 'not knowing whether'. In our case, and assuming that P corresponds to the strongest value in the rank order of the form ⟨⟨Strong | Weak⟩⟩ and compatible with "all exceptions categorically excluded", we have: (i) the speaker does not know whether P is true, (ii) the speaker does not know whether $\neg P$ is true, and (iii) the speaker also considers both P and $\neg P$ as live possibilities compatible with the speaker's knowledge.[40]

With these ingredients in place, adding the two speaker-oriented inferences to our original rank order (59), its final formulation is given below.

[40]Traditionally, and based on Grice (e.g., Grice 1975, 1989, i.a.), ignorance inferences are derived by means of maxims of conversation (quantity and quality), based on general principles of conversation and cooperative behavior of interlocutors. Ignorance inferences are taken to consist of uncertainty and possibility inferences, where the possibility inference is derived from uncertainty. Recently, however, Degano et al. (2025) have argued for a possibility inference without the uncertainty inference. Due the limits on this chapter, I cannot here address the debates concerning ignorance inferences in more detail.

(65) Rank order (final version): $\langle\langle$[gen]P | [va]P $\rangle\rangle$
where,
[gen]P: compatible with no exceptions
[va]P: incompatible with no exceptions

(i) Asserting that [gen]P holds, given that the speaker could have instead asserted that [va]P holds, may trigger
certainty inference: '$\Box_{\mathrm{SP}(c)}(P)$'
unless the context indicates otherwise.

(ii) Asserting that [va]P holds, given that the speaker could have instead asserted that [gen]P holds, triggers one of the following inferences depending on context:

(a) certainty inference: $\Box_{\mathrm{SP}(c)}\neg(P)$

(b) ignorance inference:
$\neg\Box_{\mathrm{SP}(c)}(P) \wedge \neg\Box_{\mathrm{SP}(c)}\neg(P) \leftrightarrow \Diamond_{\mathrm{SP}(c)}(P) \wedge \Diamond_{\mathrm{SP}(c)}\neg(P)$
(following Hintikka 1962)

In effect, generic sentences formally marked with the generic morpheme *-va-* are endowed with two modal (intensional) components: one that is tied to their generic, law-like, or gnomic force, which it shares with the null generic operator gen, and the other which involves reasoning with exceptions and counterexamples, which may be characterized in terms of the speaker-oriented doxastic (and epistemic) modality, in interaction with general pragmatic principles of interpretation.[41]

5.2 Supporting Evidence: The va operator with overt restrictors

The properties of the generic marker *-va-* are relevant to discussions of generic sentences in the context of ceteris paribus (cp) generalizations, given that, as

[41]The speaker-oriented modality that concerns the speaker's commitment to the strength of the generalization, which is based on the speaker's information state about exceptions or counterexamples to it can be seen as related to observations in the generic literature that generic sentences track not only (relatively) objective facts about the world, but also a variety of subjective factors, including the speaker's information state, beliefs about the facts of the world, and a variety of attitudes that the speaker may take towards them. As has been observed above, some generics track what we consider to be particularly salient, striking, dangerous or impressive facts (e.g., *Sharks attack swimmers*), which, for instance, Leslie (2008) integrates into her cognitive approach to generic sentences, or they track psychologically salient partitions of the kind (Cohen 2004), or reflect judgments based on stereotypes (e.g., *Asian people are good at mathematics*).

I propose, its use is driven by reasoning with exceptions, counterexamples, or various restricting conditions which cannot be ignored or tolerated. Philosophers draw on generic sentences in order to advance our understanding of the nature and status of CP generalizations in special sciences (see e.g. Nickel 2016 and references therein). A line is drawn between exceptions to the CP generalization that are compatible with its truth and counterexamples which are not.

The distribution of *-va-* over different types of generic sentences taps into this distinction, as has been illustrated by a number of examples above. To what has already been observed, we may add that in contrast to non-generic verbs, there are contexts in which specifically generic verbs marked with *-va-* may be odd or dispreferred with no overt restrictor, but perfectly acceptable if they are licensed by an appropriate overt restrictor. This can be shown in generic sentences with kind reference that describe what are taken to be characterizing properties of a kind to which there are tolerable exceptions compatible with their truth, but which would refute the corresponding universal generalization. Consider the following contrast:

(66) EXCEPTIONS: non-barking dogs (negative counterinstances, see Leslie 2008)
Context: 'What sound do dogs make?'
a. Psi štěkají$^{\mathrm{IPFV}}$. [=(1a)]
dogs bark.3PL.PRS
'Dogs bark.'
b. (#) Psi štěkávají.
dogs bark.GEN.3PL.PRS
'Dogs tend to bark [when… / if… / other things being equal / unless prevented].'

In this context, the preferred and the most natural answer is (66a) with the verb without *-va-*. (66a) and its closest English correspondent in this context *Dogs bark* instantiates a prototypical generic sentence (see Section 2): namely, they have no overt restrictor or quantificational element, their bare plural subject term denotes a kind whose characterizing property is denoted by a verb form that is the least marked form in their respective aspect and tense system, i.e., they exhibit the minimal marking tendency of genericity with respect to tense and aspect (Dahl 1995).[42]

[42]In languages with tense and/or aspect systems, prototypical generic sentences exhibit the "minimal marking tendency" of genericity with respect to tense and aspect, as Dahl (1995)

Both (66a) and *Dogs bark* are expected and felicitous answers to the question at hand, because it calls for the basic information about one particular characterizing property of dogs that can be *truly* predicated of dogs as a whole kind, based on our shared kind-related expectations about what it means to be a (realistically) possible dog, by default, typically, or generally, in any possible world; if something is a dog, it will likely bark. Negative counterinstances (in the sense of Leslie 2008), i.e. dogs failing to bark (e.g., breeds like greyhounds and individual dogs that do not bark for a variety of reasons like an injury), and all kinds of disturbing or facilitating factors which may be potentially infinite, are largely irrelevant in this context.

Contrary to this, by using *-va-* in (66b) the speaker shifts focus to exceptions, counterexamples, (possible) restricting conditions on the application of the generalization, implying that (i) members of the kind DOG are not homogeneous with respect to the characterizing property of barking, or (ii) do not bark in all the appropriate situations in which not exceptional barking dogs might be expected to bark. This straightforwardly follows from Semantic Property 2 in (26). By using the marked alternative (66b) with *-va-*, when (66a) without it is preferred as an answer to 'What sound do dogs make?', the speaker might be (mis)understood as conveying the ignorance inference that they are uncertain whether barking is a *characterizing* property of dogs. But this would be odd, because it would be inconsistent with our general knowledge that barking is a *characterizing* property of dogs, and the speaker is expected to know this, as well as that there exceptional dogs that fail to bark (negative counterinstances), which we tend to ignore.

Or, and most likely, the speaker, by using the generic *-va-*, may convey the certainty inference that they know that there are other exceptions (apart from the tolerable negative counterinstances which we set aside) that are noteworthy, relevant, striking or possibly not known to the hearer in a given context, which cannot be ignored. And so the speaker intends to draw attention to such exceptional individuals, perhaps to show off his knowledge of dogs. But if so, it is odd for the speaker not to explicitly specify what these exceptions are and why they are worthy of being highlighted. This is also the reason, I submit, why (66b) without an overt restrictor, and out of the blue, is odd or dispreferred.

puts it: namely, their main verb is either devoid of any tense and aspect marking (e.g., *bark* in *Dogs bark*) or corresponds to the least marked form in the verb system. In Slavic languages it is the primary (im)perfective verb; it is a verb form with no derivational morphemes, overt aspect marking, but possibly only marked with tense inflectional morphology. The Czech sentence (66a) on its generic interpretation is headed by the imperfective simplex *štěkají* 'bark.3PL.PRS', which is marked for tense, number and person, but has no marker of imperfective aspect.

Once an overt appropriate restrictor is added to the unrestricted (66b), we get a restricted generic sentence in which the generic marker *-va-* is fully felicitous. With an overt restrictor, the generic verb marked with *-va-* and its unmarked counterpart without *-va-* are substitutable salva veritate:

(67) a. Psi {štěkají[IPFV] / štěkávají}, s výjimkou některých
dogs bark.3PL.PRS bark.GEN.3PL.PRS with exception some
druhů psů jako chrti.
breeds dogs like greyhounds
'Dogs tend to bark, except for some dog breeds like greyhounds.'

b. Psi {štěkají[IPFV] / štěkávají} na ty, které neznají.
dogs bark.3PL.PRS bark.GEN.3PL.PRS at those whom NEG.know
'Dogs bark at those whom they don't know.'

In (67a) the overt restrictor specifies a subkind of dogs, greyhounds, that are explicitly exempt from the generalization. In (67b), the restrictor constrains a subset of situations in which stages of prototypical, unexceptional dogs generally bark.

These two types of overt qualifications are to be expected, if we assume that the generic morpheme *-va-* functions as a kind of generic quantifier, and generic sentences with kind reference involve a "double generalization" (in terms of Carlson 2011). For instance, *Dogs bark* is true by virtue of (i) *individual dogs* that realize the kind DOG having the disposition of barking (unless they belong to a non-barking kind like greyhounds), and also by virtue of (ii) *particular situations* of barking by a stage of an individual dog.

On the quantificational theory of generic sentences (see Section 2), the relevant quantifier in their logical form is either the covert GEN operator or an overt generic Q-adverb (e.g., *usually, typically*). And also, as I argue, the generic VA operator, introduced by the generic marker *-va-*. All three – GEN, a Q-Adverb, and VA – have either a covert restrictor or an overt restrictor.[43] Overt restrictors of generic quantifiers constitute a large and a heterogeneous class which includes time/place adverbials (e.g., *in/every winter*), conditional *if/when*-clauses, restrictive modifiers, speaker-oriented epistemics concerning the degree of commitment to the generalization (e.g., *in my opinion*), exceptive constructions (*except/but/besides/apart from X*), and a variety of ceteris paribus clauses that specify (causally) disturbing or facilitating factors.

[43]When there is no overt restrictor, as in (66a) *Dogs bark*, the context is standardly assumed to restrict the quantification to contextually appropriate situations. Roughly, (66a) may then be paraphrased as 'Generally, for situations $C(s)$ that are appropriate for dogs to bark, dogs bark in s'.

As we have also seen, the generic marker *-va-* has perfectly acceptable and felicitous uses in sentences with no overt restrictor. If so, the speaker uses it to either ensure the truth of a generic sentence in the face of counterexamples that would falsify the corresponding sentence without it, all else being equal (see Section 4.1.2), or to ensure plausible deniability by preempting pragmatic strengthening to universal/exceptionless generalization (see Section 4.1.3). A case in point illustrating this use of *-va-* is (62b). Here, *-va-* may be used to convey one of the two inferences depending on context. *Either* the speaker conveys the certainty inference that they know that there are counterexamples to the generalization that would refute the corresponding exceptionless generalization, *or* the ignorance inference that they lack enough information for making a stronger claim corresponding to an exceptionless generalization, e.g., they have incomplete beliefs about various possible disturbing factors that may refute that stronger claim.

When the generic marker *-va-* has perfectly acceptable and felicitous uses in sentences with no overt restrictor, it might be thought of as assuming a function that is comparable to that of a CP-condition, besides introducing a generic quantifier. This property clearly sets *-va-* apart from both the covert GEN operator, and also from overt Q-adverbs. The term *ceteris paribus* covers different kinds of phenomena, and has different interpretations. Here, *-va-* may be thought of as roughly amounting to an exclusive and indefinite (or indeterminate) CP-condition, e.g., "other things being equal", "unless prevented", "in the absence of disturbing factors", and the like. The exclusion-clause, as Cartwright (1983) suggests, may always be reformulated as a clause which requires that certain truth conditions for the exclusive CP law hold, namely those which exclude disturbing factors: "the literal translation is 'other things being equal', yet it would be more apt to read 'ceteris paribus' as 'other things being right'" (Cartwright 1983: 45). Generally, the number of possible disturbing or enabling factors is potentially infinite, and therefore, the restricting condition in application of the generalization cannot always be strictly and fully specified by means of an overt restrictor.

6 Conclusion and further implications

The Czech marker *-va-*, which, as all agree, is separate from the imperfectivizing suffix, is, as I argue, best treated as a marker of genericity (*pace* e.g., Dahl 1995). It is not a marker of iterativity, frequentativity, multiplicativity, in short a mere plurality of situations, or habituality, and as such a marker that is subsumed under tense and/or imperfective aspect, contrary to what is commonly assumed. One of the prima facie reasons why it is wrong to classify and label it as an iterative marker is the simple fact that it is incompatible with iterative adverbials like ‘three times’, used to count particular episodic situations that are not a part of a larger pattern.

It has been shown that this marker has properties that prohibit its subsumption under tense and that do not neatly fit under imperfectivity, either in Slavic languages or as it is generally understood cross-linguistically (see e.g., Comrie 1976). Neither can its meaning be reduced to ‘most’ or ‘usually’ (contrary to Dahl (1995), i.a.), because it imposes no requirement on a specific quantity of situations or instances counting as evidence for the truth of the generalization. By the same token, its meaning cannot be reduced to any other single quantifier or expression of quantity no matter how modalized or probabilistic it might be.

Among the novel key insights of this study is that the marker *-va-* introduces a generic quantifier into the logical representation of sentences, which, similarly to the null generic operator GEN, has a modal (intensional) import that is tied to its predictive, law-like, force, and hence requires access to other possible worlds than just the actual one. Second, it exhibits variable binding properties that are akin to GEN. Third, generic sentences formally marked *-va-* differ from generic sentences without this marker in so far as their truth conditions and felicity depend on the speaker’s reasoning with exceptions. Specifically, they reflect the speaker’s commitment to the strength of the generalization calibrated in terms of exceptions that the speaker believes it may have or has, possibly also including counterexamples that cannot be ignored. This additional modal component may be characterized in terms of epistemic and doxastic modality, and its integration into the meaning of sentences also exploits general pragmatic principles of interpretation.

An exciting domain of study for future research is the exploration of the cognate generic markers *-va-*, their common citation form across Slavic languages (see e.g., Genis et al. 2021, i.a.), in other Slavic languages, namely, in Slovak, where it is also productive, in Polish, where it is less productive, but common enough, and also in other Slavic languages where its occurrences are treated as a part of a lexicalized combination with a verb base, as in Russian, for instance. Hypothesizing that the properties that are here identified for the Czech marker *-va-* may plausibly also characterize its cognates in other Slavic languages would be at least a concrete point of departure for such (comparative) studies.

Turning to cross-linguistic consequences beyond Czech, and Slavic languages, the properties of the Czech marker *-va-* preclude its assimilation to markers that merely encode a plurality of situations (at a given world/time), i.e., markers that are variously labeled as iterative, frequentative, pluractional, habitual, and the like, in traditional descriptive grammars or contemporary linguistic studies.

The empirical and theoretical arguments presented here in support of the claim that the marker *-va-* in Czech, and also in other Slavic languages where it is attested in comparable examples, is a generic marker might serve as a blueprint for their exploration. As observed above, Dahl (1995) after all takes the Czech marker *-va-* as a paradigm example of a whole class of markers in a number of typologically distinct languages that he labels "habitual tense-aspect" markers, and which, according to him, are not generic markers (the list of languages, is given Dahl (1995: 421, fn. 8) and above). To the extent that his claims concerning the Czech marker *-va-* are invalid, as I showed, this raises at least some doubts whether other members of Dahl's class that the Czech marker *-va-* is supposed to represent are all indeed "habitual tense-aspect" markers, as Dahl claims. In short, we may pose the following question for future research:

(68) What is the status of so-called "habitual tense-aspect" markers (in terms of Dahl 1995, i.a.) with respect to the categories of the tense and aspect systems? Do they have (at least some of the) formal and semantic properties that have been here identified for the Czech marker *-va-* which make it a plausible candidate for a generic marker?

Markers that are treated as habitual markers in Tlingit (Cable 2022), Modern Hebrew (Boneh & Doron 2010), African-American Vernacular English (Green 2000), just to name a few, are reported to introduce the realization inference just like the Czech generic morpheme *-va-* does. It is an open empirical question whether they also carry the uncancellable inference that the generalization requires (at least the possibility of) exceptions or counterexamples, given that these two meaning components might be seen as related in the case of the Czech generic marker *-va-*. What is also striking is that such previous studies focus on sentences that describe habits in the strict sense, i.e., regularities of action of ordinary animate individuals, mostly human agents, such as *Fred smokes.* It is unclear whether such so-called "habitual" markers can also be used to mark generic sentences that express regularities that have nothing to do with habits proper, just like the Czech marker *-va-* does.

Perhaps future research will show that at least some markers that are referred to as habitual, and also as iterative, frequentative, multiplicative, and the like, are best classified as markers of genericity. If so, it would further bolster the claim that markers of genericity are not as rare as is thought (*pace* Dahl 1995), and invalidate the widespread traditional belief that they do not exist in natural languages (*pace* Leslie 2008, Leslie & Gelmann 2012, Johnston & Leslie 2012, Rhodes et al. 2018, Cohen 2022, Schiller 2023, i.a.) This would put the semantic episodic/generic distinction on a firmer empirical ground as a semantic distinction with direct grammatical reflexes, in addition to its indirect reflexes in the tense and aspect systems of natural languages, and in a variety of semantic and pragmatic phenomena. Consequently, this would add further arguments for the independence of genericity from the categories of the TMA system in natural languages, and specifically from tense and imperfective aspect. So far such arguments have mainly been of semantic/pragmatic nature, or based on reasoning strategies involved in generalizations which are developed in philosophy, cognitive psychology and computer science (see e.g., Carlson 1977, 1982, Krifka et al. 1995, Pelletier & Asher 1997).

Assuming that the Czech marker *-va-*, and plausibly also the Slavic marker *-va-*, is a generic marker that introduces a generic operator into the logical representation whose properties overlap with, but are not identical to, those of the null generic operator GEN, brings us to the fundamental question in the domain of genericity posed at the outset (see also Carlson 1977, 1995): Can we provide a unified semantic analysis for all generic sentences? In Czech, generic sentences fall into two main types: one formally marked with *-va-*, and the other formally unmarked. In the latter case, and similarly as in English, the generic (and habitual) reading is a function of the combined meanings of the subject and predicate interacting with context, and it can be represented by means of a tripartite logical structure with the GEN operator. Recall that in all Slavic languages, generic sentences that are not formally marked with the generic *-va-* are headed by primary or secondary imperfectives (marked for imperfectivity with the imperfectivizing suffix), and in West Slavic languages also commonly by perfective verbs.

These two main formal means for the expression of characterizing genericity, one formally marked with the morpheme *-va-* for genericity and the other formally unmarked in this respect, are not neatly aligned with the inductive and the rules-and-regulations model of genericity (Carlson 1995), respectively. The generic morpheme *-va-* is naturally used in generic sentences that provide the best examples in support of the inductive model of genericity, namely generics that express descriptive (weak) generalizations, including habits in the strict sense (i.e., regularities of action ascribed to ordinary animate individuals, as in (13)). However, it is important to emphasize that the Czech generic *-va-* is also used in generic sentences that best fit the rules-and-regulations model of genericity, including generics with a normative reading. In short, its use cuts across the inductive versus rules-and-regulations distinction, and also across the related descriptive versus normative distinction.

Tentatively, it may be concluded that this leaves open the question about the feasibility of a unified semantic analysis for all generic sentences in Czech, and perhaps also in other languages that have a dedicated marker of genericity that is comparable to that in Czech.

Abbreviations

1	first person	NEG	negation
2	second person	NOM	nominative
3	third person	PFV	perfective
ACC	accusative	PL	plural
AUX	auxiliary	PREF	prefix
CP	ceteris paribus	PRS	present tense
DAT	dative	PST	past tense
GEN	generic	REFL	reflexive
INF	infinitive	SG	singular
IPFV	imperfective	VA	morpheme *-va-*
NCI	negative concord item		

Acknowledgements

I would like to thank Greg Carlson and Manfred Krifka for their suggestions, and inspiration, as well as the audiences at the *Formal Diachronic Semantics 6* conference (University of Cologne, Germany), *The 6th International Conference on Grammar and Text* (School of Social Sciences and Humanities of NOVA University Lisbon, Portugal), *The 20th Szklarska Poręba Workshop on the Roots of Pragmasemantics* (Szklarska Poręba, Poland), *The workshop on Formal Semantics and Its Interfaces* (Zhejiang University, Hangzhou, China), and *The 19th Conference on Formal Approaches to Slavic Linguistics* (University of Maryland, USA) for all the valuable criticism and suggestions. All opinions, omissions, and errors remain my own.

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Chapter 5

Cross-Slavic aspect, passives, and temporal definiteness

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The chapter discusses cross-Slavic variation in aspect use in four different contexts (sequences of single events, habituality, historical present, factual imperfectives), with a particular focus on Czech vs. Russian. A new context is added to the mix, namely passives (participial and reflexive), and it is shown that Czech and Russian differ in that Czech, but not Russian, regularly derives both types of passives from both aspects, with predictable aspectual meanings. I summarise the few existing formal-semantic accounts of cross-Slavic aspectual variation, which all have in common that they apply some concept of definiteness to the verbal and/or temporal domain. Finally, I outline a general research programme to exploit parallels between individuals, events, and times regarding definiteness.

1 Introduction

All Slavic languages have a grammatical category of aspect: each verb form is either imperfective (IPFV) or perfective (PFV) (in a given contexts), and in many environments one or the other aspect is obligatory. It is therefore commonly assumed that the lexical meaning of a given verb can be described using either aspect, so that for many lexical verb meanings we get an ASPECTUAL PAIR of IPFV and PFV forms.[1] All Slavic languages use primarily prefixes and suffixes

[1] I am abstracting away from so-called biaspectual verbs, which are interpreted as IPFV or PFV in a given context, such as Czech *akceptovat* 'accept' or Russian *kaznit'* 'execute' (see, e.g., Janda 2007 for discussion of Russian).

Berit Gehrke. 2026. Cross-Slavic aspect, passives, and temporal definiteness. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 231–304. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078577

on the verb to make aspectual distinctions.[2] In (1) we see typical examples of aspectual pairs from Czech and Russian, which are either derived by prefixing a morphologically SIMPLE IPFV to derive a PFV partner, or by suffixing a prefixed or a morphologically simple PFV to derive an IPFV partner (the latter forms are descriptively labeled SECONDARY IMPERFECTIVES, SIs).[3]

(1) a. IPFV *psát* > PFV *na-psat* 'write'
PFV *pode-psat* > IPFV *pode-pis-ova-t* 'sign'
PFV *dát* > IPFV *dá-va-t* 'give' (Czech)

b. IPFV *pisat'* > PFV *na-pisat'* 'write'
PFV *pod-pisat'* > IPFV *pod-pis-yva-t'* 'sign'
PFV *dat'* > IPFV *da-va-t'* 'give' (Russian)

Thus, there is no uniform morphological marking of either aspect: there are PFVs that contain a prefix or not (e.g. PFV 'write' vs. PFV 'give' in (1)), as well as IPFVs that are morphologically simple (e.g. IPFV 'write') or complex, as in the case of SIs (e.g. IPFV 'sign'). Nevertheless, native speaker intuitions clearly group such verb forms into either the IPFV or the PFV group, and there are diagnostics to show which group a verb form belongs to. For example, in Czech and Russian, only IPFVs can form periphrastic future forms and can combine with phase verbs, as illustrated for Russian in (2) (see, e.g., Schoorlemmer 1995, Filip 1999, Borik 2006, Gehrke 2008 for discussion and further diagnostics).

(2)

Ona	{budet /	načala}	{pisat'	/ podpisyvat'	/
she.NOM	will	began.PFV	write.IPFV.INF	under.write.SI.INF	

*napisat'	/ *podpisat'}	pis'mo.
on.write.PFV.INF	under.write.PFV.INF	letter.ACC

(Intended:) 'She {will / started to} {write / sign} a/the letter.' (Russian)

[2]In addition to the aspectual system described here (illustrated with examples from Russian and Czech), Bulgarian, Macedonian, Sorbian, and some Serbian dialects have aorist and imperfect forms, which make aspectual contributions that are orthogonal to the IPFV-PFV-distinction and similar to what we find in, e.g., Spanish and French (see, e.g., de Swart 1998). To my knowledge, there is no formal-semantic account of the interaction between these aspectual tense forms and (I)PFV in these languages. The other Slavic languages lost these distinctions and only have one past tense form, which is historically related to the perfect participle (the so-called *l*-participle) and combines with the auxiliary 'be' in some languages (e.g. Czech) or is a stand-alone synthetic past tense form in others (e.g. Russian).

[3]I set aside less common morphological means to derive an aspectual partner, as well as suppletive forms. For similar data from some other Slavic languages, see, for instance, Arsenijević (2006), Žaucer (2009), Kwapiszewski (2022).

Despite the overall morphological commonalities in the aspect systems of all Slavic languages, descriptive Slavicists early on noted cross-Slavic variation in aspect use, particularly between Czech and Russian (Dokulil 1948, Křížková 1955, Bareš 1956, Bondarko 1959, Ivančev 1961, Širokova 1971), and these two languages have been the focus of subsequent descriptive work (e.g. Eckert 1984, Stunová 1993, Petruxina 2000, with the latter also taking into account Bulgarian, Polish, and Slovak). An investigation into the differences between ten Slavic languages in various contexts is provided by Dickey (2000), who proposes a cognitive semantic account with two contrasting aspect types, a Western type with its prototype Czech, and an Eastern type, with Russian as its prototype. Subsequently, additional corpus studies broadened the empirical picture (Gehrke 2002, von Waldenfels 2012, 2014, Alvestad 2013, Dübbers 2015, Klimek-Jankowska 2022).

The descriptive observations about cross-Slavic differences in the use of (I)PFV are by now widely acknowledged, but only recently a few formal-semantic accounts have been proposed aiming at capturing the differences. These almost exclusively deal with differences in so-called general-factual contexts (see Section 2) (Alvestad 2013, Mueller-Reichau 2018b, Klimek-Jankowska 2022), with the exception of Mueller-Reichau (2025), who includes descriptions of ongoing and habitual events. A general theory to account for the differences, be it semantic or pragmatic or a combination of both, is still absent.[4]

The aim of this chapter is three-fold. First, after an outline of formal-semantic background assumptions in Section 2, Section 3 provides an overview of empirical investigations into cross-Slavic differences in the use of (I)PFVs in various contexts, with focus on Czech vs. Russian. Second, Section 4 adds an empirical domain that is missing almost entirely from the discussion, namely the domain of passives, for which we observe further aspectual differences between Czech and Russian.[5] Third, Section 5 summarises existing formal-semantic accounts of cross-Slavic aspectual variation, which all have in common that they employ some concept of temporal definiteness. Section 6 takes stock and outlines a general research programme to explore parallels between the nominal and verbal domain with respect to definiteness.

[4]Since this chapter is part of a volume on the semantics of Slavic languages, I mostly set aside potential (morpho-)syntactic considerations, although capturing the full picture of the variation should ultimately also take syntax into account.

[5]I say "almost" because a recent corpus study (Wiemer et al. 2023) investigates the synchrony and diachrony of Polish and Russian past passive participles from the point of view of aspectual forms and readings.

2 Formal-semantic background assumptions

This section briefly outlines formal-semantic background assumptions, relevant for later sections. These include the semantics of aspect (more detailed discussion can be found in Mueller-Reichau 2026 [this volume] and Tatevosov 2026 [this volume]; see also Grønn 2026 [this volume] on Slavic tense), from a cross-linguistic perspective and including cross-linguistic variation, the so-called general-factual use of Slavic IPFVs, as well as the notions of definiteness and specificity in the nominal domain (more detailed discussion can be found in Borik 2026 [this volume]). The latter will serve as a point of departure for the discussion of proposals that use the notion of temporal definiteness to account for cross-Slavic variation in aspect use.

2.1 The semantics of aspect and cross-linguistics variation

Throughout the chapter, I assume that there is a distinction between different event types at the level of lexical or inner aspect (related to telicity, resultativity, stativity, change-of-state), on the one hand, and grammatical or outer aspect, which operates on top of event structures of various complexity (e.g. Smith 1991, Filip 1999; for further discussion see Gehrke 2008).[6] From a descriptive point of view, it has been observed that Slavic PFV forms appear in descriptions of "completed" events (however one might want to formally characterise the notion of completion), whereas IPFV forms appear in the description of ongoing or durative events, regular circumstances,[7] and (at least in some Slavic languages) iterative and habitual events (e.g. Isačenko 1962, Filip 1999, Borik 2006, for Czech or Russian). Cross-linguistically, these are common meanings associated with grammatical aspect (see, e.g., Deo 2015 for more recent discussion), which is why it is standardly assumed that Slavic (I)PFVs are instances of grammatical aspect. On the other hand, there are (morphosyntactic) proposals in the literature that relate this distinction to inner aspect (e.g. Arsenijević 2006, Łazorczyk 2010), but at least for Czech and Russian it has been shown convincingly that semantically, (I)PFV is distinct from (a)telicity (see, e.g., Filip 1999, Borik 2006, Gehrke 2008).[8]

[6]I use the term EVENT as a cover term for both events and states (or for what Vendler 1957 called states, activities, accomplishments, achievements).

[7]A special case of regular circumstances are generic statements; I will not address generic sentences in this chapter, but see Filip (2026 [this volume]) for discussion.

[8]A theoretical option that to my knowledge has not been explored formally could be that Slavic languages vary in this respect in that in some Slavic languages (I)PFV is an inner-aspectual distinction, whereas in others it is an outer-aspectual one (see Stunová 1993 for descriptive generalisations along this line, concerning Czech vs. Russian aspect; see also Breu 2000, Scholze 2008, who argue, informally, that the colloquial Upper Sorbian aspect system is based on "ter-

Cross-linguistically, it is common to analyse the semantics of (I)PFV as expressing a temporal relation between the event time (E) (or situation time; see Klein 1994, 1995) and a reference time (R) (or topic/assertion time; see Klein 1994, 1995), building on Reichenbach (1947). More precisely, (I)PFV have been treated as aspectual operators that take as input a predicate of events and turn it into a predicate of times, by introducing a reference time that gets temporally related to the run time of the event (represented as $\tau(e)$, following Krifka 1998) (3).

(3) a. $[\![\text{PFV}]\!] = \lambda P \lambda t \exists e[\tau(e) \subseteq t \wedge P(e)]$
    b. $[\![\text{IPFV}]\!] = \lambda P \lambda t \exists e[t \subseteq \tau(e) \wedge P(e)]$

With the PFV aspect (3a), the run time of the event is included in the reference time t, leading to an external perspective on the event which can therefore be viewed in its entirety or totality; with the IPFV (3b), the reference time is included in the event time, leading to an internal perspective on the described event.[9]

Broadly speaking and over-simplifying to a certain extent, there are at least two ways in which the semantic account of a given aspect can vary, which have been used in explaining cross-linguistic variation in grammatical aspect.[10] One

minativity"). For example, Łazorczyk (2010) as a proponent of the inner-aspectual analysis brings forward data from Bulgarian to argue that the aorist-imperfect distinction (recall footnote 2) belongs to the realm of outer aspect, so that (I)PFV must contribute to the level of inner aspect. It could very well be that in the presence of aorist/imperfect (and possibly other contributing factors), this language uses (I)PFV differently than Slavic languages that lack the aorist/imperfect distinction. In the remainder of the chapter, I will mostly focus on Czech vs. Russian, for which I view the arguments for treating (I)PFV as related to grammatical aspect convincing. A separate issue, which I return to below, is whether (I)PFV forms directly or only indirectly correlate with grammatical aspect meaning (or syntax).

[9] See, for example, Kratzer (1998), Ferreira (2016) for further discussion. In (3), I use a simplified extensional representation; to fully grasp the semantics especially of IPFVs it might be necessary to take into account intensions and possible worlds, but I will abstract away from this. There are also modal accounts, in particular for the IPFV, going back to at least Dowty (1979). Furthermore, there are accounts in particular of Slavic (I)PFV in terms of partial vs. total events (e.g. Filip 1999, Altshuler 2014) or in terms of linking the PFV to maximality (e.g. Filip 2008); I view these accounts as special instances of the more general account in (3).

[10] A third way is proposed in Arregui et al. (2014), who argue that the variation in the interpretation of imperfectivity in Slavic, Romance, and Jê additionally derives from a difference in the modal bases employed by the IPFV operators in these languages. An interesting and sophisticated theoretical proposal of the cross-linguistic variation as to whether an IPFV operator in a language can express the (narrower) progressive reading (e.g. English) or (broader) imperfective reading (e.g. Gujarati) is proposed in Deo (2015).

There are also possible (morpho-)syntactic accounts that I will not address in detail here, since I will be mostly concerned with semantic approaches. For example, Biskup (2023), following Tatevosov (2011, 2015) (see below), assumes that grammatical aspect operators are silent operators not directly linked to verbal aspectual morphology, but their presence is determined by the syntactic operation Agree; variation in the ordering or merging of particular morphemes could then lead to differences in aspect behaviour, both across and within Slavic languages.

point of variation concerns proper inclusion (⊂) vs. improper inclusion (⊆). For example, Altshuler (2014) argues that the English progressive, an instance of IPFV, involves proper inclusion of the reference time in the event time, whereas the Russian IPFV involves improper inclusion, and that this accounts for empirical differences between the two languages in the use of IPFVs.

Another point of variation has to do with markedness: In the opposition between IPFV and PFV one aspect in a given language might be the marked, the other the unmarked member in the opposition. This could involve morphological markedness, or markedness could be seen as more abstract, concerning the (wider or narrower) meaning spectrum of a given form. For example, the English progressive (IPFV) is taken to be the morphologically and also semantically marked aspect, whereas there is no morphological marking of the PFV; this leads to a wideheld assumption that, e.g., the simple past expresses a PFV semantics (see, e.g., Smith 1991, Klein 1994), albeit unmarked.[11] Alternatively, one could view non-progressive forms as expressing no grammatical aspect (see, e.g., de Swart 1998 for such a view, as well as Minor et al. 2023 for recent psycholinguistic evidence that points in this direction), and a PFV semantics to come about simply due to the overall context as well as due to the input, e.g. whether the underlying predicate is eventive/stative, telic/atelic, etc.; the absence of IPFV marking would then only play a contributing factor (e.g. it could be an anti-presupposition effect), but it would not be the decisive factor. In this chapter, I side with the latter view. For Slavic languages, on the other hand, it is common to assume that the PFV is the (semantically) marked member of the opposition, so that IPFVs either involve the negation of the semantics of the PFV (see, e.g., Borik 2006 for such an account of Russian), or can express either PFV or IPFV semantics (e.g. Grønn 2015 for Russian, whose account is discussed in more detail in Section 2.2.2).[12]

This latter point also relates to the general question whether a given form (e.g. Slavic (I)PFV forms) always comes with the same semantics, or whether there can be form-meaning mismatches. Given the markedness assumptions outlined above, it is common to assume (at least implicitly) that for the English progressive and the Slavic PFV the forms involved correlate respectively with an IPFV and PFV semantics (but see Paslawska & von Stechow 2012 for a general account of the Ukrainian tense-aspect system that fully dissociates form and meaning). For Slavic IPFVs, in contrast, there are authors that explicitly argue that IPFV forms do not necessarily correspond to an IPFV semantics (for Russian, see Borik

[11] A question that arises in this context is why the same does not hold for the simple present.

[12] Morphologically, it is clear from the examples in (1) that one cannot take either aspect as marked/unmarked: in some cases the PFV is morphologically more complex than its PFV partner, in others it is the other way around (see also discussion in Jakobson 1966).

2006, Grønn 2004, 2015), and others that (implicitly or explicitly) assume a form-meaning-correspondence (see e.g. Filip 1999, Gehrke 2022, 2023 for Czech and Russian, and Altshuler 2014 for Russian). Tatevosov (2011, 2015, 2026 [this volume]), in turn, argues that there is no direct correlation between form and meaning, building on insights from Klein (1994, 1995): What we call (I)PFV forms (in Russian, but presumably in Slavic more generally) does not directly correspond to (I)PFV semantic operators, given that aspectual morphology appears directly on the verb, whereas grammatical aspect is located above the VP. Nothing I say here is in principle incompatible with this view, to which I am fully sympathetic; once we reach the level of grammatical aspect, a given IPFV form could still uniformly lead to an IPFV semantics.[13] Throughout the chapter, I use (I)PFV for the forms that appear in contexts that convey a typical (I)PFV semantics.

There is a particular use of Slavic IPFVs that at first sight seems problematic for accounts that directly relate (I)PFV forms with an (I)PFV semantics (in a given sentence). This use has first been described for Bulgarian and Russian by Maslov (1959), who termed it GENERAL-FACTUAL. It is peculiar from a cross-linguistic perspective because factual IPFVs are found for the description of seemingly "completed" events (among others), in which one would expect PFV forms but for some reason the IPFV is still used. Subsequently there has been a lot of work on in particular Russian factual IPFVs, both from a descriptive and a formal perspective (e.g. Glovinskaja 1981, 1989, Padučeva 1996, Mehlig 2001, 2013, Grønn 2004, 2015, Mueller-Reichau 2018a, Gehrke 2023), but to this date there is no consensus as to the precise analysis. Since this IPFV use has played a pivotal role in semantic accounts of (in particular Russian) aspect and is a prominent point of variation within the Slavic languages, the following section is devoted to it.

2.2 The (general-)factual IPFV

In factual contexts, IPFVs can appear to describe bounded, "completed" events.[14] In such contexts, it is often assumed that the IPFV is in ASPECTUAL COMPETITION with the PFV (a term that goes back to at least Mathesius 1938), because both can (often) be used interchangeably with only subtle differences that are hard

[13] For example, building on Tatevosov's work, Mueller-Reichau (2026 [this volume]) proposes a system that derive Russian aspect semantics from the morphological input and the overall context.

[14] Again, the term "event completion" remains at an intuitive level. The traditional literature also discerns subtypes of the general-factual with intuitively non-completed events (e.g. Glovinskaja 1981, 1989, Padučeva 1996). In formal accounts of factual IPFVs (e.g. Grønn 2004), these subtypes are usually not addressed, leading to the somewhat distorted impression that general-factual contexts always involve "completed" events (see Gehrke 2023 for further discussion).

to pin down. This section first describes two commonly assumed subtypes of factuals (existential, presuppositional); since most of the literature on factual IPFVs deals with Russian, I use Russian examples to illustrate. I then outline a semantic proposal for Russian factual IPFVs (Grønn 2004 et seq.), which has been used in accounts of cross-Slavic aspect variation, to which I return in Section 5.

2.2.1 Existential and presuppositional factuals

The literature on Russian aspect distinguishes at least two subtypes of factual IPFVs, the existential type (Padučeva 1996, Grønn 2004) and what Grønn calls the presuppositional type (the "actional" type in Padučeva 1996). The EXISTENTIAL IPFV is illustrated in (4) (corpus example from Grønn 2004: 180; my glosses).

(4)

Ne	bylo	somnenij,	čto	ja	prežde	vstrečal	ee.	
not	was.IPFV.3SG.N	doubts.GEN	that	I	before	met.SI.SG.M	her	

'There was no doubt that I had met her before.' (Russian)

In this example, the (male)[15] speaker states that he had a meeting with a female person in the past. From the context it is clear that a meeting was "completed" and happened in the past; nevertheless the IPFV is used. Existential IPFVs usually involve stress on the verb form and can be paraphrased as 'There has been/is/will be (at least) one event of this type.' (following the idea that existential IPFVs involve event types or kinds; see Mehlig 2001, 2013, Mueller-Reichau & Gehrke 2015). The paraphrase for (4) would therefore be 'There has been at least one event of the type "meet her"'. The exact time when this event happened, and also whether it happened more than once, remains unspecified. In some existential contexts, Russian has to use the IPFV and cannot switch to the PFV, for example in combination with the temporal adverb *kogda-nibud'* 'ever', as illustrated in (5) (adapted from Dickey 2000: 104).

(5)

Ty	kogda-nibud'	{*prygnul	/ prygal}	s	tramplina?
you.NOM	ever	jumped.PFV	jumped.IPFV	off	diving.board

'Have you ever jumped off a diving board?' (Russian)

In Section 2.3, we will see that *nibud'*-marked indefinite pronouns express scopal non-specificity, and I will get back to such adverbs in Section 3.4 and Section 6.

[15]Czech and Russian past tense forms display gender and number agreement with the subject (e.g. in the singular -∅ for masculine, *-a* for feminine, *-o* for neuter), because they originate from a participle. I add this to the glosses only where relevant.

What examples like these show is that the general impression about the interchangeability of IPFV and PFV in factual (in this case existential) contexts that is sometimes created by the literature should be taken with a grain of salt (see also Dübbers 2015).

The PRESUPPOSITIONAL IPFV is illustrated in (6) (from Glovinskaja 1981: 108; my glosses and translation).

(6) Zimnij Dvorec stroil Rastrelli.
winter.ADJ.ACC palace.ACC built.IPFV Rastrelli.NOM
'It was Rastrelli who built the Winter Palace.' (Russian)

In this example, we have a "completed" event in the past, the building of the Winter Palace (which now houses the Hermitage Museum in Saint Petersburg). It is known that this event happened exactly once and also when it happened, so there is no temporal non-specificity, unlike what we have with the existential IPFV. Grønn (2004) labels this IPFV use presuppositional because it arises when the existence of the event in question is given in or derivable from the context, hence it is presupposed (and in Grønn's account backgrounded, following Geurts & van der Sandt 1997). An utterance with a presuppositional IPFV adds further information about this presupposed event, and this new information is in focus. A suitable paraphrase is 'The (already mentioned or contextually retrievable) event was/is/will be such and such.' In (6), context or world knowledge presupposes the existence of the event 'build (the) Winter Palace' (as objects of verbs of creation tend to do, but it could also have been talked about in the previous context); the new information is that the architect of the building was Rastrelli, which is where the focus lies. This IPFV use is usually accompanied by a particular information structure (see also Borik & Gehrke 2018 for further discussion); in our examples the presupposed (backgrounded) material appears sentence-initially and unstressed (the building of the Winter Palace) and the new information in focus is Rastrelli, in sentence-final position, resulting in a non-canonical OVS order.[16]

There is a common assumption in the literature that factual IPFVs do not just occur in Russian but also in other Slavic languages, and this is independent of whether the authors in question assume a single factual meaning or several subtypes; nevertheless, its use in Russian is reported to be more frequent than in, e.g., Czech (e.g. Dickey 2000, Alvestad 2013, Dübbers 2015, Mueller-Reichau 2018b, Klimek-Jankowska 2022), as we will see in Section 3.4. In contrast, it is

[16]Russian, like other Slavic languages, is a "free word order" language with canonical SVO order; deviations from this canonical order are primarily information-structurally motivated (see Jasinskaja & Šimík In press for discussion).

argued in Gehrke (2022) that Czech only makes use of presuppositional, but not of existential IPFVs; I will come back to this in Section 5.5. Let us turn to Grønn's (2004 et seq.) account.

2.2.2 "Fake" IPFVs and the Aspect Neutralisation Rule in Russian

Following the traditional view that the Russian IPFV is semantically unmarked, Grønn (2004) employs a weak IPFV semantics, which merely requires overlap between the event time and the reference time ($e \circ t$) (building on Klein 1995). This rather weak semantics gets pragmatically strengthened to a "true" IPFV (the reference time is part of the event time), or to an actual PFV semantics (the event time is part of the reference time), which, he argues, we find with factual IPFVs. Grønn takes into account the role of information structure to characterise the contexts in which strengthening happens in one or the other direction. This account is a precursor to Grønn (2015), in which it is proposed that IPFV forms can express both IPFV and PFV semantics, as in (7).

(7) a. $[\![\text{PFV}]\!] = \lambda t \lambda e.e \subseteq t$
    b. $[\![\text{IPFV}_{\text{ongoing}}]\!] = \lambda t \lambda e.t \subseteq e$
    c. $[\![\text{IPFV}_{\text{factual}}]\!] = \lambda t \lambda e.e \subseteq t$ "Fake" IPFV

Grønn calls the IPFV that has the same semantics as the PFV in (7c) a "FAKE" IPFV. He proposes that the existence of $\text{IPFV}_{\text{factual}}$ alongside the PFV leads to aspectual competition, and also here he follows the Slavistic tradition. In the default case the PFV is argued to appear but in certain contexts the $\text{IPFV}_{\text{factual}}$ wins the competition. This is proposed to give rise to the presuppositional IPFV in cases where narrative progression is to be avoided (under the assumption that the PFV always leads to narrative progression; see below), and to the existential IPFV when the reference time is too large for the PFV semantics to be informative.

Starting from the well-known observation that tenses can be both quantificational and referential, Grønn & von Stechow (2016) spell out a research programme that draws parallels between events and times, on the one hand, and the semantics of bare nominals in articleless languages (e.g. Russian), on the other. In particular, they treat tenses (R-S relations), aspects (E-R relations), and verbal predicates as relational predicates. Further information about times and events, given for example by adverbials, is added via predicate modification. Covert definite and indefinite operators turn events and times into dynamic generalised quantifiers, which, respectively, are anaphoric to a previous referent, maximally presupposing given information, or introduce a new referent. Following Grønn

(2004, 2015), Russian IPFV forms have either an IPFV or a PFV ("fake" IPFV) semantics. Covertly, on top, we get a definite or an indefinite event. Indefinite, complete events, on the other hand, are regularly referred to by PFVs, to ensure narrative progression, which is analysed as a pragmatic effect ("be orderly").

Employing this system, Grønn (2015) proposes that presuppositional IPFVs involve a definite event and a definite reference time, whereas existential IPFVs come with an indefinite event and an indefinite reference time. In order to account for why the IPFV is used in presuppositional contexts, Grønn & von Stechow (2016) propose the rule in (8), which builds on the general idea that the IPFV is the unmarked member of the aspectual opposition.

(8) Aspect Neutralisation Rule (see Grønn & von Stechow 2016)
When a semantically PFV aspect is definite/anaphoric, it is morphologically neutralised to IPFV.

This system is the point of departure for Alvestad's (2013) account of cross-Slavic differences in aspect use in imperatives, which I will outline in Section 5.2. The proposal to treat factual IPFVs as "fake" and not "true" IPFVs has been challenged by Gehrke (2023), and I return to this alternative account in Section 5.5. Finally, Section 6 draws more general parallels between the nominal and verbal domain with respect to definiteness. In the following, let us move to formal-semantic assumptions about definiteness and specificity in the nominal domain.

2.3 Definiteness and specificity

This section briefly summarises semantic approaches to definiteness in the nominal domain (for more detailed discussion see Borik 2026 [this volume]), in as far as it is relevant for the notion of "temporal definiteness" that will play a role in later sections (see also Section 2.2.2). (In)definites appear both in argument as well as predicative position, and it is common to take the argument use as basic and to derive the predicative use from it. A widespread approach to singular definites takes them to presuppose uniqueness and existence (following Frege 1892, Strawson 1950), while plural definites presuppose maximality (following Sharvy 1980). For example, Heim (2011) spells out the semantics of (in)definiteness (not necessarily of *the/a*) in (9), generalising maximality to both singular and plural definites.

(9) a. $[\![+\textsc{def}]\!] = \lambda P : \exists x \forall y[\textsc{max}(P)(y) \leftrightarrow x = y].\iota x.\textsc{max}(P)(x)$
b. $[\![-\textsc{def}]\!] = \lambda P.\lambda Q.\exists x[P(x) \wedge Q(x)]$

Heim proposes that indefinites and definites form a scale, with indefinites being logically weaker. She argues that in languages that have a definite article, such as English, the indefinite cannot be used in definite contexts, due to an anti-uniqueness implication with indefinites, which arises in competition with definites, assuming the principle of Maximise Presupposition.

Coppock & Beaver (2015), in turn, take the predicative use as basic and propose that (singular) definites presuppose uniqueness but not existence; existence only comes in through a covert type shift of definites in argument position. Yet in other approaches definiteness is understood in terms of familiarity: (in)definites are associated with discourse referents, and with definites the referent is anaphorically linked to a previously introduced discourse referent (Kamp 1981, Heim 1982; see also Grønn & von Stechow 2016, as outlined in Section 2.2.2). Finally, there is also a distinction between strong definites, which satisfy uniqueness, and weak definites (e.g. *go to the supermarket*), which do not and which, e.g., Aguilar-Guevara & Zwarts (2010) treat as kind-referring.

In addition, there is another notion that is relevant, namely specificity (see also Geist 2026 [this volume]). Also this notion has different uses in the literature. For example, Geist (2010) distinguishes between epistemic specificity (whether or not the speaker can identify the referent) and scopal specificity (whether an indefinite has wide or narrow scope with respect to other quantifiers or operators). She shows that Russian, which lacks articles, has several series of indefinite pronouns that are specialised to signal epistemic specificity (the *koe*-series, as opposed to the epistemically non-specific *-to*-series) or scopal non-specificity (the *-nibud'*-series). The particular morphemes are added to *wh*-items (e.g. *kto* 'who') to derive an indefinite pronoun (*koe-kto, kto-to, kto-nibud'* '(different types of) some/anyone'), and they are not limited to the nominal domain, as we already saw with *kogda-nibud'* 'ever' in (5), which is based on the *wh*-word 'when'.

All of the notions discussed in this section, apart from weak definites, have been transferred from the nominal to the verbal or sentential domain, and they will come up again in one or the other account of cross-Slavic variation in aspect use, summarised in Section 5; some have already been addressed in Section 2.2.2. With these theoretical background assumptions in hand, let us turn to the empirical description of cross-Slavic variation in aspect use in selected contexts.

3 Cross-Slavic variation in aspect use

This section describes empirical generalisations about cross-Slavic variation in the use of aspect, focusing on Czech and Russian and on particular contexts for illustration: sequences of single events, habitual events, the historical present, and factual contexts. First, though, let me start with a quantitative picture that emerges from the literature on differences in aspect use between Czech and Russian. Table 1 summarises the findings for some contexts.[17]

Table 1: Some aspectual differences between Czech and Russian

	Czech	Russian
Chains of single events (past)	IPFV, PFV	(almost excl.) PFV
Iterativity, habituality (past, present)	IPFV, PFV	(almost excl.) IPFV
Historical present	IPFV, PFV	(almost excl.) IPFV
Running instructions & commentaries	IPFV, PFV	(almost excl.) IPFV

The first, more coarse-grained picture that emerges from this table is that there are contexts in which Czech allows for the use of both aspects, whereas Russian uses just one aspect, and except for the first context it is the IPFV that is quasi-obligatory in Russian.[18] Such differences have therefore also been framed in terms of the obligatory use of a particular aspect in Russian, vs. the optional use of a particular aspect in Czech, in a particular context (e.g. Bondarko 1959, Křížková 1961, Širokova 1971, Petruxina 2000). However, it is not clear what it means for a grammatical aspect to be optional, as this suggests some kind of arbitrariness, or at least that aspect use in Czech is just a matter of choice, which is usually found with lexical, but not with grammatical categories. In other words, at this coarse-grained level, where we simply count (I)PFV forms in particular

[17] Sequences of single events are discussed in Ivančev (1961), Eckert (1984), Mønnesland (1984), Stunová (1993), Dickey (2000), Petruxina (2000), Gehrke (2002, 2022), Barentsen (2008), Fortuin & Kamphuis (2015), iterativity and habituality in Eckert (1984), Stunová (1993), Dickey (2000), Kresin (2000), Gehrke (2002, 2022), Dübbers (2015), Fortuin & Kamphuis (2015), the historical present in Křížková (1955), Bondarko (1958, 1959), Petruxina (1983), Stunová (1993), Dickey (2000), Fortuin & Kamphuis (2015), running instructions and commentaries in Dickey (2000).

Further differences between Czech and Russian (and sometimes also other Slavic languages) have been described for imperatives (Dokulil 1948, Eckert 1984, Benacchio 2010, von Waldenfels 2012, 2014, Alvestad 2013), motion verbs (Eckert 1991, Gehrke 2002, 2022), various prefixes and suffixes (Nübler 1992, Petruxina 2000, Dickey 2001, 2005, 2011, Dickey & Hutcheson 2003), contexts involving negation (Dickey & Kresin 2009, von Waldenfels 2014, Dübbers 2015), nominalisations (Dickey 2000, Biskup 2023, Gehrke to appear), as well as factual contexts (Dickey 2000, Gehrke 2002, 2022, Alvestad 2013, Dübbers 2015, Fortuin & Kamphuis 2015, Mueller-Reichau 2018b, Klimek-Jankowska 2022); I will address the latter in more detail in Section 2.2.

[18] The motivation for adding “almost excl(usively)” in Table 1 will be addressed in Section 3.1–Section 3.3.

contexts (which is also something one could do statistically, see, e.g., von Waldenfels 2012, 2014, Dübbers 2015, Klimek-Jankowska 2022), noting these differences is of course important, but a quantitative analysis cannot be the endpoint to understanding the differences; we have to take it as the starting point for a detailed qualitative analysis. In the following, I focus on differences in four particular contexts, each time trying to explore the circumstances which motivate the use of one or the other aspect, and thus to give a positive characterisation of the reasons for the occurrence of a given aspect form. I will pay special attention to the verb type employed, the importance of which has also been stressed in some of the previous empirical research (e.g. Eckert 1984, Stunová 1993, Gehrke 2002).

3.1 Chains of single events

In the descriptions of chains of single events, Russian almost obligatorily uses the PFV to give rise to a sequence of event (SOE) interpretation, whereas we find both aspects in Czech (Ivančev 1961, Eckert 1984, Mønnesland 1984, Stunová 1993, Dickey 2000, Petruxina 2000, Gehrke 2002, 2022, Barentsen 2008, Fortuin & Kamphuis 2015). For example, if the last event in a chain has a clear temporal onset but then evolves further, we have a case that is commonly labeled INGRESSIVITY. In such cases, it is plausible that there is tension between using the PFV to mark the temporal onset and to explicitly mark SOE, on the one hand, and the IPFV, on the other, to mark the process of the evolving event. It turns out that Russian consistently employs the first strategy (PFV), leaving the evolution of the process implicit, while Czech regularly goes for the second strategy (IPFV), so that in this language ingressivity is contextually derived, but not marked on the verb form. This systematic difference in aspect use was first noted by Ivančev (1961), and one of his examples is given in (10) (from Ivančev 1961: 36; Czech original by Božena Němcová; my own glosses and translations; relevant verb forms (here and in the following longer examples) italicised).

(10) a. ... zvolna si *sedl* vedle mne a Josefa, *položil*
slowly REFL down.sat.PFV next.to me and Josef put.PFV
hlavu do dlaně a *díval* na mne.
head.ACC to palm and looked.IPFV on me
'He slowly sat down next to me and Josef, put his head in his palm and looked at me.' (Czech)

b. ... on tixo *sel* vozle menja i Iozefa, *sklonil*
he quietly down.sat.PFV near me and Josef tilted.PFV

golovu na ruki i *stal* *smotret'* na menja.
head.ACC on hands and began.PFV watch.IPFV.INF on me
'He quietly sat down near me and Josef, put his head on his hands and started watching me.' (Russian)

In this example the last event in the chain of events is described by an activity predicate, 'look/watch'. In Czech, a simple IPFV is used and the circumstance that the watching follows the preceding events (which, in turn, are described by PFVs in both languages) is understood only contextually. In Russian, in contrast, the translator uses the PFV phase verb *stat'* 'begin' in combination with the (IPFV) infinitive of 'watch', and thus explicitly signals the SOE reading.

In other examples of this sort, one can find the so-called ingressive prefix *za-*, which marks the onset of an otherwise unbounded event (see Isačenko 1962). This prefix (in this use) is productively used in Russian but arguably non-existent in Czech (see also Ivančev 1961, Dickey 2000, Petruxina 2000).[19] One such case is illustrated with the Russian original from Mixail Bulgakov's *Master i Margarita* and its Czech translation in (11) (Gehrke 2022: 21) (see also Gehrke 2002).

(11) a. On *pomolčal* nekotoroe vremja v smjatenii, *vsmatrivajas'*
he PO.was.silent.PFV some time in confusion watch.IPFV.AP
v lunu, plyvuščuju za rešetkoj, i *zagovoril*: [...]
in moon.ACC swimming.ACC behind bars and ZA.spoke.PFV
'He was silent for a while, confused, watching the moon swimming behind the bars, and said: ...' (Russian)

b. Chvíli zaraženě *mlčel*, *sledoval* plující
while.ACC confused.ADV was.silent.IPFV followed.IPFV swimming.ACC
měsíc za mříží, a pak se *zeptal*: [...]
moon.ACC behind bars and then REFL inquired.PFV
'He was silent for a while, confused, followed the moon swimming behind the bars, and then inquired: ...' (Czech)

[19] As pointed out by Petr Biskup (p.c.), some examples are mentioned in the literature that could be used to argue for the existence of ingressive *za-* in Czech, such as *zelenat se – za-zelenat se* 'be green – suddenly become green' (Nádeníček 2011; translated from German). There are several reasons for arguing against the view that these are truly ingressive prefixes, rather than taking them to be homophonous. First, the meaning expressed with ingressives is one of a process evolving, whereas Czech *za*-verbs of the type above express inchoativity (a sudden onset) (see also Isačenko 1962 for a discussion of ingressivity vs. inchoativity). Second, truly ingressive *za*-verbs in Russian do not derive secondary IPFVs. However, there are SIs of Czech inchoative *za*-verbs (e.g. *za-zelená-va-t* 'to start to become green'); similar inchoative aspectual pairs are found in Russian (e.g. *za-bole(-va)-t'* 'become sick'). Third, in parallel corpora there is usually no Czech *za*-correspondence to Russian ingressive *za-* (see, e.g. Petruxina 2000, Gehrke 2002). I therefore maintain that there is no ingressive *za-* in Czech (see also Gehrke 2022 for further discussion).

In the Russian example, the last event in the chain is described by a PFV verb of saying, containing ingressive *za-*. The Czech translator did not opt for a direct translation but chose a different verbal predicate, 'inquire'. The example illustrates further systematic differences between Czech and Russian in the context of chains of events, as argued by Gehrke (2022): Whereas Russian uses finite PFV forms for a SOE reading, Czech can also employ IPFV forms, in particular in descriptions of states and activities, or of accomplishments of a longer duration. Russian uses PFVs even with activity and state predicates. For example, PFV 'be silent' in (11a) contains the delimitative prefix *po-*, which temporally bounds an otherwise unbounded activity (see Isačenko 1962). Whereas Russian delimitative *po-* is quite productive, Czech *po-* retains its spatial meaning 'a bit' and is usually not found in the same contexts (see, e.g., Gehrke 2002, 2022, Dickey & Hutcheson 2003, Dickey 2011).[20] The flip side to this is that Russian uses non-finite verb forms, in this case the adverbial participle (AP) *vsmatrivajas'* '(lit.) in-watching', to avoid a SOE reading, whereas Czech uses finite verb forms throughout. Thus, SOE or the absence thereof in this language is, again, only contextually induced and tied to neither (I)PFV nor (non-)finite verb forms.[21]

As a final illustration of systematic differences between Czech and Russian in the context of the description of chains of single events, let us look at a Czech original from Bohumil Hrabal's short story *Jetel růžák* and its Russian translation in (12) (discussed in Gehrke 2022: 21f.) (see also Gehrke 2002).

(12) a. Když *přišlo* pozdní jaro, když *bylo* léto,
when came.PFV late.NOM spring.NOM when was.IPFV summer.NOM
když se *setmělo* a *byla* sobota, *přešel*
when REFL got.dark.PFV and was.IPFV Saturday.NOM across.went.PFV
jsem osvětlený most, pak *zahnul* k mlýnu a
AUX.1SG illuminated.ACC bridge.ACC then off.bent.PFV to mill.DAT and
podle Staré rybárny jsem *kráčel* kolem
past old.ACC fish.restaurant.ADJ.ACC AUX.1SG strolled.IPFV around

[20] Some examples of purely temporally delimiting *po-* are also discussed in the literature on Czech (e.g. *po-přemýšlel* 'thought.PFV (for a while)' and *po-hovořil* 'chatted.PFV (for a while)' in Součková 2004; see also Biskup 2025) but they are far less frequent than in Russian. In the corpus data discussed in Gehrke (2002), for instance, there was not a single instance of delimitative *po-* in Czech, but an abundance of Russian delimitative *po*-verbs.

[21] Throughout the chapter, I call *l*-participles, the only past tense form in both languages, finite verb forms, contrasting them with non-finite verb forms of the adverbial (or other) participle type, discussed in this section. Strictly speaking, Czech *l*-participles are non-finite forms in a periphrastic verb form in combination with the auxiliary 'be'. Finiteness and its role for the Russian, as opposed to the Czech aspect system, is further discussed in Gehrke (to appear).

plotu farní zahrady.
fence parish.GEN yard.GEN
'When late spring arrived, when it was summer, when it got dark and it was Saturday, I crossed the illuminated bridge, then turned to the mill and past the Old Fisherman and strolled along the fence of the churchyard.' (Czech)

b. Kogda vesnja *približalas'* k koncu, kogda *bylo* uže
when spring approached.SI to end.DAT, when was.IPFV already
počti leto, odnaždy v subbotnie sumerki ja
almost summer.NOM once in Saturday.ADJ.PL twilights I
perešel osveščennyj most, a potom *svernul*
across.went.PFV illuminated.ACC bridge.ACC and then off.bent.PFV
k mel'nice i *zašagal* mimo starogo 'Rybnogo podvor'ja'
to mill.DAT and ZA.strolled.PFV past old fish.ADJ inn
vdol' ogrady cerkovnogo sada.
along fence church.ADJ.GEN yard.GEN
'When spring was coming to its end, when it was already almost summer, one Saturday evening I crossed the illuminated bridge, and then turned to the mill and started strolling past the old Fisherman's Inn along the fence of the churchyard.' (Russian)

The first four finite verb forms in the Czech original example describe backgrounded events that set the scene for the following passage, which contains a chain of three events that temporally follow one another ('cross the bridge', 'turn to the mill', 'stroll past the inn along the fence'). The scene setting is done in Czech alternating between PFV achievement predicates ('arrive', 'get dark') and IPFV states ('be'). In the Russian translation, on the other hand, only two finite verb forms are used, for the first two events setting the scene, and both appear in the IPFV ('approach.SI', 'be.IPFV'). The other two scene-setting events ('get dark', 'be Saturday') are translated non-verbally (lit. 'in Saturday twilight'), and they are backgrounded to the chain of the three following events, the start of which is explicitly marked in Russian (by *odnaždy* 'once'), but not in Czech. Both languages use PFV accomplishment predicates for the description of the first two events in this chain, but the initial temporal bound of the third event, which temporally follows the second one and is described by the activity predicate 'stroll', is marked explicitly only in Russian, by ingressive *za-*, but remains to be deduced from the context in Czech.

In sum, aspect use in Czech chains of single events is largely governed by the types of predicates employed (IPFVs for states and activities, primarily PFVs for accomplishments and achievements) and also whether the narrator wants to focus on the duration of a process, in which case the IPFV also appears with accomplishments. In contrast, in Russian a SOE interpretation (with single events) requires a PFV finite verb form, independently of the verb class. Let us then turn to iterative and habitual contexts.

3.2 Iterativity, habituality

The literature on Russian aspect opposes the description of single events (*edi-ničnost'*) with *kratnost'*, which I translate as REPEATABILITY.[22] The consensus in the literature on Russian is that repeatability requires the IPFV (e.g. Padučeva 1996, Zaliznjak & Šmelev 1997), whereas this is not the case in Czech (e.g. Eckert 1984, Stunová 1993, Dickey 2000, Kresin 2000, Gehrke 2002, 2022, Dübbers 2015, Fortuin & Kamphuis 2015). From a theoretical point of view, the use of IPFVs in such contexts has been captured by treating the Russian IPFV in these cases as quantifying over sets of plural events, as in Kagan (2008), Altshuler (2012, 2014) (the latter following Ferreira 2005).

Russian can use the PFV in iterative contexts when the number of repetitions is known and the stretch of repetitions is presented as one whole. This PFV use has been labeled the SUMMATION (*summarnoe*) reading and is illustrated in (13) (from Zaliznjak & Šmelev 1997: 19; my glosses and translation).

(13) Ona tri raza postučala v dver'.
she three times knocked.PFV in door.ACC
'She knocked on the door three times.' (Russian)

Furthermore, in iterative and habitual contexts, Czech productively uses the suffix *-va* to derive FREQUENTATIVES from all kinds of simple IPFV verbs (14a), as well as from SIs (14b) (example adapted from Biskup 2023) (see also Kopečný 1962, Petr 1986, Filip & Carlson 1997, Filip 2026 [this volume]).[23]

(14) a. IPFV *mít* 'have' > IPFV *mí-va-t* 'have.FREQ' (Czech)
b. SI *vy-pis-ova-t* 'excerpt' > *vy-pis-ová-va-t* 'excerpt.FREQ'

[22] The Russian terms are abstract nominalisations related to the adjectives for 'single' and 'multiple', respectively.

[23] I use Petr's (1986) term for frequentatives, but there is no established term; see Filip (2026 [this volume]) for recent discussion.

The derivation of frequentatives is not productive anymore in Russian, apart from a few remaining lexical items (e.g. *byvat'* 'be.freq'), and it is not possible from SIs. Some Czech researchers even consider frequentatives to represent a third category of aspect, in addition to IPFV and PFV (see discussion in Kopečný 1962). Furthermore, the frequentative suffix *-va* is commonly treated as homophonous to one of the imperfectivising suffixes Czech employs in SIs, and the fact that it attaches to a verb form that is already IPFV is taken as an argument in favour of the homophony analysis. I will stay agnostic as to whether synchronically we are dealing with homophony or with the same suffix. Diachronically, both the Czech and the Russian imperfectivising suffix derives from the frequentative suffix, and this is the only remaining productive SI suffix in Russian.

Let me leave strictly iterative contexts aside and concentrate on habitual ones. The descriptive literature sometimes distinguishes a micro-level/event (for each repetition) from a macro-level/event (see, e.g., Eckert 1984, Stunová 1993), and in the following I employ these terms descriptively. For example, while at the micro-level an event can be bounded (e.g. because it appears in a habitual chain of events), we can have non-boundedness at the macro-level, because we are dealing with a habitual discourse. This tension, again, potentially leads to variation in aspect use, namely by using the PFV to signal boundedness at the micro-level, but the IPFV to signal non-boundedness at the macro-level. In habitual contexts, we find both aspects in Czech, arguably conditioned by the same considerations as with single events, whereas Russian almost exclusively uses the IPFV; the Russian IPFV is quasi-obligatory in both present and past tense habitual contexts, with the only exception of the stylistically marked so-called vivid-exemplifying use of the PFV, which I will come back to in the discussion of (16).

For illustration, let us look at the Russian example in (15), from Mixail Bulgakov's *Rokovye jajca*, and its Czech translation (adapted from Gehrke 2002: 87).

(15) a. Mnogie iz 30 tysjač mexaničeskix ėkipažej, *begavšie*
many.nom of 30 thousand mechanical wagons run.ipfv.indet.pap
v 28-m godu po Moskve, *proskakivali* po ulice
in 28th year along Moscow through.jumped.si along street
Gercena, *šurša* po gladkim torcam, i čerez
Gercen.gen rustle.ipfv.ap over smooth pavement and through
každuju minutu s gulom i skrežetom *skatyvalsja* s
every minute with roar and crunch down.rolled.si.refl from
Gercena k Moxovoj tramvaj 16, 22, 48 ili 53-go maršruta.
Gercen to Moxovaja tram.nom 16, 22, 48 or 53th.gen line.gen
'Many of the 30.000 mechanical wagons, running in Moscow in 1928,

sped through Gercen street, rustling over the smooth pavement, and every minute Tram lines 16, 22, 48 or 53 rolled down from Gercen street to Moxovaja street, roaring and crunching.' (Russian)

b. Mnohé z třiceti tisíc drožek, které v osmadvacátém
many.NOM out.of thirty thousand carriages which in 28th
jezdily po Moskvě, *proklouzly* Gercenovou ulicí
drove.INDET.IPFV along Moscow through.slid.PFV Gercen.ADJ street
a *zasvištěly* na hladkém dřevěném dláždění; každou
and swished.PFV on smooth wooden pavement every.ACC
minutu se s řinkotem a skřípěním *přehnala* od
minute.ACC REFL with rattle and crunching past.chased.PFV from
Gercenovy ulice k Mechové tramvaj číslo 16, 22, 48 nebo 53.
Gercen.ADJ street to Mechová tram.NOM number 16, 22, 48 or 53
'Many of the 30.000 carriages that drove in Moscow in 1928 slid through Gercen street and swished on the smooth pavement. Every minute Tram no. 16, 22, 48 or 53 chased by from Gercen street to Mechová street, rattling and crunching.' (Czech)

This passage describes the regular public transport in Moscow in the late 1920s: wagons quickly move along Gercen Street, ass they rustle over the smooth pavement, and every minute a tramline passes through. Since the passage is habitual, Russian has to use the IPFV, whereas we find the PFV for the Czech descriptions of all foregrounded events; even the backgrounded event that is described by a non-finite adverbial participle in Russian ('rustling') is referred to with a Czech finite PFV form in a main clause ('swished'). Furthermore note that all Russian IPFVs are SIs. Thus, we get the impression, that the morphology is used to both render the quickness and "completion" of each micro-event (by a prefix) as well as the habituality and thus unboundedness of the macro-event (by the imperfectivising suffix). In Czech, on the other hand, habituality is deducible almost exclusively from the context, and the only verb form that makes it obvious that we are dealing with a habitual passage is the indeterminate IPFV motion verb *jezdily* in the beginning of this passage (I will come back to this below).

There is one exception to the rule that Russian requires IPFVs in habitual contexts, namely the VIVID-EXEMPLIFYING use of PFV present tense forms. Such present tense forms occur independently of whether the habitual passage is in the past or present, so these present tense forms do not necessarily express a present tense meaning. For illustration let us look at the Czech example in (16), from Milan Kundera's *Žert*, and its Russian translation, discussed in Gehrke (2022: 29).

(16) a. […] v poledne jsme *neměli* čas ani poobědvat,
in noon AUX.1PL NEG.had.IPFV time even have.lunch.PFV.INF
snědli jsme na sekretariátě ČSM dvě suché housky a pak
ate.PFV AUX.1PL on secretariat ČSM two dry rolls and then
jsme se zase třeba celý den *neviděli*,
AUX.1PL REFL again maybe whole.ACC day.ACC NEG.saw.IPFV
čekávala jsem na Pavla kolem půlnoci […]
waited.IPFV.FREQ.F AUX.1SG on Pavel around midnight
'At noon, we did not even have time to have lunch, we ate two dry rolls at the secretariat of the ČSM [Czechoslovak Youth Union] and then again maybe did not see each other the whole day, I used to wait for Pavel around midnight.' (Czech)

b. […] v polden' nam ne *xvatalo* vremeni daže
in noon us.DAT not sufficed.IPFV time.GEN even
poobedat', *s"edim*, *byvalo*, na
have.lunch.PFV.INF eat.PFV.1PL.PRS was.IPFV.FREQ.3SG.N on
sekretariate dve suxie bulki, a potom snova počti celyj
secretariat two dry rolls and then again almost whole.ACC
den' ne *vidimsja*, *ždala* ja Pavla *obyčno*
day.ACC not see.IPFV.1PL.PRS.REFL waited.IPFV.F I Pavel.ACC usually
k polunoči […]
to midnight
'At noon we did not even have time to have lunch, we used to eat two dry rolls at the secretariat and then not see each other almost the whole day, I usually waited for Pavel until midnight.'
(Russian)

This passage describes the daily sequential routine that the female narrator had with her partner Pavel. It starts with a negated stative expression ('not have time') that explains the shortness of the first event in the sequence (a quick lunch), followed by not seeing each other during the day, and then her waiting for Pavel around/until midnight. In the Czech original, all four verb forms are finite past tense forms; the one that describes the reason ('not have time') is in the IPFV because it describes a state. The Russian translator chose a different lexical item, but also an IPFV. In Czech, the first event of the habitual chain ('eat two dry rolls') is described by a PFV verb form, followed by an IPFV negated stative description ('not see each other'). The PFV verb form appears because at the micro-level two rolls were finished, rather quickly, and contextually it does not make sense to focus on the duration. Up until here the Czech verb forms by themselves do no indicate that the passage is habitual, and apart from the adverbs 'again maybe' these three verb forms could also be used in the description of single events. Only the last form, frequentative *čeká-va-la*, explicitly signals habituality.

Things are different in Russian when the chain starts: the translator switched to two present tense forms in the vivid-exemplifying use (PFV 'eat', IPFV 'see'), and this tense switch is accompanied by the addition of the habituality marker *byvalo* 'it used to be', which is absent from the Czech original. It is commonly assumed that the switch to the vivid-exemplifying use has to be accompanied by expressions like *byvalo* (see, e.g. Zaliznjak & Šmelev 1997). The vivid-exemplifying use of the PFV present is obviously a stylistic device; the translator could also have stayed in the past tense, in which case, however, the PFV would not have been possible. For the last verb form the translator switched back to the past tense and translated the frequentative Czech verb for 'waited' with a simple IPFV verb 'waited', as Russian cannot derive a frequentative from this kind of verb (**žda-va-la*); however, they added the adverb 'usually' to render the habitual nature of the Czech verb form, even though there is no such adverb in the Czech original.

Let us see what these examples illustrate in general, as argued in more detail in Gehrke (2022). In Russian, the finite verb forms are explicitly marked for two things: a) event sequencing and/or event completion (on a par with what would be the case in the description of single events, as we saw in §3.1), and b) habituality, by additionally imperfectivising the verb forms, which is the main difference between chains of single vs. habitual events in this language. In Czech, on the other hand, only few verb forms make aspect use in habitual contexts different from single events: the indeterminate motion verb *jezdily* in (15b) and the frequentative verb *čekávala* in (16a). Both types of verbs are common means in Czech to signal habituality (see Eckert 1991 for indeterminate motion verbs and Filip & Carlson 1997 for frequentatives). These verb forms appear once in a passage, to mark it as habitual; other verb forms in the same passage display the same kind of aspect use as with single events: "completed" and/or quick events are described by PFVs, stative events or events of some duration are described by IPFVs. Thus, in habitual contexts, Czech uses aspect more or less the same as in the description of single events (see Eckert 1984 for the same conclusion).[24]

[24]For example, only a small fraction (about 7%) of the past tense forms in Czech habitual contexts in Gehrke's (2002) corpus analysis explicitly mark habituality: Out of about 500 past tense forms, these were 16 frequentatives, 3 indeterminate verbs of motion, 3 SI verbs of motion (which sometimes also mark habituality, according to Eckert 1991), 9 SIs that (at least formally) with a suffix that is formally identical to frequentatives (*-va*), and 3 SIs with other suffixes (all exclusively in the Czech translation of Sergej Dovlatov's *Zapiski nadziratelja*).

This of course does not mean that Czech does not use the IPFV in habitual contexts. Rather, habituality is not directly marked on the verb form in most of the cases, apart from a few specialised IPFVs once in a longer passage. In Gehrke (2022), it is therefore argued that the use of the IPFV in Czech habitual contexts can be explained by the same reasoning that explains its use in the description of single events: for atelic states and activities, as well as for accomplishments with a focus on their duration. Furthermore, nothing is said about the verb forms in Czech being the only options, it might very well be that some PFVs could be replaced by IPFVs, this would have to be checked on a case-by-case basis. It is just that as a generalisation on the data taken into account, the conclusion is that whenever there is no need to use an IPFV (for atelic events or for events of a certain duration) Czech simply uses the PFV; this is also in line with the intuitions reported by Czech native speakers that investigated differences between Czech and Russian, such as Eckert (1984) and Stunová (1993). And this is exactly where Russian differs: in Russian one has to use the IPFV, the PFV would be ungrammatical in habitual contexts. Let us then turn to contexts that involve the historical (or narrative) present.

3.3 Historical present

The historical present is a stylistic device in narrative texts to describe events using present tense forms (even if the events happened in the past), to make them appear more vividly, as unfolding before one's eyes (see also Anand & Toosarvandani 2019 for recent formal discussion). From a theoretical and cross-linguistic point of view, it is commonly assumed that with a regular present tense semantics, the R(eference Time) coincides with the S(peech Time) (NOW). Since NOW is a point and not an interval, but since the truth-conditions for "completed" events require intervals to be evaluated, there is a common idea that ("true") present tense and PFV are semantically incompatible. In both Czech and Russian, for example, present tense morphology on PFV verbs is (often) interpreted as reference to events in the future, rather than in the present. Hence, one could expect that in the historical present, which also seems to describe events as if evolving before our eyes, only the IPFV is used. On the other hand, if the present tense in the historical present is semantically not a "true" present tense, we would not have this expectation. While I refrain from giving a semantic account of the historical present, we seem to observe two different situations in Czech and Russian. In particular, Russian exclusively uses the IPFV in historical present contexts (except for the vivid-exemplifying use of the PFV; recall (16)), whereas we find both aspects in Czech (Křížková 1955, Bondarko 1959, Petruxina 1983, Stunová 1993, Dickey 2000, Fortuin & Kamphuis 2015).

Let me illustrate with the example in (17) from Stunová (1993: 187; my English glosses and translations, context abbreviated), from Karel Čapek's novel *Krakatit*.

(17) Context: 'Indeed, nothing compares to the beauty of a summer morning, but Prokop looks down to the ground, smiles as far as he is able to do so, and wanders through small gates to the river.'

a. Tam *objeví* – ale u druhého břehu – poupata
there appear.PFV.PRS.3PL but at other bank buds.NOM
leknínů; tu zhrdaje vším nebezpečím se
water.lilies.GEN here disregarding all danger REFL
svlékne, *vrhne* se do hustého slizu zátoky, *pořeže*
undresses.PFV throws.PFV REFL in thick slime bay.GEN cuts.PFV
si nohy o nějakou zákeřnou osřici a *vrací* se s
REFL legs about some insidious sedge and returns.IPFV REFL with
náručí leknínů.
armful water.lilies.GEN
'There appear – but on the other river bank – water lily buds; disregarding all dangers, he undresses, throws himself into the thick slime of the bay, cuts his legs on some insidious sedge and returns with an armful of water lilies.' (Czech)

b. I tam, tol'ko u protivopoložnogo berega, *obnaruživaet* butony
and there only at opposite bank finds.SI buds.ACC
kuvšinok; prenebregaja vsemi opasnostjami, on *snimaet*
water.lilies.GEN disregard.SI.AP all dangers he off.takes.SI
plat'je i *brosaetsja* v gustuju sliz' zavodi, *ranit* nogi
clothes and throws.SI.REFL in thick slime bay.GEN injures.IPFV legs
o kakie-to kovarnye ostrye list'ja, no *vozvraščaetsja* s
about some insidious sharp leaves but returns.SI with
oxapkoj cvetov.
armful flowers.GEN
'And there, just on the opposite river bank, he finds water lily buds; disregarding all dangers, he takes off his clothes and throws himself into the thick slime of the bay, injures his legs on some insidious sharp leaves, but returns with an armful of flowers.' (Russian)

This passage begins with setting the scene of the protagonist Prokop walking to the river, the Czech and Russian wordings I left out; both languages use the

IPFV present here, arguably for different reasons though: In Czech we are dealing with states, activities or the dwelling on the process of an accomplishment, in Russian the historical present requires IPFVs. This scene-setting is followed by a sequence of events: water lilies becoming visible, Prokop undressing, throwing himself into the river, cutting or injuring himself, and returning. In Russian, we find only IPFV present tense forms, but apart from the return from the river, all Czech present tense forms are PFV. Stunová states that discovering flowers is a momentaneous event, so an IPFV would not be suitable; describing the undressing with an IPFV, she argues, would suggest that it took a long time, IPFV in-throwing would have the effect of an eye-witness report, and IPFV cutting would suggest intentionality, all not desired effects in this context. She furthermore states that the event of returning from the river is described by an IPFV in Czech because we get the impression that Prokop is on his way, rather than arriving (the same is stated for the first event of wandering to the river).

In sum, aspect use in Czech historical present contexts does not differ much from other contexts we have discussed so far: IPFVs appear with states, activities and accomplishments of a longer duration, PFVs appear in the other cases. In contrast, Russian has to use the IPFV throughout, and so the overall picture is quite similar to aspect use in habitual contexts, as addressed in the previous section. Finally, let us turn to factual contexts.

3.4 Factual contexts

As mentioned in Section 2.2, the existential IPFV appears in contexts in which the precise time at which the event occurred is not relevant or not known, or it might have happened more than once. Padučeva (1996), for instance, argues for Russian that potential repeatability (what she calls *kratnost'*, as outlined in the beginning of Section 3.2) is a requirement for the existential IPFV. Recall from Section 2.2 that with scopally non-specific temporal expressions like *kogda-nibud'* 'ever' (see also Section 2.3), Russian has to use the IPFV. This is different in Czech, as can be seen in (18) (adapted from Klimek-Jankowska 2022: 10) (see also discussion in Dickey 2000, Gehrke 2002, 2022, Mueller-Reichau 2018b).[25,26]

[25] Some of the examples from Klimek-Jankowska (2022) discussed in this section are adjusted qua glosses, translations, and scientific transliterations for Russian.

[26] A Czech reviewer observes that he finds the "universal *-koli(v)*-words" bad with the PFV *ztratil* in (18) and would use the existential *někdy* instead or adjust the sentence. I keep the Czech judgments reported in Klimek-Jankowska (2022) here, but I will come back to this point in Section 6.

(18) a. {Ztratil / $^{??}$ztrácel} jsi kdykoliv klíče?
lost.PFV lost.IPFV AUX.2SG ever keys
'Have you ever lost keys?' (Czech)

b. Ty kogda-libo {*poterjal / terjal} ključi?
you ever lost.PFV lost.IPFV keys
'Have you ever lost keys?' (Russian)

In this example, Czech has a preference for the PFV aspect, and one could argue that this is so because losing keys is a punctual event that does not involve a process or a duration (or even intentionality); Russian, on the other hand, has to use the IPFV because of the non-specific temporal adverb *kogda-libo* 'ever'.

Dickey (2000), who does not differentiate between different subtypes of factual IPFVs, argues that factual IPFVs are incompatible with achievement predicates in Czech but not in Russian, and the examples he provides in (19) (from Dickey 2000: 99 & 101; my glosses and translations) illustrate the existential IPFV.[27]

(19) a. Jako dítě jsem jednou {spadl / *padal} z toho stromu.
as child AUX.1SG once fell.PFV fell.IPFV from this tree
'As a child, I once fell from this tree.' (Czech)

b. Ja pomnju, v detstve odnaždy ja {upal / padal} s ėtogo dereva.
I remember in childhood once I fell.PFV fell.IPFV from this tree
'I remember, in my childhood I once fell from this tree.' (Russian)

Mueller-Reichau (2018b), in turn, argues that it is not about achievements, but about necessarily unique events; he provides examples parallel to those in (19), only with an accomplishment VP, 'fell our single tree', in which Czech has to use the PFV as well.

Klimek-Jankowska (2022) obtained data from Czech, Polish, and Russian, using questionnaires in which the informants had to fill in the missing verbs in both existential and presuppositional factual contexts. Existential contexts are further divided into neutral and resultative ones, presuppositional ones into strongly and weakly resultative ones and also whether focus is on the initiator or on the result state of the event. Independently of the context, all presuppositional examples

[27] A Czech reviewer notes that (19a) should be modified to reflect the information structure: Since *toho* 'this.GEN' brings about givenness, *z toho stromu* 'from this tree' should be placed before *jednou* 'once'. This does not affect the judgments about (I)PFVs though.

involving verbs of creation (e.g. 'cook', 'build') were treated as strongly resultative, because reference to created objects is argued to involve the presupposition of a result state; in contrast, weakly resultatives are assumed to involve other "accomplishment" predicates (e.g., 'iron' and 'wash').

(20) (adapted from Klimek-Jankowska 2022: 16; relevant verb forms italicised) is an example that is argued to involve a resultative existential.[28]

(20) a. Vidím, že kytka na římse zvadla. Určitě
see.IPFV.1SG that flowers on window.sill wilted.PFV surely
jsi je dnes *zalíval*?
are.IPFV.2SG them today watered.SI
'I see that the flowers on the window sill have wilted. Are you sure that you watered them today?' (Czech)

b. Ja vižu, čto cvety na podokonnike zasoxli. Ty uveren,
I see.IPFV.1SG that flowers on window.sill wilted.PFV you certain
čto *polival* ix segodnja?
that watered.SI them today
'I see that the flowers on the window sill have wilted. Are you certain that you watered them today?' (Russian)

Both languages can use the IPFV in this context, and Klimek-Jankowska takes this as an argument that both languages make use of existential IPFVs. On the other hand, it is not obvious to me that this example necessarily involves resultativity; it could also be a question about whether the process of watering took place, in which case the use of IPFVs in Czech could be motivated by the process reading of IPFVs. I will come back to this in Section 5.5.

The example in (21) (adapted from Klimek-Jankowska 2022: 17) is argued to illustrate strongly resultative presuppositional IPFVs, with focus on the result.

(21) a. Ta placka je skvělá, Maryšo. Z jakých ingrediencí
this.F cake.F is delicious Maryša.VOC out.of which ingredients
jsi ji *pekla*?
AUX.2SG her baked.IPFV
'Your cake is delicious, Maryša, What ingredients did you bake it with?' (Czech)

[28]A Czech reviewer notes that there is a discrepancy in (20a) between singular *kytka* 'flower' and the plural glosses 'flowers'. I kept the example as it is in the cited paper.

b. Kakoj vkusnyj pirog, Marysia! Iz čego ty jego *pekla*?
how delicious cake Marysia out.of what you it baked.IPFV
'What a delicious cake, Marysia! What did you bake it with?'
(Russian)

This example involves a presuppositional IPFV, because the baking event is presupposed by the object that came into existence through this event, the cake, which the previous context talked about. Klimek-Jankowska argues that focus is on the result, because the question is concerned with properties of the result of the baking event (the cake). Alternatively, it could be argued that a question about the ingredients of a baking event is more concerned with the process of baking, rather than with the result, so that there is no focus on the result.

Finally, (22) (adapted from Klimek-Jankowska 2022: 19) is presented as an example for a weak resultative presuppositional IPFV, with focus on the initiator.

(22) a. Vidím, že tvé kolo je nakonec funkční. Právě také
see.IPFV.1SG that your bike is finally working just also
hledám odborníky. Můžeš mi říct, kdo ti ho
look.for.IPFV.1SG specialists can.2SG me tell who you.DAT it.ACC
opravoval?
repaired.SI
'I see that your bike is finally working. I am also looking for specialists. Can you tell me who repaired it for you?' (Czech)

b. Ja vižu, čto tebe nakonec-to počinili velosiped.
I see.IPFV.1SG that you.DAT finally repaired.PFV.3PL bike
Ja tože išču mastera. Podskaži, kto tebe jego
I also look.for.IPFV.1SG specialist tell.PFV.IMP.SG who you.DAT it.ACC
činil?
repaired.IPFV
'I see that they finally repaired your bike. I am also looking for a specialist. Tell me, who repaired it for you?' (Russian)

Again, both languages can employ the IPFV, and in Russian the presupposed event is even referred to by the same PFV lexeme in the first sentence of the context (*počinili* 'repaired.PFV'), whereas in Czech the repairing event can be accommodated in a context that is preceded by a question about a bike repair specialist. Since we are not dealing with a verb of creation but with a verb that describes an event (of repairing) that the object in question (the bike) can in principle undergo multiple times, Klimek-Jankowska labels this a weakly resultative context; it is furthermore argued to involve focus on the initiator, because the question is about the person that initiated the event.

Based on the data that she obtained, Klimek-Jankowska (2022) provides a statistical analysis that yields the following significant results. In all factual contexts, Russian employs more IPFVs than Polish and Czech. In neutral existential contexts all three languages use significantly more IPFVs than in the other contexts, whereas in resultative existential contexts more PFVs were used than in neutral ones. Furthermore, only Czech and Polish can also use the PFV in neutral existential contexts with a temporal adverb 'ever', as we already saw in (18). There were no statistically significant differences between weakly and strongly resultative presuppositional contexts, nor between focus on initiator or result, for either of the languages. I will therefore not come back to these details in the discussion of Klimek-Jankowska's (2022) account in Section 5.4.

In the three examples in (20)–(22), both Czech and Russian use the IPFV, so Klimek-Jankowska (2022) assumes that both languages use the IPFV here precisely because we are dealing with (different types of) factual contexts. On the other hand, all her examples involve verbs of creation (a subtype of incremental theme verbs) or predicates that she groups under the label "accomplishment". However, it is not clear that incremental theme verbs are accomplishments to begin with and not just activities (see, e.g., Kennedy 2012), or what Rappaport Hovav & Levin (2010) would label manner verbs (in contrast to result verbs), and also some of the other verbal predicates she employed could be argued to involve activities, rather than accomplishments.[29] So it might very well be that Czech uses the IPFV in some of these examples for the same reason it uses IPFVs in other contexts: to focus on the process of a durative event. I will come back to this point in Section 5.5. Let us then turn to a new empirical domain, for which cross-Slavic differences in aspect use have not been well-described yet: passives.

4 Passives

This section addresses differences in aspect use between Czech and Russian with different types of passives. Passives are a typical syntactic topic, with many subtopics which are orthogonal to aspect use, so in this chapter I concentrate only on those subtopics that might stand in direct relation to the semantics of

[29]The translations of the complete list of the verbs used in Czech and Russian are as follows: 'eat', 'mow', 'renovate', 'drill', 'clean', 'water', 'take out', 'bring', 'milk', 'vacuum', 'build', 'sculpt', 'write', 'paint', 'sew', 'embroider', 'bake', 'sign', 'comb', 'wash', 'cook', 'iron', 'cut'.

aspect.[30] In this context, one might even wonder why aspect would play a role at all, given the standard assumption that passives and actives do not differ truth-conditionally. Thus, one might expect aspect to function the same in passives as in actives. It turns out that this is true for Czech, but not for Russian. In the following, I first address general cross-linguistic assumptions about passives to then turn to a comparison between Czech and Russian.

4.1 Different types of passives, cross-linguistically

From a morphological point of view, different constructions have been labeled passives or taken to express a verbal passive meaning/syntax (on which see below). For Slavic languages, two morphological passive types are relevant. The first type is the PARTICIPIAL PASSIVE, which is a periphrastic verb form made up of an auxiliary in combination with a past passive participle (PPP), as it is found in, for instance, Germanic and Romance languages. The second type is the REFLEXIVE PASSIVE, not found in Germanic languages, but in, e.g., Romance: a reflexively marked verb form that is interpreted on a par with a passive construction (among other readings reflexive forms can have, such as reflexive, reciprocal, inchoative/anticausative, impersonal, middle; see, e.g., Babby 2009, Medová 2009, Fehrmann et al. 2010 for discussion of Slavic). Examples from Czech for both types of passives are given in (23) (from Karlík 2017, my glosses and translation).

(23) a. Píše se stížnost.
writes.IPFV REFL complaint.NOM

b. Je psána stížnost.
is write.IPFV.PPP complaint

'A complaint is (being) written.' (Czech)

Syntactically and semantically it is common for participial passives to distinguish between verbal and adjectival passives, or between eventive and stative passives.[31] Within the adjectival or stative type, a further distinction is assumed to

[30]As we will see below, a passive reading in Slavic languages is found either with past passive participles (PPPs) or with reflexively marked verb forms. The literature on these is usually not (primarily) concerned with aspect. For further information on passives, not related to aspect, see Veselovská & Karlík (2004), Karlík (2017, 2020), Taraldsen Medová & Wiland (2017), Caha & Taraldsen Medová (2020) for Czech PPPs, Schoorlemmer (1995), Paslawska & von Stechow (2003), Borik (2013, 2014) for Russian PPPs, Babby (2009) for Russian more generally, Medová (2009) for Czech reflexives, and Fehrmann et al. (2010) for Slavic reflexives.

[31]In many works these two dichotomies are used synonymously, even though they are not straightforwardly synonymous (there are stative verbs and possibly also eventive adjectives).

hold between target state and resultant state participles (following Kratzer 2000), based for example on modifiability by 'still', which is only possible with the former. Kratzer argues that target state participles are derived from category-neutral stems that make available both an event and a target state argument (~ accomplishments and achievements), either by lexical or phrasal stativisation. Resultant state participles, in turn, are proposed to additionally involve a perfect operator (Kratzer labels this the "job-done" reading) and to necessarily be the result of phrasal stativisation. Another influential distinction among adjectival/stative PPPs is made by Embick (2004) who differentiates between resultatives (~ Kratzer's phrasal target and resultant state participles) and statives (~ Kratzer's lexical target state participles).

In this chapter, I assume that a verbal passive is a canonical passive that involves an operation on the argument structure of a verbal predicate, suppressing the external argument, which optionally surfaces in a 'by'-phrase, and promoting the internal argument to subject position. The input requirement to a canonical verbal passive are therefore verbs with external and internal arguments. In contrast, I take an adjectival passive to be a copular construction in which a copula combines with an adjectival or adjectivised PPP. There is crosslinguistic variation as to whether adjectival passives allow for all kinds of 'by'-phrases or whether there are restrictions on their availability (e.g. German, English vs. Greek; see Alexiadou et al. 2014). For example, in German, in which a verbal PPP combines with the auxiliary *werden* 'become' and an adjectival PPP with the copula *sein* 'be', the verbal passive allows for all kinds of 'by'-phrases, but the adjectival passive can only appear with non-referential 'by'-phrases that access an event kind but not an event token (24) (following Gehrke 2015).

(24) (German)
  a. Das Bild wurde {von Marta / von einem Kind} gemalt.
     the picture became by Marta by a child paint.PPP
     'The picture was/has been painted by {Marta / a child}.' VERBAL PPP
  b. Das Bild war {#von Marta / von einem Kind} gemalt.
     the picture was by Marta by a child paint.PPP
     ~ 'The picture was painted in a childish manner.' ADJECTIVAL PPP

Additional tests to distinguish verbal/eventive from adjectival/stative PPPs involve the possibility of spatiotemporal modification of the underlying event (only

Since adjectival passives involve adjectivisation of a PPP, as argued below, but reflexive verb forms are finite verb forms that cannot be adjectivised, this distinction has, to my knowledge, not been addressed and is probably not relevant for reflexive passives.

with verbal/eventive PPPs), or the compatibility with adjectival morphology (if at all, only with adjectival/stative PPPs). I will come back to some of these tests in the discussion of Czech and Russian PPPs below.

For languages like Czech and Russian it is additionally important that PPPs in predicative position can appear in two forms, i.e. with long or short agreement morphology (in the following: LONG VS. SHORT FORM), e.g. Czech *otevřena/otevřená* / Russian *otkryta/otkrytaja* ‘open(ed).(NOM.)FEM.SG’. (Predicative) long forms are commonly assumed to be adjectival (e.g. Veselovská & Karlík 2004, Borik 2013); I will set them aside here and focus only on predicative short form PPPs.

4.2 Russian passives

Russian PARTICIPIAL PASSIVES (i.e. predicative short form PPPs in combination with ‘be’) have been argued to exclusively be adjectival by Paslawska & von Stechow (2003) (in analogy to their Greek counterparts; cf. Anagnostopoulou 2003), or to be ambiguous between a verbal and adjectival passive reading (Schoorlemmer 1995, Borik 2013, Borik & Gehrke 2018), and by form alone it is impossible to distinguish between the two. Based on the following evidence, I side with the latter view in this chapter: Any type of ‘by’-phrase, which is an NP in instrumental case, is available, and it is possible to locate the underlying event in space and time, e.g. by temporal adverbials. For example in (25) (adapted from Borik 2013: 122), both the underlying event and the (result) state can be targeted by temporal modifiers, whereas with adjectival/stative passives, only the state itself should be available for such modification.

(25) Vorota (byli) otkryty storožem rovno v 6 utra na 2 časa.
gates were open.PFV.PPP watchman.INSTR exactly in 6 morning on 2 hours
‘The/A gate was opened by a/the watchman exactly at 6 in the morning for two hours.’

The auxiliary *byt’* ‘be’ that combines with the PPP appears with past (*byl(a/o/i)*) or future (*budu* etc.) tense marking, or it is zero (phonologically empty) in the present tense.[32] Unlike the copula *byt’*, which can appear with frequentative marking (e.g. *by-va-t’*), the passive auxiliary in Russian does not allow it.

[32]Throughout I gloss passive auxiliaries as forms of ‘be’, whereas I use AUX to gloss the auxiliary that appears in Czech past tense forms (see Section 4.3.1 on the Czech counterparts).

There is a wideheld assumption that Russian PPPs can only be derived from PFV verbs (e.g. Paslawska & von Stechow 2003), but there are counterexamples to this claim. Based on a corpus study, Borik & Gehrke (2018) conclude that compositional IPFV PPPs with a predictable meaning exist, with both verbal and adjectival passive readings, but that they are quantitatively few and there are further restrictions.[33] In particular, in contemporary Russian, IPFV PPPs are only derived from simple but not from secondary IPFVs (26) (from Borik & Gehrke 2018: 54) and they cannot express a process reading (see also Knjazev 2007).

(26) Storož {otkryval / otkryl} vorota.
watchman.NOM opened.SI opened.PFV gates.ACC
'A/The watchman was opening/opened a/the gate.'

a. Vorota byli otkryty storožem.
gates.NOM were open.PFV.PPP watchman.INSTR
'A/The gate was opened by a/the watchman.'

b. * Vorota byli otkryvany storožem.
gates.NOM were open.SI.PPP watchman.INSTR
Intended: 'A/The gate was opened by a/the watchman.'

Borik & Gehrke hypothesise that IPFV PPPs can only have factual readings; this is arguably due to the strictly resultative meaning associated with Russian PPPs more generally (see also Schoorlemmer 1995, who dubs this the "Perfect Effect"). Relevant examples are given in (27) (from Borik & Gehrke 2018: 58 & 60).

(27) a. Bylo pito, bylo edeno, byli slezy prolity.
was drink.IPFV.PPP.3SG.N was eat.IPFV.PPP.3SG.N were tears pour.PFV.PPP.PL
'(Things) were drunk, (things) were eaten, tears were shed.' EXISTENTIAL

b. Pisano ėto bylo Dostoevskim v 1871 godu […]
write.IPFV.PPP.3SG.N that was Dostoevskij.INSTR in 1871 year
'That was written by Dostoevskij in 1871.' PRESUPPOSITIONAL

In a recent corpus study on IPFV PPPs in Russian and Polish, Wiemer et al. (2023) confirm the generalisations concerning the absence of a process reading and of

[33] Similar examples are mentioned in Schoorlemmer (1995), who nevertheless concludes that "there is a finite set of imperfective verbs that occurs in syntactic passives in Russian. This set differs per speaker and reduces to zero for many." (Schoorlemmer 1995: 226).

SI PPPs for Russian. Taking into account also diachronic data, they show that SI PPPs and other IPFV readings cannot be found anymore in Russian texts from the 19th century onwards, so this is a rather recent development.[34]

As for Russian REFLEXIVE PASSIVES, it has been argued by Fehrmann et al. (2010) that 'by'-phrases are generally available with these. However, there is, again, the received view that there is an aspectual restriction, and this time it is commonly assumed that only IPFVs can derive reflexive passives. PFV counterexamples are discussed in the literature (e.g. Schoorlemmer 1995, Fehrmann et al. 2010), but to my knowledge no systematic investigation into this issue has been conducted. The relevant reflexive counterparts to (26), including the availability of a 'by'-phrase, are given in (28) (from Borik & Gehrke 2018: 54).

(28) a. Vorota otkryvalis' storožem.
gates.NOM opened.SI.REFL watchman.INSTR
'The/A gate was (being) opened by a/the watchman.'

b. * Vorota otkrylis' storožem.
gates.NOM opened.PFV.REFL watchman.INSTR
Intended: 'The/A gate was (being) opened by a/the watchman.'

Thus, we can draw the interim conclusion for Russian hat the functional IPFV-PFV distinction is defect with PPPs, and possibly also with reflexive passives.

4.3 Czech passives

To my knowledge, there is nothing in the literature on Czech (verbal) passives that would suggest that the semantic role aspect plays is different than in actives. The literature on Czech reflexive and participial passives has examples with both aspects (e.g. Veselovská & Karlík 2004, Medová 2009, Fehrmann et al. 2010, Karlík 2017, 2020, Caha & Taraldsen Medová 2020). This could be due to the implicit assumption (as also suggested by Denisa Lenertová, Radek Šimík, p.c.) that (I)PFV works exactly the same with passives as it does with actives. This seems to be corroborated by my data investigation, to which I turn shortly. First, however, some general words about Czech passives.

4.3.1 Participial passives

Czech participial passives also combine a form of 'be' with a PPP, so again, by form alone it seems impossible to distinguish a verbal from an adjectival passive.

[34]In contrast, Wiemer et al. (2023) show that contemporary Polish still has SI PPPs and all IPFV readings are possible. We will see in Section 4.4 that the same holds for Czech.

To my knowledge, there is no research on the formal semantics of Czech PPPs or on the types of 'by'-phrases we might find, and the literature is primarily concerned with morphosyntactic issues.

Veselovská & Karlík (2004) propose that there are three derivational options for Czech PPPs: a) lexical derivation of an adjective, b) syntactic derivation of an adjective (in adjectival/stative passives), c) postsyntactic derivation of an adjective at PF (in verbal/eventive passives); arguments for adjectivisation come from the adjectival agreement on the PPP. The first two are argued to always appear in the long form and to be interpreted (at LF) as stative/adjectival (expressed by the feature [STATE]); the latter in turn is assumed to always involve a short form and to be interpreted as eventive/verbal (expressed by the feature [ACTIVITY]). Evidence for the verbal vs. adjectival nature of the relevant PPPs comes from standard diagnostics, such as the availability of 'by'-phrases, spatiotemporal and manner modifiers with verbal/eventive PPPs as opposed to adjectival 'un'-prefixation and the (albeit limited) gradability of stative/adjectival PPPs.

The passive auxiliary is proposed to be inserted in *v* and to be distinct both from the past auxiliary (which is inserted in T) and from copular/existential 'be' (inserted in V). The passive auxiliary (like copular/existential 'be' but unlike the past auxiliary) can combine with frequentative marking, which is argued to be the only functional Aspect marking in Czech ((I)PFV are treated as features on V); this marking can additionally appear on the PPP (29) (adapted from Veselovská & Karlík 2004: 174), but more often not on the PPP alone (30) (adapted from Veselovská & Karlík 2004: 188).[35,36]

(29) a. Já bývám {chválen / chváliván}.
I am.FREQ praise.IPFV.PPP praise.IPFV.FREQ.PPP
'I am being praised (repeatedly).'

b. Já jsem {chválen / chváliván}.
I am praise.IPFV.PPP praise.IPFV.FREQ.PPP
'I am praised (repeatedly).'

[35] With past auxiliaries it is only the *l*-participle that can bear frequentative marking (see op. cited for relevant examples). Recall from Section 4.2 that in Russian the passive auxiliary cannot bear frequentative marking, unlike its Czech counterpart, but like Czech past auxiliaries.

[36] In the examples from Veselovská & Karlík (2004), I keep the translations provided by the authors, but it is not entirely clear whether their use of progressive marking (or absence thereof) is supposed to reflect a difference in the aspectual or passive readings of the Czech examples, or whether the choice is not meaningful at all. For example, the progressive could be intended to either mark an imperfective reading (in any tense), or to mark an eventive/verbal (as opposed to stative/adjectival) passive reading (in the present tense). To avoid any bias caused by a potential English translation, I do not provide any translations in examples from Czech sources, e.g. those taken from Karlík (2017).

(30) a. * Diktát byl psáván každý pátek.
dication.NOM was written.IPFV.FREQ.PPP every Friday
Intended: 'Dictations were being written every Friday.'

b. * Ten test byl opisováván docela pravidelně.
this.NOM test.NOM was copy.SI.FREQ.PPP rather regularly
Intended: 'This test was being copied rather regularly.'

Furthermore, the passive auxiliary can appear in past (*byl(a/o/i/y)*), present (*jsem* etc.), or future tense (*budu* etc.).[37]

Veselovská & Karlík (2004) argue that Aspect is interpretable only on short form PPPs, as in, e.g., (31) (adapted from Veselovská & Karlík 2004: 224), and their discussion quite generally suggests that we are dealing with the typical aspectual readings we also find in the active.

(31) Pět vojaků bylo {(z)raněno / (z)raňováno}.
five soldiers.GEN was.3SG.N wound.PFV.PPP.3SG.N wound.SI.PPP.3SG.N
'Five soldiers were {wounded / repeatedly wounded}.'

In the discussion of PFV short form PPPs in predicative position, Veselovská & Karlík (2004) state that many Czech speakers view this form as archaic (see also Biskup 2019: 96, for "stative"-interpreted short form PPPs). They furthermore argue that there is also variation in the tense options on the auxiliary, and that in general both the auxiliary and the PPP contribute to the overall temporal-aspectual interpretation of the sentence. In particular, they claim that past and future auxiliaries in combination with PFV short form PPPs tend to be interpreted eventively, while the present tense auxiliary is dispreferred with the short form (32) (adapted from Veselovská & Karlík 2004: 225) (a long form is used instead; see op. cited for examples).[38]

(32) Okno {??je / bylo / bude} otevřeno policií.
window.NOM is was will.be open.IPFV.PPP police.INSTR
'The window {is/was/will be} opened by the police.'

Moreover, the PFV short form expresses resultativity, with the result preceding or following the moment of speech, "without inferring anything about the present state or action" (33) (adapted from Veselovská & Karlík 2004: 225).

[37]To be precise, Veselovská & Karlík (2004) argue that the future examples involve the future auxiliary in T in combination with an infinitival zero passive auxiliary in *v*.

[38]We will see shortly that PFV short form PPPs with a present tense auxiliary are discussed in the literature elsewhere, so it is not clear that the claim reported here is shared by everyone.

(33) a. Hala {byla / bude} uklizena špatně.
hall.NOM was will.be clean.PFV.PPP badly
'The hall {was / will be} cleaned badly.'

b. Hala je uklizena špatně.
hall.NOM is clean.PFV.PPP badly
'The hall has (already) been cleaned badly.' (not: 'is being cleaned')

In sum, according to Veselovská & Karlík (2004), all short form PPPs are unambiguously verbal/eventive (and adjectival only postsyntactically, at PF), with PFV PPPs in combination with a present tense auxiliary always being interpreted resultatively. The discussion of short form PPPs in other papers suggests that there might still be an ambiguity between adjectival/stative and verbal/eventive passive readings, at least for PFV PPPs. For example, Caha & Taraldsen Medová (2020) argue for the syntactic three-way syncretism of Czech PPPs in (34).

(34) a. Eventive passive: [Init [Proc Res]]
b. 'Intermediate' passive (R-state): [Proc Res]
c. Purely stative passive (T-state): [Res]

Their account is spelled out in a nanosyntactic framework that incorporates Kratzer's (2000) distinction between resultant-state and target state participles (R-states and T-states in (34)), and Ramchand's (2008b) syntactic decomposition of events into maximally (an) Init(iator Phrase), (a) Proc(ess Phrase), and (a) Res(ult phrase). For example, (35) (adapted from Caha & Taraldsen Medová 2020: 119) is argued to be ambiguous between an adjectival and a verbal passive reading.

(35) Pokoj byl včera uklizen.
room.NOM was yesterday tidy.PFV.PPP
'The room was tidy/tidied yesterday.'
(i) adjectival passive (the state of the room being tidy held yesterday)
(ii) verbal passive (the event took place yesterday)

To argue for the distinction in (34), they employ similar tests as Veselovská & Karlík (2004) for short vs. long form PPPs, only this time Caha & Taraldsen Medová (2020) contrast short form PPPs. For example, with adjectival PPPs (their "purely stative" PPPs) negation directly combines with the PPP (36a), whereas with verbal (eventive) PPPs sentential negation combines with the auxiliary 'be' (36b) (both examples adapted from Caha & Taraldsen Medová 2020: 120).

(36) a. Pokoj byl včera neuklizen.
room.NOM was yesterday NEG.tidy.PFV.PPP
'Yesterday, the room was untidy.'

b. Pokoj nebyl včera uklizen.
room.NOM NEG.was yesterday tidy.PFV.PPP
'The room was not tidied yesterday.'

Thus, I conclude that Czech (I)PFV short forms regularly appear in eventive/verbal passives, and the limited discussion of the role of aspect in the literature suggests that we get regular (I)PFV readings. At least PFV short form PPPs can also get a stative/adjectival passive reading, so we will have to see whether this is restricted to PFVs and possibly also to the tense on the auxiliary.

4.3.2 Reflexive passives

To my knowledge, it is not addressed in the literature whether the role of aspect in Czech reflexive passives is the same as in actives, so the discussion in this section will be brief and will only concern 'by'-phrases. Unlike what we find with Czech participial passives, it has been argued that Czech reflexive passives do not allow for 'by'-phrases (e.g. Fehrmann et al. 2010, Karlík 2017). Karlík (2017), for example provides the contrast in (37) (my glosses).

(37) a. Škola je stavěna (zedníky).
school.NOM is build.IPFV.PPP mason.INSTR.PL

b. Škola se staví (*zedníky).
school.NOM REFL builds.IPFV mason.INSTR.PL

From their cross-Slavic investigation of reflexive constructions Fehrmann et al. (2010) conclude that there is variation in this respect: Belarusian, Bulgarian, Russian, Ukrainian, and Upper Sorbian allow for 'by'-phrases with reflexive passives, whereas BCMS, Czech, Polish, Slovak, and Slovenian do not. For Czech and Russian this is illustrated in (38) (adapted from Fehrmann et al. 2010: 210f.).[39]

(38) a. Šaty se právě šijí (*babičkou).
dress.NOM.PL REFL right.now sew.IPFV.3PL grandmother.INSTR
'The dress is being made right now.' ('by'-phrase impossible) (Czech)

[39]The referential status of the 'by'-phrases is not discussed at all; while the Czech example in (38) most likely has a referential/definite 'by'-phrase, the Russian one could be interpreted non-referentially (or generically), in which case it might not be the best example for a 'by'-phrase as a diagnostics for a verbal passive reading.

b. Dom stroitsja (plotnikami).
house.NOM builds.IPFV.REFL carpenters.INSTR
'The house is being built (by carpenters).' (Russian)

Fehrmann et al. (2010) note further differences with reflexive constructions between Slavic languages in general, which for reasons of space I cannot go into, and this leads to their proposing two different types of reflexive morphemes in Slavic languages, which are in complementary distribution (some languages employ just one, others employ either but in different reflexive contexts): the argument-blocking one converts any argument into a semantically unbound variable, which allows further specification by a 'by'-phrase if the external argument of a transitive verb is involved; the argument-binding one always binds the highest argument, which then obligatorily gets interpreted as arbitrary human so that further specification by a 'by'-phrase is not possible.

Even though reflexive passives in Czech are commonly assumed to be passives (and probably verbal passives), their incompatibility with 'by'-phrases is suspicious (at least from a Germanic point of view) and might call into question the claim that we are really dealing with a "true" verbal passive and not with some kind of impersonal construction. On the other hand, the two native speaker consultants I worked with for the empirical investigation presented in the following section both note that reflexive passives are much more frequent than participial passives, in particular in colloquial use. So for now I will assume that this is a passive construction, but this question deserves further investigation.

4.4 Empirical investigation: Aspect in Czech passives

In order to get a clear idea of whether the role of aspect in Czech passives is the same as in actives, I took examples discussed in Karlík (2017), as well as (36) from Caha & Taraldsen Medová (2020), and consulted two native speakers, asking them the following questions:[40] Are reflexive and participial passives interchangeable? Are IPFV and PFV interchangeable? What happens when we switch to a different tense? Which (I)PFV readings does one get (process, general, habitual; completed, one-time, perfect/job done)? Since Karlík (2017) is written in Czech, I added my own glosses but not translations; these are derivable from the speakers' judgments reported for the examples discussed here.

Let us start with the judgments I obtained for (36). The native speakers agreed that we are dealing with an adjective in (36a), and one added that it would be

[40]Thanks to Petra Charvátová and Denisa Lenertová; both are Moravian speakers (as is Karlík), so potential disagreement in judgments is probably not due to dialectal differences.

incompatible with a 'by'-phrase. This is the same pattern we find in, e.g., German, for which it is commonly assumed that 'un'-prefixation is incompatible with phrasal adjectivisation, which in turn is necessary for a 'by'-phrase to be possible (see, e.g., Rapp 1996, Gehrke 2013). Furthermore, according to my consultants, the PFV *uklizen* can be replaced by the IPFV *uklízen* (a SI) in (36b), but not in (36a). This could indicate that adjectival/stative short form PPPs, at least those that are not derived phrasally, are incompatible with a SI input or maybe even with an IPFV input in general, but this would have to be tested further.[41] The meaning difference we get for (36b) if we change the aspect is that the tidying up was not started or finished (PFV), or that it was not started (IPFV), and this is what we would expect with a regular (I)PFV semantics.

Let us then turn to Karlík's (2017) examples and start with (39), which involves the IPFV activity predicate 'praise'.

(39) Žák je chválen (učitelem).
student.NOM is praise.IPFV.PPP teacher.INSTR

My consultants reported that with the IPFV PPP in (39) we get an ongoing or a regular reading; a perfect reading (job done), which would indicate that we are dealing with an adjectival PPP, was not available for this example. The IPFV PPP can also be exchanged by PFV *pochválen* to give rise to a regular reading, or to a "resultative perfect" reading (Petr Biskup, p.c.; see also discussion around (33b)). A habitual reading was stated to become available if the auxiliary is changed to frequentative *bývat*, and this is possible with both IPFV and PFV PPPs (*bývá (po)chválen*).

Furthermore, without the 'by'-phrase a reflexive passive is also possible in (39) (even if the reflexive interpretation is more prominent), in both aspects and both present and past tense (*se (po)chválí/-lil*). With the IPFV present we get an ongoing or general reading, with the PFV present either a future orientation or a generic reading (as part of a daily routine), with the IPFV past a process or general reading, and with the PFV past a one-time reading. Thus, based on the judgments concerning reflexive passives with the activity predicate in (39), (I)PFVs come with the same readings we also expect from active verb forms (as we have seen in Section 3). With the participial passive in combination with

[41]The fact that the SI PPP cannot be the complement to 'seem' (i) (adapted from Biskup 2019: 99) could also point in this direction.

(i) * Pokoj se zdá být uklízen.
room.NOM REFL seems be.INF clean.SI.PPP
Intended: 'The room seems to be cleaned.'

present tense 'be' we also seem to get typical IPFV meanings with the IPFV PPP (ongoing, regular), but a somewhat less typical picture with PFV PPPs (regular or "perfect"). As we will see, this is different in all the other examples, so this might be due to the combination of an activity predicate and present tense 'be'.

Let us then turn to (40), which arguably involves an achievement predicate (or at least an accomplishment predicate), PFV 'decide'.

(40) O tom bylo rozhodnuto (delegáty) včera.
about that was decide.PFV.PPP delegates.INSTR yesterday

My consultants stated that such an example is odd with *často* 'often' (instead of 'yesterday'), but with the frequentatively marked auxiliary *bývalo* in combination with *často* and the PFV PPP it is fully acceptable. Exchanging the PFV PPP with the IPFV *rozhodováno* (a SI) has the effect that there was some deliberation but that it was not finished; with *často* also a habitual reading is possible. Finally, without the 'by'-phrase the reflexive passive is acceptable in this context as well: with a PFV present we get a future orientation, for example in combination with the temporal adverbial *zítra* 'tomorrow' (*se rozhodne zítra*), and with the PFV past a one-time interpretation (*se rozhodlo {včera / ??často}*);[42] with the IPFV past (*se rozhodovalo*), we get either a process or a habitual interpretation (depending on the context and possibly additional adverbials). I conclude that we get the same (I)PFV readings we would get in active contexts.

Let us move on to (41), parts of which already appeared in (23) and which involves an IPFV verb of creation.

(41) a. Píše se stížnost. / Právě se píše
writes.IPFV REFL complaint.NOM just REFL writes.IPFV
stížnost.
complaint.NOM

b. Je psána stížnost. / právě psaná stížnost
is write.IPFV.PPP complaint just write.IPFV.PPP.LF complaint

Both examples allow for either a regular (without 'just') or an ongoing (with or without 'just') interpretation, again typical IPFV readings. Changing (41b) to PFV *je napsána* leads to a finished-reading only (in the sense that the job is done). Karlík's (2017) second example in (41b) involves an attributive (and therefore long

[42] As pointed out by Petr Biskup (p.c.), the reflexive PFV *rozhodlo se* is also possible in the iterative interpretation, it is just that *často* needs to have some element in its nuclear scope, which is why it is out in the example above.

form) PPP, but my consultants agreed that a predicative short form PPP is possible as well (*Právě je psána stížnost.* 'right-now is written.IPFV complaint'); this time also the IPFV PPP gives rise to a job done-reading. As noted in Section 4.1 this is a typical adjectival passive reading, so here it is interesting that we get this kind of reading with the PPP in either aspect (where the IPFV is a simple one), even though in previous examples (e.g. (39)) we did not get it with the (secondary) IPFV. Karlík (2020) argues that the adverb 'just' facilitates an adjectival/stative passive reading; in addition, maybe also the difference in word order might play a role: In (39) we have a canonical subject-initial order, but in (41) the syntactic subject appears sentence-finally/postverbally. Whether word order plays a role for the kinds of readings we get with participial passives needs to be explored further.[43] Finally, it was reported that frequentative 'be' in combination with a PFV PPP (*bývá napsána*) was slightly odd but with the right context one could get the reading that events of this type are regularly finished.

Let us return to (37), repeated in (42), which involves an IPFV verb of creation.

(42) a. Škola je stavěna (zedníky).
school.NOM is build.IPFV.PPP mason.INSTR.PL

b. Škola se staví (*zedníky).
school.NOM REFL builds.IPFV mason.INSTR.PL

Here, both IPFV passives get the ongoing reading as the first reading. With the right context, for example if the subject 'school' appears in the plural, a habitual reading is also possible. With the PFV reflexive passive in the present tense we get reference to a result in the future. With the PFV participial passive and present tense 'be', we get the job done-reading; so it might additionally be important to have present tense 'be' for this adjectival passive reading to arise. The frequentative auxiliary in combination with the PFV PPP (*bývá postavena*) is acceptable with a plural subject ('school buildings are regularly finished'). Finally, in PFV past, both reflexive and participial passives get the "completed" reading, and in the IPFV past, both passives get an ongoing reading. With plural subjects, a habitual reading is possible with both passives in the past and both aspects, so just as we saw in Section 3.2, habituality seems to depend more on the context than on aspect, unless the auxiliary is marked for it. The unavailability of the 'by'-phrase with the reflexive passive was confirmed for both aspects and all tenses.

[43]In contrast, Petr Biskup (p.c.) cannot get a "stative" (job-done) reading with the IPFV PPP (*Právě je psána stížnost.*) and states that it is only eventive for him, independently of the word order or the context.

Karlík (2017) notes that there are a few counterexamples to the claim that reflexive passives cannot combine with 'by'-phrases, such as (43).

(43) V USA se prezident volí všemi občany.
in USA REFL president.NOM votes all.INSTR.PL citizen.INSTR.PL
'In the US, the president is elected by all citizens.'

Only one of my consultants accepted this example, but stated that it did not feel equivalent to the PPP version with the instrumental NP; so maybe in (43) we do not have a true 'by'-phrase but just the means by which an election is done; the consultant furthermore speculated that it might also be necessary to have 'all' here and that it is stated like a rule. The same speaker observed that the IPFV past reflexive passive would also be "kind of ok" but not the PFV one.

In sum, Czech aspect fulfils its typical functions in both types of passives, as it does in actives: With the PFV we get a completed or finished reading with both types of passives. With the IPFV, we get an ongoing or regular interpretation. In the right context both (simple) IPFV and PFV passives can be interpreted as habitual, especially the participial one with the frequentatively marked auxiliary. As for adjectival passives it is possible that lexical target state participles might only be derived from PFVs (recall the discussion of (36)) but that with the right context (which could be related to tense, adverbs, word order, or other factors) both IPFV and PFV participles can be interpreted at least as resultant state participles (giving rise to the job done-reading); this would have to be investigated further but would support the claims in Caha & Taraldsen Medová (2020) and go against the proposal by Veselovská & Karlík (2004), according to which all short form PPPs are eventive/verbal (in syntax proper). Finally, reflexive passives most likely do not allow for 'by'-phrases.

4.5 Summary: Czech vs. Russian passives

This section has shown that Czech and Russian differ in another domain, which has not received a lot of attention in the literature on cross-Slavic aspectual variation, namely passives. In Czech, both participial and reflexive passives are derived from all kinds of (I)PFVs (including SIs) and express the same (I)PFV readings we find with actives. PFV (short form) PPPs can also get a stative/adjectival reading. In contrast, Russian reflexive passives are restricted to IPFVs, whereas PPPs are regularly derived from PFVs, and there are severe restrictions on IPFVs: there are no SI PPPs, and we do not get the full range of IPFV meanings. In particular, there is no process reading, but maybe only factual readings.

Thus, the category of aspect is fully functional in Czech passives, but defective in Russian passives. An interesting parallel can be drawn to nominalisations in Czech *-ní/-tí* / Russian *-nie/-tie*. These morphologically share the *-n/-t-* component with PPPs, and they can also be regularly derived from both (I)PFVs in Czech, with regular (I)PFV readings, but only from one or the other in Russian (though this time also from SIs), with no predictable aspectual meaning (see, e.g., Dickey 2000, Biskup 2023). Data like these seem to suggest that Russian aspect is not fully functional in non-finite contexts, and this question is further explored in Gehrke (to appear). A formal-semantic account of cross-Slavic variation in passives, as well as in nominalisations, has to be left for future research.

In the following, I address existing formal-semantic accounts of cross-Slavic aspectual differences in other domains. These primarily focus on factual contexts, but (implicitly or explicitly) assume their accounts to be more general.

5 Accounting for the differences

In this section, I address formal semantic accounts of cross-Slavic differences in aspect use. First, however, I discuss Dickey's (2000) cognitive semantic proposal that PFVs require temporal definiteness in Russian but not in Czech, an idea that is further explored more formally in Mueller-Reichau (2018b, 2025) (Section 5.3) and Klimek-Jankowska (2022) (Section 5.4). Grønn's (2004) influential account of Russian factual IPFVs, which was outlined in Section 2.2.2, is taken as a point of departure for two proposals dealing with differences between Czech and Russian, among others (Alvestad 2013, Mueller-Reichau 2018b, 2025) (Section 5.2–Section 5.3). Finally, in Section 5.5, I call into question both the widespread assumption that Czech has existential IPFVs, as well as the analysis of factual IPFVs as "fake" IPFVs.

5.1 Dickey (2000, 2015)

Based on data from the literature and native consultants, Dickey (2000) describes differences in the use of (I)PFVs between ten Slavic languages in the following contexts: habituals, factuals, historical present, running instructions and commentaries, coincidence (performatives), sequences of events and ingressivity, verbal nouns. Some of these have already been presented in Table 1 and described in detail in Section 3.1–Section 3.4. Given these differences, Dickey (2000) argues that in the Slavic aspectual system of the western type, with its prototype Czech, the meaning of the PFV is totality, and the meaning of the IPFV quantitative temporal indefiniteness. In contrast, in the system of the eastern type, with its prototype Russian, the semantics of the PFV is proposed to be temporal definiteness and of the IPFV qualitative temporal indefiniteness. Following Leinonen (1982), TEMPORAL DEFINITENESS holds when a situation is "uniquely locatable in a context, contiguous in time to qualitatively different states of affairs" (Dickey 2000: 19f.), QUANTITATIVE TEMPORAL INDEFINITENESS involves "assignability of a situation to several points in time" (Dickey 2000: 107), and QUALITATIVE TEMPORAL INDEFINITENESS "the non-assignment of a situation to a single, unique point in time" (Dickey 2000: 108). Dickey furthermore argues that Serbo-Croatian and Polish belong to transitional zones, tending towards the western or eastern type, respectively.

Dickey's (2000) typology is further refined in Dickey (2015), who characterises the distinctions between the western and eastern type as in Table 2.

Table 2: "Slavic East-West Division" (adapted from Dickey 2015)

	WEST: Czech, Slovak, Sorbian, Slovenian	EAST: Russian, Ukrainian, Belarusian
Functional scope of PFV	maximal	minimal
IPFV general-factual	minimal usage	maximal usage
Productive delimitative *po-*	no	yes
Productive distributive *po-*	yes	no
Préverbe vide	*s-/z-*	*po-*

Dickey (2015) argues that the western PFV has maximal functional scope because it is also used in present and past tense habitual contexts, the historical present, and running instructions, among others, which are all contexts which require the IPFV in the eastern type (recall also Section 3.2 & Section 3.3). Furthermore, Dickey claims that the western type makes minimal, the eastern type maximal use of factual IPFVs (recall Section 3.4). Further distinctions between the two types are assumed to be in the choice of what Dickey labels "préverbes vides", a term for prefixes with a purely perfectivising function. In particular, he focuses on *po-*, which is argued to be partially grammaticalised as such a perfectivising prefix in the eastern type, but to retain its surface-contact meaning in the western type, where its distributive use is also productive.[44] The western

[44]Relevant Russian examples are, for instance, delimitative *pomolčal* 'PO-was-silent.PFV' in (11) and distributive *postučala* 'PO-knocked.PFV' in (13).

type, in turn, is argued to employ *s-/z-* as partially grammaticalised perfectivising prefixes (on variation with *po-* and *s-/z-* see also Dickey & Hutcheson 2003, Dickey 2005, 2011).[45]

Dickey (2015) correlates the cross-Slavic aspectual variation with the diachronic developments and different contact situations of the languages involved: Whereas the western type had maximal contact with German, the eastern type had varying levels of Finno-Ugric language contact. The transitional zones, in turn, had either contact with German and East Slavic (Polish), or at various levels with German, Romance, and some also with Turkish (BCMS).

The notion of temporal definiteness is not formally defined in Dickey's works, and this raises several questions: Are we dealing with definiteness of the event (Dickey speaks of "situations"), or with definiteness of a particular time (e.g. the event time, or the reference time)? What kind of definiteness is involved: familiarity (anaphoricity), uniqueness (situational or world knowledge)? Dickey speaks of uniqueness, but other formal notions could be explored. For example, it is also possible that we are dealing with specificity, or with determinacy, which in the nominal domain just means that the nominal denotes an individual of type *e* (see Coppock & Beaver 2015). All these are questions that one would have to explore if one wanted to formalise Dickey's notion of temporal definiteness. The few works aiming at formalising this, to which I turn now, have so far only scratched the surface and exploring various avenues in this respect can be fruitful.

5.2 Alvestad (2013)

Alvestad (2013) investigates the use of (I)PFVs in imperatives in twelve Slavic languages, using the ParaSol corpus (von Waldenfels & Meyer 2006) (see also von Waldenfels 2012). The cross-Slavic variation in aspect use she observes in imperatives is illustrated in (44) (see Alvestad 2013: 312).

(44) Choice of IPFV in imperatives
Russian (60%) > Belarusian (59%) > Ukrainian (58%) > Bulgarian (48%) > Polish (47%) > Serbian, Croatian (45%) > Macedonian (44%) > Upper Sorbian (43%) > Slovak (33%) > Czech (31%) > Slovenian (29%)

She argues that the use of IPFVs in imperatives is a kind of general-factual use, and she builds on Grønn's (2004) proposal for Russian factual IPFVs, as well as

[45] While Dickey (2000) treats Bulgarian and Macedonian as belonging to the eastern type, he observes some deviations from this type in Dickey (2015) in that the functional scope of PFV is argued to be narrower and the use of factual IPFVs greater, but not maximal.

Grønn & von Stechow's (2010) research programme (which is further developed in Grønn & von Stechow 2016, as outlined in Section 2.2.2). Her general proposal is summarised in Table 3 (cf. Alvestad 2013: 229).

Table 3: Slavic Tense and Aspect (see Alvestad 2013: 229)

UNIQUE-DEF. T & INDEF. ASP Prediction: Morphological PFV	INDEF. T & INDEF. ASP Prediction: Aspect neutralisation, existential fake IPFV
DEF. T & DEF. ASP Prediction: Aspect neutralisation, presuppositional fake IPFV	INDEF. T & DEF. ASP Prediction: Aspect neutralisation, presuppositional fake IPFV

In particular, Alvestad (2013) argues that the PFV is always indefinite, in the sense that it introduces a new eventive discourse referent. IPFV, on the other hand, is argued to be compatible with both (in)definite events. Similarly, Tense can involve a definite or indefinite reference time (new vs. anaphoric or unique). In Table 3, "unique-def" involves uniqueness, whereas the other instances of "def" involve anaphoric definiteness.

Inspired by Grønn & von Stechow's (2010) research programme, she revises their Aspect Neutralisation Rule in (8) as follows:

(45) ASPECT NEUTRALISATION RULE (revised) (see Alvestad 2013: 230)
   a. When a semantically perfective aspect is definite/anaphoric, it is morphologically neutralised to IPFV. This holds irrespective of whether the tense is indefinite or definite. When this rule is adhered to, we see an instance of the presuppositional type fake IPFV.
   b. When a semantically perfective aspect is indefinite AND the tense is indefinite, the aspect is morphologically neutralised to IPFV. When this rule is adhered to, we see an instance of the existential type fake IPFV.

Her account of aspect use differences is primarily pragmatic as she argues that "Slavic languages adhere to the Aspect Neutralization Rule to varying degrees" and views "Russian as the most 'law-abiding' language" (Alvestad 2013: 312).

The proposal by Alvestad (2013), as well as the one by Grønn (2015), Grønn & von Stechow (2016), raise several questions. How is the system with covert (in)definite operators restricted, i.e. when do we get one and when the other? Why is the reference time with presuppositional IPFVs necessarily definite, as argued by Grønn & von Stechow (2016), whereas it can be either for Alvestad (2013)? The reference time could also be treated as indefinite with each clause introducing a new reference time into the discourse, as argued by, for instance, Gehrke (2023) (I will come back to this in Section 5.5). Furthermore, why do PFVs always involve indefinite events (Alvestad 2013)? In Mueller-Reichau (2018b), which I address in the following section, PFV events are always definite (unique). Why is only Russian "law-abiding" (Alvestad 2013), and what exactly triggers the choice of one or the other aspect in languages with more optionality, such as Czech? Finally, two different notions of definiteness play a role in Alvestad's (2013) account: uniqueness of Tense, i.e. the reference time (requiring the PFV, in case Aspect, i.e. the event, is indefinite), as opposed to anaphoricity of Tense/reference time and Aspect/the event (leading to the use of a presuppositional IPFV, no matter whether Tense/the reference time is definite or indefinite); these are quite different notions of definiteness and it would be good to figure out when and why we use one or the other.

5.3 Mueller-Reichau (2018b, 2025)

Mueller-Reichau (2018b) provides a semantic account of cross-Slavic differences in existential factual contexts, in which Polish and Czech use the PFV but Russian the IPFV, mainly discussing data from the literature and consulting speakers for additional data points. Building on Grønn (2004), he treats the IPFV as denoting an underspecified relation between E and R in all three languages (46).

(46) $[\![\text{IPFV}]\!] = \lambda P \lambda t \exists e[P(e) \wedge e \circ t]$

He proposes that the differences between the three languages lie in the semantics of the PFV: In all three, PFVs involve event uniqueness, but in Russian there is an additional requirement of target state validity, see (47).

(47) a. $[\![\text{PFV}_{\text{POL/CZ}}]\!] = \lambda P \lambda t \exists e[P(e) \wedge e \subseteq t \wedge \neg\exists e'[P(e') \wedge e' \neq e]]$

b. $[\![\text{PFV}_{\text{RU}}]\!] =$
$\lambda P \lambda t \exists e[P(e) \wedge e \subseteq t \wedge \neg\exists e'[P(e') \wedge e' \neq e] \wedge f_{\text{END}}(t) \subseteq f_{\text{TARGET}}(e)]$

The denotation of the PFV requires that the event time is part of or equal to the reference time ($e \subseteq t$) in all three languages, which, as we saw, is a cross-linguistically common account of the semantics of PFVs. However, there is an additional requirement for the event to be unique, which is expressed in the second conjunct for both PFV operators in (47) (there is no event e', of which the property holds as well and which is not identical to e). For the Russian PFV, Mueller-Reichau adds another conjunct ($f_{\text{END}}(t) \subseteq f_{\text{TARGET}}(e)$), which is meant to capture the intuition of target state validity, as it "requires the reference time to end when the target state is in force" (Mueller-Reichau 2018b: 305).[46] Mueller-Reichau argues that the requirement of target state validity prevents the Russian PFV to occur in any existential context (including contexts in which the result state has been reversed), whereas its absence allows Czech and Polish to use the PFV in these contexts.

While the differences between Czech and Russian (and possibly other languages) might very well be captured by a difference in the conditions on the PFV, while maintaining the same semantics for the IPFV, there is an empirical problem with the proposal that the PFV in both languages requires event uniqueness. Czech quite regularly employs PFVs in contexts in which the events are not unique, namely in iterative and habitual contexts, as we saw in Section 3.2. Furthermore, the notion of target state validity should only apply to predicates that already come with a target state (in the sense of, e.g., Kratzer 2000), which are predicates that involve changes of states (accomplishments and achievements). But what about predicates without target states, which in Russian can be prefixed by, e.g., delimitative *po-* or ingressive *za-* to become PFV (recall discussion in Section 3.1)?

Mueller-Reichau's (2018b) proposal is further refined in Mueller-Reichau (2025), where he adds other contexts as well as colloquial Upper Sorbian (CUS) data discussed in the literature. Like Czech and unlike Russian, CUS can use the PFV in iterative and habitual contexts (e.g. with 'often'), as illustrated in (48) (attributed to Breu 2000).

(48) Wón je husto jenož jednu knihu předał.
he AUX often only one book sold.PFV
'He often sold only one book.' (Colloquial Upper Sorbian)

Unlike both Czech and Russian, CUS can also, rather surprisingly, use the PFV to describe an ongoing event (49) (attributed to Breu 2000).

(49) Jurij jo rune jen text šełožił, hdyž sym ja nutř šišoł.
Jurij AUX now a text translated.PFV when AUX I in came.PFV
'When I came in, Jurij was translating a text.' (Colloquial Upper Sorbian)

[46] Mueller-Reichau (2018b) argues, following Mittwoch (2008), that target state validity by itself already requires uniqueness and completion so that the first two conjuncts in the definition of the Russian PFV could be omitted. For ease of comparison I leave them here though.

To account for the differences in aspect use between the three languages, Mueller-Reichau (2025) replaces his previous definition of event uniqueness by determinacy. In particular, he builds on the notion of a path, as used in, e.g, Krifka (1998), Zwarts (2005), Gehrke (2008), with the relevant notion of a determinate event predicate, which involves unidimensionality, directedness, and boundedness (for the formalisation of these notions see Mueller-Reichau 2025).

Mueller-Reichau (2025) employs the same weak semantics for IPFVs in all three languages that we saw in (46). What is new now and what directly translates the idea that the IPFV is the unmarked member of the aspectual opposition, is that this semantics is also part of the semantics of the marked PFV operators, which adds further requirements in conjuncts (50).

(50) a. $[\![\mathrm{PFV}_{\mathrm{CUS}}]\!] = \lambda P \lambda t \exists e.P(e) \wedge \textsc{det}(P) \wedge t \circ \tau(e)$

b. $[\![\mathrm{PFV}_{\mathrm{CZ}}]\!] = \lambda P \lambda t \exists e.P(e) \wedge \textsc{det}(P) \wedge t \circ \tau(e) \wedge f_{\mathrm{END}}(\tau(e)) \subseteq t$

c. $[\![\mathrm{PFV}_{\mathrm{RU}}]\!] =$
$\lambda P \lambda t \exists e.P(e) \wedge \textsc{det}(P) \wedge t \circ \tau(e) \wedge f_{\mathrm{END}}(\tau(e)) \subseteq t \wedge f_{\mathrm{END}}(\tau(e)) \subseteq f_{\mathrm{TARGET}}(e)$

In all three languages, the event predicate has to additionally involve a determinate path ($\textsc{det}(P)$), in Czech and Russian the end of the temporal trace has to additionally be included in the reference time, and only in Russian, target state validity is also necessary, building on Mueller-Reichau (2018b).

While this proposal solves the issue of event uniqueness for Czech (and presumably also CUS), by replacing it with determinacy, it is now less clear why Russian PFVs require single events. In addition, the question about target state validity remains, and further questions arise. One would be how to integrate Russian ingressives (recall the discussion of the prefix *za-* in Section 3.1), which are PFV but with which it is not the end but the beginning of the temporal trace that is picked up by the ingressive prefix. Furthermore, the only difference between PFV and IPFV in CUS is the nature of the event path, which is argued to have to be directed, unidimensional, and bounded with PFVs. However, the idea of directed, unidimensional, bounded paths playing a role for the interpretation of events is commonly thought of as characterising the predicates as telic (see, e.g., Krifka 1998), which in turn belongs to the level of inner, rather than outer aspect. Is it the case, then, that the CUS (I)PFV distinction is an inner-aspectual one, as also indicated by the discussion in Breu (2000) and Scholze (2008), who argue informally that CUS aspect is based on "terminativity" (recall fn. 8)?

5.4 Klimek-Jankowska (2022)

In her account of the differences in aspect use in factual contexts between Czech, Polish, and Russian, Klimek-Jankowska (2022) builds on Ramchand (2004, 2008a), in assuming that the "First phase syntax" (~ *v*P/VP; see also Section 4.3.1) introduces an event variable, and grammatical aspect introduces a time variable, which is crucially an instant, rather than an interval (as otherwise commonly assumed); the event variable and the temporal variable are related by the temporal trace function $\tau(e)$. In Ramchand's proposal, Russian PFV events introduce a definite reference time R, which is a "a single unique moment" (Ramchand 2004) or "a specific moment" (Ramchand 2008a). With accomplishments (Ramchand's procP/resP-syntax), R must be within both subevents (process and result state), and since it is an instant it has to be the transition itself. Russian IPFV events, in contrast, are argued to introduce an indefinite reference time, i.e. an arbitrary moment within the temporal trace of the event; with accomplishment predicates this would be an arbitrary time within the process. Finally, Tense is argued to bind the time variable and to relate it to S.

Klimek-Jankowska (2022) proposes more generally that R can be (in)definite either with respect to the temporal trace of the event (at the micro-level), or with respect to the utterance time (at the macro-level), building on earlier work (Klimek-Jankowska 2020). Since with presuppositional IPFVs, the result is assumed to be presupposed, Klimek-Jankowska (2020) argues that the resultee is part of the conversation and event completion is inferred. In the case of existential IPFVs that combine with 'once' or 'ever', for instance, she argues that we are dealing with indefiniteness with respect to the utterance time. More generally, the constellation in existential factual contexts leads to an aspectual competition, in which speakers can choose to go for the PFV because there is definiteness with respect to the temporal trace (at AspP), or for the IPFV because there is indefiniteness with respect to the utterance time (at TP). Klimek-Jankowska (2020) assumes that both options are in principle available, because the verb form competes for lexical insertion at the CP level.

Klimek-Jankowska (2022) argues that temporal (in)definiteness at the macro-level should be understood in terms of temporal specificity, but throughout the paper she uses these terms interchangeably. She proposes that existential contexts involve temporal indefiniteness at the macro-level and underspecification for definiteness at the micro-level. This leads to the following cross-Slavic variation. In Western Polish and Czech, we find a preference for definiteness with respect to the temporal trace/AspP with accomplishments, and this is obligatory with achievements. This is argued to be so because achievements are instantaneous and the time variable can only be located at the time instant at which the change of state happens. In Eastern Polish and Russian, on the other hand, there is a preference for definiteness with respect to the utterance time/TP, and in some cases this is obligatory in Russian (e.g. with 'ever', recall discussion in Section 2.2). It is argued that the Russian (but not the Polish or Czech) PFV has to be anchored to a specific temporal location on the timeline. With presuppositional contexts, there is a relation to the earlier discourse and we are dealing with "pragmatic specificity". Therefore the PFV is also possible in Russian, but more so in Czech and Polish, and it is again a matter of speakers' choice which aspect is used.[47]

Klimek-Jankowska's (2022) account raises a number of questions, some of which have already been raised for the other accounts. If it is a matter of speakers' choice, what regulates the choice? In addition, if it is a matter of speakers' choice, it seems to be a pragmatic account, but at various points it is stated that in some cases the use of a particular aspect is obligatory. Shouldn't obligatoriness be reflected in the semantics of (I)PFV? It is also not clear whether we are dealing with definiteness (and then uniqueness or anaphoricity) or with specificity? Finally, if it is true that with presuppositional IPFVs, the result is presupposed, the resultee is part of the conversation, and event completion is inferred, doesn't that come close to accomplishments under a process reading, in which, granted, event completion is *not* inferred, but at least it is conceptually given and it is equally not in focus (as Klimek-Jankowska argues for presuppositional IPFVs without focus on the result)? Considerations like these support the idea presented in Gehrke (2022) that presuppositional IPFVs are but a subcase of process/durative IPFVs. I address this proposal in the following section, in which I also call into question whether Czech has existential IPFVs to begin with.

5.5 Gehrke (2022, 2023)

Based on the corpus study in Gehrke (2002), I concluded in Gehrke (2022) that aspect use in the description of single vs. repeated (e.g. habitual) events in Czech is almost identical, except for a few verb forms that are specialised to mark entire passages as habitual (recall discussion in Section 3.2). Therefore the Czech IPFVs that are used in these contexts are not motivated by the fact that we are dealing with non-single events, but appear for other reasons, e.g. focus on the process,

[47]This proposal is reminiscent of Petruxina's (2000: 59f.) idea that aspect choice is conditioned by two, sometimes conflicting factors: the "objective" nature of the event (i.e. completed or not) vs. the "subjective" assessment of the temporal contour of the event by the speaker, which gets resolved differently in, e.g., Czech and Russian.

duration or state. This is different in Russian, where many verb forms in habitual contexts are additionally imperfectivised, so that the IPFV in such contexts is primarily motivated by the fact that we are not dealing with a single event.

If, in addition, Padučeva (1996) is correct in assuming that the use of Russian IPFVs in existential contexts arises because we are dealing with potential repeatability, then the IPFV appears for the same reason it appears in habitual contexts: we are dealing with (potentially) plural events. This reasoning leads to the hypothesis in Gehrke (2002, 2022) that the existential IPFV is but a special case of the use of IPFVs in the description of non-single events. Coupled with the observation above that Czech does not use the IPFV to signal habituality (instead it is motivated by other IPFV readings), this leads to the assumption that Czech lacks existenial IPFVs, contra the received view. Or, in other words, the use of Czech IPFVs in existential contexts is not motivated by the existential context. For example, in the discussion of (20) in Section 3.4, which Klimek-Jankowska (2022) assumes to involve an existential IPFV also in Czech, I argued that the Czech IPFV could instead be motivated by the process use of IPFVs, as we are dealing with the question whether the process of watering has taken place. Given that many of her Czech examples involve incremental theme verbs, which I take to be activity predicates by themselves, following Rappaport Hovav & Levin (2010), Kennedy (2012), it is empirically difficult to distinguish the two views. We would need to find clear resultative examples instead, but I do not see such Czech IPFV examples discussed in the literature.[48]

Things are different with the presuppositional IPFV. Under the analysis of presuppositional IPFVs in Gehrke (2023), their use leads to a zooming in on the reference time of a previously introduced or contextually retrievable event; thus, the IPFV in such contexts involves a partitive (a "true") IPFV semantics, contra Grønn (2015). This analysis makes presuppositional IPFVs just a special case of

[48]Petr Biskup (p.c.) suggests that in the following exchange between a waiter (W) and a customer (C), the IPFV in the customer's reply is existential:

(i) W: Mám tady nějaká piva.
have.1SG here some beers
'I have some beers here (that still need to be paid).'

C: To pivo jsem platil.
DEM beer.ACC AUX.1SG paid.IPFV
'I've paid this beer already.'

However, I am not convinced that this is an existential use ('this beer was involved in a paying event'), since it could also be analysed as presuppositional ('with respect to this beer, the paying event (retrievable from the discourse) happened'). We would have to analyse the prosody and information structure to resolve this issue.

the use of IPFVs for ongoing events, as hypothesised in Gehrke (2022). Since both Czech and Russian use the IPFV to describe ongoing events, both languages are expected to use the IPFV in presuppositional factual contexts.

Let me illustrate the account of Russian presuppositional IPFVs as "true" imperfectives with the example in (51) (adapted from Gehrke 2023).

(51) Zaplatili. *Plačeny* byli naličnymi šest' tysjač rublej.
paid.PFV.3PL pay.IPFV.PPP were in.cash six thousand Rubles
'They paid. It was paid 6.000 Rubles in cash.'

In this example, the PFV verb form *zaplatili* '(they) paid' in the first sentence introduces a "completed" paying event. The presuppositional IPFV PPP *plačeny* 'paid' in the second sentence links back to this already introduced event. The marked word order and the most natural way to read this example indicate a marked information structure, a hallmark of the presuppositional IPFV: The paying event appears in the beginning of the (second) sentence and is backgrounded, focus lies on the sentence-final subject and (probably also on) the modifier ('6.000 Rubles (in cash)'). The DRT analysis (Kamp & Reyle 1993) of this example proposed in Gehrke (2023) is given in (52).[49]

(52) $[e_1, e_2, t_1, t_2, n, x \mid \text{PAY}(e_1), \tau(e_1) \subset t_1, t_1 < n, \text{PAY}(e_2), \text{THEME}(e_2, x),$
$6.000\text{R}(x), \text{IN CASH}(e_2), e_2 = e_1, t_2 \subset \tau(e_2), t_2 < n]$

Under this analysis, the first sentence introduces a new eventive discourse referent e_1, and its run time, $\tau(e_1)$, is included in the reference time t_1 (PFV semantics), which is before *n*(ow) (past tense semantics). The presuppositional IPFV in the second sentence introduces a new paying event e_2, which – due to the information structural cues – is anaphorically linked to e_1, i.e. $e_2 = e_1$. The new information in focus is about e_2, and since e_2 is identical to e_1 it is also about e_1: the theme of e_2 is '6.000 Rubles' and this was paid 'in cash' (an event modifier). The IPFV semantics specifies that there is a new reference time, t_2, that is included in the run time of e_2, $\tau(e_2)$; past tense indicates that t_2 is before n.

How does this analysis still capture the intuition that the paying event is completed, if the presuppositional IPFV is analysed as involving IPFV semantics? I argue that event completion information is already given in the first sentence about e_1 (its run time is included in the first reference time t_1). Since e_2 equals e_1,

[49]A linear notation for discourse representation structures (DRSs) is employed, where discourse referents are written on the left-hand side, before | (in a traditional DRS they appear at the top of the DRS), and the conditions on these discourse referents are listed to the right of |, separated by commas (which in a different notation can be translated as conjuncts).

the actual event of paying remains completed. Furthermore, the second reference time, t_2, is included in the run time of e_2, and therefore it is also included in the run time of e_1 (since e_2 is identical to e_1). By transitivity, t_2 must also be included in the first reference time, t_1. The effect of the presuppositional IPFV, then, is that it is used to zoom in on a narrower reference time within the bigger reference time introduced in the first sentence; the link between the two reference times t_1 and t_2 is only indirect, via the events involved, but it can still be made. The assertion that the sentence with the presuppositional IPFV makes, then, is only for part of the bigger reference time (and therefore also only for part of the actual event), and this is what is captured by the IPFV semantics.

To conclude, I argued that presuppositional and existential IPFVs are just a subcase of the canonical IPFV readings (ongoing process or potentially plural events), as suggested in Gehrke (2022). This proposal goes directly against Grønn's (2015) notion of "fake" IPFVs and furthermore makes the rather dubious Aspect Neutralisation Rules employed by Grønn & von Stechow (2010, 2016) and Alvestad (2013) obsolete; this is a welcome result particularly for Czech, in which the optionality of the rule is rather problematic. Whether or not Czech has an IPFV use that is motivated by the existential factual context alone needs to be further explored, but judging from the data discussed in the literature there are no clear cases that would suggest this. However, both languages have presuppositional IPFVs, under the assumption that this is just a special case of the process reading of IPFVs. The empirical question that has to be further explored in this case is whether achievements (or necessarily unique events, as argued by Mueller-Reichau 2018b) can appear in presuppositional IPFVs, in both languages. To my knowledge, achievements have been explicitly addressed in the literature only for existential but not for presuppositional contexts.

The following and final section takes stock and zooms out again, focusing on the notion of temporal definiteness that played a role in all the accounts of cross-Slavic variation in aspect use discussed in this chapter. I outline a general research programme that draws parallels between the nominal and verbal domain with respect to definiteness.

6 Taking stock: Temporal definiteness

Let us take stock and review how different authors employ the notion of definiteness or specificity in the verbal and sentential domain to capture the semantics of aspect and cross-Slavic differences in aspect use. As already mentioned in Section 2.3, various definitions of definiteness and specificity play a role, as well as definite/specific events and times. Dickey's (2000) account uses the notion of temporal definiteness in the sense of uniqueness: a situation is uniquely locatable in a context (Russian PFV), it can be assigned to several points in time (Czech IPFV), or it is not assigned to a single, unique point in time (Russian IPFV). From a formal-semantic point of view, this could be translated as either the event time or (maybe more likely) the reference time being (non-)unique.

Klimek-Jankowska (2022), in turn, proposes that PFVs in the North Slavic languages she investigates involve a definite or specific reference time point (as proposed for Russian by Ramchand 2008a). She does not make precise what theory of definiteness she employs (uniqueness, familiarity, or other), and the notions "definite" and "specific" are used interchangeably; from the general discussion, I assume that familiarity might be closer to what she has in mind. To account for the cross-Slavic aspect variation, she proposes that in Czech and Western Polish, the PFV is used when the reference time is definite/specific with respect to the temporal trace of the event (which is what Ramchand assumed for Russian), whereas in Russian and Eastern Polish, the IPFV is used when the reference time is indefinite/non-specific with respect to the speech time. The former is a standard aspect relation (E-R), whereas the R-S relation is commonly taken to express tense (e.g. Reichenbach 1947, Klein 1994). This account, then, would suggest that Russian and Eastern Polish aspect is not a standard aspect but closer to a tense.[50]

Mueller-Reichau (2018b, 2025) also assumes definiteness for PFVs and locates the point of variation in additional conditions for PFVs that vary across the Slavic languages he is interested in. In particular, Mueller-Reichau (2018b) argues that PFVs in Czech, Polish, and Russian involve reference to unique events; only Russian PFVs additionally require target state validity. Mueller-Reichau (2025), in turn, does not employ uniqueness anymore but assumes that PFVs in Czech, Sorbian, Russian involve determinate event paths, so this again appears to be a switch from uniqueness to specificity. Additional requirements on Czech and Russian PFVs that are not directly related to definiteness are assumed to account for the cross-Slavic variation.

The accounts discussed so far do not agree on the kind of definiteness theory we need, and whether it is the event itself that is definite/specific or the reference time. This distinction is explicitly addressed in Grønn & von Stechow's (2016) research programme, which I had outlined in Section 2.2.2. Drawing parallels to articleless languages in the way one could capture definiteness in the nominal

[50]In Gehrke (to appear), I explore what it could mean that Russian aspect is more like a tense, and how the interaction of aspect with finiteness plays a crucial role in Russian, but not in Czech. A non-standard aspectual semantics in terms of the S-R relation, albeit different from the one outlined here, has also been proposed by Borik (2006) for Russian.

domain in the absence of articles, they propose that also aspects and tenses do not directly correspond to (in)definite events and reference times,[51] but only express a relational semantics. Whether we are dealing with an indefinite or definite event or reference time is determined by covert (in)definite operators, which are proposed to be dynamic generalised quantifiers that either introduce a new event or time (indefinite) or are anaphoric to a previous event or time (maximally presupposing given information). The exact mechanisms that lead to the insertion of a definite or indefinite operator are not fully worked out, but as a general research programme this looks interesting, since it opens up a whole area of parallels one can draw to the nominal domain. I will come back to this in a bit.

Embedded within this research programme is Grønn's (2015) account of Russian aspect. He proposes that existential IPFVs have the semantics of a "fake" IPFV (essentially a PFV semantics), which arises due to an indefinite event and an indefinite reference time; therefore, the reference time is too large for the PFV semantics to be informative and the "fake" IPFV is inserted instead. Presuppositional IPFVs, in turn, are proposed to involve a definite event and a definite reference time; the "fake" IPFV comes into play because narrative progression, commonly associated with the PFV (which involves an indefinite event), is to be avoided. Grønn & von Stechow (2016) explicitly state that whenever a "semantically PFV aspect" (i.e. E is included in R) is definite/anaphoric, it gets morphologically neutralised to IPFV (recall their Aspect Neutralisation Rule from Section 2.2.2).

Intuitively it makes sense to assume that with existential IPFVs both the event and the reference time are indefinite: they assert the existence of an event (recall the paraphrase 'There is an event of this type'), and it is either not known when the event happened, suggesting that the reference time is indefinite and non-specific,[52] or the event could have happened more than once, suggesting that the reference time is non-unique and this could be captured as an anti-uniqueness implicature (recall discussion in Section 2.3). Intuitively, it also makes sense to think of the event with presuppositional IPFVs as definite: it is anaphorically linked to a previously introduced or contextually accommodatable discourse referent. It is less clear to me though why the reference time has to be definite as well; I will come back to this below.

Alvestad's (2013) account is also embedded in the kind of research programme outlined above, but she makes assumptions for Slavic that are sometimes differ-

[51] This point is also already made in Klein (1995).

[52] Recall from Section 2.2 that Russian scopally non-specific temporal adverbs of the *-nibud'*-series, such as *kogda-nibud'* even require the existential IPFV.

ent from Grønn's (2015). Just like Grønn, she takes the PFV to involve an indefinite event; additionally she specifies that the reference time has to be unique. While there is some discussion about the difference between uniqueness and anaphoricity, it is not made explicit enough why Alvestad uses uniqueness only for tense with PFVs, but anaphoricity everywhere else. At first sight, it might make intuitive sense: we have a unique (i.e. only one) reference time. However, this proposal faces the same issue outlined in the discussion of Mueller-Reichau (2018b) in Section 5.3: If this is a general proposal for all Slavic languages, why would Czech ever use PFVs in habitual contexts (recall discussion in Section 3.2)? Also like Grønn, Alvestad assumes that with existential IPFVs both the event and the reference time are indefinite. Where her account differs is with respect to presuppositional IPFVs: while the event is definite-anaphoric as well, she now assumes that the reference time can be either definite-anaphoric or indefinite.

Gehrke's (2023) proposal for presuppositional IPFVs outlined in (52) in the previous section does not explicitly discuss definite events and times, but the account itself makes clear that both start out as indefinite: each (finite) verb form introduces a new event and a new reference time. Due to the information-structural cues, the newly introduced event is anaphorically linked to a previously introduced event ($e_2 = e_1$), and this brings about the impression that we are dealing with a definite. The reference time stays indefinite, but because of the anaphoric link between the two events, it indirectly has to be part of the previously introduced event's reference time. Furthermore, there is no "fake" IPFV (Gehrke 2022, 2023): Presuppositional IPFVs involve an IPFV semantics (R is included in E) and a certain information-structurally marked discourse, whereas existential IPFVs involve potential repeatability, and plural events in Russian (but not in Czech) require the IPFV. Obviously, this proposals calls for close exploration of the role of discourse in a dynamic framework (see also Altshuler 2012), but we can dispense with any type of aspect neutralisation rule (as those in Section 2.2.2 and Section 5.2).

Let us go back to the point of departure for Grønn & von Stechow (2016), then, namely the idea that we can draw parallels to the way (in)definite interpretations arise in the absence of articles in articleless languages. While there are accounts that employ covert (in)definite operators or covert type shifts of the relevant kind also for these languages (e.g. Chierchia 1998), there are recent proposals that question whether bare nominals in articleless languages should ever (semantically) be analysed as definites, even if we get the impression that contextually they are definite. For example, Heim (2011) proposes that in articleless languages all bare nominals (in argument position) are existential, i.e. indefinite.

They are also acceptable in definite contexts because these languages lack a definite article that could block them (i.e. there is no competition with definites and thus no anti-uniqueness implicature; recall discussion in Section 2.3).

Heim's (2011) line of reasoning is picked up in recent work on the semantics of bare nominals in Russian (Šimík & Demian 2020, Seres & Borik 2021) (see also discussion in Borik 2026 [this volume]). Seres & Borik show that Russian bare nominals can appear fairly freely in both definite and indefinite contexts, and other factors, such as topicality and the overall context only indirectly contribute to the impression that we get a definite interpretation. As Seres & Borik (2021: 340) put it: "The perceived definiteness in Russian is analysed as a pragmatic effect (not as a result of a covert type-shift), which has the following sources: ontological uniqueness, topicality, and familiarity/anaphoricity." Šimík & Demian, in turn, show experimentally that factors that have been described to correlate with a definite (in the sense of unique/maximal) interpretation of Russian bare nominals (word order, prosody, number) are rather weak, and that the nominals in such contexts do not behave like German definites. They conclude that their data rather support Heim's (2011) proposal that Russian bare nominals are always indefinite.

So what if we end up with the following, which could serve as a research programme for future investigations. Aspects, tenses, VPs, NPs are predicates (following Grønn & von Stechow 2016 for events/times and Coppock & Beaver 2015 for nominals); additional information about them (e.g. provided by adverbials) are added via predicate modification. At the relevant syntactic positions we get existential closure over the respective variables. In the absence of overt determiners, all events and times are indefinite, just like bare nominals (following Heim 2011, Šimík & Demian 2020, Seres & Borik 2021); the impression of a definite interpretation is only due to the context (including information structural cues) but there is no iota (or similar) shift. Note, again, that the idea that events and times are always indefinite is already implicit in Gehrke's (2023) account of presuppositional IPFVs.

If we want to fully exploit parallels to definiteness in the nominal domain when building a theory of aspect (or tense), we will have to exploit the full spectrum: different types of both definites (unique, familiar, anaphoric, weak vs. strong, kind reference, etc.) and indefinites, including a discussion of specificity, free choice, ignorance implicatures, and so on. Furthermore, we need to take into account the contribution of adverbials that further specify the event and the reference time, information structure and context more generally. For example, the different Russian indefinite series briefly mentioned in Section 2.3 (e.g. *kto-nibud'/to* 'someone') also exist in the temporal domain (e.g. *kogda-nibud'/to* 'sometime'), which provides additional information about the reference time (epistemically or scopally (non-)specific, and similar). In the nominal domain, Geist (2010) shows that the *-nibud'*-series is always scopally non-specific and needs to be licensed, e.g. by a quantifier or a modal operator (see also Geist 2026 [this volume]). Similarly, Pereltsvaig (2008) claims that the *-nibud'*-series needs to covary within the scope of an operator or quantifier. In the verbal domain, we saw in Section 2.2 that *kogda-nibud'* requires the IPFV; we are certainly not dealing with a single time/event, in fact, the reference time is non-referential in this case. Does that mean that there is covariation with a silent generic or other operator?

Czech (as well as other Slavic languages) has comparable indefinite series, though the forms and functions differ. For example, Aguilar-Guevara et al. (2011) observe that the *koli*-series primarily expresses free choice (see also Aloni 2022), the *si*-series epistemic indefiniteness; *ně*-prefixed *wh*-words are treated as plain indefinites, e.g. *někdo* (based on 'who') 'someone', *někdy* (based on 'when') 'sometime'. Recall from the discussion of (18) in Section 3.4, fn. 26, that a Czech native speaker doubted the preference of the Czech PFV in the context of the "universal *-koli(v)*-words" (the example contained *kdykoliv* 'ever') and would use the "existential" *někdy* instead. So if the reference time is a free choice indefinite, this leads to a preference for the Czech IPFV (for this speaker, but maybe not for the informants of Klimek-Jankowska 2022), but when it is a plain existential indefinite, the PFV is the first choice (in that particular example)?

Furthermore, Klimek-Jankowska (2022: 25) discusses the Russian corpus example in (53) as potentially problematic for both Mueller-Reichau's (2018b) account of PFVs as involving unique events, as well as Gehrke's (2023) account of existential IPFVs being motivated by potential repeatability:

(53) Ėto byla vešč' lučšaja iz vsex veščej, kotorye ja kogda-libo sozdal.
this was thing best of all things which I when-LIBO created.PFV
'It was the best thing of all the things I had ever created.' (Russian)

At first sight, this indeed seems to be a problem for both accounts. According to Daria Seres (p.c.), (53) is acceptable, but we cannot replace *kogda-libo* by *kogda-nibud'*, the temporal adverb that is usually found in discussions of Russian existential IPFVs. So we have to explore what the difference is between *kogda-libo* and *kogda-nibud'*, both of which are regularly translated as 'ever'. More generally, according to Haspelmath's (1997) semantic-map approach to indefinite pronouns, the *-nibud'*-series occurs in the contexts irrealis, question, and conditional antecedent; these are a proper subset of the functions of *-libo*, which additionally appears in indirect negation and comparatives. In (53), we are dealing with a comparative context, so whatever the semantics here is makes it different enough from the contexts which require the Russian IPFV, and the PFV can be used.

Setting aside stylistic differences between *-nibud'* and *-libo*, which are also commonly noted in the literature, we see for example that the native speakers that Ward (1977) consulted could always replace *-nibud'* by *-libo*, but not always the other way around. Ward proposes that *-libo* presupposes or implies the existence of a set: "There exists or can exist a set of *x*'s but it is not asserted that there exists a particular member of that set such that that member can or does participate in the event" (Ward 1977: 465). In contrast, *-nibud'* "leaves the existence of a set unmarked" (Ward 1977: 467). So it seems that the existence presupposition of *-libo* is compatible with the Russian PFV but its absence is not. These and similar considerations, which so far only scratch the surface, would have to be explored further to get at the full picture.

Finally, let us return to Dickey's (2000) proposal that the Russian PFV expresses temporal definiteness, whereas the Czech PFV expresses totality. Traditionally, it is common to view the semantics of PFV in (many) Slavic languages in terms of totality; for example Filip (1999) argues that the "Slavic" PFV maps events of any type to total events, to represent them "as integrated wholes (i.e, in their totality, as single indivisible wholes)" (Filip 1999: 184). In Filip (2008), she argues, again for Slavic in general, that PFVs involve a maximalisation operator on events, and a similar idea is formalised in Altshuler (2014) who proposes that PFV operators cross-linguistically require "a maximal stage of an event in the extension of the VP that it combines with" (Altshuler 2014: 762). So what if this general proposal for PFVs cross-linguistically does not hold for Russian but only for Czech? And what if, as some of the accounts discussed here, the Russian PFV is truly definite, comparable to definite articles in the nominal domain? More generally, it is possible that certain tenses and aspects (in some languages) come with uniqueness (or other) presuppositions, leading to competition with other aspects/tenses, similar to the competition that Heim (2011) and others assume for the nominal domain (see e.g. Zhao 2022 for recent work on the competition between perfect and past in various languages). This is where various approaches discussed in this chapter could sneak back in and be refined accordingly.

Acknowledgments

Thanks to Petra Charvátová, Denisa Lenertová, Barbora Schnelle, and Radek Šimík for help with the Czech data, to Petr Biskup for invaluable feedback on the first draft of this chapter, to two anonymous reviewers, and to Hans Robert Mehlig for wonderful discussion of all things related to aspect over the years.

Abbreviations

1	first person	LF	long form
2	second person	M	masculine
3	third person	N	neuter
ACC	accusative	NEG	negation
ADJ	adjective	NOM	nominative
ADV	adverb	PAP	past active participle
AP	adverbial participle	PL	plural
AUX	auxiliary	PFV	perfective
CUS	Colloquial Upper Sorbian	PO	delimitative *po-*
DAT	dative	POSSREFL	possessive reflexive
DEM	demonstrative	PPP	past passive participle
E	event time	PRS	present tense
FEM	feminine	R	reference time
FREQ	frequentative	REFL	reflexive
GEN	genitive	SG	singular
IMP	imperative	SI	secondary IPFV
INDET	indterminate	S	speech time
INF	infinitive	SOE	sequence of events
INSTR	instrumental	VOC	vocative
IPFV	imperfective	ZA	ingressive *za-*

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Chapter 6

Indefiniteness and specificity in Slavic languages

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This chapter focuses on the expression of indefiniteness and specificity in Slavic. Although Slavic languages do not have a general unmarked indefinite article like Germanic or Romance languages, they employ several markers of indefiniteness such as indefinite pronouns. Indefinite pronouns have been commonly assumed to mark different types of specificity. I summarize formal approaches to some indefiniteness markers in Russian and other Slavic languages and argue that specificity distinctions marked by different indefiniteness markers can be captured by the concept of referential anchoring. This chapter also gives an overview of research on emerging indefinite articles in different Slavic languages and provides a case study on the emergence of the indefinite determiner *odin* 'one' in Russian. By applying the notion of referential anchoring to the analysis of *odin*, I show that this notion is useful as a general framework for modelling the contribution of indefiniteness markers of different kinds.

1 Expression of Indefiniteness

The category of indefiniteness has traditionally been understood as a category complementary to definiteness. Although definite NPs have some feature +definite that qualifies them for a definite interpretation, indefinite NPs should lack this feature and are interpreted as −definite. There is a debate in the literature about what this distinguishing feature should be (see the overview in Abbott 2006, Heim 2011, Brasoveanu & Farkas 2016, Borik 2026 [this volume]). Two main suggestions have been made: the traditionally assumed feature is uniqueness (Russell 1905, Roberts 2003), another one is familiarity (Christophersen 1939, Kamp 1981, Heim 1982).

Ljudmila Geist. 2026. Indefiniteness and specificity in Slavic languages. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 305–345. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078579

In the uniqueness approach, the definite article is assumed to indicate the existence of one and only one entity satisfying the descriptive condition. The indefinite article, by contrast, does not signal non-uniqueness. Rather, it indicates neutrality with respect to uniqueness in the sense that it requires at least one entity to fit the description. In the familiarity approach to definiteness, indefinite NPs are assumed to introduce a new referent into the discourse, while definites have an already familiar discourse referent. Whereas the referent of the indefinite is unknown to the hearer, it may be known to the speaker. Although novelty of referents and neutrality with respect to uniqueness introduced by indefinites are distinct concepts, they often go together.

The category of indefiniteness has traditionally been understood as a category contributed by indefinite articles like *a* in English. The formal representation of the English indefinite article has been debated. In the classical accounts of indefiniteness since Russell and Frege, which will be used as a starting point in this chapter, it has been assumed that indefinite NPs make an existential contribution to the truth conditions of clauses, i.e., they can be translated as existential quantifiers (Russell 1905). For instance, the sentence in (1) can be assigned the logical form in (1b).

(1) a. Gertrud bought a bicycle.
  b. $\exists x[\textsc{bicycle}(x) \wedge \textsc{bought}(\textsc{Gertrud}, x)]$

(1) can be paraphrased as 'There is something that is a bicycle and that was bought by Gertrud'. This logical representation of indefiniteness by existential quantification has been challenged by cross-linguistic studies that show that languages have more than one expression of indefiniteness. To capture the large variety of markers of indefinites available in different languages, the representation of indefiniteness as an existential quantification has been enriched.

This chapter focuses on indefiniteness in Slavic languages. As is well known, Slavic languages do not have a general, all-purpose, or unmarked indefinite article unlike Germanic or Romance languages. However, Haspelmath (1997) shows in his influential typological study that even languages without indefinite articles have several explicit markers of indefiniteness such as indefinite pronouns. Indefinite pronouns often consist of two morphemes: a bare indefinite root, the same as the corresponding interrogative pronoun or a generic noun like English *thing* or *body*, and a suffix or prefix acting as a series marker, cf. English *some-*, *ever-*, and *any*-series. Used as determiners, indefinite pronouns may indicate more fine-grained distinctions than those marked by the indefinite article in English. Haspelmath (1997) identifies nine different functions that indefinite pronouns may display in different languages and orders them in an implicational map. He provides such implicational maps for indefinite pronouns in some Slavic languages as well. Table 1 shows examples from three languages for comparison.

The comparison between the three Slavic languages reveals differences with respect to the number and use of pronominal series: In Bulgarian and Polish all functions are covered by three pronominal series, while in Russian the same functions are shared between seven pronominal series.

The nine functions are identified on the basis of four distinctions. The first one concerns the question whether the indefinite can have a negative interpretation under the scope of direct and/or indirect negation: e.g., Russian and Polish have a *ni*-series that must be licensed under the scope of direct negation (i.e. sentence negation). The second one is the specific vs. non-specific distinction which separates the two first functions on the left of the table from the rest. An expression can be assumed to be (scopally) specific if the speaker presupposes the existence and unique identifiability of its referent. The referent introduced by a specific indefinite is established in the discourse and is not affected by operators in the clause such as question and conditional or by modal operators subsumed under the term "irrealis" in Table 1. The third distinction concerns the type of context in which the indefinite with non-specific meaning is licensed. As shown in Table 1, the Russian *-libo, -nibud'* and *by-to-ni-bylo*-series cover different contexts in this area. The last distinction has to do with the identifiability of the referent of the indefinite expression by the speaker. It is applied only to scopally specific indefinites and divides them into those whose referent is known by the speaker and those whose referent is unknown even to the speaker, not only to the hearer. The former group is sometimes called epistemically specific and the latter one epistemically non-specific.

A look at Haspelmath's implicational maps for Russian in comparison to Polish and Bulgarian reveals that Russian has the largest inventory of indefinite pronouns and seems to be the only language that has a special pronominal series for the leftmost function on the implicational map, which Haspelmath calls "specific known".

Overall, Haspelmath (1997) gives a useful overview over the possible distinctions of indefinite pronouns, however, his survey does not provide enough empirical information about individual pronominal series. Furthermore, other indefiniteness markers such as the indefinite article are not considered. Slavic languages are particularly interesting for in-depth studies, since first, they provide a wide range of series of indefiniteness markers, and second, an indefinite article is emerging out of the numeral for 'one'. In the last decades, there have been some comprehensive studies on indefinite pronouns and on emerging indefinite

Table 1: Indefiniteness markers in Bulgarian, Polish and Russian based on Haspelmath (1997: 268, 271, 273); SK = specific known, SU = specific uknown, IRR = irrealis non-specific, Q = question, COND = conditional, CMPR = comparative, FC = free choice, IN = indirect negation, DN = direct negation

Bulgarian									
nja-	SK	SU	IRR	Q	COND				
-toidae					COND	CMPR	FC	IN	
ni-								IN	DN
Polish									
-ś	SK	SU	IRR	Q	COND			IN	
-kolwiek				Q	COND	CMPR	FC	IN	
ni-									DN
Russian									
koe-	SK								
-to		SU	IRR	Q	COND				
-nibud'			IRR	Q	COND				
-libo			IRR	Q	COND	CMPR		IN	
-by to ni bylo					COND	CMPR		IN	
ljuboj/ugodno							FC		
ni-									DN

articles in Slavic languages. These studies do not only contribute to a more differentiated view on the types of indefiniteness markers and their functions but also further develop formal tools to capture the fine-grained distinctions within the concept of indefiniteness. In some analyses, the representation of indefinites as existential quantifiers has been enriched while in others it has been questioned and replaced by other formal devices such as choice functions of different sorts. The goal of this chapter is to review research on indefinite pronouns and indefinite articles in the Slavic languages.

As with any overview, certain difficult decisions had to be made concerning the selection of the material. To give justice to the formal complexity of papers analysing the functions of indefinites I decided to narrow the thematic scope of this article to indefinite expressions that cover functions on the left and middle part of Haspelmath's implicational map. This is a serious limitation, because important work has also been carried out on indefinite expressions covering different functions on the right side of the map such as negative pronouns and free choice items. However, I found that it is impossible to adequately present this influential work and at the same time give a detailed survey of other types of indefinite expressions. Thus I will not discuss several prominent comprehensive contributions on negative pronouns and negative polarity (see Dočekal 2026 [this volume] for a survey; see Błaszczak 2001, 2003 on Polish; Dočekal 2011 on Czech; Progovac 1993 on Serbo-Croatian; Pereltsvaig 2000, 2004 on Russian) and the significant work on free choice indefinites in Czech – Strachoňová (2016). I urge the reader to consult these independently. Another limitation concerns the limitation in time. This chapter considers relevant literature till 2019, when it was written, but see the Epilogue for recent literature.

The chapter is structured as follows. In Section 2 I discuss two notions of specificity which are used to describe the differences between three types of indefiniteness markers in Russian. Section 3 presents analytical approaches to these types of indefinites provided in the literature. This section also introduces the idea of referential anchoring as a unified formal mechanism for capturing different readings of indefinite pronouns. In Section 4 I address research on indefinite pronouns used as indefinite determiners in different Slavic languages. Section 5 summarizes research on emerging indefinite articles in different Slavic languages. I also present a case study on the indefinite determiner *odin* in Russian and show how the concept of referential anchoring used in the formal treatment of indefinite pronouns in the previous section can be applied in the analysis of the meaning of *odin*. Section 6 concludes.

2 Two types of specificity and three specificity markers in Russian

2.1 Two types of specificity

This subsection introduces two types of specificity which will be used to analyse indefiniteness markers in Russian in the next subsection.

Specificity is a nominal category which is independent of indefiniteness. While indefiniteness is a discourse pragmatic property of NPs, specificity is a referential category which, following Partee (1970) and von Heusinger (2002), can be assumed to apply not only to indefinite but also to definite NPs. In English, indefiniteness and specificity can be expressed separately by complex determiners such as *a certain*, where the indefinite article indicates indefiniteness and the adjective *certain* specificity. But in many cases, specificity is not explicitly marked and indefinite NPs are ambiguous, cf. the specific and non-specific interpretation of the indefinite *a student* in the classic example from Fodor & Sag (1982) in (2).

(2) *A student in the syntax class* cheated on the final exam.
   a. Specific interpretation of *a student in the syntax class* implies that the speaker has a particular referent in mind.
   b. Non-specific interpretation of *a student in the syntax class* implies that the speaker has no particular referent in mind.

The type of specificity that distinguishes the reading in (2a) from reading (2b) has been called EPISTEMIC SPECIFICITY (Farkas 2002) since it concerns speaker's knowledge and speaker's ignorance (or indifference) about the referent of the indefinite. This type of specificity emerges in contexts without any operator. According to Farkas (1995) epistemic specificity requires the speaker (or another relevant agent) to have a uniquely identifiable individual in mind as the referent of the indefinite.[1]

Another type of specificity, the so-called SCOPAL SPECIFICITY, is attested in distributive contexts with extensional quantifiers such as *every* but also with strong intensional operators such as modals and *look for*; see (3).

(3) a. Scopally specific interpretation of *a bicycle*:
   Gertrud is looking for *a bicycle*. It disappeared on Friday.

[1]Note that an epistemically specific indefinite is not the same as an epistemic indefinite. The term EPISTEMIC INDEFINITE refers to a subtype of epistemically specific indefinites, namely those which signal speaker's ignorance about the referent (see Section 4).

b. Scopally non-specific interpretation of *a bicycle*:
Gertrud is looking for *a bicycle*. She can't find one for my size.

If the value of the indefinite *a bicycle* is fixed independently of the domain of the operator expression *looking for*, the indefinite is interpreted as specific, that is, taking wide scope (3a). If the value of the indefinite is dependent on the domain of the operator, as in (3b), the referent receives a non-specific interpretation, that is, it takes narrow scope. The scopally specific NP may be epistemically specific (i.e., identifiable by the speaker) or not. Under a scopally non-specific interpretation the referent of the NP is not identifiable by the speaker and hence the NP is also epistemically non-specific. Thus, while scopally non-specific NPs are also epistemically non-specific, scopally specific NPs may be epistemically specific or not.

Since scopally specific indefinites require additional contextual information for their interpretation and have wide scope with respect to operators in the clause, they cannot be adequately represented as plain existential quantifiers. Without any additional requirements, the existential quantifier representation is only appropriate for scopally non-specific NPs that take narrow scope. In his Quantifier Raising account May (1977) tried to preserve the uniform treatment of indefinites as existential quantifiers even for scopally specific NPs. He assumes that specific indefinites are moved by Quantifier Raising to a position above the intensional operator to take wide scope. However, Fodor & Sag (1982) showed that the Quantifier Raising mechanism does not work for all types of specific indefinites and suggested an ambiguity approach, according to which the indefinite article in English is ambiguous between the existential quantifier (non-specific use) and an indexical expression (specific use). Subsequent investigation of indefiniteness markers led to the conclusion that the binary distinction specific/non-specific is not sufficient to capture the fine-grained distinctions indicated by different indefiniteness markers. New devices such as choice functions (Winter 1997, Kratzer 1998) and a domain restriction of existential quantifiers (Portner 2002, Schwarzschild 2002), among others, were introduced to provide a more adequate representation of different types of specific indefinites.

To conclude, the interpretation of indefinites can vary at least with respect to two parameters: the identifiability of the referent by the speaker (epistemic specificity) and the scope relative to other operators in the clause (scopal specificity).[2] In the next section I will show how these specificity notions have been used to explain the differences between pronominal series in Russian.

[2]Besides these two specificity distinctions, von Heusinger (2011) identifies further specificity distinctions concerning partitivity, topicality, referentiality, noteworthiness and discourse prominence. Since these further specificity notions did not prominently play a role in the literature on indefinites in Slavic languages, I will not discuss them in this chapter.

2.2 Three specificity markers in Russian

As mentioned above, Russian has more pronominal series than e.g. Bulgarian and Polish, thus the language can make more subtle distinctions, which has been challenging linguists for many years. This subsection focuses on three pronominal series in Russian: the *koe-*, *-to* and *-nibud'*-series that attracted the most attention in the literature. I will show that these series specify the reading of the indefinite with respect to the two types of specificity introduced above. I will rely on the main empirical generalizations about these pronominal series agreed upon in the literature (Dahl 1970, Padučeva 1985/2010, Haspelmath 1997, Pereltsvaig 2004, 2008, Yanovich 2005, Geist 2008, Kagan 2011, Onea & Geist 2011; a.o.) First I will compare the wide-scope indefiniteness marker *koe-* with *-to*, then turn to the narrow scope indefinites with *-nibud'*.

2.2.1 Epistemic specificity: Russian *koe-* vs. *-to*

2.2.1.5 Speaker identifiability

The two indefiniteness markers *koe-* and *-to* differ in identifiability. The difference in interpretation of *a student* in Fodor & Sag's 1982 original example in (2) above can be disambiguated in Russian by the use of the appropriate indefinite pronoun as in (4).[3]

(4) a. *Koe-kakoj* student spisyval na ėkzamene.
KOE-which student cheated on exam
'A student [known to the speaker] cheated on the exam.'

b. *Kakoj-to* student spisyval na ėkzamene.
which-TO student cheated on exam
'A student [not known to the speaker] cheated on the exam.'
(Geist 2008: 156)

(5) Continuation compatible with (4b) but not with (4a):
Ja pytajus' vyjasnit', kto ėto byl.
I try find.out who this was
'I am trying to figure out who it was.'

According to Haspelmath (1997: 45), among others, the *koe*-series indicates that the speaker has a particular referent in mind, i.e., the referent of the DP is in some sense anchored to the speaker. By using the *-to*-series, by contrast, the speaker conveys that he/she cannot identify the referent. This difference shows

[3] All examples in this chapter are in English or Russian, unless explicitly indicated otherwise. The absence of an explicit source indicates that the example has been constructed by myself.

up in the combination with the continuation in (5), which indicates that the speaker cannot uniquely identify the referent. (5) can only be combined with (4b) containing *-to* and not with (4a) containing *koe-*. The fact that *-to*-items indicate non-identifiability of the referent by the speaker is revealed by the infelicity of (6).

(6) # Ja choču spet' *kakoj-to* romans.
I want sing which-TO romantic.song
'Now I'm singing you a particular romantic song [I do not know which one it is].' (Padučeva 2010: 211)

The singer's statement in (6) implies that he/she already knows what romantic song he/she wants to sing, i.e., the song must be identifiable by him/her. However, *kakoj-to* indicates that the singer cannot identify it, leading to a contradiction. Furthermore, Kagan (2011) shows that if the indefinite description contains modifiers which contribute additional information, the *-to*-marker is less acceptable. These observations support the assumption that referents of indefinites with *-to* are not speaker-identifiable. Thus, the *koe*-series indicates what has been called epistemic specificity while the *-to*-series marks epistemic non-specificity.

2.2.1.6 Scope under intensional operators

Indefinites with *-to* take wide scope with respect to intensional operators contributed by the future tense or intensional predicates such as *iskat'* 'look for' and *chotet'* 'want' (Ioup 1977, Pereltsvaig 2000). *Koe*-indefinites behave similarly to *-to*-indefinites in this respect (Geist 2008, Kagan 2011).

(7) Igor' chočet ženit'sja na {*koe-kakoj* / *kakoj-to*} studentke.
Igor wants marry on KOE-which which-TO student
'Igor wants to marry a [specific] student.' (Geist 2008: 155)

a. Continuation compatible with (7) → wide scope
On znakom s nej dva goda.
he known with her two years
'He has known her for two years.'

b. Continuation not compatible with (7) → narrow scope
On poka ni s kem ne poznakomilsja.
he still NI with who NEG met
'He hasn't met anybody yet.'

2.2.1.7 Scope under extensional operators

In contexts with extensional quantifiers, such as with universal quantifiers, *koe-* always indicates wide scope. Indefinites with *-to* are more flexible. They may take wide scope, like *koe*-indefinites:

(8) Petja každyj raz provožaet {*koe-kakuju* / *kakuju-to*} ženščinu.
Petja every time escorts KOE-which which-TO woman
'Petja always escorts a certain woman.'
Intended reading: the same woman → wide scope (Kagan 2011: 56)

However, *-to*-indefinites may also have narrow scope.

(9) Petja každyj raz nachodit {#*koe-kakoe* / *kakoe-to*} opravdanie.
Petja every time finds KOE-which which-TO excuse
'Petja every time finds some excuse.'
Intended reading: different excuses → narrow scope (Kagan 2011: 56)

Under the most natural reading of (9), Petja finds a different excuse every time. Thus, *kakoe-to* triggers a distributive or co-varying/functional interpretation (see more on the functional interpretation of *-to* in the next subsection). This interpretation is not available for *koe-kakoe* in (9).

2.2.2 Scopal specificity: Russian *-to* vs. *-nibud'*

In this section we will introduce the narrow-scope marker *-nibud'* and characterize it with respect to the parameters used in the previous subsection. We will compare it with *-to*, leaving *koe-* aside for simplicity.

2.2.2.8 Speaker (non-)identifiability

-Nibud' differs from the other two indefiniteness markers since it can only occur in the scope of some operators, e.g. the opaque predicate *chotet'* 'want' as in (10). Since it is impossible to use *-nibud'*-indefinites in a simple declarative sentence, they can be called polarity determiners (Pereltsvaig 2000) and are dependent indefinites (Farkas 1997, Farkas 2002, Brasoveanu & Farkas 2011). Dependent indefinites are assumed to require an appropriate licensor and to never take widest scope. The continuation indicating the non-identifiability of the referent by the speaker, which is compatible with the *-to*-series, is also compatible with *-nibud'*, hence both indicate epistemic non-specificity. However, the *-to* marker is compatible with an interpretation of (10) that the student is identifiable to Igor, i.e.,

the student may be identifiable to a discourse participant other than the speaker (Kagan 2011). In contrast, the *-nibud'* marker indicates the non-identifiability of the individual to the speaker, or to any other individual for that matter.

(10)

Igor'	chočet	ženit'sja	na	{*kakoj-to*	/	*kakoj-nibud'*}	studentke.
Igor	wants	marry	on	which-TO		which-NIBUD'	student

'Igor wants to marry a student.' (Geist 2008: 155)

2.2.2.9 Scope under operators and quantifiers

In (10) the *nibud'*- variant can be continued by 'he didn't get to know any student' indicating narrow scope with respect to the modal. Indefinites accompanied by *-nibud'* must have the narrowest possible scope also with respect to quantifiers, cf. (11). The wide scope reading is not available. *-Nibud'*-indefinites have been assumed to be non-specific (see Dahl 1970).

(11)

Každyj	student	v	našej	gruppe	voschiščaetsja	*kakim-nibud'*	professorom.
every	student	in	our	group	admires	which-NIBUD'	professor

'Every student in our group admires a professor.'

a. Unavailable: the same professor for all students → wide scope
b. Available: different professors → narrow scope

Since *-to*-indefinites can also take narrow scope relative to universal quantifiers, the question arises what distinguishes them from *-nibud'*. Onea & Geist (2011) note that the difference lies in the type of dependency between the indefinite marked with *-to* or *-nibud'*, and the quantifier expression, see (12).

(12) a.

Každyj	rebënok	polučit	na	Roždestvo	{#*kakoj-nibud'*	/	*kakoj-to*}	podarok.
every	child	will.get	for	Christmas	which-NIBUD'		which-TO	gift

'Every child will get a certain gift for Christmas.'

b. A felicitous continuation to (12a)

A	imenno	tot,	kotoryj	on	ožidaet.
and	namely	that	which	he	expect

'Namely the one which he expected.'

The natural interpretation of (12a) is that gifts are distributed to all the children in the context. The bound variable pronoun *on* 'he' in the continuation forces a systematic relation: different instances of the gifts must co-vary with different children. The function which assigns gifts to children is made explicit in the continuation (12b): children get not just any gift but each child receives the one he/she expected.

The function requires a systematic choice of value for the variable introduced by the indefinite NP. Functions of this type that can be readily accessed or named have been called natural functions (Chierchia 1993: 212) or systematic functions. Interpretations which rely on systematic functions are known as functional readings (see Kratzer 1998, Schwarzschild 2002; a.o.). In the context of such a natural or systematic function *-to* is preferred over *-nibud'*.

Onea & Geist (2011) also provide experimental evidence for the hypothesis that the functional dependency or co-variation established by *to-* is systematic and often contextually determined, while the dependency established by *-nibud'* is non-systematic and not determined. Thus, the semantic characteristics of the *koe-*, *-to*, and *-nibud'* markers of indefiniteness can be summarized as in Table 2.

Table 2: Properites of *koe-*, *-to*, and *-nibud'*

Parameter	*koe-*	*-to*	*-nibud'*
referent identifiability by the speaker	yes	no	no
scope with respect to intensional operators	wide	wide	narrow
scope with respect to extensional operators	wide	wide or narrow (functional)	narrow (non-functional)

3 Analytical approaches to indefiniteness markers in Russian

Different theoretical approaches have been used to capture the meaning of pronominal indefiniteness markers in Russian. In Section 3.1 I will introduce different analyses of *koe-*, *-to*, and *-nibud'* in the literature. In Section 3.2 I will present the formal concept of referential anchoring as a unified formal device for capturing three pronominal series. In Section 3.3 I discuss new experimental findings from acceptability studies which challenge previous analyses.

3.1 Approaches by Kagan (2011), Gorishneva & Zimmermann (2011), and Yanovich (2005)

Kagan (2011) develops a formal analysis of epistemic specificity indicated by *koe-* and *-to* in terms of speaker identifiability, which is formalized as a restriction on the set of possible worlds that represent the speaker's view of reality. More specifically, speaker identifiability is analyzed as a condition on the relative scope of an existential quantifier that ranges over individuals and a universal quantifier that quantifies over a set of possible worlds introduced in the context. Kagan captures the difference between *koe-* and *-to* in terms of felicity conditions on their use.

(13) Felicity Condition Imposed by *koe-* and *-to* Items (Kagan 2011: 71, 74)

Let S be a sentence that is uttered by speaker A which embeds an NP containing a *koe-* or *-to* item. Let P be the property contributed by the content of the NP, and let Q be another property. Then S is felicitous iff

a. For *koe-*:
$\exists y \forall w[w \in E_{A,w^0} \rightarrow P(y,w) \wedge Q(y,w)]$

b. For *-to*:
$\neg\exists y \forall w[w \in E_{A,w^0} \rightarrow P(y,w) \wedge Q(y,w)]$

The modal base E is the set of worlds that represents the speaker's beliefs about the actual world w^0. The condition for *koe-* in (13a) means that an NP referent is speaker-identifiable if and only if there is an individual who has the properties ascribed to its referent in every possible world within the speaker's epistemic base E. The condition for *-to* in (13b) is a negation of (13a). It ensures that the speaker cannot identify an individual with properties P and Q. *-To* indicates that the properties may be satisfied by different individuals in different worlds.

Kagan (2011) assumes that this difference in identifiability between *koe-* and *-to* is based on CONVENTIONAL IMPLICATURES in the sense of Potts (2007). In his system conventional implicatures constitute a meaning dimension which is independent of the so-called at-issue meaning, or "what is said." Conventional implicatures are secondary speaker-oriented entailments. Kagan (2011) assumes that *koe-* encodes the conventional implicature that the speaker can identify the referent, while *-to* encodes the conventional implicature that the speaker is not able to identify the referent.

One of the advantages of this approach is that it unifies epistemic specificity in terms of speaker identifiability with scopal specificity. In both cases, specificity is treated as a condition on scope, with the specific reading corresponding to the wide scope of the existential operator and the non-specific reading to the negated wide scope (narrow and intermediate scope) of the existential operator. Thus, epistemic specificity contributed by *koe-* vs. *-to* can be considered as a special case of scopal specificity.

Gorishneva & Zimmermann (2011) suggest an analysis of *koe-, -to,* and *-nibud'* that enriches the existential quantifier representation by relating it to modal subjects such as the speaker or another salient subject, and by this to mental models, similarly to Kagan (2011).

Yanovich (2005) analyzes *-to* and *-nibud'* as choice-functional variables (Kratzer 1998; among many others); see the representation of *-to*:

(14) $[\![\text{-to}]\!] = \lambda P\, f(P)$, where the speaker knows f (Yanovich 2005: 320)

In (14) the choice function f takes some set as an argument and returns an individual member of this set. Under a choice-functional representation, indefinite NPs are analyzed as *e*-type expressions. The choice function should either stay free and be supplied by the context (Kratzer 1998) or be existentially closed at some compositional level (Reinhart 1997). In the former case, the wide scope readings with *-to* are accounted for, in the latter the narrow scope reading. However, the fact that *-to* can never have narrow scope with respect to intensional operators is not explained by this analysis.

Since *-nibud'* can only occur in the scope of a licenser, this dependency on the licenser should be captured. To do that, Yanovich uses a Skolemized choice function. A Skolemization operation adds the licenser argument to a choice function making it of type $\langle e, \langle et, e\rangle\rangle$. Thus a choice from the set denoted by the argument of the choice function is relativized to that argument: the function can pick different individuals from the set for different values of the licensing variable.

(15) $[\![\text{-nibud'}]\!] = \lambda x \lambda P\, f(x, P)$, where the speaker knows f (Yanovich 2005: 316)

The variable x is an implicit Skolem argument of the indefinite. Yanovich relies on the descriptive generalization also made in Pereltsvaig (2008), namely that the Skolem argument x (the "domain variable" in Pereltsvaig's terms) bound by a quantifier can be of different types: object, time and world. Onea & Geist (2011), however, show that e.g. object-level variables are not appropriate domain variables for *-nibud'*-indefinites. Informants accept sentences with *-nibud'* and a universal quantifier ranging over objects only if some modal operator is also available in the clause.

(16) {*(Verojatno)} každyj mal'čik poceloval *kakuju-nibud'* odnoklassnicu.
probable every boy kissed which-NIBUD' classmate
(Intended:) '(Probably), every boy kissed some classmate.'

To account for this, the domain variable x in the representation of indefinites with *-nibud'* must be restricted to possible worlds or situations, as suggested by Onea & Geist (2011).

Since Yanovich (2005) only analyses the pronominal series *-to* and *-nibud'*, it is not clear whether other series can be accounted for with the same formal device. How can the difference in identifiability between *-to* and *koe-* be captured in the choice-functional approach? This question left open by Yanovich (2005) is answered in Geist (2008), who adjusts the choice-functional approach in order to capture differences between the three pronominal series *koe-*, *-to*, and *-nibud'*.

3.2 Specificity as referential anchoring

The discussion of the three pronominal series in Russian in this chapter makes it clear that specificity has a rather fine-grained structure that cannot be adequately described via the binary feature specific/non-specific. It seems more promising to consider that the referent introduced by the indefinite NP can depend on different expressions in the clause such as quantifiers, modals or even discourse participants such as the speaker. In order to account for such dependencies Geist (2008) and Onea & Geist (2011) use the notion of REFERENTIAL ANCHORING introduced by von Heusinger (2002). According to the concept of referential anchoring, the referent of specific and non-specific indefinites is referentially linked to some item in the discourse, its anchor. Referential anchoring can be modeled as a Skolemized or parameterized choice function in the sense of Kratzer (1998) and as used by Yanovich (2005) for the analysis of *-nibud'* (see Section 3.1). Skolemized choice functions, as shown in (17a), were used in Geist (2008) to analyse all three types of indefinites in Russian. In (17a), f is a free function variable, representing a contextually salient partial function from individuals (the subscripted anchor variable x) to choice functions. The Skolemized choice function f_x takes some set as an argument and returns an individual member of this set functionally dependent on the argument x. As the representational format of choice functions leads to problems discussed in the literature, Onea & Geist (2011) suggest another formalism for capturing referential anchoring, which preserves the original representation of indefinites as existential quantifiers; see (17b).

(17) a. $\lambda P\, f_x(P)$ (Geist 2008: 159)
b. $\lambda P \lambda Q \exists y[P(y) \wedge f(x) = y \wedge Q(y)]$ (Onea & Geist 2011: 221)

In (17b) referential anchoring is treated as a domain narrowing mechanism: the functional dependency $f(x) = y$ is integrated into the restrictor of an existential quantifier, narrowing down the domain of quantification of the indefinite to a singleton (note that f is just a function). The anchor variable can be specified as a discourse participant, e.g. the speaker, identified with the referent of another phrase in the clause or bound by a quantifier. It can be assumed that every indefinite DP takes scope below its anchor. The representation in (17b) can be taken as a general format for the representation of pronominal series expressing not only scopal non-specificity, as suggested by Yanovich (2005), but also epistemic specificity and non-specificity. For the semantic analysis of specificity markers, Onea & Geist assume that indefinites are underspecified with respect to the effects of specificity, but explicit markers such as indefinite pronouns may fix different specific readings by imposing constraints on the binding of the anchor argument. They assume that *koe-* lexically encodes referential anchoring of the indefinite to the speaker by the function f that must be spelled out as "intends to refer to y at t"; see the simplified representation for *koe-* in (18).

(18) $[\![\text{koe-}]\!]^c = \lambda P \lambda Q \exists y[P(y) \wedge f(\textsc{speaker}(c)) = y \wedge Q(y)]$
(Onea & Geist 2011: 214, simplified; for a revision see Section 5.3)

The binding of the implicit argument of *koe-* to the speaker yields identifiability by the speaker and necessary wide scope. The obligatory identification of the anchor with the speaker is part of the at-issue meaning of *koe-*. The inference at stake cannot be cancelled, as shown in (19). The non-cancellability of this inference is also compatible with the analysis as a conventional implicature suggested by Kagan (2011).

(19) Maša govorila s *koe-kakim* professorom. #No ja ego ne znaju.
Mary spoke with KOE-which professor but I him NEG know
Intended: 'Mary spoke with a certain professor. But I don't know him.'
(Onea & Geist 2011: 216)

To capture the occurrences of *-nibud'* in the above-mentioned contexts Onea & Geist extend Yanovich's (2005) analysis of *-nibud'*-indefinites, which treats them as dependent on the anchor variable *s* which must be bound by a universal quantifier. In Onea & Geist (2011) this idea is implemented as in (20).

(20) $[\![\text{-nibud'}]\!] = \lambda P \lambda Q \exists y[P(y) \wedge f(s) = y \wedge Q(y)]$
(it is presupposed that situation s is not part of the actual world w^0)
(Onea & Geist 2011: 222)

-Nibud' lexicalizes the constraint that the anchor must be a situation that has some modal flavor, the specification of which the authors leave for further research. This lexical entry can account for the licensing of *-nibud'* in modal sentences, since the anchor, situation *s*, does not belong to the actual world. Furthermore the preference to use *-nibud'* in situations which do not provide a systematic function need not be added as an additional requirement but already follows from its lexical entry: functions from situations to individuals do not seem to be systematic.

Referential anchoring as domain narrowing can also be used for the representation of *-to*. Different interpretational possibilities of *-to*-indefinites can be captured.

(21) $[\![\text{-to}]\!] = \lambda P \lambda Q \exists x[P(x) \wedge f(y) = x \wedge Q(x)]$
(where y is contextually determined and it is presupposed that f is a systematic contextually determined function)
(Onea & Geist 2011: 221)

The variability of the behavior of *-to* with extensional operators can be explained as follows: If the anchor variable is bound by the extensional quantifier, the indefinite takes narrow scope and is interpreted as co-varying. If the anchor is bound by some higher discourse referent, which can be explicit as in (7) or remain implicit, the *-to*-indefinite receives a wide scope interpretation. The entry in (21) also immediately predicts that *-to* cannot get a narrow scope reading under an intensional operator such as *chotet'* 'want'. Assuming that such operators quantify over worlds or situations, *-to* cannot enter functional dependencies with such variables not only because worlds and situations were of the wrong type to be anchors for *-to*, but because functions from worlds and situations to individuals are rather non-systematic.

The question is now how to capture the fact that the referent of *-to*-indefinites is not identifiable by the speaker. The contrast between *-to* and *-koe* plays a crucial role in the explanation. Like *-to*, *koe-* involves a systematic function, in this case "intend to refer to x at t". Thus, *koe-* contrasts with *-to* in speaker identifiability. The inference that the referent of *-to*-indefinites cannot be identifiable by the speaker can be derived as a conversational implicature. Since *koe-* lexically signals that the speaker is the referential anchor, it can be considered to be more informative. Therefore, if *-to* is used, the hearer can infer that the conditions for *koe-*, namely speaker anchoring, are not met. As was assumed in Geist (2008), the non-identifiability of *-to* referents is a standard scalar implicature since *koe-* logically implies *-to*. This implicature can be cancelled or reinforced as in (22).

(22) Igor' govoril s *kakoj-to* ženščinoj.
Igor spoke with which-TO woman
'Igor spoke with some woman.'

a. Reinforcement
Ja dejstvitel'no ne znaju kto ėto byl.
I really NEG know who it was
'I really don't know who it was.'

b. Cancellability
Mne kažetsja, ja eë znaju.
me seems I her know
'It seems to me that I know her.'

3.3 Completing the empirical picture: challenges for analytical approaches

The empirical data for the analyses of the three pronominal series described in the previous sections are mainly based on the intuition of native speakers and on introspection with the exception of the experimental investigation of functional readings with *-to* and *-nibud'*-indefinites in Onea & Geist (2011). Martí & Ionin (2019) conducted further experiments in which they tested the scopal and functional properties of the Russian indefinites with *koe-* and *-to* in comparison to *-nibud'*. Their findings complete and modify the empirical picture assumed and accounted for in the literature we described in the previous sections.

Martí & Ionin test the interpretation of indefinites in so-called scope islands established by e.g. relative clauses in sentences with two universal quantifiers like (23). It has been observed that indefinites occurring in such scope islands may receive exceptional wide scope, i.e. their scope may reach outside the syntactic island. The authors call contexts with islands such as that in (23) long-distance contexts (as opposed to local contexts like in our example (12), where no island intervenes between the surface position of the indefinite and the position at which it seems to take scope).

(23) Every student read every article [that a (*certain*) professor recommended].

a. 'There is a professor x such that for every student y, y read every article that x recommended.' [WSR]

b. 'For every student y, there is a (potentially different) professor x, such that y read every article that x recommended.' [ISR]

c. 'For every student y, y read every article that was professor-recommended.' [NSR]

(Martí & Ionin 2019: ex. (1))

To test the availability of the wide scope reading (WSR), intermediate scope reading (ISR) and narrow scope reading (NSR) for *koe-* and *-to*-indefinites, Martí & Ionin use Russian counterparts of long-distance contexts of (23) such as (24). The authors also test these indefinites in local contexts without scope islands e.g., like our example (12). *-Nibud'*-indefinites are taken for comparison in both cases. Martí & Ionin assume, following previous literature, that *-nibud'*-indefinites unambiguously take narrow scope and have non-functional interpretation.

(24) Každyj student pročital každuju knigu, [kotoruju porekomendoval
every student read.PST every book which recommended
{*koe-kakoj* / *kakoj-to* / *kakoj-nibud'*} professor.]
KOE-which which-TO which-NIBUD' professor
'Every student read every book that some professor recommended.'

(Martí & Ionin 2019: ex. (6))

The most important findings of these experimental studies with respect to the literature are:

1. The intuition that *koe*-indefinites can receive widest possible scope is confirmed, i.e. *koe*-indefinites can be considered as exceptional wide-scope indefinites, but, contra previous literature, both ISRs and NSRs (in local or long-distance contexts) are available for *koe*-indefinites when they have functional readings (when the context supports a function).

2. *-To*-indefinites, like *koe*-indefinites, allow for exceptional wide scope readings and functional NSRs, both already mentioned in the literature. Contra the assumptions in the literature, *-to*-indefinites also allow for non-functional NSRs and for functional as well as non-functional ISRs, although, functional readings still are preferred over non-functional ones. All in all, these results suggest that *-to*-indefinites are unmarked indefinites.

These findings require an update of the analytical analyses introduced above. Below I show how the referential anchoring approach sketched in Section 3.2 can capture the experimental results. To account for functional readings of *koe*-indefinites, it can be assumed that in addition to *koe*-$_1$, which binds the anchor to the speaker, there is *koe-*$_2$, whose anchor may be bound by c-commanding quantifiers and it is presupposed that the function introduced by *koe-*$_2$ is a systematic contextually determined function.

To account for the fact that *-to*-indefinites can receive all possible readings, this indefinite can be treated as an unconstrained indefinite: the function in the denotation of the indefinite is not specified as systematic or not and there is only one restriction on the possible anchor: it cannot be the speaker. If it were, *koe-*$_1$ would be preferred over *-to*. Any other option such as binding by other discourse participants or quantifiers in the clause should be available. What remains to be empirically tested is, what is the difference between functional readings with *koe-* vs. *-to*-indefinites, and how this difference can be formally accounted for.

To sum up, in the last decades progress was made in the empirical and theoretical description of indefinite pronouns expressing identifiability of the referent by the speaker and their scopal behavior in Russian, a language which, according to Haspelmath (1997), displays a larger range of pronominal series than Polish, Serbian, and Bulgarian. While Kagan (2011) focuses on *koe-* and *-to*, Yanovich on *-to* and *-nibud'*, all three pronominal series are compared in Geist (2008), Gorishneva & Zimmermann (2011), and Onea & Geist (2011). For a formal analysis, different devices have been used in the literature: Yanovich (2005) and Geist (2008) use choice functions to model specificity distinctions of indefinite pronouns. Kagan (2011), Gorishneva & Zimmermann (2011), and Onea & Geist (2011) preserve the original representation of indefinites as existential quantifiers. Kagan (2011) captures the fine-grained distinctions between the indefiniteness markers by pragmatic restrictions on the existential quantifier in terms of felicity conditions, while Gorishneva & Zimmermann (2011) relate them to the mental model of the speaker or another salient subject. Finally, Onea & Geist (2011) account for the differences using referential anchoring as a domain narrowing device. Referential anchoring is an operation that establishes a functional dependency between the referent of an indefinite DP and some other discourse item, its anchor. The major advantage of this view is that the referential anchor, modeled as an implicit argument, allows for interaction both with quantifier expressions and with discourse participants. The restrictions on the type of referential anchor may be encoded in the lexical entry of the specificity marker, or derived pragmatically from contrasts to other possible markers. Thus, the mechanism of referential anchoring provides a useful representational format for an adequate analysis of different specificity markers and can also capture the experimental findings in Martí & Ionin (2019).

4 Epistemic specificity in other Slavic languages

Indefinites signaling epistemic state of the speaker similar to the Russian *-to* and *koe*-series exist in different semantic and morphological flavours in many Slavic languages. Some important contributions in this domain were made by Šimík (2015) for Czech, Richtarcikova (2015) for Slovak, and Koev (2017) for Bulgarian. This work focused on so-called EPISTEMIC INDEFINITES (EIs), which signal ignorance of the speaker about the identity of the referent.

In Czech epistemic indefinites formed by the suffix *-si* attached to wh-words express, beyond standard indefinite meaning, ignorance about the precise identity of the referent. Šimík (2015) relies on the theory of referent identification suggested by Aloni & Port (2013) who argue that the speaker can always identify the referent in one way or another. However, the method of identification is sometimes not the one that is contextually required for knowing the referent. Among the most common methods that are available to the speaker are visual identification, identification by reported evidence, by naming, or by description. The fact that the speaker decides to use the EI, in spite of being able to identify the referent, suggests that she does not possess the knowledge required to identify the referent in some contextually more relevant way (Šimík 2015).

Like the Russian *-to*-series, the Czech *-si*-series takes wide scope with respect to intensional quantifiers and root and deontic modals. However, they differ from the *-to*-series in that they can only have a wide scope interpretation with respect to extensional quantifiers.

(25) Každý měl za úkol kontaktovat *kohosi* z Prahy.
everybody had for task contact.INF who.SI from Prague
'Everybody was supposed to contact somebody from Prague.' [$\exists > \forall$]
(Šimík 2015: 429)

Šimík points to the interesting fact that Czech EIs are unacceptable when they cooccur with epistemic modals as shown in (26) and *nějakém* must be used instead. The same "epistemic clash" seems to apply to the Russian *-to*-series, which in an epistemic context should be replaced by *-nibud'*.

(26) * Tom určitě spí na *jakémsi* gauči.
Tom surely sleep.3SG on which.SI couch
Intended: 'He must be sleeping on some couch (I don't know which one).' (Šimík 2015: 436)

This clash is explained in Šimík (2015) in the following way: Epistemic modals contribute an evidential presupposition which entails ignorance. The EIs with *-si* also imply ignorance. They are licensed if the ignorance they express is non-trivial (Aloni & Port 2013). They are ruled out under epistemic modals because their ignorance implication in this context is trivial. Šimík (2015) uses the classical representational format of indefinites as existential quantifiers to formally describe this "epistemic clash".

Richtarcikova (2015) analyzes two pronominal series with the affixes *voľa-* and *-si* that encode lack of knowledge about the identity of the referent in Slovak. These EIs can also contribute indifference similarly to free choice items (FCIs) with *-hoci* or *-koľvek*; see (27).

(27) Zobni si {*voľačo* / *čosi* / *hocičo* / *čokoľvek*} nech
peck REFL VOĽA-what what-SI HOCI-what what-KOĽVEK so
neodpadneš.
NEG.faint
'Eat something-or-other (EIs) / something-anything (FCIs) so that you don't faint.' (Richtarcikova 2015: 399)

However, in contrast to free choice items, which can be paraphrased as 'anything' and are felicitous in a context where the hearer is invited to consider all the options presented, the EIs are rather paraphrased as 'something-or-other' and convey a gentle suggestion without pointing to different options. EIs are less restricted in their distribution and may occur in episodic contexts without modal operators.

Richtarcikova proposes an analysis of the two Slovak EIs in the framework of Alternatives-and-Exhaustification based on Chierchia (2013). Chierchia develops a unified analysis for the complete range of polarity phenomena such as negative polarity items, minimizers, N-words but also universal and existential free choice items (FCIs). The two EIs in Slovak are considered by Richtarcikova as a special type of existential FCIs.

Although the idea of a unified polarity system developed in the Alternatives-and-Exhaustification framework is appealing, this theory is very complex. It involves a large system of different parameters posited along two dimensions: modes of exhaustification, and types of alternatives that get factored in. Richtarcikova uses these parameters to compare the Slovak EIs to other items that have already been analyzed with the help of Chierchia's parameters such as German *irgendein*, Spanish *algún*, and Italian *un qualche*. An interesting result of her study is that the Slovak EIs do not pattern with any of these items. She concludes that the complex theory of Alternatives-and-Exhaustification is unable to completely capture the observed characteristics of Slovak EIs.

One understudied variety of EIs is analysed in Koev (2017); see (28) from Bulgarian. He calls items like *edi-koja si* in (28) quotational indefinites (QIs). QIs in Bulgarian are made up of three parts: the indefiniteness marker *edi-* (based on *edin* 'one'), a wh-stem (e.g., *koj* 'who.M' or *kakvo* 'what'), and the *si* marker probably diachronically related to **sit* '(it) be' (Haspelmath 1997: 271). QIs with similar properties are attested in Japanese wh-doublets, such as *dare-dare*, English placeholder words like *whatshisface* or *so-and-so* and German indefinites of the form *der-und-der*.

(28) Ivan se ženi-l za *edi-koja si* student-ka.
Ivan REFL go.out-EV to EDI-which SI student-F
'Ivan married some student.'

a. The student Ivan married was mentioned to the speaker in a previous conversation.
b. The speaker does not know which student Ivan married.

(Koev 2017: 393)

Edi-koja si in (28) has an indefinite-like meaning, since it cannot refer back to an already given salient antecedent but rather establishes a new discourse referent. Beside the propositional meaning the sentence implies that the speaker heard a referential description of the student married by Ivan in a previous conversation. This reportative implication is due to the presence of *edi-koja si*, since substituting it with the regular indefinite *njakoj* 'someone.M' would remove the implication. The QI also conveys ignorance towards the identity of the referent by the speaker: *edi-koja si* indicates that the speaker does not remember which student Ivan married although she was told it.

The semantic and contextual properties of QIs are described in Koev (2017) as follows: First, QIs differ from other indefinites since they have a mixed semantics: they range over linguistic expressions rather than individuals. Second, QIs serve reportative functions; they existentially quantify over quoted speech. Third, QIs impose restrictions on the type of expressions they range over, that is, QIs referring to persons or things can only range over referential expressions within this class, e.g. proper names, definite descriptions, or demonstratives, but not over quantificational or indefinite expressions. For that reason QIs are understood as referring to specific individuals. To account for these properties of QIs in Bulgarian, Koev develops a formal proposal which builds on the previous work on QIs in other languages and is grounded in a two-dimensional semantics for quotation (Potts 2005).

To conclude, the research on indefinite pronominal series in Slavic languages has made progress in the precise formal analyses of single pronominal series and uncovered important interactions between indefiniteness, modality and reportativity. It has extended the typology of indefinites. An additional pronominal series of epistemic indefinites known from Germanic languages and Japanese, the subtype of quotational indefinites, was identified in Bulgarian. For the analysis of epistemic indefinites, different formal devices were introduced and tested. Interestingly, the approach proposed by Chierchia (2013), which proved effective in the description of indefinite pronouns and polarity sensitive expressions in Romance and Germanic languages, appeared to be not sufficiently fine-grained to account for the interpretive subtleties of EIs in Slovak, thus these pronouns provide a challenge for Chierchia's theory.

5 Emergence of indefinite articles

To understand the system of indefiniteness in Slavic languages it is necessary to also take into account explicit markers of indefiniteness other than indefinite pronouns. Slavic languages do not have a well-established indefinite article. However, literature on indefiniteness in these points to the fact that Slavic languages have been developing an indefinite article out of the numeral 'one' (henceforth ONE). In Section 5.1 I summarize work on the emerging indefinite articles in Macedonian (Weiss 2004), Bulgarian (Geist 2013, Gorishneva 2011, 2013), Polish (Hwaszcz & Kędzierska 2018), Croatian (Belaj & Matovac 2015) and Resian (Runić 2019). Section 5.2 focuses on the so-called intensifying use of the indefinite article. In Section 5.3 I present my own case study on the functions of the emerging indefinite article *odin* 'one' in Russian, based on observations in the previous literature (Ionin 2013, Padučeva 2010; among others). For the formal analysis of *odin* I employ the mechanism of referential anchoring used in the analysis of indefinite markers in Russian in Section 3.2.

5.1 Functional stages of the indefinite article

To illustrate the fact that the numeral 'one' is acquiring the function of an indefiniteness marker, contexts such as (29) have been used (Ionin 2013). In such contexts, the regular numeral use of ONE would be odd because cardinality is not at issue. Starting with Bulgarian, it can be shown that the counterpart of

ONE in this language can be used as an indefiniteness marker. To distinguish the indefinite use from the use as a numeral I will use the subscript I (ONE$_{\text{I}}$which) for "indefinite" in the glosses.

(29) Maria se omâži za *edin* lingvist.
Mary REFL married PREP ONE$_{\text{I}}$which linguist
'Mary married a linguist.' (Bulgarian; Geist 2013: 125)

Here the use of *edin* as a numeral would be infelicitous, because in our culture a woman can marry only one single man. The question 'How many linguists did she marry last year?', which highlights the cardinality of linguists, is inappropriate. Thus, *edin* indicates indefinite reference, i.e., it is used to introduce a new referent into the discourse. The indefiniteness marker is always destressed, while numerals can bear stress.

Other criteria that have been proposed for singling out the numeral function from other functions of counterparts of ONE in Slavic languages (e.g. Geist 2013 for Bulgarian; Šmelev 2002 for Russian; Weiss 2004 for Macedonian *eden*) are (i) combination with focus particles and (ii) restrictions on morphological number. According to the first criterion, only numerals can be combined with focus particles or modifiers which emphasize cardinality, while indefinite articles cannot. That *edin* in (29) is not a numeral but rather an indefinite article is confirmed in (30), where its combination with the numeral restrictor *točno* 'exactly' is excluded.

(30) Maša se omâži za (*točno) *edin* lingvist.
Masha REFL married PREP exactly ONE$_{\text{I}}$which linguist
Intended: 'Masha married a linguist.' (Bulgarian; Geist 2013: 125)

According to the second criterion, if the counterpart of ONE can be combined with count nouns in the plural, it is not a numeral, since to indicate a cardinality of ONE, the numeral and the noun must be singular. As shown in (31), *edin* as an indefiniteness marker can occur with count nouns in the plural as well.

(31) Včera vidjach *edni* amerikanci, koito poznavam.
yesterday saw.1SG ONE$_{\text{I}}$.PL Americans, whom know.1SG
'Yesterday I saw some Americans whom I know.'
(Bulgarian; Geist 2013: 129)

In the typological literature, the development of indefinite articles out of the numeral ONE has been generally accepted to proceed through the following functional stages (adapted from Givón 1981 and Heine 1997):

(32) Stages in the development of indefinite articles
(i) numeral > (ii) presentative > (iii) specific indefinite > (iv) non-specific indefinite > (v) generalized

The development from stage (i) to stage (ii) has been assumed to be triggered by discourse-pragmatic factors such as noteworthiness and referential continuity (Givón 1983, Wright & Givón 1987). At stage (ii) ONE introduces a new referent presumed to be unknown to the hearer and indicates the importance of the referent. This referent is then taken up as definite in subsequent discourse. At stage (iii) ONE does not just indicate discourse persistence of the NP referent, but rather speaker identifiability of the referent, i.e. epistemic and scopal specificity. At the next stage (iv) ONE is also used with non-specific indefinites, including the use in predicative position, in generic contexts as well as under the scope of negation or other operators (Givón 1981, Geist 2013). At the last stage (v) of generalized article, the use of ONE is extended from count nouns in the singular to other types of nouns such as plural and mass nouns. According to Wiemer (2020) among all Slavic varieties, Colloquial Upper Sorbian seems to have reached the highest degree of grammaticalization, inasmuch as *jen* 'one' is used with non-specific indefinites: It is obligatory with generic and predicative NPs (33). But since it still does not occur with plural and mass nouns, its development towards an indefinite determiner is not complete. Runić (2019) claims that the indefinite article in Resian, an endangered Slavic variety spoken in the province of Udine, has reached stage (iv) as well, since it is also used in generic contexts and with nouns having narrow scope (34).

(33) *Jen* tigor jo jene wulke zwěrjo.
ONE$_{I}$which tiger is ONE$_{I}$which big animal.NOM.SG.N
'A tiger is a big animal.' (Colloquial Upper Sorbian; Wiemer 2020)

(34) Skorë wsaka ïša ma *no* televižjun, aliböj *no* radio [...]
almost every house has ONE$_{I}$which television or ONE$_{I}$which radio
'In almost every house there is either a television or a radio [...].' [∀ > ∃]
(Resian, Runić 2019: 297)

Belaj & Matovac (2015) state that in other West Slavic and South Slavic languages ONE has reached at least stage (iii) of a specific indefinite. The hallmark of this stage is epistemic and scopal specificity. For example, in Bulgarian, NPs as aboutness topics can either be combined with a definite article, as shown in (35a), or with the indefiniteness marker *edin*, see (35b). Since topics must be specific and bare NPs, which are necessarily non-specific, cannot occur in this position, *edin* is used as a marker of epistemic specificity.

(35) a. Žena-ta ja risuva *edin* chudožnik.
woman-DEF her painted ONE$_I$which painter
'The woman was painted by a painter.'

b. {*(*Edna*)} žena ja risuva *edin* chudožnik.
ONE$_I$which woman her painted ONE$_I$which painter
'A woman was painted by a painter.'
(Bulgarian; Ivančev 1957: 515; as cited in Geist 2013)

Similarly to specific indefinites in general, Bulgarian *edin*-indefinites also have wide scope with respect to modal and other operators in the clause (Geist 2013, Gorishneva 2013). However, some non-specific uses of *edin* are also attested in the literature, indicating further development along the grammaticalization path. Geist (2013) discusses examples of the generic use of *edin* (see also Weiss 2004 on Macedonian). This use of *edin* is further examined in Gorishneva (2011). She shows that *edin* only occurs in a particular type of generic statements, those based on non-inductive reference, i.e. sentences which express rules and stereotypes; see (36):

(36) *Edna* žena vinagi šte nameri vreme za decata si.
ONE$_I$which woman always be.FUT find.PF.3SG time for children REFL
'A woman will always find time for her children.'
(Bulgarian; Gorishneva 2011: 85)

An interesting case is the intensifying use of *edin*. This use is not captured in implicational maps such as those represented in (32) above and will be discussed below.

5.2 Intensifying use of the indefinite article

The Bulgarian *edin* 'one' cannot appear with predicative NPs (37a); nevertheless, it can occur in predicative clauses like (37b):

(37) a. Toj e (**edin*) žurnalist po profesija.
he is ONE$_I$which journalist by profession
(Intended:) 'He is a journalist by profession.'

b. Ivan e *edin* glupak!
Ivan is ONE_{I}which fool
'Ivan is such a fool!' (Bulgarian; Ivanova & Koval' 1994)

Accoring to Ivanova & Koval' (1994) *edin* is felicitously used with predicate nouns such as *glupak* 'fool' or *genij* 'genius', which involve some evaluative characterizing component. The NP in (37b) has intensifying force and is correctly translated into English as such 'a (big) fool'. The use of the indefinite article in the intensifying function has also been observed in German and Greek with abstract mass nouns ('hunger', 'thirst', 'heat', etc.), which usually occur without an article. Gorishneva (2009) identifies three conditions (C1–3) for this use of the indefinite article:

C1 The predicate noun must be gradable and provide a scalar property.

Gradable nouns can be analyzed on a par with gradable adjectives as denoting a relation between an individual and a set of degrees viewed as a scale. For instance, the gradable noun *fool* can be analyzed as relating an individual to the scale of foolishness.

(38) $[\![\text{fool}]\!] = \lambda x \exists d[\text{FOOL}(d, x) \wedge d \geq d_{\text{standard}}]$ (Gorishneva 2009: 42)

C2 The indefinite article contributes a taxonomic interpretation.

The indefinite article used with a scalar noun leads to a subkind interpretation of the corresponding NP, similar to the sort reading of mass nouns caused by the use of the indefinite article in German, cf. *Reis* 'rice' vs. *ein Reis* 'one sort of rice'. Gorishneva (2009) characterizes the taxonomic relation analogously to Krifka et al. (1995: 74ff.) as follows:

(39) $\text{SUBKIND}(k', k) \wedge \text{R}(x, k') \rightarrow \text{R}(x, k)$
If k' is a subkind of kind k and the realization relation $\text{R}(x, k')$ holds, i.e., the object x is an instance of k', then x is also an instance of k. (Gorishneva 2009: 42)

C3 Exclamative prosody

The exclamative prosody indicates the remarkability of the corresponding subkind. The speaker utters an exclamative 'x is an N' where a postcopular indefinite NP denotes a subkind that is remarkable in comparison with other subkinds of k.

Gorishneva (2009) observes that languages such as Bulgarian, Greek, and German, which use the indefinite article in the intensificational function, also use the indefinite article with nouns in generic readings. From this she draws the generalization that the generic function of ONE is a precondition for its intensifying function since the generic use provides the taxonomic reference to subkinds, required by the intensifying ONE.

5.3 Case study: The emergence of the indefinite article in Russian

In this section I characterize the semantic profile of the emerging indefinite article in Russian and present my formal analysis of it in terms of referential anchoring, the mechanism proved useful for the analysis of indefiniteness markers (Section 3.2). There are some hints in the literature that Russian has been developing an indefinite article from the numeral *odin* 'one'. Padučeva (2010) notes that *odin* 'one', in addition to its use as a numeral, can be used as a marker of indefinite reference of NPs, see (40).

(40) Byl u menja *odin* prijatel'.
was at me ONE$_{\text{I}}$which friend
'Some time ago I had a friend.' (Padučeva 2010: 213)

The speaker's intension in uttering (40) is obviously not to convey the cardinality of his/her friends but rather to indicate the existence of an individual fitting the description. A formal indicator for the existence of an indefiniteness marker *odin* distinct from the numeral ONE is the plural use in (41).

(41) Ja poznakomilas' s *odnimi* amerikancami. Oni priglasili menja v New York.
I got.acquainted with ONE$_{\text{I}}$.PL Americans they invited me to New York
'I got acquainted with some Americans. They invited me to New York.'

The indefiniteness marker *odin* exhibits the characteristic behavior of specific indefinites since it takes widest scope, independently of the configuration it is in. In (44) it scopes out of syntactic islands such as relative clauses. If *odin* is modified by the intensifiers 'at least' or 'exactly', which bring out the numeral reading, it patterns with other numerals in taking narrow or wide scope. In this case it does not function as an indefinite determiner.

(42) a. Maša pročitala každuju knigu, kotoruju porekomendoval *odin* professor.
Mary read every book which recommended ONE$_{\text{I}}$which professor.NOM
'Mary read every book that was recommended by a specific professor.'
i. ✗ *každyj* > *odin*
ii. ✓ *odin* > *každyj*

b. Maša pročitala každuju knigu, kotoruju porekomendoval rovno *odin* professor.
Mary read every book which recommended exactly ONE professor.NOM
'Mary read every book that was recommended by exactly one professor.'
i. ✓ *každyj* > *odin*
ii. ✗ *odin* > *každyj* (Ionin 2013: 79)

The indefiniteness marker *odin* in (42a) is only felicitous if the speaker intends to refer to a particular professor. The narrow scope reading of the indefinite is unavailable: if the speaker does not have a particular professor in mind, *odin professor* in (42a) is undefined and the utterance is infelicitous.

Ionin observes that *odin*-indefinites carry a condition of identifiability: in order to use *odin* as indefiniteness marker the speaker must be able to identify the individual under discussion. The speaker must be able to answer the question 'Which X is it?' The response to this question names an identifying property that singles out the specific individual, distinguishing it from all other individuals in the set denoted by the NP. A further requirement is that the identifying property must be distinct from properties ascribed to the individual in the sentence. The speaker identifiability requirement can be illustrated in (43):

(43) Ja choču počitat' *odin* žurnal.
I want read ONE$_{\text{I}}$which journal
'I want to read a journal.'

a. Continuation compatible with (43):
A imenno poslednij vypusk Zdorov'ja.
and namely last issue Health
'Namely, the last number of Health.'

b. Continuation not compatible with (43):
Vsë ravno kakoj.
all same which
'It doesn't matter which one.'

The continuation in (43a) is felicitous, because it conveys additional identifying information about the referent distinct from the property of being a journal. The continuation in (43b), signaling the non-identifiability of the referent by the speaker, is excluded.

Thus, *odin* as an indefinite determiner indicates speaker identifiability and has the widest possible scope. As we saw in Section 2.2 another indefinitness marker in Russian, *koe-*, displays a similar behavior. Indefinite pronouns with *koe-* also indicate speaker identifiability of the referent. The question now is how *odin* differs from *koe-kakoj*. Padučeva (2010) notes that *koe-* has an additional meaning component, the conspiracy component, meaning that the speaker does not intend to disclose the referent of the indefinite in the discourse. Example (44) highlights this additional meaning component. Here *koe-kakoj* is incompatible with the continuation which immediately reveals the identity of the referent by naming it.

(44) *Koe-kakoj* student spisyval na ėkzamene, #a imenno Ivanov.
KOE-which student cheated on exam but namely Ivanov
Intended: 'A student [known to the speaker] cheated on the exam, namely Ivanov.'

Odin-DPs are different, cf. (45).

(45) *Odin* student spisyval na ėkzamene, a imenno Ivanov.
one student cheated on exam but namely Ivanov
'A student [known to the speaker] cheated on the exam, namely Ivanov.'

(45) violates the Gricean Cooperative Principle, in particular the Maxim of Quantity. That the speaker is unwilling to reveal the identity of the referent is a conversational implicature. It can be assumed that this implicature has become conventionalized and is now part of the conventional meaning of the lexical item *koe-*.

The question is now how to formally represent the meaning of *odin* as an indefiniteness marker. We can apply to *odin* the mechanism of referential anchoring introduced in Section 3.2, cf. (46a). For comparison I provide the meaning of the specificity marker *koe-* proposed in Section 3.2, in (46b), together with the conventional implicature discussed above.

(46) a. $[\![\text{odin}]\!]^c = \lambda P \lambda Q \exists x[P(x) \wedge f(\textsc{speaker}(c)) = x \wedge Q(x)]$
b. $[\![\text{koe-}]\!]^c = \lambda P \lambda Q \exists x[P(x) \wedge f(\textsc{speaker}(c)) = x \wedge Q(x)]$
Conventional Implicature: 'The speaker does not want to disclose the identity of x.'

(46) reveals that *odin* and *koe-* have identical core meanings: both encode existential quantification and anchoring of the referent to the speaker. However, *koe-* has an additional meaning component not available for *odin*, namely the conventional implicature that the speaker does not want to disclose the identity of the referent.

One remark should be made concerning the specification of the term "the speaker" in these representations. It has been pointed out in the literature on epistemic specificity in different languages that the specification of "the speaker" may vary depending on the type of context. In the examples we have discussed so far, it is identical to the actual producer of the utterance. However, in indirect speech, the perspective can be shifted away from the actual speaker to the speaker of embedded speech; cf. an example from the Russian National Corpus:

(47) Papa skazal, čto kostjum klouna emu odolžil *odin* prijatel' iz
father said that suit clown him loaned ONE$_{\mathrm{I}}$which friend from
cirkovogo učiliščja.
circus school
'The father said that the suit of the clown was loaned to him by a friend from circus school.'
(Russian National Corpus, E. Chaleckaya: *Čelovek po imeni beda*, 2004)

In (47) *odin prijatel'* 'one friend' refers to a person the father can identify. Thus the referent is anchored to the father – the speaker of the embedded speech act – and not to the actual speaker of the whole utterance. Like indirect speech, evidentiality markers like *javno* 'apparently' and *kažetsja* 'it seems' as well as quotative markers like *jakoby* 'ostensibly' can also shift the perspective away from the actual speaker. In general, such markers indicate the origin of the evidence the speaker possesses for a given assertion.

An alternative analysis of *odin* as an indexical is proposed by Ionin (2013), cf. (48a). Similarly to Kagan (2011) she assumes that specificity markers carry felicity conditions on their use. Ionin assumes that *odin* carries a felicity condition of speaker identifiability (48b).

(48) a. Semantics: A sentence of the form [*odin* α] β expresses a proposition only in those utterance contexts c where the speaker intends to refer to exactly one individual y which is α in c and the relevant felicity condition in (48b) is fulfilled. Then [*odin* α] β is true at an index I if y is β at I and false otherwise.

b. Pragmatics: The speaker is able to name an identifying property $\phi \in D_{\langle s,et \rangle}$ such that $\phi(w_c)(y) = 1$ and $\forall z[[\alpha(w_c)(z) = 1$ and $z \neq y] \rightarrow \phi(w_c)(y) \neq 1]$ and $\phi \neq \alpha$ and $\alpha \neq \beta$.

(Ionin 2013: 82)

Ionin's analysis can successfully capture the meaning and use of *odin* and provides a mechanism for comparison to markers of epistemic specificity in different languages that are also amenable to an analysis as indexicals. Ionin (2013) applies it to wide scope indefinites, however, since the approach allows the addition of any kind of felicity condition, it can also potentially be extended to indefinites with variable scope such as the *-to*-series but also to narrow scope indefinites like the *-nibud'*-series in Russian. I believe an advantage of the proposal based on referential anchoring over Ionin's is that it has a broader coverage, since it can capture more types of indefiniteness markers. It can account not only for indefinites with wide scope but also for indefinites with intermediate and narrow scope using the same basic formal mechanism. Thus, the mechanism of referential anchoring has a unifying force.

To conclude Section 5, I have shown that there is sufficient evidence to speak about indefinite articles in Slavic languages, even though they are still not fully developed grammatical items. The gradual development of indefinite articles out of the numeral by and large proceeds along the typologically attested path. However, at least one function, the so-called intensifying function of the indefinite article attested in Bulgarian must be added to the path. In most Slavic languages, the indefinite article has reached the stage of a specificity marker, for example, *odin* in Russian indicates scopal and epistemic specificity. The analysis in terms of referential anchoring, developed to capture differences between indefiniteness markers, can also adequately account for the meaning and use of *odin.*

6 Conclusions

In the last decades, progress has been made in the empirical and theoretical description of markers of indefiniteness and specificity in different Slavic languages. This research has uncovered important interactions between indefiniteness, modality, quotation and reportativity, and broadened our understanding of the typology of indefinites. Much empirical data have been provided on particular indefiniteness markers in single languages. The next step would be in-depth comparative studies and a unification of formal devices. An attempt for formal unification was made in this chapter. I have argued that specificity distinctions made by different indefiniteness markers can be captured by the concept of referential anchoring. According to this concept, it can be assumed that the referent of an indefinite is functionally dependent on some discourse participant or on another expression in the sentence. While the anchor must be familiar to both speaker and hearer, the content of the anchoring function must be unfamiliar to the hearer. However, the hearer has to accommodate the fact that there is a function. I have shown how this approach can account for various constraints associated with indefinite pronouns and the indefinite *odin* 'one' in Russian. My analysis suggests that the notion of referential anchoring is useful as a general framework for modelling the semantic contribution of indefinite determiners in one language. Future work will show whether this is the case cross-linguistically as well.

Epilogue

The main work on this article was finished in 2019. Since then, new studies have been conducted on the topic of indefinitenss markers and the development of indefinite articles in Slavic. Although this important work could not be integrated into this chapter I want at least to mention it: Borik et al. (2020) provide an experimental study of preverbal indefinites in Russian; Molinari (2023, 2025) offers a syntactic model for the grammaticalization of the numeral *edin* 'one' in Bulgarian; Pekelis (2023) examines the evolution of the Russian indefinite series *to* and *nibud'*, and Seres (2024) presents a corpus-based investigation of the distribution and interpretation of 'one' + N combination in six Slavic languages (Russian, Ukrainian, Czech, Serbian, Macedonian, and Bulgarian).

Abbreviations

1	first person
3	third person
DEF	definite
EDI	indefinite marker *edi-*
EV	evidential
F	feminine
FUT	future
INF	infinitive
KOE	indefinite marker *koe-*
M	masculine
NEG	negation
N	neuter
NI	indefinite marker *ni*
NIBUD'	indefinite marker *-nibud'*
NOM	nominative
ONE_I	indefinite 'one'
PF	perfect
PL	plural
PRS	present tense
PST	past tense
PREP	preposition
REFL	reflexive
SG	singular
SI	indefinite marker *-si*
TO	indefinite marker *-to*

Acknowledgements

I would like to thank three anonymous reviewers for their comments I have tried to do justice to in the final draft. Special thanks go to Radek Šimík for his valuable comments and editorial work. All remaining errors are my own. My work on this chapter was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – project number 445439335.

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Chapter 7

Tense in Slavic

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This chapter gives an overview of the topics in Slavic tense which have generated most interest in the semantic linguistic community. Data from most Slavic languages are considered. The introduction identifies some of the main issues through selected data from Ukrainian, Croatian, and Bulgarian. Section 2 traces the history of Slavic tense semantics with focus on the formalisation of the temporal domain – the interaction of tense and aspect in a compositional system.

The last two sections provide a more detailed account of what we take to be the main theoretical challenges in Slavic tense semantics: the role of temporal auxiliaries, notably the BUD-future and the BE-perfect (Section 3) and subordinate tense from complement tense to adjunct tense (Section 4). This last part centers around the role of so-called relative tenses in non-sequence-of-tense-languages.

1 Introduction

On the classic Jakobsonian view (Jakobson 1932) – that Russian tense can be reduced to ±*l* (the past tense morpheme) – everything is simple. It is up to the (im)perfective aspect to decide whether the non-past receives a present or future interpretation.

Indeed, no account of tense semantics in Slavic can ignore aspect, the other temporal category in the verbal domain. However, even if there is some truth to Jakobson's privative opposition for Russian, where the presence of the morpheme *l* in a way always produces a relative past interpretation, other Slavic languages have a richer inventory of temporal auxiliaries, something which significantly complicates this picture. A brief glance into what kind of challenging data we find in Slavic will be given below in this informal introduction through

Atle Grønn. 2026. Tense in Slavic. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 347–396. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078581

parallel translations from three sample languages (Ukrainian, Croatian, and Bulgarian).

The history of formal approaches to tense at the syntax-semantics interface in Slavic is quite recent. When did it all start? What is the core of a compositional semantics for tense and aspect in Slavic? This is the topic of Section 2, which presents a standard formalisation of tense in Ukrainian matrix clauses.

Next, Section 3 is devoted to existing analyses of temporal auxiliaries, both in the future (e.g. in Polish) and the perfect (e.g. in Macedonian, Bosnian-Croatian-Montenegrin-Serbian/BCMS, Bulgarian, and Ukrainian). And, finally, Section 4 addresses the issue of tense in complex sentences and the theoretical status of Slavic languages as traditionally being classified as non-sequence-of-tense-languages (non-SOT), languages with the so-called relative tenses. An important distinction must be made between complement tense (non-SOT in Slavic with shifted tense) and adjunct tense, where tense is often (but not always) semantically independent of the matrix tense.

Some of the major differences in the temporal domain between Germanic (here: English) and Slavic (here: Ukrainian, Croatian, and Bulgarian) can be illustrated with the following complex sentence:[1]

(1) a. When I *met* Maria, she *had* just *told* the guy whom she *would marry* that she *would have* children only after she *had finished* university.

b. Koly ja zustriv Mariju, vona ščojno skazala xlopcevi, z
when I meet.PFV.PST Maria she just say.PFV.PST guy with
jakym zbyralasja odružytysja, ščo matyme ditej
whom intend.IPFV.PST marry.PFV.INF COMP have.IPFV.FUT children
til'ky pislja toho, jak zakinčyt' universytet. (Ukrainian)
only after that how end.PFV.PRS university

c. Kad sam upoznao Mariju, upravo je
when be.PRS.AUX.1SG meet.PFV.PTCP Maria just is.PRS.AUX
bila rekla tipu za kojeg će se
be.PTCP.AUX say.PFV.PTCP guy with whom want.PRS.AUX REFL
udati da će imati djecu tek nakon
marry.PFV.INF COMP want.PRS.AUX have.IPFV.INF children only after
što bude završila fakultet. (Croatian)
COMP will.PRS.AUX end.PFV.PTCP university

[1]Whenever no source is given for an example, it is constructed by me with help from my consultants.

d. Kogato sreštnax Marija, tja tăkmo beše kazala na
when meet.PFV.PST Maria she just be.PST.AUX say.PFV.PTCP to
čoveka za kogoto štjala da se omăži, če
guy with whom want.IPFV.PTCP COMP REFL marry.PFV.INF COMP
tja šte ima deca samo sled kato
she want.PRS.AUX have.IPFV.INF children only after COMP
zavărši universitet. (Bulgarian)
ends.PFV.FUT university

The English tense system is characterised by the use of temporal auxiliaries that shift the reference time backwards (*had told, had finished*) or forwards (*would marry, would have children*).

A Slavic language like Ukrainian (1b) can do without any auxiliaries in the same context. Croatian and Bulgarian, on the other hand, display a rich variety of temporal auxiliaries, as summed up in Table 1.[2]

Table 1: Temporal constructions in English vs. three Slavic languages based on (1)

	English	Ukrainian	Croatian	Bulgarian
when-TAC	*met*	*zustriv*	*sam upoznao*	*sreštnax*
matrix	*had told*	*skazala*	*je bila rekla*	*beše kazala*
relative	*would*	*zbyralasja*	*će*	*štjala da*
complement	*would*	*matyme*	*će*	*šte*
after-TAC	*had finished*	*zakinčyt'*	*bude završila*	*zavărši*

Both Croatian and Bulgarian have a past perfect in the highest matrix clause in (1). While Bulgarian with a past aorist (≈ perfective) BE-auxiliary and a participle (*beše kazala*) is quite similar to English (*had told*), modulo the consistent difference in Slavic BE vs. Germanic HAVE, the Croatian variant is remarkable with the BE-auxiliary being itself in the form of a present perfect (*je bila ≈ is been*).

Similarly, the Croatian sentence also has a present perfect form in the WHEN-clause (*sam upoznao*). This use of present perfect as a simple past is unexpected from an English point of view, but well-known from many languages, e.g. German and French (or Czech).

Croatian and Bulgarian, furthermore, share a compound future construction with the auxiliary formed from a verb with the meaning 'want' – *će* and *šte*,

[2]TAC in Table 1 abbreviates temporal adverbial clause.

respectively, in the present form.[3] We see that Croatian also has a special "future 2" composed with the BE-auxiliary and a participle (*bude završila*). This construction can, in subordinate clauses, replace an imperfective or perfective with a future interpretation. The latter is what we find in Ukrainian (*zakinčyt'*) and Bulgarian (*zavărši*).

These brief remarks clearly demonstrate the need to further look into the role of auxiliaries in the tense system, notably versions of the compound (composite, complex, analytic) future and various shades of the perfect. This is what we will do in Section 3.

Do the temporal auxiliaries mean the same thing in English and Slavic? Almost, but not completely. A remarkable perfect meaning is found in Bulgarian, the so-called EVIDENTIAL perfect. This form is exemplified in (1d), where the participle of the forward shifter *štjala* 'wanted' occurs without the auxiliary *e* 'is' in the 3rd person singular. The absence of the auxiliary (only possible in the 3rd person) signals unequivocally an evidential interpretation, indicating that the speaker reports information he has received from Maria. We will return to this phenomenon found in Bulgarian in Section 3.

The next point to highlight in (1) is the significant discrepancy in embedded tense forms between the English sentence and the Slavic counterparts. Slavic languages are famously non-sequence-of-tense languages. Compared to the rather ad hoc SOT rules we find in English (e.g. past tense agreement everywhere in (1a)), Slavic has a simpler and cleaner system with what is traditionally referred to as RELATIVE TENSE.

The use of relative tenses is most consistent in complements under attitude verbs (Ukrainian: *skazala, ščo…* 'said that'), the most well-studied construction. Thus, the simple relative future *matyme* 'will have' in the Ukrainian example (1b) shifts the embedded reference time forward. Similarly, in Croatian and Bulgarian the present tense marking on the future auxiliaries *će* 'wants' in (1c) and *šte* 'wants' in (1d) denotes Maria's subjective now in the past, while the auxiliary itself lexically encodes the forward shift (future interpretation) of the prejacent verb (*imati*/*ima* 'have').

However, embedded tenses behave differently in different syntactic environments. Note that the relative clause in (1b) and (1d) contains a (deictic) past tense, just as in English. For the deeply embedded temporal adverbial clause (*after she had finished*), we are back to relative tense in Slavic with a semantic future (morphological present) perfective in (1b) and (1d), and a compound (relative) future in (1c).

[3]The 'desire' component in the etymology of the auxiliary is semantically bleached, perhaps not unlike the English forward shifter *will*.

In Section 4, we will discuss existing accounts of subordinate tense, possibly the hottest topic in Slavic tense at the present moment. Before embarking on the challenges presented by compound and subordinate tenses, we will next in Section 2 look at (mostly) simple tenses in (mostly) matrix clauses.

2 Formal approaches to tense

What characterises a formal approach to the semantics of tense? The main virtue of a formal approach is that it makes a clear distinction between morphology/syntax (form) on the one hand and semantics (interpretation) on the other. In Slavic linguistics, this distinction is sometimes blurred. We read about past tense (or, say, perfective aspect), but it is not always clear what the author has in mind: past tense morphology or past tense interpretation?

The distinction is crucial for a COMPOSITIONAL approach to tense (and aspect). This enterprise is potentially a strong point of Slavic linguistics, since both tense (semantically referring to external time on some time axis) and aspect (semantically referring to event internal temporality) are typically encoded explicitly in the morphology. We can therefore build a transparent compositional system in the temporal domain, where each semantic ingredient ideally corresponds to a piece of morphology.

On the standard view, semantic tense in matrix clauses relates the speaker's temporal focus, what is normally called the REFERENCE TIME, to the speech time. The anchoring of tense to the speech time is called DEICTIC tense. The semantics of aspect, on the other hand, is concerned with the internal temporal structure of the event, e.g. whether the event time is included or not in the reference time.

2.1 A brief history of time

The idea that the reference time is the glue that connects aspect and tense was popularised for Russian in Klein (1995), where the reference time is called the time of assertion. However, Wolfgang Klein was not explicit about the syntax of logical form, the compositional semantics. Although he criticised the traditional "impressionistic" approaches to Slavic tense and aspect, his own influential analysis is not really formal, but a mere translation from Russian to some meta language, whose status is unclear.[4] In Klein's work, PAST encodes the relation that

[4]There is a rich literature of non-formal analyses of Slavic tense and aspect. Most of these works are primarily concerned with aspect. A broad, more general perspective on temporality, with numerous lucid observations, is found in Padučeva (1996).

the time of assertion precedes the time of utterance (T-AST ≺ T-UT), while aspect, say PFV, encodes the relation that the time of assertion includes the time of situation (T-SIT ⊆ T-AST).[5]

Interestingly, the first formal approach to aspect in Slavic was formulated a few years earlier in Krifka (1992) with examples from Czech:

(2) a. John snědl rybu.
John eat.PFV.PST fish
'John ate fish.' (Czech)

b. John jedl rybu když Mary vztoupila.
John eat.IPFV.PST fish when Mary enter.PFV.PST
'John was eating fish when Mary entered.' (Czech)

Manfred Krifka assumed that the (im)perfective operator has scope over the complex verbal event predicate (the VP or sentence radical). He introduced two operators which modify the event description, hence functions of type $\langle\langle v,t\rangle,\langle v,t\rangle\rangle$ from sets of events to sets of events:[6]

(3) a. $[\![\text{IPFV}]\!] = \lambda P.\lambda e' \exists e[P(e) \,\&\, e' \sqsubseteq e]$ (Krifka 1992: 47)

b. $[\![\text{PFV}]\!] = \lambda P \lambda e[P(e) \,\&\, \text{QUANTIZED}(P)]$ (Krifka 1992: 50)

This is a partitive, non-temporal analysis of the imperfective/progressive aspect, which gives us access to subevents of the VP-event. The perfective aspect is also given a non-temporal analysis.[7]

So, what about time and tense? Krifka confirms, in an email on February 28, 2022, that he did not go beyond aspect in his early work. "I think that I assumed that tense would outscope aspect, but it was not the point of the paper."

In a purely syntactic approach to Russian tense and aspect, Yadroff (1996) proposed that tense indeed outscopes aspect. This is now universally accepted in the field. We will see below how such a system works in practice for Ukrainian.

[5]Klein's "situation time" is the time at which the situation described by the (verbal) predicate takes place. In event-based approaches, this notion is replaced by the event time, as we will do below.

[6]We use the following primitive semantic types: e – individuals; i – times; v – events/states; s – worlds; t – truth values. We assume an intensional semantics with a world parameter w and an assignment function g. For readability of the formulae, we will skip these indices and other details not pertaining to our discussion of tense proper.

[7]QUANTIZED is a property of predicates which encodes the idea that the event is completed, has a set terminal point. For more on aspect and event structures, see Tatevosov (2026 [this volume]).

One reason Krifka neglected tense (apart from the fact that his aspectual analysis, unlike Klein 1995, was non-temporal) was maybe because he focused on matrix clauses, and there is no a priori reason to believe that Slavic tense behaves differently from, say, English. What we know about matrix tense in English holds in general for tense in natural languages.

In the theoretical literature, consensus has been reached on certain general points. Thus, it can be shown that the old Montagovian system, where tenses shift the EVALUATION TIME merely in the meta-language (through the manipulation of a single index) is empirically inadequate. We need to talk about times in the object language, see e.g. Kusumoto (2005).

Here is a complicated natural example which illustrates this point:[8]

(4) a. His twin called after him to hurry up, and he must have done so, because a second later, he had gone. (English)

b.

Bliznakăt	mu	izvika	sled	nego	da	pobărza	i	toj
twin	him	shout.PFV.PST	after	him	COMP	hurry.PFV.PRS	and	he

javno	beše	bărzal,	zaštoto	sekunda	po-kăsno
clearly	be.PST.AUX	hurry.IPFV.PTCP	because	second	later

be	izčeznal.
be.PST.AUX	disappear.PFV.PTCP

(Bulgarian)

There is no way to account for anaphoric temporal expressions like *po-kăsno* 'later' in natural language if one cannot keep track of times encountered earlier in the discourse. These times should definitely be represented at LF (logical form). The example (4b) is not chosen arbitrarily, since it illustrates nicely how compound tenses like the Bulgarian pluperfect interact with temporal adverbials, a topic we will return to in Section 3.[9]

Given that times and tenses should be represented syntactically at LF, we probably want to work with structures like in Figure 1 or Figure 2. Both trees take the syntax-semantics interface in the temporal domain seriously, albeit in different ways.

In practice, the movement theory in Figure 1 is often tacitly assumed in the literature; see for instance Grønn (2004) and Mueller-Reichau (2026 [this volume]). It captures the idea that semantic tenses and aspects have their own projections and are not interpreted at the verb. A past (or future) interpretation arises with

[8](J.K. Rowling, *Harry Potter and the Philosopher's Stone*), Parasol corpus data (von Waldenfels & Meyer 2006).

[9]As pointed out by my Bulgarian consultants, the adverbial *javno* 'clearly' gives an additional evidential flavour to the perfect in (4b).

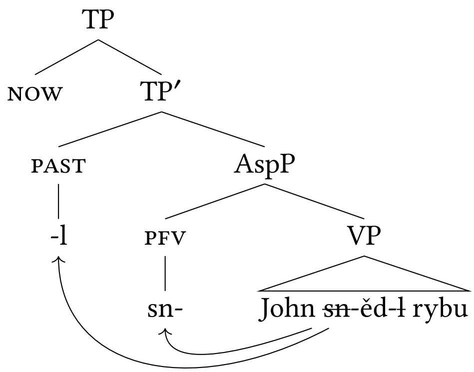

Figure 1: Movement account of (2a)

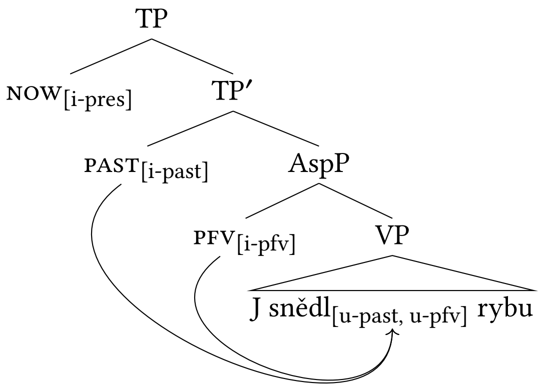

Figure 2: Feature account of (2a)

the corresponding morphology, which moves to TP′, where it gets a semantic interpretation (as an operator/temporal quantifier or temporal pronoun). In the last step, the semantic past is anchored to the evaluation time, which we can treat as a covert pronoun NOW of type *i*.

Alternatively, as in Figure 2, PAST and PFV are abstract semantic operators (with interpretable features) which license the corresponding morphology on the verb in a certain checking relation. A feature checking theory for Russian tense and aspect – from interpretable to uninterpretable features – is adopted in Mezhevich (2008) and Grønn & von Stechow (2012).

A full analysis of Slavic tense and aspect must in any case rely on certain abstract (covert) semantic operators. A clear example is the imperfective interpretation (from a null suffix) of the bare Czech verb form *jedl* '≈ was eating' in (2b). The morphological perfective present in Slavic also requires an abstract FUT at LF since there is no overt future suffix in Slavic perfectives.

We have still not said much about what a semantic PAST (or FUT) should look like, i.e. the denotation of ⟦PAST⟧ in Figure 1 and Figure 2. There are two competing approaches on the market: quantificational tenses (adopted in Section 2.2 and onward) vs. referential tenses.

The two theories come in various flavours. A simple view of the referential approach is to think of past/future tenses as denoting a temporal pronoun *then* of type *i*, which picks out a salient past/future time.

The referential/anaphoric approach is due to Barbara Partee, who proposed the following analogy between tenses and pronouns: "I will argue that the tenses have a range of uses which parallels that of the pronouns, including a contrast between deictic (demonstrative) and anaphoric use, and that this range of uses argues in favour of representing the tenses in terms of variables and not exclusively as sentence operators" (Partee 1973: 601).

Although her influence on the development of formal semantics in Slavic linguistics is incalculable, due to her regular teaching in Moscow for many years, she has never worked on tense in Russian or any other Slavic language. Her conjecture in Partee (1973) was that morphological tense, e.g. *-ed* in English or *-v/-la* in Ukrainian, could have a pronominal (referential) semantics, while temporal auxiliaries (see Section 3) typically have a quantificational semantics as existential time shifters ($\exists t$…). We note for the record that nobody has sought to explain the difference between morphological future tenses in Slavic (e.g. Ukrainian *matime* 'will have' in (1b) above) and periphrastic future tenses (alternative construction in Ukrainian: *bude mati ditej* 'will have children') in terms of this dichotomy.

The anaphoric flavour of tenses can be captured in the quantificational approach with a contextual domain restriction, a variable *C*. Some of the more elaborate quantificational and referential tense theories may ultimately be simply notational variants of each other, or maybe we need both denotations co-existing in one single system in order to account for different phenomena, as argued in Grønn (2015). Are there any language-specific, Slavic-internal, reasons to choose one approach over the other?

For Slavic, I am only aware of one non-trivial empirical argument put forward for the existence of referential tenses in a static system, i.e. when we consider tenses only at the sentence level.[10] Sharvit (2014) argues that Polish must have a referential past tense on the basis of examples from BEFORE-adjuncts:

(5) Ania przyszła na przyjęcie [zanim Marcin przyszedł.]
Ania come.PFV.PST to party before Marcin come.PFV.PST
'Ania came to the party before Marcin came.' (Polish; Sharvit 2014: 2)

[10] See Grønn (2015) for a contextual, dynamic (as opposed to static) approach to tense and aspect.

Sharvit (2014) makes a valid point that the complement of the temporal preposition/connective should denote a definite time. This is transparent in Polish: *zanim* 'before that (time)'. Accordingly, she claims (with sophisticated arguments), there is no place for a quantificational past in the adjunct, and the past must receive a referential interpretation. Hence, Polish (and also English) has a referential past.[11]

However, in Grønn & von Stechow (2012) we argue that temporal adjuncts in Slavic have even more structure than assumed by Sharvit (2014). For instance, Ukrainian/Polish *pislja toho, jak* (cf. (1b))/*po tym, jak* 'after that how' points to a temporal abstraction (*jak* 'how'). In Grønn & von Stechow (2012), we suggest a way in which this allows us to have a quantificational deictic past inside the *jak*-'how'-clause while keeping the idea that BEFORE/AFTER takes a definite time as input, see also Section 4 below.

It is time now, however, to present a fully explicit compositional and truth-conditional account of tense and aspect in Slavic. For historical reasons I will choose Ukrainian data to illustrate the general architecture.

2.2 A compositional tense semantics for Ukrainian

The insights of Klein (1995) were soon to be developed into a coherent compositional theory. For non-Slavic languages, this enterprise was first carried out in Kratzer (1998) (with a referential semantics for PAST), while the first compositional account for Slavic is probably due to Paslawska & von Stechow (1999), an unpublished paper that started circulating before the new millennium.

Arnim von Stechow, who like Krifka and Klein was not a Slavicist, had been Angelika Kratzer's and Irene Heim's teacher of semantics in Konstanz in the late 1970s. His mother was Ukrainian, and he was therefore eager to look at the Slavic data when he established contact with Alla Paslawska, a professor of German from Lviv.

Their point of departure is the following simple sentence:

(6) Vona posnidala.
she eat.breakfast.PFV.PST
'She ate breakfast.' (Ukrainian; Paslawska & von Stechow 1999)

The exact location of semantic tense and aspect in the syntax is not an issue which needs to bother the semanticists. What is important is the relative hierarchy of the categories (tense above aspect), their semantic type and interpretation.

[11]According to Sharvit (2014), this distinguishes Polish from Japanese, which has a quantificational past, and therefore can never use a past in BEFORE-adjuncts.

The aspectual operator is in Paslawska & von Stechow (1999) a function from an event description, a VP of type $\langle v, t\rangle$, to a predicate of times, a set of times of type $\langle i, t\rangle$. On this view, aspect closes off the event argument ($\exists e$) and relates the run time of the event *e* to a reference time *t*:[12]

(7) $[\![\text{po-}_{\langle\langle v,t\rangle,\langle i,t\rangle\rangle}]\!] = \lambda P_{\langle v,t\rangle}\lambda t.\exists e[P(e)\,\&\,e \subseteq t]$ (perfective aspect)

The value of the reference time (Klein's assertion time) comes from tense (often in combination with temporal adverbials).[13] Past (or future) tense thus takes a predicate of times as input (the type of AspP, after aspect has done its job) and returns another predicate of times. Times are partially ordered by the relation $\prec$ 'before' and $\succ$ 'after'.[14]

(8) $[\![\text{-l}_{\langle\langle i,t\rangle,\langle i,t\rangle\rangle}]\!] = \lambda P_{\langle i,t\rangle}\lambda t.\exists t'[P(t')\,\&\,t' \prec t]$ (relative past)

On this view, the past tense is inherently relative. It can end up being deictic, but need not if it occurs embedded in a subordinate clause. The reference time, which is involved in the aspectual relation, gets existentially closed by tense ($\exists t'$), hence this is a quantificational approach.[15]

We want to end up with the following truth conditions: There is an event *e* of her eating breakfast and a time *t* before now, such that *e* is completed at *t*. Above the PAST operator, we therefore insert (by default) the covert pronoun NOW of type *i*, denoting the speech time, a distinguished variable s^*. We finally get an LF which ends in a truth value of type *t*, cf. (9) and Figure 3.

(9) $\exists e \exists t[\text{SHE EATS BREAKFAST}(e)\,\&\,t \prec s^*\,\&\,e \subseteq t]$

[12]The condition $e \subseteq t$ could be more transparently written as $\tau(e) \subseteq t$, where τ is a function that gives us the temporal trace of *e*. I will ignore this detail below.

[13]Paslawska & von Stechow (1999) contains also the first discussion of the contribution of temporal adverbials like *včora* 'yesterday' and negation in the compositional architecture.

[14]In fact, the analysis in Paslawska & von Stechow (1999) is somewhat more complicated since the authors wanted to capture the so-called perfect readings in Ukrainian (*(will) have P*), readings which – with the exception of the pluperfect – are not in any obvious way morphosyntactically encoded in this language. In the influential paper (Paslawska & von Stechow 2003), the same idea was also applied to Russian, see Mueller-Reichau (2026 [this volume]) for a discussion. However, at the doctoral disputation in Oslo of Grønn (2004), von Stechow (p.c.) rejected this part of his earlier analysis, most notably the idea that perfective morphology in East Slavic could be ambiguous between a semantic perfective operator and a semantic perfect operator. I will therefore ignore this aspect of the analysis.

[15]In their published work on Russian a few years later (Paslawska & von Stechow 2003), the authors switched to a referential tense, apparently without any deeper motivation. Indeed, the choice between a quantificational and referential semantics for past tense is often rather arbitrary.

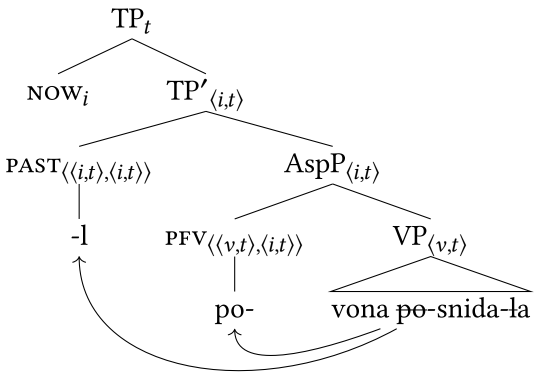

Figure 3: Composition of (6)

For comparison with the synthetic perfective past in (6) let us also introduce a synthetic imperfective future (10a), where the two temporal relations involved are reversed.[16]

(10) a. Ja perepysuvatymu stattju.
I rewrite.IPFV.FUT article
'I will rewrite the article.' (Ukrainian; Paslawska & von Stechow 1999)

b. $\exists e \exists t[\text{I REWRITE THE ARTICLE}(e)\,\&\, t \succ s^{*} \,\&\, t \subseteq e]$

(11) a. $[\![\text{uv}_{\langle\langle v,t\rangle,\langle i,t\rangle\rangle}]\!] = \lambda P_{\langle v,t\rangle} \lambda t.\exists e[P(e)\,\&\, t \subseteq e]$ (imperfective aspect)

b. $[\![\text{mu}_{\langle\langle i,t\rangle,\langle i,t\rangle\rangle}]\!] = \lambda P_{\langle i,t\rangle} \lambda t.\exists t'[P(t')\,\&\, t' \succ t]$ (relative future)

The compositionality in Figure 4 is straightforward. For matrix sentences, we always assume that the evaluation time (the highest tense) is the speech time, which is available here in the final step in the calculations of the truth conditions through a covert pronoun NOW.

Thus, following most contemporary work in tense semantics, we should carefully separate the "lower" reference time (i.e. the assertion, predication or topic time) from the "higher" evaluation time (i.e. the perspective time, temporal anchor or temporal centre). The PAST and FUTURE tenses encode relations between the reference time and the evaluation time. The present tense, on the other hand,

[16]Ukrainian has a rich system of future grams. The imperfective synthetic future analysed here is unusual in Slavic. According to Danylenko (2011), the synthetic future is derived from an underlying East Slavic periphrastic construction with *mati*. However, contrary to what one might expect, *mati* in this construction was not the HAVE-auxiliary, but a de-inceptive imperfective auxiliary with the meaning 'to take'.

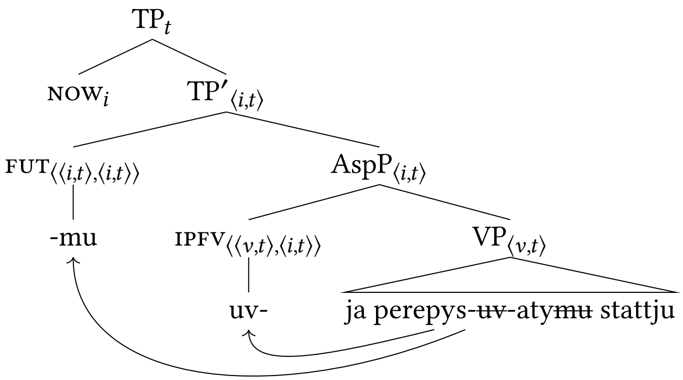

Figure 4: Composition of (10a)

is not necessarily analysed as relational. In Section 4 below on embedded tenses, I will argue for a particular view of the present tense in Slavic.

The tenses and aspects discussed above are transparent in the sense that the morphology (suffix or prefix) points to the interpretation (an operator). However, we should also mention the following case which is found in most Slavic languages:

(12) Vona posnidaje.
she eat.breakfast.PFV.PRS
'She will eat breakfast.' (Ukrainian)

A perfective with present tense morphology is coerced here into having a future tense interpretation. Hence, at LF, we assume the presence of an operator FUT with the same semantics as (11b) above, the mirror image of a semantic PAST.

The reason for this coercion into the future is common knowledge among Slavic semanticists: The present tense denotes the utterance time s^*, a very short interval, in fact a moment. However, the perfective aspect introduces the inclusion relation $e \subseteq s^*$, which does not make sense semantically. No event is small enough to be included in the utterance time.[17]

3 Compound tenses: Temporal auxiliaries

The main semantic function of temporal auxiliaries is to shift the reference time of the prejacent verb (participle or infinitive) forward or backward. They are genuine quantifiers over times of the type $\langle\langle i,t\rangle,\langle i,t\rangle\rangle$.

[17] Cf. also Todorović (2015) on the future interpretation of perfective aspect with present tense morphology in Serbian.

Typical forward (future) shifters are the auxiliaries *będzie*/*bude* in Polish/ Ukrainian and *šte*/*će* in Bulgarian/Croatian. Macedonian has a backward shifter of the Germanic/Romance type, namely *ima* 'have', but the characteristic Slavic auxiliary, both in the compound future and perfect, is 'be'.

The semantic status of the perfect BE-auxiliary is rather unclear in many Slavic languages – as it may sometimes be semantically empty – and its morpho-syntactic representation is also often reduced, a "truncated perfect". In East Slavic languages like Ukrainian the perfect auxiliary has disappeared altogether in the present tense and the *v*- or *l*-participle has been reinterpreted as a general past tense.

Consider the following architecture:

(13) Template for compound tenses:
$[_{TP}$ Tense $[_{shifter}$ Forward/Backward $[_{AspP}$ Aspect $[\nu P]]]]$

It follows from (13) that temporal shifters are tensed and therefore come with their own verbal morphology. Thus, Polish/Ukrainian *będzie*/*bude*, which is often called a "future" tense, is formally a present tense with a lexical semantics as a forward (future) shifter.

Furthermore, the temporal auxiliaries subcategorise for a complement of a particular kind, a participle or infinitive. There is some variation from construction to construction, from language to language, with not always predictable semantic effects.

There is also the question of whether the various compound tenses in some or several Slavic languages can have a more subtle semantics than simply shifting the reference time back and forth. Bulgarian stands out in this respect, and we will discuss below the seminal work of Izvorski (1997) and Pancheva (2003), who happen to be the same person working on the Bulgarian perfect before and after marriage.

3.1 Periphrastic future

The temporal auxiliaries of the stem *bud*- 'will' change the reference time of the prejacent to a future time in various Slavic languages. No Slavic language seems to have an equivalent forward shifter in the past (cf. English *would*), since the corresponding stem *byl*- is either a past existential verb, a past copula or, in some languages, a past shifter (pluperfect).

Paslawska & von Stechow (1999) provide the following pair for the periphrastic imperfective future in Ukrainian, where the first one is standard and the second dialectal.

(14) a. Ja budu vidviduvaty svoho dida.
I will.PRS visit.IPFV.INF my grandpa
'I will visit my grandpa.' (Ukrainian)

b. Ja budu vidviduvala svoho dida.
I will.PRS visit.IPFV.PTCP my grandpa
'I will visit my grandpa.' (dialectal Ukrainian)

The authors note that (14b) with a past participle does not license a semantic backshifting. Błaszczak et al. (2014: 13) make the same remark for the "tenseless" participle under the *(będ-)*-auxiliary in Polish. Recall also the Croatian periphrastic with a perfective participle *bude završila* in (1c). Despite the morphological similarities, none of these *bud*-variants correspond semantically to the Germanic future perfect of the type: *I will have visited him.*[18]

In fact, (14a) with an infinitive prejacent and (14b) are completely synonymous, indicating that the subcategorisation feature of *bud-* is semantically inert.[19] The coexistence of two synonymous future *bud*-constructions is puzzling.

As discussed extensively in Whaley (2000), the complement is always the *l*-participle in South Slavic languages, while in North Slavic the verbal complement is infinitival with imperfective marking. Polish, Kashubian, and Ukrainian dialects are exceptional in allowing for both variants.

Are there any semantic differences between the periphrastic future and the synthetic perfective present (semantically coerced into a future) apart from the overtly marked aspectual distinctions? Consider the formalization of the Ukrainian auxiliary given in Paslawska & von Stechow (1999):

(15) $[\![\text{bud-}]\!] = \lambda P_{\langle i,t\rangle} \lambda t.\exists t'[P(t') \,\&\, t' \mid\!\succ t]$ (extended relative future)

We can interpret $t' \mid\!\succ t$ as meaning that t' is an interval which has t as its left boundary (a kind of extended now towards the future). The semantics given in (15) anticipates later work on Polish.

[18] In Bulgarian, with the future-like WANT-auxiliary, the future perfect is, in principle, acceptable:

(i) Dokato pristigneš, i veče šte săm posetil djado.
until come.PFV.PRS.2SG and already want.PRS.AUX be.PRS.1SG.AUX visit.PFV.PST grandpa
'Before you come, I will already have visited grandpa.' (Bulgarian)

[19] A reviewer correctly points out that there is more to say about the aspectual marking on the prejacent. Why does the auxiliary in Ukrainian or Polish subcategorise for the imperfective aspect? Unfortunately, I do not have a good answer, at least not a short one.

According to Błaszczak et al. (2014: 2), "Notions such as 'plan', 'prearrangement', 'holding true at the moment of speaking' play a crucial role in an explanation of [...] the use/meaning [of the periphrastic future]." We can try to combine this insight for Polish with the formula for Ukrainian in (15). My very sketchy truth conditions are given in (16b), based on (16a).

(16) a. Jan będzie jadł obiad.
Jan will.PRS.AUX eat.IPFV.PTCP lunch
'Jan will eat / will be eating lunch.' (Polish; Błaszczak et al. 2014)

b. $\exists t \exists e[t \mid\succ s^* \mathbin{\&} \textsc{Jan eats lunch}(e) \mathbin{\&} e \circ t \mathbin{\&} \textsc{plan}(\textsc{Jan}, e, \textsc{lb}(t))]$

In words: The event *e* of Jan's eating lunch overlaps with the reference time *t* (the most general imperfective temporal relation). Jan has a plan at the left boundary LB of the reference time, i.e. at the utterance time s^*, of carrying out *e*.

However, Błaszczak et al. (2014: 2) presumably have in mind something different, since they claim that the future reference "can be reduced to one basic fact, namely: in all these constructions there is a perfective present tense element. The role of the perfective aspect is to forward-shift the reference time and to locate it (right) after the speech time, hence deriving the future time reference."

Their claim is that the *bud-/ będ-*stem is perfective.[20] If we take this alleged perfectivity to be inchoative, denoting the transition into an event/state in the future, we could explain why the infinitival complement must carry the imperfective aspect, as imperfective marking on infinitives under phasal verbs is characteristic of Slavic aspect. But in absence of any formal proposal along these lines, I will not go further into this hypothesis. The traditional view is that the BE-auxiliary in Slavic is aspectually inert.

The formula in (15) does not say anything about aspect. The input to the operator, the prejacent, is a predicate of times, which already contains all the aspectual information (both participles and infinitives are marked for aspect in Slavic).

A topic which has not been properly investigated is the behaviour of compound tenses in embedded contexts. Consider the periphrastic future in a relative clause under a past matrix verb:

(17) U misti vidkryly novyj akvapark, jakyj bude
in city open.PFV.PST.PL new aquapark which will.PRS.AUX
pracjuvaty ščonedili.
work.IPFV.INF every.Sunday
'In our city, they opened a new aquapark, which will be open on Sundays.'
(Ukrainian)

[20] This view is controversial and explicitly contested in Pitsch (2015).

Given an analysis of *bud-* as a relative future, one might expect a forward-shifting (future in the past) of the relative clause. But this is not what we get. The sentence can only mean that the aquapark will be open on Sundays *after* the speech time.

So we see that *bud-* with its present tense morphology gets a purely deictic interpretation, and gives us a reference time strictly after *s**. This fact goes against the established wisdom in Slavic linguistics that Slavic has relative tenses. I will return to this point in Section 4.

One final point concerns the modality of the future. Above, we have assumed that there is one actual future, the mirror image of the past. Such an analysis may work for *bud-*/*będ-*, depending on the researcher's metaphysical position.

However, the Bulgarian WANT-auxiliary, *šte* in (1d) above, has a stronger modal flavour and remains invariant throughout the whole present tense paradigm. According to Migdalski (2006: 102), the auxiliary is a modal generated above tense. All the same, even a modal must be anchored in time. It should be interesting to examine the temporal profile of *šte* (and Croatian *će*, cf. (1c)) in embedded contexts.[21]

3.2 The perfect

Most contemporary approaches to the syntax-semantics interface of European languages assume a structure as in (18), where tense, perfect and aspect are represented as functional heads.[22]

(18) Temporal template for the perfect:
$[_{\text{TP}}$ Tense $[_{\text{PerfP}}$ Perfect $[_{\text{AspP}}$ Aspect $[\nu\text{P}]]]]$

With the exception of Russian, most Slavic languages motivate this three-way distinction between tense, perfect, and aspect. The auxiliary comes in two versions; the HAVE-perfect, which we find in Macedonian (and English), and the BE-perfect. The latter clearly derives from a copula-construction. The auxiliary selection in contemporary Indo-European languages is not in any obvious way correlated with semantic (temporal) effects, but several authors report syntactic differences between HAVE- and BE-perfects, cf. Iatridou et al. (2001) and Migdalski (2006).

The semantic operator associated with the perfect modifies the reference time. As with the periphrastic future in the previous section, the perfect arguably has

[21] There is a lot of subtle variation to be discovered. For instance, my Croatian consultant strongly preferred the present tense version *će* in the relative clause in (1c) under a matrix past, where my Bulgarian and Ukrainian consultants used a past modal-temporal auxiliary.

[22] See Todorović (2016) for a syntactically oriented discussion.

the type $\langle\langle i,t\rangle,\langle i,t\rangle\rangle$. A natural parallel with the future suggests that the higher tense layer provides the temporal anchoring of the future/perfect auxiliary. At the syntax-semantics interface, this can be achieved in at least two ways: Either the morphological tense on the auxiliary moves to TP, or an abstract semantic tense in TP licenses the morphological tense feature on the auxiliary in PartP, cf. the discussion above of the two strategies in Figure 1 vs. Figure 2.

A nice and uncontroversial point is that aspect in Slavic can play a crucial role in distinguishing between the different perfect meanings in languages with an overt auxiliary and aspect marking on the participle. This was first demonstrated and formally accounted for in Pancheva (2003) for Bulgarian, which we will see below.

3.2.1 General past vs. real perfect

In Ukrainian, the present tense auxiliary has disappeared and the participle is reinterpreted as a general past (19a). For Czech, a similar situation obtains in the 3rd person (19b), but in the 1st (and 2nd) person, the auxiliary is still present, correlating with pro-drop and syntactically occurring in the second position, cf. (19c) and (19d).

(19) a. Včora ja zjiv rybu.
yesterday I eat.PFV.PST fish
'Yesterday I ate up the fish.' (Ukrainian)

b. Včera John snědl rybu.
yesterday John eat.PFV.PST fish
'Yesterday John ate up the fish.' (Czech, 3rd person)

c. Včera jsem snědl rybu.
yesterday be.PRS.1SG.AUX eat.PFV.PTCP fish
'Yesterday I ate up the fish.' (Czech, 1st person)

d. Snědl jsem rybu.
eat.PFV.PTCP be.PRS.1SG.AUX fish
'I ate up the fish.' (Czech, 1st person)

Czech appears to be a language where the auxiliary merely points to phi-feature distinctions and is reanalysed as a person and number marker. In some Slavic languages, this semantic impoverishment of the auxiliary corresponds to their morphological reduction from auxiliaries to clitics (cf. (19d)) and further to affixes (Migdalski 2006: 47). We can analyse the auxiliary in (19c)/(19d) as semantically empty, as an identity function without any temporal semantics.

Compare the general past interpretation above with a clear case of a real perfect, a pluperfect in Ukrainian:

(20) Vona bula posnidala.
she be.PST.AUX eat.breakfast.PFV.PTCP
'She had eaten breakfast.' (dialectal Ukrainian)

Note first that the *will-bud*-parallel between English and Slavic in the future construction (with present morphology on the auxiliary) does not carry over to *would* vs. *bul-*. When the auxiliary is marked with past tense, it remains a forward shifter in English, but not in Slavic.

The pluperfect in Ukrainian invites a compositional analysis with two semantic layers of past. The auxiliary subcategorises for a participle, which is informally often called a "past participle". The highest PAST at LF is obviously linked to the past marking on the auxiliary *bula*. But which piece of morphology corresponds to the backward-shifting of the PERFECT operator?

There are two possible answers. In their early work, Paslawska & von Stechow (1999: 18) claim that the second backward-shifting in the construction is due to the *l*-participle of *posnidala*: "Die Präteritumsmorphologie des Partizips besetzt T2 und bringt uns in die Vorvorzeitigkeit."[23] (T2 is the lower temporal layer, while T1 is the higher tense in their system.)

However, this is not the only possible analysis. Whaley (2000: 146f.) notes that those Slavic languages which can use the *l*-participle in the periphrastic future (notably Polish and Ukrainian) also have the pluperfect construction with the same participial form.[24] She takes this as evidence for treating the *l*-participle as unmarked for tense and semantically empty. She contrasts this with Russian, a language which does not have a pluperfect, where the *l*-participle is the carrier of past morphology with a corresponding past tense interpretation.

So, on Whaley's view, both sources of semantic pastness in the Ukrainian pluperfect stem from the auxiliary: the past marking *-l* in *bula* and the lexical meaning of the temporal auxiliary *bu-* as a backwards shifter. If the temporal shift is done by the auxiliary (and not the participle), we get a nice parallel with

[23] "The past morphology of the participle occupies T2 and brings us to the past of the past."

[24] It should be noted that the pluperfect is not commonly used in contemporary Ukrainian and Polish. Most of my consultants do recognise the construction, although they would hardly use the pluperfect themselves. According to Krzysztof Migdalski (p.c.), he mostly uses the Polish pluperfect (actively) in conditional/modal contexts, but he (passively) accepts archaic usages of the pluperfect in other contexts.

the compound future, where the forward shift clearly comes from *bud-* (and not the embedded infinitive/participle).[25]

However, the arguments remain inconclusive. Iatridou et al. (2001) argue with respect to the location of the perfect operator that we must distinguish between HAVE and BE-perfects. This is what we will do next.

3.2.2 The HAVE-perfect

The *ima–have*-perfect in Macedonian and Kashubian is the exception in Slavic (Migdalski 2006). In Iatridou et al. (2001) and Grønn & von Stechow (2020) the temporal auxiliary HAVE is analysed as a temporal quantifier of type $\langle\langle i,t\rangle,\langle i,t\rangle\rangle$. On its most simple anteriority reading, HAVE is simply the mirror image of WILL – it shifts the reference time backwards:

(21) $[\![\text{ima/have}]\!] = \lambda P_{\langle i,t\rangle}\lambda t.\exists t'[P(t') \,\&\, t' \prec t]$ (HAVE as anteriority)

The temporal auxiliary contributes the same semantics as the relative past discussed in (8) above. However, there is a difference in the morphological features. The auxiliary comes with tense morphology (here: present tense) which requires the presence at LF of a higher semantic tense (here: NOW).

Being located between aspect and tense, the perfect can be seen as a sort of second, embedded tense (Iatridou et al. 2001) that takes a set of times as input and returns a set of times. Schematically and informally, the composition goes as follows (Pancheva 2003: 285):

- Viewpoint aspect: from event time to reference time.
- Perfect: from reference time to reference time.
- Tense: from reference time to speech time.

Let us look at the analysis of a concrete example from Macedonian. The participle under *ima* invariably appears in the neuter singular:

(22) a. Petar ja ima završeno taa rabota.
Petar it have.PRS.AUX finish.PFV.PTCP that work
'Petar has finished that work.' (Macedonian; Migdalski 2006: 129)

b. $[_{\text{TP}}$ NOW$[_{\text{PerfP}}$ ima $[_{\text{AspP}}$ PFV $[_{\text{VP}}$ Petar finished work$]]]]$

[25] A reviewer agrees with this point and finds it strange that Paslawska & von Stechow (1999) treat the *l*-participle as a semantic backward-shifter in (20) and as semantically empty in the future construction in (14b).

Importantly, the perfect scopes over viewpoint aspect (here: PFV). The denotation of AspP, which is the input to the temporal shifter *ima*, is thus:

(23) $[\![\text{AspP}]\!] = \lambda t \exists e[e \subseteq t \,\&\, \text{Petar finish work}(e)]$

Applying *ima* to AspP in (23) gives us:

(24) $\lambda t' \exists t \exists e[e \subseteq t \,\&\, \text{Petar finish work}(e) \,\&\, t \prec t']$

In the next step, t' is equated to the speech time s* and we get the final truth-conditions for the sentence:

(25) $\exists t \exists e[e \subseteq t \,\&\, \text{Petar finish work}(e) \,\&\, t \prec s^*]$

The anteriority reading is available for the perfect in the right context (for instance in the pluperfect), but it is questionable whether a relative past as in (21) can account for all the perfect interpretations available in a given language (Grønn & von Stechow 2020). One could object that (25) does not capture the result state interpretation which is arguably present in (22a).

From the examples provided in Migdalski (2006), the *ima*-'have'-perfect in Macedonian and Kashubian seems to display the same range of perfect meanings found in other European languages, notably the resultative, experiential and universal. The distinction between the three readings is largely determined on the basis of the aspectual morphology/interpretation, hence the perfective marking on the participle in (22a) is responsible for the result state inference. In this respect, the inventory of perfect readings in Macedonian appears to be similar to the more common BE-perfect in other Slavic languages which we will review in more detail below.

3.3 The BE-perfect

Several Slavic languages make use of the BE-auxiliary in the perfect. However, it is often claimed that in, say, BCMS, the present perfect is used as the general narrative past tense, without the characteristic perfect readings (Migdalski 2006: 47). I will therefore mostly focus on Bulgarian below, a language where the perfect shows a great variety of semantic readings with a corresponding elaborate morpho-syntax.

As noted above, Iatridou et al. (2001) argue that HAVE and BE-perfects should be represented by two different structures. They provide data from reduced relative clauses in Bulgarian to support their view that the BE-auxiliary is semantically empty:

(26)

Zapoznax	se	s	ženata,	[napisala	knigata].
acquaint.PFV.PST.1SG	REFL	with	woman.the	write.PFV.PTCP	book.the

'I met the woman having written the book.' (Bulgarian)

Their point is that in languages with the HAVE-perfect, the auxiliary itself expresses the perfect meaning and can therefore never be dropped (cf. English **the woman written…*). So what then gives us the perfect operator in (26)? It presumably must come from the only morphology available, namely the *l*-participle *napisala* 'written', pace Whaley (2000) and Błaszczak et al. (2014).[26]

3.3.1 Different readings and denotations of the perfect

A constant challenge for the theories of the perfect is to account for the distribution of the three main readings cross-linguistically and language-internally: the RESULTATIVE, EXPERIENTIAL and UNIVERSAL perfects.

The beauty of the Slavic perfect stems from the overt marking of viewpoint aspect embedded under the perfect. The different temporal inclusion relations of the two aspectual operators play a decisive role in distinguishing between the resultative and experiential, which are often grouped together in Germanic languages as the so-called "existential perfect" in opposition to the universal perfect.

Pancheva (2003: 295) used Bulgarian data to show that the experiential and the resultative interpretations should be kept apart, since the distinction has a grammatical basis:

(27) a.

Ivan	e	stroil	pjasăčna	kula.
Ivan	be.PRS.AUX	build.IPFV.PTCP	sand	castle

'Ivan has been building a sandcastle.' (Bulgarian; "neutral" IPFV)

b.

Ivan	e	strojal	pjasăčna	kula.
Ivan	be.PRS.AUX	build.IPFV.PTCP	sand	castle

'Ivan has been building a sandcastle.' (Bulgarian; imperfect)

c.

Ivan	e	postroil	pjasăčna	kula.
Ivan	be.PRS.AUX	build.PFV.PTCP	sand	castle

'Ivan has built a sandcastle.' (Bulgarian; PFV)

The participles in the two first examples, a bare imperfective (a simplex form with an imperfective ∅-suffix) and an imperfect stem, respectively, cannot receive a resultative interpretation. The resultative must be encoded through a perfective

[26] See Grønn & von Stechow (2020) for a recent discussion.

participle (*postroil*), as in (27c). Only in the latter case can the speaker follow up with 'and now he will destroy it'.

The experiential–universal distinction can further be accounted for along the lines of Grønn (2004): The imperfect(ive) in Slavic is ambiguous between different inclusion relations: The canonical imperfective relation $t \subseteq e$ produces the universal reading in a perfect tense setting, while the complete event interpretation $e \subseteq t$ results in an experiential reading.[27] We will show how the different readings can be derived compositionally below.

A merit of Pancheva (2003) is that she could propose a uniform overall structure for the perfect, where additional grammatical components, notably aspect, make it possible to derive the three distinct readings. Her analysis is couched in the EXTENDED NOW-framework of Dowty (1979), abbreviated XN. The XN-interval and the PERFECT operator (of type $\langle\langle i,t\rangle,\langle i,t\rangle\rangle$ as before) are defined as follows (Grønn & von Stechow 2020):

(28) a. $\textsc{xn}(t,t') := t'$ is a final subinterval of t.

b. $[\![\textsc{perf-xn}]\!] = \lambda P_{\langle i,t\rangle}\lambda t'.\exists t[P(t) \,\&\, \textsc{xn}(t,t')]$ (extended now-perfect)

The semantic contribution of this perfect is to introduce the XN-interval (the perfect time span) at which the temporal predicate is true. The crucial difference from the anteriority approach in (21) above is that the XN-interval t, the perfect time span, does not properly precede the higher tense t', i.e. the speech time s^* in a present perfect, but includes it. The underlying event is therefore said to be true also at the utterance time.

A consequence of the analysis in (28b) is that modification by adverbials such as *yesterday* is predicted to be ruled out. An interval which includes the utterance time, such as XN, cannot itself be included in yesterday: $t \subseteq \textsc{yesterday} \,\&\, \textsc{xn}\,(t,s^*)$ is contradictory since it says that the utterance time is included in yesterday.

(29) a. Lo avete udito ieri. (Italian)
it have.PRS.AUX.2PL hear.PTCP yesterday

b. To ste juče čuli. (BCMS)
that be.PRS.AUX.2PL yesterday hear.PFV.PTCP

c. Nali čuxte včera. (Bulgarian)
right hear.PFV.PST.2PL yesterday

d. You heard it yesterday. (English)

[27]The experiential reading is much discussed in Slavic linguistics mostly outside the perfect, namely with respect to the factual imperfective in the simple/general past. The factual imperfective displays a whole range of complete event interpretations, see Grønn (2015), Gehrke (2026 [this volume]), and Mueller-Reichau (2026 [this volume]) for a recent discussion (mostly on Russian).

A pattern emerges that suggests a contrast between Italian/BCMS and Bulgarian/English (**have heard*). The former allow a past adverbial in a present perfect. If we assume that the present perfect is interpreted as a general past in these languages, and we use the anteriority denotation in (21), we can straightforwardly account for the acceptance of the adverbial in the structure.

A simple analysis of temporal adverbials like *yesterday* is the following:

(30) Predicate format of positional temporal adverbials, type $\langle i, t\rangle$:
$[\![\text{juče/včera}]\!] = \lambda t.t \subseteq \textsc{yesterday}$

We further assume a standard version of PREDICATE MODIFICATION so that positional adverbials (predicates of times) can restrict the tense or perfect operator (Grønn & von Stechow 2020). BCMS (and Italian) can now be analysed as follows, where the adverbial modifies AspP of type $\langle i, t\rangle$ and the auxiliary *ste* is simply a relative/general past:

(31) a. $[_{\text{TP}}\ \textsc{now}_i\ [_{\text{PerfP}}\ \text{ste}_{\langle\langle i,t\rangle,\langle i,t\rangle\rangle}\ [\text{juče}_{\langle i,t\rangle}]\ [_{\text{AspP}}\ \textsc{pfv}_{\langle\langle v,t\rangle,\langle i,t\rangle\rangle}\ [_{\text{VP}}\ \text{čuli}_{\langle v,t\rangle}]]]]$
b. $\exists t[t \prec s^* \,\&\, t \subseteq \textsc{yesterday} \,\&\, \exists e[e \subseteq t \,\&\, \textsc{you hear}(e)]]$

We note that Bulgarian in the same context (29c) uses the aorist (≈ perfective past) and English uses a simple past (29d). A natural hypothesis is that the present perfect in these languages has an XN-denotation, as in (28b), and therefore cannot co-occur with *včera*/*yesterday*.

Izvorski (1997: 232) indeed shows that the Bulgarian present perfect only allows modification by such adverbials in the special evidential form (without the auxiliary):

(32) a. Te sa došli (??včera).
they be.PRS.AUX come.PFV.PTCP yesterday
Intended: 'They have come yesterday.' (Bulgarian, perfect)

b. Te došli včera.
they come.PFV.PTCP yesterday
'They apparently came yesterday.' (Bulgarian, evidential perfect)

We will return to the evidential perfect below. For more on the present perfect puzzle, see Pancheva & von Stechow (2004) and Grønn & von Stechow (2020).

3.3.2 Resultative perfect

The resultative reading obtains when an event in the (typically recent) past brings about a result state which holds at the evaluation time. Due to the current relevance of the event at the utterance time (in the present perfect), the resultative reading obtains only with the perfective aspect in Slavic, here exemplified with Serbian:[28]

(33) a. Where have you buried the poor body? (English)

b. Gde ste sahranili jadno telo? (Serbian)
where be.PRS.AUX bury.PFV.PTCP poor body

In Slavic languages, this reading is compatible with both the conservative relative past semantics and an XN-semantics for the perfect. In the first case, we end up with the temporal relations $e \subseteq t \,\&\, t \prec s^*$, while with the XN-configuration we get $e \subseteq t \,\&\, \mathrm{XN}(t, s^*)$. In both cases, there is a subinterval of t that contains a VP-eventuality. Hence, the event is completed before the utterance time and its resultant state obtains at the utterance time (given reasonable pragmatic considerations).

In the presence of certain temporal modifiers of the perfect time span, the XN-approach may be preferable, cf. the following example.[29]

(34) a. Since the good people disfigured him, he has become cruel and hard. (English)

b. S tex por kak dobrye ljudi izurodovali ego, on
from those times COMP good people disfigure.PFV.PST him he
stal žestok i čerstv. (Russian)
become.PFV.PST cruel and hard

c. Otkato dobri xora sa go obezobrazili, e
since good people be.PRS.AUX him disfigure.PFV.PST be.PRS.AUX
stanal žestok i koravosărdečen. (Bulgarian)
become.PFV.PTCP hard and cruel

d. Od tog vremena kako su ga dobri ljudi
since those times COMP be.PRS.AUX him good people
unakazili, on je postao okrutan i
disfigure.PFV.PST he be.PRS.AUX become.PFV.PST hard and
bezdušan. (Croatian)
cruel

What interests us here are the translations. Since there is no perfect in modern Russian (Bulgakov's original), the verbs in the matrix and temporal clause of (34b) carry both perfective aspect and past tense morphology.

[28] Parasol corpus data (Umberto Eco, *The Name of the Rose*) from Grønn & von Stechow (2020).
[29] Parasol corpus data (Mixail Bulgakov, *The Master and Margarita*) from Grønn & von Stechow (2020).

The SINCE-clause gives us a set of times *after* and *including* a definite time (the complement of the temporal preposition). This is transparent in Croatian *od tog vremena kako* 'from those times when'. We will see in Section 4 how we compositionally derive temporal adverbial clauses in Slavic. Let us for now just call the definite time which is input to *od* 'since', for ϕ.

The importance of SINCE-clauses for the perfect is that they provide the left boundary (LB) of XN, as in the formalisation in (35). The formula is intended to capture the construction in Slavic, except for the Russian original.

(35) a. $[_{\text{TP}}\ \text{NOW}_i\ [_{\text{XN}}\ \text{je}_{\langle\langle i,t\rangle,\langle i,t\rangle\rangle}\ [\text{od}_{\langle\langle i\rangle,\langle i,t\rangle\rangle}\ \phi_i]\ [_{\text{AspP}}\ \text{PF}_{\langle\langle v,t\rangle,\langle i,t\rangle\rangle}\ [_{\text{VP}}\ \text{postao}_{\langle v,t\rangle}]]]]$
b. $\exists t[\text{XN}(t,s^*)\ \&\ \text{LB}(t,\phi)\ \&\ \exists e[e \subseteq t\ \&\ \text{BECOME CRUEL}(e)]]$

3.3.3 Experiential perfect

Experiential perfects signal the existence of at least one past eventuality. The underlying VP can be telic or atelic. For telic VPs in languages like English, it is difficult to come up with criteria that would separate the experiential from the resultative. However, in Slavic, the resultative-experiential distinction is grammaticalised and should follow from a proper theory of aspect.

Pancheva (2003: 297f.) illustrates the contrast through (in)compatibility with various adverbials:

(36) Maria e {pristignala / *pristigala} sega.
Maria be.PRS.AUX arrive.PFV.PTCP arrive.IPFV.PTCP now
'Maria has arrived now.' (Bulgarian)

The adverbial *sega* 'now' forces the resultative reading and is only compatible with perfective aspect.

(37) Maria e {pristigala / *pristignala} v polunošt i predi.
Maria be.PRS.AUX arrive.IPFV.PTCP arrive.PFV.PTCP in midnight and earlier
'Maria has arrived at midnight before as well.' (Bulgarian)

The indefinite existential quantification over times *predi* 'before' is, on the contrary, only compatible with the imperfective aspect and produces an experiential reading.

The simplest account of these data is to assume that the imperfective has a complete event interpretation $e \subseteq t$ in complementary distribution with the perfective.[30] One characteristic reading of the so-called (general-)factual Slavic imperfective – in past tense and elsewhere – is precisely related to indefinite-existential event quantification (Grønn 2015).[31]

These data further invite a pragmatic account along two lines: aspectual competition (PFV vs. IPFV; Grønn 2004) and tense competition (present perfect vs. other past tenses available in the language; Zhao 2022).

3.3.4 Universal perfect

The XN-theory of the perfect was designed by Dowty (1979) to account for the universal reading in English, so there is no surprise that the theory is well suited to handle the universal perfect also in Slavic.

Here are two characteristic examples from Bulgarian:

(38) a. Marija vinagi e običala Ivan.
Maria always be.PRS.AUX love.IPFV.PTCP Ivan
'Maria has always loved Ivan.' (Bulgarian; Iatridou et al. 2001: 172)

b. Ivan cjala sutrin e stroil pjasăčna kula.
Ivan all morning be.PRS.AUX build.IPFV.PTCP sand castle
'Ivan has been building a sandcastle all morning.'
(Bulgarian; Pancheva 2003: 296)

The key to the universal reading is the imperfective interpretation of the participle in combination with an adverbial whose denotation includes the utterance time (*vinagi* 'always', *cjala sutrin* 'all morning').

The imperfective has its canonical meaning $t \subseteq e$ in this construction, hence does not present the eventuality as completed at the reference time t (= XN). Since the eventuality (state) holds at a superinterval of t, it necessarily continues beyond the right boundary of the XN-interval, that is beyond the utterance time s^*.

A convenient outcome of this analysis is that it accounts for the universal reading independently of the meaning of the perfect itself (Pancheva 2003). In this light, let us look at the universal–existential-distinction – the so-called U vs. E-readings – in Slavic.[32]

[30] The analysis in Pancheva (2003) is slightly more complicated since she assumes that Bulgarian has three aspects with three different temporal inclusion relations. We ignore her so-called "neutral aspect" here, since imperfect(ive) and neutral participles have a similar semantic effect with respect to the perfect construction.

[31] See also Grønn (2004), Gehrke (2026 [this volume]), and Mueller-Reichau (2026 [this volume]).

[32] Parasol corpus data (J.K. Rowling, *Harry Potter and the Philosopher's Stone*) from Grønn & von Stechow (2020).

(39) a. The nation's owls have been behaving very unusually today. (English)

b. [...] su se naše nacionalne sove danas ponašale
be.PRS.AUX REFL our national owls today behave.IPFV.PTCP
vrlo neobično. (BCMS)
very unusually

c. [...] dnes sovite v stranata sa projavili mnogo
today owls.the in country.the be.PRS.AUX show.PFV.PTCP very
stranno povedenie. (Bulgarian)
strange behaviour

The two corpus translations are truth-conditionally not equivalent. Only the BCMS translation in (39b), with an imperfective participle, conveys the universal reading of the English original. The Bulgarian translation (39c) with the perfective participle instantiates an existential-resultative reading. Let us see how this is captured by the theory outlined above.

The adverbial *dnes*/*danas* 'today' modifies XN by restricting the interval to the time of the utterance day which precedes and includes the speech time. It must accordingly scope above aspect and below the perfect, which existentially closes off XN. For the U-reading we have an imperfective aspectual relation below *today*, forcing the owl's unusual behaviour to continue to hold at the utterance time. For both English (-*ing* in (39a)) and BCMS, this aspect is overtly marked.

In the Bulgarian E-reading of (39c), the embedded perfective predicate is said to hold at some subinterval of XN, i.e. the owls misbehave only for some time during the part of today that precedes the utterance time. This is a proper past interpretation due to a slightly inaccurate translation of the continuous interpretation in the English original.

Here are the relevant truth-conditions, spelled out with event(ualities) and viewpoint aspect:

(40) a. $\exists t[\textsc{xn}(t, s^*) \,\&\, t \subseteq \textsc{today} \,\&\, \exists e[\textsc{owls' behaviour}(e) \,\&\, t \subseteq e]]$ (U-reading)

b. $\exists t[\textsc{xn}(t, s^*) \,\&\, t \subseteq \textsc{today} \,\&\, \exists e[\textsc{owls' behaviour}(e) \,\&\, e \subseteq t]]$ (E-reading)

It follows from this account that the universal reading of the present perfect is predicted to compete with a simple present tense. How this competition unfolds is an open issue in Slavic, but the patterns below suggest that this issue deserves further attention:[33]

[33]Parasol corpus data (J.K. Rowling, *Harry Potter and the Philosopher's Stone*) from Grønn & von Stechow (2020).

(41) a. How long have I been in here? (English U-perfect)

b. Ot kolko vreme săm tuk? (Bulgarian present tense)
from how.much time be.PRS.1SG here

c. Koliko sam već ovde? (Serbian present tense)
how.long be.PRS.1SG already here

The English original has a present perfect with a durative measure adverbial *how long* and the underlying VP must hold both at the utterance time and for some time extending backwards into the past, but the Slavic corpus translations both prefer a simple present for this scenario, cf. Grønn & von Stechow (2020). I am not aware of any accounts of the semantics of the present tense in Slavic which could shed light on this issue.

3.3.5 Past perfect

The past perfect (pluperfect) consists of two nested layers of "pastness": The semantic past associated with the past tense morphology of the auxiliary and, depending on the author (as we saw above), either a backward-shifting operator lexically expressed by the auxiliary or a past operator originating with the participle.

The good old anteriority theory (relative past) in (21) also seems to account straightforwardly for past BE-perfects, as shown in Grønn & von Stechow (2020). In interaction with the two past layers, temporal adverbials may create an ambiguity, as illustrated here for Bulgarian with the auxiliary in the aorist past:[34]

(42) Marija beše došla v 6 časov.
Mary be.PFV.PST.AUX leave.PFV.PTCP at six o'clock
'Mary had left at six o'clock.' (Bulgarian)

In the formulae, we assume for simplicity that the perfect operator originates somehow with the BE-auxiliary. I am here only committed to the semantic types, not the labelling of syntactic categories.

(43) $[\textsc{now}_i\ [_{it}\ \textsc{past}_{\langle it\rangle,\langle it\rangle}\ [_{it}\ \lambda_1\ \textsc{at}\ 6(t_1)]\ [_{it}\ \textsc{be}_{\langle it\rangle,\langle it\rangle}\ [_{it}\ \textsc{pfv}_{\langle vt\rangle,\langle it\rangle}\ [_{vt}\ \textsc{leave}]]]]]$
$(\exists t \prec s^*)\, t$ AT 6 O'CLOCK & $(\exists t' \prec t)(\exists e)$ MARY LEAVES(e) & $e \subseteq t'$

(44) $[\textsc{now}_i\ [_{it}\ \textsc{past}_{\langle it\rangle,\langle it\rangle}\ [_{it}\ \textsc{be}_{\langle it\rangle,\langle it\rangle}\ [_{it}\ \lambda_1\ \textsc{at}\ 6(t_1)]\ [_{it}\ \textsc{pfv}_{\langle vt\rangle,\langle it\rangle}\ [_{vt}\ \textsc{leave}]]]]]$
$(\exists t \prec s^*)(\exists t' \prec t)\, t'$ AT 6 O'CLOCK & $(\exists e)$ MARY LEAVES(e) & $e \subseteq t'$

[34] The pluperfect is quite rare in contemporary BCMS, hence we once more draw on data from Bulgarian.

The adverbial of type $\langle i, t \rangle$ can modify either the higher (tense) time or the lower (perfect) time. (43) is a past-time modification: the leaving is before six. (44) is a perfect-time modification: the leaving is at six.

Typically, the context disambiguates the adverbial. Thus, in the following authentic example, we obviously have a low temporal adverbial which modifies the perfect time span at which the state holds.[35]

(45) a. In the first forty days a boy had been with him. (English)

b. Prez părvite četirideset dni be vzimal edno
in first.the forty days be.PFV.PST.AUX take.IPFV.PTCP a
momče săs sebe si. (Bulgarian)
boy with him REFL

The Slavic perfect is a rich field, and we cannot do justice to all possible tense combinations. Migdalski (2006: 98) provides examples with a double participle construction in BCMS and Bulgarian.[36] In the pair of sentences below, the temporal auxiliary is itself in the perfect:

(46) a. Ja sam bio pročitao knjigu.
I be.PRS.AUX.1SG be.AUX.PTCP read.PFV.PTCP book
'I had finished reading the book.' (BCMS)

b. Az săm bil četjal knigata.
I be.PRS.AUX.1SG be.AUX.PTCP read.IPFV.PTCP book.the
'I am said to have been reading the book.' (Bulgarian)

The constructions in the two languages are similar on the surface, but the meaning is different. The BCMS example (46a) is semantically a standard pluperfect as discussed above, while the Bulgarian construction in (46b) is an evidential perfect. This brings us to the next section.

3.3.6 Evidential perfect

The evidential perfect is also known as the indirect perfect or the inferential/renarrated/unwitnessed mood, hence the peculiar translation of (46b) above as 'I am said to have been reading the book'. The speaker has only indirect evidence for the event. Another typical translation could be 'I had apparently read the book'.

[35] RuN-Euro corpus data (Grønn 2009) (Ernest Hemingway, *The Old Man and the Sea*) from Grønn & von Stechow (2020).

[36] Recall the Croatian *je bila rekla* 'had said' in (1c).

The evidential perfect is found in Bulgarian and Macedonian. As for Macedonian, the presence in the tense system of the HAVE-auxiliary *ima*, as we saw above (21), changes the distribution of the BE-perfect. The HAVE-perfect is gradually taking over the traditional functions of the BE-perfect to such an extent that only the evidential functions remain for the latter, according to Graves (2000) and Lindstedt (2000).

The situation is different in Bulgarian where the traditional BE-perfect happily coexists with the evidential perfect. In the 3rd person the two paradigms are formally distinct, with pragmatic/semantic effects:

(47) Ajnstajn (#e) posetil Prinstăn.
Einstein be.PRS.AUX visit.PFV.PTCP Princeton
'Einstein apparently visited Princeton.' (Bulgarian; Izvorski 1997: 233)

In the standard present perfect, the auxiliary *e* 'is' must be used. However, in both English and Bulgarian, the so-called Einstein-sentences sound bad in the present perfect since Einstein is dead and should not be ascribed any present perfect-like properties at the utterance time (often referred to as a "life-time effect" of the present perfect). Still, in the 3rd person, the perfect participle can be used in the Einstein-scenario *without* the auxiliary. In this case, the sentence receives some kind of hearsay interpretation and is perfectly fine with Einstein as the subject.

Thus, certain pragmatic and/or semantic restrictions of the standard perfect disappear in the evidential construction. Still, the two constructions on the surface share a common core (the *l*-participle), and they are formally indistinguishable in the 1st and 2nd persons. Semantically, the two constructions appear to have in common the idea that a past event is evaluated from a distance, either through its current result state (the temporal perfect) or through the speaker's second-hand information or reasoning at the utterance time (the evidential perfect) (Gvozdanović 1995).

In Izvorski's (Pancheva's) influential study, the first formal account of this phenomenon in Bulgarian, she tried to unify the temporal present perfect (direct evidence) and the epistemic evidential perfect (indirect evidence) in a Kratzerian framework (Izvorski 1997: 225).

Her formalisation of the evidential perfect as a universal epistemic modal splits into two parts (here simplified):

(48) Assertion: p is necessary in view of the speaker's knowledge state.
Presupposition: The speaker has indirect evidence for p.

To bring about a parallel with the temporal perfect, she assumes an analysis of the latter as a resultative stativiser.[37] The reasoning goes as follows:

The present perfect in the TEMPORAL domain asserts that the consequent state of a past eventuality, CS(*e*), holds at the utterance time of the speaker. This supposedly echoes the evidential perfect in the MODAL domain which asserts that the proposition *p* holds in (all) the worlds epistemically accessible for the speaker.

Furthermore, the present perfect in the TEMPORAL domain does not entail that the underlying eventuality *e* holds at the utterance time of the speaker. And, in a similar vein, the evidential perfect in the MODAL domain does not entail that the proposition *p* is true in the world of the speaker (Izvorski 1997: 226).

The parallel above breaks down at one place, namely in the universal quantification over epistemically accessible worlds (the present perfect does not involve universal temporal quantification).

Izvorski's analysis raises many questions, and she concludes: "Further formalization is needed [...] Still, I believe that even formulating these questions is an important step towards our better understanding of the links between temporal and modal categories" (Izvorski 1997: 235). And 30 years later, evidentiality is indeed a hot topic in semantics.

The "evidential perfect" may turn out to be a misnomer, and the parallels with the temporal perfect seem to be more or less abandoned in contemporary work, such as Koev (2017), who also argues against a modal account of the Bulgarian evidential marker. Koev (2011) claims that evidential marking in fact entails the scope proposition:

(49) Ivan celunal Marija.
Ivan kiss.PFV.PTCP Maria
'(I was told that) Ivan kissed Maria.' (Bulgarian; Koev 2011: 116)

Pace Izvorski (1997), a sentence like (49) allegedly entails that Ivan kissed Maria. Thus, the evidential is no longer a modal. Instead Koev (2011) develops a purely temporal account. The temporal ontology is enriched with a new concept: the LEARNING EVENT, i.e. the event/time when the speaker acquired the relevant evidence for the claim. A paraphrase of (49) which reflects this idea, is: 'Ivan kissed Maria, as I learned later.' (Koev 2011: 124). The learning event is after the described event, but still before the speech event. The radical new idea is that

[37] As discussed in Grønn & von Stechow (2020), this is a common strategy in the perfect literature. However, in her later work (Pancheva 2003), as we saw above, Izvorski/Pancheva prefers another approach to analyse the temporal perfect, viz. the XN-theory.

the overall PST-marking in the construction (through the *l*-participle) is linked to the pastness of the overt event with respect to the covert learning event.

In Smirnova (2013), a temporal analysis of the learning time was developed further in a combined temporal-modal account of the Bulgarian evidential. In his own later work, Koev (2017) proposed a purely pragmatic account. The learning event is still central to his analysis, but evidentiality is here derived through pragmatic implicatures.

In a recent study, Karagjosova (2021) challenges earlier classifications of the data, and shows that more fine-grained interpretations of evidential marking in Bulgarian (e.g. mirativity) also depend on factors such as the choice of aorist vs. imperfect stems of the participle, often ignored in the previous accounts.

I conclude that there is much activity and little consensus in the research community concerning evidentiality in Bulgarian. This is exciting news for those who want to delve into the Bulgarian paradigms.

4 Subordinate tense

Subordinate tense is a vast topic. The main question it raises is under which conditions the embedded tense is semantically and/or syntactically dependent on matrix tense. When the subordinate tense is independent, it is interpreted with respect to the speech time s^* and behaves (more or less) like a standard matrix tense.

A central issue is the dichotomy between SEQUENCE-OF-TENSE (SOT) languages and non-sequence-of-tense (NON-SOT) languages. It is generally assumed that the Slavic languages belong to the latter group. In fact, Russian is the language par excellence used in general linguistics to illustrate NON-SOT. The literature has mostly focused on accounting for the phenomenon of SOT in English, which from a compositional point of view is challenging since it seems to violate the expected one-to-one correspondence between form and meaning.

4.1 Complement tense

Consider a standard example with an embedded stative predicate under an ATTITUDE verb:

(50) a. Zlata skazala, ščo Porošenko je prezydentom. (Ukrainian)
Zlata say.PFV.PST COMP Porošenko be.PRS president
b. Zlata said that Porošenko was president. (English)

Both constructions – "present-under-past" in Ukrainian and "past-under-past" in English – receive a simultaneous interpretation, but Ukrainian apparently represents the most natural form–meaning mapping since both the matrix past of the *verbum dicendi* and the embedded present tense in (50a) are interpreted locally, within their own clause.

This difference between Germanic sot-languages (English) and Slavic non-sot-languages (Ukrainian) is tentatively illustrated in Figure 5.

past Zlata skazala ščo (Zlata's now) Porošenko je prezydentom.
(local tenses; Ukr.)

past Zlata said that Porošenko was president.
(tense agreement; Eng.)

Figure 5: Local tenses vs. tense agreement – schematic analysis of (50a)/(50b)

This contrast in complement tense was given a semantically motivated explanation – the sot-parameter – in Grønn & von Stechow (2010). In English, tense morphology is not in a one-to-one relation with semantics due to temporal agreement: the past tense morphology (*was*) in the complement is semantically void and somehow agrees with the past tense in the matrix. This phenomenon requires a special mechanism, such as feature transmission from a higher tense operator into a non-local binding domain (Grønn & von Stechow 2010). But this problem posed by sot-languages need not concern us here.

In non-sot-languages, tense morphology and tense semantics work locally in tandem. In Ukrainian, an embedded past under a matrix past will therefore produce a backward-shifted interpretation:

(51) Myxajlo skazav, ščo Porošenko buv prezydentom.
Myxajlo say.pfv.pst comp Porošenko be.pst president
'Myxajlo said that Porošenko had been president.' (Ukrainian)

Similar data are of course found everywhere in Slavic, cf. a minimal pair from Polish:

(52) a. Ania powiedziała, że Marcin płacze.
Ania say.pfv.pst comp Marcin cry.ipfv.prs
'Ania said that Marcin was crying.' (Polish; Kusumoto 1999: 86)

PAST Myxajlo skazav ščo (Myxajlo's NOW) Porošenko PAST buv prezydentom.

Figure 6: Backward-shifting of the past in Ukrainian – schematic analysis of (51)

b. Ania powiedziała, że Marcin płakał
Ania say.PFV.PST COMP Marcin cry.IPFV.PST
'Ania said that Marcin had been crying.' (Polish; Kusumoto 1999: 86)

In the discussion of SOT vs. NON-SOT, most semanticists agree on the role played by the attitude verb, whose lexical semantics can be assumed, for our purposes, to be the same for all languages. Importantly, the intensional attitude embeds a temporal proposition. Accordingly, the complex type of attitude verbs is $\langle s\langle i,t\rangle, i\langle e,t\rangle\rangle$ (Grønn & von Stechow 2010):

(53) $[\![\text{skazav/powiedziała/said}]\!] = \lambda w \lambda P_{\langle s,\langle i,t\rangle\rangle} \lambda t \lambda y.$
$\forall w_1 \forall t_1 [(w_1, t_1)$ is compatible with everything y believes of (w,t) in w at time $t \rightarrow P(w_1)(t_1)]$

The semantics of these verbs involves quantification over time–world pairs, the time and world the attitude holder (matrix subject) locates herself at, her belief times/worlds. We can think of these times as the attitude holder's SUBJECTIVE NOW.[38]

A temporal proposition (the complement clause) is then the input to the attitude verb. It follows from the semantic types that the highest time of the embedded clause must be a temporal abstract, i.e. λ-bound. Let us examine in more detail the compositionality involved.

We start with the easy case, the backward-shifting in (51) and (52b). We assume that the embedded past has the semantics we defined earlier for a relative past in Ukrainian, cf. (8) above. The TP′ after application of (8) in Figure 3 in Section 2.2 has almost the right semantic type: We just have to abstract over the world parameter, which we have ignored so far since we have not been concerned with intensionality.

So, after the application of the relative past inside the embedded clause, we get (54).

[38]The formula in (53) ignores for simplicity the event variable associated with the *verbum dicendi.*

(54) $[\![\text{TP}'_{\langle s,it\rangle}]\!] = \lambda w \lambda t.\exists e \exists t'[\textsc{Porošenko is president}(e) \text{ in } w \,\&\, t' \prec t \,\&\, t' \subseteq e]$

When we saturate all the arguments of (53) (the embedded temporal proposition in (54), matrix tense and the matrix subject), we end up with the truth-conditions in (55).

(55) $[\![(51)]\!] = \lambda w \exists t[t \prec s^* \,\&\, \forall w_1 \forall t_1[(w_1, t_1)$ is compatible with everything Myxajlo believes of (w, t) in w at $t \rightarrow$
$\exists e \exists t_2[\textsc{Porošenko is president}(e) \text{ in } w_1 \,\&\, t_2 \prec t_1 \,\&\, t_2 \subseteq e)]]$

Informally, we tend to think that the embedded reference time precedes the reference time or event time of the saying. However, this is not what happens, and our formulae capture this correctly. The embedded time t_2 precedes the shifted evaluation time t_1, viz. Myxajlo's subjective now. This point is crucial for the theory developed in my works with von Stechow. We will soon see why.

4.2 Relative present?

How do we account for our initial Ukrainan example (50a) and the similar one in Polish (52a)? Characteristically for NON-SOT, we have a simultaneous interpretation with an embedded present.

The standard solution is the introduction of a semantic RELATIVE PRESENT in the complement at LF, cf. Ogihara (1996), Kusumoto (1999), Schlenker (1999), Ogihara & Sharvit (2012), Sharvit (2014), Vostrikova (2018), and others. In Grønn & von Stechow (2010: 133), we assumed the simplest format for this relative present:

(56) Relative present (to be revised!)
$[\![\textsc{pres}]\!] = \lambda P_{\langle i,t\rangle} \lambda t.P(t)$

Semantically PRES is identity, i.e. void. The trivial relative present merely serves the purpose of allowing the embedded clause to have present tense morphology and effortlessly inherit the reference time of the matrix. The result is a simultaneous interpretation.

But here comes a problem. If a language has the PRES in (56), we expect the relative present to be used everywhere when a simultaneous interpretation is needed, e.g. in the tense of the complement in (50a), but not only there. Japanese is indeed a language with such a relative present, see Ogihara (1996) and Kusumoto (1999). But what about Slavic?

Some partly forgotten, challenging data were brought to the attention of the research community in handouts and talks by Natalia Kondrashova in the early

1990s, cf. Kondrashova (1999). Here is a Ukrainian version of the problematic data:

(57) Marijka zustrila ljudynu, jaka {#plače / plakala}.
Maria meet.PFV.PST person who cry.IPFV.PRS cry.IPFV.PST
'Maria met a man who cried.' (Ukrainian)

We want to express the simultaneity of the two reference times: the crying is simultaneous with the meeting. The configuration with an embedded relative present makes perfect sense semantically, but the present morphology is blocked in the relative clause. (It can only have a deictic interpretation: 'Mary met in the past somebody who is crying *now*'.) Instead, both Slavic and English have to use a past-under-past in extensional relative clauses.

In Grønn & von Stechow (2012), we abandoned the relative present in (56). We wanted to capture the observation that embedded present morphology with a simultaneous, dependent interpretation is licensed only in intensional contexts, e.g. under attitude verbs. These facts have also been noted by Schlenker (1999) and others, but the question remains how a general tense theory for Slavic can be set up so that it avoids overgeneration.

Babyonyshev & Matushansky (2006) introduced a syntactic shifting condition:

(58) A shifted indexical is licensed by a c-commanding attitude report predicate. (Babyonyshev & Matushansky 2006)

On this view, the present tense in Slavic is semantically a shiftable INDEXICAL referring either to the utterance time or the time of the reported attitude/speech act (with the additional syntactic requirement above). In the feature account of Grønn & von Stechow (2012), we added a new clause to the SOT-parameter:

(59) Intensional temporal quantifiers of SOT-languages do not license present tense morphology. (Grønn & von Stechow 2012)

By contraposition, Slavic languages are non-SOT and intensional temporal quantifiers (e.g. *skazav/powiedziała* 'said' in (53)) license present tense morphology in the complement clause.

In Section 4.5, we will see how the principles established in Babyonyshev & Matushansky (2006) and Grønn & von Stechow (2012) can be generalised to account for true relative tenses such as the past and future.

4.3 Competing accounts

All the authors mentioned in the previous section acknowledge the importance of the SOT-dichotomy, but there also exists a quite recent, alternative approach, developed in Altshuler (2016). This theory is fundamentally different from Grønn & von Stechow (2012) in most thinkable ways, as illustrated in Table 2. For comparison we consider a past-under-past, cf. (51) above.

Table 2: Why past-under-past is backshifted in Slavic complements.

	Grønn & von Stechow (2012)	Altshuler (2016)
SOT-parameter (Eng. vs. Rus.)	✓	✗
Semantic explanation	✓	✗
Relative present (Rus.)	✗	✓
Pragmatic explanation	✗	✓
Conclusion	Russian is NON-SOT	Cessation implicature of PAST in Russian

The point of departure for the alternative account is an old idea formulated by, among others, Klein (1995). Klein pointed out that a past imperfective/stative, which in our notation produces the two conditions $t \prec s^* \ \& \ t \subseteq e$, does not semantically encode any information about whether or not the eventuality also obtains at or after the utterance time. All we know is that the time we are focusing on, the assertion time (reference time), is included in the temporal trace of the event/state in question, but the pure semantics remains silent about the state of affairs obtaining after the assertion time.

Daniel Altshuler develops this idea further. Arguably, the use of a present tense is semantically more informative if the eventuality actually obtains at the utterance time (or, in Slavic, at the local evaluation time). Hence, if the speaker uses the alternative past tense, the hearer will pragmatically infer that a semantic present should be false. Accordingly, the interpretation of the clause is strengthened to a proper past eventuality. What interests us here, is how this reasoning plays out in embedded clauses. Now comes the point where Altshuler (2016) and Grønn & von Stechow (2012) radically depart in their respective analyses, although the authors agree on the data.

According to Altshuler, Russian, unlike English, has a purely relative present (the standard view disputed by Grønn/von Stechow). In English, the present tense is always deictic, referring to the utterance time, and therefore unavailable in complements under past attitudes. This in turn means that an embedded English past does not compete with the present form and is not strengthened to a proper past interpretation of the eventuality. An embedded past in English can therefore be interpreted as more or less simultaneous with the matrix past.

In Russian, however, the viable alternative to the past, viz. the relative present, triggers a pragmatic competition between the two forms also in embedded contexts. Here, the relative present would pick out the attitude holder's subjective now. Accordingly, the past is strengthened to a backshifted past under a past attitude verb. This pragmatic effect is called a CESSATION IMPLICATURE. In Altshuler's own words, "properties of the present tense in a given language will effect whether or not an embedded past tense triggers a cessation implicature" (Altshuler 2016: 135f.).[39]

4.4 *de re* past in Slavic

Given the relative past interpretation of a past-under-past in Slavic complement tense (say, under *verba dicendi*), constructions like the following are puzzling:

(60) a. Darina bačyla, jak Mykola plakav.
Darina see.IPFV.PST how Mykola cry.IPFV.PST
'Darina saw that Mykola was crying.' (Ukrainian)

b. There was a crying *e* by Mykola in the past & Darina saw *e* in the past.

c. $(\exists t \prec s^*)(\exists e)[t \subseteq e \,\&\, \text{CRYING}(\text{MYKOLA}, e)] \,\&$
$(\exists t_1 \prec s^*)(\exists e_1)[t_1 \subseteq e_1 \,\&\, \text{SEE}(\text{DARINA}, e_1, e)]$
(Grønn & von Stechow 2010)

Despite the embedded past, the sentence is interpreted as DIRECT PERCEPTION, hence the event perceived is simultaneous with the act of perception. Arguably, the complementizer *jak* 'how' (in combination with the past) is used to encode direct perception of the matrix subject, although the embedded past event is reported from the PERSPECTIVE OF THE SPEAKER. The latter observation means that the event is described by the speaker independently of the subject's belief state. The matrix perception verb is therefore not an attitude verb in the context of (60a).

In (60b)/(60c) the event description in the complement is accordingly brought to a transparent *de re* position with respect to the seeing-predicate. Since the

[39] A recent experimental study of cessation implicatures in Polish casts doubts on associating (NON-)SOT phenomena with implicatures (Mucha et al. 2023).

two past tenses are independent of each other, the analysis cannot explicitly encode simultaneity. Such cases of a (pseudo-)simultaneous past-under-past do not therefore show that Slavic has sot, pace Altshuler (2004) and Altshuler (2008).[40]

For simultaneous interpretations of past-under-past with attitude verbs proper, the same *de re* strategy is also available according to Vostrikova (2018), who addresses the interaction of Russian complement tense with temporal adverbials:

(61) V 2016 Tanja skazala, čto togda Putin byl prezidentom.
in 2016 Tanja say.pfv.pst that then Putin be.pst president
'In 2016 Tanja said that Putin was the president of Russia then.'
(Russian; Vostrikova 2018)

The author shows that *togda* 'then' patterns with an embedded past, while *sejčas* 'now' in the complement only cooccurs with the present tense. The explanation for the first case, exemplified in (61), is that both the adverbial, the indexical shifter *togda* 'then', and the complement tense must be interpreted *de re*, as independent of the matrix tense: "*togda* carries a presupposition that the time intervals it picks are not equal to the evaluation time. This presupposition is strong enough to make *togda* incompatible with the relative present, however, at the same time weak enough to be compatible with the simultaneous reading of past-under-past. The reason for this is that the simultaneous reading of past-under-past in Russian is derived via a *de re* construal and the meaning resulting from this construal does not require that the two intervals are equal, it is enough for them to simply overlap." (Vostrikova 2018: 475)

4.5 Tense in adjuncts

A theory of subordinate tense has to distinguish between complement tense and adjunct tense, where the latter includes both relative clauses and temporal adverbial clauses. A non-deictic, shifted embedded present (or past, future) is normally blocked in Slavic adjuncts under a past matrix verb, which suggests that adjunct tense in Slavic, unlike complement tense, is mostly independent of the matrix, hence deictic (Kubota et al. 2011).

The temporal organisation in Slavic adjuncts is thus fairly simple since tense (semantics and morphology) operates locally, with adjunct and matrix tense being independent of each other. Still, consider once more this past-under-past, repeated from above:

[40] For more on perception verbs in Russian, including a discussion of indirect perception and present tense complements, see Khomitsevich (2007) and Grønn (2023).

(62) Marijka zustrila ljudynu, jaka plakala.
Maria meet.PFV.PST person who cry.IPFV.PST
'Masha met a man who (had) cried.' (Ukrainian)

In fact, (62) is ambiguous with respect to the temporal relation between the relative clause and the matrix. A deictic past in the relative clause predicts the ambiguity: If all we know is that the time of the crying is before the speech time, the crying can be before (in principle even after) or overlapping with the meeting. The deictic analysis only requires the adjunct tense to precede the speech time.

As pointed out by a reviewer, the deictic analysis predicts that a past adjunct could be forward-shifted with respect to the matrix event.[41] This prediction is borne out:

(63) Poznał kobietę, która została jego żoną.
meet.PFV.PST.M.SG woman who become.PFV.PST his wife
'He met a woman who became his wife.' (Polish; reviewer's example)

In Grønn & von Stechow (2012), we entertain the possibility that the highest tense in the adjunct is ANAPHORIC, a temporal pronoun Tpro ($\approx$ *then*). For (63), we would still have to say that Tpro is bound by the deictic NOW. However, Tpro allows us in principle to account for the backward-shifting in (62) (cf. the English pluperfect) by binding Tpro to a temporal variable of the matrix clause (the reference time for the matrix meeting), while still having a local precedence relation $t \prec$ Tpro in the adjunct.

Note that even if Tpro is the evaluation time of the adjunct, the embedded past is still accounted for locally. This seems to be an absolute constraint for embedded Slavic tense, a defining characteristic of being NON-SOT.

However, an unrestricted Tpro overgenerates, so the safest solution is to treat extensional adjuncts such as (62) and (63) as deictic. In the latter case, the tense has to be deictic (either through a default/covert pronoun NOW or Tpro coindexed with matrix NOW). This being said, for adjunct tense embedded under attitudes and modals we definitely need something like Tpro (Grønn & von Stechow 2012). We argue that the highest tense in the adjunct must be ANAPHORIC in most intensional contexts.

Concerning relative clauses, they have type $\langle e, t\rangle$. This means that the highest tense in a relative clause cannot be a temporal abstraction (λt) as in complement clauses (Kusumoto 1999). Hence, intensional temporal quantifiers in a matrix clause do not quantify over the evaluation time of a relative clause, but the

[41]The marriage must still be before the speech time in (63).

highest tense in the relative clause may still be anaphorically bound by a time in the matrix. In fact, it turns out that Tpro in a deeply embedded intensional context in Russian (and presumably in Slavic in general) is always bound by the modalised evaluation time of the matrix (Grønn 2025).

Here is a relevant Russian example of a shifted *budet* 'will' in a deeply embedded relative clause:

(64) Vanja podumal, čto Maša rodit syna, u kotorogo
Vanja think.PFV.PST COMP Masha give.birth.PFV.PRS son by whom
budut golubye glaza kak i u otca.
be.PRS blue eyes like and by father
'John thought that Mary would give birth to a son who had blue eyes like his father.' (Russian; Grønn & von Stechow 2012)

The adjunct tense in (64) cannot be deictic since Mary may never have a child in our world, and if she indeed had a boy with blue eyes, it could have been before the speech time. Accordingly, the interpretation of the deeply embedded adjunct is not dependent on the speaker's speech time, but on the subjective now of the attitude holder (Vanja). The final truth conditions given below ignore aspects/events and work only with times for perspicuity.

(65) $\lambda w.(\exists t \prec s^*)$ John thinks in w at t $[\lambda w_1 \lambda t_1.(\exists t_2 \succ t_1)(\exists x)$
[Mary gives birth to x in w_1 at t_2 & boy(x) &
$(\exists t_3 \succ t_1)[x$ has blue eyes at t_3 in $w_1]]]$

The first crucial point is that the input to the attitude verb *podumal* 'thought' is a temporal proposition of the form $\lambda w_1 \lambda t_1$..., while the forward-shifter *budet* in the embedded adjunct gives us $\exists t_3 \succ \text{Tpro}_i$. Tpro_i can only be bound by Vanja's subjective now, that is t_1. If we bind Tpro_i to t_2, the time of the birth, we violate the constraint saying that the possible binders for Tpro_i are NOW or a MODALISED evaluation time (Grønn 2025). But we also get the truth conditions wrong because we then say that the boy will not be born with blue eyes, but will only get blue eyes after birth ($\# t_3 \succ t_2$).

The reasoning above applies to other cases of a modalised matrix and other types of adjuncts, such as temporal adverbial clauses. Thus, the higher intensional context provides a temporal perspective for the embedded clauses in (66a):[42]

[42]RuN-Euro parallel corpus data from Grønn & von Stechow (2012).

(66) a. The wedding would take place in May, before the cargo boats headed south. (English)

b. Svad'ba dolžna byla sostojat'sja v mae, do togo, kak
wedding must be.PST take.place.IPFV.INF in May before that how
karbasy ujdut na jug. (Russian)
ships go.PFV.PRS to south

The entire *do togo, kak*-'before'-construction is embedded under the necessity modal. The construction leaves it open whether the wedding – and/or departure of the boats – took place at all. Furthermore, the departure of the boats could have been before the speech time. Thus, the embedded semantic future (morphology: perfective present) cannot be deictic, but must be bound.

The forward-shift is with respect to the time of the matrix modal. Here I simply give the truth conditions for the interested reader:[43]

(67) $[\![(66b)]\!] = \lambda w.(\exists t \prec s^*)[(\forall w_1)(\forall t_1)[\text{ACCESSIBLE}_{w,t}(w_1, t_1) \rightarrow$ $[\text{WEDDING TAKES PLACE}(w_1)(t_1) \,\&\, t_1 \prec \text{the earliest}$ $t_2 : (\exists t_3 \succ t_1)[\, t_3 = t_2 \,\&\, \text{BOATS LEAVE}(w_1, t_3)]]]]]$
(Grønn & von Stechow 2012)

The important thing to note is that the relative future interpretation of the adjunct translates into $\exists t_3 \succ \text{Tpro}_i$. Following the discussion above, Tpro must be bound by the perspective time of the modal, hence $\exists t_3 \succ t_1$.

In Grønn (2025), I argue that the phenomenon of RELATIVE TENSE in Russian (Slavic) is always dependent on intensionality. The evaluation time of the embedded tense is shifted to the modalised perspective of the matrix.

So the "problem" discussed in Section 4.2 concerning the status of a relative present in Slavic is more general: Whenever a so-called relative tense interpretation is found in Slavic (or at least in Russian) – be that past, future or present – there is a perspective shift from the embedded tense to a very specific modalised time in the matrix.

5 Conclusion

The study of Slavic tense in formal semantics does not have a long history, mainly because of the dominant role played by aspect in Slavic grammar. It is difficult to

[43] I must refer to Grønn & von Stechow (2012) for a discussion of the internal semantic composition of temporal adverbial clauses in Russian (*do/posle togo kak*... 'before/after that how...') and their counterparts in other Slavic languages.

give a proper semantics for tense in Slavic without also proposing a full formal analysis of Slavic aspect.

The future seems bright in the sense that there are many interesting topics for Slavic tense semanticists to work on. We need a better understanding of the data in individual Slavic languages, notably concerning compound tenses (temporal auxiliaries) and complex sentences (subordinate tense), the two main areas of research which I have focused on here. A question naturally occurs at this point: How do temporal auxiliaries in various Slavic languages behave in subordinate tenses?

Ideally, the notational practices and frameworks used should be more uniform in order to enhance comparison between different accounts. Some of the issues ahead, questions which I have merely alluded to in the presentation above, will only be resolved in close interaction with the research community outside Slavic linguistics, e.g. questions such as referential vs. quantificational tense, the integration of temporal adverbials of various sorts in the temporal calculus, the issue of cross-sentential temporal anaphora, the role of pragmatic competition (e.g. for the perfect vs. past/present), the role of temporal features (interpretable/uninterpretable) at the syntax-semantics interface.

What will always keep semanticists busy is the study of TAME (tense, aspect, mood, and evidentiality) and how these categories/grams interact in the verbal domain. Here I have focused on a formal account of tense with a superficial but hopefully consistent view on aspect. There is, however, much more to be said about tense in interaction with mood and evidentiality in Slavic.

Abbreviations

1	first person
2	second person
AUX	auxiliary
COMP	complementizer
FUT	future tense
INF	infinitive
IPFV	imperfective aspect
PFV	perfective aspect
M	masculine
PL	plural
PRS	present tense
PST	past tense
PTCP	participle
REFL	reflexive
SG	singular
SOT	sequence of tense
TAC	temporal adverbial clause
XN	extended now

Acknowledgments

I am grateful to the reviewers and editors for their generous comments. I thank my consultants Irena Altarac-Romano (BCMS), Kjetil Rå Hauge (Bulgarian), Božo Kolov (Bulgarian), Krzysztof Migdalski (Polish), Ljiljana Šarić (Croatian), as well as my numerous friends and teachers of Ukrainian, notably Natali Žytova, Inesa, and Zorjana. A special thanks to Irena Altarac-Romano for proofreading the manuscript.

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Chapter 8

The morphosemantics of Russian aspect

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Compositional semantic analyses of Slavic aspect face the challenge that no clear morphological exponents of perfective or imperfective can be identified on the verb. Against this background, the present article, concentrating on the case of Russian, pursues two goals. The first is to provide an overview of formal semantic analyses that answer the question of which parts of verbal morphology carry which meaning. The second aim is to present a new formal semantic analysis, stated within the theoretical framework of discourse representation theory (DRT), which makes do with a single aspectual null operator. Unlike previous theories of this kind, the new approach relies on state change (rather than quantization) as the relevant lexical aspectual feature, and it presupposes time intervals (rather than time points) as relevant temporal units in the model. It is shown in detail how the new proposal can explain the range of (im)perfective readings identified in traditional Russian aspectology.

1 Introduction

A lot of literature has been devoted to the form and function of Russian verbal aspect.[1] Let me therefore emphasize that in this chapter, I will be concerned with the relationship between morphology and semantics of Russian aspect exclusively from a compositional semantic perspective. The following discussion will concentrate only on studies that approach the topic from this theoretical

[1]This chapter is about theories on Russian aspect only. Whether and in how far the results of the discussion presented below carry over to other Slavic languages remains to be seen. Although the aspectual systems of the Slavic languages resemble each other from a coarse-grained perspective, they differ in many relevant details. Inner-Slavic aspectual variation can be found in semantics as well as in morphosyntax (e.g. Dickey 2000, Petruchina 2012).

Olav Mueller-Reichau. 2026. The morphosemantics of Russian aspect. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 397–459. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078583

perspective. Even with this confinement, such a survey must necessarily be selective.

Within formal (compositional) semantics, Russian aspect is usually modelled as an operation over verbal meanings. Approaches disagree as to whether the syntactic constituent carrying the "verbal meaning" is a VP or a V (see Rothstein 2020 for recent discussion). Another parameter with respect to which theories differ from each other relates to morphology. As is well known, there are no uncontroversial perfective or imperfective morphemes in Russian, that is to say, there are no necessary and sufficient markers of perfective or imperfective meaning.[2] Prefixes, which are not infrequently associated with perfectivity, appear also in imperfective forms (1a), and the secondary imperfective suffix, which seems to suggest itself as exponent of imperfectivity, shows up in perfective forms as well (1b).

(1) a. otkryt'$^{\text{PFV}}$ 'open' – otkryvat'$^{\text{IPFV}}$ 'open'
    b. otkryvat'$^{\text{IPFV}}$ – zaotkryvat'$^{\text{PFV}}$ 'start opening'

From a formal semantic perspective, the frequently made assumption that verbal prefixes mark perfectivity faces the problem of how to deal with imperfective verbs like *otkryvat'* in (1). To claim for these that the prefix does not introduce the perfective meaning that it carries in *otkryt'* would run against basic assumptions of compositionality. This problem is maybe even more apparent in aspectual triplets such as (2):

(2) a. čitat'$^{\text{IPFV}}$ 'read' – pročitat'$^{\text{PFV}}$ 'read (through)'
    b. pročitat'$^{\text{PFV}}$ – pročityvat'$^{\text{IPFV}}$ 'read (through)'

If one wanted to maintain that prefixes introduce perfective operators, one would have to come up with a semantics for the imperfective suffix YVA that, operating on a perfective meaning as input, somehow gets rid of the completedness entailment added by the prefix beforehand.[3] As we will see below, proposals along these lines have been made, notably by Zucchi (1999) and Grønn (2004).

The alternative is to free prefixes from expressing perfectivity. This position, which was first argued for in Filip (1992), has widely become accepted in formal Slavic linguistics, and it is also prominently advocated in non-formal aspectology (e.g. Plungjan 2011). If prefixes cannot be considered as exponents of perfectivity,

[2]To the exception of semelfactive *nu*, from which I abstract away here.

[3]I use "YVA" as a handy term to refer to the operator the application of which manifests itself morphologically as suffixation. Subject to laws of morphonology, it can show up as *-yva, -iva, -va, -á*, among further options (e.g. Švedova 1980, Zaliznjak & Šmelev 1997).

no overt structure will be left that could serve this function, and there seems to be no way around the conclusion that perfectivity has zero exponence.

A second often made assumption is that the secondary imperfective suffix is a marker of imperfectivity, hence its name. This position faces two theoretical challenges. The first relates to verbs like *zaotkryvat'* in (1b). Why should a perfective form contain an imperfective marker? The second is the absence of imperfectives like *zakrikivat'*:

(3) a. kričat'$^{\text{IPFV}}$ 'scream' – zakričat'$^{\text{PFV}}$ 'start screaming'
b. zakričat'$^{\text{PFV}}$ – *zakrikivat'$^{\text{IPFV}}$ 'start screaming'

If the secondary imperfective suffix was an imperfectivity marker, why should it not be used to form an imperfective form of the perfective *zakričat'*? Even if one could somehow show that the meaning 'to be starting to scream' was semantically ill-formed (which I believe it is not), there is definitely no conceptual obstacle for an iterative construal, i.e. for a meaning like 'to repeatedly start screaming'. Yet the form *zakrikivat'* is ungrammatical (Tatevosov 2013: 63). In contrast to that, the imperfective form *zapevat'* from perfective *zapet'* 'start singing' is perfectly fine.

To sum up so far, every compositional semantic approach to Russian aspect has to address the problem that Russian verb morphology does not seem to provide clear markers that could be made responsible for introducing perfective or imperfective meaning components into semantic structure. As Schoorlemmer (1995: 153) puts it: "aspectual morphology in Russian presents a problem from whatever angle you look at it".

In the sections to follow, I will briefly recapitulate formal semantic approaches that explicitly address the "aspectual morphology problem". Given this scope, I have to set aside a lot of important work in Russian aspectology. This concerns not only the whole cosmos of non-formal theories, but also those formal semantic studies that present meanings for perfective and imperfective aspect, but remain silent as to the morphological source of aspect information.

As for the latter, to name just two, I will not go into the details of Borik (2006) and Paslawska & von Stechow (2003). Borik (2006: 166–167) arrives at the conclusion that a Russian perfective form expresses the two conditions of (i) non-overlap of reference time and speech time, and (ii) inclusion of event time in reference time.[4] The Russian verb is thereby treated as a ready-made word, which

[4]Different authors use different terminology to designate what Borik (2006), following the tradition of Reichenbach (1947), refers to as reference time. In the exposition below, I will use topic time throughout, indicating where the authors discussed themselves say reference time or assertion time.

the author acknowledges right from the start: "issues of aspectual morphology [...] will not be thoroughly discussed in this work" (Borik 2006: 3). The same could be said about Paslawska & von Stechow (2003). In their seminal study, the authors take it as given that Russian verbs bear morphological aspects, perfective or imperfective, that license semantic aspects. Perfective verbs license INCLUDES, which requires that the run time of the event must fall into (or equal) the topic time interval, or POST, which says that the topic time interval has to start after the event time is over (see Paslawska & von Stechow 2003: 322). Both these works consider the imperfective to be semantically unmarked, standing in a privative opposition to the perfective (see Paslawska & von Stechow 2003: 336, Borik 2006: 186).

The "aspectual morphology problem" does not imply that it would be difficult to establish the aspect of a given verb. There is a long-standing set of diagnostics, which I reproduce here in the version of Ramchand (2008: 1698). As pointed out in Zinova (2021: 22), most of these tests provide contexts from which perfectives are excluded:[5]

(4) A perfective predicate...
  a. ... cannot get simple ongoing interpretations in the present tense
  b. ... cannot be used as the complement of a phasal verb such as *načat'* 'to begin', *prodolžat'* 'to continue' or *končit'* 'to finish'
  c. ... cannot form a present participle
  d. ... combines with another perfective predicate to form non-overlapping events in narrative discourse

The present chapter consists of two parts. In the first (§2) I will discuss different proposals that have been made to deal with the morphosemantics of Russian aspect. Specifically, I will present (hopefully fair) summaries of the studies by Zucchi (1999), Grønn (2008b), Filip (2000, 2008), Altshuler (2014), Tatevosov (2011, 2015, 2017), Bohnemeyer & Swift (2004) and Ramchand (2004, 2008).

In the second part (§3) I will outline my own proposal, which combines insights from the theories discussed before. In a nutshell, my approach is time-relational with topic times (aka reference times, see fn. 4) being modelled as

[5]Note that generalization (4d) seems premature in view of examples like (i), where the three events do overlap (they start simultaneously), see Dickey (2000: 224):

(i) Zagudeli, zavorčali, zakričali.
start.hooting.PST.PFV start.grumbling.PST.PFV start.shouting.PST.PFV
'They started hooting, grumbling and shouting.'

intervals: it relies on the notion of state change, it makes do with a single default aspect operator, and it treats YVA as applying below the aspect operator. It should be noted that I will exclude semelfactives from consideration, that I will not say much about habituals and that I will only scratch the surface of external prefixation towards the end of this chapter.

2 Theories of Russian aspect

2.1 Zucchi (1999)

The examples that Zucchi (1999) uses to explain his theory are the simple imperfective *pisat'* 'write', the prefixed perfectives *napisat'* 'write down' and *vypisat'* 'write out', and the suffixed imperfective *vypisyvat'* 'write out'.[6] In a nutshell, the semantics of a prefix (like *na-* in *napisat'* or *vy-* in *vypisat'*) is a function mapping a predicate of complete and incomplete events onto a predicate of complete events, while the imperfective suffix *-yva* (in *vypisyvat'*) semantically maps a predicate of complete events onto a predicate of complete and incomplete events.

Zucchi (1999) demonstrates that, in principle, both Landman's (1992) and Parsons's (1990) theories may be used to formalize this basic idea. Whichever theory one chooses, however, at a certain point it will be necessary to adopt some component of the opposing theory in order to arrive at a proper account of the Russian facts. Below, I will recapitulate the Parsons-style semantic translations, indicating where the Landman component must be added.

The semantics assumed for the imperfective verb *pisat'* is given in (5).[7]

(5) $[_{\text{V[+ipfv]}}\ \text{pisat'}] \Rightarrow \lambda Q \lambda x \lambda e[\textsc{writing}(e) \wedge \textsc{agent}(e,x) \wedge \textsc{theme'}(e,Q)]$

The prefix *na-* translates as follows:

[6] Zucchi's (1999) analysis of Russian aspect serves to illustrate the more general theoretical point that the proper modeling of imperfectivity calls for the integration of two competing theories in the field, Landman (1992) and Parsons (1990). This is not the space to go into this. Instead, I will only give an outline of the specific proposal regarding Russian aspect semantics.

[7] The variable Q denotes a generalized quantifier intension. The reason is that "if John is writing a letter, the letter comes into existence only when the writing event is completed" (Zucchi 1999: 198). Since the events denoted by the imperfective form *pisat'* need not be complete, the direct object of *pisat'* 'write' should be assigned an intensional meaning (if something, say a novel or a confession, is incompletely written in some world, it will not exist in that world). This is reflected in the prime notation of the thematic role relation THEME', which is absent in AGENT. Note that I will not take over the Zucchi-style formalization. The reader will unavoidably note deviations in metalinguistic representation between the authors discussed in this chapter and my personal practice. Zucchi's abbreviation for imperfective aspect, for instance, is *impfv*, Grønn's uses *ipf*, and I prefer *ipfv*.

(6) $[_{\text{V[+pfv]}}$ na- $[_{\text{V[+ipfv]}}\ \alpha\ ]] \Rightarrow \lambda Q\lambda x\lambda e\lambda t[\alpha'(Q)(x)(e) \wedge \textsc{cul}(e, t, {}^{\wedge}\alpha'(Q)(x))]$

Applying this meaning of *na-* to the meaning of *pisat'* will yield (7).

(7) $[_{\text{V[+pfv]}}$ napisat'$] \Rightarrow \lambda Q\lambda x\lambda e\lambda t[\textsc{writing}(e) \wedge \textsc{agent}(e, x) \wedge \textsc{theme}'(e, Q) \wedge \textsc{cul}(e, t, {}^{\wedge}\lambda e[\textsc{writing}(e) \wedge \textsc{agent}(e, x) \wedge \textsc{theme}'(e, Q)])]$

As can be seen, the meaning of *napisat'* is correctly derived as perfective, for which in Parsons' (1990) theory the predicate CUL is responsible.[8] The culmination condition is fed into semantic composition by the prefix *na-*.

So far, a sentence with *napisat'* will come out as having an intensional theme participant. This is incorrect, however, because culminated writing events entail the existence of a specific product of writing. To close this gap, Zucchi (1999) adds the following meaning postulate ("The Writing Principle"):

(8) $\forall e\forall t\forall x\forall Q[\textsc{cul}(e, t, {}^{\wedge}\lambda e[\textsc{writing}(e) \wedge \textsc{agent}(e, x) \wedge \textsc{theme}'(e, Q)]) \rightarrow (\textsc{theme}'(e, Q) \leftrightarrow {}^{\vee}Q\ \lambda y(\textsc{theme}^{*}(e, y)))]$

As for *vy-* and *vypisat'*, Zucchi (1999) refrains from giving precise semantic translations, being aware of the complexity of the task. Instead, he only presents the meaning postulate in (9). It tells us that the truth of a sentence with *vypisat'* as a predicate will always entail the culmination of an event at *t* relative to the event type of writing-out:

(9) $\forall e\forall x\forall Q[\textsc{write-out}(Q)(x)(e) \rightarrow \textsc{cul}(e, t, {}^{\wedge}\textsc{write-out}'(Q)(x))]]$

Thus, *vypisat'* will impose on the interpretation the culmination condition, which was added to the meaning of *vypisat'* by the prefix *vy-*. Now the question arises: how to state the semantics of the suffix *-yva* such that it will, so to speak, get rid of the culmination interpretation again? A sentence based on the predicate *vypisyvat'* does not require the event to culminate, after all. The solution is found in the theory of Landman (1992), whose imperfective operator PROG relates actual events to possible events. Zucchi (1999) proposes to incorporate this feature into Parsons's (1990) theory. Specifically, since imperfectivity is taken care of by the predicate HOLD in Parsons (1990), it is the content of HOLD that has to be redefined accordingly (see Zucchi 1999: 194):

(10) $[\![\textsc{hold}(e, t, {}^{\wedge}P)]\!]_{w,g} = 1$ iff $\exists e'\exists w'\exists t'$ such that $\langle e', w'\rangle \in \textsc{con}(g(e), w, [\![{}^{\wedge}P]\!]_{w,g})$ and $[\![\textsc{cul}(e, t, {}^{\wedge}P)]\!]_{w',g[e/e',t/t']} = 1$

[8] Zucchi (1999) argues that CUL expresses the requirement of an eventuality *e* culminating at a time *t* relative to an eventuality type. On Parsons's (1990) original account, CUL is merely a binary relation between *e* and *t*.

This says, roughly, that the relation HOLD will hold of an event *e*, a time *t* and a property in some world *w* if *e* culminates at *t* relative to some other world w' in which there is an event e' that traverses along a continuation branch of *e*.[9]

With this recalibration of HOLD, the meaning of *-yva* is now stated as follows:

(11) $[_{\mathrm{V[+impfv]}}[_{\mathrm{V[+pfv]}}\ \alpha]\ \text{-yva}] \Rightarrow \lambda Q \lambda x \lambda e \lambda t[\textsc{hold}(e, t, {}^{\wedge}\alpha'(Q)(x))]$

Zucchi (1999: 197) writes that "the absence of isolated forms like *pisyvat'* tells us that the imperfective suffix *-yva* takes perfective forms like *vy-pisat'* as inputs". This is, in fact, not true, as *pisyvat'* does exist in Russian, see (47) below. In defence of his analysis, the author could argue that *-yva* in *pisyvat'* is not the same as *-yva* in *vypisyvat'*, despite their homonomy. A more serious issue arises in view of perfective verbs that result from further prefixing an already prefixed perfective. Examples are *perezapisat'* 'rerecord', *podrastajat'* 'melt away a little bit' or *donapisat'* 'finish writing down', see Tatevosov (2013: 52). Without further modification, Zucchi's (1999) theory would predict such forms to be impossible because the output of the first prefixation yields the wrong input for the second prefixation.

2.2 Grønn (2004)

Grønn (2004) subscribes to the common view that aspect operates over verbal meanings to relate the topic time (which he calls assertion time) to the time of the event. In his particular proposal there are two aspectual operators, PF and IPF, that map the set of events delivered by the VP onto a set of (topic) times.[10]

(12) a. PFV $\Rightarrow \lambda P \lambda t[e|P(e), e \subseteq t]$
b. IPFV $\Rightarrow \lambda P \lambda t[e|P(e), e \circ t]$

Perhaps most innovative is Grønn's (2004) underspecification analysis of the IPF as introducing the overlap relation $e \circ t$, which is inspired by Klein (1995). This relation is general enough to subsume the event time being included in the topic time ($e \subseteq t$) and the event time including the topic time ($t \subseteq e$) as special cases. In this way, it is warranted that the theory allows for interpretations ranging from

[9] Here I will not go into the complicated issue of how a continuation branch is to be defined, but see Zucchi (1999: 214) and, of course, Landman (1992).

[10] Letting PF introduce the relation $e \subseteq t$ is nowadays standard (Klein 1994). It is "the notion of temporal inclusion, the central concept for the correct temporal interpretation of perfective morphology, as many researchers believe" (Paslawska & von Stechow 2003: 314). But see Borik (2006) for an entirely different solution.

progressive readings ($t \subset e$) to general-factual readings ($e \subset t$). More on that below.

As far as aspectual coding is concerned, Grønn (2004) takes the following stand. At the level of VP, which in this conception includes internal as well as external nominal arguments, the (projection of the) simplex verb *čitat'* is still void of aspectual meaning (13a). It may get "aspectualized" by the zero operator "$\text{Ipf}_{\varnothing}$" (Grønn 2004: 53), consider (13b). Alternatively, it may undergo prefixation by the perfective operator *pro-*, as shown in (13c). Secondary imperfective *-yva* is treated as an operator that takes a set of times, declares a topic time as existing and requires it to overlap a new (topic) time. This way it produces a new property of (topic) times (13d). The input set of times may be provided by (13c), for instance. This will result in (13e) as the meaning of *pročityvat'*.

(13) a. $[\![\text{čitat'}]\!] = \lambda e[\ |\, \textsc{read}(e)]$

b. $\text{Ipf}_{\varnothing}([\![\text{čitat'}]\!]) = \lambda t[e\,|\,\textsc{read}(e), e \circ t]$

c. $[\![\text{pro-}]\!]([\![\text{čitat'}]\!]) = \lambda t[e\,|\,\textsc{read}(e), e \subseteq t]$

d. $[\![\text{-yva}]\!] = \lambda Q \lambda t_1[t_2\,|\,Q(t_2), t_1 \circ t_2]$

e. $[\![\text{pročityvat'}]\!] = \lambda t_1[t_2, e\,|\,\textsc{read}(e), e \subseteq t_2, t_1 \circ t_2]$

Taken together this means that, in Grønn's system, semantic perfectivity ($e \subseteq t$) is carried by overt morphemes (verbal prefixes), while semantic imperfectivity ($e \circ t$) is signalled by a zero morpheme or, in the case of secondary imperfectives, by an overt morpheme (*-yva*), which in effect returns $e \subseteq t$ to $e \circ t$.[11]

Grønn's (2004) main goal is to develop an analysis that is capable of dealing with general-factual imperfectives. (14) is the prime example to illustrate what general-factuals are (e.g. Forsyth 1970, Comrie 1976, Klein 1995). While the perfective (14a) will obligatorily refer to a completed event of reading *War and Peace*, the imperfective (14b) may, depending on context, be read as referring to an ongoing event (14b-i) or as likewise reporting on a completed reading of *War and Peace* (14b-ii). The latter reading is general-factual.[12]

[11]To complete the picture of Russian aspectual morphology, only two additional assumptions need to be made: (i) Certain prefixes, such as *vy-* in *vygljadet'* or *za-* in *zaviset'*, do not participate in the aspectual coding system; (ii) Certain roots, such as *bros-* or *reš-* have the perfective operator built into their lexical meaning. These additions can clearly not be counted as a drawback because every theory of aspectual coding in Russian has to live with these two limited sets of exceptions (see Zaliznjak & Šmelev 1997).

[12](14b) still allows for other readings, which I ignore here. The completed event interpretation will actualize if sentence stress falls on the verb, for instance.

(14) a. Ja pročital Vojnu i mir.
I read.PST.PFV war and peace
'I read *War and Peace* (to the end).'

b. Ja čital Vojnu i mir.
I read.PST.IPFV war and peace
i. 'I was reading *War and Peace*.'
ii. 'I read *War and Peace*.'

Given the way perfective and imperfective operators are set up above in (12), the VP headed by the perfective verb in (14a) will give rise to the AspP in (15a), whereas the VP headed by the imperfective verb in (14b) will result in (15b).

(15) a. AspP $\Rightarrow \lambda t[e \,|\, \textsc{speaker's reading of War and Peace}(e), e \subseteq t]$
b. AspP $\Rightarrow \lambda t[e \,|\, \textsc{speaker's reading of War and Peace}(e), e \circ t]$

If reference to an ongoing (incomplete at topic time) event is intended, the imperfective will be the only choice because ongoingness is incompatible with the "perfective condition" $e \subseteq t$, but compatible with the "imperfective condition" $e \circ t$. When reference to a completed event is intended, however, both aspects compete for being used, because completedness is semantically reconcilable with both $e \subseteq t$ and $e \circ t$. Here the question arises as to which form speakers should choose in the particular case.

According to Grønn (2004), the possibility of factual imperfectives arises from a division of labor between the two aspectual categories, whereby "Pf[v] is drawn towards the culmination of the event" (Grønn 2004: 61).[13] But just why is perfective aspect drawn towards the culmination? Grønn offers a partial answer that captures "some core cases of aspectual competition" (Grønn 2004: 234). These core cases are those where the VP supplies a target state in the sense of Parsons (1990) and Kratzer (2000). The idea is that in these particular cases the semantics of the perfective strengthens to a more specific meaning: "Perfective morphology instructs us to invoke the inclusion relation $e \subseteq t$, while the presence of a well-defined target state actualizes an additional condition: $f_{\text{end}}(t) \subseteq f_{\text{target}}(e)$" (Grønn 2004: 234). Thus, the perfective operator (12a) is flanked by the one in (16):

(16) PFV $\Rightarrow \lambda P \lambda t[e \,|\, P(e), f_{\text{end}}(t) \subseteq f_{\text{target}}(e)/\text{if defined}]$

[13]The reader should note that Grønn's (2004) approach is actually much more complex than I can do justice to in this chapter. For instance, I ignore the important case of presuppositional imperfectives and the way the author accounts for them.

Accordingly, if the VP-property provides a well-defined target state ("if defined"), the topic time will have to end when the target state is in force. This aspectual configuration is called TARGET STATE VALIDITY.

Associating the condition of target state validity with perfective aspect formalizes the traditional view of perfectivity as resultativity, making a number of correct predictions. For instance, the proposal explains bidirectional imperfectives (also called "annulment of result" in the literature): the non-use of perfective aspect signals that target state validity is not what the speaker wants to convey, triggering the implicature on the side of the hearer that the topic time extends beyond the end of the target state; the topic time ends at some point of time when the target state is no longer in force. A prerequisite for this implicature to arise is that the target state, as described by the predicate, is reversible. Consider (17). This dialogue comes from a web forum where foreign language learners ask questions to native speakers. Here, a Japanese Russian learner (A) asks and a speaker of Russian (B) answers her question.

(17) A: "Ne otkryvaj okno. Ja uže otkryvala ego 5 minut
not open.IMP.IPFV window I already open.PST.IPFV it 5 minutes
nazad." Počemu "otkryvala" a ne "otkryla"??
before why open.PST.IPFV but not open.PST.PFV
'*Ne otkryvaj okno. Ja uže otkryvala ego 5 minut nazad.* (= Don't open the window. I already opened it five minutes ago.) – why *otkryvala* and not *otkryla*??'

B: "Otkryla" – okno vse ešče otkryto. "Otkryvala" –
open.PST.PFV window all still open open.PST.IPFV
okno bylo otkryto kakoe-to vremja, no uže zakryto.
window was open some time but already closed
'*Otkryla* – the window is still open. *Otkryvala* – the window was open for some time but is already closed again.' (ru.hinative.com)

Sometimes, the imperfective can even be found in examples where the target state *is* in force at evaluation time. Such cases are covered by Grønn's (2004) explanation, because here too, the particular conditions of the target state, although valid, are irrelevant to the speaker's communicative goal. The following example is illustrative. What the speaker is primarily interested in communicating is not that the pancake is flipped to the other side (which it is, the result has not been annulled), but rather that the addressee does not need to spring into action (see Mueller-Reichau 2018: 289 for some discussion):

(18) Ne nado, ja uže perevoračival blin.
not necessary I already turn.PST.IPFV pancake
'No need. I already turned the pancake.' (constructed)

What the theory cannot explain is the use of general-factual imperfectives outside of the "core cases", that is to say, when the event denoted by the VP involves no target state. A case in point is (14b), because the state of having read a book does not count as a target state in this strict sense (see Grønn 2004: 232).[14] Since the VP-predicate does not have a target state, the alternative meanings that the two aspectual operators make available are those stated in (12). Now, given these two options, Grician reasoning suggests that the semantically more specific perfective operator is to be preferred over the imperfective one. This pragmatic tie-breaker is sometimes called "semantic blocking" (Kiparsky 2005, Deo 2012). Here is how it is stated in Dowty (1980: 32):

(19) *Semantic blocking:* If a language has two (equally simple) types of syntactic structures A and B, such that A is ambiguous between meanings X and Y while B has only meaning X, speakers should reserve structure A for communicating meaning Y (since B would have been available for communicating X unambiguously and would have been chosen if X is what was intended).

Imagine Ivan wanted to tell Mary that he is one who read Tolstoy's *War and Peace*. His language, Russian, makes available the two alternative syntactic structures in (15). According to (19), (15a) should win over (15b). We would expect Ivan to pronounce (14a). In fact, however, Ivan will utter (14b). Thus, the classic example is left unexplained.

It should be noted that this critical remark relates only to Grønn (2004). In a series of subsequent papers, the author developed a pragmatic account to fill exactly this gap. For reasons of space, I cannot discuss this add-on analysis, but simply refer the reader to Grønn (2006, 2008a,b).

[14] A reviewer doubts that the predicate *read War and Peace* does not qualify as a target state predicate in Parsons's (1990) conception. Consider the following quote, however: "If I throw a ball onto the roof, the target state of this event is the ball's being on the roof, a state that may or may not last for a long time. What I am calling the Resultant-state is different; [...] it is a state that cannot cease holding at some later time" (Parsons 1990: 235). The state of me having read *War and Peace* is not a "state that may or may not last for a long time", but one that will necessarily hold forever after.

2.3 Filip (2000, 2008, 2017)

According to Filip (2008), for a verb to be perfective means that it must involve in its semantics a covert operator called MAX_E. The effect of MAX_E is to narrow down the set of events in the denotation of the verbal predicate to a subset of it, i.e. to the set of maximal events in the predicate's denotation. The maximalization operator was first introduced in Filip & Rothstein (2006), with central assumptions foreshadowed by Filip (2000).[15]

Technically, MAX_E "is a monadic operator, such that $\mathrm{MAX}_E(\Sigma) \subset \Sigma$, which maps sets of partially ordered events Σ onto sets of maximal events" (Filip 2008: 219). The events in the input set of MAX_E have to be partially ordered, because were it otherwise, i.e. within a set of unordered events, no maximal events could be determined.

To model the partial ordering of the events, Filip exploits the stage-of relation between events proposed in Landman (1992). The basic idea is that events may be ordered with respect to whether they constitute developmental stages of each other. All those events that are stages of each other will automatically form a partial order.

The formal definition of the stage-of relation is as follows (after Filip & Rothstein 2006):

(20) If e_1 and e_2 are events and e_1 is a stage of e_2 ($e_1 \preccurlyeq e_2$) then:
   a. 'Part-of': $e_1 \leq e_2, e_1$ is part of e_2 (and hence $\tau(e_1) \subseteq \tau(e_2)$).
   b. Cross-temporal identity: e_1 and e_2 share the same essence: they count intuitively as the same event or process at different times.
   c. Kinesis: e_1 and e_2 are qualitatively distinguishable, e_1 is an earlier version of e_2, e_1 grows into e_2.

As can be seen, for the stage-of relation to hold among events, there has to be a criterion for "sameness at different times"; there has to be something that sup-

[15] Filip (2017: 182) claims "to individuate what is intuitively 'a single event seen as an unanalysed whole' (Comrie 1976, Dahl 1985) relative to a predicate *P* and a particular context". Indeed, given that "MAX_E singles out the largest unique event stage in a poset of eventuality stages in the denotation of *P*" (Filip 2017: 182), the asserted *P*-event cannot be but "whole" because only largest event stages are fed to the existential quantifier, which in Filip's system is introduced by tense operators applying later on, i.e. after the aspectual operator. Moreover, the asserted event will be "unanalysed" in the sense that the issue of whether an existential claim is made about developmental substages of the event does not arise because non-maximal stages have been filtered out beforehand.

plies a common identity to let one event be an "earlier version" of another. This something is a scale, relative to which the events are measured.[16]

The scale on which events are ordered as stages of each other has to have an upper bound – otherwise the operator MAX_E would lack the criterion of what event stage to count as maximal. Which bounded scale is relevant for interpretation in a particular case is the result of the interplay of lexical semantics, semantic composition and context, with verbal prefixes playing a central role. In the following brief recapitulation of what is outlined in that respect in Filip (2008), I will focus exclusively on Russian data.

Simplex perfective verbs like Russian *dat'* 'give', *kupit'* 'buy' or *skazat'* 'say' have MAX_E built in their semantic representation. This implies that these verbs lexically provide some criterion of what counts as a maximal giving, buying or saying event. Simple perfectives, in other words, supply an upper-bounded scale by themselves. The provided criterion (*maximality condition*) is not absolute, however, because which point on the scale serves as the upper bound in a given case may still be subject to some variation. Since they provide a maximality condition, the lexical meanings of these verbs are possible inputs for MAX_E, which applies to turn the maximality condition into a maximality requirement.

Simple imperfective verbs like *pisat'* 'write' or *govorit'* 'say', by contrast, provide no maximality condition. As a consequence, their lexical meanings do not lend themselves as inputs to MAX_E. To adopt to the input requirements of MAX_E, these verbs typically undergo prefixation:

> When applied to verb predicates at a lexical ('pre-functional') level, prefixes add meaning components that contribute to specifying a criterion for ordering of events in their denotation. In this way, prefixes contribute to licensing the application of MAX_E. (Filip 2008: 244)

Thus, prefixes do not mark perfectivity, but prepare verbal meanings for the possibility of being perfective.

Pay attention to the fact that there is no imperfective operator in Filip's system, which is made explicit in the following quote:[17]

[16]This aspect of Filip's theory, scalarity, has been elaborated on in Kagan (2012, 2015).

[17]The author seems to have changed her mind on that issue, as in Filip (2000), the perfective operator $\lambda P \lambda e[P(e) \wedge \text{TOT}(P)]$ still has an imperfective counterpart (TOT is the precursor of MAX_E) in $\lambda P \lambda e[P(e) \wedge \text{PART}(P)]$. Also in Braginsky & Rothstein (2005: 12) we read that "following Filip 2000 and Filip & Rothstein 2006, the perfective/imperfective distinction is a non-privative distinction, with both aspects introduc[ing] grammatical operators", a passage missing in Braginsky & Rothstein (2008).

> [W]e assume that only the perfective verb has MAX_E in its logical representation, while the imperfective verb [...] lacks it. Implicit in this proposal is the traditional Jakobsonian view on which perfectivity is the marked category in the privative aspectual opposition, and imperfective unmarked. (Filip 2008: 247)

Recall that MAX_E is conceived of as an operator that maps sets of (partially ordered) events onto sets of (maximal) events. It follows that MAX_E does not change the semantic type of its input meaning. Accordingly, there is indeed no need to come up with an imperfective aspect operator in addition to MAX_E. Perfective verbs denote sets of maximal event stages, whereas imperfective verbs denote sets of all kinds of event stages, including maximal stages.

One prediction that follows from Filip's (2008) assumptions is that the use of a perfective will always be preferred over the use of an imperfective form if reference to a maximal event is intended by the speaker. This follows from (19), which would be violated otherwise. The prediction does not seem to be borne out by the Russian data, however. Recall (18) from above. It appears difficult to argue that the speaker in (18) did not want to remind the hearer of a complete turning of the pancake, so we may safely conclude that maximality is intended. The verb, nevertheless, is imperfective. How is this possible, given that perfectives are specialized in expressing maximality? In other words, why should the form denoting maximal events be dispreferred in a context like (18)? As far as I can see, the theory of Filip (2008) leaves this question open.[18]

2.4 Altshuler (2014)

Altshuler's (2014) theory builds on Filip (2008), but unlike her (see above 2.3) Altshuler posits two covert aspectual operators for Russian.[19] Both the imperfective and the perfective operators are modelled as partitive operators denoting functions from events to event stages (Altshuler 2014: 738). What distinguishes the perfective operator from the imperfective one is that the former produces forms

[18] Note that if the semantic content of perfective aspect was resultativity, however spelled out formally, the non-use of a perfective verb in (18) would find a quite natural explanation: focus on the result would lead to a conflict because the speaker intends to focus on polarity. The point behind this is that the notion of maximality does not by itself imply emphasis on the result, which is why maximality is in harmony with a context like (18).

[19] Altshuler's (2014) conclusions about Russian are part of a broader typology of aspectual operators that aims at accounting for cross-linguistic variation in what perfective and imperfective aspects may convey.

denoting maximal stages of events described by the VP, whereas the latter produces forms that denote arbitrary stages of events described by the VP, including maximal and non-maximal stages. This matches Zucchi's (1999) proposal for the Russian secondary imperfective suffix (see §2.1). Limitation to maximal stages is what characterizes perfective operators. Let us have a look at the details of the proposal.

Altshuler's theoretical point of departure is Landman (1992). In that work, the English progressive aspect is analysed as imposing on sentence interpretation the requirement that the event referred to in the actual world w^* is a stage of an event having the property P (i.e. the property denoted by the VP) in some 'near enough' world w. The aspectual operator introduced by an English progressive form is given in (21a). (21b) is Altshuler's (2014) definition of what it means for an event to be a stage of another event (see also (20)):

(21) a. Op $\rightarrow \lambda P \lambda e' \exists e \exists w[\text{STAGE}(e', e, w^*, w, P)]$

b. $[\![\text{STAGE}(e', e, w^*, w, P)]\!]^{M,g} = 1$ iff (i)–(iv) holds:

i. the history of $g(w)$ is the same as the history of $g(w^*)$ up to and including $\tau(g(e'))$

ii. $g(w)$ is a reasonable option for $g(e')$ in $g(w^*)$

iii. $[\![P]\!]^{M,g}(e, w) = 1$

iv. $g(e') \subseteq g(e)$

Altshuler (2014) argues contra Landman (1992) that the operator presented in (21) does not really correspond to the English progressive operator. The reason is condition (21b-iv). Stated as it is, this condition allows for the event stage to be maximal, i.e. match the entire event in w. As far as the English progressive is concerned, however, this requirement is clearly too liberal, because verb forms of progressive morphology never denote maximal events (see already Filip 2000: 53). The English progressive aspect is therefore more accurately captured by replacing (21b-iv) with $g(e') \subset g(e)$.

Now Altshuler goes on to propose that the aspectual operator stated in (21) actually captures the semantics of the Russian imperfective, because of the Russian imperfective's capacity to express general-factual interpretations. We already saw in (14b) that Russian imperfective verbs may be used with reference to completed/maximal events.

Which interpretation is eventually realized depends on whether the context narrows $g(e') \subseteq g(e)$ down to $g(e') \subset g(e)$, giving rise to the progressive reading, or down to $g(e') = g(e)$, giving rise to the completive (general-factual) reading. A context for (14b) that will suggest $g(e') = g(e)$ is given in (22):

(22) Vse sčitali ego obrazovannym čelovekom. On
everyone take.for.PST.IPFV him educated person he
čital "Vojnu i mir".
read.PST.IPFV War and peace
'Everyone took him for a knowledgeable person. He had read *War and Peace*.'

The Russian perfective operator, as already said, comes with a maximal stage requirement. This means that "for all events e'', if e'' properly contains the VP-event part denoted by e' and is at least a sub-part of the VP-event denoted by e, then e'' does not satisfy the description denoted by the VP in w^*" (Altshuler 2014: 761). This condition rules out any event in the actual world w^* that would be true of the VP-property at the same time containing the event denoted by the sentence as a less developed stage.

(23) a. PFV $\rightarrow \lambda P \lambda e' \exists e \exists w[\text{MAXSTAGE}(e', e, w^*, w, P)]$
b. $[\![\text{MAXSTAGE}(e', e, w^*, w, P)]\!]^{M,g} = 1$ iff (i)–(v) holds:
i.-iv. the same as in (21)
v. $\forall e''[(g(e') \subset e'' \wedge e'' \subseteq g(e) \rightarrow [\![P]\!]^{M,g}(e'', w^*) = 0]$

So far the gist of the theory. The problem that it faces is addressed by the author himself (Altshuler 2014: 765ff.). Recall that imperfectives are analysed such that they may be used to refer to maximal events, whereas for perfectives the reference to maximal events is mandatory. This raises the question: Why would an imperfective sentence implicate maximalization when its perfective counterpart entails it?

To be concrete, let us look at (14) and assume with Altshuler that the perfective *pročital* has a more specific meaning than the imperfective *čital*. Since both meanings fit, *čital* should be subject to semantic blocking according to (19). Aware of that problem, Altshuler (2014: 767) speculates that in those contexts where the imperfective is used to denote completed (and hence maximal) events the perfective alternative might be ruled out because its meaning does *not* fit the context. He presents two cases to validate that idea. Let me discuss at least one of these.

In (24) the sentence with the relevant predicate is preceded by two sentences having perfective prediates (*povernulas'* 'turn over', *utonili* 'drown'), but a chain-of-events interpretation is excluded for conceptual reasons. Altshuler (2014) proposes that the perfective *pročitali* is inappropriate in this case because "it has been well documented that the perfective 'moves the narrative forward' " (Altshuler 2014: 769).

(24) Mne prišlos', čto my v lodke, potom ona povernulas', i
I.DAT dream.PST.PFV that we in boat then she turn.over.PST.PFV and
vse krome nas utonuli. My {čitali / ??pročitali}
all except we.GEN drown.PST.PFV we read.PST.IPFV read.PST.PFV
knigu pro Titanik, i ėto nas spaslo.
book about T. and this we.GEN rescue.PST.PFV
'I dreamed that we were in a boat, then it turned over, and everyone except us drowned. We had read a book about the Titanic and this saved us.'

Fair enough, but just why does the perfective move the narrative forward? For Altshuler's explanation to work, there should be some ingredient in the perfective operator, but not in the imperfective operator, which triggers the chain-of-events interpretation. In his actual theory, however, there is no such element: the meaning of a perfective verb form differs from the meaning of an imperfective one merely in that it lacks non-maximal stages in its extension. Another open question is: Why should the perfective, conceived of as in (23), not fit the context in (22), where there is no chain-of-events embedding?[20]

2.5 Tatevosov (2011, 2015, 2017)

In an unpublished but well-cited paper, Tatevosov (2017) carefully recapitulates the theory of Russian aspect that he developed in earlier papers (notably Tatevosov 2011, 2015). Central is the notion of "aspectual invariance", a term coined to refer to the fact that in Russian and other Slavic languages, the aspectual value of a sentence stands and falls with the choice of the particular verb form. Once the form of the verb is set, so is the aspectual value of the sentence. This direct match strongly suggests the conclusion that Slavic verb forms inherently bear aspectual meanings, i.e. that perfectivity is coded in the lexical entry of a Russian verb – in compliance with traditional approaches.

Tatevosov (2017) argues that the conclusion that Russian verbs bear aspect as a lexical feature is not without alternative. The phenomenon of aspectual invariance is also explicable under the assumption that the perfective operator

[20] Grønn (2015: 187) points to this gap regarding the imperfective by saying that "[the] question for Altshuler (2014) [...] is how the weak partitive meaning is pragmatically strengthened to denote either a proper subpart of the event (<) or the complete event (=)". At this place, I would like to draw the reader's attention to Gyarmathy & Altshuler (2020), a paper in which the authors aim at answering precisely this question.

and the imperfective operator apply higher in syntax, above the verb phrase.[21] Accordingly, the verb phrase (and, thus, the verb) are still aspectless.

Assuming this has two important consequences. First, verbal prefixes and the secondary imperfective suffix (which are often identified as aspectual markers, see the introduction of this chapter) cannot have aspectual content, as they figure below the syntactic level where aspect enters semantic composition. Consequently, Tatevosov (2017) refers to them as "aspectual morphology" in inverted commas. Secondly, perfective and imperfective meanings must be introduced by zero exponents. The latter consequence creates a new problem: Assume a compositional semantic approach to Russian aspect, in which there are two aspectual operators, PF and IPF. Assume, furthermore, that both PF and IPF have zero exponence. How can one tell from a given form whether the zero operator PF or the zero operator IPF is present in semantic structure?[22]

To cope with this problem, Tatevosov (2017) proposes that (i) the perfective operator and the imperfective operator take inputs of two different semantic types, and (ii) the "aspectual morphology" of the verb determines a particular semantic type. It follows that a verb will be selected by an aspectual operator only if the input conditions of the latter agree with the semantic type of the verb. Since the perfective operator and the imperfective operator select for different semantic types, aspectual invariance is taken care of without directly attributing aspectual meanings to lexical verbs.

Now Tatevosov goes on to show that his new account of aspectual invariance is not merely an alternative to the received view, but in fact superior to it. In two papers, Tatevosov (2011) and Tatevosov (2015), the author has laid out arguments that aim at falsifying the assumption that lexical verbs in Russian carry aspectual meanings. For reasons of space, I will not be able to review the presented arguments in full detail, but let me indicate at least one piece of evidence contra to the assumption that it is the Russian verb by itself that expresses perfective aspect. The reader is referred to the above-mentioned papers for a more detailed exposition. Consider (25):

(25) napisanie pis'ma v moment moego prichoda
writing letter.GEN in moment my.GEN coming.GEN
'writing a letter at the moment of my coming'

[21] Specifically above *v*P; to not overload the chapter with technicalities, I abstract away from the distinction between *v*P and VP. It should be noted, though, that Tatevosov's actual theory is much more detailed than my brief recapitulation of it.

[22] This question is also thrown up (but left unaddressed) by Altshuler's (2014) theory.

The deverbal noun *napisanie* is derived from the stem *napis(a)-* that, if combined with a verbal ending, will create a perfective verb. Nevertheless, as (25) shows, the event referred to by *napisanie* may properly include the topic time (the moment of my coming). Since this temporal configuration is usually attributed to the imperfective aspect, (25) is difficult to reconcile with the idea that *napis(a)-* expresses the perfective aspect.

According to Tatevosov (2017), aspectual information comes into play when (by themselves aspectless) verbs serve as complements of aspectual operators. There are two aspectual operators, both having zero exponence: "The generalization that semantic aspects are structurally dissociated from 'aspectual morphology' forces us to conclude that they are phonologically silent" (Tatevosov 2017: 26). The operators are shown in (26).

(26) a. $[\![\text{IPFV}]\!] = \lambda P_{\langle v,t\rangle}.\lambda t \exists e[P(e) \wedge \tau(e) \circ t]$

b. $[\![\text{PFV}]\!] = \lambda R_{\langle v,\langle v,t\rangle\rangle}.\lambda t \exists e \exists s[R(s)(e) \wedge \tau(e) \circ t \wedge \tau(s) \circ t]$

As can be read from this, and as noted above, the perfective operator and the imperfective operator are conceived of as being of different semantic types. More specifically, they differ with respect to their input requirements, but yield the same type of output meaning. The imperfective operator applies to a property of events, whereas the perfective operator applies to a relation between an event and a state. Both operators map their input meanings onto a property of times.

To sum up: According to Tatevosov's theory, Russian verbs are not lexically coded for aspect. By virtue of their lexical semantics verbs merely determine the meaning of the verb phrase as being either a relation between events and states, or a property of events.[23] It is this difference in event structural meaning that the two aspectual operators, which enter at the syntactic level of AspP, are sensitive to.

Most Russian simplex verbs (*pisat'* 'write', *čitat'* 'read', *idti* 'go', ...) denote properties of events. As such, they sanction the application of IPFV, which yields an imperfective meaning at AspP, imposing on interpretation the condition that the run time of the event has to overlap the topic time (which Tatevosov calls reference time), see (26a). Other simplex verbs (*brosit'* 'throw', *dat'* 'give', *kupit'* 'buy', ...) express 2-state contents, i.e. a relation between events and states, which is why they invite the application of PFV. The result is a perfective meaning at

[23]Tatevosov calls his analysis "neo-Kleinian" because the former meanings correspond to 2-state contents in the conception of Klein (1995), the latter meanings correspond to Klein's 1-state contents.

AspP, which requires that the reference time overlaps the run time of the event as well as the time interval at which the state is in force, see (26b).

Under the assumption that "all lexical prefixes are result-state inducing operators" (Tatevosov 2017: fn.21), the majority of Russian prefixed verbs (*napisat'* 'write down', *výčitat'* 'subtract', *vyjti* 'exit', …) will denote relations between events and states. Given this, they sanction the application of PFV, which correctly predicts that these verbs cannot but express perfective meanings.[24] The situation will change when YVA appears: secondary imperfective morphemes are considered to be "eventizers", following Paslawska & von Stechow (2003: 345). Applying to relations between events and states (2-state predicates), they existentially bind the state argument to produce a property of events (1-state predicate). This way verbs like *otkryvat'*, *podpisyvat'* etc. are analysed as having a semantics that is suitable only to the imperfective operator (26a).

Tatevosov's (2017) theory is certainly good news for all those who prefer a syntactic approach to Slavic aspect. It should be noted, however, that it builds on premises that not everyone might want to subscribe to.[25] Prima facie, aspectual invariance remains a strong point in favor of lexicalist approaches to Slavic aspect (see Rothstein 2020 for some recent discussion).

2.6 Bohnemeyer & Swift (2004)

Bohnemeyer & Swift (2004) develop a formal account of the cross-linguistic grammatical coding strategy for which the term "factative" has recently been coined (Shluinsky 2012, Arkadiev & Shluinsky 2015). A language has a factative system if there is a grammatically relevant two-way classification within its verbal lexicon such that elements of one class are assigned perfective aspect by default, and elements of the other class are assigned imperfective aspect by default, and any aspectual value other than those assigned by default has to be overtly marked. Russian is analysed as a case in point where, according to Bohnemeyer & Swift (2004), the lexical partition is based on the opposition between telic and atelic predicates, whereby *telicity* is understood in the sense of Krifka (1998) as quantization. Russian verbal prefixes are treated as telicizers, and the secondary imperfective suffixation as overt aspectual marking (Bohnemeyer & Swift 2004: 274, more on that below).[26]

[24] Recall that there are some exceptional prefixes, see footnote 11 in this regard.

[25] An anonymous reviewer finds the argument based on deverbal nouns unconvincing: "Why should the aspect meaning of deverbal nouns be the same as those of their morphologically related verbs, just because the two share the same stem?"

[26] Other types of factative aspect marking are based on the oppositions process predicate vs. state-change predicate (e.g. Yukatec) and stative predicate vs. dynamic predicate (e.g. English), see Bohnemeyer & Swift (2004); Shluinsky (2012).

The ideal of telicity-dependent aspect coding is summarized in the following table (Bohnemeyer & Swift 2004: 266):

Table 1: Telicity-dependent aspect coding

	Predicate	
Viewpoint	Atelic	Telic
Imperfective	∅	Overtly expressed
Perfective	Overtly expressed	∅

To formalize the alignment between perfectivity and telicity on the one hand, and imperfectivity and atelicity on the other hand, Bohnemeyer & Swift (2004) propose that there is a default assignment mechanism for aspectual interpretations which is driven by the principle of *event realization*. Event realization requires that some part of the event e, e', has to temporally overlap the topic time (t_{top}), and that part of e must fall under the event predicate P, just like e does:

(27) *Event realization* (Bohnemeyer & Swift 2004: 286)
A predicate P is realized by event e at topic time t_{top} iff at least the run time of a subevent e' of e that also falls under the denotation of P is included in t_{top}.

In formal terms, the definition of event realization is stated as in (28), see Bohnemeyer & Swift (2004: 286).

(28) *Event realization, formally*
$\forall P, t_{top}, e \subseteq E\,.\,\mathrm{REAL}_E(P, t_{top}, e) \leftrightarrow \exists e'.P(e') \wedge e' \leq_E e \wedge \tau(e') \leq_T t_{top}$

With (28) pinning down the content of event realization, the aspectual zero operator is then stated as follows:

(29) $\mathrm{DASP} \Rightarrow \lambda P \lambda t_{top} \exists e\,.\,\mathrm{REAL}_E(P, t_{top}, e)$

Let us now trace the workings of DASP. The operator is stated such that, with telic predicates as input, it will output the perfective viewpoint aspect, and with atelic predicates, it will output the imperfective viewpoint aspect.

Telic predicates, to begin with, do not have the subinterval property. Therefore, to meet (28), e' must be identical to e. If e' was some proper part of e, and if the predicate P was telic, P could not be true of e and of e'. It follows that, with a telic predicate, no part of e smaller than e itself can be included in the topic time

t_{top}, i.e. realize P. Accordingly, telic predicates will have to express the temporal relation $\tau(e) \leq_T t_{top}$.

Unlike telic predicates, atelic predicates have the subinterval property. This makes a crucial difference because now the principle of event realization allows t_{top} to include not only the whole event e, but also a subinterval of e. Bohnemeyer & Swift (2004) argue that the first option ($e' = e$) is pragmatically blocked for atelic predicates because this interpretation is what the alternatives, telic predicates, are specialized for expressing. Above I referred to this blocking mechanism as *semantic blocking*, see (19).[27]

Sentences based on zero-coded telic predicates are thus predicted to be used to express inclusion of the eventuality time in topic time, whereas sentences based on zero-coded atelic predicates are predicted to be used to express inclusion of the topic time in the time of the eventuality. This way, one default operator manages to derive the two meanings that Bohnemeyer & Swift (2004: 280) assume with many for the two viewpoint aspects:

(30) a. PFV $\Rightarrow \lambda P \lambda t_{top} \exists e \,.\, P(e) \wedge \tau(e) \leq_T t_{top}$

b. IPFV $\Rightarrow \lambda P \lambda t_{top} \exists e \,.\, P(e) \wedge t_{top} <_T \tau(e)$

There are several problems with this approach. One issue, which has been pointed to by a reviewer, is acknowledged by the authors themselves as a potential problem in a footnote (fn. 12): there are prefixed perfectives in Russian that have the subinterval property (see Filip 1999). Accordingly, these predicates cannot be accounted for within a theory of telicity-dependent aspectual reference. A solution might be to analyse these prefixes as overt aspectual markers, which does not seem too far-fetched (see Dickey 2011, and my own exposition below).

A second issue is that we certainly would not want the implicature to proper inclusion with atelic predicates to always be drawn in the absence of perfective marking. Otherwise, we could not explain general-factual uses of imperfective verbs in Russian. But just why should this implicature not be drawn in cases of general-factuals?

Moreover, no concrete analysis is offered for the secondary imperfective suffix. The most explicit passage on this is the following:

[27] "[T]ruth-conditionally, the interpretation of DASP with homogeneously divisive predicates is vague regarding perfectivity. But in this case, all else being equal, a scalar implicature licensed by Grice's (1975) first maxime of Quantity (Q1, "make your contribution as informative as is required", or Levinson's (2000) equivalent 'Q-heuristic' ("What isn't said, isn't")) will assign an imperfective reading to DASP due to the absence of perfective marking" (Bohnemeyer & Swift 2004: 287–288).

> [V]erbs encoding telic predicates are zero-marked for perfective aspect: if they produce imperfective forms at all, they require the suffix *-iv/-yv* for this purpose. In contrast, atelic predicates without overt aspect marking are compatible with both imperfective and perfective interpretations. (Bohnemeyer & Swift 2004: 274)

The "contrast" mentioned in this quote seems to imply that Bohnemeyer & Swift (2004) view the secondary imperfective suffix as an overt aspectual marker that directly introduces the imperfective operator (30b). If so, there will again be a problem with Russian general-factual imperfectives, as a reviewer correctly pointed out, because secondary imperfectives may actualize general-factual readings, see (18) as an example.[28]

The fact that Bohnemeyer & Swift (2004) cannot deal with general-factuals raises the question: is it possible to fix this shortcoming while at the same time keeping the elegance of the default aspect theory? In §3 I will affirm this question, proposing that default aspect in Russian should be based not on quantization, but on state change. A default aspect approach to Russian aspect that is based on state change has been proposed before, notably by Ramchand (2008), to which I turn now.

2.7 Ramchand (2008)

Ramchand (2008: 1696) aims at "understanding exactly what it means to be 'perfective"'. Her starting point is the distinction between internal/lexical and external/superlexical verbal prefixes (Ramchand 2004, Romanova 2004, Svenonius 2004, Tatevosov 2007, Gehrke 2008). Internal prefixes apply at the lexical level, which is modelled as the syntactic phase at which an event description is built (Ramchand 2004: 1694). Their semantic impact is that they introduce a transition into the meaning of the base they attach to or, more technically, they specify a result phrase within the event structure (see Ramchand 2004, 2008 for details). External prefixes, by contrast, apply outside of the first (lexical) phase, in the Spec Asp position. Unlike internal prefixes, "[t]hese prefixes do not seem to change the meaning of the lexical root, but add an identifiable extra bit of information relating to how the event progresses" (Ramchand 2008: 1695).

Perfective verbs may result from internal as well as from external prefixation. (31a) shows examples of the former type of perfectives, (31b) shows examples of the latter type (Ramchand 2008: 1692–1695):

[28]Note that already Klein (1995) ignores the capacity of secondary imperfectives to express general-factuals (see Grønn 2004: 53ff. for discussion).

(31) a. *vbit'* 'knock in', *vytjanut'* 'pull out', *zavernut'* 'roll up', *ubrat'* 'tidy away'
b. *popit'* 'drink a little', *zaplakat'* 'burst into tears', *dočitat'* 'finish reading', *nabrat'* 'gather lots of something'

The difference in the word-internal makeup between the forms in (31a) and those in (31b) produces syntagmatic differences. Perfectives formed by means of internal prefixation (31a) are generally compatible with inclusive temporal adverbials like *za dva časa* 'within two hours', which indicates that the respective predicates are telic. In contrast to that, perfectives formed by external prefixes (31b) show "a much muddier picture" (Ramchand 2008: 1697) regarding compatibility with inclusive temporal adverbials. Ramchand (2008) concludes that perfectivity must not be equated with telicity (Filip 1992, 1999; Gehrke 2008; Borik 2006, among many others), and she argues that the observed differences between the two classes of perfectives follow from the different structural positions of internal and external prefixes. Since internal prefixes apply inside of the first (lexical) phase, they semantically contribute to the construction of a telic event description, whereas external prefixes, which apply outside of the first phase, "interact with an already fully constructed event description" (Ramchand 2008: 1696), which may be telic or atelic.

If perfectivity is not telicity, what is it? Ramchand (2008) reasons that the correct answer should account for the fact that "the event-structure properties of the verb phrases created by Russian prefixation are clearly different from each other, but they nevertheless uniformly pass the diagnostics for perfectivity" (Ramchand 2008: 1698). I provided the diagnostics mentioned in this quote above in (4).

Ramchand (2008) seeks an answer to the question of why all of the verb forms in (31) behave alike with respect to these diagnostics. She starts off with the following assumptions:

(32) a. The *v*P is the syntactic constituent within which an event description is formed; the *v*P accordingly denotes a set of events.
b. Aspectual meaning is a function that, taking *v*P-meanings as input, introduces a topic time (which Ramchand calls assertion time) and relates it to the event characterized by the *v*P.
c. This relationship between the topic time and the event is mediated via the temporal trace function (Krifka 1992, 1998), which maps an event onto its run time.

On the basis of these theoretical assumptions, which so far are fairly standard, Ramchand (2008) proposes a zero aspect operator close in spirit to the DASP-operator proposed by Bohnemeyer & Swift (2004). Like DASP, Ramchand's (2008) operator is stated such that it will, depending on input, produce perfective as well as imperfective meanings. Here it is:

(33) $[\![\text{Asp}_{\varnothing}]\!] = \lambda P \lambda t \exists e\, [P(e)\, \&\, t \in \tau(e)]$

Similarities notwithstanding, the particular working principle of (33) differs fundamentally from the way DASP works, as it is based on two additional theoretical assumptions which are specific to Ramchand's analysis:

(34) a. The topic time is a moment rather than an interval (this can be seen from that the aspectual configuration in (33) is $t \in \tau(e)$ rather than $t \subseteq \tau(e)$).
b. Any valid relationship between the topic time and the event should meet the requirement that every subevent of the event must anchor to tense.

Against the background of (32) and (33), the two assumptions in (34) interact in the following way: if the *v*P describes a process event, i.e. if there are no subevents, any moment of the run time of the process qualifies for being the topic time moment, because for any chosen moment (34b) will be satisfied. As a consequence, with process verbs like *pisat'* 'write' or *pit'* 'drink', the topic time moment is non-unique.

Things change if the *v*P describes the transition of a process into a result state, as do predicates like *zapisat'* 'record' or *vypit'* 'drink up'. In these cases, there is only one moment that the default aspect operator (33) can pick out without violating (34b). Hence, the topic time can only be the single moment that is at the same time part of the process and part of the result state. It follows that, with prefixed verbs like *zapisat'* or *vypit'*, only one moment qualifies for being the topic time. The topic time will be unique.

This way, Ramchand (2008) arrives at an answer to her initial question. To be perfective means to have a necessarily unique topic time moment. This, according to Ramchand (2008), is the essence of perfectivity in Russian. Such a meaning is called "definite". A non-unique topic time moment is called "indefinite" and viewed as the core of imperfectivity.

What remains to be done is to define the semantics of external prefixes as also involving a unique topic time moment. Here is Ramchand's (2008) proposal for ingressive *za-* and delimitative *po-*:

(35) a. ⟦za-⟧ = $\lambda P \lambda t.P(t) \wedge t$ occurs at the onset of the temporal trace
b. ⟦po-⟧ = $\lambda P \lambda t.P(t) \wedge t$ is a specific moment a short way in to the temporal trace

With respect to secondary imperfectives, Ramchand (2008: 1704) proposes a second operator that competes with (33) for realising the aspectual head:

(36) $[\![\mathrm{Asp}_{\mathrm{ivaj}}]\!] = \lambda P \lambda t \exists e\, [P(e)\ \&\ \exists e'\ [\textsc{subevent}(e', e)\ \&\ \textsc{proc}(e')\ \&\ t \in \tau(e')]]$

Unlike (33), (36) is introduced by overt morphology. Since (36) has a more specific meaning than (33), it will be the preferred choice when *v*P delivers an event description involving subevent structure.

The specific representation (36) is problematic because it only derives the progressive reading, while secondary imperfectives in fact allow for the whole range of imperfective interpretations. This is acknowledged in a footnote, where Ramchand (2004: 1704) sketches how (36) might be elaborated on to also cover habitual and iterative readings. What remains unexplained is how (36) could serve as the basis for general-factual imperfective readings, as above in (18), for instance.

This is even more problematic in view of the fact that imperfectives lacking secondary imperfective suffixes, i.e. simple imperfectives, display the general-factual reading as well, as we saw in (14): (36) only applies to a *v*P that contains a result phrase (Ramchand 2008: 1704). Even if it were somehow possible to modify $[\![\mathrm{Asp}_{\mathrm{ivaj}}]\!]$ in such a way that it would also account for general-factuals, the solution would not carry over to (14). Not being a secondary imperfective, the imperfective *čital* 'read' in (14b) should be derived via the zero operator (33). If it counted as containing a result phrase (say as a consequence of aspectual composition with the direct object *Vojnu i mir* 'War and peace'), the predicted interpretation would be perfective/definite because the topic time moment should match the point where the process and the result state abut. If it counted as a process predicate, (36) would not be applicable.

Above, we saw that Ramchand (2008) conceives of the topic time (her assertion time) as a temporal instant rather than an interval. This assumption is crucial as it allows us to derive definite topic times for perfective predicates, but it comes at a price. Empirically, it leads to trouble with factual imperfectives. Theoretically, it opposes to the otherwise widely accepted departure from (Reichenbachian) reference time points in favor of reference/topic time intervals, as argued for at length in Klein (1994), see also Paslawska & von Stechow (2003: 314). This raises the question of whether it could be possible to develop a default aspect approach much in Ramchand's spirit that treats topic times as intervals. In the next section, I will do precisely that.

3 Default aspect based on state change

In this section, I will outline a theory of default aspect for Russian. Overall, I will follow Bohnemeyer & Swift (2004), although with a crucial difference. As we saw in §2.6, the feature on which Bohnemeyer & Swift (2004) base their default aspectual operator is telicity, understood in terms of quantization. I reject this theoretical decision, arguing that replacing telicity with state change produces better results in explaining the distribution of forms across contexts. In this respect, I support Ramchand's (2008) approach. In another respect, however, my proposal will differ from Ramchand's. In line with the received view I take the topic time to be a temporal interval.

In §3.1 I will first outline some background assumptions that I take for granted. These concern the notions of state change, 1-state and 2-state predicates, as well as the framework that I use. In §3.2 I show how VP-meanings are built given these assumptions. §3.3 continues with the semantic composition above the level of VP. It is at this stage that I take secondary imperfectivization to be operative. In §3.4 I will present my modified version of Bohnemeyer & Swift's (2004) DASP-operator. Then I will move on to demonstrate the working of DASP when it applies to 1-state predicates (§3.5), 2-state predicates (§3.6), and 2-state predicates involving secondary imperfective morphology (§3.7). Having shown that DASP so far correctly distributes Russian verb forms among contexts, I will finally bring into the picture external prefixes (§3.8). Focusing on delimitative *po-* and ingressive *za-*, I will explain how these prefixes apply after DASP to produce perfectives from atelic verb phrases. It should be noted that the external location of external prefixes, their application after DASP, makes an approach based on a state change possible in the first place. I will conclude with a summary in §3.9.

3.1 Some preliminaries

In Klein (1995), a lexical distinction is drawn, and claimed to be aspectually relevant in Russian, between predicates describing and not describing the transition, or switch, from one state to another state. The author calls the former 2-state predicates, and the latter 1-state predicates. The two states involved in the description of a 2-state predicate are dubbed "source state" and "target state" (see also Klein 1994).

Adjusted to the framework that I use, I take Kleinian 2-state predicates to be predicates that have two thematic eventuality arguments (more precisely below). In contrast to that, 1-state predicates have only one thematic eventuality argument. The notion of thematic argument that I use here deserves some comment.

To represent the systematic composition of meaning accompanying successive increase of morphosyntactic complexity, I use DRT (Kamp & Reyle 1993, Kamp et al. 2011, Geurts et al. 2020). Specifically, I resort to the version of Farkas & de Swart (2003). These authors draw a distinction between thematic arguments and discourse referents, which is absent in the original DRT-version that was presented in Kamp & Reyle (1993). The basic idea is that predicates bring with them thematic arguments that show up in the condition set of the DRS under construction, but that do not yet have the status of discourse referents, which is why they do not appear in the DRS-universe. It is only at a higher syntactic stage that thematic arguments may be instantiated, a technical term meaning that they are substituted by discourse referents, and as such show up in the universe of the DRS. Against this theoretical background, I translate Kleinian 2-state predicates as predicates introducing two thematic eventuality arguments, and 1-state predicates as predicates introducing one thematic eventuality argument.[29]

Farkas & de Swart (2003) are concerned with nominal projections, and the syntactic level at which thematic arguments are instantiated in their system is the DP. The respective mechanism is called D-instantiation accordingly.[30] I am concerned with verbal projections, and I will refer to the respective syntactic level as InstP here. Overall, I assume the syntactic structure Figure 1 to underlie semantic composition.

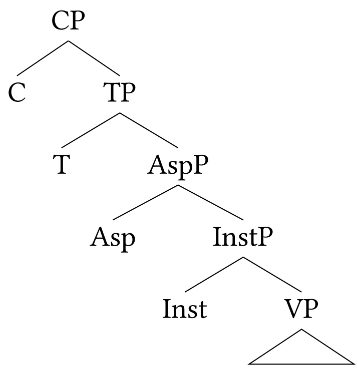


Figure 1: Syntactic background assumptions

The VP is the syntactic domain within which a predicate of eventualities is formed. On the semantic side, this corresponds to the representation of the pred-

[29]See Tatevosov (2017) for a similar move (§2.5).

[30]Besides that, Farkas & de Swart (2003) foresee a mechanism of A-instantiation, which is largely irrelevant to the purposes of the present chapter.

icate and its thematic argument(s) in the condition set of the DRS under construction. This holds for the minimal case, the case of an isolated intransitive verb. If the verb is accompanied by (VP-internal) arguments or adverbials, these will feed the DRS with additional conditions and, perhaps, discourse referents.

InstP is the domain within which thematic eventuality arguments are, or are not, D-instantiated in the sense of Farkas & de Swart (2003). Like determiners in the nominal projection, linguistic elements serving as Inst head may, depending on their lexical semantics, impose additional restrictions on the interpretation of the instantiated argument, aka discourse referent. Below, I will argue that Inst may host either a default zero head, which triggers instantiation of all of the thematic eventuality arguments of VP, or the secondary imperfective suffix, which imposes special constraints.

AspP is the domain within which a predicate of topic times is formed. The topic time relates in this or that way to the run time of the eventuality assigned to the discourse referent(s) supplied by InstP. I will show that, in order to model the distribution of Russian verbs over contexts correctly, it suffices to posit a single covert operator in Asp, similar in spirit to Bohnemeyer & Swift's (2004) DASP-operator.

3.2 Building VP-meanings

The construction of a DRS proceeds bottom-up. To start off, let us assume that the V-constituent is formed by a verb made up of the stem *pis-* (e.g. *pisal* 'read'). Processing V will lead to the construction of the DRS in (37). I assume that the verbal predicate *pis-* introduces one thematic eventuality argument and two thematic object arguments.[31]

(37)
$$\begin{array}{|c|}\hline \\ \hline \textsc{write}(\underset{\cdot}{e}) \\ \textsf{AUTHOR}(\underset{\cdot}{e},\underset{\cdot}{x}) \\ \textsf{TEXT}(\underset{\cdot}{e},\underset{\cdot}{y}) \\ \hline \end{array}$$

If there is no direct object (and if there are no adverbial modifiers) this DRS will be passed on to VP. As can be seen, at the level of VP, the arguments of

[31]The frame write.v of the English FrameNet project, with its sense being described as 'compose a text in writing' contains no less than 10 frame elements. Note that the two that I take to be syntactically relevant as thematic arguments, 'Text' and 'Author', are by far the most frequently realized (see: https://framenet.icsi.berkeley.edu). Note also that I abstract away from the word sense 'doing a painting' that *pis(a)-* is likewise associated with.

the predicate do not have the status of discourse referents (the DRS-universe is empty). I use subscript dots to indicate that an argument in the condition set CON_K has not yet been addressed by Inst, assuming that every thematic argument must be addressed by Inst during the course of derivation.[32]

Now let our VP include a direct object, *pis'mo* 'letter.ACC', as illustrated in Figure 2.

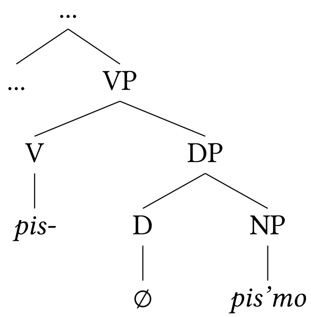


Figure 2: transitive VP, non-prefixed V

Starting from the bottom again, the NP leads to the construction of the following DRS:

(38)

LETTER($\underset{\cdot}{z}$)

We move on to resolve the DP node. Russian does not have articles as D-elements. In order not to complicate the chapter with discussions irrelevant to our main concern, I simply stipulate a zero determiner, as shown in (2), which D-instantiates the thematic argument of the nominal, creating the DRS in (39).

(39)

z
LETTER(z)

Climbing up the tree we arrive at VP. Principles of linking theory (which I ignore here) advice us to identify the direct object argument z with the text argument y of (37), so that we get:

[32]This mere notational convention spares using different kinds of letters for thematic arguments and discourse referents, as Farkas & de Swart (2003) do (see footnote 43).

(40)

$$\begin{array}{|c|}\hline z \\ \hline \textsc{write}(\underset{.}{e}) \\ \text{AUTHOR}(\underset{.}{e},\underset{.}{x}) \\ \text{TEXT}(\underset{.}{e},z) \\ \textsc{letter}(z) \\ \hline \end{array}$$

Verbs formed by the stem *pis-* are 1-state predicates. The attribute "1-state" refers to the fact that the predicate introduces only one thematic eventuality argument, see (37) and (40). Verbs formed by stems like *bros-* 'throw' or *podpis-* 'sign' are 2-state predicates. They have a complex event structure, which manifests itself in that such predicates introduce two thematic eventuality arguments. Let a verb having the stem *podpis-* (e.g. *podpisal* 'sign') form the V-constituent. I will notate the DRS resulting from processing V in the following way:[33]

(41)

$$\begin{array}{|c|}\hline \\ \hline \textsc{sign}(\underset{.}{e}_1) \\ \textsc{signed}(\underset{.}{e}_2) \\ \text{CAUSE}(\underset{.}{e}_1,\underset{.}{e}_2) \\ \text{AUTHOR}(\underset{.}{e}_1,\underset{.}{x}) \\ \text{TEXT}(\underset{.}{e}_1,\underset{.}{y}) \\ \hline \end{array}$$

As is well known, prefixed verbs like *podpisat'* are transitive, that is, the syntactic realization of a direct object is mandatory for them. Let the direct object constituent be $[_{\text{DP}}\ [_{\text{D}}\ \varnothing][_{\text{NP}}\ \textit{pis'mo}]]$, on analogy to (2). Following the same compositional steps as there, we arrive at the following DRS for the VP of a 2-state predicate:

(42)

$$\begin{array}{|c|}\hline z \\ \hline \textsc{sign}(\underset{.}{e}_1) \\ \textsc{signed}(\underset{.}{e}_2) \\ \text{CAUSE}(\underset{.}{e}_1,\underset{.}{e}_2) \\ \text{AUTHOR}(\underset{.}{e}_1,\underset{.}{x}) \\ \text{TEXT}(\underset{.}{e}_1,z) \\ \textsc{letter}(z) \\ \hline \end{array}$$

[33] I do not believe that it is possible to state a general compositional rule that governs the derivation of prefixed stems like *podpis-*. Internal prefixation is a lexical word formation process and well-known for being largely non-compositional. But see Kagan (2015) and Biskup (2019) for interesting attempts.

So this is how VP-meanings are derived. What we have to bear in mind for what follows are two things: (i) there are VP-meanings that involve two eventuality arguments besides VP-meanings that involve only one eventuality argument, and (ii) eventuality arguments are not yet instantiated at this syntactic stage.

3.3 Instantiation and secondary imperfectives

The projection above VP takes care of instantiation. I conceive of InstP as the verbal counterpart of DP in the nominal domain, the level at which the thematic arguments undergo D-instantiation, which "replaces the thematic argument of the descriptive NP by the discourse referent introduced by the D" (Farkas & de Swart 2003: 35). Adjusted to the case of verbal projections, this amounts to saying that instantiation replaces the thematic argument of the descriptive VP by the discourse referent introduced by the "verbal D", which I call Inst. Farkas & de Swart (2003: 34) point out that "[d]eterminers differ from one another with respect to further restrictions they impose on the interpretation of the discourse referent they introduce". I propose that, with respect to the Russian verb, there are two possible "verbal determiners", DINST and YVA.

The operator DINST is phonologically zero. It will head InstP whenever there is no overt instantiator expressed on the verb. In other words, it is a default operator. When it applies, it will instruct to substitute every thematic eventuality argument that there is by a discourse referent.

Let the VP-constituent be [$_{\text{VP}}$ [$_{\text{V}}$ *pis-*] [$_{\text{DP}}$ [$_{\text{D}}$ ∅][$_{\text{NP}}$ *pi'smo*]]]. The DRS it gives rise to is given above in (40). Furthermore, let there be no phonological material in Inst, meaning that DINST serves as an instantiator. Resolution of InstP will then result in the following DRS:

(43)

e z
WRITE(e) AUTHOR(e,$\underset{\cdot}{x}$) TEXT(e,z) LETTER(z)

By the same token, if the VP is [$_{\text{VP}}$ [$_{\text{V}}$ *podpis-*] [$_{\text{DP}}$ [$_{\text{D}}$ ∅][$_{\text{NP}}$ *pis'mo*]]], with the corresponding DRS being (42), the impact of DINST will lead us to construct (44).

(44)
$$\begin{array}{|c|}\hline e_1\ e_2\ z \\ \hline \textsc{sign}(e_1) \\ \textsc{signed}(e_2) \\ \text{CAUSE}(e_1,e_2) \\ \text{AUTHOR}(e_1,x) \\ \text{TEXT}(e_1,z) \\ \textsc{letter}(z) \\ \hline \end{array}$$

Now, it is time to turn to YVA. I propose that in the grammar of Russian YVA is the alternative to the default instantiator DINST.

YVA is ambiguous in that it is associated with two alternative construction rules:[34]

(45) *Construction rules for YVA* (first version)

a. Substitute the first thematic eventuality argument of the input DRS by a discourse referent.

b. Substitute the first thematic eventuality argument of the input DRS by a discourse referent, and introduce for it a condition PLURAL(e) in Con_K.

This, of course, deserves some comment. To begin with, what should count as the first thematic eventuality argument in (45a)? Since causation implies precedence (46a), we can define firstness in a purely temporal sense as in (46b).

(46) Let e be a variable over thematic eventuality arguments.

a. $\forall i(1 \leq i < n)$: $\text{CAUSE}(e_i, e_{i+1}) \rightarrow \text{PRECEDES}(e_i, e_{i+1})$

b. $\forall e$: $\text{FIRST}(e) \rightarrow \neg\exists e'$: $\text{PRECEDES}(e', e)$

Note that this definition of firstness allows for the activation of (45a) in those cases where there is only one thematic eventuality argument in the input DRS (i.e. where the V-constituent is a 1-state predicate). If we wanted to exclude this possibility for empirical reasons, we would have to recalibrate this part of the theory. Such empirical reasons do, in fact, exist. It is known that YVA may apply to a stem like *pis-*, but if it does, the result will always be a pluractional:

[34] By "input DRS" in (45) I mean the DRS that results from the resolution of the VP. Recall from above that substitution includes the introduction of a new discourse referent in U_K.

(47) Stiški ja pisyval i v junosti, no v bol’šom količestve
poems.DIM I write.PST.IPFV also in youth but in big quantity
ėtim delom zanjalsja uže v nynešnem tysjačeletii.
this activity engage.PST.PFV already in current millennium
‘I used to write little poems in my youth, but on a large scale I started engaging in this activity only in this millennium.’ (ritminme.ru)

This suggests that (45a) should be no option for projections starting from 1-state predicates. In such cases, we would like (45b) to be the only choice. Therefore, to make (45a) more selective, I build in a presuppositional component:

(48) *Construction rules for YVA*
   a. On condition that there are two thematic eventuality arguments in the input DRS: Substitute the first thematic eventuality argument by a discourse referent.
   b. Substitute the first thematic eventuality argument of the input DRS by a discourse referent, and introduce for it a condition PLURAL(e) in Con_K.

Note that the presupposition that there are two thematic eventuality arguments is limited to (48a). Given the way we defined firstness above, this implies that (48a) may be activated only with 2-state predicates, whereas (48b) may be activated with 2-state predicates, but also with 1-state predicates.

Let us now see how this works, with the syntactic structure stated in Figure 3 for a sentence involving the VP *podpis- pis’mo* ‘sign letter.ACC’.

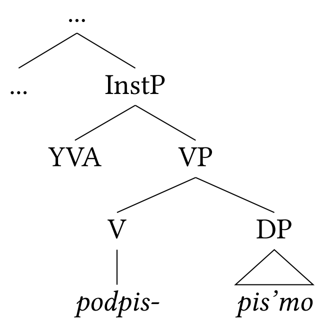


Figure 3: transitive VP, prefixed V

To resolve InstP, we can apply the construction rule (48a). This yields the DRS in (49). Note that, although there is no discourse referent for e_2, this thematic argument has now been addressed by Inst, the only unaddressed thematic argument at this stage being x.

(49)
$$
\begin{array}{|c|}
\hline
e_1 \ z \\
\hline
\textsc{sign}(e_1) \\
\textsc{signed}(e_2) \\
\mathrm{CAUSE}(e_1,e_2) \\
\mathrm{AUTHOR}(e_1,\underset{\cdot}{x}) \\
\textsc{text}(e_1,z) \\
\textsc{letter}(z) \\
\hline
\end{array}
$$

If we apply construction rule (48b) instead, we will get (50).[35]

(50)
$$
\begin{array}{|c|}
\hline
e_1 \ z \\
\hline
\textsc{sign}(e_1) \\
\textsc{signed}(e_2) \\
\mathrm{CAUSE}(e_1,e_2) \\
\mathrm{AUTHOR}(e_1,\underset{\cdot}{x}) \\
\mathrm{TEXT}(e_1,z) \\
\mathrm{PLURAL}(e_1) \\
\textsc{letter}(z) \\
\hline
\end{array}
$$

Now, let us change the input DRS. Let the syntactic structure be like the one in Figure 4, exemplified by (47).

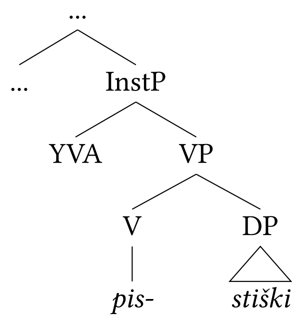


Figure 4: transitive VP, suffixed non-prefixed V

In this case, there will be only one possibility for resolving the node InstP, because the presupposition of (48a) is not met. The DRS that we obtain by processing InstP will accordingly be (51).

[35]In the remainder of this chapter I will not pursue construction rule (48b) any further for space reasons. In other words, I ignore habitual imperfectives.

(51)

$e\ z$
WRITE(e)
AUTHOR(e,$\underset{\cdot}{x}$)
TEXT(e,z)
PLURAL(e)
LITTLE POEMS(z)

I wish to emphasize that my way of treating YVA described above is by no means new. It is, in fact, strongly inspired by Tatevosov (2017). As we saw in §2.5, Tatevosov analyses the secondary imperfective morpheme as an operator mapping 2-state predicates onto 1-state predicates, referring to Paslawska & von Stechow (2003). Crucial to Tatevosov's proposal is the idea that the semantic contribution of YVA is computed *before* aspectual meanings come into play at the level of AspP, the level at which topic times are related to eventuality times. This idea of low YVA is also advocated by Ramchand and Minor in an unpublished paper in which the authors build on Ramchand (2018).[36] Let us now turn to AspP.

3.4 The Russian default aspect operator

The core of my proposal is that Russian has only one aspectual operator at AspP which correctly assigns "perfective meanings" to forms that we know to be perfective (from well-known tests like those in (4)) and "imperfective meanings" to forms that we know to be imperfective. As noted above, important precursors are to be found in Bohnemeyer & Swift (2004) and Ramchand (2008). My approach differs from the former in that I exploit the notion of state change instead of quantization/telicity for feeding the default aspect operator, and it differs from the latter in that I make use of time intervals instead of time points for modelling the topic time. The syntactic representation is thus as shown in Figure 5.

To resolve AspP, we have to follow the instructions stated in the construction rule associated with DASP.[37]

(52) *Construction rule for DASP*

a. Introduce in U_K: a new topic time discourse referent t_{top}

b. Introduce in Con_K: OVL(e_{fin}, t_{top})

[36]The paper entitled "Stativity vs. Homogeneity: Similarities and Differences between the English Progressive and the Russian Imperfective" was presented as part of the 2019 Workshop Formal Approaches to Russian Linguistics (FARL) 3 at Moscow State University.

[37]"OVL" is for the temporal overlap relation.

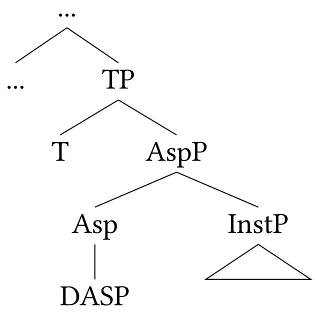


Figure 5: AspP: introduction of DASP

c. If there is only one discourse referent e in the input DRS, then introduce in Con_K: $e = e_{\text{fin}}$

d. If there are two discourse referents e_1 and e_2 in the input DRS, then introduce in Con_K: $e_2 = e_{\text{fin}}$

Let us now see what this amounts to for 1-state and 2-state predicates.

3.5 DASP applied to 1-state predicates

Assume a sentence, the VP of which is headed by a 1-state predicate. Our toy example above was *pisal* 'write.PST'. Assume furthermore that there is a direct object NP *pis'mo* 'letter', and that DINST serves as an instantiator. As for InstP, this will give us the DRS in (43). When resolving the AspP node, we will have to update this DRS along the lines of (52). The resulting meaning will be (53).

(53)

$t_{top}\ e\ z$
WRITE(e)
OVL(e_{fin},t_{top})
$e = e_{fin}$
AUTHOR(e,$\underset{\cdot}{x}$)
TEXT(e,z)
LETTER(z)

According to the basic tenet of DRT, a sentence will be true in a model whenever there is an embedding function f that (i) assigns to each discourse referent in U_K an individual in the model, and that (ii) verifies all conditions in Con_K (e.g. Kamp & Reyle 1993, Geurts et al. 2020). (53) clearly does not represent the meaning of a sentence, but the meaning of an AspP. Note, however, that subsequent resolution of the nodes up to CP will leave the aspectual relation OVL(e, t_{top}) unchanged (it will add tense conditions, a subject discourse referent and maybe more). Therefore, we can already tell from (53) that an embedding function for the sentence (whatever the sentence may look like in the end) will have to verify the condition OVL(e, t_{top}). This condition is compatible with several scenarios of how the topic time and the eventuality overlap in the model. More precisely, it is compatible with exactly four logical possibilities. These are indicated in Figure 6 (solid lines symbolize the run time of $f(e)$; /// stands for the interval $f(t_{top})$).

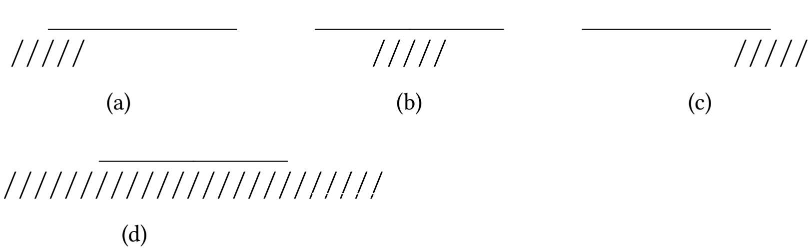


Figure 6: possible scenarios for (53)

I will now discuss the four scenarios in Figure 6 one by one, providing evidence for each option.

3.5.1 Ingressive reading (Fig. 6, sce. (a))

Scenario (a) shows the transfer from a time where the eventuality assigned to *e* is not yet running to a time where it is running. Accordingly, if we stick to sentences with a verb having the stem *pis-*, the respective interpretation should amount to the onset of a writing.

More often than not, however, the verb *pisat'* by itself will not be found to express 'start writing'. Instead, speakers use a phase verb plus an imperfective infinitive as its complement, as in *načal pisat'*, or an ingressive prefix *za-*, as in *zakuril*:

(54) Orlov načal pisat' pis'mo, a Grejg
O. begin.PST.PFV write.INF.IPFV letter whereas G.
zakuril svoju trubku …
start.smoke.PST.PFV REFL pipe
'Orlov started writing a letter, and Grejg lighted his pipe …' (NKRJa)

Nevertheless, an ingressive reading for Russian 1-state verbs can be attested. In the following example, it is the verb *plakat'* 'to cry' that is used in this way.[38]

(55) Menja posetilo takoe oščuščenie sčast'ja i odnovremenno
me catch.up.PST.PFV such feeling luck and simultaneously
otčajanija, čto ja pribežal v artističeskuju, zapersja i
desperation that I run.to.PST.PFV in backstage.room lock.PST.PFV and
plakal.
cry.PST.IPFV
'I got such a feeling of luck and desperation that I ran to the backstage room, locked myself up and started to cry.' (NKRJa)

3.5.2 Progressive reading (Fig. 6, sce. (b))

DASP applying to (the meaning of InstP projected from) a 1-state predicate also generates scenario (b) as possible interpretation. This interpretation corresponds to what is usually called progressive reading. The topic time is properly included in the time of the eventuality, giving rise to the internal viewpoint effect (see Smith 1991). (56) shows an example:

(56) Džen sidela na kuchne i pisala ėlektronnoe pis'mo,
J. sit.PST.IPFV in kitchen and write.PST.IPFV electronic letter
kogda Mėtt prišel domoj s raboty.
when M. come.PST.PFV home from work
'Jane was sitting in the kitchen and was writing an email, when Matt came home from work.' (e-libra.ru)

In this example, the topic time is supplied by the temporal clause, corresponding to the run time of the event of Matt's coming home. This event is temporally included in the event of Jane's writing an email, thus instantiating the interpretation depicted in scenario (b) of Figure 6.

3.5.3 Egressive reading (Fig. 6, sce. (c))

In scenario (c), we encounter another relation between the topic time and the eventuality time that (53) allows for. Here, the topic time exceeds the right edge of the run time of the eventuality assigned to the discourse referent *e*. An interpretation in line with this scenario will thus convey the message that the final stage of the eventuality has been left behind.

[38]Thanks to two reviewers for telling me that such examples do exist.

This case represents the mirror image of 3.5.1. And just like in that case, Russian provides special means serving the function of expressing the interpretation relevant to the issue. For one thing, Russian makes systematic use of the phase verb *končit'* 'finish' in combination with imperfective infinitives like, for instance, *pisat'*. Moreover, the completive prefix *do-* may productively be used to form perfective verbs like *dopisat'* 'finish writing'.

Nevertheless, 1-state predicates may be used to convey the egressive reading in the appropriate setting. Witness the compatibility of the form *čital* with *do konca* 'to the end'.[39]

(57) … a potom už ne mog prervat' čtenie,
and then already not can.PST.IPFV interrupt.INF.PFV reading
čital do konca, sdelal liš' nebol'šoj pereryv, čtoby
read.PST.IPFV until end do.PST.PFV only small pause to
podkrepit'sja rjumočkoj.
draw.strength.INF.PFV vodka.shot
'…and from then I could not stop reading, I read to the end, made only a small pause to regain energy by having a vodka.' (NKRJa)

3.5.4 General-factual reading (Fig. 6, sce. (d))

Last but not least, (53) allows for the interpretation (d) of Figure 6. In this case, the topic time properly includes the whole time of the eventuality. Recall that we deal only with 1-state predicates in this section, so the eventuality has to be either a state or a process. This scenario implies two changes: the first leading from the non-existence of the eventuality to its existence, the second from its existence to its non-existence.

The interpretation given in scenario (d) of Figure 6 is well-attested. It is referred to as the "general-factual" meaning in the traditional aspectological literature. (58) shows an example discussed in Paslawska & von Stechow (2003: 343) (see Zaliznjak & Šmelev 1997: 25 for more examples):[40]

[39] I thank a reviewer for bringing this up to me. In the first version of this chapter I held the view that the egressive interpretation of a simple imperfective would totally be blocked by the availability of the respective phase verb construction.

[40] In (22) I presented a variation of it. Note that, unlike the predicate *read War and Peace*, the predicate *read Lenin* is atelic (1-state).

(58) Vse sčitali ego obrazovannym čelovekom. On
everyone consider.PST.IPFV him educated person he
čital Lenina.
read.PST.IPFV Lenin
'Everyone took him for a knowledgeable person. He had read Lenin.'

3.6 DASP applied to 2-state predicates

The previous section was about the spectrum of readings that DASP will give rise to when applying to the meaning of an InstP that is based on a 1-state predicate. Now I move on to investigate the range of interpretations that DASP produces when it operates over 2-state meanings. For the purpose of illustration, I will use the same example that I used above: this section discusses the case of verbs having the stem *podpis-*, §3.7 will be about verbs having the same stem prolonged by YVA, i.e. *podpisyv-*.

Processing InstP with [$_{VP}$ [$_{V}$ *podpis-*] [$_{DP}$ [$_{D}$ ∅][$_{NP}$ *pis'mo*]]] as VP will result in the DRS given in (44). To compute the meaning of AspP, we have to update it in line with the construction rule for DASP, (52). The result is shown in (59).

(59)
$$\begin{array}{|c|}\hline t_{top}\ e_1\ e_2\ z \\ \hline \textsc{sign}(e_1) \\ \textsc{signed}(e_2) \\ \mathrm{CAUSE}(e_1,e_2) \\ \mathrm{OVL}(e_{fin},t_{top}) \\ e_2 = e_{fin} \\ \mathrm{AUTHOR}(e_1,\underset{\cdot}{x}) \\ \mathrm{TEXT}(e_1,z) \\ \textsc{letter}(z) \\ \hline \end{array}$$

Given (59), the scenarios in Figure 7 exhaust the range of interpretations that the verb *podpisat'* may in principle give rise to. They represent all of the logically possible relations that the relation $\mathrm{OVL}(e_2, t_{top})$ allows for (the vertical line symbolizes the temporal point where $f(e_1)$ and $f(e_2)$ abut).

As before, I will now briefly discuss these configurations one by one.

3.6.1 Concrete-factual reading (Fig. 7, sce. (a))

Following Russian aspectological tradition, I call the first reading to be discussed CONCRETE FACTUAL. The content of this label is described in the Academy Grammar (Švedova 1980: 604) as reference to a single situation presented as a concrete

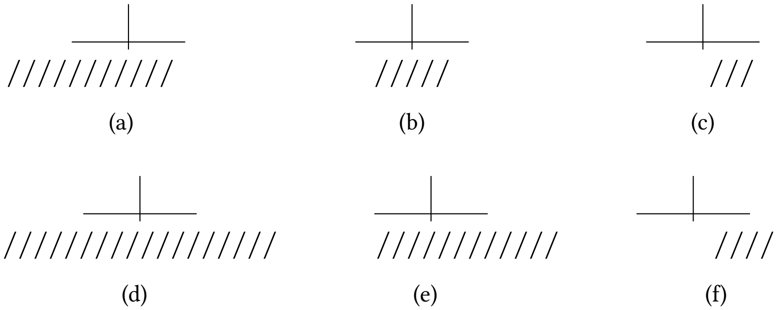


Figure 7: possible scenarios for (59)

whole fact limited by a boundary ("konkretnyj celostnyj fakt, ograničennyj predelom"). This is just what is depicted in scenario (a) of Figure 7: The "whole fact" corresponds to the full inclusion of the eventuality assigned to e_1 in the topic time. The "limitation by a boundary" corresponds to that the topic time ends within the time of the eventuality assigned to e_2, which implies a change from $f(e_1)$ to $f(e_2)$. A classic example with three such changes forming a chain of events is given in (60).

(60) Prišel, uvidel, pobedil.
come.PST.PFV see.PST.PFV defeat.PST.PFV
'Veni, vidi, vici.'

3.6.2 Culmination-in-focus (Fig. 7, sce. (b))

The interpretation depicted in Figure 7 as scenario (b) differs from scenario (a) (of Figure 7) in that the start of the eventuality assigned to e_1 precedes the start of the topic time interval. Since the onset of the first eventuality lies outside of "the time for which an assertion is made" (Klein 1995: 687), it will not fall within the scope of what the utterance is about. Such a reading, which I will call CULMINATION-IN-FOCUS for lack of a better term, is pragmatically licenced as an answer to an (implicit or explicit) question that queries whether $f(e_2)$ has, or has not, been reached within t_{top}, presupposing that $f(e_1)$ is running. Imagine a situation where two friends are reading a book together, turning one page after the other. One is about to turn the next page. Here the following dialogue fits in (from Plungjan 2001: 145).[41]

[41] An anonymous reviewer points out that the theory seems to overgenerate at this point. In a context where Peter has been reading *War and Peace* for months, and is already near the

(61) A: Pročital?
read.PST.PFV
'Are you done with reading?'

B: Da, pročital.
yes read.PST.PFV
'Yes, I am.'

It should be noted that in the Russian Academy Grammar what I call culmination-in-focus here is grouped together with the reading in §3.6.1 under the heading "konkretno-faktičeskij tip upotreblenija" (Švedova 1980: 605). The same holds for the interpretation to be discussed in the next section, which is likewise treated as a special case of concrete-factual (see Švedova 1980: 606).

3.6.3 Perfect reading (Fig. 7, sce. (c))

Scenario (c) of Figure 7 shows the interpretation known as perfect reading ("perfektnoe značenie"). What is characteristic of the perfect reading is that the moment of change to $f(e_2)$ lies before the beginning of the topic time, and that the topic time ends when $f(e_2)$ is still in force. The respective utterance will accordingly be about the holding of the target state, i.e. the eventuality assigned to e_2. The following example is from Švedova (1980).

(62) On postarel, raspolnel i obrjuzg.
he grow.old.PST.PFV fatten.up.PST.PFV and bloat.PST.PFV
'He is old, fat and bloated now.'

Two things are worth noting with respect to (62). For one thing, the example shows that perfectivity does not rule out the possibility that the topic time and the utterance time overlap (contra Borik 2006). Moreover, the example also shows that perfectivity does not necessarily enforce a chain-of-events interpretation (contra to what is stated above in (4d), see also fn. 5).

end, and the speaker thinks that Peter will finish it tomorrow, there is no culmination-in-focus interpretation available for *Petja zavtra pročitaet "Vojnu i mir"* (Peter tomorrow read-through War-and-Peace). Indeed, the correct form to convey the intended message would be *dočitaet*. Completive *do-* has a presuppositional meaning (described in Kagan 2015) which is well-suited for culmination-in-focus contexts. In such contexts, perfectives formed by means of *do-* seem to win over perfectives formed by means of "empty" prefixes (like *pro-* in *pročitat'* or *na-* in *napisat'*). Why that should be so is an intriguing topic for future research.

3.6.4 Pluperfect readings (Fig. 7, sce. (d), (e), and (f))

Besides the interpretations discussed so far, DASP allows for three more options, depicted in Figure 7 as the scenarios (d), (e), and (f). Also in these cases, an overlap of the topic time and the run time of the eventuality assigned to e_2 is warranted. We observe systematic relationships: (d) differs from (a) in that the right edge of the topic time goes beyond the end of the run time of $f(e_2)$, and so differ (e) and (f) from (b) and (c), respectively. Below, I argue that these interpretations represent pluperfect, or past perfect, readings (Borik 2006: 133).

Later on in the derivation, when the semantic contribution of T is calculated at TP, the topic time will be related to the time of the utterance. If the tense is past, the topic time interval should be located before the utterance time. Since this also includes the end of the topic time, the configurations (d)–(f) leave no room for the utterance time to fall within the time of the event. This is in contrast to the scenarios (a)–(c) where, under the past tense, the utterance time may be located after the end of the topic time within the time of $f(e_2)$.

This dissociation of the eventuality time from the utterance time entails a transposition of the relevance center of the utterance, as the consequences of the change that the event describes, i.e. the conditions of $f(e_2)$, must now be understood as being relevant to some time prior to the utterance time. The scenarios depicted in (d)–(f) imply, in other words, a secondary evaluation time for the event in addition to the ultimate evaluation time, which is the utterance time.

Comrie (1997) points out that speakers often use the phase particle *uže* 'already' together with a perfective verb in order to express pluperfect meanings, for which there is no specialized grammatical construction in Russian (see also Paslawska & von Stechow 2003: 309). This is not by chance. The semantics of *uže* is such that it contrasts two temporal phases with each other: a phase at which the relevant property (the one delivered by the predicate with which the particle combines) is asserted to hold, and a later phase at which the same property is presupposed to hold (see Ippolito 2007). Applied to our case, the relevant property is the condition of $f(e_2)$. With these conditions being presupposed (and thus expected) to hold at utterance time, it is asserted that they in fact, unexpectedly, held at some earlier time. This "earlier time" (the new evaluation time) is usually introduced by explicit linguistic material. In Paslawska & von Stechow's (2003) example (63), for instance, it is explicated by the temporal adverbial *v vosem' časov*:

(63) V vosem' časov, Maša uže vyšla.
in eight hours M. already go.out.pst.pfv
'At eight, Masha had already left.'

While (63) illustrates the reading (d), (64) exemplifies the reading (e) of Figure 7. This latter scenario is tantamount to the pluperfect version of the "culimination-in-focus" reading (§3.6.2).

(64) Kogda vrač prišel, ona uže rodila.
when doctor come.pst.pfv she already give.birth.pst.pfv
'When the doctor arrived, she had already given birth.' (babyblog.ru)

Here, the earlier evaluation time, relative to which the birth is interpreted, is supplied by the subordinate clause. Finally, (65) shows a pluperfect construal of the perfect reading described in §3.6.3:

(65) V 2012 godu ja uže ustala ot tennisa i rešila
in 2012 year I already get.tired.pst.pfv of tennis and decide.pst.pfv
zakončit'.
finish.inf.pfv
'In the year 2012 I was already tired of tennis and decided to finish with it.' (sports.ru)

By uttering (65), the speaker informs the hearer that she is tired of playing tennis and that this state was held earlier than the hearer presumably expected, namely already in 2012.

3.7 DASP applied to a 2-state-predicate with the second eventuality argument uninstantiated

Recall from above (§3.1) that I treat the secondary imperfective marker YVA as a determiner element, which imposes on the two thematic eventuality arguments of a 2-state predicate the constraint that only the first of these, the source state argument, will be substituted by a discourse referent.[42]

If V is realized by some form of the verbal lexeme *podpisat'*, and if Inst is realized by YVA, the DRS which is constructed by resolving InstP (the verbal counterpart of DP in the nominal projection) will be (49). Now let DASP apply to

[42]In fact, that is only one of two meanings that YVA may have, see (48). For space reasons, I cannot explore the theoretical consequences of the second, pluralizing construction rule (48b) in this chapter. I will have to leave that for future research.

(49) to form the meaning of AspP. Recall that I propose in this chapter that DASP will be operative in *every* full verbal projection in Russian. The construction rule for DASP is stated in (52). It will give us (66).

(66)

$$\begin{array}{|c|}\hline t_{top}\ e_1\ z \\ \hline \textsc{sign}(e_1) \\ \textsc{signed}(e_2) \\ \text{CAUSE}(e_1,e_2) \\ \text{OVL}(e_{fin},t_{top}) \\ e_1 = e_{fin} \\ \text{AUTHOR}(e_1,\underset{.}{x}) \\ \text{TEXT}(e_1,z) \\ \textsc{letter}(z) \\ \hline \end{array}$$

Given this meaning, the relation between the topic time and the run time of the eventuality assigned to e_1 may manifest itself in one of the ways shown in Figure 8.

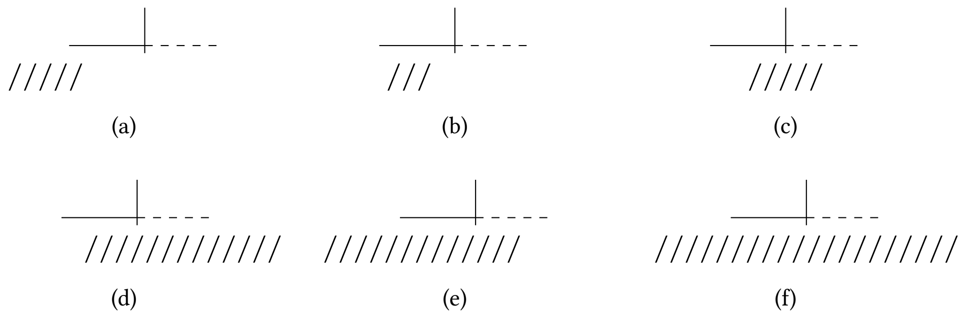


Figure 8: possible scenarios for (66)

In each of the scenarios of Figure 8, the topic time /// overlaps the run time of the eventuality assigned to the final discourse referent, which, due to the impact of YVA, is now e_1. The dotted lines indicate that the respective target state argument e_2 is an implicit argument in (66).

Just what does it mean for an argument to be implicit? Farkas & de Swart (2003: 61) discuss example (67) in this regard. In that example, the implicit argument is the agent of the breaking event.

(67) The vase was broken.

The authors propose that implicit arguments are represented by uninstantiated thematic arguments in final DRSs. As the final DRS for (67) they, accordingly, provide (68).[43]

(68)

u
VASE(u) BREAK(x,u)

The reader who is familiar with standard DRT might want to object that the DRS (68) is not proper because it seems to involve a free discourse referent x. This problem does not arise, however, as x is not a discourse referent, but a thematic argument. Unlike discourse referents, thematic arguments are explicitly allowed to be free in final DRSs (Farkas & de Swart 2003: 60). Of course, this modification of standard DRT calls for clarification of what it means for a condition involving a thematic argument to be verified in a model. To this end, Farkas & de Swart (2003: 63) make the following proposal:

(69) Let i be a variable over the elements in $\{1, \dots, n\}$. A function f verifies a condition of the form $P(a_1, \dots, a_n)$ relative to a model M iff there is a sequence $\langle e_1, \dots, e_n \rangle \in E^n$, such that $\langle e_1, \dots, e_n \rangle \in I(P)$, and if a_i is a discourse referent, $e_i = f(a_i)$, and if a_i is a thematic argument, e_i is some element in E.

Following this example, I will assume that a DRS with an uninstantiated eventuality argument e_2 can be embedded if there is some element in E (the domain of entities in M) that corresponds to e_2. Importantly, nothing in definition (69) requires $f(e_2)$ to belong to the same world as $f(e_1)$.[44] It is certainly true that the entity corresponding to the implicit argument x in (68), the agent of the breaking event, must live in the same world as the entity assigned to the discourse referent u, the broken vase. But this, I claim, is a pragmatic effect. It results from the fact that the existence of a consequent state (the vase being broken) presupposes the existence of a causing event (the breaking involving a causer) in the same world. In the cases of interest to us, the situation is different because of

[43] Farkas & de Swart (2003) use different letters for thematic arguments (x, y, z, ...) and discourse referents (u, v, w, ...). To not confuse the reader with too many different kinds of letters, I decided to use the same letters for thematic arguments and discourse referents in my semantic representations. Whether or not an argument is instantiated can unequivocally be told from whether or not it appears in U_K (see footnote 32).

[44] Note a potential source of confusion: the letter "e" stands for entities in the model in (69), whereas in the main text it stands for elements in the DRS (thematic event arguments or discourse referents).

the flipped temporal order of the eventualities: the existence of a causing event (the signing of a document) does not presuppose the existence of a consequent state (the document being signed). The causing event may conceivably be interrupted half-way before it has caused the consequent state. (70) shows an attested example illustrating this case.

(70) 71-letnij akter v moment DTP perechodil ulicu, kogda na nego
71-year.old actor in moment accident cross.PST.IPFV street when on him
naechal avtomobil'.
drive.at.PST.PFV car
'In the moment of the accident, the 71 year old actor was crossing the street when he was hit by a car.' (dni.ru)

So this is how the problem of the famous imperfective paradox (Dowty 1980, Landman 1992) is addressed in the present approach. Let us now look at the scenarios of Figure 8 one by one.

3.7.1 Ingressive reading (Fig. 8, sce. (a))

In scenario (a) of Figure 8, we revisit the ingressive reading, this time based on a 2-state predicate. Again, it should be noted that this reading is preferably expressed by means of the respective phase verb construction (e.g. *načal otkryvat'*) or, however rare these cases may be, by ingressive *za-* attaching to an otherwise imperfective base, as exemplified by *zaotkryval* in (71) (from Tatevosov 2015: 471).

(71) … chrustnuli rebra, vydavilsja poslednij vozduch iz
crack.PST.PFV ribs squeeze.out.PST.PFV final air from
legkich, i mal'čiška zaotkryval rot, kak ryba.
lungs and boy.DIM start.open.PST.PFV mouth like fish
'… ribs cracked, the last air was squeezed out of the lungs and the little boy started opening his mouth like a fish.'

Besides that, one can also find attestations of bare 2-state imperfectives which are used in contexts where it is the onset of the event described that is at issue. Consider the following: it contains an imperfective verb morphologically derived from the otherwise perfective *nasvistet'* 'whistle a piece of music'.

(72) On umylsja, pereodelsja i nasvistyval
he wash.oneself.PST.PFV change.clothes.PST.PFV and whistle.PST.IPFV
čto-to veseloe.
something cheerful
'He washed himself, changed clothes and started to whistle something cheerful.' (NKRJa)

3.7.2 Progressive reading (Fig. 8, sce. (b))

Secondary imperfectives often actualize the progressive reading, represented in Figure 8 by scenario (b). Recall from above that, due to the status of e_2 as an implicit/uninstantiated argument, the verifying embedding for the final DRS built from (66) does not require the entities corresponding to e_1 and e_2 to belong to the same world.

We already saw one example of the progressive reading in (70). Here is another one.

(73) Kogda ja podpisyval kontrakt, ja znal, čto 'Dinamo'
when I sign.PST.IPFV contract I know.PST.IPFV that D.
nachoditsja daleko ne v lučšem položenii.
is far not in best state
'While I was signing the contract I knew that Dynamo was anything but in good condition.' (vtbrussia.ru)

3.7.3 An impossible reading (Fig. 8, sce. (c))

Scenario (c) in Figure 8 shows another way the topic time may overlap with the time of the eventuality assigned to e_1. I claim, however, that this configuration will never realize with secondary imperfectives, and here is why.

Note that the reading depicted in (c) wants the topic time to end within the time interval of the eventuality corresponding to e_2. This implies that the properties of the eventuality that is caused by $f(e_1)$ are of primary relevance to what the respective sentence conveys (*target state relevance*, see Grønn 2004). This is arguably not reconcilable with the fact that the argument of the caused eventuality is an implicit argument. To put it in a slogan: at-issue content should not be implicitly coded. Therefore, an accommodation process is activated:

(74) *Derived instantiation*
If the topic time assigned to t_{top} begins or ends within some eventuality corresponding to an argument in Con_K, that eventuality requires a discourse

referent in U_K. If there is none, such a discourse referent will be accommodated.

Now see what this amounts to. After the accommodation of a discourse referent e_2 in (66), the picture in (c) should be adjusted accordingly (the dotted line should be replaced by a straight line). What we end up with, then, is the same interpretation as the one depicted in scenario (b) of Figure 7 from above. Since scenario (b) of Figure 7 is taken care of by the same verb without secondary imperfective morphology, we arrive at a situation in which we have two candidate verbs for the same interpretation, *podpisat'* and *podpisyvat'*. One of these is structurally more complex than the other. In such a situation, the pragmatic mechanism of *morphological blocking* (Kiparsky 2005, Deo 2012) will filter out the more complex expression.[45]

Applied to our case, the impact of morphological blocking is that it will prevent secondary imperfectives from expressing the culmination-in-focus reading (scenario (b) of Figure 7). Being equivalent to scenario (b) of Figure 7, scenario (c) of Figure 8 turns out to be no interpretive option for verbs like, for instance, *podpisyvat'*, *vybrasyvat'* or *otkryvat'*.[46]

3.7.4 Annulment-of-result reading (Fig. 8, sce. (d))

Now let us consider Figure 8, scenario (d). The topic time does not end within the time of the eventuality corresponding to e_2, but extends beyond its offset.[47] If tense is past, the end of the target state eventuality must be understood as being located prior to the utterance time. So far, the interpretation depicted in this scenario matches the one of scenario (e) in Figure 7.

Moreover, a sentence expressing scenario (d) of Figure 8 will never be about the whole complex event made up of one eventuality causing the other because the initial phase of the causing eventuality lies outside of the topic time. This is also in line with scenario (e) of Figure 7.

The crucial difference between scenario (d) of Figure 8 and scenario (e) of Figure 7 relates to the absence of a discourse referent for e_2 in the DRS generating

[45]This morphological blocking mechanism is often stated as an economy constraint saying that, all other things being equal, less complex expressions are preferred over more complex expressions (e.g. Le Bruyn 2007).

[46]Looking at scenario (c) of Figure 8, a reviewer wonders whether my approach predicts secondary imperfectives to allow for the perfect reading. No, it does not. The perfect reading falls into the range of interpretative options for perfectives (see §3.6.3). Therefore, the use of the structurally more complex secondary imperfective in this function will also be ruled out by morphological blocking.

[47]This is why (74) does not apply.

the former, because the argument e_2 is an uninstantiated argument in (66). In §3.7.3 I stated that implicitly coded information can only have not-at-issue status. So the question arises: Is there a way to understand a sentence that is about a caused state that is not at issue? Yes, there is. What one needs to do is to draw the implicature that the target state, which should belong to what the utterance is about, is cancelled within the limits of the topic time (see Grønn 2004: 236; see also §2.2). This way, what the utterance is about is no longer the mere change from the source state to the target state, but rather the double change from the source state to the target state and again away from the target state. Of course, such an interpretation should be supported by the lexical semantics of the predicate (which should be such that the target state is cancellable under normal circumstances). I claim, in other words, that the reading of scenario (d) of Figure 8 is attested by the much-discussed annulment-of-result imperfectives (see Rassudova 1982, Padučeva 1996). Recall (17) as an illustrative example.

3.7.5 Another impossible reading (Fig. 8, sce. (e))

In Figure 8, scenario (e) shares with scenario (c) the feature that the topic time ends within the interval of the caused eventuality, with the argument of the caused eventuality being uninstantiated. As argued in (74), such a scenario calls for accommodation of a discourse referent e_2. Since this will in effect make scenario (e) equivalent to scenario (a) of Figure 7, and since this latter interpretation is taken care of by a less complex verb (the same verb without secondary imperfective morphology), scenario (e) of Figure 8 turns out to be no interpretive option for a secondary imperfective, due to morphological blocking.

3.7.6 General-factual reading (Fig. 8, sce. (f))

The final reading to be discussed is shown in scenario (f) of Figure 8, which resembles scenario (d) of Figure 7 in many respects. Indeed, the two scenarios match in every detail, except for the way the information about the caused event is conveyed. In both cases, the topic time properly includes the intervals of the causing eventuality and the caused eventuality. The difference is that, while the argument e_2 of the DRS belonging to scenario (d) of Figure 7 is instantiated, the argument e_2 of the DRS belonging to scenario (f) of Figure 8 is uninstantiated.

Expecting that implicitly coded information provides not-at-issue content, we are entitled to draw the implicature that the specific conditions of the caused eventuality (target state) are irrelevant to the speaker's message. Since this is the core characteristics of general-factuals (Švedova 1980: 611; Grønn 2004), we arrive at the interpretation that we missed in Bohnemeyer & Swift's (2004) theory, i.e. the one which is expressed by sentences such as (18).

What *is* relevant instead of the target state conditions are the consequences of a rule in the background knowledge of the interlocutors, which is pragmatically pointed at by stating that an event of the respective kind has taken place. In (18), for instance, the relevant background rule follows from the fact that pancake recipes foresee only one flip of the pancake. Therefore, if a pancake has already been flipped, there is no need to do that again; see Mueller-Reichau (2018) for more on the relevance of presupposed rules for general-factual interpretations.

3.8 Briefly on external prefixation

The distinction between internal and external prefixes is well established in current research on the Slavic verb (see §2.7). The most detailed stocktaking of Russian prefixation has probably been carried out in Tatevosov (2013). According to that study, external prefixes subclassify into left-peripheral, selectionally restricted, and positionally restricted prefixes. Here I can only scratch the surface and address selectionally restricted prefixes, in particular ingressive *za-* and delimitative *po-*.

Selectionally restricted prefixes are so-called because of their limited combinatorics, as they successfully combine with an imperfective base only. The result of their application is a perfective. Being aspect switchers, so to speak, they play a central role in the aspectual system of Russian. As pointed out by Dickey (2006), they fill a systemic gap by deriving perfectives from atelic (or as I would say: 1-state) predicates.

Following once again Ramchand (2008), I treat Russian ingressive *za-* and delimitative *po-* as operating over the meaning of AspP, see (35). Note that the high application of external prefixes is essential to an analysis that considers perfective and imperfective meanings produced by DASP to be dependent on the presence or absence of state change.

3.8.1 Ingressive *za-*

Letting ingressive *za-* apply after DASP only seems consequent in view of its selectional restriction to imperfective inputs. Its contribution of meaning may be stated in terms of the following construction rule:[48]

[48] P stands for the property of the morphological base to which *za-* attaches, the function $t_{init}(e)$ maps an eventuality onto its initial moment; the function $t_{fin}(e)$ maps an eventuality onto its final moment.

(75) Construction rule for ingressive *za-*
- Introduce in U_K: a new eventuality discourse referent e'
- Introduce in Con_K: $\supset\subset(e', e)$
- Introduce in Con_K: $\text{OVL}(t_{top}, e')$
- Introduce in Con_K: $\neg\, \text{P}(e')$
- Introduce in Con_K: $\neg\, \text{OVL}(t_{top}, t_{init}(e'))$
- Introduce in Con_K: $\neg\, \text{OVL}(t_{top}, t_{fin}(e))$

Take a sentence whose VP is formed by the 1-state verb *kuril* 'he smoked'. According to what was said above, resolution of the AspP-node will give us the DRS (76).

(76)

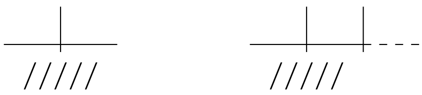
t_{top} e
$\text{SMOKE}(e)$
$\text{OVL}(e,t_{top})$
$\text{SMOKER}(e,\underset{\cdot}{x})$

Now let there be ingressive *za-* attached as an adjunct to AspP. On account of the construction rule in (75), the DRS resulting from processing the higher AspP *zakuril* will look as follows:

(77)

t_{top} e e'
$\text{SMOKE}(e)$
$\text{OVL}(e,t_{top})$
$\text{SMOKER}(e,\underset{\cdot}{x})$
$\supset\subset(e',e)$
$\text{OVL}(t_{top},e')$
$\neg\, \text{SMOKE}(e')$
$\neg\, \text{OVL}(t_{top},t_{init}(e'))$
$\neg\, \text{OVL}(t_{top},t_{fin}(e))$

This DRS will give rise to the readings depicted in Figure 9. Note that it is the second eventuality alone that satisfies the property described by the VP (due to condition "$\neg$ smoke(e')"). The respective sentence is about a change from a situation of not smoking to a situation of smoking. (54) shows such a sentence. As we saw in (71), moreover, ingressive *za-* may also be used to prefix secondary imperfectives. This will give rise to an interpretation according to scenario (b).



(a) (b)

Figure 9: possible scenarios for (77)

3.8.2 Delimitative *po-*

The second selectionally restricted prefix that I want to discuss here is delimitative *po-*. (78) shows a representative example.

(78) A ja vyšel, pokuril i vzjal taksi.
and 1SG go.out.PST.PFV smoke.a.bit.PST.PFV and take.PST.PFV taxi
'And I went out, smoked a bit and took a taxi.' (NKRJa)

To account for cases like *pokuril* in (78), I propose the following construction rule:

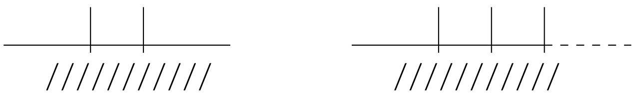
(79) Construction rule for delimitative *po-*
- Introduce in U_K: a new eventuality discourse referent e'
- Introduce in U_K: a new eventuality discourse referent e''
- Introduce in Con_K: $\supset\subset (e', e)$
- Introduce in Con_K: $\supset\subset (e, e'')$
- Introduce in Con_K: $\text{OVL}(t_{top}, e')$
- Introduce in Con_K: $\text{OVL}(t_{top}, e'')$
- Introduce in Con_K: $\neg$ P(e')
- Introduce in Con_K: $\neg$ P(e'')
- Introduce in Con_K: $\neg\ \text{OVL}(t_{top}, t_{init}(e'))$
- Introduce in Con_K: $\neg\ \text{OVL}(t_{top}, t_{fin}(e''))$

Following the same procedure as above, the resolution of the first AspP-node will give us again (76). To this, we let delimitative *po-* attach. The construction rule (79) will lead us to the meaning (80) for the higher AspP *pokuril*:

(80)

t_{top} *e e' e"*
SMOKE(e)
OVL(e,t_{top})
SMOKER(e,$\underset{\cdot}{x}$)
⊃⊂ (e',e)
⊃⊂(e,e'')
OVL(t_{top},e')
OVL(t_{top},e'')
¬ SMOKE(e')
¬ SMOKE(e'')
¬ OVL(t_{top},$t_{init}(e')$)
¬ OVL(t_{top},$t_{fin}(e'')$)

This DRS constrains possible interpretations to readings of the kind depicted in Figure 10.



(a) (b)

Figure 10: possible scenarios for (80)

As can be seen, we arrive at an analysis that treats *po*-delimitatives as expressing two subsequent changes, with the intermediate eventuality falling under the description of the 1-state predicate to which the prefix attaches. Such an analysis of Russian *po*-delimitatives as expressing "3-state-contents" is not new. It has been proposed before by Šatunovskij (2009: 313–314).

Delimitative *po*- may also apply to secondary imperfectives, on condition that the eventuality leading to the state change (i.e. $f(e_1)$) is homogeneous (see Mehlig 2006 for details). This results in the reading depicted in scenario (b) of Figure 10.

3.8.3 Loose ends

Besides positionally-restricted and left-peripheral prefixations, I also ignore other selectionally-restricted prefixes noted by Tatevosov (2013: 49), such as distributive *pere*- and cumulative *na*-. Since their semantic impact goes beyond the manipulation of temporal relations, it is more complicated to state appropriate construction rules for these prefixations. So I refrain from doing that here, but I want to mention egressive *ot*-, the antipode of ingressive *za*-. Consider (81), which is among the top ten phrases of 2020 in Russia.[49] In this example, egressive *ot-* attaches to the state verb *imet'* 'have'. The resulting perfective *otymet'* describes a change from having something to not having something.

(81) Žizn' otymela smysl.
life cease.to.have.PST.PFV sense
'Life no longer makes sense.'

The interpretation of this sentence will look like scenario (a) of Figure 9, whereby this time it is the first eventuality alone that satisfies the property of the base, P:

(82) Construction rule for egressive *ot-*
- Introduce in U_K: a new eventuality discourse referent e'
- Introduce in Con_K: $\supset\subset (e, e')$
- Introduce in Con_K: $\textsc{ovl}(t_{top}, e')$
- Introduce in Con_K: $\neg$ P(e')
- Introduce in Con_K: $\neg$ $\textsc{ovl}(t_{top}, t_{init}(e))$
- Introduce in Con_K: $\neg$ $\textsc{ovl}(t_{top}, t_{fin}(e'))$

Let me end by pointing to a problem that my way of treating selectionally-restricted (SR) prefixes faces. It is well-known that many perfectives that are derived that way (i.e. by attachment of a SR-prefix) do not undergo secondary imperfectivization. This is, so far, in line with the theory presented above in which YVA applies before DASP and SR-prefixes apply after DASP. There are, however, also cases where SR-prefixes do allow for secondary imperfectivization, contra to what my analysis is predicting. Prominent examples are *zapevat'* 'start singing' and *zakurivat'* 'start smoking'. According to Zaliznjak et al. (2015), the possibility of deriving such imperfectives increases with the frequency of the respective perfective. Given this, I am tempted to argue that in cases like *zapevat'* or *zakurivat'*, the respective prefixed verb has been lexicalized, i.e. reinterpreted by speakers as an instance of lexical prefixation. But this certainly deserves more exploration.

3.9 Summary

Above, I presented a theory that links verb forms to the range of aspectual interpretations that they can actualize in Russian texts. In that approach, a single zero operator manages to correctly distribute perfective and imperfective forms among contexts.

[49] https://echo.msk.ru/blog/epsht.m/2759832-echo/

The focus of the analysis was on the lexical stage of Russian verb formation, the domain of lexical/internal prefixes, and so-called secondary imperfective suffixes. In the end, I added some discussion of selected external prefixes. I gave a DRT-analysis in which verbal prefixes and suffixes do not by themselves express perfective or imperfective meanings (semantic aspects). Instead, these affixes are used to prepare, so to speak, the input of the aspect operator. To this end, the story told is in line with conclusions drawn in works such as, inter alia, Filip (2003) or Tatevosov (2017).

The aspect operator DASP is conceived of, in the spirit of Klein (1994), as establishing a relation between topic time and eventuality time. DASP is stated such that the eventuality assigned to the final discourse referent has to overlap topic time. Crucial is the assumption that secondary imperfective YVA marks the second argument of a 2-state predicate as an implicit argument, meaning that it has no discourse referent on its own.

This gives us precisely what we observe: 2-state predicates without secondary imperfective morphology yield perfective meanings, while 1-state predicates and 2-state predicates with secondary imperfective morphology yield imperfective meanings. "Perfective meanings" are meanings where the utterance is about the conditions of the instantiated target state. In the canonical case the topic time ends when the target state is in force (*target state validity*, see Grønn 2004). A variation on the theme are pluperfect readings, where the part of the topic time that goes beyond the end of the target state is dislocated from the utterance time. "Imperfective meanings" are all those meanings that do not entail target state validity.

As for external prefixes like delimitative *po-* or ingressive *za-*, I have proposed that these come into play after DASP has applied (in line with Ramchand 2004, 2008). One might want to call this the second (syntactic) cycle of Russian verb formation. Accordingly, the semantics of these prefixes has to be stated such that they apply to already "aspectualized" meanings. These operators, in other words, take the output of DASP as input, mapping properties of times onto properties of times. Their output is a perfective meaning inasmuch as the topic time is required to end within the time of the final instantiated eventuality.

Acknowledgements

I wish to thank Berit Gehrke, Radek Šimík and three anonymous reviewers for their very helpful comments. Many of the thoughts expressed here are the product of extremely constructive discussions with Hana Filip. I am also grateful to the audiences at the conference Formal Approaches to Russian Linguistics 3 (2019, Moscow State University), and at the Slavicus Expert Meeting on Aspect in Slavic (2022, University of Wrocław) for criticism and comments. Special thanks go to: Daniel Altshuler, Hanna Bazanava, Petr Biskup, Marcel Börner, Tom Elvins, Dorota Klimek-Jankowska, Serge Minor, Michael Müller, Gillian Ramchand, Bożena Rozwadowska, Iuliia Shcherbina, Yulia Sorokina, Maria Sulimova, Sergei Tatevosov, Marcin Wągiel, Yulia Zinova, Karolina Zuchewicz. This chapter was submitted in February 2020. I regret that work published after that date, including my own, could not be considered.

Abbreviations

1	1st person	INF	infinitive
ACC	accusative	IPFV	imperfective
DAT	dative	PFV	perfective
DIM	diminutive	PST	past
GEN	genitive	REFL	reflexive pronoun
IMP	imperative	SG	singular

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Chapter 9

Polar question semantics and bias: Lessons from Slavic/Czech

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This chapter is concerned with the semantics and pragmatics of polar questions, i.e. questions which solicit a 'yes' or 'no' answer. It provides a survey of the main empirical and theoretical issues surrounding polar questions while concentrating on Slavic languages, and especially Czech. My main goals are (i) to provide an accessible introduction into the area of polar question meaning and bias, building primarily on evidence from Czech, (ii) to provide an overview of how polar questions are formally encoded in different Slavic languages, and (iii) to provide an analysis of Czech negative polar questions, with special attention paid to so-called outer (aka "pleonastic") negation and to the particle *náhodou* lit. 'by chance'.

1 Introduction

Polar questions, like *Does Jane play golf?*, are utterances which solicit a 'yes' or 'no' response from the addressee (e.g. *Yes / (Yes,) She does (play golf)*). Despite the apparent simplicity of these constructions, polar questions represent a lively research topic in all areas of grammar – from prosody and morphosyntax to semantics and pragmatics. Polar questions are vital for our understanding of direct and indirect speech acts and have sparked controversies about how speech acts should be represented in the grammar. The function of polar questions is rarely simply to ask a question (indicate the way in which one is ignorant) in order to solicit an answer from the addressee (become knowledgeable); in addition, polar questions often convey the speaker's attitude towards one or both of the possible answers. These attitudes are called biases and have a crucial impact on the ways in which polar questions are formally encoded.

Radek Šimík. 2026. Polar question semantics and bias: Lessons from Slavic/Czech. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 461–520. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078585

This chapter is primarily about the semantics and pragmatics of polar questions in Slavic languages. Slavic polar questions represent an empirical area which is largely unexplored – at least within formal approaches to natural language meaning – and is at the same time very rich in the variety of formal means used for expressing such questions. In addition to prosodic strategies, which are to a varying extent present in all the Slavic languages, some Slavic languages resort to word order alternations (subject–verb/auxiliary inversion), others to particles (clause-initial or encliticized), and yet others to some combination of the two. This immediately raises the question of how these strategies are mapped to meaning – whether to the core question meaning (asking and soliciting a response) or to the various biases.

Since formal-semantic work on Slavic polar questions has only just begun to gain momentum, this chapter is less of a survey of existing accounts and more of an incentive for future research. My goals are (i) to provide an accessible introduction into the area of polar question meaning and bias, building primarily on evidence from Czech, in comparison to other Slavic languages, as well as better-studied languages such as English, (ii) to provide an overview of how polar questions are formally encoded in different Slavic languages, and (iii) to provide an analysis of Czech negative polar questions, with special attention paid to so-called outer (aka "pleonastic") negation and to the particle *náhodou* lit. 'by chance'.

The chapter is organized as follows. In Section 2 the notion of a polar question is introduced in more detail, building on Czech data, and is contrasted with related notions and constructions such as alternative questions, wh-questions, or embedded interrogatives. Section 3 concentrates on the notion of question bias. Using novel data patterns from Czech, it illustrates the core strategies of encoding bias (esp. negative or declarative polar questions) and briefly sketches the main (competing) approaches to modelling biases in formal semantics and pragmatics. The strategies of encoding polar questions in Slavic languages are discussed in Section 4, where I also discuss existing formal-semantic and recent empirical work on Slavic polar questions. The section contributes novel data from a diverse set of Slavic languages (collected via informal elicitation from native-speaker linguists) and focuses on the default (unbiased) forms of polar questions. My analysis of Czech polar questions with outer negation and with the particle *náhodou* is presented in Section 5. The main take-home message from this part of the chapter is that outer negation in Czech – and potentially Slavic more generally – has a significantly more underspecified meaning – and a correspondingly broader distribution – than their kin in better-studied languages, esp. English and German. Section 6 presents some conclusions.

2 Polar questions and related constructions

A POLAR QUESTION (PQ), also called a "yes–no question", is an utterance which invites the addressee to provide one of two answers – either an affirmative ('yes') or a negative ('no').[1] A PQ is exemplified by (1a), an example from Czech, in which PQs commonly involve a combination of verb-initial word order (see Staňková 2023 and the discussion of example (7) below) and a dedicated utterance-final intonation contour (either rising or rise-falling; Daneš 1957, Veroňková 2002).[2] As opposed to a STATEMENT, exemplified in (1b), it does not have truth conditions, i.e. it cannot be judged as true or false. Ever since Hamblin (1973), the most common way of modelling the lack of truth conditions in questions is to give them the denotation of the set of propositions which constitute the question's possible answers. The PQ in (1a) would thus denote the set (2a)/(2b), or, for brevity's sake, (2c).[3,4]

(1) a. Snížila digitalizace počet úředníků?
decreased digitization.NOM number.ACC officials.GEN
'Has digitization decreased the number of officials?'

b. Digitalizace snížila počet úředníků.
digitization.NOM decreased number.ACC officials.GEN
'Digitization has decreased the number of officials.'

(2) Hamblin (1973)-style denotation of (1a)

a. {'digitization has decreased the number of officials',
'digitization has not decreased the number of officials'}

b. {λw[digitization has decreased the number of officials in w],
λw[digitization has not decreased the number of officials in w]}

c. $\{p, \neg p\}$

[1] The meaning of response particles is a closely related topic which goes beyond the scope of this chapter. For some recent work on Slavic languages, see Esipova (2021), Geist & Repp (2023), Hrdinková & Šimík (2025).

[2] Unless stated otherwise, all examples are in Czech and English, and the meaning and felicity of all the Czech examples is supplied by the author. The author is aware that there is speaker variation and takes the acceptability/felicity/interpretation judgements to be hypotheses about a more reliable and replicable state of affairs.

[3] A PROPOSITION is standardly defined as a function that characterizes the set of possible worlds in which it is true, whence (2b).

[4] Accessible introductions to (the theories of) question semantics include Hagstrom (2003) and Cross & Roelofsen (2014).

Polar questions need to be distinguished from two other major types of question – WH-QUESTIONS, exemplified in (3a), and ALTERNATIVE QUESTIONS, exemplified in (3b) and (3c). Unlike PQs, wh- and alternative questions typically involve falling intonation and generally cannot be answered by plain response particles (such as *ano* 'yes' or *ne* 'no'). In addition, wh-questions obligatorily exhibit a wh-word or phrase (*jak moc* 'how much' in (3a)) and alternative questions an exclusive disjunction (*nebo* 'or' in (3b)/(3c)). Like PQs, wh- and alternative questions solicit an answer from the addressee, lack truth conditions, and are typically modelled by means of sets of propositions. The set denoted by wh-questions, (4a), is systematically tied to the denotation of the wh-phrase, often has a cardinality larger than 2, and can in principle be infinite; alternative questions denote a set whose cardinality matches the number of explicitly mentioned disjuncts, which happens to be 2 in both (3b) and (3c); cf. (4b) and (4c), respectively.

(3) a. Jak moc snížila digitalizace počet úředníků?
how much decreased digitization.NOM number.ACC officials.GEN
'How much did digitization decrease the number of officials?'

b. Snížila digitalizace počet úředníků významně, nebo zanedbatelně?
decreased digitization.NOM number.ACC officials.GEN significantly or negligibly
'Has digitization decreased the number of officials significantly or negligibly?'

c. Snížila digitalizace počet úředníků, nebo ne?
decreased digitization.NOM number.ACC officials.GEN or not
'Has digitization decreased the number of officials or not?'

(4) Hamblin (1973)-style denotation of wh- and alternative questions

a. {'digitization has decreased the number of officials by 10 %',
'digitization has decreased the number of officials by 20 %',
'digitization has decreased the number of officials by 30 %',
…}

b. {'digitization has decreased the number of officials significantly',
'digitization has decreased the number of officials negligibly'}

c. {'digitization has decreased the number of officials',
'digitization has not decreased the number of officials'}

The identity of (2) and (4c) – the denotations of the PQ (1a) and the alternative question (3c), respectively – indicates the limits of a Hamblin-style semantics for modelling the meaning of the two question types. Given that PQs and alternative questions differ not only in formal, but also in meaning-related respects (related to exhaustivity and discourse-pragmatic factors; see Bolinger 1978, Biezma 2009, Biezma & Rawlins 2012), their semantics, or at least pragmatics, must be different in some way. As will be discussed in Section 3.3.1, differences between alternative and polar questions have motivated modifications to a Hamblin-style semantics of polar questions.

The insufficient specification – or perhaps underspecification – of Hamblin's (1973) semantics is further evident from the fact that the PQ in (1a) is semantically indistinguishable from its negated counterpart (5a). Assuming, for the sake of the argument, that (5a) involves canonical negation (see below for more discussion), its semantics is captured in (5b), and more schematically in (5c), which, building on the axioms of propositional logic and set theory, highlights the identity with (2c). This identity is an oversimplification: above and beyond posing a question (whose meaning could well be captured by (5b)/(5c)), negative PQs are known to convey a bias – or even more biases of a different kind – towards one of the two possible answers. The notion of bias and the kinds of context that negative PQs can appear in will be discussed in detail in Section 3.

(5) a. Nesnížila digitalizace počet úředníků?
NEG.decreased digitization.NOM number.ACC officials.GEN
'Hasn't digitization decreased the number of officials?'

b. {'digitization has not decreased the number of officials',
'it is not the case that digitization has not decreased the number of officials'}

c. $\{\neg p, \neg\neg p\} = \{\neg p, p\} = \{p, \neg p\}$ = (2c)

Negative PQs are not the only questions that are considered biased. Bias also characterizes DECLARATIVE POLAR QUESTIONS (Gunlogson 2002; henceforth also referred to as DeclPQs) – utterances which solicit answers, lack truth conditions, and typically have a PQ-dedicated intonation, but at the same time exhibit a declarative (morpho)syntax, as illustrated in (6), an example with a canonical SVO word order. Given their core meaning, declarative PQs represent yet another type of utterance which can be assigned the Hamblin-style semantics above. And again, this meaning does not capture the added value of declarative PQs, namely their specific bias profile (Gärtner & Gyuris 2017).

(6) Digitalizace {snížila / nesnížila} počet úředníků?
digitization.NOM decreased NEG.decreased number.ACC officials.GEN
'Digitization {has / hasn't} decreased the number of officials?'

Declarative PQs stand in formal and semantic/pragmatic opposition to INTERROGATIVE POLAR QUESTIONS (henceforth also INTERPQs), which exhibit dedicated interrogative (morpho)syntax, such as verb-first in Czech (see (1a)) or the interrogative particle *li* in Bulgarian (Rivero 1993, Rudin et al. 1999; see Section 4), and are – or at least can be – construed as unbiased (neutral).

A brief digression on the form of Czech interrogative polar questions is in order. It is common to assume that Czech does not have a dedicated interrogative grammatical form and that it encodes polar questions by intonation alone (see e.g. Veroňková 2002; but cf. Staňková 2023 or the feature GB260 of Grambank, Skirgård et al. 2023, which draws on Naughton 2005). This conviction rests on the observation that verb-initial clauses can function both as statements and as questions and are thus not indicative of interrogativity; see (7a) (final punctuation is intentionally left out). Nevertheless, this observation is limited: as is clear from (7b), if the subject is overt, a verb-initial root clause can only be construed as a question. For this reason, all the Czech examples in this chapter involve an overt subject.[5]

(7) a. Viděla tam svého syna
saw.SG.F there REFL.POSS son.ACC
  i. ✓ ‘She saw her son there.’
  ii. ✓ ‘Did she see her son there?’

b. Viděla tam Hana svého syna
saw.SG.F there Hana.NOM REFL.POSS son.ACC
  i. ✗ ‘Hana saw her son there.’
  ii. ✓ ‘Did Hana see her son there?’

Coming back to our main discussion: If there are declarative (polar) questions, are there also “interrogative statements”? That is, are the grammatical form of a sentence and its ILLOCUTIONARY FORCE in principle independent of one another? What comes close to “interrogative statements” are RHETORICAL POLAR QUESTIONS (Biezma & Rawlins 2017). As illustrated in (8A), a PQ can indeed be used rhetorically, i.e. without soliciting an answer or in fact conveying the answer

[5]The generalization is actually even more complex than that. Verb-initial root clauses with an overt subject can be construed as statements (i) if the subject is focused and, at the same time, (ii) if there is no other non-pronominal constituent than the subject. To the best of my knowledge, these morphosyntactic constraints have yet to receive proper attention in the literature.

itself. The rhetorical meaning can be, but does not have to be supported by particles, such as *snad* in (8A).[6] Also, the rhetorical reading does not need any specific intonational support. That the PQ below can have the force of a statement is supported by the felicity of the continuation in (8B).

(8) A: Snížila (snad) digitalizace počet úředníků?
decreased SNAD digitization.NOM number.ACC officials.GEN
'Has digitization decreased the number of officials?' / 'Digitization hasn't decreased the number of officials, as you surely know.'

B: Máte pravdu.
have.2SG.HON truth.ACC
'You're right. [It didn't.]'

Finally, let us note that there are interrogative clauses which lack any illocutionary force whatsoever. These are called EMBEDDED POLAR INTERROGATIVES (also known as "embedded" or "indirect polar questions") and are illustrated by (9b). Embedded polar interrogatives are introduced by dedicated complementizers, such as *jestli* or *zda* 'whether' in Czech, and are selected by responsive predicates (embedding declarative or interrogative clauses; e.g. 'know', 'find out') or rogative predicates (embedding only interrogative clauses; e.g. 'wonder', 'ask').[7] According to Karttunen's (1977) influential proposal, embedded interrogatives denote the set of true answers (i.e. not all possible answers, as argued by Hamblin 1973).[8] In the case of polar interrogatives, this corresponds to the singleton set containing the true answer – either affirmative or negative. The logical forms in (10) capture the truth conditions of (9), relying on Dayal's (1996) idea that embedded interrogatives denote a definite description of the proposition which corresponds to the true answer. The version of (9) with *nevěděl* 'NEG.knew' is true iff there is a unique true proposition in the set $\{\lambda w$[digitization has decreased the number of officials in w], λw[digitization hasn't decreased the number of officials in w]$\}$ ($\{p, \neg p\}$ for short) and Mirek didn't know the proposition; see (10a). The version with *zajímalo* 'wondered' can be derived from the previous one by decomposing the attitude predicate into 'want to know'; (10b).

[6] An in-depth corpus study of the particles *snad* and *náhodou* in PQs is available in Chodounská et al. (2025).

[7] Predicates which only embed declarative clauses (e.g. *hope*) are called anti-rogative; for recent discussion and references, see Theiler et al. (2019), Uegaki & Sudo (2019).

[8] For an approach in terms of possible worlds partition and its empirical consequences, see Groenendijk & Stokhof (1984).

(9) a. {Mirek nevěděl / Mirka zajímalo / … },
Mirek.NOM NEG.knew.SG.M Mirek.ACC interested.SG.N
'Mirek didn't know / Mirek wondered / …'

b. {jestli / zda} digitalizace (ne-) snížila počet
whether whether digitization NEG decreased number.ACC
úředníků.
officials.GEN
'whether digitization has(n't) decreased the number of officials.'

(10) a. $\neg\textsc{know}\big(w_0\big)\big(\textsc{Mirek}\big)\big(\iota q\, q(w_0) = 1 \wedge q \in \{p, \neg p\}\big)$
b. $\textsc{want to know}\big(w_0\big)\big(\textsc{Mirek}\big)\big(\iota q\, q(w_0) = 1 \wedge q \in \{p, \neg p\}\big)$

In summary, polar questions are utterances which solicit 'yes' or 'no' answers. They do not have truth conditions themselves, but are systematically related to them via their possible or true answers, which – collected in a set – constitute their denotation. Polar questions must be distinguished from related constructions, esp. wh-, alternative, or rhetorical questions. In this chapter, I concentrate on polar questions (including their different forms, e.g. interrogative vs. declarative, positive vs. negative), to some extent on embedded polar interrogatives, but leave other question types aside.

3 Bias in polar questions

3.1 Basic notions

By posing a (non-rhetorical) polar question (PQ), the speaker solicits an answer from the addressee. This is modelled by the set-of-propositions denotation of (polar) questions: the speaker does not contribute information, but rather highlights the kind of information that he is missing. By putting forth a set of propositions, he offers the addressee an issue to be resolved – referred to as the (current) question under discussion (Roberts 1996/2012; Farkas & Bruce 2010) – and expects her to supply the information needed.[9]

Plain information-seeking of this kind, however, is not always – and perhaps even *mostly not* – the only motivation for asking a PQ (van Rooij & Šafářová 2003, AnderBois 2019, Repp & Geist 2025; among many others). One frequently discussed case involves a situation where the speaker asks a PQ because, on the one hand, he holds a prior belief that *p*, but this belief has recently been

[9]For ease of pronominal reference, I treat the speaker as a male and the addressee as a female.

put into doubt by evidence that $\neg p$. Asking a PQ denoting $\{p, \neg p\}$ can thus be motivated by resolving two conflicting (contradictory) sources of information – the speaker's private EPISTEMIC BIAS and the contextual and public EVIDENTIAL BIAS (Sudo 2013, building on Ladd 1981, Büring & Gunlogson 2000, Romero & Han 2004; a.o.).[10]

It has been argued that the speaker's intention (called QUESTION CONCERN by Repp & Geist 2025) to double-check the epistemic or the evidential bias can motivate the use of different PQ grammatical forms or particles. The use of high vs. low negation in English PQs (*Aren't you Swedish?* vs. *Are you not Swedish?*) or the use of the particles *razve* vs. *neuželi* in Russian PQs are cases in point: the former devices have been claimed to double-check the epistemic or evidential bias, while the latter are supposed to be specialized for double-checking the evidential bias (Romero & Han 2004, Repp 2013, Romero 2015, Repp & Geist 2025).

Not all PQs are posed with the primary aim of resolving the issue $\{p, \neg p\}$ or a conflict between two information sources each pointing to a different state of affairs (p vs. $\neg p$). These – what we could call POLARITY-SEEKING PQs (a term from Esipova & Romero 2023) and CONFLICT-RESOLVING PQs, respectively – can be contrasted with EXPLANATION-SEEKING PQs (Tomioka & Kim 2014, Esipova & Romero 2023; see also Korotkova 2023).[11] An explanation-seeking PQ puts forth a proposition p (in the form of a question) as a possible explanation (cause) of an observed effect. If, for instance, Max does not show up for a planned meeting, it could be that he is sick (p), had other, unexpected obligations (q), or got stuck in a traffic jam (r). Asking *Is he sick?* in this scenario contributes not primarily to resolving the issue $\{p, \neg p\}$ but to resolving the issue $\{p, q, r, \ldots\}$, which corresponds to the causal question *Why did Max not show up?*[12] Explanation-

[10] This terminological distinction, introduced by Sudo (2013), is currently the most popular one, including in the literature on Slavic PQs (see, e.g., Korotkova 2023, Staňková 2023, Repp & Geist 2025). Oppositions used in the same or a similar sense include speaker vs. addressee bias (Rudin 2022), speaker vs. contextual bias (Northrup 2014), and speaker vs. evidential bias (Onoeva & Staňková 2025). Bias has also often been referred to as "presumption", and biased questions as "presumptive questions" (e.g. Rakić 1984).

[11] Explanation-seeking questions correspond to what Logvinova (2022) calls "inference-based questions", following Bolinger (1978) and Mehlig (1991).

[12] As an anonymous reviewer noticed, this begs the question whether explanation-seeking PQs are not more on a par with wh-questions than with PQs. My answer is that they have properties of both question types. Besides raising an issue corresponding to a why-question (a property of wh-questions) and at the same time proposing a possible answer (a property of PQs), responding with a plain 'no' to an explanation-seeking PQ (as opposed to a polarity-seeking one) does not provide a complete (exhaustive) answer – leaving the issue $\{q, r\}$ unresolved – and is therefore felt not to be cooperative. (This corresponds to a response like 'Dave didn't' in reaction to the wh-question 'Who came?'.)

seeking PQs count as evidentially biased (Max's absence being the evidence at issue), but do not necessarily convey prior epistemic bias.[13] Explanation-seeking PQs are briefly discussed in Section 4.2.3 in connection with accent-placement in Russian PQs.

Finally, information-seeking is also not the primary concern in situations in which PQs are used to convey requests (*Would you pass me the knife?*), offers (*Do you want a glass of water?*), suggestions (*Shall we go to the movies?*), or the like. In these cases, resolving the issue $\{p, \neg p\}$ is closely tied to a subsequent action performed by the addressee, speaker, or both (i.e. a "perlocution"; Austin 1962). In contrast to explanation-seeking PQs, requests and offers are typically not accompanied by evidential bias, but may involve a speaker's private bias, which could be called PREFERENTIAL BIAS, expressing a preference, hope, or in some cases dispreference/fear that the prejacent of the question (p) is true and will have a corresponding extra-linguistic effect.[14]

In what follows, I provide some basic illustrations of how bias correlates with formal properties, especially word order and negation. For a more comprehensive and exhaustive approach to bias, building on Sudo's (2013) epistemic vs. evidential distinction and on the idea that a bias can either be oblitatorily present, obligatorily absent, or optionally present, see esp. Gärtner & Gyuris (2017). A recent survey on biased polar questions can be found in Romero (2024).

3.2 Expressing bias: the case of Czech

An unbiased PQ is hard to come by. To this end, Repp & Geist (2025) claim that "there is no such thing as a truly neutral [polar] question" (p. 112).[15] As observed by Büring & Gunlogson (2000), even a plain positive PQ is biased against negative contextual evidence. Uttering *Is it sunny?* in a situation where the addressee has entered the house wearing a dripping raincoat is infelicitous. It is blocked by more suitable forms, such as *Is it raining?*, *Is it not sunny?*, or the corresponding

[13] An anonymous reviewer wonders whether the presence of evidential bias without epistemic bias is equated with the explanation-seeking nature of a PQ. I would indeed hypothesize so. For a more detailed discussion, see Section 3.2.

[14] The newly introduced term PREFERENTIAL BIAS builds on Uegaki & Sudo's (2019) term "preferential predicate", which is intended to cover the corresponding attitude predicates ('hope', 'fear', etc.). In Sudo's (2013) approach, the preferential bias would be considered a subtype of the broadly conceived epistemic bias (together with bouletic or deontic bias; see also Reese & Asher 2010).

[15] Zimmerling (2023) holds an opposing view, namely that "[u]nbiased polar questions are possible and widespread" and that bias, understood as the speaker's considering p more likely than $\neg p$ (or conversely), is only "encoded lexically or by morphosyntax or prosody" (p. 120).

declarative PQs – *It is raining? / It is not sunny?*, all of which are more suitable to reflect the evidence pointing to rain. In this subsection, I will show how PQ polarity (negative/NPQ vs. positive/PPQ) and word order (interrogative/InterPQ vs. declarative/DeclPQ) interact to produce different bias patterns in Czech.

Example (11) shows that in the absence of any prominent evidential or epistemic bias, Czech makes use of an interrogative PPQ. It should be pointed out that the expectation that Marek cleaned the third floor, stemming from the cleaning plan, shared between both discourse participants, and one that could be considered a case of deontic bias (what should be the case according to the plan), does not license any one of the PQ forms that are more specific than the default InterPPQ.[16]

(11) *Scenario:* On a busy day, two hotel cleaning service coordinators (A and B), sitting in their office, are taking note of the progress made thus far. A is about to make a tick for the whole third floor, which Marek was responsible for. B is informed about the progress. A asks:

a. Uklidil Marek třetí patro? — InterPPQ
cleaned Marek third floor
'Has Marek cleaned the third floor?'

b. # Neuklidil Marek třetí patro? — InterNPQ
NEG.cleaned Marek third floor
'Hasn't Marek cleaned the third floor?'

c. # Marek (ne-) uklidil třetí patro? — DeclPQ
Marek NEG cleaned third floor
'Marek has(n't) cleaned the third floor?'

In (12) the speaker (A) is privately epistemically biased towards the positive prejacent (that Marek also cleaned the third floor). Until A's question, the addressee (B) had not been aware of this bias. Moreover, there is no shared evidential bias for either p or $\neg p$. In this scenario, A can use an InterPPQ or InterNPQ; a DeclPQ sounds infelicitous, irrespective of its polarity.[17] Intuitively, the epistemic

[16]The "cleaning" examples that follow make use of the verb *uklidit* 'clean.PFV' or *uklízet* 'clean.IPFV'. If the imperfective is used, it is in the "general-factual" sense, i.e. one that is compatible with a telicity/completedness inference (see Grønn 2004, Mueller-Reichau 2018a,b, Mueller-Reichau 2026 [this volume], Gehrke 2026 [this volume]). I invite any Czech reader to switch the (im)perfective forms of the verb whenever they consider the alternative more suitable.

[17]Declarative PQs can be used if they are motivated by information-structural considerations; see the discussion around example (18).

bias is communicated more prominently in InterNPQ than in InterPPQ. Suppose that, in fact, Marek has not cleaned the third floor. In that case, B could interpret the InterPPQ as suggesting that A holds a false belief that Marek had both the second and the third floor in his regular cleaning plan. The InterNPQ, on the other hand, quite clearly gives away the implication that A considers it likely that Marek cleaned the third floor not on the basis of the regular plan, but on the basis of more specific information available to A. The negation thus contributes an extra modal/epistemic inference which goes beyond the stereotypical and shared expectation of what could have happened. Finally, it should be noted that the positive epistemic bias of the InterNPQ does not have to be matched by a negative evidential bias. In other words, no conflict between biases is required for the licensing of Czech InterNPQs. This is in line with the experimental results on Czech reported in Staňková (2023), and with the properties of high negation in other languages (Ladd 1981, Büring & Gunlogson 2000, Romero & Han 2004, Hartung 2009, Gyuris 2017, Domaneschi et al. 2017).[18]

(12) *Scenario:* On a busy day, two hotel cleaning service coordinators (A and B), sitting in their office, are taking note of the progress made thus far. A is about to make a tick for the whole third floor. Marek was responsible for the second floor (and he cleaned that), but earlier that day A asked Marek if he could also exceptionally take care of the third floor. B was not informed about A's request to Marek. B is informed about the progress. A asks:

a. Uklízel Marek i TŘETÍ patro? — InterPPQ
   cleaned Marek also third floor
   'Has Marek cleaned the third floor too?'

b. Neuklízel Marek i TŘETÍ patro? — InterNPQ
   NEG.cleaned Marek also third floor
   'Hasn't Marek cleaned the third floor too?'

c. # Marek (ne-) uklízel i TŘETÍ patro? — DeclPQ
   Marek NEG cleaned also third floor
   'Marek has(n't) cleaned the third floor too?'

The felicity of the InterNPQ is reminiscent of what has been observed for outer negation in English or German (Ladd 1981, Büring & Gunlogson 2000, Romero

[18] An anonymous reviewer wonders whether the stress on *třetí* 'third' (instead of the default stress on *patro* 'floor') interacts with the bias profile in any way. I do not think so. It merely conveys that the background – cleaning another floor than the third one – is salient in the context. The epistemic bias is derived from the prejacent, not the background or the focus alternatives.

& Han 2004, Repp 2013; a.o.). At this point, it should be noted that the formal repertoire of negative PQs is richer in English (and German) than it is in Czech. In English, negation in InterPQs can either be preposed together with the auxiliary, see (13a), or it can occupy its base position, (13b). I follow AnderBois (2019) (among others) in referring to this distinction as high vs. low negation. This formal distinction is complemented by a meaning-related or "functional" distinction – outer vs. inner negation – see (14), and argued to be diagnosed by the use of negative polarity items (NPIs; e.g. *either*, diagnosing inner negation) or positive polarity items (PPIs; e.g. *too*, diagnosing outer negation).[19,20]

(13) *The form of negation in English interrogative PQs*
  a. *Hasn't* Max cleaned the third floor? high neg
  b. *Has* Max *not* cleaned the third floor? low neg

(14) *The function of negation in English interrogative PQs*
  a. *Hasn't* Max cleaned the third floor *too*? outer neg
  b. *Hasn't* Max cleaned the third floor *either*? inner neg
  c. *Has* Max *not* cleaned the third floor {*either*/#*too*}? inner neg

In Czech and Slavic languages more generally, negation is prefixed or procliticized on the verb. Since an utterance becomes interrogative by preposing the verb in Czech, negation is necessarily preposed too. There is thus no high vs. low

[19]The widely entertained idea that English high negation is ambiguous between the outer and inner reading (Ladd 1981, Romero & Han 2004, Repp 2013, Romero 2015) has recently been contested; see Sailor (2013), AnderBois (2019), Goodhue (2022). See also Quirk et al. (1985: 809), who report a register difference: high negation is preferred in spoken language, while low negation is rather formal. (My thanks to Mitkovska & Bužarovska 2024: 141 for this piece of information.)

[20]I apply the term "function" to the outer vs. inner property of negation as a convenient way of remaining agnostic as to what exactly constitutes the difference – whether it is syntax, semantics, pragmatics, or some combination of these. Similarly, I do not wish to go deeply into the (non-)licensing of NPIs or PPIs in (negative) PQs. Staňková (2023), Staňková & Šimík (2025) follow Repp (2006) in analyzing outer negation as fundamentally different from classical propositional negation (see Dočekal 2026 [this volume]), which in turn leads to the non-licensing of NPIs (and compatibility with PPIs); see also Section 3.3.2. For alternative accounts of NPI (non-)licensing in (N)PQs, see, e.g., van Rooij (2003), Nicolae (2013), Trinh (2023). An anonymous reviewer points out that the non-licensing of NPIs (and "rescuing" of PPIs) could also be due to a "flip-flop" effect (which is more general and not specific to PQs), whereby two stacked NPI licensors (in our case the Q operator and the negation) lead to NPI anti-licensing (see, e.g., Homer 2021). A deeper discussion of NPIs and PPIs in (negative) PQs goes beyond the scope of this chapter.

difference in Czech InterNPQs. The closest one can get to replicating the high vs. low difference is by using interrogative vs. declarative PQs. When it comes to the outer vs. inner status, Staňková (2023) provided experimental evidence (from naturalness ratings) that interrogative NPQs primarily involve outer negation, while declarative NPQs are in principle ambiguous between inner and outer negation (with a preference for the former). The semantic status of the negation was diagnosed by indefinite determiners, contrasting the PPI *nějaký* 'some' and the negative concord item (NCI) *žádný* 'any/no'; example (15) provides an illustration of this pattern, with the judgement corresponding to the experimental findings of Staňková (2023).

(15) *The function of negation in Czech PQs*

a. Nekoupila si Linda {*nějakou* / ?*žádnou*} knížku? OUTER
NEG.bought REFL Linda DET.PPI DET.NCI book
'Didn't Linda buy a book?'

b. Linda si nekoupila *nějakou* knížku? OUTER
Linda REFL NEG.bought DET.PPI book
'Linda didn't buy a book?'

c. Linda si nekoupila *žádnou* knížku? INNER
Linda REFL NEG.bought DET.NCI book
'Linda didn't buy a book?'

Let me conclude this brief digression by noting that, despite its absence in Czech, the high vs. low negation contrast might well be present in interrogative PQs in those Slavic languages in which verb position is not the only interrogative strategy; a good example of that is Serbian (see Section 4).

We have thus far seen that Czech InterNPQs correlate with the presence of the speaker's private epistemic bias and noted that in this respect Czech matches the behaviour of high negation PQs in English.[21] The scenario in example (16) introduces a conflict between two biases: the speaker's epistemic bias for p and an evidential bias for $\neg p$. The latter stems from B's information that Marek cleaned the second floor in conjunction with the stereotypical expectation that he would clean a single floor. Since the second and third floor are contrasted here, *třetí* 'third' carries the nuclear accent in A's question. As (16) shows, this type of conflict-resolving PQ is necessarily negated, but the choice of the interrogative vs. declarative strategy is not crucial. The presence of evidential bias thus

[21]In Section 5 I will show that this is not a *necessary* property of Czech InterNPQs. See also the discussion of example (17).

makes the DeclPQ strategy available. Intuitively, there is a slight pragmatic difference between the interrogative and the declarative strategy: the InterNPQ is concerned (using Repp & Geist's 2025 terminology) about the speaker's positive epistemic bias and the DeclNPQ about the evidence, i.e. about what B reports. This matches what has been reported for English high vs. low negation: the former double-checks the epistemic bias, the latter the evidential bias (Romero & Han 2004, Repp 2013).[22]

(16) *Scenario:* On a busy day two hotel cleaning service coordinators (A and B), sitting in their office, are taking note of the progress made. B informs A that Marek has finished cleaning the second floor. A is slightly surprised as he believed that Marek was responsible for the third floor and it's common for one person to take care of a single floor. A asks:

a. # Uklízel Marek TŘETÍ patro? — InterPPQ
cleaned Marek third floor
'Has Marek cleaned the third floor?'

b. Neuklízel Marek TŘETÍ patro? — InterNPQ
NEG.cleaned Marek third floor
'Hasn't Marek cleaned the third floor?'

c. # Marek uklízel TŘETÍ patro? — DeclPPQ
Marek cleaned third floor
'Marek has cleaned the third floor?'

d. Marek neuklízel TŘETÍ patro? — DeclNPQ
Marek NEG.cleaned third floor
'Marek hasn't cleaned the third floor?'

A different pattern emerges when one considers a case with evidential bias but no prior epistemic bias (or at least no prior bias the speaker would intend to convey). Scenario (17) involves a situation-based evidential bias for *p*. The bias arises as a plausible explanation for the observed effect, namely the earbud found on the floor. In other words, the PQs under (17) are explanation-seeking questions in the sense introduced in Section 3.1.[23] The first thing to observe is that the

[22] I add that example (16a) would be felicitous if the additive particle *i* 'also' preceded *třetí* 'third' and/or the question started with the (additive) conjunction *a* 'and'. The intuition is that in that case, the speaker would disregard his concern about the evidence (he simply accepts it as a fact not to be questioned) and ask a plain information-seeking question (conveying a weak epistemic bias for the positive prejacent) about the third floor.

[23] The explanation – that Marek cleaned the third floor – is a plausible answer to the implicit question that the finding gives rise to, namely 'How can an earbud like the ones Marek has have appeared here?'

change in the polarity of the evidential bias – from $\neg p$ in (16) to p in (17) – correlates with the change in the polarity of the DeclPQ; in (17), only DeclPPQ is available, matching the evidential bias for p.[24] What is striking is that the scenario licenses, or in fact prefers, the use of negation in interrogative PQs: the InterNPQ feels more appropriate than InterPPQ. In other words, an InterNPQ is not specialized in conveying prior epistemic bias; if such a bias is absent (or not intended to be expressed), it is also compatible with expressing an evidential bias which corresponds to the suspected explanation of an observed effect. As indicated by the translation of (17b) (with thanks to Daniel Goodhue, p.c.), English does not use negation in this case, suggesting that the distribution (and meaning) of Czech InterNPQs is broader than that of English high negation PQs. In Section 5 we will see more examples where high negation is not licensed in English, but its correlate in Czech (InterNPQ) is.

(17) *Scenario:* Two hotel cleaning service coordinators (A and B) are walking around the hotel and inspecting the progress made. A has no idea about the cleaning plan for today (esp. who is responsible for what), only B does. When on the third floor, A and B find an earbud that A suspects belongs to Marek, one of the cleaners. A asks:

a. # Uklízel Marek třetí PATRO? — InterPPQ
cleaned Marek third floor
Intended: 'Has Marek cleaned the third floor?'

b. Neuklízel Marek třetí PATRO? — InterNPQ
NEG.cleaned Marek third floor
'Has Marek cleaned the third floor?'

c. Marek uklízel třetí PATRO? — DeclPPQ
Marek cleaned third floor
'Marek has cleaned the third floor?'

d. # Marek neuklízel třetí PATRO? — DeclNPQ
Marek NEG.cleaned third floor
'Marek hasn't cleaned the third floor?'

The data discussed so far are consistent with the generalization that DeclPQs are specialized for conveying evidential bias (Büring & Gunlogson 2000, Gunlogson 2002; a.o.). There are circumstances, however, in which Czech declarative PQs

[24] Note that despite the DeclNPQ's capacity to express outer negation (see the discussion around example (15)), the DeclNPQ in (17d) is clearly not on a par with (17b). This matches the experimental results reported in Staňková (2023), who shows that the polarity of DeclPQs matches the polarity of the evidential bias irrespective of the inner vs. outer distinction.

are completely unbiased. This is exemplified in (18), where the urge to place the contrastive topic *Marek* (which contrasts with *Lucie* by virtue of the preceding conversation of A and B) utterance-initially outweighs the need for verb movement.[25]

(18) *Scenario:* On a busy day, two hotel cleaning service coordinators (A and B), sitting in their office, are taking note of the progress made thus far. B is informed about the progress. After a brief chat about two new cleaners who have just joined the team – Lucie and Marek – A wants to take note of the final two floors and wonders who (Lucie or Marek) cleaned which floor (second or third). A starts by asking:

a. ? Uklidil Marek TŘETÍ patro? — InterPPQ
cleaned Marek third floor
'Has Marek cleaned the third floor?'

b. # Neuklidil Marek TŘETÍ patro? — InterNPQ
NEG.cleaned Marek third floor
'Hasn't Marek cleaned the third floor?'

c. Marek uklidil TŘETÍ patro? — DeclPPQ
Marek cleaned third floor
'Marek has cleaned the third floor?'

d. # Marek neuklidil TŘETÍ patro? — DeclNPQ
Marek NEG.cleaned third floor
'Marek hasn't cleaned the third floor?'

A comparable effect obtains in Czech wh-questions. While Czech is predominantly a wh-movement language – placing its wh-word clause-initially in wh-questions (Veselovská 2021) – this strong tendency can be overridden if the question involves a contrastive topic (cf. Bobaljik & Wurmbrand 2015). The difference between information-structure-induced unbiased declarative PQs and declarative wh-questions is that the former strongly prefer the declarative syntax, while in the latter, the declarative syntax is optional.

(19) *Scenario:* On a busy day, two hotel cleaning service coordinators (A and B), sitting in their office, are taking note of the progress made thus far. B is

[25] With its free word order (Jasinskaja & Šimík In press), Czech (or Slavic more generally) is especially prone to information-structure-related word order manipulations. In their spoken-corpus study on Czech, Onoeva & Staňková (2025) report a significant correlation between evidential bias and DeclPQs, but still, there are a great number of DeclPQs which Onoeva & Staňková (2025) categorized as free of evidential bias. I hypothesize that bias-free DeclPQs involve a marked information structure.

informed about the progress. She just informed A that Lucie cleaned the second floor. Now B wonders which floor Marek cleaned.

a. A které patro uklidil MAREK?
and which floor cleaned Marek
'And which floor did Marek clean?'

b. A Marek uklidil KTERÉ patro?
and Marek cleaned which floor
'And Marek cleaned which floor?'

In summary, I have shown that most PQs are not neutral information-seeking questions, but rather express an additional bias towards one (or both) of the alternatives which constitute the question meaning. The bias is either tied to the speaker's (prior) beliefs – the epistemic bias – or to the properties of the context or situation – the evidential bias. Two grammatical means are instrumental in conveying bias: negation, and the form (interrogative vs. declarative) of the PQ. I have shown that, in general, declarative PQs give rise to an evidential bias for the proposition which corresponds to the prejacent of the question. This bias can be suppressed in favour of other factors which prefer declarative syntax, in particular information-structural ones. In interrogative PQs, negation typically gives rise to a positive epistemic bias, esp. the speaker's prior belief that p. This bias can additionally be in conflict with an evidential bias for $\neg p$. Finally, I have shown that negation in Czech interrogative PQs is also licensed in the absence of prior belief, in particular in explanation-seeking questions.

In Section 4 and Section 5, I will discuss another productive way of expressing bias, namely particles, such as the Russian *razve* (Korotkova 2023, Repp & Geist 2025).[26]

3.3 Modelling bias

We have seen that a question like (20) conveys (i) that the speaker would like to resolve the issue of whether Jana went by bus or not, (20a) (see Section 2), (ii) that there is situational or contextual evidence that Jana did not go by bus, (20b), and (iii) that the speaker expected (had a prior belief) that Jana would go by bus (20c) (see Section 3.1–Section 3.2).

(20) Jana nejela autobusem? DECLNPQ
Jana.NOM NEG.went bus.INS
'Jana didn't go by bus?'

[26] Question tags, which often complement declarative PQs (*It's raining, isn't it?*), are not discussed in this chapter.

a. *Basic meaning:* {'Jana went by bus', 'Jana didn't go by bus'}
b. *Evidential bias:* 'Jana didn't go by bus'
c. *Epistemic bias:* 'Jana went by bus'

How are these meaning components encoded in the form of the question? Are they a conventional part of the meaning or do they arise as a result of conversational dynamics (i.e. as a conversational implicature)? If the former, should they be modelled as a presupposition or a conventional implicature? These questions are subject to ongoing lively debate which I cannot do justice to here (see Romero & Han 2004, Repp 2013, Northrup 2014, Trinh 2014, 2023, Krifka 2015, Romero 2015, AnderBois 2019, Goodhue 2022; a.o.). I will limit myself to sketching out the most common analytical strategies and arguments put forth in favour of them.

3.3.1 Semantic approaches

Building on the intuition that PQs are inherently biased (cf. Repp & Geist's 2025 "[...] there is no such thing as a truly neutral question"), Biezma & Rawlins (2012) (with an early precedent in Roberts 1996) proposed that a PQ does not denote the set of its possible answers (Hamblin 1973) or a possible worlds partition (Groenendijk & Stokhof 1984), but just the singleton set containing the question prejacent. The clearly irreducible fact that PQs raise an issue, i.e. that they imply a non-singleton set of propositions under consideration, is encoded in a presupposition contributed by the question operator Q; see (21).

(21) $[\![$Q [Jana.NOM NEG.went bus.INS]$]\!]^c = \{\lambda w[\text{Jana didn't go by bus in } w]\}$
defined only if ($= \{\neg p\}$)
a. $\{\neg p\} \subseteq \text{SalientAlts}(c)$, and
b. $|\{\neg p\} \cup \text{SalientAlts}(c)| > 1$

(adapted from Biezma & Rawlins 2012: 392)

Biezma & Rawlins (2012) motivate this semantics for PQs by a systematic comparison with alternative questions (see Section 2). Alternative questions ('Did Jana go by bus or not?') lay out the two alternatives explicitly and, as it were, on an equal footing (Bolinger 1978, van Rooij & Šafářová 2003). Moreover, they involve an obligatory disjunction ('or'), which is assumed to collect alternatives in a set (Alonso-Ovalle 2006). PQs, on the other hand, only explicitly mention a single alternative, which is typically biased (i.e. not equal with its polar alternative). They also often involve no overt morphology (such as the disjunction 'or') indicating the presence of non-trivial (non-singleton) alternatives. The presupposition of a

PQ brought about by the Q-operator – (21a)/(21b) – states that the singleton set denoted by the question is a subset of a non-singleton set of contextually salient alternatives. This set will often correspond to the standard Hamblin-style meaning of a PQ ($\{p, \neg p\}$), but – or so I conjecture – it can also correspond to another (implicit) question under discussion, as in explanation-seeking PQs discussed in Section 3.1.

Biezma & Rawlins' (2012) analysis thus provides a good starting point for modelling some biased PQ types. However, a comprehensive account of a more representative sample of the various bias types is, to the best of my knowledge, missing. I would also like to point out that Slavic languages might provide valuable grounds for further exploration of Biezma & Rawlins (2012), as many of them involve a question particle which is morphologically identical or at least related to a disjunction (cf. *li* in Bulgarian; *czy* in Polish; see Logvinova 2022 and Section 4). Following up on Biezma & Rawlins' logic, one could expect these questions to have the classical Hamblin-style denotation, be unbiased, and hence more on a par with alternative questions.[27]

Another type of approach in which question bias is intimately tied to question semantics has been developed in the framework of inquisitive semantics (Ciardelli et al. 2013, 2019) by Farkas & Roelofsen (2017) and AnderBois (2019), among others. The core idea of inquisitive semantics is that propositions are endowed with two kinds of meaning – informative and inquisitive content. Statements containing disjunctions such as *Jana went by bus or by train* primarily provide new information, but at the same time explicitly raise the issue concerning which means of transport Jana used. The former meaning component is informative, the latter is inquisitive. In questions, the situation is reversed: what is at issue is the inquisitive content; this, however, does not rule out the potential relevance of the informative content. Following this logic, Farkas & Roelofsen (2017) argue that declarative PQs raise the issue $\{p, \neg p\}$, but at the same time convey the speaker's commitment to (bias for) the informative content corresponding to the question's prejacent, called the "highlighted" alternative by Farkas & Roelofsen (2017). AnderBois (2019) adopts this framework and develops a particular analysis of a broader range of biased question types. Within Slavic semantics, inquisitive semantics has been applied to Serbian PQs by Todorović (2023) (see Section 4).

[27]What remains an open issue in Biezma & Rawlins' (2012) analysis is the exact contribution of the final rising intonation. Nevertheless, if Biezma & Rawlins have reasons not to unify the rising contour with the abstract Q morpheme, one could entertain the idea that rising intonation conveys the lack of speaker commitment to the truth of the PQ prejacent (see, e.g., Gunlogson 2003, 2008, Malamud & Stephenson 2015, Rudin 2022).

3.3.2 Commitment operators

The phenomenon of high/outer negation (see Section 3.2) and various PQ-related particles has led researchers to enrich the structure and conventional meaning components of PQs. The core idea of this type of approach, first explicitly formulated in Romero & Han (2004) and later refined in Repp (2006, 2009, 2013), Romero (2015), Krifka (2015, 2017), and others, is that what is at issue while uttering PQs is not just the propositional contents of the PQ prejacent (a set of possible worlds), but the degree to which the speaker and potentially addressee are committed to the truth of that content. Very informally, by asking whether p, the speaker does not (only) ask whether p is true, but how certain the addressee is that p should be added to the COMMON GROUND (CG) – the set of propositions corresponding to the mutual beliefs of the discourse participants (Stalnaker 1978).

The two main operators entertained in this type of analysis are Romero & Han's (2004) VERUM, expressing strong commitment to p, and Repp's (2006, 2009, 2013) FALSUM, expressing the lack of commitment to p; see (22).[28] In syntactic terms, these operators are "sandwiched" between the propositional contents of the utterance (TP, or possibly a lower projection within CP) and the force of the utterance. If the force hosts the question operator, the result will be a PQ which wonders not about p (or $\neg p$) directly, but about the addressee's certainty as to whether p belongs (or does not belong) to the common ground.[29]

(22) a. $[\![\text{VERUM}]\!] = \lambda p \lambda w \forall w' \in \text{EPI}_x(w)[\forall w'' \in \text{CONV}_x(w')[p \in \text{CG}_{w''}]]$

b. $[\![\text{FALSUM}]\!] = \lambda p \lambda w \forall w' \in \text{EPI}_x(w)[\forall w'' \in \text{CONV}_x(w')[p \notin \text{CG}_{w''}]]$

(where x is resolved to the addressee in a question; $\text{EPI}_x(w)$ is the set of worlds compatible with what x knows in w and $\text{CONV}_x(w)$ the set of worlds compatible with the conversation goals of x in w; CG_w is the common ground in w)

Repp's idea was that outer negation spells out FALSUM, yielding the denotation (23b) for (23a).

(23) a. Nejela Jana autobusem?
NEG.went Jana.NOM bus.INS
'Didn't Jana go by bus?'

[28] Romero (2015) and Frana & Rawlins (2019) have proposed that the commitment-related proposition is not at issue (in the technical sense; see Simons et al. 2011), i.e. either a presupposition (Frana & Rawlins 2019) or a "common ground-management content" (Romero 2015, building on Repp 2013).

[29] The operators under discussion are not limited to questions – they can also be used in statements; see, e.g., Gutzmann & Castroviejo Miró (2011), Romero (2015).

b. $\{\mathcal{P}, \neg\mathcal{P}\}$
(where $\mathcal{P} = \lambda w \forall w' \in \text{EPI}_{\text{ADDR}(c)}(w)[\forall w'' \in \text{CONV}_{\text{ADDR}(c)}(w')$
$[\lambda w[\text{Jana went by bus in } w] \notin \text{CG}_{w''}]]$ and c is the utterance context)

The use of FALSUM has two main benefits. First, it provides a natural account of the inability of outer negation to license negative polarity items (or negative concord items) within the proposition: there simply is no logical negation within p. Second, it goes some way towards modelling the bias profile of high negation PQs by making the *absence* of p in the common ground at issue. This naturally comes about if the falsity of p is contextually salient (evidential bias for $\neg p$) and, given that by asking about it the speaker expresses a concern about that falsity, he thereby implies (implicates) his epistemic bias for p. This pragmatic derivation of the bias profile of outer negation questions comes close to what Goodhue (2022) assumes for bias more generally, namely that it is a result of conversational dynamics (i.e. a conversational implicature; see Section 3.3.3). Yet it is fair to state that this is not what Repp (or the others following her) assumed; Repp (2013: 243), for instance, stipulates the bias profile of outer negation PQs separately (in terms of "discourse conditions"; cf. Gutzmann's 2015 "use-conditions"), and does not attempt to derive it from the semantics of FALSUM. Besides this problem, the VERUM/FALSUM-based approach offers no immediate account of other types of bias profiles, including cases where the positive epistemic bias is not accompanied by a negative evidential bias (see Goodhue 2022 and the discussion of example (12) above).

For an application of the VERUM/FALSUM-based approach to Slavic languages, see Staňková (2023), Repp & Geist (2025).

3.3.3 Conversational approach

Goodhue (2022) has come up with a radically different approach to deriving the positive epistemic bias in English high negation PQs (corresponding to Czech INTERNPQs).[30] He argues that the bias is a conversational implicature derived by negating the speaker's ignorance about the truth of both polar alternatives, which in turn is implied by the use of positive PQs. In other words, if the speaker were ignorant about both p and $\neg p$, it would be more useful to use a positive PQ because it would guarantee a more informative answer (if the answer happened to be negative). If the speaker uses a high negation PQ instead, it implicates that he is not ignorant, or, in other words, that he is biased. The reason why he is

[30] Goodhue (2022) argues (following Sailor 2013, AnderBois 2019) that high negation PQs are always outer negation PQs.

biased towards the positive rather than negative alternative is the unbalanced nature of the partition denoted by a high negation PQ (Romero & Han 2004, Krifka 2017).

Goodhue (2022) builds on Romero & Han (2004) in that he treats high/outer negation as negation scoping over a high epistemic operator, which he identifies with the illocutionary operator ASSERT (see, e.g., Meyer 2013), understood in terms of doxastic necessity/belief relativized to the addressee.[31] A high negation PQ then denotes the set $\{\neg\text{DOX}_a p, \text{DOX}_a p\}$ (where $\text{DOX}_a p$ is to be read as 'the addressee believes that p'). The denotation of the corresponding positive PQ, or rather the set of possible addressee's assertions of answers is $\{\text{DOX}_a p, \text{DOX}_a \neg p\}$.[32] The negative alternative in positive PQs ($\text{DOX}_a \neg p \approx$ 'the addressee believes that $\neg p$') is thus semantically stronger than the corresponding negative alternative in high negation PQs ($\neg\text{DOX}_a p \approx$ 'the addressee doesn't believe that p'). This strength relation, together with the unbalanced nature of the partition involved in high negation PQs, generates the implicature that the speaker believes the positive alternative.

4 Polar questions in Slavic languages

Despite their considerable genealogical closeness, Slavic languages have developed a broad range of strategies for forming polar questions – both cross-linguistically and intra-linguistically. What factors underlie this cross- and intra-linguistic variation? What are the semantics and pragmatics of the individual strategies? What are the implications for the general theory of polar question meaning? These questions largely remain open. In what follows, my goals are rather modest, namely (i) to concisely characterize the variation, and (ii) to survey the existing research in the area of Slavic PQ (formal) semantics and pragmatics.

4.1 Default PQ strategies

I have shown in Section 2 and Section 3 that a neutral information-seeking PQ is in fact quite hard to come by. Even simple questions like *Is it raining?* can be and often are biased. In an attempt to identify the default (unbiased) strategy

[31] High negation PQs thus inolve two illocutionary operators – ASSERT and Q, with high negation sandwiched between the two (Q > ¬ > ASSERT).

[32] Goodhue (2022) assumes that the denotation of positive PQs does not involve the speech act layer; it only denotes $\{p, \neg p\}$, which, however, does not afford entailment relations between the members of the positive and high negation PQs. See Goodhue (2022) for discussion.

of forming a PQ, some authors (Farkas & Roelofsen 2017, AnderBois 2019) have resorted to "quiz scenarios" – scenarios in which the speaker's primary concern is to provide no clue whatsoever as to which one of the polar alternatives is true, and in which an answer is solicited from the addressee. While the speaker typically knows the true answer, he acts as though he were completely ignorant. This makes the quiz scenario a good tool for diagnosing default PQ strategies. In the course of the discussion, we will see that some PQ strategies are neutral in the sense of not conveying a bias, but are obligatorily information-seeking and hence are not suitable in the quiz scenario. This will motivate a richer semantic structure of these PQ strategies.

The scenario assumed for all the PQs in this subsection is formulated in (24).

(24) *Scenario for questions (25) to (34):* A TV quiz moderator asks the following polar question, expecting a simple 'yes' or 'no' answer.

The Czech example (25) demonstrates that the only felicitous way of asking a PQ in the scenario (24) is the InterPPQ, i.e. a verb-initial PQ with an affirmative form of the verb. Given the discussion in Section 3.2, I take this result to be a certain sanity check proving that quiz questions are indeed a good way to identify unbiased PQ strategies. My intuitions about the alternative strategies – (25b) and (25c) – confirm the expectation that they are infelicitous because they convey a bias (epistemic and evidential, respectively), which is clearly inappropriate in the given scenario.[33]

(25) a. Dožil se císař Justinián I. 80 let? InterPPQ
lived.to REFL emperor Justinian I 80 years
'Did Emperor Justinian I live up to 80 years?'
b. # Nedožil se císař Justinián I. 80 let? InterNPQ
c. # Císař Justinián I. se (ne)dožil 80 let? DeclPQ

There are three other Slavic languages that behave like Czech in that they involve obligatory verb or auxiliary movement across the subject in the default PQ: Slovak, (26), Upper Sorbian, (27), and Slovenian, (28). As in Czech, adding negation is reported to trigger positive epistemic bias, and declarative questions involve evidential bias. Slovak and Upper Sorbian seem to be identical to Czech

[33]For reasons of space, I will gloss all the variants only when this is necessary for a proper understanding of the examples. The glosses for the unglossed examples can be easily reconstructed from those that are glossed. The negative morpheme in Slavic is *ne/nie/nje* and is prefixed or procliticized on the verb or the auxiliary.

in all relevant respects; the only notable difference is that Upper Sorbian fronts an auxiliary rather than the lexical verb. Slovenian optionally uses the clause-initial question particle *ali*. I further note that *ali* is reported to be incompatible with DeclPQs, (28c), independently of the context of use. The fact that *ali* is available in the default PQ in Slovenian is potentially significant because Slovenian also has two other clause-initial PQ particles – *a* (Greenberg 2006: 165; Stegovec 2017: 153) and *kaj* (lit. 'what'; Adrian Stegovec, p.c.) – the former of which is considered neutral, and yet they are reported not to be appropriate in the quiz PQ. It is an open issue which factor this boils down to, whether register (*a* is colloquial, *ali* belongs to a higher register), semantics (the use of *a* implies actual information-seeking), or the presence of a weak bias.[34]

(26) a. Dožil sa cisár Justinián 80 rokov? InterPPQ
live.to refl emperor Justinian 80 years
'Did Emperor Justinian I live up to 80 years?'

b. # Nedožil sa cisár Justinián 80 rokov? InterNPQ

c. # Cisár Justinián sa (ne)dožil 80 rokov? DeclPQ
(Slovak; Ivana Dragonová, p.c.)

(27) a. Je kejžor Justinian Wulki 80 lět wordował? InterPPQ
is.aux emperor Justinian Great 80 years become
'Did Emperor Justinian I live up to 80 years?'

b. # Njeje kejžor Justinian Wulki 80 lět wordował? InterNPQ

c. # Kejžor Justinian Wulki (nje)je 80 lět wordował? DeclPQ
(Upper Sorbian; Katja Brankačkec, p.c.)

(28) a. (Ali) Je cesar Justinijan doživel 80 let? InterPPQ
ali is.aux emperor Justinian lived.to 80 years
'Did Emperor Justinian I live up to 80 years?'

b. # (Ali) Ni cesar Justinijan doživel 80 let? InterNPQ
ali neg.is.aux emperor Justinian lived.to 80 years

c. # (*Ali) Cesar Justinijan {je/ni} doživel 80 let? DeclPQ
(Slovenian; Adrian Stegovec, p.c.)

[34]An analogous intuition is reported for the Serbian je+li strategy; see (31). If semantics is at stake (i.e. information-seeking necessary), one could hypothesize that the particles (*a/kaj* in Slovenian and *je+li* in Serbian) incorporate an implicit predicate 'wonder' (with the speaker as the attitude holder), as in performative approaches to question semantics (Lewis 1970, Krifka 2023), which would account for the information-seeking character of the question. See Section 4.2.1 for an analysis along these lines.

Ukrainian and Polish employ the clause-initial particles *čy* and *czy,* respectively; see (29) and (30).[35] This particle is obligatory in our quiz PQ. The relative position of the verb and the subject is reported to be immaterial in Ukrainian (cf. (29a) and (29b)) and preferably canonical (SV) in Polish.[36] Adding negation or leaving out *čy* in Ukrainian leads to infelicity in the quiz scenario. What is worth noting is that Polish is reported to allow for a negative PQ in the quiz scenario, as long as the PQ is introduced by *czy*. (30b) thus exemplifies a quiz question asking about Justinian not living up to 80 years old. Declarative PQs in Polish (without *czy*) are grammatical, but infelicitous in the quiz scenario because of evidential bias; (30c). A verb-initial PQ in (30d) is reported to be unacceptable (cf. the corresponding Czech example (25a)). Verb-initiality is possible in principle, but not if the verb moves across an overt subject (Dorota Klimek-Jankowska, p.c.; see also the discussion around example (7) above).

(29) a. Čy dožyv imperator Justynian Velykyj do 80 rokiv? — cyPPQ
 cy lived emperor Justinian Great to 80 years
 'Did Emperor Justinian I live up to 80 years?'

b. Čy imperator Justynian Velykyj dožyv do 80 rokiv? — cyPPQ

c. # Čy ne dožyv imperator Justynian Velykyj do 80 rokiv? — cyNPQ

d. # Čy imperator Justynian Velykyj ne dožyv do 80 rokiv? — cyNPQ

e. # Imperator Justynian Velykyj (ne) dožyv do 80 rokiv? — DeclPQ

(Ukrainian; Anastasiia Vyshnevska, p.c.)

(30) a. Czy Justynian I Wielki dożył wieku 80 lat? — czyPPQ
 czy Justinian I Great live.to age 80 years
 'Did Emperor Justinian I live up to 80 years?'

b. Czy Justynian I Wielki nie dożył wieku 80 lat? — czyNPQ
 czy Justinian I Great neg live.to age 80 years
 'Did Emperor Justinian I not live up to 80 years?'

c. # Justynian I Wielki (nie) dożył wieku 80 lat? — DeclPQ

d. * {Dożył / Nie dożył} Justynian I Wielki wieku 80 lat?

(Polish; Dorota Klimek-Jankowska, p.c.)

[35] Belarusian employs the particle *ci,* with comparable properties (see Logvinova 2022); *ci* is also used in Yiddish (Sadock & Zwicky 1985: 181).

[36] Logvinova (2022: 26) reports that if a Ukrainian PQ is explanation-seeking (in her terminology: "inference-based"), then only the SV order is available. A similar tendency is found in Polish.

Serbian, having word order alternations and multiple particles at its disposal, exhibits a rich inventory of PQ strategies (Rakić 1984, Todorović 2023).[37] According to Neda Todorović (p.c.), the only strategy fully natural in the quiz scenario involves the clause-initial particle *da* combined with the clitic *li* (optionally reduced to *l'*) and a positive form of the verbal complex; see (31a).[38] The competing strategy using the particle *je* instead of *da*, (31b), considered as neutral/unbiased in Todorović (2023), is reported to be colloquial and probably only information-seeking (see footnote 34). Negative and declarative PQs give rise to biases and are thus inappropriate in the quiz scenario.

(31) a. Da li je Justinijan I/Veliki doživeo 80. godinu? DA+LIPPQ
DA LI is.AUX Justinian I/Great lived.to 80 years
'Did Emperor Justinian I live up to 80 years?'

b. # Je l' (je) Justinijan I/Veliki doživeo 80. godinu? JE+LIPPQ
JE LI is.AUX Justinian I/Great lived.to 80 years
'Did Emperor Justinian I live up to 80 years?'

c. # Nije li Justinijan I/Veliki doživeo 80. godinu? HighNPQ
NEG.is.AUX LI Justinian I/Great lived.to 80 years
'Didn't Emperor Justinian I live up to 80 years?'

d. # Je l' Justinijan I/Veliki nije doživeo 80. godinu? LowNPQ

e. # Justinijan I/Veliki (ni)je doživeo 80. godinu? DeclPQ

(Serbian; Neda Todorović, p.c.)

Macedonian quiz PQs are preferably introduced by the particle *dali*. Quiz PQs introduced by this particle can even be negative, without giving rise to a bias (see (32a)), a situation that is reminiscent of Polish (see above). The verb-fronting strategy combined with the particle *li* is claimed to be less formal, but also possible in the quiz scenario, unless the verb is negated, in which case the PQ is reported to acquire positive epistemic bias; see (32b). The fact that *li* can be used in quiz PQs is surprising in light of existing claims (see, e.g., Rudin et al. 1999) that LIPQs in Macedonian are pragmatically marked; see Section 4.2.5 for more discussion. Macedonian, unlike Russian or Bulgarian (see below), also allows verb-first PQs without the particle *li*. These are, however, reported to be too colloquial to be natural in a quiz scenario; see (32c). Also in this case, negation contributes

[37] The intonation of Serbian PQs was investigated by Nakić & Browne (1975).

[38] This appears to be at odds with occasional reports that the DA+LIPQ strategy is a more "emphatic" version of the plain LIPQ strategy, in which *li* is preceded by a fronted auxiliary/verb (Browne 1971, Rakić 1984).

positive epistemic bias and is thus unnatural in the given scenario.[39] Besides *dali* (and *li*), Macedonian has other PQ-initial particles at its disposal, particularly *zar* and *neli*. These, however, are reported to be completely inappropriate in PQs, as they convey strong biases; see Mitkovska & Bužarovska (2024) for recent discussion.

(32) a. Dali Justinijan Veliki (ne) doživeal 80-godišna vozrast? daliPQ
DALI Justinian Great NEG lived.to 80-yearly age
'Did Emperor Justinian I (not) live up to 80 years?'

b. (#Ne) Doživeal li Justinijan Veliki 80-godišna vozrast? liPQ
NEG lived.to LI Justinian Great 80-yearly age
'Did Emperor Justinian I live up to 80 years?'

c. # {(Ne) Doživeal} Justinijan Veliki 80-godišna vozrast? v1PQ
(Macedonian; Liljana Mitkovska, Eleni Bužarovska, p.c.)

Bulgarian and Russian quiz PQs make use of the particle *li* immediately preceded by the verb; see (33) and (34).[40] Adding negation triggers epistemic bias, making the question infelicitous in the quiz scenario. PQs without *li* and with declarative order are reported not to be felicitous.[41] Despite the superficial similarity

[39] Given the availability of verb fronting in Macedonian PQs, one could wonder whether the verb can also front in daliPQs. While the order V-*dali*-S is reported to be plainly ungrammatical, the order *dali*-V-S is available. As opposed to liPQs and v1PQs, where verb fronting is arguably related to interrogativity, verb fronting in daliPQs appears to be information-structurally motivated. One argument for that is that *dali*-V-S PQs involve subject-focus (contrasting alternative subjects), which would follow under the assumption that the verb fronts altruistically (i.e. in order to render the subject focused; see Arnaudova 2001 for this type of analysis applied to the closely related Bulgarian). Another argument is that *dali*-NEG-V-S, i.e. daliNPQs with verb fronting, do not necessarily give rise to any bias in our quiz scenario and could thus be felicitously used when the contestant is asked about the life spans of different famous people. I am grateful to Liljana Mitkovska and Eleni Bužarovska for providing these crisp intuitions.

[40] See Simeonova & Kamali (2024) for a comparison between Bulgarian *li* and the clause-initial *dali*. Although Serbian *da li* is felicitous in quiz questions, Bulgarian *dali* is not, as it incorporates a 'wondering' component and has a rhetorical flavour.

[41] Zimmerling (2023) questions the Russian judgement and provides example (i) as evidence that IntonPQs are felicitous in quiz scenarios.

(i) *Quiz scenario:* The professor asks the student during an exam.
A ↗vse gadjuki jajcеživorodjaščie?
PTCL all vipers ovoviviparous
⟨What about ovoviviparous snakes:⟩ 'Are all vipers ovoviviparous?'
(Russian; Zimmerling 2023: 119; glosses added by RŠ)

Judging by the angle-bracketed prequel to the translation, the PQ in (i) has a marked informaof the two languages, there are important differences. First, while *li* in Russian is preceded by a single constituent (which happens to be the verb in (34a), but can also be the focus; see King 1994, Franks & King 2000), Bulgarian *li* always encliticizes to the *focused* constituent (which happens to be the verb in (33a)), which can in turn be preceded by a *topical* constituent (Izvorski et al. 1997, Rudin et al. 1997, 1999; cf. also Dukova-Zheleva 2010, Jordanoska & Meertens 2021 for relevant discussion on Macedonian). The notion of topicality involved here is arguably quite weak or underspecified; in our example, *Justinian* is not, for instance, contrastively topicalized (Vesela Simeonova, p.c.).[42] Another difference concerns the (apparently) declarative PQs in examples (c). While in Bulgarian, (33c) is arguably a run-of-the-mill DeclPQ, triggering the expected evidential bias, in Russian, (34c) is normally considered a default unbiased PQ, whose question force is conveyed by a dedicated pitch accent pattern (Odé 1989, Meyer 2004, Meyer & Mleinek 2006), something I reflect by calling this type intonation PQs (IntonPQ).[43] Its infelicity in the quiz scenario could thus be on a par with the infelicity reported for Slovenian *a*-PQs and Serbian *je l'*-PQs: one could conjecture that Russian IntonPQs are not biased, but inherently information-seeking.[44] This conjecture is, however, probably not on the right track, as IntonPQs can also be used rhetorically (Esipova & Romero 2023).

(33) a. Justinian doživjava li do 80-godišna vŭzrast? liPPQ
Justinian lives li to 80-yearly age
'Did Emperor Justinian I live up to 80 years?'

b. # Justinian ne doživjava li do 80-godišna vŭzrast? liNPQ

c. # Justinian (ne) doživjava do 80-godišna vŭzrast? DeclPQ
(Bulgarian; Vesela Simeonova, p.c.)

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tion structure. More specifically, the polarity focus which is at issue in the PQ (being or not being ovoviviparous) is accompanied by the topical quantifier *vse gadjuki* 'all vipers', which introduces an additional layer of contrast (either all vs. just some, or all vipers vs. other kinds of snakes). The contrastively-topical reading of the subject is further enhanced by the particle *a*, which is dedicated to encoding contrastive topic–focus structures (Jasinskaja & Zeevat 2008). For these reasons, I conjecture that in Russian, just as in Czech (see the discussion around example (18)), information structure can motivate the use of the declarative word order even in the absence of any bias.

[42] Placing the subject after *li* is in principle possible, but dispreferred in the given scenario.

[43] For a discussion of how IntonPQs could be distinguished from DeclPQs in Russian, see Meyer (2004). There are indications (present in the hitherto unpublished work Esipova & Romero 2023) that IntonPQs with prosodic prominence placed elsewhere than on the verb share some characteristics (esp. evidential bias) with DeclPQs; see Section 4.2.3 for some discussion.

[44] The hypothesized unbiased nature of IntonPQs is supported by the spoken corpus research of Onoeva & Staňková (2025), whose random sample of 500 PQ occurrences includes only 6 instances of liPQs, and the majority of PQs (338) were annotated as unbiased by the authors.

(34) a. Dožil li Justinian Velikij do 80 let? liPPQ
lived.to li Justinian Great to 80 years
'Did Emperor Justinian I live up to 80 years?'

b. # Ne dožil li Justinian Velikij do 80 let? liNPQ

c. # Justinian Velikij dožil do 80 let? IntonPQ

(Russian; Maria Onoeva, Mariia Razguliaeva, p.c.)

In summary, Slavic languages exhibit a variety of formal means of marking a default PQ: from interrogative word order (Czech, Slovak, Upper Sorbian, Slovenian, Macedonian), through utterance-initial particles (Slovenian, Serbian, Polish, Ukrainian, Macedonian), to a verb-attached particle (Bulgarian, Russian, Macedonian). Moreover, we have seen some preliminary evidence that quiz PQs are sometimes formally distinguished from genuine information-seeking (but arguably still unbiased) PQs, either by the use of a different particle (Slovenian, Serbian) or by the use of an altogether different strategy (Russian). Where (the correlate of) high negation is available (cf. Ukrainian), the positive epistemic bias tends to be reported; where declarative PQs are available (cf. Russian and Ukrainian), we see a correlation with evidential bias.

4.2 Analyses of Slavic PQ semantics and bias

The study of formal semantics and bias in Slavic PQs is only just emerging, although in some cases semantic and pragmatic aspects of Slavic PQs have caught the attention of descriptive linguists and philologists. In this section, I will briefly report on the main areas of interest and the progress made thus far.

4.2.1 Some preliminaries

Although morphosyntax is not my focus, it is worth pointing out some basic facts and assumptions about this aspect of Slavic PQs, at least in so far as these are relevant for the (compositional) semantics. As I showed in Section 4.1, many Slavic languages make use of a particle in their interrogative PQs, in particular *li* or its cognates (used in Russian, Bulgarian, Macedonian, Serbian; cf. also Slovenian *ali*) and *čy* or its cognates (Ukrainian, Belarusian, Polish).[45] It is unanimously assumed that the *li* particle is an interrogative complementizer, which either lowers onto the verb (Rivero 1993) or – and more likely so – stays in its left-peripheral position, where it hosts a preceding constituent, which is either

[45] Languages with no particle (in root questions) involve interrogative verb or auxiliary movement (Czech, Slovak, Upper Sorbian).

head-adjoined (if it is an auxiliary or verb) or located in SpecCP (if it is a phrasal constituent); see Izvorski et al. (1997), Rudin et al. (1997, 1999), Schwabe (2004); cf. Dimitrova's (2020) analogous account in terms of the Pol projection. I am not aware of any generative syntactic literature dedicated to the interrogative *čy* (or its cognates), but the null hypothesis seems to be that it is also located in C (cf. Hansen et al. 2016).[46] What is significant is that both *li* and *čy* are also employed (i) in embedded interrogatives and (ii) as disjunctions (see Logvinova 2022 for a systematic overview and references and Arsenijević 2009 and Mitrović 2021 for theoretical and diachronic implications). Both of these facts suggest that while the particles may be analyzed as having set-forming semantics (Alonso-Ovalle 2006, Mitrović 2021), they do not necessarily carry question-related illocutionary force. The set-forming semantics (following Alonso-Ovalle 2006) is sketched in (35a), and the result it should produce when applied to a proposition is given in (35b).[47] I leave aside the issue of how exactly the polar alternative to the IP 'rain' is supplied in the schematic example below. There is suggestive evidence that (polarity) focus is involved.[48] This aligns with the observation that what precedes *li* in languages that have it (e.g. Russian or Bulgarian) is focus, whether polarity focus (realized as the auxiliary or verb, corresponding to a possible short answer) or constituent focus (Rudin et al. 1997, Izvorski et al. 1997). See Han & Romero (2004) and Dimitrova (2020) for a related analysis.

(35) a. $[\![\alpha \text{ li/čy } \beta]\!] = \{\alpha, \beta\}$

b. $[\![[_{\text{CP}} \text{ li/čy/Q } [_{\text{IP}} \text{ rain}]]]\!] = \{\lambda w[\textsc{rain}(w)], \lambda w[\neg\textsc{rain}(w)]\}$

It should be noted that *li* and *čy* might be precisely the alternatives-introducing elements that Biezma & Rawlins (2012) failed to find in English PQs. As discussed in Section 3.3.1, the non-existence of an overt disjunction-like element in English was one of Biezma & Rawlins' (2012) arguments against a Hamblin-style semantics for PQs. Since this argument does not hold for (most) Slavic languages, and since the PQs discussed in Section 4.1 can be completely unbiased (used in quiz

[46] Bartosz Wiland (p.c.) draws my attention to the fact that *čy* is morphologically related to 'what' (cf. Polish *czym* 'what.INS'). If *čy* is a wh-word of sorts (cf. also Slovenian clause-initial PQ particle *kaj* 'what' and Han & Romero's 2004 wh-word-style analysis of the English *whether*; see also Haegeman 2012), it would motivate its SpecCP (rather than C) status, which in turn might explain its clause-initial positioning.

[47] In languages with no overt particle (Czech, Slovak, Upper Sorbian), C is hypothesized to host a covert Q, to which V adjoins by head movement.

[48] The idea that the Q operator (here hosted by the interrogative C head) is focus-sensitive is not new; see, e.g., Beck (2006).

scenarios), there is good reason to stick to Hamblin's (1973) semantics for Slavic default PQs.[49]

I further note that Biezma & Rawlins' (2012) analysis of PQs in terms of singleton sets containing just the question prejacent might well be the right model for declarative questions. As we saw in Section 4.1, declarative questions do not contain an overt Q particle (or a movement to one). Adopting Biezma & Rawlins' logic, we can assume that polar alternatives are not collected into a set in this case. Rather, a single alternative is under discussion – the one that is evidentially biased for – with any other relevant alternatives being just pragmatically supplied.

I finish this subsection by putting forth a hypothesis about the grammatical forms which entail information-seeking (and hence are incompatible with the quiz scenario), in particular the Serbian je+liPQ strategy, the Slovenian aPQ strategy, and possibly the Russian IntonPQ strategy. These forms are hypothesized to map to the logical form in (36), based on (35b). More specifically, the structure in (35b) is selected by a covert attitude predicate whose semantics corresponds to the semantics of 'wonder', which in turn can be modelled as 'want to know' (see the discussion of example (9) in Section 2). Since the quiz master is not genuinely interested to know whether p (in fact, he knows whether p), the logical form in (36) is not suitable for quiz PQs.

(36) $[\![[\textsc{wonder}\ [_{\mathrm{CP}}\ \mathrm{Q}\ [_{\mathrm{IP}}\ \mathrm{rain}]]]]\!]^c = \textsc{want to know}\big(w_0\big)\big(\textsc{spkr}(c)\big)\big(\iota q\, q(w_0) = 1 \wedge q \in \{\lambda w[\textsc{rain}(w)], \lambda w[\neg\textsc{rain}(w)]\}\big)$

In what follows, I briefly discuss existing formal semantic analyses of various aspects of Slavic PQs. I also include the discussion of the results of selected empirical studies about Slavic PQs.

4.2.2 Bias profiles of PQs in selected Slavic languages

Todorović's (2023) work on Serbian offers probably the most comprehensive view of bias profiles of different kinds of PQs in Slavic languages, building on the tradition of Ladd (1981), Büring & Gunlogson (2000), Romero & Han (2004), Sudo (2013), Gärtner & Gyuris (2017), and providing an analysis within the framework of inquisitive semantics, following AnderBois (2019).[50] Table 1 summarizes Todorović's (2023) results. The finding that da+liPPQs are limited to neutral

[49] Hamblin's (1973) semantics is defended for both liPQs and IntonPQs in Russian by Zimmerling (2023).

[50] For an earlier treatment of Serbian PQs, reflecting formal literature, see Rakić (1984).

contexts resonates with the quiz scenario investigation in Section 4.1; the distribution of je+liPPQs, on the other hand, is broader: they can also be used in scenarios with positive epistemic or evidential bias. PPQs are found to be incompatible with negative biases. With NPQs, low NPQs are found to have a broader distribution than high NPQs. The latter are claimed to be available only for the purpose of a resolution of the conflict between positive epistemic bias and negative evidential bias. This goes counter to the traditional observation for English, where high NPQs can also be used with neutral evidential bias (called "suggestion" scenarios); see also Section 5 and the discussion of example (12).

Table 1: Bias profile for Serbian positive polar questions (PPQ; for subtypes, see Section 4.1), high negation polar questions (HNPQ), and low negation polar questions (LNPQ) according to Todorović (2023)

		epistemic bias		
		positive	neutral	negative
evidential bias	positive		je+liPPQ	
	neutral	je+liPPQ	da+liPPQ je+liPPQ	LNPQ
	negative	LNPQ HNPQ	LNPQ	

Staňková (2023) (see also Staňková & Šimík 2025) ran naturalness rating experiments with the aim of tapping into the interactions between evidential bias (epistemic bias was not manipulated), type of negation (outer vs. inner), and different PQ strategies in Czech (interrogative vs. declarative PQs). In line with existing claims for English, Staňková found that only DeclPQs require matching evidential bias in order for them to be natural (cf. discussion around (18)); InterPQs – whether positive or negative – are insensitive to evidential bias, which, however, does not mean that they are incompatible with it. Staňková (2023) further found that InterNPQs, i.e. negative PQs with a fronted verb, map to outer negation, which is diagnosed by the inability to license negative concord items (NCIs), whereas the corresponding DeclNPQs are compatible with both inner and outer negation readings (although the former is preferred). Importantly, the inner vs. outer negation reading, which Staňková (2023) models in terms of Repp's (2006, et seq.) verum (scoping over propositional negation)

Table 2: Bias profile for Czech polar questions (based on Staňková 2023 and the discussion in Section 3.2); *conditioned by information structure (see discussion around example (18))

		EPISTEMIC BIAS		
		positive	neutral	negative
EVIDENTIAL BIAS	positive		DeclPPQ InterNPQ	DeclPPQ
	neutral	InterPPQ InterNPQ	InterPPQ DeclPPQ*	
	negative	DeclNPQ InterNPQ	DeclNPQ	

vs. FALSUM (see Section 3.3.2), has no impact on the bias profile: both inner and outer negation DeclPQs require negative evidence for naturalness. The existing findings – combining the experimental results of Staňková (2023) and the present investigation (Section 3.2) – are summarized in Table 2 (glossing over details).

Chirpanlieva et al. (2025) adapted Staňková's (2023) experiment to Russian and Polish. In Russian the authors compared liNPQs with IntonNPQs.[51] They found that all NPQs in Russian were more natural if the evidence was neutral; i.e. negative evidence is not only not required, but even dispreferred in Russian NPQs. The preference for neutral evidence was stronger in liPQs than in IntonPQs. The negation was found to be preferentially outer rather than inner in both types of PQs, although the effect was much stronger in liPQs than in IntonPQs. In Polish, the authors compared czyNPQs with DeclNPQs. They discovered that evidential bias interacts with the type of negation: while inner negation is more or less inert to the presence or absence of negative bias, outer negation is more naturally combined with neutral evidence. This interaction, or more precisely the preference for neutral evidence, is more pronounced in czyNPQs than in DeclNPQs, although only numerically so. In summary, we see a clear cross-Slavic difference between Czech on the one hand and Russian and Polish on the other. Only in Czech do we see evidence for "English-style" declarative NPQs, correlating with the presence of negative evidential bias. What superficially looks like a declarative NPQ in Russian and Polish (characterized by declarative word order and the

[51]The stimuli were written, so intonation was not controlled for. The authors' assumption was that participants apply default PQ-prosody in silent reading.

lack of interrogative particles) can easily be accompanied by a lack of negative evidence.

Onoeva & Staňková (2025) conducted a corpus study with the aim of investigating the correlations between biases and different PQ strategies in Russian and Czech. They found that LIPQs are very rare in spoken Russian – only 6 out of 500 randomly selected PQ occurrences were LIPQs; the rest were INTONPQs. For Russian, the authors found no significant correlations between different PQ formal factors (word order, negation, the presence of tags) and the PQs' bias profiles. There were only a handful of PQ occurrences with various non-default particles (*razve, neuželi, čto li*); the instances found largely confirm their reported semantics, such as a prominent epistemic bias for the negation of the prejacent with *razve* (see Section 4.2.4). For Czech, the authors found a significant correlation between DECLPQs and evidential bias and between the presence of tags and epistemic bias.

4.2.3 Accent placement in Russian INTONPQs

Russian INTONPQs – PQs with a declarative word order in which question meaning is conveyed exclusively prosodically (see Odé 1989, Meyer & Mleinek 2006) – exhibit variation with respect to accent placement and the final prosodic contour: while neutral INTONPQs exhibit rising pitch accent on the verb (with the H peak delayed, when compared to contrastive accent employed in declaratives; see Meyer & Mleinek 2006) and typically a low final tone, biased PQs can exhibit (low) pitch accent on a different constituent, such as the object, followed by a high final tone; see (37).

(37) a. Nina sdaLA$_{L+H^*}$ ėkzamen$_{L-L\%}$?
Nina passed exam
'Did Nina pass the exam?'

b. Nina sdala ėkzA$_{L^*}$men$_{H-H\%}$?
Nina passed exam
'Did Nina pass the exam?' / 'Nina passed the exam?'
(Russian; adapted from Esipova & Romero 2023)

Esipova & Romero (2023) argue that both types of PQ above denote the Hamblin-set {'Nina passed the exam', 'Nina didn't pass the exam'}, but (37b) in addition indicates the existence of a superordinate question under discussion to which the question prejacent ('Nina passed the exam') is a possible answer. (The superordinate question implies the relevance of additional polar questions [or their prejacents], such as 'Did Nina win the lottery?') In other words, Esipova & Romero

(2023) argue that non-default accent placement gives rise to explanation-seeking questions, which in turn convey evidential bias (see Section 3.1). The utterance in (37b) would thus be felicitous in scenario (38), putting forth a possible answer (in the form of a PQ) to the contextually salient superordinate question under discussion (QUD) in Figure 1. Notice that this makes PQs like (37b) functionally close to DeclPQs in English and other Slavic languages.[52]

(38) *Scenario:* Nina was due to take an exam. I just saw her cheering in the hallway.

Why is Nina cheering?

Did Nina pass the exam? Did Nina win the lottery? …

Figure 1: Questions under discussion relevant for uttering (37b) in scenario (38)

Neutral PQs such as (37a) are "root questions" (in the terminology of Esipova & Romero 2023) in the sense that they lack a superordinate QUD.

4.2.4 Russian *razve* and its kin in other Slavic languages

Besides having a rich cross- and intra-linguistic repertoire of PQ strategies (see Section 4.1), Slavic languages are also known for possessing particles specialized in conveying various biases. Probably the best-studied among these is the Russian particle *razve*, which roughly corresponds to the English *really*, but differs from it at least in that it is only licensed in polar questions, albeit only in IntonPQs (and potentially DeclPQs, if these can be reliably distinguished; cf. Meyer 2004), not liPQs; see (39a). Repp & Geist (2025) and Korotkova (2023), who recently approached *razve* from a formal-semantic perspective, agree with the traditional insight that *razve* "indicate[s] moderate surprise or doubt concerning the situational evidence" for the question prejacent (Apresjan 1980, Rathmayr 1985, Baranov 1986, Bulygina & Šmelev 1987, Mat'ko 2014; quoted from Repp & Geist 2025: 110). This mirative ('surprise') or dubitative ('doubt') flavour corresponds to what I called a conflict-resolving PQ in Section 3.1, whose purpose is

[52] Something else to note is that pitch accent on the object in IntonPQs need not indicate a narrow focus on the object (cf. Meyer & Mleinek 2006). In scenario (38), the accent on *ėkzamen* 'exam' in (37b) represents broad or at least VP focus, implicitly contrasting 'pass the exam' with 'win a lottery', etc.

to resolve a conflict between evidential bias for p (the question prejacent) and the speaker's prior epistemic bias for $\neg p$ (formulated as 'I thought that…' in (39) and below).

(39) a. Razve vy govorite po-russki?
RAZVE you speak Russian
'Do you (really) speak Russian? [I thought that you didn't.]'

b. Razve u vas net kota?
RAZVE at you NEG.be cat
'Do you (really) not have a cat? [I thought that you would have one.]'
(Russian; adapted from Korotkova 2023)

Repp & Geist (2025) build on Repp (2006, 2009, 2013) and, although they do not provide an explicit compositional semantics of *razve,* they suggest that *razve* is a "semantic epistemic operator that is compatible with FALSUM and with VERUM" (Repp & Geist 2025: 133); see the simplified logical forms in (40) ((40c) is added by me for completeness).

(40) a. [Q [FALSUM *razve* p]] OUTER NEG PQ
b. [Q [VERUM *razve* $\neg p$]] INNER NEG PQ
c. [Q [VERUM *razve* p]] POSITIVE PQ
(adapted from Repp & Geist 2025: 133)

Repp & Geist's (2025) crucial insight is that *razve* is compatible with both outer and inner negation, which amounts to saying that when associated with a negative PQ, *razve* can double-check both the negative and the positive alternative – something that the authors call "question concern" (cf. Section 3.1). The evidence for the compatibility with both kinds of negation is that both the weak NPI *eščë* 'yet' and the PPI *uže* 'already' are licensed in the scope of the negation in PQs with *razve*; see (41). NPIs are assumed to be licensed by inner negation; PPIs are only compatible with outer negation.

(41) a. Razve *eščë* ne skazal? INNER NEG
RAZVE yet NEG said
'Haven't you told it to me yet? [I thought that you had.]'
(A. I. Slapovskij, *Bolšaja kniga peremen / Volga,* 2010)

b. Razve ty *uže* ne vytaščila menja iz prošlogo? OUTER NEG
RAZVE you already NEG dragged me out.of past
'Haven't you dragged me out of the past already? [I thought you that you had.]' (Gennadij Praškevič, Aleksandr Bogdan, *Čelovek "Č",* 2001)
(Russian; adapted from Repp & Geist 2025: 112)

It is notable that the bias profile of *razve* NPQs stays constant irrespective of whether VERUM (≈ inner negation) or FALSUM (≈ outer negation) is used: they presuppose negative evidential bias and convey positive epistemic bias.[53] The rationale behind this is that FALSUM is prompted by evidence for the falsity of p (while being concerned about the corresponding epistemic bias, i.e. the truth of p; paraphrasable as 'Is it really not the case that p?') and VERUM by evidence for the truth of $\neg p$ (while being "concerned" about the evidential bias, i.e. the truth of $\neg p$; paraphrasable as 'Is it really the case that $\neg p$?'). Repp & Geist (2025) further argue that *razve* is different from *neuželi* in that the latter lexicalizes VERUM and is thus only compatible with inner negation (cf. Bulygina & Šmelev 1987).

Korotkova (2023) proposes a more nuanced approach to *razve*. She does not challenge the claim that the use of *razve* aims at resolving the evidential vs. epistemic conflict, but she notices that *razve* is picky about the relation between the evidence that constitutes the bias and the prejacent of the PQ. In particular, *razve* is only licensed if the prejacent is a likely and mutually shared abductive inference drawn from the evidence. An ABDUCTIVE INFERENCE (Douven 2021) is an inference from an observed effect (the presence of a mouse) to its cause or best explanation (the absence of a cat); see (42a). In (42b), *razve* is not licensed because the absence of mice is not a likely explanation for the absence of a cat (effect/evidence).

(42) a. *Scenario:* I am over at your house in a village and see a mouse. I ask:

Razve u vas net kota?
RAZVE at you NEG.be cat
'Do you not have a cat? [I thought that you would, like every village house.]'
Background assumption (likely mutual): The absence of cats is a very plausible explanation for the presence of mice.

b. *Scenario:* I am over at your house in a village and ask where your cat is. You tell me you don't have one. My next question is:

Razve u vas net myšej?
RAZVE at you NEG.be mice
'Do you not have mice? [I thought that you would, like every village house.]'
Background assumption (unlikely mutual): The absence of mice is a very plausible explanation for the absence of cats.

(Russian; adapted from Korotkova 2023)

[53]This aligns with Staňková's (2023) experimental findings for Czech DECLNPQs, which are natural in contexts that presuppose negative evidence, irrespective of whether the negation is outer or inner.

Korotkova (2023) also proposes a compositional treatment of *razve*, building on Biezma & Rawlins' (2012) analysis in which PQs denote singleton sets.[54] She argues that the singleton-based analysis accounts for some core properties of *razve* PQs, such that *razve* cannot be used in embedded interrogatives (which make obligatory use of *li*) or that it cannot be used in wh-interrogatives.

A particle like *razve* is not a rare find in Slavic languages (see Logvinova 2022 for a survey). Particles with a similar meaning include the Ukrainian and Belarusian *xiba*, Polish *czyż(by)*, Bulgarian *nima* (Tisheva 2003), Macedonian *zar* (Englund 1977, Mitkovska & Bužarovska 2024), Czech *co(ž)pak* (Nekula 1996, Šebestová & Malá 2016, Staňková 2023), and Serbian *zar* (Rakić 1984, Piper 2005, Todorović 2023). I am not aware of formal-semantic studies dedicated to these particles. Their precise semantic and pragmatic properties and their potential affinity to the Russian *razve* are thus yet to be explored.

4.2.5 Variation in the semantics/pragmatics of *(da)li*

In Section 4.1 we saw that the particle *li* is used in default PQs in a number of Slavic languages, either in isolation (Bulgarian, Russian, Macedonian) or in combination with *da* (Serbian *da li*, Macedonian *dali*). Despite the superficial similarity of these particles, existing literature suggests that we should be cautious about equating them.

Macedonian *li,* in contrast to Bulgarian (or Russian) *li,* has been reported to be infelicitous in neutral questions, as it is supposed to convey surprise (Rudin et al. 1999, Lazarova-Nikovska 2003) or a negative answer expectation (Englund 1977); both of these properties received experimental support in Jordanoska & Meertens (2021) (who, however, concentrated primarily on *li* preceded by constituents other than the verb). Rudin & Rudin (2022) provided further independent support for a qualitative distinction between Bulgarian and Macedonian PQ strategies. They argue that while in Bulgarian, question semantics is conveyed by *li* (see Section 4.2.1), in Macedonian it is conveyed by question intonation. The consequence is that Macedonian rising intonation is specifically question-related, while Bulgarian rising intonation only conveys a lack of speaker commitment, but not necessarily question force, as it can also be used, for instance, in imperatives (cf. Rudin 2022). It should be noted that the claims about non-neutrality of the Macedonian *li* are in obvious contrast to the data presented in Section 4.1, where *li* (preceded by the finite verb) was elicited as one of the strategies for expressing an unbiased quiz PQ. My consultants (Liljana Mitkovska and

[54]See also my suggestion in Section 4.2.1 that this might be an appropriate analysis for (Slavic) DeclPQs, to which *razve* PQs are arguably related.

Eleni Bužarovska) hypothesize that the apparent contradiction could be due to register: the most common strategy in colloquial Macedonian is arguably the particle-free intonation strategy (Englund 1977, Rudin & Rudin 2022) and the *li*-strategy is rather infrequent and formal. In and of itself, this cannot explain the intuition that *li* expresses surprise or negative answer expectation. Russian may serve as another member of a minimal pair in this case, where *li* is also rather formal and infrequent (Onoeva & Staňková 2025), but is not reported to give rise to pragmatic implications.

The particle *dali* is also subject to cross-Slavic variation. While Macedonian daliPQs and Serbian da+liPQs are neutral (Section 4.1; see also Jordanoska & Meertens 2021 and Todorović 2023), Bulgarian daliPQs incorporate a 'wondering' component and typically have a rhetorical flavour in the sense that they do not necessarily solicit a response (Simeonova & Kamali 2024).

I conclude that there is a puzzling cross-Slavic microvariation in the domain of PQ particles which awaits deeper investigation.

5 Czech negative polar questions and *náhodou*

In Section 3.2, and particularly in the discussion of example (17), I noted that Czech InterNPQs, i.e. interrogative (verb-initial) negative PQs, have a broader distribution than their English counterparts (high negation PQs): they can convey positive epistemic bias (and are compatible with negative evidential bias), but they can likewise convey positive *evidential* bias, at least in the absence of any epistemic bias. In this section, I discuss more data that bear evidence of an even broader distribution of Czech InterNPQs and which suggest a reconsideration of the semantics and pragmatics of Czech (and potentially Slavic) "high negation". What I will also bring into the discussion is the particle/adverb *náhodou* lit. 'by (any) chance', which, under its question particle guise, is licensed by outer negation (Zanon 2023 reports the same for Russian *slučaem/slučajno* 'by chance'). It is therefore a good diagnostic of outer negation, which is the kind of negation I wish to investigate.

5.1 Basic data and proposal

The set of examples in (43) illustrates the basic properties of inner/outer negation and *náhodou*. Examples (43a)/(43b) show (i) that negation in statements always has the inner reading – it licenses NCIs, but not PPIs in its scope – and (ii) that the adverb *náhodou* can occur in statements, but if it does, it has the canonical meaning, covered by concepts like ‘accidentally’ or ‘incidentally’.[55] Relying on the NCI/PPI-based test, example (43c) shows that Czech InterNPQs primarily involve outer negation and only marginally inner negation (see Staňková 2023 for experimental evidence; see also the discussion around example (15)). Example (43d) shows that once *náhodou* is added to the InterNPQ, the NCI *žádné* becomes completely ungrammatical; in other words, *náhodou* requires outer negation. Finally, example (43e) shows that *náhodou* – under its particle guise – is not licensed in positive PQs.[56]

(43) a. Max (*náhodou*) nemá {#nějaké / žádné} námitky. STATEMENT
Max by.chance NEG.has DET.PPI DET.NCI objections
(Intended:) ‘(Incidentally,) Max doesn’t have any objections.’

b. Max (*náhodou*) má {nějaké / *žádné} námitky. STATEMENT
Max by.chance has DET.PPI DET.NCI objections
(Intended:) ‘(Incidentally,) Max has some objections.’

c. Nemá Max {nějaké / ?žádné} námitky? InterNPQ
NEG.has Max DET.PPI NEG.NCI objections
‘Doesn’t Max have some/any objections?’

d. Nemá Max *náhodou* {nějaké / *žádné} námitky? InterNPQ
NEG.has Max by.chance DET.PPI NEG.NCI objections
‘Does Max have some objections, by any chance?’

e. * Má Max *náhodou* nějaké námitky? InterPPQ
has Max by.chance DET.PPI objections
Intended: ‘Does Max have some objections, by any chance?’

The hypothesis to entertain upon witnessing these data is to say that *náhodou* is licensed by Repp’s (2006, et seq.) FALSUM (see Section 3.3.2); indeed, the existence of *náhodou* could even be considered a striking piece of evidence in favour of the very existence of FALSUM. While I consider this idea to be on the right track, I will show that, in line with the broader distribution of Czech InterNPQs, the semantics of FALSUM (or, more neutrally said, high negation) is qualitatively

[55]Additionally, it can be used as a focus-sensitive adversative particle pragmatically akin to *ale* ‘but’. Although this function of *náhodou* in statements is arguably related to its question-particle function, I leave it aside for reasons of space.

[56]This is supported by corpus data (all corpus findings reported from this point on were gathered using the SYN v11 corpus of the Czech National Corpus; Křen et al. 2022). Of 100 random occurrences of *náhodou* in PQs, all involved negation; of the 6 indefinite pronoun/determiners, all were PPIs (diagnosing outer negation). Of 100 random occurrences of *náhodou* in embedded interrogatives, all involved negation; all of the 14 indefinites were PPIs.

different in Czech (and potentially Slavic) than what was proposed for English and German.

The proposed lexical entry for the Czech falsum (or, more neutrally high/outer negation) is provided in (44). The operator is defined if the attitude holder $g(1)$ (resolved to the speaker in matrix questions) considers it possible that the (positive) prejacent is true; if defined, it selects the prejacent p and returns the proposition characterizing the set of worlds w in which p is not part of the common ground.[57] (44) differs from Repp's (2006) falsum (see Section 3.3.2) in the following ways: (i) it conveys a weak commitment (epistemic possibility rather than belief), (ii) the commitment is not necessarily tied to the speaker (or addressee), but possibly also to linguistically expressed attitude holders (see below), (iii) the commitment is not at issue (as in Romero's 2015 update of Repp's semantics), and (iv) the commitment is not conventionally tied to conversational goals (in fact, a conversation might not even be at stake, as in embedded interrogatives).

(44) $[\![\textsc{falsum}_1^{\mathrm{CZ}}]\!]^g(p) = \lambda w : \exists w' \in \textsc{epi}_{g(1)}(w)[p(w') = 1] \, . \, p \notin \mathrm{CG}_w$

Assuming that falsum scopes under the alternatives-generating Q particle, the meaning of (43c)/(43d) is thus as in (45).

(45) $[\![$(43c)/(43d)$]\!]$ is defined if the speaker considers it possible (and relevant) [that Max has some objections = p]. If defined, then
$[\![$(43c)/(43d)$]\!] = \{\lambda w[p \notin \mathrm{CG}_w], \lambda w[p \in \mathrm{CG}_w]\}$

Space does not permit an explicit formal account of *náhodou.* I hypothesize, though, that its function is to "loosen" the default stereotypical ordering source (Kratzer 1991) of the epistemic modal contributed by falsum so as to include more remote (less likely) possibilities in the quantification domain of the modal. Being an (optional) domain widener modifying an existential quantifier, it comes with a requirement that the resulting proposition is stronger than the alternative without widening (Kadmon & Landman 1993, Krifka 1995). I hypothesize that the strengthening comes about via the negation contributed by falsum: ruling out the truth of a proposition in less likely worlds entails its ruling out in more likely (stereotypical) worlds (but not conversely). The idea that *náhodou* is an ordering source modifier / domain widener is supported by my native speaker intuition

[57] Modelling the epistemic implication as a presupposition may turn out to be inaccurate. The "presupposition" can certainly be new information to the addressee (hinting at its status as a conventional implicature of sorts; Romero 2015, Korotkova 2023); at the same time, however, it can be attributed to a linguistically expressed attitude holder (which in turn hints at local presupposition accommodation). I leave this issue for future research.

about the subtle meaning difference between a PQ with and without *náhodou.* In (46), by using *náhodou*, the speaker invites the addressee to consider even less evident signs of her husband's sleeping problems. The addressee's negative response (the one that is at issue) will then give the speaker more confidence that *p* indeed does not belong to the common ground than if he asked the corresponding PQ without *náhodou.*[58]

(46) Netrpí váš manžel (náhodou) nespavostí?
NEG.suffers your husband NAHODOU insomnia
'Does your husband suffer from insomnia (by any chance)?'

In the next section, I will go through some evidence that supports the proposed analysis of Czech high negation.

5.2 Evidence

The assumption that the epistemic bias expressed is weak and does not have to amount to belief (as assumed for epistemic bias in, e.g., English; see Section 3) but rather just to epistemic possibility receives support by the significantly broader distribution of InterNPQs in Czech. I use examples and data collected from the Czech National Corpus (see footnote 56); see (47). The scenarios are my brief summaries of the preceding context; the "context" in (47d) is a translation of the preceding utterance. All the examples in (47) contain *náhodou* (which testifies the outer status of negation), but would also be felicitous without it, without any major meaning difference.

(47) a. *Scenario:* The speaker finds the addressee alone in a restaurant; he had expected the addressee to be with company, playing cards, as usual on that day.
Nehráváte dneska náhodou karty?
NEG.play.2PL today NAHODOU cards
'Don't you play cards today? [I'm convinced that you normally do.]'
b. *Scenario:* The addressee shows sign of unease; the speaker is wondering about the cause:

[58]This intuition is indirectly supported by corpus data. Of 100 random occurrences of embedded polar interrogatives with *náhodou,* 19 were embedded by *napadnout* 'cross one's mind', a predicate expressing a sudden idea, suggestion, or suspicion, whose validity or relevance is up for discussion or verification; I take this to match the loosened ordering source (exploring potentially unlikely scenarios). Of 94 random occurrences of negative embedded polar interrogatives without *náhodou*, only 3 (3.2 %) were embedded by *napadnout.*

Nejsi náhodou s outěžkem, děvče?
NEG.are NAHODOU with burden girl
'Are you pregnant, by any chance, girl? [I think that you might be.]'

c. *Scenario:* The speaker and addressee are contemplating a solution to a complicated situation. At once the speaker proposes a source of inspiration by asking:

Nečetlas náhodou Deník Anny Frankové?
NEG.read.2SG NAHODOU Diary.ACC Anne Frank.GEN
'Have you read Anne Frank's Diary by any chance? [It will be relevant for what I want to say now.]'

d. *Context:* I'd like to ask Marlene about it.

Nemáte na ni náhodou aktuální telefonní číslo?
NEG.have.2 on her NAHODOU current phone number
'Do you happen to have her phone number? [I hope you do.]'

Example (47a) shows that Czech outer negation is compatible with the speaker's prior belief (often in combination with conflicting evidential bias) – the prototypical use of high negation in English (see also Section 3.2). Although this meaning is not conventionally expressed by the Czech FALSUM, I take it to be consistent with it (possibility is compatible with necessity if the possibility expression – here FALSUM^{CZ} – has no necessity competitor; Deal 2011). Example (47b) is a case of explanation-seeking – with supporting situational evidence, but without any prior or present belief. The speaker simply considers it possible – based on what she takes to be evidence – that the addressee is pregnant (see also the discussion of (17) above). Example (47c) could be called a "relevance PQ": by asking about *p* (here: 'having read Anne Frank's Diary'), he suggests that *p* is relevant for further discussion. Example (47d) represents another relatively frequent type: the speaker considers it possible that the addressee has her phone number and at the same time hopes that this might be the case.[59]

Offers represent a specific category of InterNPQ use. They involve outer negation, as is evident by the use of PPIs and speaker-related bias, but *náhodou* is unavailable in them; (48).[60] If *náhodou* is included, the PQ ceases to be an offer and changes to a factual question. I hypothesize that this might have to do with the kind of modal base that *náhodou* can modify. I leave open the question of what

[59]Together, the four types exemplified make up 94 % of all the uses of InterNPQs with *náhodou*, with the following distribution: belief 14 %, explanation-seeking 40 %, relevance 20 %, hope 20 % (based on 100 random occurrences).

[60]See Chodounská et al. (2025) for experimental evidence supporting the intuition.

examples like (48) tell us about outer negation in Czech and whether they are consistent with the analysis proposed above.

(48) Nedáš si se mnou (#náhodou) něco k obědu?
NEG.give.2SG REFL with me NAHODOU something.PPI for lunch
'Will you join me for lunch?'

Czech outer negation is not limited to matrix PQs; it is frequently attested in embedded polar interrogatives (EPIs).[61] In Czech EPIs, outer negation is not encoded by verb fronting, as the initial position is taken up by the polar interrogative complementizer *jestli*. Nevertheless, it can be easily diagnosed by the availability of PPIs, the particle *náhodou*, and the inability to license NCIs. The corpus example (49a) contains both a PPI and *náhodou*; the example in (49b) does not, but, as indicated, *náhodou* could easily be added without any significant difference in meaning. Finally, the example in (49c) shows that upon using inner negation (diagnosed here not by a polarity item, but by the strong negative evidential bias – bias to 'not wanting kids'), *náhodou* is ruled out.

(49) a. [Hjonmi] kontrolovala, jestli náhodou někdo nepřibíhá za ní
Hjonmi controlled whether NAHODOU someone.PPI NEG.run after her
'Hjonmi was checking whether there's anybody running after her'

b. taťka měl [...] pochybnosti, jestli to [náhodou] není riskantní
dad had doubts whether it NAHODOU NEG.is risky
'daddy was worried whether it is risky / that it could be risky'

c. *Scenario:* The speaker, a hospitalized patient, engages in a risky behavior, threatening her own health. She reports:

Doktorka se mě ptala, jestli [#náhodou] nechci mít děti, nebo co.
doctor REFL me asked whether NAHODOU NEG.want.1SG have kids or what
'The doctor asked me whether I don't want to have kids, or what.'

What is important is that embedded outer negation encodes bias which is not tied to the speaker, but rather to the attitude holder. The truth conditions of (49a) can thus be captured as in (50), resorting to the simplifciation that the matrix

[61]In a random sample of 63 EPIs, 22 (35 %) have a negative predicate. Of the 22, 21 arguably involve outer negation (judging by the felicity of adding *náhodou*).

predicate *kontrolovala* denotes 'want to know' (see Section 2). The denotation presupposes that Hjonmi considered the positive prejacent, i.e. that somebody is running after her, possible (or, even remotely possible – the contribution of *náhodou*) and is true iff Hjonmi wanted to know the true answer to the question whether somebody running after her is part of the common ground or not. In this case, the attitude holder is alone in the situation and hence the common ground reduces to Hjonmi's own epistemic state.

(50) $[\![(49a)]\!]$ is defined in an evaluation world w_0 if
$\exists w \in \textsc{epi}_{\textsc{Hjonmi}}(w_0)\big[\exists x[\textsc{run towards}(w)(x, \textsc{Hjonmi})]\big]$; if defined, then
$$[\![(49a)]\!] = \lambda w\big[\textsc{want to know}(w)\big(\textsc{Hjonmi}\big)\big(\iota q\, q(w) = 1 \wedge q \in \{\lambda w[p \notin \mathrm{CG}_w], \lambda w[p \in \mathrm{CG}_w\}]\big)\big],$$
where $p = \lambda w \exists x\big[\textsc{run towards}(w)(x, \textsc{Hjonmi})\big]$

The contribution of outer negation in embedded contexts is further evident from the contrast between the corpus example (51a) – a positive EPI – and its modification (51b) – a negative EPI. While (51a) is natural in a situation where the police officer reacts to a question or statement suggesting that the criminal deed under discussion was racially motivated – a case of positive evidential bias with no obligatory implication tied to the attitude holder – (51b) requires no evidential bias, but conveys a weak epistemic bias of the attitude holder (the police officer). This bias is conveyed in order to indicate that it is a possibility to be seriously considered and explored.

(51) a. Nevíme, jestli šlo o rasově motivovaný čin.
NEG.know.1PL whether went about racially motivated deed
'We don't know whether it was a racially motivated deed.'

b. Nevíme, jestli (náhodou) nešlo o rasově motivovaný čin.
NEG.know.1PL whether NAHODOU NEG.went about racially motivated deed
'We don't know whether it could have been a racially motivated deed.'

The predicates which embed outer negation EPIs are typically verbs of asking (30 %), verbs of finding out by direct observation such as 'look', 'listen', or 'verify' (26 %), and verbs of pondering such as 'think', '(not) know', or 'hesitate' (30 %). All outer negation EPIs embedded under these predicates exhibit biases comparable to the ones described for matrix contexts (see the discussion of examples (47)) and also allow for productive use of *náhodou* (although the particle occurs only in 6 % of cases in the random sample of 100 occurrences of negative EPIs).

Just like in InterNPQs, *náhodou* is infelicitous in reported offers; (52).[62] This is because, by hypothesis, offers involve a non-epistemic modal base.

(52) Loni mi volal, jestli bych s ním [#náhodou] nezazpíval
last.year me called whether sbjv.1sg with him nahodou neg.sing
tu písničku na předávání Cen Anděl [...]
dem song at ceremony Prizes Angel
'Last year he called me (to ask) whether I'd sing the song with him at the Angel Prizes ceremony [...]'

To sum up, the productive and functionally analogous use of outer negation in embedded interrogatives lends support to the analysis proposed above, in which the epistemic component of the falsum operator is relativized to an attitude holder and not just the speaker or addressee.

6 Conclusion

This chapter has surveyed the basic semantic and pragmatic properties of polar questions and, to a lesser extent, embedded polar interrogatives, with a focus on Slavic data. I have attempted to show that Slavic facts might be vital for a deeper understanding of polar question semantics and pragmatics and, more generally, of a formal-semantic theory of speech acts.

I have shown that Slavic languages make use of a variety of strategies for encoding polar questions. Despite some important emerging generalizations, I have also discussed the fact that the properties of one and the same particle can differ substantially in two closely related languages; this concerns esp. the particles *li* and *dali* in Bulgarian vs. Macedonian. All in all, most of the strategies for encoding polar questions are still poorly understood from a semantic or even a morphosyntactic perspective.

My investigation into default or neutral (unbiased) polar questions has uncovered a hitherto neglected contrast between quiz polar questions and information-seeking questions. This contrast suggests that some neutral polar questions do incorporate a 'wondering' component in their conventional semantic make-up while others do not; only the latter can be used in a quiz, where no genuine 'wondering' (about the truth of the prejacent) takes place. It remains to be seen whether the contrast between quiz and information-seeking polar questions is rooted in semantics, as I have suggested, or in register (e.g. formal vs. informal).

[62] The judgement reflects my personal intuition. Chodounská et al.'s (2025) experiments did not confirm this infelicity. (Cf. footnote 60.)

I have shown that Czech outer negation has a significantly broader distribution than its kin in English or German. I have proposed a modification of Repp's (2006 et seq.) FALSUM-based analysis to account for the broad distribution. The modification is not meant as a replacement of Repp's original account; rather, it is supposed to reflect what I believe to be a genuine cross-linguistic variation in the semantic properties of outer negation. Although there are indications that other Slavic languages might be similar to Czech in this respect (see, e.g., Mitkovska & Bužarovska 2024: 147, who claim that Macedonian high negation PQs "imply some speaker's belief about *p* ranging *from very weak to quite strong*", emphasis mine), a detailed investigation is missing. This clearly represents highly fruitful grounds for further investigation. The use of outer negation in constructions beyond (polar) questions is another important direction for future inquiry. In Czech, for instance, outer negation with very similar properties (including the licensing of the particle *náhodou*) occurs in conditionals (cf. Romero 2015), purpose clauses, and some types of embedded declarative clauses.

Finally, I have touched upon the issue of particles specialized in expressing biases. There is some new and promising work on the Russian particle *razve* (see Korotkova 2023, Repp & Geist 2025), but most particles that exist in other Slavic languages and are akin to *razve* remain understudied within formal semantics. Moreover, there are many other particles with quite different semantic properties which are awaiting proper investigation. In this chapter, I have taken the first step towards a better understanding of the semantics and pragmatics of the Czech particle *náhodou.*

Abbreviations

1	first person
2	second person
ACC	accusative
AUX	auxiliary
DEM	demonstrative
DET	determiner
F	feminine
GEN	genitive
HON	honorific
INS	instrumental
M	masculine
N	neuter
NCI	negative concord item
NEG	negation
NOM	nominative
NPI	negative polarity item
PL	plural
POSS	possessive
PPI	positive polarity item
PTCL	particle
REFL	reflexive
SBJV	subjunctive
SG	singular

Acknowledgments

The research was supported by the Czech Science Foundation (GAČR) via project 21-31488J. I am very grateful to the following colleagues (ordered alphabetically) for their invaluable feedback and/or data: Katja Brankačkec, Eleni Bužarovska, Ivana Dragonová, Ilaria Frana, Ljudmila Geist, Daniel Goodhue, Dorota Klimek-Jankowska, Natasha Korotkova, Liljana Mitkovska, Sophie Repp, Catherine Rudin, Vesela Simeonova, Adrian Stegovec, Neda Todorović, Tue Trinh, Anastasiia Vyshnevska, Bartosz Wiland, Jacek Witkoś, and Anton Zimmerling. Special thanks go to the members of the project *Modelling the question–statement opposition in Slavic languages* (a GAČR–DFG collaboration), to whom I am grateful for the opportunity to learn about this fascinating topic, namely: Roland Meyer, Maria Onoeva, Anna Staňková, Mariia Razguliaeva, Kateřina Hrdinková, Michaela Chodounská, Adéla Koštejnová, Michael Andrle, and Mihaela Chirpanlieva. Finally, I would like to thank the three anonymous reviewers of this chapter, as well as Tim Morgan, for careful proofreading. Any remaining problems and errors are my own responsbility.

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Chapter 10

Lessons from Slovenian imperatives

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Slovenian is unique among Slavic languages in allowing imperatives to occur as embedded clauses. This is used as a backdrop for an examination of issues concerning imperative meaning, use, and the relationship between imperative forms and functions. The main takeaways are that Slovenian embedded imperatives are true imperatives and that clauses with subjects beyond second person and inclusive first person can be semantically and functionally equivalent to imperatives.

1 Introduction: Imperative issues

Assigning truth-conditions to sentences is one of the foundations formal semantics rests on. But a complication arises with the notions of truth-conditions and truth-values when considering sentences beyond declaratives. Among the three major clause types: declaratives, questions, and imperatives, there is a clear contrast between declaratives and the other two. While the truth of a declarative clause can be accepted or questioned, as illustrated by the exchange in (1a) for Slovenian and English (in the translations), the same kind of response is not felicitous when the initial utterance is a question, as in (1b), or an imperative, as in (1c) (see Kaufmann 2012: 4.3 on the *That's (not) true*-test applied in (1)).[1,2]

[1]Examples are glossed using Leipzig glossing rules. Unmarked number, case, gender, and tense values are omitted from glosses unless relevant; e.g. 1, 2, 3 with no number information corresponds to first, second, and third person singular. Additional conventions: referential indexes of subjects in pro-prop languages are marked on the verb/auxiliary marked for person, clitic pronouns are glossed with just their grammatical information so they can be easily distinguished from their strong counterparts (e.g. 3M.DAT = 'him.DAT'), C = complementizer without an English counterpart, Q = question particle, and PRTC = discourse-related particle.

[2]All examples are in Slovenian (or English) unless explicitly indicated otherwise. Slovenian examples presented without a specified source are based on my own native speaker intuitions and checked with at least one more native speaker of Slovenian.

Adrian Stegovec. 2026. Lessons from Slovenian imperatives. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 521–571. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078587

(1) a. A: Igral si kitaro.
played.M AUX.2 guitar.F.ACC
'You played the guitar.'
B: Res je! / Ni res!
true AUX.3 / NEG.3 true
'That's (not) true!'

b. A: Si igral kitaro?
AUX.2 played.M guitar.F.ACC
'Did you play the guitar?'
B: #Res je! / #Ni res!
true AUX.3 / NEG.3 true
'That's (not) true!'

c. A: Igraj kitaro!
play.IMP.2 guitar.F.ACC
'Play the guitar!'
B: #Res je! / #Ni res!
true AUX.3 / NEG.3 true
'That's (not) true!'

It is thus not particularly useful to study the meaning of non-declaratives in terms of truth-conditions and truth-values. A more useful approach, in the case of questions, is to instead consider the felicity of question-answer pairs, where at least answers normally have truth-conditions (see Dayal 2016: Ch. 1 for an accessible overview; also see Šimík 2026 [this volume] for the case of polar questions in Slavic). Unfortunately, there is no immediately obvious counterpart to question-answer pairs in the case of imperatives. This is one of the reasons why the semantic contribution of imperatives is sometimes argued to be outside of the realm of truth-condition semantics (see e.g. Han 2011: 1792, and Kaufmann 2012, 2020 for discussion and arguments for an alternative view).

Imperatives further stand out when it comes to the possibility of appearing in complement clauses under attitude verbs. Declaratives and questions are fairly ubiquitous as complement clauses, restricted primarily by the choice of matrix predicate.[3] This is illustrated for Italian and English (in the translations) in (2).

(2) a. Mi hanno detto che suonavi la chitarra.
1.DAT have.3PL told that played.2 the.F guitar.F
'They told me that you were playing the guitar.'

b. Mi hanno chiesto se suonavi la chitarra.
1.DAT have.3PL asked if played.2 the.F guitar.F
'They asked me if you were playing the guitar.'

(Italian; Irene Amato p.c.)

[3] To the extent that all languages allow some kind of clausal embedding (see Pullum & Scholz 2010: §5 for an overview of potential counter-examples), declarative complement clauses are always available (although they may not always be finite: e.g. in Imbabura Quechua [Quechuan; Ecuador], only non-finite clauses can be embedded; Cole 1982: 33). In contrast, questions are cross-linguistically more restricted as complement clauses: languages that allow complement clause declaratives but not questions exist (see e.g. Zu 2018: 161 on Newari [Tibeto-Burman; Nepal]), whereas languages with the reverse configuration to my knowledge do not exist.

In the case of imperatives the state of affairs is very different, as imperatives are far from ubiquitous in this context. In fact, imperative complement clauses are rare enough cross-linguistically to have been claimed in the past to be completely unattested (see e.g. Sadock & Zwicky 1985, Han 1998, 2000). The resistance of imperatives to occur as complement clauses is illustrated in (3) for English and in (4) for Italian. Note though that it is perfectly felicitous to report a previously uttered imperative by alternative means, as illustrated in the translation of (4).[4]

(3) * They ordered you that play the guitar.
(English; based on Crnič & Trinh 2009: 125 (33b))

(4) * Ti hanno ordinato che suona la chitarra.
2.DAT have.3PL ordered that play.IMP.2 the.F guitar.F
Intended: 'They ordered you to play the guitar.'
(Italian; Irene Amato p.c.)

The incompatibility of imperatives with clausal embedding is often attributed directly to their unique meaning. The idea is that, somehow, imperatives are more intrinsically tied to their characteristic discourse function than the other two major clause types, which makes imperative clauses felicitous only as unembedded utterances (see e.g. Sadock & Zwicky 1985, Han 1998, 2000).

What complicates the plausibility of this idea is that some languages actually allow imperatives in complement clauses. Slovenian is a notable case, as shown in (5) (equivalent to the ungrammatical English and Italian examples above).

(5) Ukazali so ti, da igraj kitaro.
ordered.M.PL AUX.3PL 2.DAT that play.IMP.2 guitar.F
'They ordered you to play the guitar.'

Additionally, clauses with a non-imperative verb can express the intended meaning of an imperative in (3) and (4) even in languages that ban imperatives in complement clauses. The alternative verb form is very often an infinitive or subjunctive; this is illustrated for Italian in (6a) (infinitive) and (6b) (subjunctive).[5]

[4] Although the ungrammaticality of (3) has been used to argue that English completely disallows imperatives in complement clauses, the situation is more complicated. Imperative complement clauses are actually allowed in (colloquial) English as long as the complementizer *that* is not present/expressed (Crnič & Trinh 2009, 2011); since *that* is obligatory with finite clauses under the verb *order*, an imperative is always impossible in this case.

[5] The subjunctive is often reported as the primary alternative verb form used in Italian in this context (see e.g. Han 1998: 112 (184c)). However, Irene Amato informs me that infinitives are strongly preferred and that the subjunctive (at least in her variety) requires an overt subject in this context and seems to be further conditioned by the choice of embedding predicate. There thus seems to be speaker variation regarding the use of the subjunctive here, hence the '%'.

(6) a.

Ti	hanno	ordinato	di	suonare	la	chitarra.
2.DAT	have.3PL	ordered	of	play.INF	the.F	guitar.F

b.

% Ti	hanno	ordinato	che	suoni	la	chitarra.
2.DAT	have.3PL	ordered	that	play.SBJV.2	the.F	guitar.F

'They ordered you to play the guitar.'

(Italian; Irene Amato p.c.)

The question here is whether such clauses have the same semantic and pragmatic contribution as imperatives. If they do, the ban against imperative complement clauses can be seen as a morphosyntactic restriction rather than a ban against complement clauses with "imperative meaning", whatever that turns out to be.

What makes this question difficult to answer is that, even outside of cases like (6), clauses without imperative verbs can have the discourse function of imperatives: for example, modalized declaratives like *You must play the guitar!* (Ninan 2005, Kaufmann 2012, i.a.). Pinning down the characteristic discourse function of imperatives is also not a straightforward task, since they can be used for a number of distinct speech acts beyond orders or commands, such as: warnings, invitations, and wishes, to name a few. In short, it is not obvious whether imperatives have a semantics and discourse function that clearly sets them apart from other clause types, nor is their discourse function easily definable.

The issue of imperative complement clauses and their counterparts in languages where they are prohibited can also be approached from a flipped perspective. That is, it could be that Slovenian complement clause imperatives are only apparent imperatives, which use an imperative verb form without the semantics and discourse function of true imperatives. If this is the case, then Slovenian does not pose a problem for the approach to the complement clause ban that attributes it specifically to the characteristic discourse function of imperatives.

Slovenian offers a unique opportunity to explore the relationship between imperative forms and functions, and the core meaning of imperatives. Slovenian not only allows imperative verbs in embedded clauses, it also uses a construction distinct from imperatives for subjects beyond second and first person inclusive that seems to otherwise have all the same characteristic semantic and pragmatic properties as imperatives.

In order to examine this issue, I first review the complicated relationship between imperative forms and functions in Section 2. I then establish in Section 3 the existence of embedded imperatives in Slovenian, focusing on its outlier status in Slavic and the Slovenian clauses in question being truly imperative. In Section 4, I argue that the construction used in Slovenian for imperative-like functions outside second and first person inclusive subjects should be considered a natural class together with imperatives. In Section 5, I review the main types of theories of imperative meaning by putting them to the test against the Slovenian data, and discuss some modifications that can be made to them to accommodate for Slovenian. Section 6 concludes the chapter.

2 Setting the stage: Imperative forms and functions

To properly tackle the issue of embedded imperatives in Slovenian we need to first review the complex relationship between imperative functions and imperative forms. There is a wide range of speech acts that imperatives can be used for that goes beyond the canonical ones, such as commands and directions (Section 2.1), and on the flip side the speech acts imperatives can be used for are not limited to imperatives, although some alternative constructions can only be used for those speech acts when the use of an imperative is somehow prohibited (Section 2.2). In fact, there seems to be a double dissociation between imperative forms and functions that arises both language internally and cross-linguistically – this makes it necessary to clearly establish the difference between imperative verbs, imperative clauses, and imperative speech acts (Section 2.3).

2.1 The functional diversity of imperatives

Imperatives can be used for speech acts that fit a fairly standard characterization of directive speech acts (see Searle 1976: 11 and Section 2.2), such as orders, requests, and warnings, where the speaker attempts to make the addressee act in a particular way. Slovenian imperatives are no exception, as illustrated in (7).

(7) a. Nehaj se smejati! Ukazujem ti! ORDER
stop.IMP.2 REFL laugh.INF ordering.1 2.DAT
'Stop laughing! I'm ordering you!'

b. Pomagaj mi rešiti križanko. REQUEST
help.IMP.2 1.DAT solve.INF crossword.F.ACC
'Help me solve the crossword puzzle.'

c. Ne igraj se z ognjem! WARNING
NEG play.IMP.2 REFL with fire.INST
'Don't play with fire!'

However, it is well known that imperatives can be used for a much wider range of speech acts that may not fit as straightforwardly under the umbrella of directive speech acts (see Schmerling 1982, Davies 1986, Wilson & Sperber 1988, Han 1999, Kaufmann 2020, i.a.). An example of this kind of use of imperatives is with what Condoravdi & Lauer (2012) call SPEAKER DISINTERESTED ADVICE: a speech act with which the speaker only provides information on how to achieve something without attempting to make the addressee act in a particular way. An imperative used like this in Slovenian is illustrated in the exchange in (8).

(8) A: Če bom delal potico, kako naredim, da se ne sprime?
if will.1 make.M potica.F.ACC how make.1 that REFL not stick.3
'If I make potica, how do I get it not to stick?'

B: Model namasti in potresi z moko.
model.M.ACC grease.IMP.2 and sprinkle.IMP.2 with flour.F.INST
'Grease the mold and sprinkle it with flour.' ADVICE

Another class of speech acts that imperatives can be used for which falls outside typical directive speech acts are PERMISSIONS (see Portner 2007, Kaufmann 2012, Oikonomou 2016), which can be broadly described as giving the addressee additional permissible options rather than constraining their actions. These can be divided into more specific speech acts based on how clear it is that the addressee desires to carry out the action, and how much the speaker is reluctant to allow for the addressee to carry it out. For example, with an INVITATION such as (9), there is a perceived bias against the addressee carrying out a desired action and the speaker explicitly negates this prohibition with an imperative.

(9) Vzemi še en kos torte. INVITATION
take.IMP.2PL also one.M.ACC piece.M.ACC cake.F.GEN
'Take another slice of the cake.'

Conversely, in exchanges such as (10), an example of ACQUIESCENCE (von Fintel & Iatridou 2017), the imperative conveys that the speaker will merely not object to the addressee carrying out the action if the addressee chooses to do so.

(10) A: A lahko dobim še en kos?
Q can get.1 also one.M.ACC piece.M.ACC
'Can I get another slice?'

B: Pojej celo torto, če hočeš ACQUIESCENCE
eat.IMP.2 whole.F.ACC cake.F.ACC if want.2
'Eat the whole cake, if you want.'

Lastly, in instances of CONCESSION such as (11), the speaker and addressee disagree regarding the latter's desire to carry out the action, and the imperative signals the speaker allowing it to be carried out with a high degree of reluctance.

(11) Pa pojej kar celo prekleto torto! CONCESSION
PRTC eat.IMP.2 what whole.F.ACC damn.F.ACC cake.F.ACC
'Well, eat the whole damn cake then!'

The clearest example of non-directive speech acts imperatives can be used for are WISHES (Kaufmann 2012), exemplified for Slovenian with (12). With wishes the addressee (or anyone else) may objectively not be able to make the speaker's desired state of events come true. Thus, wishes cannot be viewed as a speaker's attempt to make the addressee do something or make something happen.

(12) Hitro se pozdravi! WISH
fast REFL heal.IMP.2
'Get well soon!'

Following Condoravdi & Lauer (2012), I refer to the ability of imperatives to be used for this wide variety of speech acts – directive and beyond – as FUNCTIONAL DIVERSITY (in Schwager 2006 and Kaufmann 2012 the term used for the same concept is FUNCTIONAL INHOMOGENEITY).[6] Due to its universality (see Kaufmann 2020), I will take functional diversity – and the specific speech acts imperatives can be used for – as a key identifying trait of imperative clauses.[7]

2.2 What is a surrogate imperative form?

We saw that imperatives can be used for speech acts beyond directive speech acts. The flip side of this is that the speech acts imperatives can be used for are not limited to imperatives. Most importantly, not even directive speech acts, defined as in (13), are exclusive to imperatives. Consider (14) in relation to (15).

(13) *Directive Speech Act*
An attempt by the speaker to get the addressee to do something.
(based on Searle 1969: 72, 1976: 11)

(14) Close the window! (imperative)

(15) a. You should close the window! (modalized declarative)

[6] Imperatives can be used for additional distinct speech acts not illustrated in this section; see Kaufmann (2012: Ch. 4–5), Kaufmann (2020: §3.2), and the references above. Crucially, Slovenian imperatives can (to my knowledge) be used for all those additional speech acts as well.

[7] Exceptions do exist, but only in the sense that an imperative may not be used for a non-directive speech act when another construction must be used for that specific speech act. For example, in Greek subjunctive clauses are used for wishes instead of imperatives (Oikonomou 2016); interestingly, subjunctives are also surrogate imperatives in Greek (cf. Section 2.2).

b. Can you close the window? (question)

c. That window won't close itself. (declarative)

Like the imperative in (14), the three sentences in (15) can be used as directive speech acts where the speaker attempts to make the addressee open the window.

Note that in (14)–(15) all four options are matrix clauses that can be used as a directive speech act. This is in contrast with cases where an imperative cannot be used for seemingly morphosyntactic reasons, forcing the use of an alternative verb form or construction. A well known case is the BAN ON NEGATIVE IMPERATIVES (Zanuttini 1991, 1997, Rivero 1994, Rivero & Terzi 1995, Han 1998, 2000, i.a.), where in a number of languages an alternative verb form must be used instead of the expected imperative in negative imperative clauses.

The restriction can be even more specific. For example, in many Slavic languages including Bosnian/Croatian/Serbian (BCS hereafter), an imperative verb is prohibited with negation only in the perfective aspect.[8] This is shown in (16), where a negative imperative is possible with imperfective aspect (see (16a)), but not with perfective aspect (see (16b)). In the latter case, the construction in (16c), which Despić (2020) dubs the ANALYTIC IMPERATIVE, must be used, where imperative inflection surfaces on the negation and the verb is an infinitive.

(16) a. Ne jedi tu jabuku!
not eat.IMP.2 that.F.ACC apple.F.ACC
'Don't eat that apple!' (~ 'Don't keep eating that apple.')

b. * Ne pojedi tu jabuku!
not PFV.eat.IMP.2 that.F.ACC apple.F.ACC
Intended: 'Don't eat that apple!' (~ 'Don't eat up that apple.')

c. Nemoj pojesti tu jabuku!
not.IMP.2 PFV.eat.INF that.F.ACC apple.F.ACC
'Don't eat that apple!' (~ 'Don't eat up that apple.')
(BCS; Despić 2020: 175)

I will use the term SURROGATE IMPERATIVES (following Zanuttini 1991, 1997, Rivero 1994) for verb forms and constructions that must be used to perform any of the characteristic imperative speech acts when an imperative is unavailable. Surrogate imperatives may be required for a number of reasons, including purely paradigmatic ones. For example, in Slovenian, the paradigm of imperative verbs

[8] Different varieties of BCS do not seem to differ much with respect to the phenomena discussed in this chapter (although see footnote 11). However, for the sake of consistency, I use the names of the varieties used by the original authors when discussing their work or using their examples.

includes 1st person non-singular subjects and all 2nd person subjects, while English only has imperative verbs for 2nd person subjects. Thus, a surrogate imperative form is used in English with 1st person plural subjects: *Let's all go!*

As discussed in the introduction, one grammatical context in which surrogate imperatives are usually required is in complement clauses under attitude verbs (Zanuttini 1991, 1997, Rivero 1994, Han 1998, 2000, i.a.). This will be considered in detail below in relation to the availability of complement clause imperatives in Slovenian, as no surrogate imperative is required then, even though it is commonly assumed that such contexts are inherently incompatible with imperatives and should thus universally force the use of surrogate imperatives.

2.3 Imperative speech acts, clauses, and verb forms

It should be clear by now that the relationship between the speech acts imperatives are used for and imperatives is not a one-to-one relationship. As we saw in the previous section, clauses that are clearly not imperatives can be used for the same types of speech acts. This is even more obvious when we consider languages that lack dedicated imperative verb forms or constructions. One such example is Ngalakan [Gunwinyguan; Australia], where future tense verbs or the unmarked present verb forms must be used for speech acts where imperatives are used in other languages, as shown in (17) with a present tense verb.

(17) ŋiñ-waken ṛere-kaʔ
2SG-return.PRS camp-ALL
a. 'Go home!'
b. 'You are going home.' (Ngalakan; Aikhenvald 2010: 39)

In addition to present and future forms, verbs unmarked for tense, verbs encoding potential and intentional modalities, or irrealis verb forms can be used instead of imperatives in languages that lack them (Aikhenvald 2010: 38–44).

Even in a language that does not lack imperative verbs, other specialized constructions may be employed for speech acts otherwise canonically performed with imperatives. For example, as illustrated in (18), in Russian (see e.g. Fortuin 2001: 441–449, Aikhenvald 2010: 281) and Slovenian, a matrix infinitival clause can be used to express an order, request, or warning – the core directive functions.

(18) a. Otnesti ee von!
take.INF her.ACC out
'Take her away!' (Russian; Aikhenvald 2010: 281)

b. Otpret usta!
open.INF mouth.F.ACC
'Open your mouth!' (Slovenian)

Note that these are not surrogate imperative forms, as they are used alongside, not instead of imperatives. Although, in languages where matrix infinitives can be used for some subset of speech acts that imperatives are typically used for, they often have restrictions on their use distinct from imperatives (see Aikhenvald 2010: 281–283 for additional examples and discussion).[9]

More importantly, concerning the relationship between imperative forms and imperative functions, imperative verbs can be used for clearly non-imperative functions, beyond even those that were subsumed above under the label of functional diversity. For example, in Russian an imperative verb may be used in conditional constructions like (19) (see e.g. Jakab 2005, Aikhenvald 2010: 237).

(19) Bud' on p'janim, on pel by gromče.
be.IMP.2 he drunk.INST he sang MOD louder
'If he were drunk, he would sing more loudly.' (Russian; Jakab 2005: 301)

Interestingly, the imperative verb in these constructions is always in the 2nd person singular forms, regardless of the subject (in (19) the subject is 3rd person).

A more surprising environment where imperative verbs are used without any imperative function are so-called EXOCENTRIC VERB-NOUN COMPOUNDS (Progovac 2012). In these compounds, neither the verb not the noun are the head of the compound, but the verb very often appears in imperative form, as shown in (20) with Serbian examples (see Progovac 2012 for more examples from other Slavic, Germanic, Romance, and several non-Indo-European languages). Additionally, as the Slovenian examples in (21) show, imperative verbs in exocentric compounds are not limited to verb-noun configurations.

(20) a. cepi-dlaka
split.IMP.2-hair
'hairsplitter'

b. deri-koža
rip.IMP.2-skin
'person who rips you off' (Serbian; Progovac 2012: 51)

[9]These restrictions are not limited to types of speech acts they can be used for. For example, Pak et al. (2024) show that directive infinitives in Italian are incompatible with most indexical elements appearing appearing inside of them. Restrictions like this point towards directive infinitives not being merely imperatives in disguise.

(21) a. reci piši
say.IMP.2 write.IMP.2
'surprising (amount or fact)'

b. ne-bodi-ga-treba
not-be.IMP.2-3M.ACC-need
'a good-for-nothing' (Slovenian)

In fact, there are several more examples cross-linguistically, where imperative verbs are used for obviously non-imperative functions, very often involving modality of some kind (see Aikhenvald 2010: Ch. 7).

Given the nature of the discussion in the continuation of the chapter, it is important to acknowledge the double dissociation between imperative forms (clauses using imperative verbs or dedicated imperative constructions) and imperative functions established in this section and summarized in Table 1.

Table 1: Dissociation between imperative forms and functions

	imperative form	non-imperative form
imperative function	CANONICAL IMPERATIVES	• modalized declaratives • surrogate imperatives • present forms (Ngalakan) • directive infinitives …
non-imperative function	• conditionals (Russian) • exocentric VN compounds • various modal uses …	

To avoid confusion below, I use the term IMPERATIVE without a modifier to refer to canonical imperatives, which are imperative in both form and function. For speech acts imperatives can perform, including not strictly directive ones (see Section 2.1), I will use the term IMPERATIVE SPEECH ACTS. Finally, I use IMPERATIVE VERB/FORM and IMPERATIVE CLAUSE to refer respectively to a dedicated verb paradigm/construction used in canonical imperatives, and to a clause used for the full range of imperative speech acts. This will be particularly important in relation to the hypothesis, presented below, that a clause can be imperative without containing an imperative verb in the case of surrogate imperatives.

3 A closer look at embedded imperatives

This section is primarily concerned with the existence of imperative verbs in embedded clauses, both cross-linguistically (Section 3.1) and specifically in Slovenian (Section 3.2). The latter will be contrasted with other Slavic languages, which all prohibit imperative verbs in embedded clauses (Section 3.3). In the last part of this section (Section 3.4), I establish that Slovenian embedded imperatives are not merely superficially imperative, but pattern with canonical imperatives in terms of their characteristic semantic and pragmatic behavior.

3.1 Embedded imperatives exist

Due to their absence in the well-studied major languages of Europe, embedded imperatives have been claimed to be universally unavailable in the past (see e.g. Sadock & Zwicky 1985, Han 1998, 2000). However, what exactly should be encompassed by this ban is often not made explicit, so let us first consider a simple syntactic conception of embedding, where any imperative clause properly contained within another syntactic constituent counts as an embedded clause.

As Kaufmann (2020) points out, due to the quite general availability of conjunction and disjunction of imperatives (see (22a)), coordination is usually considered excluded from the ban on imperative embedding, and imperatives as consequents of conditionals (see (22b)) are also excluded for similar reasons.

(22) a. Call your grandmother and/or fill the birdfeeder, please.
b. If you get lost, call me. (English; Kaufmann 2020)

Available analyses of clausal coordination and conditionals leave enough room to count the imperatives in (22) as matrix clauses. For example, since coordination retains the syntactic distribution of the coordinated constituents and arguably their category, coordinated matrix clauses together still count as a matrix clause. Similarly, if-clauses can be analyzed as adjuncts to the consequent clause (see Bhatt & Pancheva 2006: 646–647), which would mean that the imperative clause in (22b) still counts as a matrix clause. Thus, to capture what I believe is the core idea behind the ban on embedded imperatives, a stricter definition of embedded imperatives is necessary: an imperative clause properly contained in another clause or a noun phrase counts as an embedded imperative.

Crucially, even under this stricter definition, which works for the cases normally considered to demonstrate the ban on embedded imperatives (e.g. (3)–(4)), counterexamples can be found when the language sample is extended to include non-European languages, dead languages, colloquial varieties, and understudied languages. Some notable examples of languages that permit imperatives to occur in a variety of embedded clauses are: Korean (Portner 2007, Pak et al. 2008), Japanese (Oshima 2006, Schwager 2006), Old Scandinavian (Rögnvaldsson 1998), Ancient Greek (Medeiros 2013), colloquial German (Schwager 2006, Kaufmann & Poschmann 2013), Mbyá [Tupi-Guarani; Argentina, Brazil, Paraguay, and Uruguay] (Thomas 2014), Chukchi [Chukotko-Kamchatkan; Russia] (Naumov 2018), Mari [Uralic; Russia] (Burukina 2024), and even colloquial English (Crnič & Trinh 2009, 2011), although the last case is somewhat controversial.

This increased awareness of the typological availability of embedded imperatives has helped establish a new consensus where, while imperative embedding is not as liberal as the embedding of other clause types, it does exists as an option in many languages (see Kaufmann 2020 for a brief overview). What is most important for present purposes, however, is that embedded imperatives can also be found right under our noses, in a well-studied Slavic language: Slovenian.

3.2 Embedded imperatives in Slovenian

In Slovenian imperatives are not limited to matrix clauses and liberally appear as embedded clauses (Sheppard & Golden 2002, Dvořák 2005, Rus 2005, Dvořák & Zimmermann 2008, Stegovec & Kaufmann 2015, Štarkl 2023). Imperative verbs can occur in Slovenian at least inside complement clauses of verbs of speech, adnominal complement clauses, argument clauses, and (restrictive) relative clauses, as long as certain semantic requirements are met (more on this below).

A basic example of an imperative complement clause is illustrated in (23). The baseline matrix imperative clause in (23a) can be reported as indirect speech under a verb of speech, as in (23b), using the *da* complementizer, which is used also with finite indicative complement clauses.[10] Aside from the presence of the complementizer and the resulting change in the position of the 2nd position clitics (in this case *mi*), the matrix and embedded imperative clauses are identical.

(23) a. Kupi mi avto.
buy.IMP.2 1.DAT car.M.ACC
'Buy me a car.'

b. Rekla ti je, da mi kupi avto.
said.F 2.DAT AUX.3 that 1.DAT buy.IMP.2 car.M.ACC
'She told you that you should buy me a car.'
(based on Sheppard & Golden 2002: 251)

[10] In (23b) and below, I use modal constructions with *should* in translations when imperatives are impossible or unnatural in English, as this is usually the most natural way to paraphrase them.

Since the most common environment in which embedded imperatives occur in Slovenian is as complement clauses under verbs of speech, one could say that these are in fact direct quotations. It is quite easy to show why this is not the case. The first issue for this analysis is that the complementizer *da* is required with embedded imperatives in Slovenian (see (23b)), which would make this the only context in which *da* can be used with a direct quotation.

Furthermore, a 3rd person pronoun inside the embedded imperative can be co-referential with the subject of the matrix clause in (24a), which should not be possible with a direct quotation, as the subject is the original speaker and thus the pronoun should be a 1st person one in a direct quotation. Similarly, a pronoun inside the imperative clause can be a variable bound by a quantifier in the matrix clause, as in (24b), which should be impossible with a direct quotation.

(24) a. Rekla ti je$_i$, da ji$_i$ kupi avto.
said.F 2.DAT AUX.3 that 3F.DAT buy.IMP.2 car.M.ACC
'She$_i$ told you that you should buy her$_i$ a car!'
(cf. She$_i$ told you: "You should buy her$_{k,*i}$/me$_i$ a car.")

b. Noben$_i$ ti ni prosil, da ga$_i$ povabi.
no one 2.DAT NEG.3 asked.M that 3M.DAT invite.IMP.2
'No one$_i$ asked you to invite him/them$_i$.'

Another way in which embedded imperatives pattern with other complement clauses is in that they allow syntactic extraction out of them, as shown in (25a) with wh-movement, and (25b) with focus movement (the focused constituent is in italics).

(25) a. Koga$_i$ sem rekel, da pokliči t_i?
who AUX.1 said.M that call.IMP.2
'Who did I say you should call?'

b. *Markota$_i$* sem rekel, da pokliči t_i.
Marko.ACC AUX.1 said.M that call.IMP.2
'It was Marko that said you should call.'
(Stegovec & Kaufmann 2015: 643)

It is also not the case that embedded imperatives under verbs of speech can only report previously uttered imperatives. An utterance like (26a) can later be reported with an imperative, as in (26b), where the addressee is 'Peter' from (26a).

(26) a. Peter bi moral poslušati.
Peter would should.M listen.INF
'Peter should listen.'

b. Rekel je, da poslušaj.
said.M AUX.3 that listen.IMP.2
'He said that you should listen.' (Stegovec & Kaufmann 2015: 643)

Lastly, embedded imperatives in Slovenian are not limited to speech reports. For example, they can occur in relative clauses, as in (27), and even in embedded questions, as in (28). Neither involve reporting a previously uttered imperative.

(27) To je film, ki si ga oglej čimprej.
this AUX.3 film.M which REFL.DAT 3M.ACC watch.IMP.2 a.s.a.p
'This is a film which you should see as soon as you can.'
(Sheppard & Golden 2002: 251)

(28) a. Boš že zvedel, ko bo čas, kam se postavi.
will.2 already find.out.M when will.3 time where REFL put.IMP.2
'You'll find out when it's time where to stand.'

b. Boš videl, ko prideš v mesto, kam zavij.
will.2 see.M when arrive.2 in town where turn.IMP.2
'You'll see when you get into town where to turn.'

What I present here is only a sample of the evidence against the direct quotation analysis; more evidence is discussed in the numerous previous works on embedded imperatives in Slovenian (see Sheppard & Golden 2002, Dvořák 2005, Rus 2005, Dvořák & Zimmermann 2008, Stegovec & Kaufmann 2015). However, what has been established so far is only that imperative verb forms can appear in embedded clauses in Slovenian. Whether or not this makes these clauses imperative in the same sense as their matrix counterparts still needs to be established. But before we get to that, it is important to situate Slovenian inside the broader Slavic picture when it comes to embedded imperatives.

3.3 The lack of embedded imperatives across Slavic

In the language most closely related to Slovenian, BCS, imperatives are excluded from complement clauses under verbs of speech, which is shown in (29b) (they are also excluded from all other embedded environments where they are possible in Slovenian).[11] Thus, in order to report the matrix imperative in (29a) as indirect speech, the verb must surface in its indicative form, as shown in (29c).[12]

[11] Some Kajkavian dialects of Croatian are an exception, as they allow imperatives in complement clauses (Galić 2019). These are also the dialects that are geographically closest to Slovenian.

[12] This does not mean that the verb is actually indicative. Serbian specifically is known to realize as indicative of a number of different constructions (see e.g. Zec 1987, Kaufmann et al. 2023).

(29) a. Kupi mi auto!
buy.IMP.2 1.DAT car.M/N.ACC
'Buy me a car!'

b. * Rekla ti je da mi kupi auto!
said.F 2.DAT AUX.3 that 1.DAT buy.IMP.2 car.M/N.ACC

c. Rekla ti je da mi kupiš auto!
said.F 2.DAT AUX.3 that 1.DAT buy.2 car.M/N.ACC
'She told you that you should buy me a car!'
(BCS; Ivana Jovović p.c.)

In another language from the South Slavic branch, Macedonian, we see the same kind of pattern arising with the imperative verb being substituted for an indicative verb the indirect speech report, as illustrated in (30).

(30) a. Kupi mi avtomobil!
buy.IMP.2 1.DAT car.M.ACC
'Buy me a car!'

b. * Ti kaźa da mi kupi avtomobil!
2.DAT said.3 C.SBJV 1.DAT buy.IMP.2 car.M.ACC

c. Ti kaźa da mi kupiš avtomobil!
2.DAT said.3 C.SBJV 1.DAT buy.2 car.M.ACC
'(S)he told you that you should buy me a car!'
(Macedonian; Metodi Efremov p.c.)

The picture across the other branches of Slavic more or less matches what we saw in BCS and Macedonian, with differences mainly arising regarding the types of verb forms and clause types used to replace imperatives in embedded environments. Czech nicely illustrates this, allowing multiple options in the embedded clause where the imperative is not possible. As seen in (31), imperatives cannot be complement clauses regardless of the choice of complementizer.

(31) a. Kup mi auto!
buy.IMP.2 1.DAT car.M.ACC
'Buy me a car!'

b. * Řekla ti, {že / ať / aby} kup mi auto.
said.F 2.DAT that C.IMP C.SBJV buy.IMP.2 1.DAT car.M.ACC
Intended: 'She told you that you should buy me a car.'
(Czech; Radek Šimík p.c.)

In order to report (31a), one must either use a *periphrastic imperative*, formed with a special type of complementizer and a present indicative verb, as in (32a), of a subjunctive complementizer and past participle verb, as in (32b).

(32) a. Řekla ti, ať mi koupíš auto.
said.F 2.DAT C.IMP 1SG.DAT buy.2 car.M.ACC

b. Řekla ti, abys mi koupil auto.
said.F 2.DAT C.SBJV.2 1.DAT buy.PTCP.M car.M.ACC
'She told you that you should buy me a car.'
(Czech; Radek Šimík p.c.)

Polish is similar in that it exhibits the ban on embedded imperatives, as shown in (33), and uses more than one strategy to report imperatives: a subjunctive complementizer with a past participle verb, as in (33c) (cf. Czech (32b)), or an infinitival complement, as in (34), although the later requires changing the matrix verb from *powiedzieć* to *kazać*, as only the latter takes infinitival complements.

(33) a. Kup mi samochód!
buy.IMP.2 1.DAT car.M.ACC
'Buy me a car!'

b. * Powiedziała ci że kup mi samochód!
said.F 2.DAT that buy.IMP.2 2.DAT car.M.ACC

c. Powiedziała ci żebyś kupił mi samochód!
said.F 2.DAT C.SBJV.2 buy.PTCP.M 2.DAT car.M.ACC
'She told you that you should buy me a car.'
(Polish; Marcin Dadan p.c.)

(34) Kazała ci kupić mi samochód!
told.F 2.DAT buy.INF 2.DAT car.M.ACC
'She told you to buy me a car.' (Polish; Marcin Dadan p.c.)

Using a subjunctive complementizer and past participle verb instead of the embedded imperative is also the strategy used in Russian, as illustrated in (35), as is also the use of an infinitival complement when the matrix predicate allows it.

(35) a. Kupi mne mašinu!
buy.IMP.2 me.DAT car.F.ACC
'Buy me a car!'

b. * Ona skazala tebe, {čto / čtoby} (ty) kupi mne mašinu!
she said.F you.DAT that C.SBJV you buy.IMP.2 me.DAT car.F.ACC

c. Ona skazala tebe, čtoby ty kupil mne mašinu!
she said.F you.DAT C.SBJV you buy.PTCP.M me.DAT car.F.ACC
'She told you that you should buy me a car.'
(Russian; p.c. Ksenia Bogomolets, Irina Burukina)

Although I have only presented examples with complement clauses under verbs of speech for reasons of space, Slovenian is unique among Slavic languages in allowing imperatives in embedded clauses. In the embedded environments where they are prohibited, surrogate imperative constructions must be employed.

Given the pervasiveness of the use of surrogate imperatives in embedded contexts across Slavic, one could entertain the possibility that Slovenian is no different. Given the double dissociation between imperative forms and functions established in Section 2.3, one could say that none of the complement clauses used to report imperatives across Slavic, including Slovenian, are not actually imperative in the semantic sense. As an anonymous reviewer suggests, it is conceivable that Slovenian has a verbal mood that is obligatory in imperative clauses, but also available in other clause types, including non-imperative embedded clauses. This would mean that Slovenian actually has no dedicated imperative forms, but also that embedded imperatives are not really imperatives in Slovenian.

In order to address this possibility, I review in the next section what are considered to be semantic and pragmatic hallmarks of imperative clauses that go beyond their characteristic discourse function, and show that embedded imperatives in Slovenian are not only imperative in their form, but also in their characteristic semantic and pragmatic contribution.

3.4 Are embedded imperatives truly imperative?

As discussed in the introduction, unlike declaratives, imperatives resist accepting or questioning their truth: i.e. the *That's (not) true*-test. While applying this test in the case of embedded imperatives is not as straightforward as with matrix imperatives, a contrast can be detected in Slovenian when an embedded imperative is compared to an alternative construction that can also be used to report previous imperative utterances in speech reports.

Consider the two complement clauses in (36a) and (36b), where the former uses an imperative verb and the latter a necessity modal and infinitive verb. Both can be used to report a previous imperative utterance, but differ in terms of what kind of responses from the addressee they are compatible with. Thus, with the embedded imperative in (36a), an addressee may only question the truth of someone uttering the imperative (e.g. with (37a)), but it feels very unnatural to question the truth of the embedded clause (e.g. with (37b)); in this case, that the addressee is truly obligated to tidy up their room. In contrast, both responses in (37) are completely felicitous in the case of the embedded modal construction in (36b).

(36) a.

Rekel	je,	da	pospravi	sobo.
said	AUX.3	that	tidy.up.IMP.2	room.F.ACC

b.

Rekel	je,	da	moraš	pospraviti	sobo.
said	AUX.3	that	must.2	tidy.up.INF	room.F.ACC

'He said that you should tidy up your room.'

(37) a.

Ni	res,	da	je	to	rekel.
NEG.3	true	that	AUX.3	this.N.ACC	said.M

'It's not true that he said that.'

b.

Ni	res,	da	moram	pospraviti	sobo.
NEG.3	true	that	must.1	tidy.up.INF	room.F.ACC

'It's not true that I should tidy up my room.'

An objection to this contrast could be that what is negated in (37b) does not exactly correspond to the meaning of the complement clause in (36a). That is a fair objection, but also related to the point I am trying to make, which is that embedded imperatives in Slovenian are more than some special modal or subjunctive construction. More importantly, it is also telling that when the response to either (36a) or (36b) is a simple 'That's not true!', the response is ambiguous in the case of (36b), but not in the case of (36a), where it seems not possible to construct a reading where only the truth of the complement clause is rejected.

Another characteristic property of imperatives is that they commit the speaker to endorsing the action expressed by them (Kaufmann 2012, Condoravdi & Lauer 2012, 2017).[13] Thus, acts of DISTANCING by the speaker are infelicitous, as shown in (38a). Note that with the modal expression in (38b), which can otherwise also be used performatively like an imperative, the distancing act is allowed.

[13]The discussion of speaker commitment is simplified here for expository purposes (for more detailed discussion, see Kaufmann 2012, 2016, Condoravdi & Lauer 2012), mostly regarding how it relates to the functional diversity of imperatives (see Section 2.1). For example, a concession signals that the speaker actually wishes for the opposite of the prejacent, and with an advice a speaker may not have any personal interest in the prejacent coming true. What is important is that speaker distancing (see below) remains infelicitous across the different speech acts, so the issue is how to correctly characterize what exactly the speaker publicly commits to when an imperative is uttered such that it precludes distancing follow ups; see Kaufmann (2012, 2016) and Condoravdi & Lauer (2012) for different approaches to the issue, as well as Section 5.3.

(38) a. Leave! #But I don't want you to leave.

b. You should leave. But I don't want you to leave.
(English; based on Kaufmann 2012: 159, 2020: 18, Stegovec 2019: 70)

The distancing ban is observed in Slovenian, as shown in (39a), although with an embedded imperative like (39b) distancing by the actual speaker is possible.

(39) a. Pojdi! #Ampak nočem da greš.
go.IMP.2 but NEG.want.1 that go.2
'Go! #But I don't want you to go.'

b. Rekel je, da pojdi, ampak nočem, da greš.
said.M AUX.3 that go.IMP.2 but NEG.want.1 that go.2
'He said you should go! But I don't want you to go.'
(based on Stegovec 2019: 59–60)

This does not mean that embedded imperatives do not exhibit the distancing ban. As shown in (40), distancing by the original speaker is not possible (Stegovec & Kaufmann 2015). Although this is not the analysis I ultimately endorse, the pattern can be viewed as the speaker endorsement component of imperative meaning behaving like a SHIFTED INDEXICAL (Schlenker 2003) in that it is anchored to the original context of the utterance rather than the actual one.[14]

(40) Rekel je, da pojdi #ampak, da noče, da greš.
said.M AUX.3 that go.IMP.2 but that NEG.want.3 that go.2
'He said you should go but that he doesn't want you to go.'
(based on Stegovec 2019: 60)

An anonymous reviewer suggests (40) merely shows that there can be no distancing by the original speaker in the original utterance, so this carries over to any speech reports of the original utterance. One complication under this view is that embedded imperatives do not have to report utterances that were originally imperatives (see (26)), and another is that there exist conditions on the use of embedded imperatives that go beyond those imposed on them in the original context, as we will see with the final property of imperatives I will consider.

[14]Interestingly, under this characterization of the speaker distancing pattern, all other indexical elements in the embedded clause are interpreted with respect to the actual context and are thus not shifted, which would suggest that the SHIFT TOGETHER principle (Anand & Nevins 2004) is violated in this case (see Stegovec & Kaufmann 2015). Depending on how important one considers the universality of shift together, this can be taken as evidence against a shifted indexical analysis of the speaker endorsement component.

The use of imperatives is crucially restricted to contexts where the action they describe is a possible action. For example, a matrix imperative like (41a) cannot be felicitously uttered in case the addressee had already made a right or left turn, thus settling the decision problem that the imperative is meant to resolve (Kaufmann 2012: 159–162). This restriction carries over to embedded imperatives, which means that (41b) feels awkward if used in a situation where a turn had already been made. In contrast, the same awkwardness does not arise in the same situation if the embedded clause is a modal+infinitive construction, as in (41c).

(41) *Scenario:* A person in the passenger seat is reminding the driver of the path that the GPS app on the passenger's phone indicated they should take.

a. Zavij levo.
   turn.IMP.2 left
   'Turn left.'

b. Zemljevid je pokazal, da zavij levo.
   map.M AUX.3 showed.M that turn.IMP.2 left

c. Zemljevid je pokazal, da moraš zaviti levo.
   map.M AUX.3 showed.M that should.2 turn.INF left
   'The map indicated that you should turn left.'

Note here, in relation to the discussion around (40), that the decision problem must be unresolved at the point when (41b) is uttered for it to be felicitous (see also Stegovec & Kaufmann 2015: 647). This means that it is not sufficient to say that all the conditions on the felicitous use of embedded imperatives are tied to the original utterance they report. Otherwise (41b) would be felicitous as a report after a turn had already been taken, as long as the map instructed the speech act participants to do so before the turn could be taken – but this is not the case.

To conclude this section, I should note that even though I only considered embedded imperatives in speech reports, imperatives in Slovenian also retain these characteristic properties in other embedded contexts, such as relative clauses (for the latter, see Kaufmann & Stegovec 2019). This leads me to conclude that embedded imperatives in Slovenian are not merely imperative in their form, but also in their characteristic semantic contribution. In the next section, I take things even further and argue that a paradigmatically restricted surrogate imperative form in Slovenian also meets the semantic and pragmatic criteria reviewed in this section in spite of not having an imperative form.

4 The status of surrogate imperatives in Slovenian

Slovenian uses a surrogate imperative construction in both matrix and embedded clauses to perform imperative speech acts with subjects that are not 2p and inclusive 1p (Section 4.1). This construction, in complementary distribution with imperative verbs, has the same characteristic restrictions on its use as canonical imperatives, which suggests that it forms a natural class of directive clauses together with imperatives (Section 4.2). This connection between the two types of clauses also reveals new insights into the phenomenon of subject obviation (Section 4.3), which in turn allows us to reevaluate the ban on imperative questions as well as the exceptions to the ban observed in Slovenian (Section 4.4).

4.1 Beyond addressee subjects

Canonically, imperative verbs are defined as dedicated verb forms used for directing the addressee to carry out the described action. As such, they are limited to 2nd person subjects and inflection (if they are inflected at all). But the role of the addressee in imperatives can be separated into two roles: (i) the individual meant to perform the action described by the verb – the subject, and (ii) the individual meant to make sure that the action described is realized – the controller (Belnap et al. 2001, Kaufmann 2020). In a canonical imperative these roles align, but the question is whether they have to: is it an inherent requirement of imperatives that addressees must be involved as subjects, controllers, or both?

Many recent analyses of imperatives assume that the addressee must at least be the controller in an imperative speech act (Portner 2004, 2007, Zanuttini 2008, Zanuttini et al. 2012, i.a.), which allows for non-2p subjects, but means the addressee must still be involved as the controller. This view is supported by the behavior of English imperatives with overt non-2p subjects, like (42a), where quantification is nonetheless understood to range over the addressees in that context. More importantly, an example like (42b), where a 2p pronoun can be bound by the quantifier, suggests the connection to the addressee is more than a pragmatic principle (see Zanuttini 2008, Zanuttini et al. 2012).

(42) a. Nobody move, this is a robbery!

b. Everyone$_i$ put your$_i$ hands up!

(English; based on Zanuttini 2008: 190–191)

The privileged role of the addressee is also key in the proposal of Speas & Tenny (2003), where imperatives are a clause type directly tied to the addressee perspective, as well as Searle's definition of directive speech acts (see Section 2.2), where the addressee is always the controller (in the current terminology). The "addressee perspective" position of Speas & Tenny has been widely criticized (see e.g. Gärtner & Steinbach 2006, Zu 2018), and I provide further arguments against it (based on observations from Stegovec 2017, 2019) in Sections 4.4 and 5.4.

In contrast, I will not dispute the claim that the addressee is at least prioritized as a controller or subject in canonical imperatives. I will merely show that some clauses with main verbs outside of the canonical imperative paradigm can be part of the same natural class as imperatives in terms of their characteristic semantic and pragmatic properties. This claim is going to be based on data from a case of paradigmatically restricted surrogate imperatives in Slovenian.

Slovenian imperative verbs are restricted to 2p subjects (imperatives in a narrow sense) and inclusive 1p subjects (or exhortatives; see Zanuttini 2008, Zanuttini et al. 2012). The imperative paradigm is thus restricted to clauses where the subject refers to an individual or group that includes the addressee. In order to perform a directive speech act where the directed individual or group does not include the addressee, an alternative construction must be used in Slovenian, which I call the *naj*-subjunctive (also called the optative construction or the analytic imperative); see Table 2 for a summary of the distribution of imperative verbs and *naj*-subjunctives in relation to the person of the subject.

Table 2: Paradigm of directives in Slovenian

Help!	singular	dual	plural
1p (excl)	**naj* pomaga-m 'I should help!'	**naj* pomaga-va 'we two should help!'	**naj* pomaga-mo 'we should help!'
1p (incl)	———	pomaga-*j*-va 'let's us two help!'	pomaga-*j*-mo 'let's help!'
2p	pomaga-*j* 'help!'	pomaga-*j*-ta 'help (you two)!'	pomaga-*j*-te 'help (you all)!'
3p	*naj* pomaga '(s)he should help!'	*naj* pomaga-ta 'them two should help!'	*naj* pomaga-jo 'they should help!'

The *naj*-subjunctive involves the uninflected particle *naj*, which gives it its name, and a main verb from the present tense paradigm inflected for person and number. Note that in matrix contexts the *naj*-subjunctive cannot be used with exclusive 1p subjects; we will see below that this is not a paradigmatic restriction.

Imperatives and *naj*-subjunctives pattern together in all aspects concerning their meaning and use, with the exception of the subjects they are used with. The first indication of this is that a *naj*-subjunctive is the most accurate way to report an imperative as indirect speech in the scenario in (43), where the original speaker becomes the addressee, and vice versa in (44), where the referent of the 3P subject in the original utterance becomes the addressee.

(43) a. Pero$_i$ ⇒ Luka$_k$: Poberi$_k$ me$_i$ ob osmih.
pick.up.IMP.2 1.ACC at eight.LOC
'Pick$_k$ me$_i$ up at eight!'

b. Luka$_k$ ⇒ Pero$_i$: Rekel si$_i$ da naj te$_i$ poberem$_k$ ob osmih.
said.M AUX.2 that SBJV 2.ACC pick.up.1 at eight.LOC
'You$_i$ said that I$_k$ should pick you$_i$ up at eight!'

(44) a. Pero$_i$ ⇒ Tone$_j$: Naj me$_i$ pobere$_k$ ob osmih.
SBJV 1.ACC pick.up.3 at eight.LOC
'He$_k$ should pick me$_i$ up at eight!'

b. Tone$_j$ ⇒ Luka$_k$: Rekel je$_i$ da ga$_i$ poberi$_k$ ob osmih.
said.M AUX.3 that 3M.ACC pick.up.IMP.2 at eight.LOC
'He$_i$ said that you$_k$ should pick him$_i$ up at eight!'

Note also that the *naj*-subjunctive in (43b) has a 1P singular subject, which means the lack of exclusive 1P directives in Table 2 is not a paradigmatic gap, but rather a contextual restriction on the subjects of directives that seems to arise in matrix environments only. I return to this restriction in Section 4.3.

Crucially, the parallelism in their usage is not limited to directive speech act. *Naj*-subjunctive can be used across the same range of speech acts subsumed under functional diversity. This is illustrated in (45) for advices, concessions, and wishes with examples matching the imperative ones from Section 2.1: (45a) corresponds to (8), (45b) corresponds to (11), and (45c) corresponds to (12).

(45) a. Naj potrese model z moko. ADVICE
SBJV sprinkle.3 model.M.ACC with flour.F.INST
'(S)he should sprinkle the mold with flour.'

b. Pa naj poje kar celo torto! CONCESSION
PRTC SBJV eat.3 what whole.F.ACC cake.F.ACC
'Well, (s)he can eat the whole cake then!'

c. Naj se hitro pozdravi! WISH
SBJV REFL fast heal.3
'May (s)he get well soon!'

Imperatives and *naj*-subjunctives can thus be used for the same speech acts even though the subjects of *naj*-subjunctives never refer to the addressee or a group containing the addressee. This is comparable to directive subjunctives in Italian and 3p imperatives in Bhojpuri [Indo-Aryan, India/Nepal] discussed by Zanuttini et al. (2012), in that the addressee is not expressed by the subject. Zanuttini et al. posit that such constructions still involve the addressee in the role of what I call the controller, and can all be roughly paraphrased as 'You see to it that X', where X is the action described by the clause in question. However, *naj*-subjunctives can easily express wishes/curses like (46a) or performative declarations like (46b), where the controller cannot be the addressee, as they would have no ability to ensure that the desired state of affairs comes true. In fact, even imperatives can be used in acts of conjuring or calling into existence like the Biblical (47a), which can also be paraphrased as *naj*-subjunctives like (47b).[15]

(46) a. Naj te strela udari!
sbjv 2.acc lightning hit.3
'I hope/wish you get hit by lightning!'

b. Naj se igre začnejo!
sbjv refl games begin.3pl
'Let the games begin!'

(47) a. Bodi svetloba!
be.imp.2 light
'Let there be light'

b. Naj bo nebo in naj bo zemlja!
sbjv will.be.3 sky and sbjv will.be.3 earth
'Let there be sky and let there be earth!'

Wishes and similar speech acts illustrated in (46)–(47) at the very least complicate the picture that the addressee must always be the controller, and Zanuttini et al. acknowledge this, leaving open the issue of what they call "true optative meanings". What is important for the argument at hand is that once again we see parallelism between imperatives and *naj*-subjunctives in this respect as well.

To sum up, *naj*-subjunctives can be used for all the same speech acts as imperatives in both matrix and embedded contexts (although only select examples were shown for space reasons). However, given the double dissociation between imperative form and and function (see Section 2.3), one could extend this to *naj*-subjunctives and suggest that they are some kind of modalized declarative (cf.

[15]The imperative is from the official translation, while the *naj*-subjunctive is used in colloquial paraphrases of the Bible, like the song *'Osmi dan'* [The Eighth Day] by the band *Pankrti.*

May you get hit by lightning!). However, as we will see in the next section, *naj*-subjunctives pattern with imperatives rather than modal expressions also when it comes to characteristic semantic and pragmatic behavior.

4.2 Speaker distancing and *naj*-subjunctives

Recall from Section 3.4 that the use of an imperative commits the speaker to endorsing the action expressed by the imperative, which prohibits any follow ups expressing a distancing from the speaker regarding the endorsement. Crucially, *naj*-subjunctives show exactly the same pattern with respect to distancing as imperatives, which is illustrated in (48): no speaker distancing is allowed in matrix contexts (see (48a)), while in embedded contexts, distancing is allowed for the actual speaker (see (48b)), but not the original speaker (see (48c)).

(48) a. Naj grejo! #Ampak nočem da grejo.
SBJV go.3PL but NEG.want.1 that go.3PL
'They should go! #But I don't want them to go.'

b. Rekel je, da naj grejo, ampak nočem, da grejo.
said.M AUX.3 that SBJV go.3L but NEG.want.1 that go.3PL
'He said they should go, but I don't want them to go.'

c. Rekel je, da naj grejo #ampak, da noče, da grejo.
said.M AUX.3 that SBJV go.3PL but that NEG.want.3 that go.3PL
'He said they should go but he doesn't want them to go.'
(based on Stegovec 2019: 60)

The speaker endorsement component of imperative meaning is thus present also with the surrogate *naj*-subjunctive forms across matrix and embedded contexts.

Furthermore, although I do not present the relevant examples here for space reasons, the restriction observed with imperatives concerning possible actions in the actual context (see (41) in Section 3.4), also holds for *naj*-subjunctives, as do all other restrictions on the use of imperatives I am aware of. For this reason, I refer to imperatives and *naj*-subjunctives together as DIRECTIVES or DIRECTIVE CLAUSES for the rest of this chapter, operating under the working hypothesis that the two types of clauses form a natural class that is obscured by superficial differences. As we will see in the following section, there is an additional restriction on the subjects of imperatives and *naj*-imperatives that can be only fully understood once the two types of clauses are considered together as a natural class.

4.3 Subject obviation

This section establishes the existence of SUBJECT OBVIATION (Bouchard 1982, Picallo 1985, Kempchinsky 1986, 2009, Farkas 1992b, i.a.) in Slovenian directives. The canonical case of subject obviation is observed with Romance subjunctive clauses and refers to the restriction where the subject of an embedded subjunctive clause cannot co-refer with the matrix subject, as shown in (49).

(49) Queremos$_i$ que {ganen$_k$ / *ganemos$_i$}.
want.1PL that win.SBJV.3PL win.SBJV.1PL
(Intended:) 'We$_i$ want them$_k$ / us$_i$ to win.' (Spanish; Quer 2006: 662)

Subject obviation is also found in several Slavic languages despite the lack of subjunctive verbal paradigms (see Sočanac 2017). Example (50) illustrates this for Serbian (see Zec 1987, Farkas 1992b, Sočanac 2017, Kaufmann et al. 2023).

(50) Ivan$_i$ je naredio da ode$_{k,*i}$.
Ivan AUX.3 ordered.M that leave.3
'Ivan$_i$ ordered him$_k$ to leave.'
(Unavailable: 'Ivan$_i$ ordered himself$_i$ to leave.') (Serbian; Sočanac 2017: 85)

Subject obviation is crucially also observed in Slovenian with *naj*-subjunctives and imperatives in complement clauses (Stegovec 2017, 2019), as shown in (51).

(51) a. Rekel je$_i$, da naj igra$_{k,*i}$ bolje.
said.M AUX.3 that SBJV play.3 better
'He$_i$ said that he$_k$ should play better.'
(Unavailable: 'He$_i$ said that he$_i$ should play better.')

b. * Rekel si$_i$, da igraj$_i$ bolje.
said.M AUX.2 that play.IMP.2 better
Intended: 'You$_i$ said that you$_i$ should play better.'
(based on Stegovec 2019: 51–52)

The lack of matrix *naj*-subjunctives with exclusive 1P subjects can now also be reevaluated from the perspective of subject obviation. As shown in (52), an embedded *naj*-subjunctive with a 1P subject is only grammatical if the matrix subject is not also 1P due to the subject obviation effect.

(52) Rekel {je$_i$ / *sem$_k$}, da naj igram$_k$ bolje.
said.M AUX.3 AUX.1 that SBJV play.1 better
(Intended:) 'He$_i$ / I$_k$ said that I$_k$ should play better.'
(based on Stegovec 2019: 52)

What we see with matrix *naj*-subjunctives can also be considered subject obviation: in (52) the subject of the *naj*-subjunctive may not refer back to the original speaker of the directive (the matrix subject), and in a matrix *naj*-subjunctive like (53), the subject also may not refer to the speaker of the directive.[16]

(53) * Naj igram(o) bolje!
SBJV play.1(PL) better
Intended: 'I/We should play better.' (based on Stegovec 2019: 55)

Note that the obviation effect cannot be merely due to the oddness of the act of self-direction. In a scenario where a speaker may direct oneself, for example by uttering an imperative when looking in a mirror, it is still infelicitous to report that with (52) or for the speaker to instead utter (53). Conversely, the same scenario can be reported felicitously with a modal+infinitive construction like (54a) and the speaker may also felicitously utter (54b) to encourage oneself.

(54) a. Rekel sem$_k$, da moram$_k$ bolje igrati.
said.M AUX.1 that must.1 better play.INF
'I$_k$ said that I$_k$ should play better.'

b. Moram bolje igrati!
must.1 better play.INF
'I should play better!' (based on Stegovec 2019: 52)

The obviation effect is thus a grammatical effect limited to specific types of matrix and complement clauses, such as subjunctives and imperatives, and we can summarize the classic and matrix versions of subject obviation as follows:

(55) SUBJECT OBVIATION
The subject of the complement clause may not co-refer with the subject of the matrix attitude verb.

(56) MATRIX SUBJECT OBVIATION
The matrix subject may not refer to the speaker.

[16]The restriction can actually be lifted in a small set of contexts, such as when the speaker shifts the responsibility of ensuring the prejacent solely to the addressee: e.g. *You have the alarm, I need you to wake me up* (see Stegovec 2019: 74n33 and also Oikonomou 2016: 167–169 on Greek), or *I better be the first one on the list tomorrow (when dissatisfied with my position on the waiting list)* (Kaufmann et al. 2023: 16). Another example is when ensuring the prejacent is not up to the speaker or the addressee (Ema Štarkl p.c.): e.g. *I better win at some point (when playing Uno).* These in fact add further support for the analysis of subject obviation presented in Section 5.4, where it comes about as a binding restriction against a syntactically present perspectival center: they all exemplify a perspective shift similar to the one observed in questions (cf. Section 4.4).

In fact, they can be unified into a single phenomenon (Stegovec 2017, 2019) given that matrix subjects and speakers are in each case the attitude holder:

(57) Generalized (Subject) Obviation
The subject may not co-refer with the attitude holder.

We will see next that even in matrix directives the attitude holder can be changed from the speaker to the addressee, which in turn changes the obviation pattern in the same way it changes in embedded directives with the change of the matrix subject. More importantly, the phenomenon that will be examined from this perspective is the unavailability of imperative questions, which means it will open up the possibility to re-examine the traditional view of core clause types.

4.4 Interrogative flip and implications for clause typing

Theories of clause types, such as Sadock & Zwicky's (1985), which see the major clause types – declaratives, questions, and imperatives – as mutually exclusive, predict that no clause can be an imperative and a question at the same time. At first glance, that prediction seems to hold not only for Slovenian, as illustrated in (58) for both polar and constituent questions, but seemingly universally.

(58) a. * Igraj(te) bolje?
play.IMP.2(PL) REFL
Intended: 'Should you (all) play better?'

b. * Kaj igraj(te)?
what play.IMP.2(PL)
Intended: 'What should you (all) play?' (based on Stegovec 2019: 73)

In Slovenian, the examples in (58a) and (58b) are identified as questions by their characteristic question intonation and, in the case of (58b), the fronted wh-phrase. Additionally, they are meant to be interpreted as seeking information from the addressee, as indicated by the suggested intended meanings.

Importantly, the absoluteness of the ban against imperative questions has been challenged, as imperatives may occur in echo or rhetorical questions (Kaufmann & Poschmann 2013), and even so-called RISING IMPERATIVES in English (Rudin 2018), which are imperatives used with rising question intonation, although they are not true information seeking questions.[17] This means that the clause type in-

[17]Rudin (2018) argues that rising imperatives lack any sort of speaker commitment (see Section 3.4). This is interesting from the perspective of what happens to speaker commitment in Slovenian imperative directives, where the commitment requirement shifts to the addressee, also responsible for the change in the subject obviation pattern (Stegovec 2017, 2019). The contexts in which subject obviation can be lifted discussed in footnote 16 are also relevant here.

compatibility may still hold for imperatives and questions, although it must be limited to true information seeking questions. As we will see, even this weaker position is still too strong, as imperatives in Slovenian may under the right circumstances occur in true information seeking questions.

To substantiate this we must first go back to *naj*-subjunctives, which cannot have exclusive 1p subjects in matrix contexts due to obviation (blocked co-reference between the subject and the speaker). Note though that unlike imperatives, *naj*-subjunctives are perfectly acceptable as questions, as shown in (59), and furthermore that as matrix questions they allow exclusive 1p subjects.

(59) a. Naj igram(o) bolje?
SBJV play.1(PL) better
'Should I/we play better?'

b. Kaj naj igram(o)?
what SBJV play.1(PL)
'What should I/we play?' (based on Stegovec 2019: 72)

This is actually expected from the perspective of generalized obviation, where a change in attitude holder predicts a change in the obviation pattern even in matrix directives. This is because with other matrix phenomena sensitive to perspective and the identity of the attitude holder, information seeking questions switch the attitude holder from speaker to addressee (Garrett 2001, Faller 2002, Speas & Tenny 2003, Zu 2018); this is commonly known as the INTERROGATIVE FLIP (Tenny 2006). This means that in the questions in (59), there is no co-reference between the 1p subjects and the attitude holder – the addressee.

If the restriction against 1p matrix *naj*-directives is indeed due to the prohibited co-reference between the speaker and the subject, this also opens up the possibility that the unacceptability of the imperative questions in (58) is not a matter of conflicting clause types, but actually a conspiracy involving generalized obviation, interrogative flip, and paradigmatic restrictions on imperative subjects. Namely, in questions the attitude holder is the addressee, and an imperative subject in Slovenian is either 2p or inclusive 1p. This means the imperative subject has to be co-referential with the attitude holder in information seeking questions, which is a configuration excluded by generalized obviation.

If this alternative view of the absence of imperative questions is correct, we expect imperative questions to be possible if somehow the attitude holder can be made to be anything but the addressee. I will show next that this indeed becomes possible in SEQUENTIAL SCOPE MARKING QUESTIONS (Dayal 2000), which are sequences of constituent questions like (60) that despite being two matrix questions pattern in many respects with long distance questions.[18]

(60) What do you think? What should we buy? (Dayal 2016: 39)

In other words, the sequence of questions in (60) is closer in meaning to the long distance question in (61a) than the superficially similar sequence in (61b).

(61) a. What do you think (that) we should buy?
 b. Where did you go? What did you buy? (Dayal 2016: 39)

This can be shown by considering the appropriate answers to the questions. The scope marking question and the long distance question both anticipate a single answer like (61a) rather than two answers, while the only felicitous response to the sequence of questions in (61b) are the two answers in (62b).

(62) a. I think we should buy cheese. (answer to (60) and (61a))
 b. I went to the store. I bought cheese. (answer to (61b))
 (based on Dayal 2016: 39–40)

Slovenian also allows for sequential scope marking questions, with interesting consequences for the availability of imperative questions. In a context like (63) a question like (63a), with wh-extraction out of an imperative complement clause, is felicitous – which already qualifies as a type of imperative question. More importantly, a scope marking question sequence like (63b) is also felicitous.

(63) *Scenario:* Paula sends me to the liquor store to buy some drinks for her party, and I go there with Marcin. By the time we get there, I've forgotten what I'm supposed to buy, so I call Paula on the phone to ask her. Marcin, who didn't hear our phone call, can then ask me:
 a. Kaj je rekla, da kupi? (long distance wh-extraction)
 what AUX.3 said.F that buy.IMP.2
 'What did she say you should buy?'
 b. Kaj je rekla? Kaj kupi? (scope marking)
 what AUX.3 said.F what buy.IMP.2
 'What did she say? What should you buy?' (Stegovec 2017: 155)

[18] Note that not all scope marking questions are like this. In some languages, like German, scope marking questions do involve syntactic embedding; see Dayal (1993, 2000) for discussion.

Crucially, the second question in (63b) is not an embedded clause. For example, like in comparable English examples, variable binding across the two questions is impossible (see (64a)),[19] unlike with long distance wh-questions (see (64b)).

(64) a. Kaj je vsak$_i$ rekel? Kaj mora$_{*i,k}$ kupiti?
what AUX.3 everyone said.M what must.3 buy.INF
'What did everyone$_i$ say? What should he$_{*i,k}$ buy?'

b. Kaj je vsak$_i$ rekel, da mora$_i$ kupiti?
what AUX.3 everyone said.M that must.3 buy.INF
'What did everyone$_i$ say he$_i$ should buy?'

The second question in (63b) must therefore be a *bona fide* information seeking matrix imperative question. If no clause could simultaneously be an information seeking question and an imperative, as expected from Sadock & Zwicky's (1985) theory, then (63b) should not be possible. But, if questions in other contexts cannot be imperatives because of the interrogative flip combined with generalized obviation, then this is exactly what we expect, as the attitude holder in the second question in (63b) is not the addressee but the subject of the first question.

This also predicts that in sequential scope marking configurations, co-reference between the subject of the first question and the subject of the second question, when the latter is a directive, should never be possible. This prediction is borne out, as we see with the *naj*-subjunctive in (65a) and the imperative in (65b).

(65) a. Kaj je$_i$ rekla? Kaj naj kupi$_{k,*i}$?
what AUX.3 said.F what SBJV buy.3
'What did she$_i$ say? What should she$_{k,*i}$ buy?'

b. * Kaj si$_i$ rekla? Kaj kupi$_i$?
what AUX.2 said.F what buy.IMP.2
'What did you$_i$ say? What should you$_i$ buy?' (Stegovec 2017: 168)

In conclusion, we again see here imperatives and *naj*-subjunctives behaving as a natural class. The generalized obviation analysis of the ban on imperative questions is contingent on this: in the baseline matrix exclusive 1P directives (*naj*-subjunctives) are disallowed and inclusive 1P and 2P directives (imperatives) are allowed, while in questions the former are allowed and the latter are disallowed.

[19] Ema Štarkl (p.c.) finds the variable-bound interpretation of (64a) possible. This means there could be variation across dialects concerning the matrix/embedded status of the second clause in scope marking configurations comparable to what is observed cross-linguistically (Dayal 1993, 2000): Ema speaks an eastern dialect (Celje), while I speak a western dialect (Nova Gorica). Since this is not a major point in this chapter, I leave exploring this possibility for future work.

More importantly, the existence of imperative questions calls into question a mutual exclusion theory of major clause types, as well as analyses such as Speas & Tenny's (2003) which directly link major clause types to either a speaker or an addressee perspective, and we saw that at least imperatives are flexible with respect to perspective (i.e. the change in attitude holder). All these generalizations will crucially have to be taken into account when considering the appropriate formal analysis of the Slovenian data in the next section.

5 Consequences for the analysis of imperatives

Before we can tackle the significance of the Slovenian data from the previous two sections for the analysis of imperatives, we must consider what types of theories of imperative semantics we have at our disposal (Section 5.1). I will then suggest that the empirical generalizations from Slovenian most naturally fit a performative modal theory of imperatives (Section 5.2), which will form the basis of the final analysis based on centered modal operators (Section 5.3) and binding restrictions induced by syntactically encoded perspectival centers (Section 5.4).

5.1 A brief overview of analyses of imperative semantics

In this section, I provide a brief overview of theories of imperative meaning, focusing on properties relevant for the phenomena discussed in this chapter. For more comprehensive overviews, the reader is directed to: Kaufmann (2012, 2020, 2024) and Charlow (2014). In my discussion, I follow the categorization of theories suggested by von Fintel & Iatridou (2017) and expanded by Kaufmann (2018), considering first what they call MINIMAL THEORIES of imperatives.

In short, minimal theories (e.g. Portner 2004, 2007, Barker 2012, von Fintel & Iatridou 2017) treat imperatives as non-modal objects and rely on imperative specific discourse principles to derive the canonical discourse functions of imperatives. The "minimal" part here refers to the simple denotational semantics imperatives receive in such theories; for example, in Portner's (2007) approach, imperatives merely denote a property restricted to the addressee, as in (66).

(66) $[\![\text{leave!}]\!] = \lambda x \,:\, x \text{ is the addressee} \,.\, x \text{ leaves}$

On its own, this minimal denotation cannot explain why imperatives, compared to delaratives, have the unique functions they do, so the trade-off is that the dynamic semantics/pragmatics associated with imperatives must be enriched. In other words, what is needed is a more elaborate theory of the CONTEXT CHANGE potential (Heim 1982, Groenendijk & Stokhof 1991, Kamp & Reyle 1993) of imperatives. Such a theory must explain what aspect of context imperatives specifically change upon their use and how they accomplish this.

Portner (2004, 2007), the proponent of perhaps the most well known minimal theory of imperatives, couches his theory in a version of the classic theory of major clause types. Each of the three major clause types has a distinct semantic type: declaratives are propositions, questions are sets of propositions, and finally imperatives are properties. The characteristic discourse functions each of them can have are then modeled in terms of three basic discourse processes of context updating: (i) declaratives update information about the world (the Common Ground; Stalnaker 1978, 2002), (ii) questions update information about the issues to be resolved (the Question Stack/Set; Ginzburg 1995a,b, Roberts 2012), and (iii) imperatives specifically update information about what to do, which is encoded in the To-Do List of each discourse participant (Portner 2004, 2007). A To-Do List is pretty much what it sounds like: a list of what the discourse participant has to do. It is modeled in Portner's theory as a discourse object, specifically: an ordering source (see Kaufmann 2020 for discussion of similar discourse objects postulated in other theories of imperatives). As an ordering source, the To-Do list is a set of propositions, in this case created by saturating the properties that imperatives denote (see (66)) with the entity associated with the To-Do List.

One of the main advantages of minimal theories is that they can easily model the functional diversity of imperatives. This is because of the underspecified denotational content of imperatives, which is by itself not limited to any particular discourse function, but which is supplemented by a richer discourse component that adjusts the canonical force of imperatives (updating To-Do Lists) to different subtypes of imperative uses. See also von Fintel & Iatridou (2017) on how a minimal theory can straightforwardly model acquiescence and indifference uses, which can be rather tricky to account for in other theories.

However, minimal theories also face a few unique challenges when it comes to Slovenian. The first challenge is imperatives in indirect speech reports. Recall that in a minimal theory imperative force comes about only post-compositionally when an imperative is uttered. But when an imperative is inside an indirect speech report, what is uttered is a declarative, so the theory has to be complicated to also handle imperative complement clauses (see Portner 2007 on applying his To-Do List theory to Korean imperatives in indirect speech reports). A bigger challenge is posed by imperative questions. Minimal theories need some way of linking imperatives and questions to appropriate discourse processes, but how does that work if a clause is both a question and an imperative? Portner (2004, 2007) specifically bases the linking procedure on the three major clause types (of distinct semantic types), and as discussed in Section 4.4, imperative questions are highly problematic for three-way major clause type theories.

Although it is entirely possible that a minimal theory can be adjusted to accommodate for Slovenian, we will see that the remaining two types of theories can achieve this with very minor adjustments. The first are STRONG THEORIES (e.g. Schwager 2006, Kaufmann 2012, Grosz 2011, Condoravdi & Lauer 2012), which seek to derive the canonical meaning and function of imperatives from a modal layer in the semantics of the imperative and general, as opposed to imperative-specific, discourse principles. The trade-off relationship between the denotational component and discourse principles is reversed here with respect to minimal theories: the denotation of imperatives is enriched with a modal component, while the discourse principles governing the effect of imperatives on the context are the same general principles that are used with declaratives. The other group is INTERMEDIATE THEORIES (e.g. Portner 1997, Roberts 2015, Oikonomou 2016), which differ from strong theories mainly in not treating imperative marking as the realization of a specific modal element, but rather assume that imperative marking is merely licensed by modal operators. This allows more flexibility in modeling the different types of modality imperatives can express.

Since the fine grained differences in the predictions made by different strong and intermediate theories do not play a role in the phenomena discussed in this chapter, I only discuss one specific strong theory in detail below. Namely, Kaufmann's (2012) PERFORMATIVE MODAL ANALYSIS which not only avoids the challenges that minimal theories have with embedded imperatives and imperative questions, but also provides a good basis for deriving the generalized obviation effect.

5.2 The performative modal analysis

The performative modal approach of Kaufmann (2012) (building on Lewis 1979a, Schwager 2006, Ninan 2005) takes as a starting point the fact that modal expressions like (67a) can be used performatively just like imperatives; cf. (67b).

(67) a. You must call me!
b. Call me! (Kaufmann 2012: 45, 91)

The proposal is that at least at the at-issue level imperatives are modal expressions (modeled in the approach of Kratzer 1981, 1991, 2012), so the imperative operator (IMP), hypothesized to scope over the prejacent and be expressed as imperative marking, is equivalent in its denotation to a necessity modal like *must*:[20]

(68) $[\![\text{must}]\!]^c = [\![\text{IMP}]\!]^c = \lambda f\,.\,\lambda g\,.\,\lambda p\,.\,\lambda w\,.\,(\forall w' \in O(f,g,w))[p(w')]$

a. f = modal base (the body of information)

b. g = ordering source (criteria for comparing worlds compliant with f)

c. where $O(f,g,w)$ is defined as the set of worlds conforming to f at w (i.e., in $\bigcap f(w)$) that are the best according to g at w.

Crucially a modal expression like (67a) can, but does not have to, be used performatively, whereas an imperative counterpart like (67b) can only be used performatively. Kaufmann (2012) argues that this asymmetry is due to a difference at the non-at-issue level: imperatives, but not plain modal expressions, trigger presuppositions that are only satisfied if the expression is used in a performative way. The presuppositions associated with imperatives are listed in (69) (based on the informal phrasing used in Kaufmann & Stegovec 2019).

(69) a. PRACTICALITY: Imperatives are used to address decision problems, specifically, the question of 'what should the addressee do?'

b. ANSWERHOOD: Imperatives have to provide an answer to such a contextually given decision problem, that is, the state of affairs named in an imperative has to single out as optimal a course of action from a set of chooseable alternatives.

c. ENDORSEMENT: Imperatives commit the speaker to the endorsement of that choice (speakers cannot use imperatives to single out actions as optimal according to other sources they might disagree with).

Practicality presupposes that at the moment of utterance a decision problem is unresolved (e.g. 'the addressee leaves' or 'the addressee doesn't leave'), while Answerhood presupposes that the imperative provides an answer to the decision problem. Together these prevent imperatives from being felicitous when a decision problem has already been resolved (see (41) in Section 3.4). Related to this, the Endorsement presupposition is responsible for the speaker commitment effect and consequently the ban on speaker distancing (see (38)–(40) in Section 3.4).

[20] Although not crucial for the analysis I adopt later, the issue of the modal force is a major point of discussion in the literature on modal analyses of imperatives. For example, Kaufmann (2012) argues that, in the absence of explicit modification, the modal force of imperative operators is always universal, Oikonomou (2016), on the other hand, argues that even unmodified imperatives have existential force, Medeiros (2013) argues that imperatives are weak necessity modals, while Grosz (2011) takes them to be variable in quantificational force.

In sum, if any of the presuppositions is not met, using the imperative results in presupposition failure and thus infelicity. The need to meet all three presuppositions is also what limits imperatives to uses where a *That's (not) true* follow-up is infelicitous, creating the impression that imperatives lack truth-values (see Kaufmann 2012: 4.3 for discussion). This contrasts with Portner's (2004, 2007) approach, where the lack of truth-values is taken at face value and is one of the reasons imperatives are taken to be properties rather than propositions.

For current purposes, what is most important is that explaining embedded uses of imperatives becomes fairly straightforward under the performative modal analysis, as it basically amounts to the embedding of a modal expression and the performative aspect can be retained in embedded contexts due to it being part of the prosuppositional component of meaning (see Kaufmann 2012: Ch. 6; 2020: 5.1; Stegovec & Kaufmann 2015 for more detailed discussion). However, what still remains unexplained in relation to Slovenian is the generalized subject obviation effect, and related to this, how this analysis extends to *naj*-subjunctives.

5.3 Centered modality and individual anchors

The analysis presented in this section is a condensed version of the proposals in Stegovec (2017, 2019). I will postpone discussion of the broader implications of the analysis for the questions raised in the chapter's introduction until Section 6, focusing here mostly on a simplified version of the key technical assumptions.

The analysis in question primarily builds on the parallelism between speaker distancing and subject obviation. Namely, the individual with which the subject of the directive may not co-refer in the case of the obviation effect is also the individual to whom the distancing ban applies to:

(70) a. *Matrix directives*
no speaker distancing, no co-reference with speaker
b. *Embedded directives*
no matrix subject distancing, no co-reference with matrix subject

One way in which this parallel can be interpreted is that the individual to whom the imperative commitments are anchored is syntactically encoded as a pronoun in directive clauses and this allows it to interact with the binding principles. Since the presence of this pronoun must somehow be tied to the characteristic semantic and pragmatic properties of imperatives, a natural way to enforce its presence is to make it a requirement of the imperative modal operator.

The gist of the idea is that the modal component of imperatives is a bit more complex than in corresponding modal expressions, and more parallel to what has been proposed for the semantics of subjunctive mood. For example, Quer (1998, 2001) suggests that the semantics of subjunctives involves a shift in the model of evaluation of the proposition, where truth is relativized to models within a context and to individuals – INDIVIDUAL ANCHORS (see also Farkas 1992a, Giannakidou 1998 for related ideas). In matrix contexts, the individual anchor is the speaker, while in embedded contexts the individual anchor is the matrix subject. The proposal is that directive semantics (unifying imperatives and *naj*-subjunctives) is the result of a directive modal operator (DIR), whose semantic type requires it to combine with an individual type element (i.e. the individual anchor). The denotation of the DIR operator is presented in (71)–(72), which you will note is only a minor modification of the denotation of IMP from (68).

(71) $[\![\text{DIR}]\!]^c = \lambda f \,.\, \lambda g \,.\, \lambda p \,.\, \lambda x \,.\, \lambda w \,.\, (\forall w' \in O(f_x, g_x, w))[p(w')]$

(72) a. CENTERED MODAL BASE:
f_x = the body of information available to x
b. CENTERED ORDERING SOURCE:
g_x = criteria for comparing worlds compliant with f_x *endorsed* by x
c. where $O(f_x, g_x, w)$ is defined as the set of worlds conforming to f_x at w that are the best according to g_x at w.

The key changes from IMP are: (i) the operator is not limited to imperatives, (ii) the modal base and ordering source are centered to an individual variable x, (iii) Endorsement is part of the ordering source. The latter capitalizes on the idea that subject obviation and the distancing ban are connected, and also the difference observed in Slovenian embedded directives between speaker distancing and the restriction to unresolved decision problems (cf. Practicality & Answerhood in (69)). Recall from Section 3.4 that while speaker commitment is always tied to the original speaker (i.e. the attitude holder), the unresolved decision problem requirement is tied to the actual context. In other words, the Practicality and Answerhood presuppositions must be met at the point of utterance even in a speech report, but Endoresement will always be tied to the original speaker.

It should be noted that the idea of centered modality is not novel (see among many others Hacquard 2006, 2010, Stephenson 2007), but the difference is that in the case of directives the individual to which the conversational modal backgrounds are centered must be syntactically realized in the same way an argument or a verb would be. Specifically, the individual variable x must be filled in via lambda abstraction, which is argued to be achieved via a silent PRO element.

5.4 Deriving generalized subject obviation

The introduction of PRO into the structure has significant consequences for binding possibilities, ultimately yielding generalized subject obviation, which puts this analysis in the realm of analyses that derive subject obviation as a syntactic binding restriction (e.g. Picallo 1985, Kempchinsky 1986, 2009, Rizzi 1990, Progovac 1993, Bianchi 2001).[21] However, this particular approach draws an explicit parallel between control infinitives and embedded directives, the PRO in a directive is controlled by the attitude holder, as illustrated in (73a), just as it is controlled in an infinitival control complement clause, illustrated in (73b).[22]

(73) a. He$_i$ said [$_{\text{CP}}$ that [PRO$_i$ [DIR [$_{\text{TP}}$ ⟨*you*$_k$⟩ [$_{\text{VP}}$ leave.IMP.2]]]]]
b. He$_i$ expected [$_{\text{CP}}$ [$_{\text{TP}}$ PRO$_i$ [to [$_{\text{VP}}$ win]]]]

As the PRO in (73a) is not an argument in the traditional sense, like PRO in (73b), and it essentially encodes the attitude holder, or the person whose perspective is being reported, this type of control configuration may be called PERSPECTIVAL CONTROL and the silent pronoun in question a PERSPECTIVAL PRO.

The result of having perspectival PRO in the structure is also that it is a potential binder for the principles of BINDING THEORY (Chomsky 1981). In this case, what is relevant is PRINCIPLE B, which in the case of control infinitives like (74a) prevents co-reference between PRO and an object pronoun within its binding domain (indicated by the wavy underline). By analogy, the perspectival PRO in the embedded imperative in (74b) induces a Principle B violation when co-referential with the subject pronoun.[23]

(74) a. * He$_i$ promised [PRO$_i$ to shave him$_i$]
b. * He$_i$ said [that PRO$_i$ DIR [⟨*you*$_i$⟩ help.IMP him]]

[21] Kaufmann (2019) also assumes the individual anchor is part of directive semantics, but not realized syntactically, deriving obviation instead as a discourse effect. Interestingly, some languages do seem to overtly realize a version of perspectival PRO; see Burukina (2023) on Mari.

[22] Although I leave out the details for space reasons, the mechanism by which the attitude holder ends up controlling PRO in Stegovec (2017, 2019) is SELF-ASCRIPTION of a property denoted by the complement clause (following Lewis 1979b, Chierchia 1987, Pearson 2012, 2016).

[23] The exact nature of the binding domain is not crucial for the account as long as the CP is comprised of two binding domains: one containing the subject and objects, and the other containing the subject and perspectival PRO. In Stegovec (2019) this is implemented in terms of Principle B being phase-bound, the two domains being the *v*P and the CP phase respectively (Chomsky 2000). The privileged status of the subject as being in both binding domains can be attributed to its base position at the edge of the *v*P phase which makes it visible from the CP phase.

The principle is similar with matrix directives, the difference is only in how PRO ends up being controlled. Without a matrix attitude verb, this takes place by way of ATTITUDE OPERATORS (see e.g. Pearson 2012), causing PRO to refer to the speaker in non-questions (via the COMMIT operator) and the addressee in questions (via the ASK operator). This results in plain directives disallowing exclusive 1P subjects (co-reference between speaker and subject pronoun), as illustrated in (75), and directives in questions disallowing 2P and inclusive 1P subjects (co-reference between addressee and subject pronoun), as illustrated in (76).

(75) a. [COMMIT(speaker$_i$) [PRO$_i$ DIR [⟨*you*$_k$⟩ leave.IMP]]] (*Pojdi!*)

b. * [COMMIT(speaker$_i$) [PRO$_i$ DIR [⟨*I*$_i$⟩ leave.1]]] (**Naj grem!*)

(76) a. [ASK(addressee$_k$) [PRO$_k$ DIR [*pro*$_i$ leave.1]] (*Naj grem?*)

b. * [ASK(addressee$_k$) [PRO$_k$ DIR [*pro*$_k$ leave.IMP.2]] (**Pojdi?*)

Recall also that directives are imperatives in Slovenian only with 2P and inclusive 1P subjects. Because of this a binding violation will always occur with imperative questions. As discussed above in Section 4.4, this makes impossible the use of imperatives in matrix questions. However, it also predicts that the ban will be voided even in questions should the ASK operator be absent, since then the perspectival PRO may also refer to entities other than the addressee. This is, in fact, what we see in sequential scope marking questions.

Because the questions in a sequential scope marking sequence are not in a syntactic subordination relation, one must adopt a INDIRECT DEPENDENCY analysis of scope marking Dayal (1993, 2000, 2016). In this analysis the two questions combine only at the level of semantics through standard functional application at the level of sets of (centered) propositions: the two question CPs combine with each other as shown in Figure 1, with CP_1 quantifying over CP_2, as in (77).

(77) $[\![CP_3]\!] = \lambda T_{\langle\langle e,\langle e,t\rangle\rangle,t\rangle} . [\![CP_1]\!] ([\![CP_2]\!])$

Without going into unnecessary details, the gist of the analysis is that Figure 1 has more in common semantically with embedded questions than with matrix ones.[24] The way CP_1 quantifies over CP_2 is analogous to how the attitude verb *say* quantifies over clausal complements in (73)–(74), which is also how the attitude holder comes to control PRO (more precisely it self-identifies as PRO; see

[24]For a more detailed discussion of this analysis see Stegovec (2017), and for an accessible overview of the indirect dependency analysis of scope marking questions in the context of a broader theory of the semantics of questions see Dayal (2016: Ch. 2).

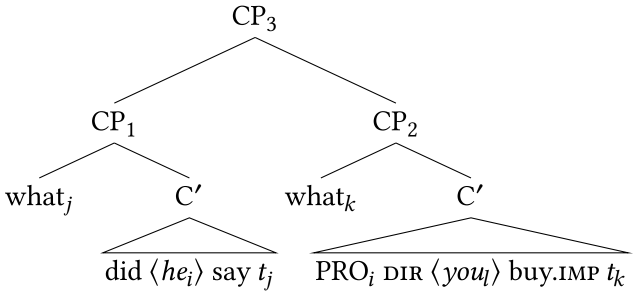

Figure 1: An INDIRECT DEPENDENCY analysis of wh-scope marking

footnote 22 and Lewis 1979b, Chierchia 1987, Pearson 2012, 2016). The result is that PRO in the second question co-refers with the subject of the first question, just as it does with the matrix subject in embedded directives. This contrasts with matrix questions, where ASK causes PRO to always refer to the addressee.

In sum, the updated performative modal analysis, though the assumption that the centered modal operator requires the presence of a perspectival PRO, can connect a number of seemingly unconnected empirical generalizations: generalized subject obviation, speaker distancing asymmetries, the absence of imperative questions, as well as the existence of imperative questions in very specific contexts. More importantly, other than the introduction of perspectival PRO, the other assumptions are all based on independently proposed analyses of control infinitives, speech act/attitude operators, and scope marking questions.

6 Conclusion, implications, and open questions

Imperatives in Slovenian can occur in unexpected contexts, such as embedded clauses and, under certain conditions, even information seeking questions. I argued that these clauses are not merely superficially imperative, but that they retain the characteristic semantic and pragmatic properties of imperatives, such as the speaker distancing ban, failure to pass the *That's (not) true*-test, and their use being restricted to contexts where they provide a solution to an unresolved decision problem. If my assessment is on the right track, this calls for a re-evaluation of some widely held beliefs about imperatives. For example, it cannot be that the characteristic meaning of imperatives is directly tied to them being limited to matrix contexts. Similarly, if imperatives are one of the main clause types, then the existence of imperative questions tells us that a theory of clause types cannot be based on the mutual exclusivity of the main clause types.

However, my discussion of embedded imperatives was mostly limited to imperatives as complement clauses under attitude verbs. Another key instance of embedded imperatives in Slovenian is imperatives in relative clauses. For a detailed discussion of the unique restrictions observed with imperative relative clauses in Slovenian, the reader is directed to Kaufmann & Stegovec (2019).

The other main claim was that *naj*-subjunctives in Slovenian behave semantically and pragmatically like imperatives. The only difference is that they are paradigmatically limited to subjects that imperatives cannot be used with: exclusive 1p and 3p subjects. Importantly, the full pattern of generalized subject obviation can only be observed once imperatives and *naj*-subjunctives are considered together. I argued based on this that they should be analyzed as involving the same modal operator as the source of their characteristic meaning.

This proposal opens up the possibility that other surrogate imperative constructions may only superficially differ from imperatives. That is, they may involve the same kind of modal operator as imperatives, but differ in their morphosyntactic realization. The first place to look for more evidence for this position would be by examining the surrogate imperatives used in complement clauses in languages where embedded imperatives are unavailable, such as the majority of other Slavic languages (see Section 3.3). In fact, this kind of position has already been independently proposed in the past by Kempchinsky (1986, 2009) for Spanish, Sočanac (2017) and Kaufmann et al. (2023) for Serbian, and Sočanac (2017) for Slavic more broadly. The prediction is that such clauses should also retain the aspects of imperative meaning that I reviewed in Section 3.4.

Of course, one can also take the exact opposite analytical path, and use the cross-linguistic rarity of embedded imperatives, coupled with subjunctives often substituting imperatives in this context, to argue that Slovenian embedded imperatives are actually underlyingly a type of subjunctive clause (as argued by Štarkl 2023). Under this type of analysis one then has to explain how subjunctives can attain the characteristic semantic and pragmatic properties of imperatives in the relevant contexts. If I am correct, then embedded imperatives may be cross-linguistically more pervasive than previously thought, they are merely disguised as surrogate forms. If the alternative analysis is correct, then many imperative clauses might actually be only wearing an imperative disguise. The hope is that more extensive cross-linguistic comparative research on the subject can help us develop a way to better tease apart these two analytical options.

Abbreviations

1	first person	LOC	locative
2	second person	M	masculine
3	third person	MOD	modal
ALL	allative	N	neuter
AUX	auxiliary	NEG	negation
ACC	accusative	PFV	perfective
C	complementizer	PL	plural
DAT	dative	PRTC	particle
DIR	directive	PTCP	participle
F	feminine	Q	question marker
GEN	genitive	PRS	present tense
IMP	imperative	REFL	reflexive
INF	infinitive	SBJV	subjunctive
INST	instrumental	SG	singular

Acknowledgments

I would like to thank three anonymous reviewers and the editors of this volume, Berit Gehrke and Radek Šimík, for their valuable comments, feedback, and patience. Additionally, I would like to thank Paula Fenger, Ema Štarkl, and Magda Kaufmann for discussion, comments, and suggestions at different stages of this project. Magda in particular has helped me greatly over the years, as a teacher, co-author, and author of many works on imperatives that keep informing my work on the subject. All remaining errors in the chapter are my own.

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Chapter 11

Event structure and derivational morphology: A few reflections on the system of Russian

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This chapter explores the event structure of simplex imperfective, lexically prefixed perfective, and secondary imperfective verbs and their extended projections. It pursues two related goals. One is to make a case, building on Wolfgang Klein's work, for a higher subevental complexity of prefixed configurations, be they perfective or secondary imperfective, compared to simplex configurations. The other goal is to investigate the typology of lexical prefixes which are responsible for a result state being part of the event structure. The basic divide is between the verbs where the denotation of a stem undergoes a non-compositional shift after prefixation, and the verbs where this does not happen. The latter, compositional lexically prefixed verbs, are further classified according to the argument structure of a prefix: Monadic prefixes take a single individual argument, the holder of a result state, while dyadic prefixes denote a relation between two entities, the figure and the ground. Among the verbs with monadic prefixes, two important classes are isolated. One contains verbs with argument-sharing prefixes, that is, prefixes whose individual argument is identical to the internal argument of the verb stem. The other contains verbs where the two entities are distinct. Finally, argument-sharing prefixes can be characterized as either restrictive or non-restrictive. The former introduce result states only compatible with a restricted subset of eventualities from the original extension of a verb stem. The latter denote states that can be brought about by every such an eventuality; they are also known under the name "pure perfectivizing" or "empty" prefixes.

Sergei Tatevosov. 2026. Event structure and derivational morphology: A few reflections on the system of Russian. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 573–653. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078589

1 Event structure and derivational morphology

The goal of this chapter is two-fold. First, it aims at providing a critical overview of the existing literature on the event structure of various types of verbal predicates in Slavic languages, with special attention to the contribution of verbal derivational morphology to the subevental make-up of an eventuality description. Second, it will focus on two specific topics in the realm of event structure. One is event-structural correlates of (lexical) prefixation and secondary imperfectivization. The other is the contribution of lexical prefixes to the argument structure and interpretation of the entire predicate.

Throughout the discussion I will consistently keep three theoretical notions apart. ARGUMENT STRUCTURE of a predicate comprises information about its syntactically projected nominal or clausal arguments and their thematic relations to the event argument. EVENT STRUCTURE defines whether an eventuality that falls under an eventuality description is simplex or complex, that is, decomposed into subevental components, what descriptive properties of each component are, and how they are related to each other. To identify the event structure of a predicate, semanticists normally use diagnostics based on the observations that some subevental components can participate in certain processes or undergo certain operations independently from others.

Both notions should be kept conceptually and representationally distinct from each other and from EVENTUALITY TYPE, a partition of predicates into non-overlapping classes like states, activities, accomplishments, and achievements, to take Vendler's (1957) or Dowty's (1979) systems as a point of departure.[1] Even though there are clear correlations between these dimensions of the meaning of a predicate, they cannot be fully reduced to each other. The same argument structure can correspond to different eventuality types. For example, transitive verbs with the agent and theme can be either telic (e.g. *open*) or atelic (e.g. *push*). The same eventuality type can be associated with different event structures: both *wipe the floor* and *break the window* are telic, yet the latter, unlike the former, is event-structurally complex (Levin & Rappaport Hovav 1995, Rappaport Hovav & Levin 1998 and much further literature).

As Levin & Rappaport Hovav and Rappaport Hovav & Levin indicate, event-structural complexity of predicates like *break the window* has several grammatical manifestations. Such predicates disallow indefinite object alternation (cf. (1a)

[1] Membership in such classes is defined by second-order properties of sentences like truth at a point, properties of predicates like the subinterval property (in temporal semantics; Bennett & Partee 1972, Taylor 1977 and subsequent literature) or cumulativity and quantization (in event semantics; Krifka 1989, 1992, 1998).

and (1b)), are unavailable in configurations with non-subcategorized objects (cf. (2a) and (2b)) and do not readily accept expressions that specify the result state of an eventuality (cf. (3a) and (3b)) (examples from Rappaport Hovav & Levin 1998: 102–103).

(1) a. * John broke.
    b. Terry wiped.

(2) a. * The clumsy child broke the beauty out of the vase.
    b. The child rubbed the tiredness out of his eyes.

(3) a. * Kelly broke the dishes off the table.
    b. Kelly swept the leaves off the sidewalk.

Putting technical details aside, Levin & Rappaport Hovav and Rappaport Hovav & Levin attribute the difference between result predicates like *break* and manner predicates like *wipe* to the fact that the semantic representation ("lexical conceptual structure") of the former but not of the latter contains a result state.

The eventuality type of a predicate, on the other hand, does not reduce to event structure. A predicate does not have to have a result state to be telic. *Wipe the table* can be telic without there being a result state, either contributed by the verb or a resultative expression. I will take up issues surrounding event structural complexity in due order.

The subsequent discussion is based on Russian material. Obviously, cross-linguistic variation within the Slavic family is substantial, both in terms of the structure and semantic characteristics of the verbal system. Nevertheless, I hope that the most basic generalizations I will try to establish in what follows for Russian would mostly extend to other Slavic languages, with relevant qualifications and adjustments.[2]

These minimal clarifications having been made, in the next section I will offer an overview of the verbal derivational system of Russian. Relying on this background, in Section 3–Section 4 I will address two central topics of this chapter. Section 3 presents the set of data that is best accounted for if one assumes that (verb phrases projected by) lexically prefixed perfective verbs and secondary imperfective verbs contain a result state in their semantic representation, but their simplex imperfective counterparts do not. In Section 4, a more detailed typology of lexical prefixes is set up: the entire set of lexical prefixes is divided into subclasses defined by a number of criteria, including their compositionality, argument structure and type of interaction with a verb stem.

[2] See Gehrke (2026 [this volume]) for extensive discussion of the cross-Slavic variation in the aspectual domain.

2 Derivational morphology

This section focuses on two topics. First, it aims at surveying basic phenomena in the domain of derivational morphology of Russian. Second, it provides an overview of the existing literature where these phenomena are addressed from various perspectives. I will start with an outline of the Russian verbal system, which can be skipped by the reader familiar with this material. See also a recent survey in Borik (2024) for a detailed and comprehensive overview of the topic.

2.1 Verb structure and interpretation

A finite morphological verb in Russian consists of the three elements in (4).

(4) Verb stem – (thematic element) – inflection

A few examples are given in (5).

(5) a. del-aj-eš
do-TH-NPST.2SG
'you are doing$^{\text{IPFV}}$ / you do$^{\text{IPFV}}$'

b. pis-a-l
write-TH-PST.M
'I (masc.) / you (masc.) / he was writing$^{\text{IPFV}}$ / would write$^{\text{IPFV}}$'

c. ris-ova-l-i
paint-TH-PST-PL
'they were painting$^{\text{IPFV}}$ / would paint$^{\text{IPFV}}$'

d. sox-n-et
dry-TH-NPST.3SG
'it is drying$^{\text{IPFV}}$ / dries$^{\text{IPFV}}$'

e. ljub-i-l-a
like-TH-PST-F
'I (fem.) / you (fem.) / she liked$^{\text{IPFV}}$'

As (5b), (5c), and (5e) indicate, in the past tense, which is diachronically a participle, the verb is inflected for number and gender; in the non-past tense, for person and number, as in (5a) and (5d). Following Jakobson (1948), many morphologists (Halle 1963, Lightner 1965 and much subsequent literature) decompose person-number morphemes into a tense morpheme and person-number morpheme proper. Under such an analysis *-eš* of *delaeš* in (5a), for example, would be treated as involving *-e-* of the non-past/present tense and *-š* of the 2nd person singular.

Thematic elements define morphological classes of Russian verbs; similar classes can be found across other Slavic languages as well. Thus, *del-a-* 'do, make' in (5a) and *ris-ova-* 'draw, paint' in (5c) fall under different classes, since they combine with different thematic elements, *-a-* and *-ova-*.[3] Thematic elements are part of the structure of the vast majority of verbs; a very limited number of "athematic" verbs merge inflectional morphology directly on top of the root, cf. *nes-l-a* 'she carried'. For ease of exposition, in the remainder of this chapter thematic elements will be treated as part of the stem in the example sentences and morpheme-by-morpheme interlinear glosses.

At least some of the thematic elements can be thought of as the realizations of grammatical morphemes that derive a verbal configuration from category-neutral roots in the style of Marantz (1997), as in (6), assuming that the verbalizing element is *v* that takes (the projection of) a root as its complement; see Jabłońska (2007), Biskup (2019), Mišmaš & Simonović (2022), Milosavljević & Arsenijević (2022), among others, for further discussion of the patterns attested in a number of Slavic languages.[4]

(6) THEMATIC ELEMENTS AS VERBALIZERS
$[_{v\mathrm{P}}$ … *i* … $[_{\surd\mathrm{P}}$ … *ljub-* …]]

The verbs in (5) contain a thematic element but no further instances of derivational morphology. In what follows I will call such verbs DERIVATIONALLY MINIMAL, or SIMPLEX. In most Slavic languages, Russian among them, derivationally minimal verbs are subject to the following descriptive generalization that relates a verb and an aspectual configuration it can occur in:

(7) SIMPLEX VERBS
Most derivationally minimal verbs can only occur in grammatically imper-

[3] In the past and non-past subparadigms, thematic elements normally surface in different forms, cf. *del*-[aj]-*eš* and *del*-[a]-*l*, *ris*-[uj]-*eš* and *ris*-[ova]-*l*. There is a tradition going back to Jakobson (1948), which assumes a derivational account of this variation, whereby the surface allomorphs are derived from the same underlying representation (e.g. Halle 1963, Lightner 1965 and subsequent work, Halle & Matushansky 2004, Itkin 2007). Other morphologists tend to treat "stem allomorphs" as lexically stored and account for their distribution across subparadigms by invoking an independent mechanism (e.g. Brown 1998, Rubach & Booij 2001).

[4] Matushansky (2009, 2021, 2024), relying mostly on evidence from the morphology of secondary imperfectivization, offers compelling arguments that "thematic elements" are not structurally homogeneous: only a proper subset of them can be analyzed as verbalizers; other "thematic elements" attach higher up in the structure.

fective clauses (“simplex imperfective verbs”). A limited number of derivationally minimal verbs can only occur in perfective clauses (“simplex perfective verbs”).[5]

(8) offers an example of the simplex imperfective verb *pisa-t’* ‘write’:

(8) Kogda Nadja voš-l-a, Volodja pisa-l pis’mo.
when Nadja walk.in-PST-F Volodja write-PST.M letter.ACC

a. ‘When Nadja walked in, Volodja was writing a letter.’
$\tau(\text{WALK IN}) \subseteq_T \tau(\text{WRITE})$

b. Unavailable: ‘When Nadja walked in, Volodja wrote a letter / had already written a letter.’
$\tau(\text{WALK IN}) <_T \tau(\text{WRITE}); \tau(\text{WALK IN}) >_T \tau(\text{WRITE})$

The sentence in (8) comes out imperfective. It warrants the episodic interpretation in (8a), where the topic time specified by a temporal clause, the interval in which Nadja’s walking in occurs, is included into the time of writing a letter.[6] This is the progressive construal. The perfective interpretation, marked by the non-overlap of walking in and (culminated) writing of a letter in (8b), is unavailable.

The sentence in (9) with the simplex perfective verb *reši-t’* ‘solve’ shows the opposite range of interpretations:

(9) Kogda Nadja voš-l-a, Volodja reši-l zadač-u.
when Nadja walk.in-PST-F Volodja solve-PST.M quiz-ACC

a. ‘When Nadja walked in, Volodja solved a quiz / had already solved a quiz.’
$\tau(\text{WALK IN}) <_T \tau(\text{SOLVE}); {}^{?}\tau(\text{WALK IN}) >_T \tau(\text{SOLVE});$
${}^{?}\tau(\text{WALK IN}) =_T \tau(\text{SOLVE})$

b. Unavailable: ‘When Nadja walked in, Volodja was solving a quiz.’
$\tau(\text{WALK IN}) \subseteq_T \tau(\text{SOLVE})$

[5] A number of verbs, labeled “bi-aspectual verbs”, are not subject to (7) and are available in both perfective and imperfective configurations. Such verbs, e.g. *issledova-t’* ‘study, explore’ or *atakova-t’* ‘attack’ are typically thought of as lexically ambiguous; I know of no account that can derive “bi-aspectuality” from other characteristics of a verb. For the purposes of this chapter I put bi-aspectual verbs aside. For a comprehensive overview of existing theories of Russian aspect see Mueller-Reichau (2026 [this volume]).

[6] In the quasi-formal notation in (8)–(9), τ is the temporal trace function, $<_T$ and $\subseteq_T$ are the relations of temporal precedence and temporal inclusion, respectively; $\tau(\text{WALK IN}) <_T \tau(\text{SOLVE})$ thus reads as “the time of walking in temporally precedes the time of solving”.

The preferred interpretation of (9) is the one where a complete culminated solving of the quiz follows Nadja's walking in. An alternative interpretation, less easily available in the null context, is the past-over-past reading where the temporal sequencing is reverse. Yet one more option is a complete temporal overlap of walking in and solving a quiz. These readings are shown in (9a). Crucially, (9) cannot have the progressive construal in (9b).

The characterizing property of the aspectual system of most Slavic languages is that all inflected forms of the verbs like *pisa-t'* 'write' must occur in imperfective clauses while the forms of *reši-t'* are restricted to perfective clauses. This ASPECTUAL INVARIANCE has lead aspectologists to the suggestion, standardly accepted since the 19th century, that grammatical aspect is a lexically specified property of the verb in Slavic, *pisa-t'* 'write' being lexically imperfective, *reši-t'* lexically perfective. Tatevosov (2011, 2015b, 2021) argues that this view is fundamentally misguided, and that "aspectual morphology" and "aspectual interpretation" are located at a substantial structural distance, as shown in (10).

(10) VERB PHRASES ARE ASPECTLESS

   a. […. IPFV … [$_{\mathrm{VP}}$ … *pisa-* …]]
   b. […. PFV … [$_{\mathrm{VP}}$ … *reši-* …]]

Generalizing over (10a)–(10b), I will be assuming that at the point of derivation where the verb, either derivationally minimal or not, is fully assembled from the pieces of morphology, no semantic aspect is part of its meaning. This assumption is made explicit in Section 3–Section 4, where all VPs/*v*Ps are treated as aspectless predicates of events.[7]

However, in order to minimize introducing new terminology I will continue to follow the tradition in using *imperfective verbs* and *perfective verbs* as descriptive terms. The reader should keep in mind, however, that *an (im)perfective verb* should be taken as a shorthand for *a verb that can only occur in an (im)perfective configuration if it undergoes no further morphological derivation.*

Simplex verbs can be combined with various pieces of derivational morphology, characterized below.

2.2 The ingredients

Russian, as well as other Slavic languages, has an extensive inventory of verbal prefixes, listed in (11). (11), according to Švedova (1980: §850), forms an exhaustive list:

[7]For an apporoach along similar lines see Mueller-Reichau (2026 [this volume]).

(11) Verbal prefixes in Russian
v(o)-, *v(o)z(o)-*, *vy-*, *de(z)-*, *dis-*, *do-*, *za-*, *iz(o)-*, *na-*, *nad(o)-*, *nedo-*, *niz(o)-*, *o-*, *ob(o)-*, *ot(o)-*, *pere-*, *po-*, *pod(o)-*, *pre-*, *pred(o)-*, *pri-*, *pro-*, *raz(o)-*, *re-*, *s(o)-*, *u-*

Suffixal derivational morphology comes in two pieces: the so-called secondary imperfective morpheme -*yva-* and the semelfactive morpheme -*nu-*. The latter is illustrated in (12).

(12) Semelfactives

a. tolk-a-t'
push-th-inf
'push, give pushes' (ipfv)

b. tolk-nu-t'
push-smfct-inf
'give a push' (pfv)

c. tolk-a-nu-t'
push-th-smfct-inf
'give a push' (pfv)

Morphologically, the semelfactive allows for two options: -*nu-* either merges directly with the root, as in (12b), or attaches on top of the thematic element, as in (12c). In both cases, semantically, a derived predicate denotes atomic pushing eventualities, whereas the non-derived 'push' in (12a) has both atoms and their sums in its extension.

In terms of grammatical aspect, the semelfactive is subject to the following generalization:[8]

(13) Aspect of the semelfactive

a. The semelfactive affix requires its complement to be imperfective.

b. The semelfactive is perfective.

[8]The semelfactive morpheme -*nu-* should be distinguished from the thematic element -*nu-* which is part of the derivation of a number of denominal verbs, e.g. *soxnu-t'* 'dry'. Semantically, the latter are not subject to (13) and fall under the simplex imperfective class. Morphologically, the difference is manifested in the past tense, where the thematic element -*nu-*, unlike the semelfactive -*nu-*, has no surface realization, cf. *sox-(*nu)-l-a* 'she dried' and *tolk-*(nu)-l-a* 'she gave a push'.

According to (13a), the semelfactive must attach to verb stems like *tolka-* 'push' in (12a), which come out imperfective if they stay derivationally minimal. According to (13b), the *nu*-verbs themselves can only occur in perfective configurations. This is what happens in (12b)–(12c).

The semelfactive, restrictions on its derivation, differences between semelfactives like (12b) and (12c) remain one of the least explored topics in Slavic aspectology. For more information about the semelfactive the reader can refer to Markman (2008), Makarova & Janda (2009), Gribanova (2013), Taraldsen Medová & Wiland (2019), Matushansky (2024), and literature therein.

In the rest of this section I will mostly focus on prefixation and secondary imperfectivization. Prefixes in Slavic languages are frequently described as "perfecitivizing". Such a description should not be taken too literally, however. The output of prefixation must be a perfective verb, as captured by the generalization in (14).

(14) Aspect of prefixed verbs
Prefixed verbs that undergo no further morphological derivation are perfective.

However, although the output of prefixation is always perfective, the input to prefixation does not have to be imperfective. (15a)–(15d) are illustrations of a prefix being merged with a perfective verb stem:[9]

(15) Prefixes in combination with a perfective verb

a. [ot- [reši]^{PFV}]^{PFV} -t'
   LP solve INF
   'dismiss^{PFV}'

b. [pod- [kupi]^{PFV}]^{PFV} -t'
   LP buy INF
   'bribe^{PFV}'

c. [pere- [za- pisa]^{PFV}]^{PFV} -t'
   SLP LP write INF
   'record^{PFV} again'

d. [pod- [za- by]^{PFV}]^{PFV} -t'
   SLP LP be INF
   'forget^{PFV} partially'

[9]Throughout the chapter LP and SLP stand for "lexical prefixes" and "superlexical prefixes", discussed in more detail in Section 2.3 and subsequent sections.

In (15a)–(15b), the prefixes *ot-* and *pod-* combine with simplex perfective verbs *reši-t'* 'solve' and *kupi-t'* 'buy'. In (15c)–(15d), *pere-* and *pod-* are affixed to *za-pisa-t'* 'record' and *za-by-t'* 'forget', which are perfective by virtue of being prefixed and not further suffixed. Therefore, in (15a)–(15d) no "perfectivization" occurs, since the input to prefixation is perfective already.

(15c)–(15d) independently illustrate another prominent property of the Slavic verbal derivational morphology: MULTIPLE PREFIXATION. While (15c)–(15d) contain two prefixes each, (16a) is an example of a verb stem with triple prefixation. (16b) shows its derivation in a step-wise manner.

(16) TRIPLE PREFIXATION

a. do-pere-za-pisa-t'
   SLP-SLP-LP-write-INF
   'complete recording again'

b. *pisa-* 'write'
   → *za-pisa-* 'record'
   → *pere-za-pisa-* 'record again'
   → *do-pere-za-pisa-* 'complete recording again'

Constraints on multiple prefixation in Russian have been explored in Tatevosov (2009, 2013), Mueller-Reichau (2021), among others.

The secondary imperfective morpheme *-yva-* outputs an imperfective verb:[10]

(17) ASPECT OF SECONDARY IMPERFECTIVE VERBS
yva-verbs that undergo no further morphological derivation are imperfective.

The examples in (18a)–(18d) show the imperfective counterparts of the simplex perfective verb *reši-t'* 'solve', prefixed perfective verb *pro-čita-t'* 'read', and perfective verbs from (15b) and (15d). Note that the secondary imperfective morpheme has three surface realizations: *-va-* after vowel-final stems, but *-yva-* or stressed *-á-* after consonant final stems. Throughout the chapter the label "YVA"

[10]The final segment of *-yva-* can be identified with the thematic element *-a(j)-* found in a number of non-derived stems, e.g. *dela-t'* 'do, make', see, specifically, Isačenko (1960): 176–181. For better readability, this information is left implicit in the morpheme-by-morpheme interlinear translation.

is used to cover all of them in the morpheme-by-morpheme interlinear translation.[11]

(18) The -*yva*- morpheme in combination with perfective verbs

a. [[reš]^{PFV} -a]^{IPFV} -t’
   solve yva inf
   ‘solve^{IPFV}’

b. [[pro- čit]^{PFV} -yva]^{IPFV} -t’
   lp read yva inf
   ‘read^{IPFV}’

c. [pod- kup]^{PFV} -a]^{IPFV} -t’
   lp buy yva inf
   ‘bribe^{IPFV}’

d. [[pod- za- by]^{PFV} -va]^{IPFV} -t’
   slp lp be yva inf
   ‘forget^{IPFV} partially’

One of the issues in the derivational morphology of Slavic that does not seem to have been completely settled comes from the verbs like (19a)–(19c) derived by the affixation of -*yva*- to an imperfective stem:

(19) The -*yva*- morpheme in combination with imperfective verbs

a. [[siž]^{IPFV} -iva]^{IPFV} -t’
   sit yva inf
   ‘sit occasionally’

b. [[igr]^{IPFV} -yva]^{IPFV} -t’
   play yva inf
   ‘play occasionally’

c. [[pis]^{IPFV} -yva]^{IPFV} -t’
   write yva inf
   ‘write occasionally’

[11] Matushansky (2009) offers a theory that derives all the three allomorphs from the same underlying representation, /ʊ/, “the third yer”, followed by the thematic element. In verb stems like *(pod)-kup-á-t’* in (18c), /ʊ/ gets deleted, and its presence in the derivation can only be detected by the obligatory stress on the following thematic element. An extension of this theory is offered in Matushansky 2021; an alternative phonological theory has been developed by Enguehard (2017).

Filip & Carlson (1997) discuss Czech counterparts of (19a)–(19c) and argue for a theory that assumes two distinct *yva*'s: the imperfective one, combining with perfective stems, as in (18a)–(18d), and the generic one, which only attaches to the imperfective stems, as in (19a)–(19c).

Given the distribution of the two *yva*'s across aspectual stems ([…]^{PFV} + "imperfective *yva*" vs. […]^{IPFV} + "generic *yva*") and the fact that one of the readings of the "imperfective *yva*" is the habitual / generic one, it looks desirable to reduce them to the same semantics. At the moment, however, no compelling semantic theory unifying uses like (18) and (19) has been proposed.

2.3 Lexical and superlexical prefixes: an overview

In (11), 26 phonological strings that appear in the left periphery of a verb stem are listed, some with phonologically conditioned variants like *iz-* and *izo-*. The actual number of prefixal lexical items may be considerably higher, since many elements from (11) come in different varieties, normally identified as "meanings" / "uses" of a prefix. One example is the prefix *pere-* in (20) (see also Kagan 2015: 118 ff. and literature therein):

(20) a. Distributive *pere-*
pere-lovi-t' vsex vrag-ov
slp-catch-inf all.acc enemy-acc.pl
'catch all the enemies one after another'

b. Repetitive *pere-*
pere-pisa-t' stat-ju
slp-write-inf article-acc
're-write the article'

c. *Pere-* of excess
pere-jes-t'
slp-eat-inf
'overeat'

d. *Pere-* of comparison
pere-ži-t' dedušk-u
slp-live-inf grandfather-acc
'outlive the grandfather'

(20a)–(20d) show the readings of *pere-* most readily accessible in the null context for a particular combination of *pere-* and a non-derived stem: distributive *pere-*, (20a), repetitive *pere-*, (20b), *pere-* of excess, (20c), and *pere-* of comparison, (20d).

Different readings are not complementarily distributed over lexical verbs. One can easily find examples where the same combination allows for several interpretations:

(21) pere-čita-t'
slp-read-inf

a. 'read one by one' (distributive)
b. 're-read' (repetitive)
c. 'read too much' (excess)

In a similar fashion, the grammar of Russian (Švedova 1980) lists 5 meanings for *iz-*, 8 meanings for *ot-*, 9 meanings for *pod-*, 10 meanings for *do-*, and so on.

The question about examples like (20)–(21) is whether different "meanings" can be reduced to the same underlying semantic representation. There have been many attempts at a unification within different approaches and frameworks, which target some or all of the meanings of different prefixes. One of the most recent studies is Kagan (2015), who offered a degree semantic account for 19 prefixes.

For obvious reasons, such a unification is highly desirable. If every submeaning of a prefix needs to be listed, the amount of distinct prefixal lexical items starts exceeding several hundred, which makes the acquisition task for a learner of a Slavic language too complicated. However, one can be somewhat skeptical as to whether existing accounts can offer a comprehensive, complete, and fully explicit semantic theory of prefixation where every prefix in (11) is assigned a general underlying meaning that gives rise to the attested surface diversity.[12]

Moreover, there are serious empirical reasons to believe that at least some of the meanings of the prefixes in (11) are irreducible to others. Specifically, there is robust evidence that prefixes show radically different morphosyntactic distribution and interpretational properties under different meanings.

Starting at least from Babko-Malaya (1999), it has been recognized that Slavic prefixes can be divided into two types normally referred to as lexical, or internal, and superlexical, or external prefixes; see e.g. Babko-Malaya (1999), Ramchand (2004), Romanova (2004, 2007), Svenonius (2004, 2008), Di Sciullo & Slabakova (2005), Gehrke (2008), Tatevosov (2008, 2009, 2013), Žaucer (2009), Gribanova (2013, 2015), Tolskaya (2015), Biskup (2019) on Russian; Progovac (2002), Milićević

[12]Remarkably, Kagan's (2015:123) account for *pere-* does not cover the repetitive *pere-* in (20b). There is no obvious way of subsuming its interpretation under the general format she proposes, in which *pere-* indicates that the two relevant interval degrees stay in the inclusion relation.

(2004), Arsenijević (2006, 2007a,b, 2012) on Serbo-Croatian; Istratkova (2004), Di Sciullo & Slabakova (2005), Slabakova (1997, 2005), Markova (2011) on Bulgarian; Gehrke (2008), Biskup (2009, 2012, 2015, 2019) on Czech; and Žaucer (2009, 2010, 2012, 2013) on Slovenian.[13]

The distribution of prefixal items in (11) across these two classes can be captured by the following generalization:

(22) Lexical prefixes (LPs) and superlexical prefixes (SLPs)
Every phonological string from (11) can be a lexical prefix. Some of the phonological strings in (11) can also be superlexical prefixes.

According to (22), many prefixes have both lexical and superlexical manifestations. Different authors (Babko-Malaya 1999, Svenonius 2004, Romanova 2007, Gehrke 2008, Tolskaya 2007, 2015, Tatevosov 2009, 2013, Biskup 2019) identify overlapping but not identical inventories of SLPs. The most inclusive list is shown in (23).

(23) Superlexical prefixes
do- (completive), *za-* (inceptive), *iz(o)-* (totality), *na-* (cumulative) *ot-* (terminative), *pere-* (distributive), *pere-* (repetitive), *pere-* (excessive), *po-* (delimitative), *po-* (distributive), *pod(o)-* (attenuative), *pri-* (attenuative), *pro-* (perdurative), *raz(o)-* (evolutive)

As (22)–(23) suggest, a prefix can only be characterized as lexical or superlexical under a particular construal. For instance, *po-* is superlexical as a delimitative or distributive prefix, but lexical elsewhere. Similarly, while the varieties of *pere*'s in (20)–(21) are all superlexical, one can easily find instances of the lexical *pere-*. It is for this reason that completely unifying the prefixal meanings and treating every prefix as a single lexical item may ultimately prove impossible. See Section 4.3 for further discussion.

2.4 Lexical and superlexical prefixes: a few diagnostics

LPs have a number of characterizing properties that distinguish them from SLPs. Simplifying somewhat, one can group these properties into three clusters: semantics, argument structure, morphosyntax.

[13]Babko-Malaya's (1999) observations about Russian parallel in many significant respects Di Sciullo's (1997) work on external and internal prefixation in Romance, specifically, in French and Italian.

On the morphosyntactic side, the key observation is that in case of multiple prefixation, SLPs occur outside of LPs, never inside. A few illustrations appear in (24), where bracketing indicates morphological constituency.

(24) SLP OUTSIDE LP

a. [na- [[za- bi] -va]] -t' *za-na-bi-va-t'
SLP LP hit YVA INF
'hammer a lot of (e.g., nails)'

b. [po- [[o- pis] -yva]] -t' *o-po-pis-yva-t'
SLP LP write YVA INF
'describe for a while'

c. [za- [[vz- dyx] -a]] -t' *v(o)z-za-dyx-a-t'
SLP LP breath YVA INF
'start sighing'

d. [po- [[vy -bras] -yva]] -t' *vy-po-bras-yva-t'
SLP LP throw YVA INF
'throw out one by one'

e. [do- [so- bra]] -t' *so-do-bra-t'
SLP LP take INF
'finish collecting / assembling'

f. [pere- [za- pusti]] -t' *za-pere-pusti-t'
SLP LP let INF
're-start'

(24a)–(24f) exemplify the cumulative *na-*, delimitative *po-*, inceptive *za-*, distributive *po-*, completive *do-* and repetitive *pere-*. All of them merge with a stem which a lexical prefix is part of. The right column in (24) represents attempts to merge the prefixes in the reverse order, which invariably produce an ungrammatical output.

Semantically, as Babko-Malaya (1999: 76) points out, SLPs tend to have "systematic and predicable meanings"; LPs, in contrast, are associated with "idiosyncratic meanings". Svenonius (2004: 230) indicates that SLPs contribute "temporal or quantizing" meanings, whereas the contribution of LPs is normally spatial or resultative meanings. Similar descriptions can be found in other work on prefixation.[14]

[14] To do justice to the earlier work on Slavic verbs, it should be pointed out that similar observations had started circulating in the literature long before Babko-Malaya's work; see e.g. the discussion of prefixal "qualifiers" and "modifiers" in Isačenko (1960): 222–224.

Another set of observations that tell LPs and SLPs apart concern the interaction of prefixation and argument structure of a non-derived stem. SLPs do not add an argument to the root verb or affect thematic properties of an object (Romanova 2004, 2007; Ramchand 2004), do not license unselected objects (Babko-Malaya 1999, Svenonius 2004, Romanova 2007; but see Žaucer 2009, 2010, 2012, and have no consequences for the indefinite object alternation (Babko-Malaya 1999, Svenonius 2004). LPs can have the opposite properties.[15]

(25) represents a typical case where a lexically prefixed unergative verb acquires the internal argument position, (25b), but with a superlexical prefix the same stem stays intransitive, (25c):

(25) a. rabota-t' (*den'gi) — SIMPLEX STEM
work-INF money.ACC
'work'

b. za-rabota-t' den'gi — LP
LP-work-INF money.ACC
'earn money'

c. za-rabota-t' (*den'gi) — SLP
SLP-work-INF money.ACC
'start working'

As an illustration of how LPs, unlike SLPs, manipulate thematic properties of arguments and project arguments not subcategorized for by a stem, consider (26)–(28).

(26) NON-PREFIXED
rubi-t' {derevj-a / drov-a / ??plennogo}
chop-INF trees-ACC firewood-ACC captive.ACC
(Intended:) 'chop {trees / firewood / a captive}'

The verb *rubi-t'* 'chop' subcategorizes for an affected or effected object ('trees' and 'firewood' in (26)) and cannot be interpreted as a verb of killing (as in 'hack a captive').

Two effects of lexical prefixation are shown in (27a)–(27b).

[15]As Biskup (2019: 20–23) emphasizes, SLPs can nevertheless impose certain semantic and syntactic restrictions on their arguments. For instance, distributive SLPs need a cumulative (i.e. plural or mass) argument. The cumulative *na-* has been argued to require the internal argument not be a fully projected DP, but a smaller nominal constituent, QP (Pereltsvaig 2006).

(27) LPs

a. s-rubi-t' {derevj-a / *drov-a / *plenn-ogo}
lp-chop-inf trees-acc firewood-acc captive-acc
(Intended:) 'chop down {the trees / the firewood / the captive}'

b. za-rubi-t' {*derevj-a / *drov-a / plenn-ogo}
lp-chop-inf trees-acc firewood-acc captive-acc
(Intended:) 'slash {the trees / the firewood / the captive}'

In (27a), the prefix narrows down the range of options available in (26): unlike *rubi-t'*, *s-rubi-t'* selects for an affected object 'trees', but not for an effected object 'firewood'. Just like *rubi-t'*, *s-rubi-t'* is not a verb of killing (as in 'hack a captive'). In (27b), prefixation of the lexical *za-* forces the derived verb to take an unselected object 'the captive', while both types of arguments the stem *rubi-* subcategorizes for become unavailable. I will address the argument structure of lexically prefixed verbs in more detail in Section 4.

As before, superlexical prefixation has no consequences for the thematic properties of the internal argument projected by the stem. (28) with the cumulative *na-* exhibits the same range of options as (26).

(28) SLP
na-rubi-t' {derevj-ev / drov / *plenn-yx}
slp-chop-inf trees-gen.pl firewood.gen captive-gen.pl
(Intended:) 'chop (down) a quantity of {trees / firewood / captives}'

Finally, (29) shows that lexical prefixation effectively blocks the indefinite object alternation typically available for simplex imperfective verbs.

(29) Indefinite object alternation

a. Dima čita-l (knig-u).
Dima read-pst.m book-acc
'Dima was reading (a book).'

b. Dima pro-čita-l *(knig-u).
Dima lp-read-pst.m book-acc
'Dima read a book.'

c. Dima po-čita-l (knig-u).
Dima slp-read-pst.m book-acc
'Dima spent some time reading (a book).'

The simplex imperfective verb *čita-t'* 'read' in (29a) can occur in a clause with no syntactically projected direct object, which describes the agent's activity ('Dima was involved in reading').[16] This option is preserved in (29c) with the delimitative superlexical *po-*, but not in (29b) with the lexical *pro-* (see, among others, Babko-Malaya 1999, Dimitrova-Vulchanova 1996).[17] If (29b) without the object NP can be interpreted at all, it either involves object ellipsis (*Žora pro-čita-l knigu, i Dima pro-čita-l ~~knigu~~.*) or a null pronoun in the object position (— 'Who read Das Kapital$_i$?' — *Dima pro-čita-l pro$_i$.*).[18]

Since the early work on the lexical-superlexical distinction, there has been no general agreement as to how this phenomenon should be accounted for. For a theory of prefixation, the entire set of questions about mechanisms of and constraints on word formation needs to be settled, including the central question of whether it happens in the syntax or in a separate module of the grammar.

Putting aside Slavic academic traditions (e.g. Švedova 1980, Grzegorczykowa et al. 1998, Štícha 2018, a.o.), the majority of existing literature on the LP/SLP divide assumes a syntactic, anti-lexicalist view of prefixation, and it is the view that I will focus on in what follows. One of the relatively rare examples of the opposite approach is Spencer & Zaretskaya (1998) who distinguish between lexical prefixes and "Aktionsart prefixes" (≈ SLPs in a different terminology) and assume that at least the former occur at the level of lexical conceptual structure.[19]

[16] Verbs like *čita-t'* 'read' that allow for indefinite object alternation are all manner verbs in terms of Rappaport Hovav & Levin (1998) and elsewhere. Overall, Russian seems to conform to Rappaport Hovav & Levin's observation that being a manner verb is a necessary, but not a sufficient condition for the alternation.

[17] The interaction between superlexical prefixation and indefinite object alternation requires further study. There certainly are SLPs that pattern with the delimitative *po-* in (29) in not affecting the alternation. This is true of the inceptive *za-* (*za-čita-t'* 'start reading'), perdurative *pro-* (*pro-čita-t' ves' den'* 'spend the entire day reading'), terminative *ot-* (*ot-čita-t'* 'finish reading'), and, with some reservations, *pere-* of excess (*pere-čita-t'* 'read too much'), even though not all speakers find these particular prefixed verbs natural. The judgments about the completive *do-* are less clear, since for some completive verbs the alternation seems to be available easier than for others.

[18] Care should be taken when applying other alleged diagnostics of the lexical or superlexical status of a prefix mentioned in the literature, e.g. their ability to undergo secondary imperfectivization (Ramchand 2004, Gribanova 2013), form event nominals (Babko-Malaya 1999, Svenonius 2004) and stative/resultative participles or head infinitival subjects (Gehrke 2008: 165). Upon closer scrutiny, one can observe that none of these diagnostics works quite the way it is supposed to work; for instance, there is massive evidence that only a proper subset of SLP configurations is incompatible with secondary imperfectivization. See Tatevosov (2013) for an extensive discussion of the relevant constraints on secondary imperfectivization and nominalization.

[19] Spencer & Zaretskaya (1998) argue that a prefix is a "core predicate" in the sense that it determines the event structure of a verb, whereas the verb stem should be treated as a modifier to this structure.

Simplifying somewhat, one can identify two parameters of variation within anti-lexicalist theories. One has to do with projection properties shown in Table 1, which includes minimal references to the work where a particular theoretical position is assumed or argued for.

Table 1: The morphosyntax of Slavic prefixes

minimal projections	heads	Svenonius (2004) (LPs) Slabakova (2005) (LPs) Žaucer (2009) Biskup (2019)
	head-adjuncts	Babko-Malaya (1999)
maximal projections	specifiers	Svenonius (2004) (SLPs) Svenonius (2008) (all)
	adjuncts	Slabakova (2005) (SLPs)
concord morphology		Arsenijević (2012)

The dominant view of prefixation is that prefixes are heads that combine with verb stems via head movement or a similar mechanism. One major departure from this view, advanced in the work by Peter Svenonius, is that either some or all prefixes are maximal projections. Svenonius (2004, 2008) discusses a number of arguments supporting this position, including compelling evidence that prefix + stem combinations exhibit phonological effects characteristic of word boundaries rather than morpheme boundaries. A different alternative is offered by Arsenijević (2012: and elsewhere), who argues that prefixes are not syntactically represented at all and are to be analyzed as phonological signatures of the agreement relations between V, Asp and a preposition.

The other, and more significant, major parameter of theoretical variation is established by existing assumptions about the location of prefixes in the hierarchical structure. Table 2 represents a number of options attested in the literature.

Most proposals rely on the structural asymmetry between LPs and SLPs, and many on a particular type of asymmetry in (30), where LPs merge within the complement of V, while SLPs appear outside of VP.

(30) [… SLP … [$_{\text{VP}}$ … [… LP …]]]

Svenonius (2004) argues, specifically, that LPs originate in the R(esult) projection, whereas SLPs adjoin to a functional projection above VP. This explains a bunch

Table 2: The origin of Slavic prefixes

LPs	within complement of V	Svenonius (2004) Asbury et al. (2007) Romanova (2007) Gehrke (2008) Žaucer (2009)
	adjoined to V	Babko-Malaya (1999)
	within FP outside VP	Slabakova (2005) Wiland (2012)
SLPs	within FP outside highest projection of LP/VP	Babko-Malaya (1999) Svenonius (2004) Slabakova (2005) Gehrke (2008) Gribanova (2013, 2015)
	within complement of V (like LPs)	Žaucer (2009, 2010, 2012)

of properties that separate the two classes: LPs are correctly predicted to occur inside of SLPs, to have impact on the argument structure, to develop idiomatic meanings easier, and so on.

However, virtually all logically possible alternatives to (30) can be found in the literature. Some authors assign a different position to LPs; cf. Babko-Malaya's (1999) head-adjunction and Slabakova's (2005) analysis in Table 2. Others argue for a different location of SLPs. Žaucer (2009, 2010, 2012), for example, develops a set of carefully designed arguments suggesting that SLPs, just like LPs, originate under V. One of the arguments is based on the observation that at least some superlexicals introduce a result state into the semantic representation of a predicate. If result state descriptions are only generated below V, Žaucer's conjecture follows. A theory along the lines of (30) needs to assume that SLPs can introduce result states from outside of VP. This obviously disrupts a correspondence between the layers of structure and types of eventuality descriptions that form their denotations, which is taken to be a desideratum in a number of theories, e.g. Ramchand's (2008) First Phase Syntax, where all result states come out as denotations of R(esult)P/resP.

A separate question is how SLPs, if they are VP-external, are located with respect to *v* and Asp. Many authors assume, much in line with traditional Slavic linguistics, that "aspectual morphology" contributes directly to the computation of semantic aspect and originate within AspP, i.e. outside of *v*P. Tatevosov (2011, 2015b, 2021), however, builds a chain of arguments that pieces of aspectual morphology are *v*P-internal and modify event structure, while true semantic aspectual operators in Asp are phonologically silent. Proposals that treat aspectual morphology as concord morphology (e.g. Arsenijević 2012 and subsequent work) assume that prefixes are lower than Asp, too.

Assessing far-reaching predictions of these proposals would take us too far from the central topic of this chapter, however. Having in mind the alternatives in Table 2, in what follows I will be assuming the architecture of the verbal domain along the lines of (30).

In the next section, event-structural correlates of lexical prefixation and secondary imperfectivization are surveyed.

3 Lexical prefixation, secondary imperfectivization, and event structure

The goal of this section is to establish a number of generalizations about the event structure of morphologically SIMPLEX IMPERFECTIVE VERBS, VERBS DERIVED BY LEXICAL PREFIXATION, and SECONDARY IMPERFECTIVE VERBS, shown in Table 3.

Table 3: Simplex Imperfective, Perfective, and Secondary Imperfective verbs

Verb meaning	Simplex Imperfective	Perfective	Secondary Imperfective
'write'	*pisa-t'* write-INF	*na-pisa-t'* LP-write-INF	–
'sign'	–	*pod-pisa-t'* LP-write-INF	*pod-pis-yva-t'* LP-write-YVA-INF
'read'	*čita-t'* read-INF	*pro-čita-t'* LP-read-INF	*pro-čit-yva-t'* LP-read-YVA-INF

Table 3 illustrates three most basic types of relationship between simplex imperfective, lexically prefixed perfective and secondary imperfective verbs. *Na-pisa-t'* 'write$^{\text{PFV}}$' as opposed to *pisa-t'* 'write$^{\text{IPFV}}$' is frequently referred to in the literature as an instance of "pure perfectivizing", or "empty" prefixation: the only perceivable contribution of the prefix *na-* in *na-pisa-t'* 'write$^{\text{PFV}}$', as Babko-Malaya (1999: 51) puts it, "is to indicate that the process denoted by the verb is completed". The word "empty" here reflects the intuition that prefixation does not produce any change in the lexical meaning of a verb. *Pisa-t'* 'write$^{\text{IPFV}}$' and *na-pisa-t'* 'write$^{\text{PFV}}$' are thus considered aspectual partners that form ASPECTUAL PAIR.

Pod-pisa-t' 'sign$^{\text{PFV}}$' illustrates "resultative prefixation" where there is more to the meaning of the prefixed verb than "perfectivization". Intuitively, *pod-* derives a different lexical item from *pisa-t'* 'write$^{\text{IPFV}}$', *pod-pisa-t'* 'sign$^{\text{PFV}}$'. An imperfective partner of *pod-pisa-t'* 'sign$^{\text{PFV}}$' is *pod-pis-yva-t'* 'sign$^{\text{IPFV}}$', the product of secondary imperfecitivization.[20]

In Section 4 I will discuss a more refined typology of lexical prefixes, not restricted to "empty" (*na-* in *na-pisa-t'*; *pro-* in *pro-čita-t'*) and "resultative" ones (*pod-* in *pod-pisa-t'*). For the purposes of this section these two will suffice, however.

Whereas most perfective verbs derived by "empty" prefixes, *na-pisa-t'* 'write$^{\text{PFV}}$' in Table 3 among them, do not undergo secondary imperfectivization, some do. This happens to *pro-čita-t'* 'read$^{\text{PFV}}$', which has both simplex and secondary imperfective counterparts, *čita-t'* and *pro-čit-yva-t'* 'read$^{\text{IPFV}}$'. Derivationally related items like *čita-t'* 'read$^{\text{IPFV}}$' / *pro-čita-t'* 'read$^{\text{PFV}}$' / *pro-čit-yva-t'* 'read$^{\text{IPFV}}$' are frequently referred to in the literature as ASPECTUAL TRIPLES. In general, it is not predictable if a simplex verb forms a pair like *pisa-t'* 'write$^{\text{IPFV}}$' or a triple like *čita-t'* 'read$^{\text{IPFV}}$'.[21]

In what follows we will see that event-structurally, lexically prefixed perfective verbs, "pure perfectivizing" and "resultative" verbs alike, differ systematically from simplex, but not from secondary imperfective verbs.

3.1 Simplex imperfective and prefixed perfective verbs

The first generalization about the event structure projected by the verbs like those in Table 3 comes in (31).

[20] A terminological note is due. The notion of "resultative" prefixation (e.g. *pod-pisa-t'*) as opposed to "pure perfectivizing" prefixation (e.g. *na-pisa-t'* 'write$^{\text{PFV}}$') goes back to Babko-Malaya (1999): 51-56. To avoid confusion, one should keep in mind that there is a different tradition (e.g. Isačenko 1960) where the term "resultative" has a broader meaning and covers both *na-pisa-t'* 'write$^{\text{PFV}}$' and *pod-pisa-t'* 'sign$^{\text{PFV}}$'. Within this tradition, there is no designated term reserved for verbs like *pod-pisa-t'* 'sign$^{\text{PFV}}$'.

[21] Aspectual triples are not available in every Slavic language. Klimek-Jankowska et al. (2025) indicate that triples are systematically absent in West Slavic languages, as opposed to East Slavic languages, and put forward the hypothesis that this cross-linguistic difference can be reduced to the range of syntactic positions available for the secondary imperfective morpheme.

(31) EVENT STRUCTURE OF SIMPLEX IMPERFECTIVE AND LEXICALLY PREFIXED PERFECTIVE VERBS
The event structure of simplex imperfective verbs lacks a result state. Lexically prefixed verbs are associated with a more complex event structure containing a result state.

According to (31), simplex imperfective and lexically prefixed perfective verbs manifest different degrees of subevental complexity, the latter having one subevent more than the former. A natural way of making (31) explicit is to translate it into event-semantic representations of the format in (32)–(33).

(32) VPS PROJECTED BY SIMPLEX IMPERFECTIVE VERBS: A predicate of events, simplex event structure
$\lambda e. \dots e \dots$

(33) VPS PROJECTED BY PREFIXED PERFECTIVE VERBS: A relation between events and states, complex event structure
$\lambda s.\lambda e. \dots e \dots \wedge \dots s \dots \wedge R(s)(e)$

(32) is a predicate of eventualities, whereas (33) is a relation between events and (result) states. The relation R in (33) is typically conceived of as a causal relation. (32)–(33) can be thus thought of as approximating what has been labeled activities and accomplishments in Dowty's (1979 and much further work) line of inquiry, with its decompositional treatment of accomplishments.[22] Since then, in the literature configurations like (32) or similar have been sometimes referred to as an activity event structure, while those like (33) as an accomplishment event structure. I will refrain from using this terminology in what follows to avoid confusion with accomplishments and achievements as eventuality types. Rather, (32)–(33) will be merely identified as a simplex and a complex event structure, respectively.

The uninflected tenseless and aspectless VPs from (34a) and (35a) would have the denotations in (34b) and (35b).[23]

[22](32) and (33) correspond to Dowty's (1979) activity and accomplishment templates, DO(α, $\pi(\alpha, \dots)$) and DO(α, $\pi(\alpha, \dots)$) CAUSE (BECOME$\rho(\beta, \dots)$), respectively. The major differences are the following: Dowty takes the external argument to be part of the representation, employs temporal semantics rather than event semantics, makes use of the BECOME operator, and assigns more structure to the process component.

[23]The representations in (34b) and (35b) assume that external arguments are introduced by a functional head outside of VP and rely on the Davidsonian association of internal arguments with events (Kratzer 1996, 2003 and much further work). As a simplification, nominal arguments are represented in (34b) and (35b) as individual constants.

(34) SIMPLEX IMPERFECTIVE VERBS: A predicate of events; simplex event structure

a. Volodja pisa-l pis'm-o.
Volodja write-PST.M letter-ACC
'Volodja was writing a letter.'

b. $[\![[_{\text{VP}}\ \textit{pisa- pis'mo}]]\!] = \lambda e.\textsc{write}(\textsc{letter})(e)$

(35) LEXICALLY PREFIXED VERBS: A relation between events and states; complex event structure

a. Volodja na-pisa-l pis'm-o.
Volodja LP-write-PST.M letter-ACC
'Volodja wrote a letter.'

b. $[\![[_{\text{VP}}\ \textit{na-pisa- pis'mo}]]\!] =$
$\lambda s.\lambda e.\textsc{write}(\textsc{letter})(e) \wedge \textsf{CAUSE}(s)(e) \wedge \textsc{written}(\textsc{letter})(s)$

Essentially, (34b) represents the simplex event structure in (32), whereas (35b) has the complex structure in (33).[24] Much in the spirit of Klein (1995), *pisa-* 'write' and other non-stative simplex imperfectives project VPs that denote predicates of eventualities (of type $\langle v, t \rangle$) (cf. Klein's 1-state descriptions). The predicate in (34b) has writing events in its extension, whose theme is a letter.

The representation of (35b) contains, in addition, a target state of being written brought about by a writing event. Prefixed stems like *na-pisa-* thus denote relations between events and states (of type $\langle v, \langle v, t \rangle \rangle$; Klein's 2-state descriptions). The eventive part of (35b) corresponds to a "source state" in Klein's (1995) terminology; the state argument ranges over "target states". A holder of that state is identical to the theme of writing. The target state component is represented as "WRITTEN(LETTER)(s)", where "WRITTEN" is of type $\langle e, \langle v, t \rangle \rangle$, a relation between individuals and states. An analysis of the simplex perfective verbs like *da-t'* 'give[PFV], a verb of transfer of possession, would be similar, possibly with minor adjustments.

As we have already seen in the previous section, in the extensive work on prefixation in Slavic languages (Babko-Malaya 1999, Ramchand 2004, Romanova 2004, Svenonius 2004, 2008, Arsenijević 2007a,b, Gehrke 2008, Tatevosov 2009, 2013, Žaucer 2009, 2010, 2013, Biskup 2015, 2019), it has been independently argued that the role of prefixes like *na-* in (35a) is to project a structure interpreted

[24]The eventive element of (35b) can be further decomposed into the activity and change-of-state, yielding a three-component decomposition, as in Padučeva (1996), Ramchand (2008) or Beavers & Koontz-Garboden (2020). As far as I can see, nothing in what follows depends on this possible refinement.

as a target state description. (35b), which differs from (34b) in the presence of a lexical prefix, makes this fully explicit.

There are a number of empirical arguments for (34b) and (35b) based on converging evidence for higher event-structural complexity of lexically prefixed verbs. The body of literature on predicate decomposition starting from Dowty (1979) (see von Stechow 1996, Rappaport Hovav & Levin 1998, Rapp & von Stechow 1999, Kratzer 2000, Rothstein 2004, Ramchand 2008, among many others) offers a number of diagnostics that allows us to tell the two configurations apart. The list includes, specifically, ARGUMENT REALIZATION PATTERNS, SCOPE OF ADVERBIALS like ‘almost’ and ‘again’, INTERPRETATION UNDER NEGATION.

The first piece of evidence comes from argument realization. As Rappaport Hovav & Levin (1998) observe, verbs that show the indefinite object alternation (e.g. *sweep*) are all associated with the simplex event structure. If a verb has the structure of accomplishments (e.g., *break*), the alternation is unavailable:

(36) a. John swept the floor.
b. John swept.

(37) a. John broke the window.
b. # John broke.

As we have already seen in the previous section, simplex imperfective and lexically prefixed perfective verbs in Slavic differ in exactly the same way. Examples (29a)–(29b) are repeated as (38a)–(38b).

(38) INDEFINITE OBJECT ALTERNATION
a. Dima čita-l (knig-u).
Dima read-PST.M book-ACC
‘Dima was reading (a book).’
b. Dima pro-čita-l *(knig-u).
Dima LP-read-PST.M book-ACC
‘Dima read a book.’

The prefixed verb in (38b) is ungrammatical without the direct object, while its non-prefixed counterpart in (38a) is readily available. This is exactly what one would expect, given Rappaport Hovav & Levin’s (1998) generalization, if prefixed verbs introduce accomplishment event structures whereas non-prefixed verbs have the structure of activities.

Rappaport Hovav & Levin (1998) propose that the ungrammaticality of (37b) is due to a principle, operational in the grammar of natural languages, that requires syntactic realization of the structural aspects of the meaning of lexical items. For *break*, associated with a complex event structure, but not for *sweep*, the internal argument is structural. It is construed as the holder of a result state of being broken and as such must be syntactically projected. The internal argument of *sweep* is only licensed as an argument of the lexical constant of the verb, which makes its syntactic realization optional.

The Slavic pattern in (38a)–(38b) is addressed in detail in Babko-Malaya (1999: 64–68); Dimitrova-Vulchanova (1996), Biskup (2019). Despite being very different as to the specifics of how the structure of prefixed configurations is organized, they seem to fundamentally agree that obligatoriness of the syntactic realization of the object is induced by the prefix, which introduces a result state description.

Another set of diagnostics is based on a further prediction derivable from an analysis along the lines of (32)–(33). If an event is decomposed into subevents, one should expect to find semantic operations that can target some of the subevents, but not the others. Applying to subeventally more complex structures like (35b), such operations should yield a different range of interpretations compared to simplex structures like (34b).

One piece of evidence for (35b) comes from the distribution of adverbials like 'again' (Dowty 1979, von Stechow 1996, Beck 2005, Lechner et al. 2015) and 'almost' (Rapp & von Stechow 1999). With the complex event structure, both are known to be able to manifest varying scopal possibilities illustrated informally in (39a)–(39b). On the low reading in (39a), the scope of 'almost' and 'again' only includes the result state. On the "high" reading in (39b), it extends to the entire eventuality description including the process component.

(39) a. PROCESS 'almost' / 'again' [RESULT STATE]
b. 'almost' / 'again' [PROCESS RESULT STATE]

Combining with a simplex event structure that lacks a result state altogether, 'almost' and 'again' are expected not to show this type of ambiguity. In (40), their position is outside of the only component of the event structure, the process component. The only available reading of 'almost' and 'again' should thus correspond to the "high" reading in (39b).

(40) 'almost' / 'again' [PROCESS]

Consider 'almost'. Intuitively, 'almost' indicates that the scope proposition is false in the evaluation world, but a close-by scalar alternative is true (e.g., Penka 2006). Compare the discourses in (41a) and (41b).

(41) a. *Context*: — Volodja, how long will it take you to finish this simple letter? You've been working on it for an hour!
— Počti na-pisa-l! Ešče dve minut-y.
almost LP-write-PST.M more two minute-GEN.SG
'I almost wrote it. Two more minutes.'

b. *Context*: — We need to ask the dean for discretionary funds. — You know,
ja počti na-pisa-l pis'm-o dekanu, no pereduma-l.
I almost LP-write-PST.M letter-ACC dean-DAT but change.mind-PST.M
'I almost wrote a message to the dean (about that), but then changed my mind.'

(41a) contextually entails that the the agent's writing activity has occurred, but the culmination and a subsequent result state have not. (41a) thus instantiates (39a): the scope of 'almost' only includes the result state. In the null context, this interpretation of *počti na-pisa-t'* 'almost write' is preferred.

Another interpretation is observed in (41b): The agent comes close to performing an entire event, including its eventive component, but changes his mind. No part of a writing-a-message eventuality occurs in the evaluation world. The scope of 'almost' in (41b) includes both components of an eventuality, in line with (39b).

These examples together show, therefore, that exactly as (39) predicts, 'almost' can take scope over the entire complex eventuality, (41b), or over the result state component only, (41a). This provides us with a piece of evidence for (35b), a semantic representation containing an activity and a result state as distinct elements of event structure.

When it comes to the event structure of simplex verbs like *pisa-t'* 'write$^{\text{IPFV}}$', a few precautions should be taken. As we have already seen, such verbs can only occur in imperfective clauses, which, on the progressive reading, come with the inference that the culmination is not included into the topic time. Consider (42).

(42) Volodja počti pisa-l pism-o.
Volodja almost write-PST.M letter-ACC
'Volodja was almost writing a letter.'

(42) is unambiguous, indicating that the agent is close to writing a letter in the evaluation world but no writing activity occurs (yet). However, the lack of ambiguity can have more than one explanation. (42) may be unambiguous because simplex imperfective verbs do not have a result state component, as (34b) suggests. But there can be another reason as well: *pisa-t'* 'write' in (42) does have a

result state in its semantic representation, but this state is not reached in the evaluation world because of the progressive, which renders the low reading of *počti pisa-l* 'was almost writing', parallel to that of *počti na-pisa-l* in (41a), unavailable. Progressive sentences like (42) are therefore uninformative with respect to the event structure of simplex verbs.

Fortunately, there is a way of getting around this difficulty. The imperfective in Russian is available in a few environments where it can describe a culminating eventuality, thus being truth-conditionally equivalent to the perfective. (In the traditional literature on Slavic aspect this is known as ASPECTUAL COMPETITION between the perfective and the imperfective, see, e.g. Maslov 1984; configurations, in which this happens are sometimes called "Maslov's contexts"). In such an environment, if simplex imperfectives are as subeventally complex as lexically prefixed perfectives, they should exhibit the same range of interpretations as observed in (41a)–(41b).

One configuration of this type is served by Historical Present (Schlenker 2004, Eckardt 2015, a.m.o.). Consider (43).

(43) *Context*: Volodja comes home, sits down at his desk, takes a pen and paper, …
piš-et pis'm-o i lož-it-sja spa-t'.
write-NPST.3SG letter-ACC and lie.down-NPST.3SG-REFL sleep-INF
'writes a letter, and goes to bed'.

The narrative in (43) describes a sequence of eventualities each of which, including 'write a letter', reaches a culmination; see Gehrke (2026 [this volume]: section 3.3) for an overview of the Historical Present in Czech and Russian. This guarantees that the Historical Present provides a right environment for examining the scopal possibilities of 'almost' with a simplex verb. In (44), the same discourse is modified in order to accommodate 'almost'.

(44) *Context*: Volodja comes home, sits down at his desk, takes a pen and paper, …
i počti piš-et pis'm-o, kogda razda-ët-sja
and almost write-NPST.3SG letter-ACC when give.out-NPST.3SG-REFL
zvonok v dver'.
ring in door.ACC
'and almost writes a letter, when the doorbell rings.'

The only reading available to *počti* 'almost' in (44) is the "high" reading where no activity occurs in the evaluation world. But unlike with the progressive in

(42), this cannot be attributed to the fact that 'writes' in (44) cannot describe a culminating eventuality. As (43) shows, it can. Therefore, the only reason for there being no "low" reading of 'almost' in (44) is that *pisa-t'* 'write' genuinely lacks a result state that *počti* 'almost' can take scope over independently of the eventive component. Therefore, (44) argues for (40) and, in this way, for the semantic representation of simplex imperfective verbs along the lines of (34b).

Another one of "Maslov's contexts" in which the imperfective is considered aspectually parallel to the perfective is provided by habitual sentences. Consider (45).

(45) *Context*: In the morning, Volodja gets up early, sits down at his desk, reads newspapers, ...
piš-et pis'm-o komu-nibud' iz druz-ej i
write-NPST.3SG letter-ACC who.DAT-INDEF from friend-GEN.PL and
id-et zavtraka-t'.
go-NPST.3SG have.breakfast-INF
'writes a letter to one of his friends and goes to breakfast.'

(45) describes a regularity verified by multiple occurrences of episodic writing eventualities. Crucially, (45) conveys that each of these eventualities culminates. Therefore, *počti pisa-t'* 'almost write' within the scope of the habitual provides us with the same environment as Historical Present. If *pisa-t'* 'write$^{\mathrm{IPFV}}$' is subeventally complex, *počti pisa-t'* 'almost write' is expected to be ambiguous. But it is not: just as (44), the sentence in (46) can only mean that no writing a letter activity occurs.

(46) *Context*: Arriving in Geneva, Volodja unsuccessfully tries to establish regular correspondence with friends. Every morning after breakfast he sits down at his desk, takes a pen and paper, ...
počti piš-et pis'm-o komu-nibud' iz druz-ej, no
almost write-NPST.3SG letter-ACC who.DAT-INDEF from friend-GEN.PL but
objazatel'no zasypa-et.
obligatorily fall.asleep-NPST.3SG
'almost writes a letter to one of his friends, but always falls asleep.'

Evidence from the Historical Present and habitual sentences therefore converges: no ambiguity with a decomposition adverb can be detected. Because of the strict parallelism of the two environments, in what follows I will only cite examples with Historical Present.

The same pattern can be brought out by *opjat'* 'again'. For the sake of space, I will refer the reader to Tatevosov (2016: 493–497) and not reproduce corresponding examples here.

Before moving on, let me briefly address one more potential complication with the range of interpretations available for decomposition adverbs. As Berit Gehrke (p.c.) points out, there is a class of verbs that are subeventally complex, but nevertheless do not allow for 'almost' and 'again' to show scope ambiguities. These are ditransitive verbs discussed in detail in Bondarenko (2018):

(47) Maša opjat' otda-l-a Vas-e knig-u.
Maša again give-PST-F Vasja-DAT book-ACC
'Maša gave Vasja the book, and she had given it to him before.' (repetitive)
Unavailable: 'Maša gave Vasja the book, and he had had it before.' (restitutive)
(adapted from Bondarenko 2018: 28)

Bondarenko argues extensively that this pattern cannot be derived from any special properties of *opjat'* 'again' in Russian. She finds independent arguments that the semantic representation of ditransitives does contain a result state. She concludes, therefore, that the reason for the unavailability of the restitutive reading is syntactic: The result state subevent is not properly represented in the syntax by a phrase that could be scoped over by 'again' independently from the causing subevent. If this scenario is empirically real for ditransitives, the question is whether it can be extended to simplex imperfective verbs. That is, can it be the case that *počti* 'almost' is unambiguous with *pisa-t'* 'write$^{\text{IPFV}}$' because *pisa-t'* 'write$^{\text{IPFV}}$' is like ditransitives: It is subeventally complex, but it does not project a result state denoting phrase that would serve as a landing site for an adverbial? One argument for a negative answer to this question may be based on the interpretation of *-n/t-* passive participles. According to Paslawska & von Stechow (2003), such participles are stativizers: They take a relation between events and states and return a property of states (48).

(48) $[\![\text{-}n/t\text{-}]\!] = \lambda R_{\langle v,\langle v,t\rangle\rangle}.\lambda s.\exists e[R(s)(e)]$

With ditransitives, these participles are interpreted just as one would expect given (48): (49) describes a result state that holds of the theme at the reference time.

(49) V dannyj moment knig-a otda-n-a Vase.
in given moment book-NOM give-N/T-F Vasja-DAT
'At the moment the book is in a state of having been given to Vasja.'

(49) shows that what matters for the interpretation of the *-n/t-* morphology, unlike for 'almost', is the presence of the result state in the semantic representation, and not its syntactic manifestation. With (49) in mind, consider the minimal pair in (50)–(51):

(50)
V	dannyj	moment	statj-a	nakonec	na-pisa-n-a.
in	given	moment	article-NOM	finally	LP-write-N/T-F

'At the moment the paper is finally in a state of having been written.'

(51)
* V	dannyj	moment	statj-a	nakonec	pisa-n-a.
in	given	moment	article-NOM	finally	write-N/T-F

Intended: 'At the moment the paper is finally a state of having been written.'

(51) suggests that the passive participle of *pisa-t'* 'write$^{\text{IPFV}}$' is impossible as a description of a result state, unlike its prefixed counterpart in (50). (Note that "imperfective" non-prefixed PPs like *pisa-n-a* are available in other semantic environments, Borik & Gehrke 2018, Gehrke 2026 [this volume], but there they cannot be analyzed in terms of the stativizer in (48).) If simplex verbs had a complex event structure of the $\langle v, \langle v, t\rangle\rangle$ type, the *-n/t-* semantics in (48) should have been able to apply to it just like to (49) and (50). The fact that it does not happen seems to reinforce the conclusion that such verbs denote a simplex event structure and are not subject to Bondarenko's (2018) scenario for ditransitives.

Now consider interpretations of the prefixed and non-prefixed verbs under negation.

(52) PREFIXED VERB UNDER NEGATION: ambiguous

Volodja	ne	na-pisa-l	stat'j-u.
Volodja	not	LP-write-PST.M	article-ACC

'Volodja did not write the article.'

a. No writing activity has been performed.
b. No writing activity has been completed.

(52) exhibits at least two readings. On (52a), the sentence says that there was no writing an article activity and a target state of having been written has never been attained. On (52b), the existence of the target state is negated, too, but that of the activity is not. On the analysis in (35b), this ambiguity is expected: since there are two subevental components connected by the conjunction, falsity of any or

both of the conjuncts makes the whole negated sentence true, as indicated in the simplified paraphrase in (53).[25]

(53) It is not the case that [there is a writing an article activity and a target state of having been written].

As with 'almost', one should make sure that a simplex imperfective verb is examined in an environment where it is compatible with a culminated construal. (54) is another narrative in the Historical Present.

(54) *Context*: Volodja comes home, sits down at his desk, takes a pen and paper, ...
no ne piš-et pis'm-o, kak obyčno, a
but not LP-write-NPST.3SG letter-ACC as usual but
na-bras-yva-et nebrežnyj portret Nad-i.
LP-throw-YVA-NPST.3SG sketchy portrait.ACC Nadja-GEN
'but does not write a letter, as usual, but draws a sketch of Nadja's portrait.'

(54) is not ambiguous in the way (52) is. It entails that no writing activity occurs. Again, (34b) predicts exactly this. Being based on the predicate of reading events with no target state component, all (54) says is (55).

(55) It is not the case that [there was a writing a letter activity].

Crucially, as we have already seen in (43), telic predicates can describe culminated events in the Historical Present. This guarantees that if a target state was part of the representation of the simplex verb *pisa-t'* 'write$^{\text{IPFV}}$', (54) would have had the same range of interpretations as (52). The fact that this does not happen, once again, provides us with an argument for less subevental complexity of non-prefixed verbs.

This conclusion is reinforced by the behavior of simplex verbs in one more configuration where the imperfective is compatible with the culmination being reached in the evaluation world. Consider (56).

(56) NON-PREFIXED STEM UNDER NEGATION: unambiguous
Volodja ni razu ne pisa-l roman.
Volodja not once not write-PST.M novel.ACC
'Volodja has never written a novel.'

[25]The third logically possible reading (the target state holds but the activity does not) is presumably unavailable for an independent reason: a state of having been written cannot come about without a causal input from an activity.

a. No writing activity has ever been performed.
b. Unavailable: No writing activity has ever been completed.

In (56), the imperfective clause is taken under the general-factual construal. The general-factual reading has been extensively discussed in the literature (see Mehlig 1981, 1995, Padučeva 1996, Grønn 2004, 2015, Altshuler 2012, Arregui et al. 2014, Gehrke 2026 [this volume], among others) and analyzed in a variety of ways. Crucially, under this interpretation, a culminating interpretation is readily available for telic verbs. Hence, the general-factual offers one more environment where event-structural characteristics of *pisa-t'* 'write^{IPFV}' and similar verbs can be examined. (56) shows the same pattern as (54), the one represented in (55). Because of the lack of subevental complexity, simplex imperfective verbs do not create a structure with a conjunction, hence are not ambiguous under negation.

Before leaving this section, I have to acknowledge one open problem that the view laid out above needs to solve in the future. In a nutshell, this view suggests that simplex imperfective verbs are predicates of events, while their prefixed counterparts appear, in addition, with a result state, the latter thus being a product of prefixation (In Section 4, I will make it a bit more explicit.)

However, lexical prefixation is possible with simplex perfective verbs, too, as e. g. in (15a)–(15b) above, repeated as (57a)–(57b). (57c)–(57d) are additional examples of the same type.

(57) Lexical prefixes in combination with a simplex perfective verb
a. [ot- [reši]^{PFV}]^{PFV} -t'
LP solve INF
'dismiss'
b. [pod- [kupi]^{PFV}]^{PFV} -t'
LP buy INF
'bribe'
c. [o- [ses]^{PFV}]^{PFV} -t'
LP sit.down INF
'settle, settle down'
d. [za- [sta]^{PFV}]^{PFV} -t'
LP become INF
'catch/find in the middle of sth.'

Such examples are problematic because simplex perfective verbs, e.g. *reši-t'* 'make decision, solve^{PFV}' in (57a), as a reviewer points out, are arguably as subeventally complex as prefixed perfective verbs discussed in the current section:

(58) Event structure for *reši-t'* 'make decision, solve$^{\text{PFV}}$'
$\lambda s.\lambda e.$ … $\textsc{decide}(e)$ … $\wedge$ … $\textsc{decided}(s)$ … $\wedge$ $\textsf{CAUSE}(s)(e)$

(58) for *reši-t'* 'make decision, solve$^{\text{PFV}}$' can be supported by the same observations as (35b) for *na-pisa-t'* 'write$^{\text{PFV}}$'. For instance, *počti reši-t'* 'almost solve' shows the same range of interpretations as *počti na-pisa-t'* 'almost write' in (41a)–(41b) above. "Lexical prefixes", an anonymous reviewer indicates, "are uniformly analyzed as introducing a result state into the event structure of a predicate. In Rappaport Hovav & Levin (1998), it is claimed that complex verbs like the English 'break' do not appear in resultative constructions precisely because they already contain a result, and adding an additional result is simply not possible (Rappaport Hovav & Levin 1998: 103)".

At the moment, I know of no compositional analysis of the verbs like (57a)–(57d). One possible way of treating them is to assume that Russian is unlike English: a result state can somehow come on top of a complex eventuality description for which its own result state has been specified. If this is the case, *ot-reši-t'* 'dismiss from office' would be associated with two result states. One is a state of being decided or solved, lexically assigned to *reši-t'* 'make decision, solve$^{\text{PFV}}$', the other, a state of being dismissed from office, would be the contribution of a prefix.

Two event structures that accommodate two result states are shown in (59a)–(59b), ignoring individual arguments. In (59a) a single eventuality brings about two different states that fall under $\lambda s.\textsc{decided}(s)$ and $\lambda s.\textsc{dismissed}(s)$. In (59b), an eventuality consisting of eventive and stative parts, decide and decided, brings about an additional state, one from the extension of $\lambda s.\textsc{dismissed}(s)$.

(59) Two hypothetical event structures for *ot-reši-t'* 'dismiss from office$^{\text{PFV}}$'
a. $\lambda s.\lambda s'.\lambda e.$ … $\textsc{decide}(e)$ … $\wedge$ … $\textsc{decided}(s)$ … $\wedge$ … $\textsc{dismissed}(s')$ … $\wedge$ $\textsf{CAUSE}(s)(e)$ $\wedge$ $\textsf{CAUSE}(s')(e)$
b. $\lambda s.\lambda e.\textsc{dismissed}(s) \wedge \textsf{CAUSE}(s)(e) \wedge \exists e' \exists s' [e = e' \oplus s'$ … $\wedge$ … $\textsc{decide}(e')$ … $\wedge$ … $\textsc{decided}(s') \wedge \textsf{CAUSE}(s')(e')]$

These analytical options, apart from stipulating irreducible cross-linguistic variation between Russian and English, seem to bring up quite a lot of difficult questions that need to be addressed for an analysis to be minimally consistent.

A general question for (59a) is whether an eventuality can culminate in more than one result state and whether there are other semantic environments where this phenomenon is observed. For (59b), which involves sum formation, one may want to find out what triggers this operation and how it is constrained.

Both theories need to determine how the argument structure of a prefixed verb is composed given the event structures in (59a)–(59b) and the fact each of the result states is associated with its own holder. Both (59a) and (59b) face a further empirical complication: They predict that the result state specified for a simplex perfective stem, $\lambda s.\text{DECIDED}(s)$, should be a detectable meaning component of a prefixed verb. It is at least dubious if this prediction is borne out for all such verbs. While for *ot-reši-t'* 'dismiss from office' it may be argued that a dismissal presupposes making a decision, it is much less clear if settling in (57c) involves sitting down.

Finally, an analysis has to address the observation that lexically prefixed simplex perfective verbs do not look productive: such verbs form a compact closed class of items, indicating that structures like (59a)–(59b), if they are real, are not built up freely. I will leave these issues for a further study and will not try to have them settled in what follows.

3.2 Secondary imperfective verbs

As discussed in Section 1, most secondary imperfective verbs are based on perfective ones, either prefixed or simplex (see (18) in Section 2.2). They pattern with the simplex imperfective verbs as to the range of aspectual interpretations, but event-structurally resemble the lexically prefixed perfective. The latter property is captured by the generalization in (60).

(60) Event structure of secondary imperfective verbs
The event structure of secondary imperfective verbs, like that of lexically prefixed verbs, contains a result state.

Tatevosov (2016) proposes that the secondary imperfective morpheme takes a relation between events and states generated by a prefixed perfective VP as an argument and, minimally, existentially binds the state variable, creating a derived property of events.[26] The *-yva-* morpheme is thus Paslawska & von Stechow's (2003) Eventizer, (61).

[26] Nothing in the discussion of event structure of the secondary imperfective seems to hinge on whether eventization shown in (61) exhausts the semantic contribution of the *-yva-* morpheme. In (61), possible additional components of its meaning are shown as "…". A similar idea about the contribution of the secondary imperfective, couched withing the DRT framework, is advanced in Mueller-Reichau (2026 [this volume]).

(61) $[\![\textit{-yva-}]\!] = \lambda R.\lambda e.\exists s[... \wedge R(e)(s)]$

In (62b), the denotation of the aspectless verb phrase representing part of the meaning of (62a) is a predicate of events of reading newspapers that bring about a result state of being read.

(62) The secondary imperfective of a lexically prefixed verbs: A derived predicate of events; complex event structure

a. Volodja pro-čit-yva-l gazet-y.
Volodja LP-read-YVA-PST.M newspaper-ACC.PL
'Volodja would read newspapers.'[27]

b. $[\![[\ \textit{-yva-}\ [_{\text{VP}}\ \textit{pro-čita- gazety}\ ]]]\!] =$
$\lambda e.\exists s[\textsc{read}(\textsc{newspapers})(e) \wedge \textsc{cause}(s)(e) \wedge \textsc{read}(\textsc{newspapers})(s)]$

(62b), based on (61), predicts that the secondary imperfective patterns, event-structurally, with the subeventally complex prefixed perfective in (35b) above, rather than with the simplex imperfective in (34b).

The most straightforward way of seeing that this prediction is correct is to examine the event-structural behavior of simplex and secondary imperfective elements of aspectual triples like those in Table 3 above, cf. more examples in Table 4.

Table 4: Aspectual triples

Meaning	Simplex imperfective	Prefixed perfective	Secondary imperfective
'drill'	*sverli-t'* drill-INF	*pro-sverli-t'* LP-drill-INF	*pro-sverl-iva-t'* LP-drill-YVA-INF
'drink'	*pi-t'* drink-INF	*vy-pi-t'* LP-drink-INF	*vy-pi-va-t'* LP-drink-YVA-INF
'mow'	*kosi-t'* MOW-INF	*s-kosi-t'* LP-MOW-INF	*s-kaš-iva-t'* LP-MOW-YVA-INF

The imperfective members of a triple form minimal pairs contrasting with respect to all the event-structural diagnostics discussed above.

[27] As with many secondary imperfective members of aspectual triples, the progressive reading of *pro-čit-yva-t'* 'read$^{\text{IPFV}}$' in the null context is somewhat degraded; (62a) illustrates its habitual interpretation. Nothing in what follows depends on that.

First, the secondary imperfective does not show the indefinite object alternation:

(63) Simplex / secondary imperfective and indefinite object alternation

a. Po utr-am Volodja čita-l (svež-ie
on morning-DAT.PL Volodja read-PST.M fresh-ACC.PL
gazet-y).
newspaper-ACC.PL
'In the morning, Volodja would read (fresh newspapers).'

b. Po utr-am Volodja pro-čit-yva-l *(svež-ie
on morning-DAT.PL Volodja LP-read-YVA-PST.M fresh-ACC.PL
gazet-y).
newspaper-ACC.PL
'In the morning, Volodja would read fresh newspapers.'

The simplex imperfective in (63a), just as in (38a) above, can either project a VP with the direct object or an objectless VP that describes a reading activity. As (63b) shows, if the direct object is not realized in the syntax, the sentence is ungrammatical. In this respect, the secondary imperfective patterns with the prefixed perfective in (64), see also (38b) above:

(64) Utrom Volodja pro-čita-l *(svež-ie gazet-y).
in.the.morning Volodja LP-read-PST.M fresh-ACC.PL newspaper-ACC.PL
'In the morning, Volodja read fresh newspapers.'

With respect to the range of interpretations of decomposition adverbs 'almost' and 'again' and negation, the secondary imperfective, too, patterns with the prefixed perfective rather than with the simplex imperfective. In the Historical Present configuration, parallel to (44), the secondary imperfective in (65b), but not the simplex imperfective in (65a), allows for the narrow scope of *počti* 'almost', where it is only associated with the result state:

(65) Simplex / secondary imperfective and the interpretation of *počti* 'almost'

Context: Volodja comes home, sits down at his desk, sees a letter from his friend, …

a. i počti čita-et ego, kogda razda-ët-sja
and almost read-NPST.3SG it.ACC when give.out-NPST.3SG-REFL
zvonok v dver' ...
ring in door.ACC
'and almost reads it, when the doorbell rings.'
(i) The reading a letter activity does not start.
(ii) Unavailable: The reading a letter activity is not completed.

b. i počti pro-čit-yva-et ego, kogda
and almost LP-read-YVA-NPST.3SG it.ACC when
razda-ët-sja zvonok v dver' ...
give.out-NPST.3SG-REFL ring in door.ACC
'and almost reads it, when the doorbell rings.'
(i) The reading a letter activity does not start.
(ii) The reading a letter activity is not completed.

Availability of the low reading of 'almost' separates the secondary imperfective from the simplex imperfective. In this respect the secondary imperfective again patterns together with the prefixed perfective:

(66) PREFIXED PERFECTIVE AND THE INTERPRETATION OF *POČTI* 'ALMOST'
Volodja počti pro-čita-l pis'm-o.
Volodja almost LP-read-PST.M letter-ACC
'Volodja's reading the letter is close to completion.'

In a similar way, under negation the secondary imperfective is ambiguous in a way the simplex imperfective is not. (67) illustrates that with a general factual sentence parallel to (56).

(67) SIMPLEX / SECONDARY IMPERFECTIVE UNDER NEGATION

a. Volodja ni razu ne čita-l "Kapital".
Volodja not once not read-PST.M Kapital.ACC
'Volodja has never read *Das Kapital*.'
(i) No reading activity has ever been performed.
(ii) Unavailable: No reading activity has ever been completed.

b. Volodja ni razu ne pro-čit-yva-l "Kapital".
Volodja not once not LP-read-YVA-PST.M Kapital.ACC
'Volodja has never read *Das Kapital*.'
(i) No reading activity has ever been performed.
(ii) No reading activity has ever been completed.

(62b) correctly predicts, therefore, that the secondary imperfective sentence is as ambiguous as its perfective prefixed counterpart in (35a)–(35b). A target state is introduced by the prefix once and for all. Existential binding of the state variable in (62b) does not remove the target state component from the semantic representation. Secondary imperfective verbs are associated with the same complex structure as prefixed perfective verbs, which explains why they pattern together with respect to event-structural diagnostics.[28]

In the next section, I focus on the result state component and explore its influence on the interpretation of lexically prefixed verbs in more detail.

4 Lexical prefixes, result states, and projection of arguments

4.1 Introducing the typology

The previous section offers a few pieces of empirical evidence suggesting that lexically prefixed perfective verbs show subevental complexity in contrast with

[28]A reviewer asks what happens with simplex and secondary imperfective verbs outside of aspectual triples discussed in this section. I believe that the pattern discussed for *čita-t'* and *pro-čit-yva-t'* 'write$^{\mathrm{IPFV}}$' can be replicated with other simplex and secondary imperfective verbs no matter if they are members of an aspectual triple. Consider (i) with *vy-pis-yva-t'* 'write out, copy$^{\mathrm{IPFV}}$'.

(i) *Context*: Volodja needs a quotation for his term paper; he comes home, sits down at his desk, takes a pen and a piece of paper, …

i	počti	vy-pis-yva-et	dlinnuju	citat-u	iz	knig-i,	kogda
and	almost	LP-write-YVA-NPST.3SG	long	quotation	from	book-GEN	when

razda-ët-sja	zvonok	v	dver'	…
give.out-NPST.3SG-REFL	ring	in	door.ACC	

'and almost copies a long quotation from a book, when the doorbell rings.'

a. The copying activity does not start.

b. The copying activity is not completed.

Unlike in (44) and (65a), but in parallel with (65b), the secondary imperfective is compatible with the two readings of *počti* 'almost', suggesting a complex event structure for *vy-pis-yva-t'* 'write out, copy$^{\mathrm{IPFV}}$'. There may be secondary imperfectives for which one of the two readings of 'almost' is more readily accessible than the other. For instance, for *počti pod-pis-yva-et* 'almost signing' in the same environment the low reading is more difficult to get, presumably because one has to accommodate a somewhat unnatural context in which it is relevant to convey that the agent has started writing his signature under a document, is maximally close to completion, but is not yet done. Nevertheless, I believe that (i) represents a regular and systematic pattern.

the corresponding simplex verbs. According to (35b), repeated as (68), VPs projected by such verbs denote a relation between events and (result) states:

(68) VPs PROJECTED BY LEXICALLY PREFIXED PERFECTIVE VERBS
$[\![[_{\text{VP}}\ \textit{na-pisa- pis'mo}\ ]]\!] =$
$\lambda s.\lambda e.\textsc{write}(\textsc{letter})(e) \wedge \textsc{cause}(s)(e) \wedge \textsc{written}(\textsc{letter})(s)$

The presence of a result state in the event structure of lexically prefixed verbs uniformly tells them apart from simplex imperfective verbs.

The class of lexically prefixed verbs is, however, not homogeneous. Depending on the setting of a number of semantic parameters, discussed in Babko-Malaya (1999), Romanova (2007), Tatevosov (2016), Biskup (2019), and many others, lexically prefixed verbs fall into a number of subtypes, summarized in Figure 1. The terminology used in Figure 1 (*restrictive*, *(non-)argument-sharing* [(n)AS], *monadic*, etc.) will be introduced and explained shortly.

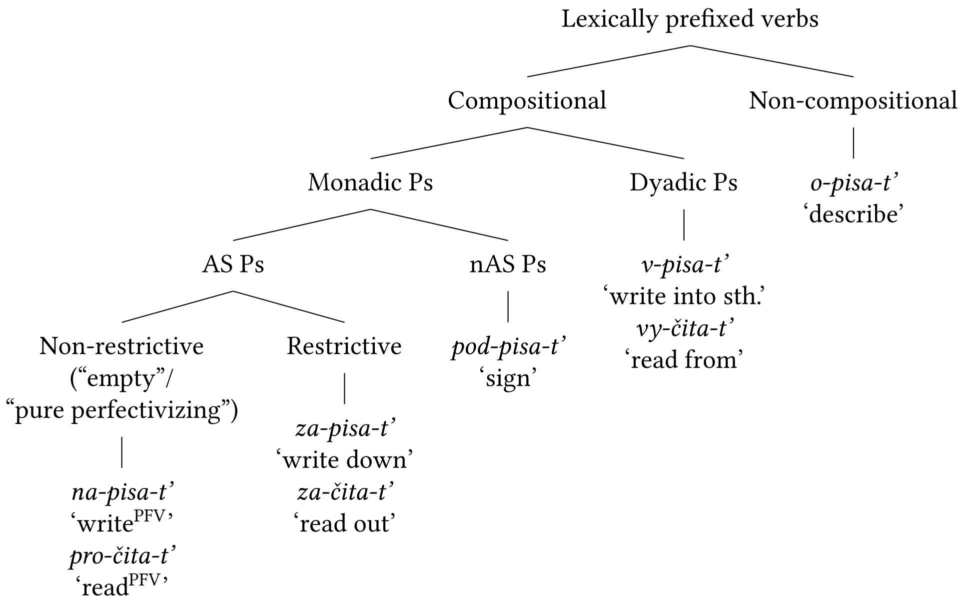

Figure 1: Typology of lexically prefixed verbs

Membership in one of the subtypes in Figure 1 has consequences for the argument structure of a prefixed verb, as well as for the perceived "semantic distance" between a simplex verb and its derived counterpart.

Throughout this section, I will mostly focus on the prefixed configurations derived from transitive lexical verbs like *pisa-t'* 'write' or *čita-t'* 'read'.[29] The generalizations discussed below are extendable to unaccusative and unergative lexical verbs as well, possibly with minor adjustments; see Romanova (2007: 70–122) for an extensive discussion. For a recent alternative to the typology in Figure 1, see Biskup (2019: ch. 3), who elaborates on and extends the system based on Bergsma et al. (2010).

As we have seen above, lexical prefixes contribute a result state into the semantic representation of a verb / VP. In what follows, I will be making a straightforward assumption about the descriptive content of predicates of such states:

(69) Content of state descriptions
Descriptive properties of result states associated with lexical prefixes in a given lexical configuration V are specific enough to differentiate states introduced by different prefixes in combination with V.

Na- in *na-pisa-t'* 'write' in (68) is understood as the description of states of having been written, $\lambda s.\textsc{written}(x)(s)$, for a fixed x. *Pod-* in *pod-pisa-t'* 'sign', on the other hand, comes with the description of different type of states, states of having been signed, $\lambda s.\textsc{signed}(x)(s)$, for a fixed x. Crucially, since the non-derived verb lacks a state argument altogether, all such information can only be the contribution of a prefix, which makes some or other version of (69) inevitable.

A separate question is how this information is computed and by what mechanism. This constitutes one of the most difficult topics in the semantic literature on Slavic verbs, which I will address in Section 4.3. Anticipating this discussion, I can mention two specific ways of thinking about such a mechanism. On the one hand, prefixes can be conceived of as collections of correspondences between verb stems and state descriptions. On this view, part of the lexical specification of *pod-* in *pod-pisa-t'* 'sign$^{\text{PFV}}$' is a function that maps the stem *pisa-* 'write' to a state of being signed, "write ↦ signed". On the other hand, one can try to assign a general meaning to every prefix (or a restricted set of meanings, if one assumes that prefixes are polysemous). Since prefixes are mostly drawn from the prepositional inventory, such a meaning can naturally be thought of in terms of spatial relations. For *pod-* that would yield 'under', the meaning of its prepositional counterpart, cf. *pod stolom* 'under the table'. The state of being signed, then, would be

[29] Throughout this section I will be abstracting away from prefixed verbs whose derivation involves both a prefix and the reflexive clitic *-sja*, e.g. *u-čita-t'-sja* 'get one's fill of reading'. See Žaucer (2009) for an analysis of this type of verbs in Slovenian.

derived as an outcome of a semantic interplay between the stem, the prefix and event structural information.

For the purposes of the current section, however, I will be abstracting away from this dichotomy. In order to keep the presentation simple, I will assume that prefixes are merely relations between individuals and states, and that descriptive properties of those states are always specified in every lexical context, in accordance with (69):

(70) a. $[\![na\text{-}]\!] = \lambda x.\lambda s.\textsc{written}(x)(s)$, in the context of *pisa-* 'write'
b. $[\![pod\text{-}]\!] = \lambda x.\lambda s.\textsc{signed}(x)(s)$, in the context of *pisa-* 'write'

As was already mentioned above, the discussion below is based on the assumption that prefixes and the secondary imperfective do not denote directly any aspectual verb stem meanings. Verb phrases, with all the relevant pieces of morphology merged with the stem, are aspectless predicates of events, relations between events and states or the like.

4.2 Compositionality

The first parameter that divides lexically prefixed verbs into two major groups is whether *only* events from the original extension of a non-prefixed verb can be found in the extension of its prefixed counterpart.[30] This parameter will be called COMPOSITIONALITY and defined as in (71).

(71) COMPOSITIONALITY
Both inside and outside of a lexically prefixed configuration, the verb stem denotes events with the same descriptive properties.

(71) tells apart prefixed verbs like (72) and (73), on the one hand, and (74), on the other.

(72) Volodja na-pisa-l pis'm-o.
Volodja LP-write-PST.M letter-ACC
'Volodja wrote a letter.'

[30]A technical clarification is due at this point. Above, it was assumed that transitive verbs denote relations between individuals and events, hence their extensions are not (characteristic functions of) sets of eventualities, but sets of ordered pairs of individuals and eventualities. An expression like "an eventuality is in the extension of a non-derived verb" is therefore to be understood as a shorthand for "an eventuality is an eventive member of an ordered pair in the extension of a non-derived verb".

(73) Volodja pod-pisa-l dogovor.
Volodja LP-write-PST.M agreement.ACC
'Volodja signed an agreement.'

(74) Volodja o-pisa-l scen-u prestupleni-ja.
Volodja LP-write-PST.M scene-ACC crime-GEN
'Volodja described the crime scene.'

Events that *na-pisa-t'* 'write' in (72) and *pod-pisa-t'* 'sign' in (73) describe are writing events, those that fall under the extension of *pisa-t'* 'write', the prefixless counterpart of (72)–(74). The agent who writes a letter or signs an agreement has to write. Both (72) and (73) thus entail (75) (abstracting away from the aspectual difference between them).

(75) Volodja (čto-to) pisa-l.
Volodja something.ACC write-PST.M
'Volodja was writing / has written (something).'

Whatever validates (75) as the description of a writing eventuality should also be part of the description of the eventive components of (72)–(73). To put it differently, events that fall under *na-pisa-t'* 'write' and *pod-pisa-t'* 'sign' form (not necessarily proper) subsets of the extension of *pisa-t'* 'write'. This is shown in (76).[31]

(76) a. $\{e : \exists s[[\![\textit{na-pisa- pis'mo}]\!](s)(e)]\} \subseteq \{e : \exists x[[\![\textit{pisa-}]\!](x)(e)]\}$
b. $\{e : \exists s[[\![\textit{pod-pisa- dogovor}]\!](s)(e)]\} \subseteq \{e : \exists x[[\![\textit{pisa-}]\!](x)(e)]\}$

In contrast, (74) fails to entail (75). (74) can but does not have to involve a writing activity: to describe an object one does not have to write; for instance, a verbal description would suffice. Therefore, the set of describing events is not a subset of writing events:

(77) $\{e : \exists s[[\![\textit{o-pisa- scenu prestuplenija}]\!](s)(e)]\} \not\subseteq \{e : \exists x[[\![\textit{pisa-}]\!](x)(e)]\}$

If the prefix only contributes a property of (result) states into a complex structure, events that bring those states about must be introduced by the verb stem. One has

[31] Note that the object of a prefixed verb derived from the simplex imperfective *pisa-t'* can but does not have to be the theme of writing. In (69), for example, the agreement is an object of signing without being an object of writing. For this reason, the right-hand part of the formulas in (76)–(77) contain the most inclusive set of writing events: the set of wiring events in which something is written. This topic will be taken up shortly, in Section 4.3.

to conclude, therefore, that the extensions of *pisa-* in *o-pisa-t'* 'describe' and of the unprefixed *pisa-* are different. After the prefixation of *o-*, the extension of *pisa-* widens: it acquires exactly those eventualities where something is described with no writing being involved.[32] There is no obvious way of deriving this widening in a compositional way. (71) captures exactly this property of lexically prefixed verbs like *o-pisa-t'* 'describe'. On *o-pisa-t'*, (71) fails.

While the eventive component of *o-pisa-t'* 'describe' is not included into that of *pisa-t'* 'write', (77), these components do have a non-empty overlap. This overlap obtains if a description is done in writing. But one can easily find prefixed verbs with no such an overlap. For example, *ot-kry-t'* 'open' is derived from *kry-t'* 'cover' by the lexical prefix *ot-* (cf. the preposition *ot* 'from'; diachronically, *ot-kry-t'* 'open' is 'from-cover'). In an opening eventuality nothing is getting covered, so for *kry-t'* 'cover' and *ot-kry-t'* 'open' (78) holds.

(78) $\{e : \exists s[[\![\textit{otkry- dver'}]\!](s)(e)]\} \cap \{e : \exists x[[\![\textit{kry-}]\!](x)(e)]\} = \varnothing$

Therefore, prefixes like *ot-* in *ot-kry-t'* 'open' or *o-* in *o-pisa-t'* 'describe' are NON-COMPOSITIONAL. They induce a non-compositional shift in the denotation of a verb stem whereby after prefixation it starts describing events that had not been there originally. COMPOSITIONAL are prefixes with which this does not happen.

In what follows, I will focus on compositional prefixes like *na-* and *pod-* in (72) and (73), for which (71) does hold.

4.3 Argument structure of a prefix

(Result) states that compositional lexical prefixes introduce into the event structure can be further divided into several classes. The most basic division is induced by whether a state involves one or two participants. Accordingly, the argument structure of a prefix can be MONADIC or DYADIC, to use Hale & Keyser's (2002) term.

(79) PARTICIPANTS OF A RESULT STATE
MONADIC PREFIXES are associated with result states with a single participant, its holder. DYADIC PREFIXES introduce states corresponding to a relation between two entities, figure and ground, and take two arguments.

[32] In addition, the extension loses those writing events that do not take an argument into a state of being described. This is, however, what *o-pisa-t'* 'describe' has in common with *pod-pisa-t'* 'sign'. Not all writing events culminate in a state of having been signed. Likewise, there are writing events irrelevant for bringing about a state of being described. I will address such "narrowing-down" effects of lexical prefixation shortly, in Section 4.3.2–Section 4.3.3. I am grateful to an anonymous reviewer who encouraged to introduce this qualification here.

An important background assumption underlying (79) is that a prefix is only considered dyadic if both of its arguments can be associated with overt material in the syntax. Otherwise, a prefix is treated as monadic. This strategy, which follows Romanova (2007), is not the only admissible one, of course. In other systems, Babko-Malaya (1999) and Biskup (2019) among them, (near) all prefixes are dyadic, but in certain cases one of the arguments is obligatorily left implicit for some or other reason.

These two approaches seem to manifest different theoretical desiderata. (79) prioritizes minimizing implicit arguments in the argument structure of a prefix whose empirical reality is difficult or impossible to assess. The alternative aims at keeping the semantics and syntax of prefixes as uniform as possible. Whether the two approaches make different empirical predictions should ultimately depend on the representational and interpretational status of implicit arguments (see e.g. Landau 2010, Bhatt & Pancheva 2017) assumed within the latter type of theory. This topic, however, requires more examination than one can find in the current literature.

Monadic prefixes fall into three subtypes characterized below in Section 4.3.1–Section 4.3.4. In Section 4.3.5, dyadic prefixes are briefly discussed.

4.3.1 Monadic prefixes and argument identity

(72)–(73) with *na-pisa-t'* 'write$^{\text{PFV}}$' and *pod-pisa-t'* 'sign$^{\text{PFV}}$' above are both examples of event structures in which the prefix is monadic. A characteristic property of verbs with monadic prefixes is that the argument corresponding to the holder of result state is typically realized in the direct object position, rather than appearing in the subject or indirect object slots.[33]

Na-pisa-t' and *pod-pisa-t'* instantiate two major subtypes of event structures with monadic prefixes. *Na-pisa-t'* projects VPs where the object is SUBCATEGORIZED for by the verb stem. This is not the case with *pod-pisa-t'*.

In (72), *pis'mo* 'letter' is construed both as the theme of writing and the holder of a result state of having been written. In contrast, *pis'mo* 'letter' in (73) is the holder of the result state of having been signed without being the theme of writing.

One more illustration appears in (80)–(81).

(80) Volodja e-l {jablok-o / #puz-o}.
Volodja eat-PST.M apple-ACC belly-ACC
(Intended:) 'Volodja was eating/ate {an apple / a belly}.'

[33] I am grateful for an anonymous reviewer for encouraging me to make this clarification.

(81) Volodja na-e-l {puz-o / *jablok-o}.
Volodja LP-eat-PST.M belly-ACC apple-ACC
(Intended:) 'Volodja acquired {a belly / an apple} by eating.'

The verb *es-t'* 'eat', when it appears in the unprefixed configuration in (80), is a verb of consumption that takes a DP denoting edible substances as its object. In (81), the object of *na-est'* 'acquire by eating', derived from *es-t'* 'eat' by the lexical prefix *na-*, is *puzo* 'belly', which is not consumed but effected in the course of eating. In (80), it can only be understood as the object of eating. Similar effects have been observed in (27a)–(27b) in Section 2, where lexical prefixes were said, descriptively, to be able to introduce non-subcategorized arguments or change the thematic properties of the object.

These observations lead to establishing two classes of monadic prefixes in (82), based on identity of the holder of a result state and the internal argument of a verb stem. I will call them ARGUMENT-SHARING (AS) PREFIXES and NON-ARGUMENT-SHARING (nAS) PREFIXES:

(82) ARGUMENT IDENTITY
a. ARGUMENT-SHARING PREFIXES: The argument of a prefix, the holder of a result state, is identical to the internal argument of the non-derived verb stem.
b. NON-ARGUMENT-SHARING PREFIXES: The holder of a result state is different from the internal argument of the non-derived verb stem.

A simple heuristic that separates the two types of prefixes is based on whether a sentence with a lexically prefixed verb entails a corresponding prefixless sentence with the same object (again, abstracting away from aspectual differences). Whereas the entailment from *na-pisa-t' pis'mo* to *pisa-t' pis'mo* goes through, that from *pod-pisa-t' pis'mo* to *pisa-t' pis'mo* does not, (83a)–(83b), or, in set-theoretic terms, (84a)–(84b).

(83) a. Nadja na-pisa-l-a pis'm-o. → Nadja pisa-l-a pis'm-o.
Nadja LP-write-PST-F letter-ACC Nadja write-PST-F letter-ACC
'Nadja wrote the letter.' entails 'Nadja was writing the letter.'
b. Nadja pod-pisa-l-a pis'm-o ↛ Nadja pisa-l-a pis'm-o
Nadja LP-write-PST-F letter-ACC Nadja write-PST-F letter-ACC
'Nadja signed the letter.' does not entail 'Nadja was writing the letter.'

(84) a. $\{e : \exists s\ [\![\textit{na-pisa- pis'mo}]\!](s)(e)\} \subseteq \{e : [\![\textit{pisa- pis'mo}]\!](e)\}$
b. $\{e : \exists s\ [\![\textit{podpisa- pis'mo}]\!](s)(e)\} \not\subseteq \{e : [\![\textit{pisa- pis'mo}]\!](e)\}$

Na-pisa-t' 'write' thus contains an argument-sharing prefix, whereas *pod-pisa-t'* 'sign' and *na-est'* 'acquire by eating' in (81) do not.

4.3.2 nAS prefixes: putting events and states together

The phenomena discussed in the previous section pose a number of questions about how a result state description projected by a prefix is integrated into a larger configuration. What is the relationship between the verb stem and the projection of a prefix? How is the argument structure of their combination computed? How can the difference between AS and nAS prefixes be accounted for?

These questions are not specific to prefixal configurations in Slavic languages. Exactly the same type of issues are brought out by other types of resultative constructions cross-linguistically, e.g., by the familiar resultatives in English and German.

Consider the resultatives in (85), which are nAS, since the objects *the pub, seine Familie* 'his family' and *the baby* are not subcategorized for by the verb.

(85) a. They drank the pub dry. (Rappaport Hovav & Levin 2001)

b. Er hat seine Familie magenkrank gekocht. (Kratzer 2005)
he AUX self's family stomach.sick cooked
'He cooked his family stomach sick.'

c. John sang the baby asleep. (Rothstein 2004)

The composition of resultative XPs like those in (85a)–(85c) has been addressed in a variety of studies (e.g. Rappaport Hovav & Levin 2001, Rothstein 2004, Kratzer 2005, to mention just a few). Even though details of specific proposals vary, there seems to be a fundamental agreement that arguments not subcategorized for by the verb stem are licensed as arguments of a result state expression like *dry*, *magenkrank* 'stomach sick', and *asleep*.

Along similar lines, Babko-Malaya (1999), Romanova (2007), Gehrke (2008), Tatevosov (2015a), Biskup (2019) all argue that in a prefixal configuration like (73) or (81) the object DP originates as an argument of a prefix. Gehrke's (2008: 164ff.) proposal, for example, assumes that lexical (= internal, in her terminology) prefixes head a functional projection that V takes as its complement, the object DP being merged in its Spec, see Figure 2.

Less uniformity across existing theories is attested as to how a result state description is integrated into a larger configuration and what happens to the internal argument of the verb stem, if present.

I will introduce a toy fragment that offers a possible treatment of nAS-prefixed verbs like *pod-pisa-t'* 'sign' in (73), repeated as (86), and discuss several proposals found in the literature against this background.

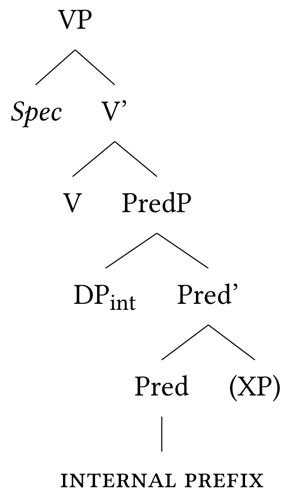

Figure 2: Proposal according to Gehrke (2008)

(86) Volodja pod-pisa-l dogovor.
Volodja LP-write-PST.M agreement.ACC
'Volodja signed an agreement.'

This fragment will be used to assess theories that vary along the two parameters in (87) and (88).

(87) Result states
  a. Result states are integrated by a special rule of construal.
  b. Result states are introduced by a special morphosyntactic device.

(88) Internal arguments
  a. Internal arguments get existentially bound.
  b. Internal arguments are not there to begin with.

First, let us assume, as before, that the semantic representation of a non-prefixed transitive verb looks like (89), taking *pisa-* 'write' as an example:

(89) $[\![pisa\text{-}]\!] = \lambda x.\lambda e.\text{WRITE}(x)(e)$

Following (69), two irreducible ingredients of the semantic representation of a monadic prefix should be a predicate of states, specifying its descriptive properties, and an individual argument construed as the holder of that state. For the prefix *pod-* in (86) this yields (90).

(90) $[\![pod\text{-}]\!] = \lambda x.\lambda s.\text{SIGNED}(x)(s)$, in the lexical context of *pisa-* 'write'

As was mentioned above for the moment I am putting aside the question of how the descriptive properties of a specific state in a specific lexical context are computed. This will be the topic of Section 4.4.

With these assumptions, a possible outline of the system that combines the denotations of the two expressions may look like (91a)–(91h).

(91) a. $[\![pod\text{-}]\!] = \lambda x.\lambda s.\text{SIGNED}(x)(s)$ (Assumption in (69))

b. $[\![pisa\text{-}]\!] = \lambda x.\lambda e.\text{WRITE}(x)(e)$ (Lexicon)

c. $[\![pis'mo\ pod\text{-}]\!] = \lambda s.\text{SIGNED}(\text{LETTER})(s)$ (FA)

d. $[\![\text{CS}]\!] = \lambda P.\lambda s.\lambda e.P(s) \wedge \text{CAUSE}(s)(e)$ (Causative Shift)

e. $[\![\text{CS [} pis'mo\ pod\text{-]}]\!] = \lambda s.\lambda e.\text{SIGNED}(\text{LETTER})(s) \wedge \text{CAUSE}(s)(e)$ (FA)

f. $[\![\exists_x]\!] = \lambda R.\lambda e.\exists x[R(x)(e)]$ (Existential Closure)

g. $[\![\exists_x\ pisa\text{-}]\!] = \lambda e.\exists x[\text{WRITE}(x)(e)]$ (FA)

h. $[\![[\exists_x\ pisa\text{-}]\ [\text{CS [} pis'mo\ pod\text{-]}]]\!]$ (Event Identification)
$= \lambda s.\lambda e.\exists x[\text{WRITE}(x)(e) \wedge \text{SIGNED}(\text{LETTER})(s) \wedge \text{CAUSE}(s)(e)]$

In (91c), the denotation of the prefix, (91a), merges with its argument, the holder of a state that falls under $\lambda s.\text{SIGNED}(x)(s)$. (For simplicity, I keep on representing DPs as individual constants. Also, I ignore the subsequent movement of the DP for case reasons.)

The resulting predicate of states undergoes Causative Shift, (91d)–(91e), which makes a relation between events and states, causally connected, out of a state description. As a consequence, states from the extension of (91c) are construed as result states, those brought about by a causing eventuality. A rationale behind Causative Shift can be thought of in terms of naturalness of a state. It captures the intuition that states like being signed described by *pod-* in examples like (86) are not found in non-derived environments across languages. While languages systematically have lexical items like *sick* or *open* whose denotations are states of being sick and open, no language that we know possesses a lexical item for states like being signed. Such states only come out as denotations of derived state descriptions, e.g. "passive participles" in Germanic or Slavic languages.[34]

[34]Koontz-Garboden & Beavers (2017: and subsequent work) make a similar case for non-derived change of state verbs whose roots introduce a state description. They argue such roots fall into two types: some come with entailments of change, being similar in that respect to a state of being signed in (86), while others ("property concept states") do not. I am grateful to an anonymous reviewer for turning my attention to this parallelism.

In (91f)–(91g), the internal argument of *pisa-t'* 'write' from (91b) undergoes existential closure, yielding a predicate of events. This captures the pattern in (76b) whereby sentences with *pod-pisa-t'* 'sign' entail that something has been written in the course of signing a letter.

Finally, the output of Causative Shift in (91e) and the verb stem denotation in (91g) combine by a version of Kratzer's (1996) Event Identification. In (91h), Event Identification restricts the eventive members of the relation in (91e) to those that fall under (91g). (91h) is thus the relation between events and states such that an event in which something has been written causes the letter to enter a state of being signed. This is exactly the meaning of (86).

A similar derivational scenario can be assumed for unergative lexically prefixed verbs like *za-rabota-t'* 'earn' in (25a)–(25b) above or *ob-rabota-t'* 'process' in (92a), derived from *rabota-t'* 'work' in (92b), where the lexical prefixation leads to transitivization.

(92) a. Komitet ob-rabota-l sto zajavok.
committee LP-work-PST.3SG hundred.ACC application.GEN.PL
'The committee processed one hundred applications.'

b. * Komitet ra-bota-l sto zajavok.
committee work-PST.3SG hundred.ACC application.GEN.PL
Intended: 'The committee was working one hundred applications.'

Arguably, the only difference between (92a) and (86) is that unergatives lack the internal argument altogether and thus denote predicates of eventualities to begin with: $\lambda e.\textsc{work}(e)$. As a result, Existential Closure in (91f) does not happen and Event Identification applies to this denotation directly. This creates a relation between working events and result states of the applications having been processed:

(93) ⟦[*rabota-*] [CS [*sto anket ob-*]]⟧ =
$\lambda s.\lambda e.\textsc{work}(e) \wedge \textsc{processed}(\textsc{one hundred applications})(s) \wedge \textsc{cause}(s)(e)$

The derivation in (91), apart from lexical information, functional application, and event identification independently argued for in the literature, makes use of two more specific mechanisms for deriving the denotation of a prefixed VP: Causative Shift and Existential Closure. These are exactly the parts of the derivation where existing theories of resultative predication vary along the two dimensions in (87)–(88).

The first dimension in (87) has to do with specific assumptions about the way the eventive and stative components of a complex description are combined. In (91), this is accomplished by Causative Shift. Establishing a relation between events and result states is an unavoidable ingredient of any theory of resultative predication. However, the relation can in be conceived of as an output of a rule of construal, as in (87a), or as the denotation of a phonologically null morpheme, as in (87b).

Some theories assume a special rule of construal. For English resultatives, for example, Rappaport Hovav & Levin (2001) propose Template Augmentation, while Rothstein (2004) makes use of the TPCONNECT relation. Beck's (2005) Principle R falls under this type of approach, too. Kratzer (2005) considers both options for the derivation of Germanic resultatives, and having discussed a number of pros and contras, opts for the morphological solution. She poses a special derivational morpheme represented in the syntax which is interpreted as introducing the relation of immediate causation.

For Slavic languages, both types of semantic theories of lexical prefixation have been proposed. Some, in line with Kratzer (2005), assume a syntactically represented (and typically silent) grammatical element. For instance, Babko-Malaya (1999: 53) suggests a morpheme that adjoins to a prefix, takes its denotation as an argument and yields a relation to which the verb stem denotation applies at the next stage of derivation. Biskup (2019: ch. 3) introduces a similar element as the denotation of a functional head located in between the verb stem and the projection of a prefix. On the other hand, Gehrke (2008) argues that Rothstein's architecture is better suited than Kratzer's to account for attested constraints on the derivation of resultative configurations cross-linguistically; see Gehrke (2008: 61–63) for specifics of her critical assessment of Kratzer (2005) and Rothstein (2004).

It is not immediately obvious, however, whether the choice between a special rule of construal or a silent functional element necessarily makes different predictions with respect to the event structure or argument structure of nAS lexically prefixed verbs. While (91d) assumes the latter, I believe that nothing in what follows should be incompatible with the alternative.

The other parameter of theoretical variation in (88a)–(88b) is how a theory treats an argument that appears as an object in an unprefixed configuration and is perceived as a subcategorized argument of a verb stem. For the transitive verb stem *pisa-*, for example, this argument is an object effected in the course of an event ('write a letter' and the like).

The sketch in (91a)–(91h) relies on the three assumptions in (94).

(94) Argument projection in nAS configurations
  a. Lexically, transitive verbs denote relations between events and (internal) arguments.
  b. The argument of a prefix is syntactically realized.
  c. The argument of a verb stem gets existentially bound.

The same or a similar set of assumptions can be found, for example in Rothstein (2004) for English resultatives and in Babko-Malaya (1999: 53) for Russian resultative lexical prefixes. A related proposal appears in Bruening (2021), where a verb can merge with a head that prevents it from projecting its internal argument and existentially binds it. (The other such head makes the implicit object definite.)

An alternative advanced by Kratzer (2005) for Germanic and Biskup (2019) for Slavic languages does not assume (94a), thus eliminating (94c). Specifically, Biskup (2019: 48-50) suggests that prefixless transitive verbs denote predicates of eventualities, while their internal arguments are introduced by the functional structure within the complement of V. Biskup's proposal amounts to treating the event / argument structure of prefixed and unprefixed configurations as being products of different derivations, unlike in (91a)–(91h), where the former is derived from the latter.

Assessing these two types of theories, one can observe that neither is exempt from certain vulnerabilities. Theories assuming that internal arguments are not true arguments of a verb stem need additional stipulations to explain selectional restrictions that stems massively impose on internal arguments, but not on external arguments (as well as other asymmetries between external and internal arguments discussed, e.g., in Kratzer 2003). Theories that assume existential closure of the internal argument (or a similar mechanism) bear the burden of finding an independent motivation for this operation. For example, Tatevosov (2015a) argues that at least for some nAS lexically prefixed verbs existential closure of the internal argument results in coercion, which prevents the derivation from outputting a predicate with an empty extension. However, it is not clear whether such considerations extend to all prefixal configurations.

Properly motivating something like the existential closure may also be substantial for preventing the system from generating ungrammatical structures in which both the internal argument of a stem and the argument of a prefix are projected. For *pod-pisa-t'* 'sign$^{\text{PFV}}$', as a reviewer points out, that would yield something like (95) with the meaning 'one's name has been written, and the agreement is consequently in a signed state'.

(95) * Volodja pod-pisa-l imja dogovor.
Volodja LP-write-PST.M name.ACC agreement.ACC
Intended: 'Volodja signed the name an agreement.'

One strategy of ruling (95) out may be to explore possible ways in which it could have come about and find a reason why such derivations are problematic. To produce (95) out of the available ingredients one may need, assuming the syntax in (96), a derivation along the lines of (97).

(96) $[_{\mathrm{VP}}$ *imja* $[_{\mathrm{V'}}$ *pisa-* [CS *pismo pod-*]]]

(97)
a. $[\![pisa\text{-}]\!] = \lambda x.\lambda e.\textsc{write}(x)(e)$ (= (91b))
b. $[\![\text{CS } [\ pis\text{'}mo\ pod\text{-}\ ]]\!] = \lambda s.\lambda e.\textsc{signed}(\textsc{letter})(s) \wedge \textsc{cause}(s)(e)$ (= (91e))
c. $[\![pisa\text{-}\ [\text{CS } [\ pis\text{'}mo\ pod\text{-}\ ]]]\!]$ (Generalized Event Identification)
$= \lambda x.\lambda s.\lambda e.\textsc{write}(x)(e) \wedge \textsc{signed}(\textsc{letter})(s) \wedge \textsc{cause}(s)(e)$
d. $[\![imja\ pisa\text{-}\ [\text{CS } [\ pis\text{'}mo\ pod\text{-}\ ]]]\!]$
$= \lambda s.\lambda e.\textsc{write}(\textsc{name})(e) \wedge \textsc{signed}(\textsc{letter})(s) \wedge \textsc{cause}(s)(e)$ (FA)

Unlike in (91), the internal argument position does not undergo existential closure in (97), but stays open. Then, to combine (97a) with (97b) one would need a rule of construal that can be thought of as a generalized version of event identification. Kratzer's (1996) rule only identifies events that fall under a particular description (of type $\langle v, t\rangle$) with events that stand in a particular relation to individuals (of type $\langle e, \langle v, t\rangle\rangle$). Generalized event identification (GEI) would identify events that are elements of any relations to any other entities:

(98) Generalized event identification (GEI)
If α is a branching node and $\{\beta, \gamma\}$ is the set of its daughters,
$[\![\beta]\!]$ is in $D_{\langle\sigma_{\alpha_1},\langle\ldots,\langle\sigma_{\alpha_n},\langle v,t\rangle\rangle\rangle\rangle}$, and $[\![\gamma]\!]$ is in $D_{\langle\sigma_{\beta_1},\langle\ldots,\langle\sigma_{\beta_k},\langle v,t\rangle\rangle\rangle\rangle}$,
then $[\![\alpha]\!] = \lambda\omega_{\alpha_1}.\ldots.\lambda\omega_{\alpha_n}.\lambda\omega_{\beta_1}.\ldots.\lambda\omega_{\beta_k}.\lambda e.\ [\![\alpha]\!](\omega_{\alpha_1})(\ldots)(\omega_{\alpha_n})(e) =$
$[\![\beta]\!](\omega_{\beta_1})(\ldots)(\omega_{\beta_k})(e) = 1$, where $\sigma_{\alpha_i}, \sigma_{\beta_j}$ and the like are types, and $\omega_{\alpha_i}, \omega_{\beta_j}$ are corresponding variables.

Now, with GEI in (98), in (97c)–(97d) we have identified events of writing the name with events that bring about a state of an agreement having been signed.

Given that (95) is drastically ungrammatical, something must be wrong with this scenario, and the question is what exactly. One can come up with a number of possible answers. GEI may not be a rule of construal that is available in natural languages in the first place. This is not implausible, given that cross-linguistically, we do not find descriptions, either morphosyntactically simplex or derived, of complex eventualities *e* such that *e* is, for example, an eventuality of Olga eating a sausage and Cathy swimming in the pool, or the like. Another reason may be that while there is nothing wrong with GEI in itself, there are output constraints that descriptions like (97d) fail to meet. For instance, one can speculate that larger syntactic configurations in which VPs like (96) occur fall short of the sources of structural case for both nominals. Also, it is not unimaginable that (97d) does not meet some or other semantic well-formedness requirement based, e.g., on fine-grainedness considerations.

As one can see, at this juncture we may take several ways to go. However, discussing this in further detail should be a topic for a separate study, which I am not attempting in what follows.

One more potentially sensitive issue with the existential closure in (91f) may have to do with verbs of creation, as a reviewer points out. 'Write' is such a verb, and its internal argument, the one that undergoes $\exists_x$ in (91g), is an effected object. Effected objects only exist at the end of an event of creation. For this reason, the reviewer indicates, existentially bound objects are only warranted in perfective sentences, which entail the culmination, but not in imperfective sentences that come with no such entailment. However, there is one aspect of (91) that may make it less problematic than initially appears. Consider first a predicate like (99).

(99) $\lambda e.\exists x.\textsc{write}(x)(e) \wedge \textsc{letter}(x)$

In (99), an object that falls under LETTER only exists at the end of writing, and (99) falls short of capturing that. (91g), repeated as (100), is crucially different in that an effected individual falls under no particular description; it is only identified as an entity staying in the write relation to an event.

(100) $\lambda e.\exists x.\textsc{write}(x)(e)$

WRITE, as any verb of creation, is incremental (e.g., Piñón 2008). This means, due to mapping to subobjects, that in any writing subevent there is, trivially, something written. This written entity may not qualify as an entity that can be described by LETTER or, in fact, by any other (nominal) predicate. But (100), unlike (99), does not require that. For there to be something written it is enough that some writing has occurred. I believe that the combination of WRITE being incremental and the lack of any descriptive content associated with the effected object circumvents the problem that would have emerged otherwise and does in fact emerge with any predicate like (99).

4.3.3 AS prefixes

Having in mind the spectrum of possibilities for nAS lexical prefixes discussed above, one can address the derivation of AS prefixal configurations. One of those are "pure perfectivizing", or "empty" prefixes like *na-* in *na-pisa-t'* 'write$^{\text{PFV}}$', or *pro-* in *pro-čita-t'* 'read$^{\text{PFV}}$', whereby the prefixation does not induce any perceivable change of the lexical meaning. With such prefixes, an entity undergoing writing or reading is identical to one attaining a state of having been written or read. In Section 3, for *na-pisa-t' pis'mo* 'write a letter' the representation repeated in (101) is assumed.

(101) $[\![[_{\text{VP}}\ \textit{na-pisa- pis'mo}]]\!]$
$= \lambda s.\lambda e.\textsc{write}(\textsc{letter})(e) \wedge \textsc{cause}(s)(e) \wedge \textsc{written}(\textsc{letter})(s)$

In (101), the fact that the letter is understood as the object of writing and the holder of a state of having been written is fully explicit: the object DP figures in (101) as an argument of both the verb and the prefix *na-*. To achieve this result, one may need to make certain assumptions about a corresponding syntactic structure and syntax-semantic interface. If, for example, a DP is allowed to merge in more than one thematic position, and if such positions are interpreted as variables bound by the topmost copy of the DP, (101) is not difficult to derive. As an example of a theory of this type one can think of Ramchand's (2008 and elsewhere) First Phase Syntax and its later developments (for a recent discussion see Svenonius 2016).

As before, (101) is not the only admissible option. In the literature, one can find two logically possible alternatives to (101). First, the object is construed as being an argument of an AS verb without being an argument of a prefix. Second, the other way around, the object is an argument of a prefix, but not of a verb.

The first approach is what Babko-Malaya (1999: 55-56) proposes for pure perfectivizing prefixes, which she keeps apart from the rest of the lexical prefixes. Rather than introducing a result state, empty prefixes take the denotation of a verb stem as an argument and modify it by contributing the entailment of "completion". However, this comes at a cost: VPs containing empty prefixes are predicted to manifest less subevental complexity than other VPs projected by lexically prefixed verbs. As discussed in Section 3, this is not the actual pattern attested in Russian and other Slavic languages. It is not clear if the analysis can be amended in order to incorporate observations from Section 3 while maintaining that the object is an argument of the stem only.

Under the second approach, the object of AS-prefixed verbs can be treated on a par with the object of nAS-prefixed verbs, that is, as an argument of a prefix only. This is the path Biskup (2019) takes, assuming no derivational asymmetry between AS and nAS configurations. Romanova's (2007) approach falls under this type, too. This implies that the internal argument of the verb stem is treated identically in both AS and nAS configurations. As was mentioned above, for Biskup (2019), this argument is absent altogether.

If, on the other hand, one assumes a derivational scenario for nAS-prefixes along the lines of (91), where the argument of the stem gets existentially bound, the derivation of a parallel AS-configuration would output the relation in (102) instead of (101).

(102) $[\![[_{\mathrm{VP}}$ *na-pisa- pis'mo*$]]\!]$
$= \lambda s.\lambda e.\exists x[\textsc{write}(x)(e) \wedge \textsc{cause}(s)(e) \wedge \textsc{written}(\textsc{letter})(s)]$

At first sight, (102) looks very wrong: the relation is predicted to have events in it where something other than a letter undergoes writing. One can speculate, however, that in our world as well as in other worlds where causal regularities resemble ours, the extensions of (101) and (102) may come out indistinguishable. The predicate of states $\lambda s.\textsc{written}(\textsc{letter})(s)$ comes, according to (69), with very specific descriptive properties. States that fall under this predicate are states of having been written, not other kinds of states that may result from writing events. World knowledge, part of which is knowledge of causal dependencies, tells us that for any x to attain such a state, x has to undergo writing, otherwise an event and a state would fail to fall under CAUSE. This restricts the events that can enter the relation in (102) to those where the theme of writing is the same as the holder of the state, hence the relation in (102) would reduce to the relation in (101).

4.3.4 Restrictive and non-restrictive AS prefixes

AS lexically prefixed verbs do not form an entirely homogeneous class. As the above discussion suggests, this class contains all the verbs with "empty" prefixes. It does not only contain such verbs, however. One can identify many prefixes that are AS without being empty.

Consider (103)–(104).

(103) Volodja pro-čita-l doklad.
Volodja LP-read-PST.M report.ACC
'Volodja read the report.'

(104) Volodja za-čita-l doklad.
Volodja LP-read-PST.M report.ACC
'Volodja read the report out loud.'

Both (103) and (104) pass the diagnostic for argument-sharedness in (83)–(84), since both entail, modulo the aspectual difference, the prefixless sentence in (105) with the same object.

(105) Volodja čita-l doklad.
Volodja read-PST.M report.ACC
'Volodja was reading the report.'

While *pro-čita-t'* 'read$^{\text{PFV}}$' in (103) is a paradigmatic example of "pure perfectivization", *za-čita-t'* in (104) has a more specific meaning: it comes with the inference that a reading of the report is aloud and, possibly, in front of an audience.

What (103)–(104) and lots of similar examples tell us can be captured in the following way. Every event of reading a report can take the report into a state associated with the prefix *pro-*, a state of having been read. No matter what specific properties a reading event has, whether reading is silent or aloud, slow or fast, thoughtful or superficial, it can culminate in a state that falls under $\lambda s.\textsc{read}(\textsc{report})(s)$. But only a subset of reading events can culminate in a state of having been read out aloud in front of the audience, associated with *za-*. Events of silent reading do not enter a causal chain where the maximal element is such a state. For *za-čita-t'* 'read out aloud' and *pro-čita-t'* 'read$^{\text{PFV}}$', therefore, the following holds:

(106) For any x,
a. $\{e : \exists s[\textsf{CAUSE}(s)(e) \wedge \textsc{read}(x)(s)]\} = \{e : \textsc{read}(x)(e)\}$
'Events that cause a state of having been read are reading events.'
b. $\{e : \exists s[\textsf{CAUSE}(s)(e) \wedge \textsc{read out aloud}(x)(s)]\} \subset \{e : \textsc{read}(x)(e)\}$
'Events that cause a state of having been read out aloud are a proper subset of reading events.'

Prefixes like *pro-* and *za-* can thus be called NON-RESTRICTIVE and RESTRICTIVE, respectively. According to (106a), the set of events that bring about a *pro*-state of having been read is identical to the entire set of culminating reading events that fall under the unprefixed verb. *Pro-* is thus non-restrictive on the extension of *čita-* 'read'. *Za-* is restrictive in the sense that events that cause a *za*-state of having been read out aloud form a restricted proper subset of reading events, (106b). For (106a)–(106b), this is represented diagrammatically in Figure 3.

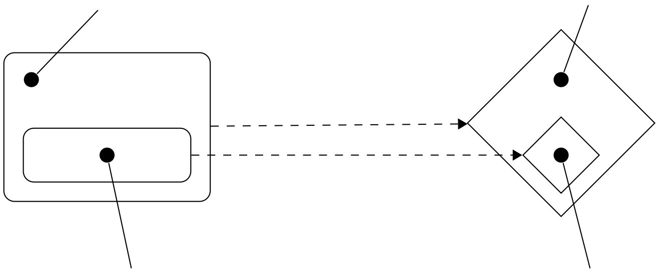

Figure 3: Process and result state components of *pro-čita-t'* 'read$^{\text{PFV}}$' and *za-čita-t'* 'read$^{\text{PFV}}$ out aloud'

It is for this reason that the restrictive AS prefixes produce intuitions that the lexical meaning of a non-derived verb has been affected by prefixation. The non-derived predicate has the entire set of reading eventualities in its extension. But when it comes to bringing about a state of having been read aloud in front of the audience, associated with the prefix *za-* in (104), only a proper subset of them is involved, a different eventuality description. Crucially, restrictive AS-prefixation is always a strengthening of the input lexical meaning, never a weakening, and the proper subset relation in (106b) captures that in a principled way. This case is thus crucially different from examples like (74) with *o-pisa-t'* 'describe' in Section 4.2. Whereas both *za-čita-t'* 'read out aloud' and *o-pisa-t'* 'describe' only make use of a proper part of events from the original extension of the unprefixed verb, *o-pisa-t'* 'describe', but not *za-čita-t'* 'read out loud' acquires additional events that have not been there before prefixation; see also footnote 32.

These considerations allow us to give the notion of "empty prefixation" the following content:

(107) Empty prefixes as non-restrictive AS prefixes

a. Empty prefixes denote result states compatible with every eventuality in the extension of a non-derived verb.

b. A lexical prefix P is empty, or non-restrictive with respect to the verb stem V iff $\forall x \forall e[[\![V]\!](x)(e) \rightarrow \Diamond \exists s[[\![P]\!](x)(s) \wedge \textsc{cause}(s)(e)]]$

(107b) generalizes what is said in (106a) about the non-restrictive prefix *pro-* in *pro-čita* 'read$^{\text{PFV}}$'. According to (107b), P is empty with respect to V just in case the following holds: if V's denotation is true of an event and an individual, this event can take this individual into a state described by P. This is not so for restrictive prefixes: for them, there are events that stay in the V-relation to an individual but cannot culminate in a P-state that would hold of the same individual. As we have just seen, not every reading event can bring about a state of having been read aloud in front of the audience described by the prefix *za-*. The $\Diamond$ operator in (107b) takes care of the fact that the extension of a non-derived verb in a world can contain events that do not culminate in that world; see Martin & Demirdache (2020) for a recent overview of predicates of non-culminating eventualities.

Note that (107b) only defines empty prefixation for a class of verbs that are relations between individuals and events, i.e. verbs that take an internal argument. This seems to make a welcome prediction: empty prefixation is characteristic of transitive and unaccusative, but not unergative verbs. The reader is referred to Romanova (2007: ch. 4) for further discussion.

According to (107a)–(107b), empty prefixes are not empty after all. Just like other lexical prefixes, they introduce a result state into the event structure of a prefixed configuration. "Emptiness", that is, the lack of a perceivable lexical meaning change resulting from prefixation, is an epiphenomenon of there being no narrowing down effect of the type represented in (106b) and Figure 3. According to (107b), for any event e and individual x, once e and x are in the relation established by a non-derived verb, x can attain a state brought about by e that falls under the denotation of an empty prefix. As we have just seen, this is what happens in case of *pro-čita-t'* 'read$^{\text{PFV}}$' as opposed to *za-čita-t'* 'read out aloud$^{\text{PFV}}$'.

This view of "emptiness" bears certain resemblance to the idea circulating in the literature since Vey (1952) and van Schooneveld (1958), see, e.g., Filip (1999: 193–194). As Gehrke (2008: 177-178) puts it, "it could be argued that 'empty' prefixes still express particular meanings in isolation, but that they contribute a redundant meaning when they combine with the verb, since they supply information that is already presupposed in the simple verb".[35]

[35]Effects of this type may not be confined to Slavic prefixation. I am grateful to an anonymous reviewer for turning my attention to the parallelism between "empty" prefixation discussed here and the role of the clitic *se* attached to verbs of consumption in Spanish (Campanini &

Not infrequently, this redundancy is understood in terms of a literal overlap of the lexical meaning of a prefix and a verb. For instance, Janda et al. (2013) suggest that the prefix *u-* indicates that the figure is moving away from the ground, and the same meaning component can be found in the description of *krast'* 'steal': stealing involves movement of a stolen entity away from its possessor. For Janda et al., *u-krast'* 'steal^{PFV}' is therefore expected to be an instance of "empty prefixation", since the former meaning is contained in the latter.

With (107), what "is already presupposed in a simple verb" can receive a slightly different characterization. It can be understood as the culmination potential of a simplex verb, the ability of every eventuality from its extension to bring an object into a certain result state. A non-restrictive prefix happens to denote exactly such a state, so its attachment has no perceivable effects on lexical semantics. It is in this sense that the non-restrictive AS prefixes supply presupposed information (even if, technically, they are not redundant, since they introduce a result state into the event structure that had not been there before).

Typically, simplex imperfective verbs have one perfective counterpart derived by "empty" prefixation. Nothing in (107) suggests, however, that this always has to be the case. In fact, one can easily find cases of AS lexical prefixation where several prefixes can be characterized as pure perfectivizing with respect to the same stem. In such a case speakers judge different prefixal counterparts of the non-derived verb synonymous. An example is *žari-t'* 'fry, roast' and its prefixed partners in (108) (Polivanova 1975).

(108) a. Volodja žari-l kartošk-u.
Volodja fry-PST.M potato-ACC
'Volodja was frying potatoes.'

b. Volodja {po-žari-l / pod-žari-l / za-žari-l / iz-žari-l}
Volodja LP-fry-PST.M LP-fry-PST.M LP-fry-PST.M LP-fry-PST.M
kartošk-u.
potato-ACC
'Volodja fried potatoes.'

The prefixed verbs *po-žari-t'*, *pod-žari-t'*, *za-žari-t'*, and *iz-žari-t'* in (108b) are all associated with result states with more or less the same descriptive properties which can be brought about by any frying eventuality described by *žari-t'* 'fry, roast' in (108a). As a result, truth-conditional differences between the four

Schäfer 2011). In *comerse una manzana* 'eat.REFL an apple', the reviewer indicates, "*se* expresses, according to Campanini & Schäfer (2011), a relation of ingestion of the theme by the external argument, but this relation is already expressed by *comer una manzana*" 'eat an apple'.

prefixed variants are not detectable, so none of them can be a candidate for the status of the only "pure perfective" aspectual partner of the non-derived verb.

On the other hand, a non-derived verb can fail to have an aspectual partner derived by empty prefixation. One typical instance of such a pattern occurs if the extension of a verb is too wide and heterogeneous to satisfy (107b). (109a)–(109b) illustrate the verb *kopa-t'* 'dig'.

(109) a. Volodja kopa-l jam-u.
Volodja dig-PST-M hole-ACC
'Volodja was digging a hole.'

b. Volodja kopa-l ogorod.
Volodja dig-PST.M vegetable.garden.ACC
'Volodja was digging up a vegetable garden.'

As (109a)–(109b) suggest, the relation between individuals and events denoted by *kopa-t'* 'dig' contains at least two types of ordered pairs: those where digging events are coupled with effected objects like *jama* 'a hole' in (109a), and those where events are related to affected objects like *ogorod* 'vegetable garden' in (109b). However, no prefix can denote result states attained by both types of entities in the course of digging. Rather, those states are distributed over different prefixes, as (110)–(111) show.

(110) a. Volodja vy-kopa-l jam-u.
Volodja LP-dig-PST.M hole-ACC
'Volodja dug a hole.'

b. * Volodja vy-kopa-l ogorod.
Volodja LP-dig-PST.M vegetable.garden.ACC
Intended: 'Volodja dug up a vegetable garden.'

(111) a. * Volodja vs-kopa-l jam-u.
Volodja LP-dig-PST.M hole-ACC
Intended: 'Volodja dug a hole.'

b. Volodja vs-kopa-l ogorod.
Volodja LP-dig-PST.M vegetable.garden.ACC
'Volodja dug up a vegetable garden.'

In (110a)–(110b), the prefix *vy-* expresses a state of having been effected as the result of digging. In (111a)–(111b), the prefix *vs-* introduces a state of having been (totally) affected by digging. If one recognizes two distinct relations *kopa-t'*$_{EFF}$ 'dig with an effected object' and *kopa-t'*$_{AFF}$ 'dig with an affected object', both being subsets of the denotation of *kopa-t'*, one can argue that *vy-* is pure perfectivizing with respect to *kopa-t'*$_{EFF}$, while *vs-* is so with respect to *kopa-t'*$_{AFF}$. Given the highly degraded status of (110b) and (111a), neither prefix can be characterized as pure perfectivizing with respect to *kopa-t'* as a whole, i.e., with respect to *kopa-t'*$_{EFF}$ ∪ *kopa-t'*$_{AFF}$.

The above observation only holds, of course, if one believes that *kopa-t'*$_{EFF}$ and *kopa-t'*$_{AFF}$ fall under the denotation of the same unprefixed verb, *kopa-t'* 'dig$^{\text{IPFV}}$'. If this is not the case, and *kopa-t'*$_{EFF}$ and *kopa-t'*$_{AFF}$ form the denotations of two distinct, but homophonous lexical items, each can be said to have its own pure perfectivizing partner. As a reviewer points out, this analysis may be warranted by "the verbs' distinct meanings, s-selectional properties, and behavior under prefixation". On the other hand, there is a reason to assume that both *vy-kopa-t'* 'dig$^{\text{PFV}}$' and *vs-kopa-t'* in (110)–(111) 'dig$^{\text{PFV}}$' are derivatives of the same unprefixed verb: distinct result states are brought about by an activity with exactly the same descriptive properties: 'to break up and move soil using a tool, a machine, or your hands', as Cambridge Dictionary puts it. If the lexical meaning of *kopa-t'* only entails these characteristics of the activity and is underspecified with respect to the type of object, the fact that we find both *kopa-t'*$_{EFF}$ and *kopa-t'*$_{AFF}$ relations as part of its denotation comes as no surprise.

Another class of non-derived verbs that fail to have a single aspectual partner derived by empty prefixation are manner of motion verbs like *lete-t'* 'fly' that do not come with any specific lexical entailments about the source and goal of motion. The AS lexically prefixed counterparts of *lete-t'* 'fly', *u-lete-t'* 'fly away' and *pri-lete-t'* 'arrive flying' introduce the state 'be outside of the deictic center as the result of flying' or 'be at the deictic center as the result of flying'. These states are too specific to be compatible with any flying event denoted by *lete-t'* 'fly', hence neither *u-lete-t'* 'fly away' nor *pri-lete-t'* 'arrive flying' come across as involving empty prefixation; both should be regarded as instances of restrictive prefixation.

4.3.5 Dyadic prefixes

Dyadic prefixes are singled out by a prominent property of theirs: verbs with such prefixes have an extra argument which is not there in the corresponding unprefixed configuration. Consider (112).

(112) Volodja v-pisa-l dopolnitel'n-oe uslovi-e v dogovor.
Volodja LP-write-PST.M additional-ACC condition-ACC in agreement.ACC
'Volodja wrote an additional condition into the agreement.'

(112) entails a result state 'an additional condition is in the agreement', with the goal PP *v dogovor* 'in the agreement' specifying the location of the condition after an eventuality culminates. A non-prefixed counterpart of (112) in (113) is highly degraded.

(113) ?? Volodja pisa-l dopolnitel'n-oe uslovi-e v dogovor.
Volodja write-PST.M additional-ACC condition-ACC in agreement.ACC
Intended: 'Volodja was writing an additional condition into the agreement.'

Verbs like *pisa-t'* 'write' under normal circumstances do not describe a movement in the physical space and for that reason do not take goal (and source) adjuncts, as in (113). Given (113), one can conclude that the goal PP in (112) can only appear as an argument of the prefix. In (112) the prefix *v*-, therefore, introduces a state of the condition being in the agreement; such a state is attained as the result of writing. The prefix can thus be treated as dyadic: one specifying a relation between its external argument ('an additional condition'; Figure) and its internal argument ('the agreement'; Ground). For various analyses of the syntactic structure projected by such prefixes see Ramchand (2004), Svenonius (2004, 2008), Romanova (2007), Biskup (2019). Figure 4 illustrates the structure of (112) along the lines of Svenonius (2004).

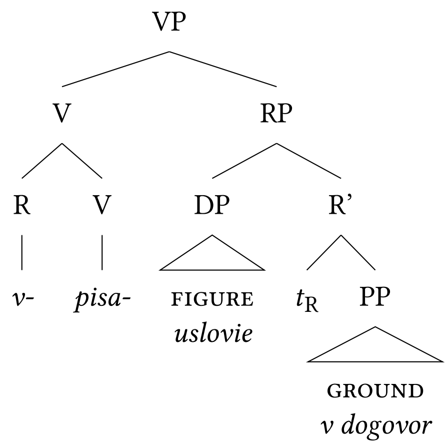

Figure 4: Proposal according to Svenonius (2004)

Another example of a dyadic lexical prefix and its unprefixed counterpart is shown in (114)–(115).

(114) Volodja is-pisa-l tetrad' stix-ami.
Volodja LP-write-PST.M notebook.ACC poem-INSTR.PL
'Volodja filled the notebook with poems.'

(115) * Volodja pisa-l tetrad' stix-ami.
Volodja write-PST.M notebook.ACC poem-INSTR.PL
Intended: 'Volodja was writing the notebook with poems.'

In (114), the direct object *tetrad'* 'notebook' enters a result state of being filled with poems. As in (113), the unprefixed counterpart of (114) shown in (115) is severely ungrammatical. However, (114) with *is-pisa-t'* 'fill sth. with sth. written' differs from (112) with *v-pisa-t'* 'write into' in one significant respect.

In (112), the theme of writing, *uslovie* 'condition', is realized as the direct object of the prefixed verb. In (114), the theme of writing are poems, which however, appears as an instrumental DP *stixami* 'with poems', not as the direct object. The direct object is *tetrad'* 'notebook', which denotes the location of the poems that come about in the course of writing. The DP *tetrad'* 'notebook' is thus Ground, the internal argument of the prefix, whereas *stixami* is the Figure, its external argument.

Romanova (2007: 110ff.) argues that this configuration should be analyzed in parallel with the verbal passive. The passive feature on the prefix does the regular job of the passive: descriptively, it induces "demotion" of the external argument, which makes the internal argument eligible for "promotion". As a result, the final position of *tetrad'* in (114) is the one where the external argument of prefixes is normally found, as, e.g., in (112). The parallelism between configurations like (114) and verbal passives is reinforced by the fact that in both cases the demoted external argument is realized as an adjunct bearing the instrumental case.[36]

This completes the survey of the typology of prefixes set up in Figure 1. Before closing this chapter, however, a few reflections on the descriptive properties of results states in light of the generalization in (69) above are due.

4.4 Descriptive properties of result states

In this section I will address a few related open questions that all have to do with how the descriptive properties of a state predicate in the denotation of a lexical

[36]I am grateful to Berit Gehrke (p.c.) who noted the parallelism between examples like (114) and the *spray/load*-alternation (as in *John loaded the hay onto the wagon* and *John loaded the wagon with the hay*). Remarkably, Larson's (1990) analysis of the alternation involves the demotion of the locatum (*the hay*) to the adjunct position and A-movement of the location (*the wagon*) to the specifier position; see Beavers (2017) for further discussion. Essentially, this is what Romanova (2007) proposes for examples like (114).

prefix are computed. The toy fragment in section Section 3.2 has been abstracted away from this issue.

To begin with, one should recognize two conflicting types of empirical evidence concerning the meaning of prefixes. On the one hand, descriptive properties of states they introduce must be specific enough for a theory to be able to capture semantic judgments about the differences between combinations of the same verb stem with different prefixes. In *na-pisa-t'* 'write$^{\text{PFV}}$' writing events lead the object to a state of having been written. In *pod-pisa-t'*, the object enters a state of having been signed. In both cases, writing events fall under *pisa-t'* 'write'. The source of the difference between the two verbs must therefore be descriptive properties of states associated with the prefixes *na-* and *pod-*. This implies that the meaning of prefixes is rather fine-grained and specific, as is pointed out in (69) above.

On the other hand, the descriptive properties of a state associated with a lexical prefix cannot be too specific, since every prefix should be able to combine with an open class of verb stems. A predicate of states of having been written cannot be lexically associated with the prefix *na-*. Otherwise if the same prefix combines with *risova-t'* 'paint' to produce *na-risova-t'* 'paint$^{\text{PFV}}$', one would get a set of painting events that bring about a state of having been written, which obviously makes no sense.

At the moment, I know of no fully articulated theory that can successfully reconcile the challenges of having a rather specific state description for any combination of a prefix and a verb while keeping the meaning of a prefix general enough for it not to be subject to severe lexical restrictions. However, the space of possibilities shaping the content of such a theory seems to be more or less clear. With a certain simplification, it can be reduced to two clear-cut strategies (and various mixed combinations of their ingredients), which can be called FUNCTIONAL strategy and INTRINSIC MEANING strategy.

The functional strategy would assume that in order to build up a state description in a specific lexical context a prefix needs access to the information about the event description denoted by the verb stem. A straightforward way of implementing this would be to treat (the projection of) a prefix as a (possibly very partial) function that takes the denotation of a verb as an argument and picks out an appropriate predicate of states.

Above, in (91), the (shifted) denotation of the projection of a prefix and that of the verb stem, repeated as (116a)–(116b), are combined by Event Identification. The output is a relation between events and states in (116c).

(116) a. The denotation of the projection of a prefix (after Causative Shift)
$\lambda s.\lambda e.\text{SIGNED}(\text{LETTER})(s) \wedge \text{CAUSE}(s)(e)$

b. The denotation of a verb stem (after existential closure of the internal argument)
$\lambda e.\exists x[\textsc{write}(x)(e)]$

c. The denotation of the VP projected by a prefixed verb
$\lambda s.\lambda e.\exists x[\textsc{write}(x)(e) \wedge \textsc{signed}(\textsc{letter})(s) \wedge \textsf{CAUSE}(s)(e)]$

The simplified fragment in (91)/(116) was abstracted away from how the descriptive properties of a state in (116a) were computed. But given the above considerations, the denotation in (116a) may need to be amended as in (117a). It would apply to (116b) yielding (117b) as the denotation of *pod-pisa-t'* 'sign' instead of (116c).

(117) a. The denotation of the projection of a prefix
$\lambda P.\lambda s.\lambda e.\textsc{pod}(P)(\textsc{letter})(s) \wedge \textsf{CAUSE}(s)(e) \wedge P(e)$

b. The denotation of the VP projected by a prefixed verb
$\lambda s.\lambda e.\exists x[\textsc{pod}(\lambda e'.\exists x[\textsc{write}(x)(e')])(\textsc{letter})(s) \wedge \textsf{CAUSE}(s)(e) \wedge \textsc{write}(x)(e)]$

In (117a), *pod-* is of type $\langle\langle v,t\rangle,\langle v,\langle v,t\rangle\rangle\rangle$. It does not name a relation between events and states directly, as in (116a). Rather, this is done by $\textsc{pod}(\lambda e'.\exists x[\textsc{write}(x)(e')])$ in (117b). Every state description created by a prefix is thus relativized to the event predicate denoted by a verb stem it combines with. Combining with the stem *pisa-* 'write', *pod-* yields a relation between individuals and states of being signed:

(118) $\textsc{pod}(\lambda e'.\exists x[\textsc{write}(x)(e')]) = \lambda x.\lambda s.\textsc{signed}(x)(s)$

The denotation of the prefix itself can be represented as the function in (119).

(119) $[\![\textit{pod-}]\!] = \lambda x.\lambda P.\lambda s.\lambda e.\textsc{pod}(P)(x)(s) \wedge \textsf{CAUSE}(s)(e) \wedge P(e)$

Technically, this should work. However, this type of theory reveals an obvious vulnerability. The meaning of a prefix is now conceived of as a mapping from event predicates to relations between individuals and states, as in (117a). For the *pod*-verbs in (120), this mapping would look like (121).

(120) a. *pod-pisa-* 'sign' ← *pod* + *pisa-* 'write'

b. *pod-dela-* 'fake' ← *pod* + *dela-* 'do, make'

c. *pod-stroi-* 'set up' ← *pod* + *stroi-* 'build'

d. *pod-kova-* 'shoe (a horse)' ← *pod* + *kova-* 'forge'

e. ...

(121) The extension of *pod*-, quasi-formally
{ *pisa*- 'write' $\mapsto \lambda x.\lambda s.\textsc{signed}(x)(s)$, *dela*- 'do, make' $\mapsto$ $\lambda x.\lambda s.\textsc{faked}(x)(s)$, *stroi*- 'build' $\mapsto \lambda x.\lambda s.\textsc{set up}(x)(s)$, *kova*- 'forge' $\mapsto$ $\lambda x.\lambda s.\textsc{shoed}(x)(s)$, … }

If nothing else is said, each combination of a prefix and a stem is unique, and the extension of every prefix starts looking like a list of such combinations. Strictly speaking, this gives us as many submeanings of a prefix as there are stems it can combine with. This may make the acquisition task for a learner of Russian and other Slavic languages rather costly. For every prefix, one needs to learn a list like (121): a collection of state descriptions which it maps verbs from its domain to. Whether one can convince oneself that this is how the system works may ultimately depend on experimental evidence from L1 and L2 acquisition and, as Berit Gehrke (p.c.) points out, on evidence from "languages with less transparent morphology, in which there are different lexical items for each of the meanings". At the moment, however, this seems to be more a prospect for a future inquiry than a working research agenda.

It should be noted, finally, that making the denotation of a verb stem an argument of a prefix, as in (119), makes it inevitable that the semantics incorporates the effects of causative shift. This is so because the verb denotation is not only used to calculate a state description, but also, as before, contributes eventualities that cause such a state. As a result, the semantics of a prefix now duplicates the event structure of the entire complex event description: all its subevental components are already present in (119). Whether this is a welcome consequence is a separate question that I am not trying to address here.

An alternative to the view just outlined is to assume that lexical prefixes have intrinsic meanings in and by themselves and to specify how they interact with the meaning of a verb stem. This interaction should not be conceived of as verb-specific and must follow from independent principles and constraints.

A prominent line of inquiry within this intrinsic meaning strategy is to explicate the meaning of a prefix in terms of properties of states defined by a spatial configuration between the figure and the ground. Of those two, one receives syntactic realization, while the other can (but does not have to) be implicit. Typically, given that the vast majority of prefixes have prepositional counterparts, the meaning of a prefix is thought of in terms of a relation established by the corresponding preposition, e.g., *na* 'on, at the surface of', *pod* 'under', *u* 'at', and so on.

For instance, the prefix *pod*- can be regarded as introducing a state of the figure being under the ground. In *pod-pisa-t'* 'sign' (Biskup 2019: 67), the figure is a signature, which has no syntactic realization. The ground is an entity under which a signature appears, and it projects as the direct object. Biskup's semantics for *pod-pisa-t' pis'mo* 'sign the letter', adjusted to the format used throughout this chapter, is shown in (122).[37]

(122) $\lambda s.\lambda e.\exists x[\textsc{under}(\textsc{letter})(x)(s) \wedge \textsc{cause}(s)(e) \wedge \textsc{write}(x)(e)]$

One can take a step further and argue that certain figure/ground configurations may not be confined to the spatial domain and extend to other domains as well, most importantly, to the temporal domain. An early exemplar of this approach can be found in Flier (1975), who points out that "verbal prefixes [...] ultimately make reference to abstract delimitation, dimension, and direction, eliciting metaphorical interpretations of these notions in nonspatial universes" (Flier 1975: 219). In Flier (1975), one can find an analysis of the four "spanned" Russian prefixes *pere-*, *pro-*, *ob-*, and *po-* along these lines. With a certain idealization, Kagan's (2015) degree semantics for a number of Russian prefixes can be regarded as a further development of this idea, one that takes degree structures to be an analytical tool that can capture semantic regularities across domains.

Another possible implementation of the intrinsic meaning strategy would be to set up a polysemy model of the semantics of lexical prefixes, whereby every prefix represents a family of meanings structured in a particular way. A prominent example of this line of thought has been developed in a series of publications by Laura Janda and colleagues of hers. In Janda et al. (2012), for example, every prefix is associated with a "radial category", where submeanings are organized into a network, centered around the prototype, as in Figure 5, which represents the semantics of the prefix *u-*.

In Figure 5, the prototype is MOVE AWAY, which manifests itself in the meaning of verbs like *u-beža-t'* 'run away'. Other submeanings that form the network in Figure 5 can be defined in terms of properties they share with the parent node(s). For instance, the link between moving away and moving downwards is established "because when an object moves away, it sinks below the horizon" (Janda et al. 2012: 249).

The intrinsic meaning strategy does not run into the acquisition problem inherent to the previous one. If a theory is equipped with an elaborated machinery of making spatial configurations, movements and paths entirely explicit, it should ultimately be able to develop a model in which the general meaning of a prefix

[37] (122) differs from Biskup's (2019) original proposal in two respects: Existentially bound implicit arguments are represented in Biskup as free variables; verb stems come with no individual arguments.

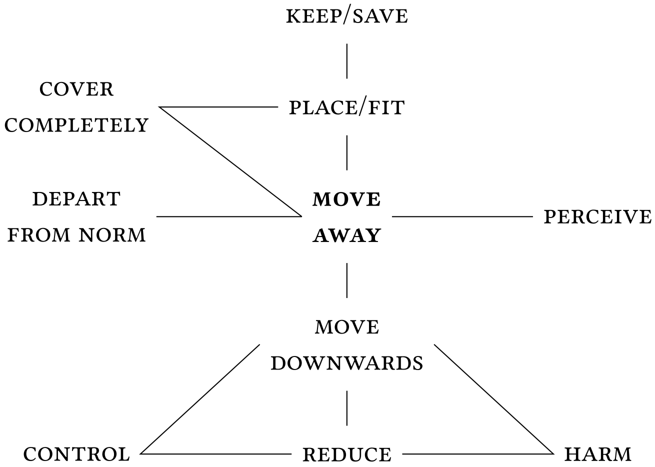

Figure 5: Radial category for the prefix *u-* (Janda et al. 2012)

interacts with the meaning of a stem in a regular and predictable way, deriving the range of event structural and argument structural characteristics of prefixed verbs discussed above.

At the moment, however, building up such a model looks more like a theoretical desideratum than like an articulated theory. One obvious difficulty in realizing this desideratum is that if a prefix is assigned a single general meaning, it can be too devoid of specific content for a theory to derive testable predictions about the descriptive properties of a result state in every lexical configuration. For instance, assume 'under' as the meaning of the prefix *pod-*. Looking at *pod-pisa-t'* 'sign = under-write', *pod-rubi-t'* 'under-cut' or *pod-žari-t'* 'under-fry' one can try to convince oneself that all of these verbs involve something happening at the bottom of or under the entity denoted by the internal argument. However, this is much less obvious in case of *pod-stroi-t'* 'set up, frame, trick = under-build' or *pod-stegnu-t'* 'hurry up, spur = under-lash'. If a car accident has been set up, can one say that the car accident ends up in the under-relation to some other entity? Failure of a theory to address questions like this makes its empirical prospects dubious.

Polysemy models typically come with a more fine-grained semantic structure that can potentially be more successful in assigning the right descriptive properties to result states. However, they face additional challenges. One is to identify and motivate the set of individual submeanings of a prefix and relations between them. For example, for Figure 5 to be fully convincing, one has to present elaborated arguments showing that the prefix *u-* has exactly 10 submeanings, that these meanings are exactly {MOVE AWAY, MOVE DOWNWARDS, CONTROL, REDUCE, HARM, ...}, as in Figure 5, and that they are connected to each other as indicated. If structures like Figure 5 are taken for granted rather than thoroughly argued for on empirical grounds, predictions of a theory would be difficult if not impossible to assess.

5 Summary

After an overview of the landscape of the derivational morphology of Russian, taken to represent several typical characteristics of the entire Slavic family, this chapter addressed two main topics. One is the event structure of simplex imperfective, lexically prefixed perfective, and secondary imperfective verbs and their extended projections. The other is lexical prefixation, with special attention to the parameters of variation across lexical prefixes.

In Section 3, relying on the idea circulating in the literature since Klein (1995), I presented a few pieces of data that argue for a higher subevental complexity of prefixed configurations, be they perfective or secondary imperfective, compared to simplex configurations. This is manifested in a number of phenomena that are typically taken to diagnose event-structural complexity: scope of decomposition adverbs, interpretation under negation, argument realization patterns. In event-semantic terms, therefore, VPs projected by prefixed perfective verbs are proposed to be analyzed as denoting relations between events and result states. Secondary imperfectives can be argued to share this event-structural make up with prefixed perfectives. Their prefixless imperfective counterparts, in contrast, lack a result state.

The results of Section 3, as well as numerous considerations found in the literature, strongly suggest that the presence of a result state in the event structure of VP can be attributed to lexical prefixes. However, lexical prefixes do not form a homogeneous class and fall into a number of subclasses, identified and motivated in Section 4. The basic divide within the entire set of lexically prefixed verbs is that between the verbs where the denotation of a stem undergoes a non-compositional shift after prefixation, and the verbs where this does not happen. The latter, compositional lexically prefixed verbs, are further classified according to the argument structure of a prefix. I proposed to identify monadic prefixes that take a single individual argument, the holder of a result state, and dyadic prefixes which denote a relation between two entities, the figure and the ground. Among the verbs with monadic prefixes, two important classes are isolated. One contains verbs with argument-sharing prefixes, that is, prefixes whose individual argument is identical to the internal argument of the verb stem. The other contains verbs where the two entities are distinct. Finally, argument sharing prefixes can be characterized as either restrictive or non-restrictive. The former introduce result states only compatible with a restricted subset of eventualities from the original extension of a verb stem. The latter denote states that can be brought about by every such an eventuality; they are also known under the name "pure perfectivizing" or "empty" prefixes.

Abbreviations

2	second person
3	third person
ACC	accusative
DAT	dative
F	feminine
GEN	genitive
INDEF	indefinite
INF	infinitive
INSTR	instrumental
IPFV	imperfective
LP	lexical prefix
M	masculine
N	neuter
N/T	past passive participle morphology
NPST	non-past
PFV	perfective
PL	plural
PST	past
REFL	reflexive
SG	singular
SLP	superlexical prefix
SMFCT	semelfactive
TH	thematic element
YVA	the *-yva-* morpheme, "secondary imperfective"

Acknowledgments

I would like to express my immense gratitude to Berit Gehrke and anonymous reviewers for their benevolent attention to this work and helpful comments. This chapter owes a lot to the conversations with Daniel Altshuler, Tanya Bondarenko, Sabine Iatridou, Hans-Robert Mehlig, Elena Viktorovna Paducheva, Roumyana Pancheva, Anna Pazelskaya, David Pesetsky, Mitya Privoznov, Olav Mueller-Reichau about syntax and semantics of aspect and many other beautiful things.

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Chapter 12

Family of exclusives in Polish

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I define the semantics of four exclusive adverbials in Polish – *tylko*, *zaledwie*, *dopiero*, and *aż* – as composed of three core components: an assertion of exclusivity, a factive presupposition and a scalar presupposition. By minimally adjusting parameters such as the direction and dimension of the scale and the size of the associated focus, different flavors of exclusivity emerge. These parameter settings give rise to temporal readings (*dopiero*), approximative and counterfactual readings (*zaledwie*), and even additive-like readings (*aż*).

1 Introduction

This chapter presents the behavior of four exclusive adverbials in Polish, *tylko*, *zaledwie*, *dopiero* and *aż*, and shows how they can be defined as a family: Each lexical entry includes a common core (assertion of exclusivity, presupposition of scalarity) with minimal differences in specific parameters (the direction and the dimension of the scale, the size of the associated focus).[1] We will start with a context where all four adverbs have a similar effect on the meaning and then use a battery of tests to identify the individual meaning components.

All of the adverbs contribute to the sentence in (1) the additional meaning that the 30th place is a weak result, i.e. the team did not perform any better than that.

[1]The glosses of the Polish particles in the examples reflect the parallels with the English counterparts. In Section 4, we will see that *tylko* is similar to *only* and *zaledwie* to *merely*. *Dopiero* has a German counterpart, *erst*, which has been named TEMPORAL *only*. *Aż* on my view is a scalar opposite of *merely*, so it has no approximate translation and is glossed here as AŻ.

Barbara Tomaszewicz-Özakın. 2026. Family of exclusives in Polish. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 655–701. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078591

(1) Zespół był słaby i zajął {tylko / zaledwie / dopiero / aż} 30 miejsce.
team was poor and achieved only merely only$_{\text{TEMP}}$ AŻ 30$^{\text{th}}$ place
'The team performed poorly and earned only 30$^{\text{th}}$ place.'

English *only* has exactly this meaning '(at least) *X* and nothing more', with 'nothing more' excluding all contextually ranked higher alternatives. Is it only a coincidence that in this particular context the four adverbials seem to mean 'only'? Could the sentence rather be saying that 'The team was ranked as low as the 30$^{\text{th}}$ place'? This meaning is similar but it does not contain the exclusive component 'nothing more'. I argue that the four adverbial particles should all be classified as exclusives, i.e., they all share the exclusive meaning component, and that this primary component is at-issue (Potts 2005, Beaver & Clark 2008, Simons et al. 2011, Roberts 2011). Each one of them, however, has its own flavor so that in most contexts they are not interchangeable as they are in (1). I identify the differences in their use that make them similar only under certain conditions. It turns out that Polish does not have one adverbial that can contribute the exclusive meaning in all contexts, the way English *only* can. Instead, each of the items places particular restrictions on the dimension of scale of the excluded alternatives. Furthermore, the direction of exclusion differs: alternatives both higher and lower on the scale can be excluded (Tomaszewicz 2012, Tomaszewicz 2013a,b). I propose lexical entries for each exclusive such that they all include a common core with minimal differences in specific parameters.

2 Exclusivity and scalarity

The adverb *only* contributes the meaning of exclusion: in contrast to (2), the sentence in (3) states that Maria met Jan and she did not meet anyone else.

(2) Maria met Jan.

(3) Maria met only Jan.

In some contexts *only* additionally contributes scalarity by implying a ranking among the excluded alternatives. Consider how in (4) we understand that the 30th place is not a good result, even though we do not know how many places were assigned in total. In (5) it is implied that 9 PM is late, and in (6)–(7) that the manager is lower on the scale than some contextually relevant alternative.

(4) Jan got only the 30th place.

(5) Jan arrived only at 9 PM.

(6) Jan is only/merely a manager.

(7) Maria talked only/merely to the manager.

In the example in (4) it is obvious that you can be ranked at a single place only, so the exclusive meaning should be paraphrased as ‘Jan got no better place than the 30th’ rather than ‘Jan got the 30th place and no other place’ in contrast to the paraphrase of the sentence in (3). These two readings of *only* have been called SCALAR and QUANTIFICATIONAL (Bonomi & Casalegno 1993, Krifka 1993, Klinedinst 2005, Riester 2006, Beaver & Clark 2008, Zeevat 2009, among others). The scalar reading of the sentence in (5) is roughly equivalent to ‘Jan arrived no sooner than 9 PM’ because the most plausible scenario is that Jan arrived once, as opposed to ‘Jan arrived at 9 PM and at no other time’, which would be the quantificational reading. In contrast to (4), in (6) Jan is a manager in addition to being other obvious things such as human, terrestrial, mortal… These properties are not ordered on the scale with respect to the property of being a manager, they are not contextually relevant. Thus, in (6), *only* evokes a contextually relevant scale and places ‘manager’ low on that scale, as illustrated in (8). The example in (7) also shows that the scale of exclusion involves only relevant alternatives, i.e., the people ranked in terms of importance in (8), because (7) is true even if Maria talked to the manager’s secretary (assuming that this conversation was irrelevant to the topic of her conversation with the manager). Note that in English *merely* is dedicated to scalar readings, (6)–(7) (Coppock & Beaver 2011, 2014).

(8) Scale for (6)–(7):
‘Jan is the president.’
‘Jan is a vice president.’
…
‘Jan is a senior manager.’
‘Jan is a manager.’
‘Jan is an assistant manager.’
…
‘Jan is an employee.’

The Polish exclusives in the example in (1), *tylko*, *zaledwie*, *dopiero* and *aż*, are all compatible with the scalar readings, but now the examples in (9)–(12) will show that Polish does not have one adverbial that can contribute the exclusive meaning in all contexts like English *only* can.[2] The quantificational reading requires *tylko*, as shown in the sentence in (9) which is the counterpart of the sentence we saw in (3). With temporal scales, on the other hand, *tylko* cannot be used, (10), unlike *only* in (5). With the scale of important people, (11)–(12), *tylko*, *zaledwie* and *dopiero* are possible, though here the manager is low on the scale, just like in (6)–(7) with *only/merely*.

(9) Maria spotkała {tylko / #zaledwie / #dopiero / #aż} Jana.
Maria met only merely $\text{only}_{\text{TEMP}}$ AŻ Jan
'Maria met only Jan and nobody else.'

(10) Jan przyjechał {dopiero / aż / #tylko / ?zaledwie} o 9-tej.
Jan arrived $\text{only}_{\text{TEMP}}$ AŻ only merely at 9
'Jan arrived only at 9 PM and no sooner.'

(11) Jan jest {tylko / zaledwie / dopiero / #aż} menedżerem.
Jan is only merely $\text{only}_{\text{TEMP}}$ AŻ manager
'Jan is only/merely a manager (so far) and nobody more important.'

(12) Maria rozmawiała {tylko / zaledwie / dopiero / #aż} z menedżerem.
Maria talked only merely $\text{only}_{\text{TEMP}}$ AŻ with manager
'Maria only/merely talked to the manager (so far) and nobody more important.'

Crucially, the scalar readings of (11)–(12) "flip" when *aż* is used: In (13) the manager is interpreted to be high on the scale of importance. The exclusive *aż* results in a reading that is similar to the effect of the addition of *even* (Polish *nawet*). In the sentence with *nawet* in (14), similarly to (13), the manager is implied to be a noteworthy person for Maria to talk to. However, *nawet* is not compatible with the continuation that Maria did not talk to anybody else, (14), in contrast to *aż* in (15). This is because *nawet*/*even* is an additive scalar particle (Horn 1969, Karttunen & Peters 1979, Rooth 1985, 1992, a.o.), whereas *aż* is an exclusive as I argued in Tomaszewicz (2012), Tomaszewicz (2013a,b).

(13) Maria rozmawiała {#tylko / #zaledwie / #dopiero / aż} z menedżerem.
Maria talked only merely $\text{only}_{\text{TEMP}}$ AŻ with manager
'Maria talked to somebody so important as the manager.'

[2] All non-English examples in this chapter are from Polish, unless indicated otherwise.

(14) Maria rozmawiała nawet z menedżerem, #ale nie rozmawiała z
Maria talked even with manager but not talked with
nikim innym.
nobody else
'Maria even talked to the manager, *[Incongruous:]* but she did not talk to anybody else.'

(15) Maria rozmawiała aż z menedżerem, ale nie rozmawiała z
Maria talked aż with manager but not talked with
nikim innym.
nobody else
'Maria talked to somebody so important as the manager, but she did not talk to anybody else.'

I proposed in Tomaszewicz (2012), Tomaszewicz (2013a,b) that *aż* is the scalar opposite of *merely*/*zaledwie*: It asserts exclusion just like *only*/*merely*, evokes scales like *merely*, but it places the mentioned alternative high on the scale. In this chapter, I provide tests to specify the meaning components for each of the exclusives shown in (1), demonstrating that the semantics contains a common core predicting the existence of scalar opposites like *aż* cross-linguistically, even though such a lexical item is unavailable in, for instance, English.

Our starting point is the basic meaning contribution of *only*/*merely*, which involves, on the one hand, exclusivity (16a)–(16b) and, on the other hand, the prejacent, which is the meaning of the sentence without *only*/*merely* (16) (Klinedinst 2005, Roberts 2006, van Rooij & Schulz 2007, Beaver & Clark 2008, a.o.).[3]

(16) Maria only talked to the manager.
Exclusive/negative contribution:
a. 'Maria did not talk to anybody else than the manager.'
QUANTIFICATIONAL READING
b. 'Maria did not talk to anybody more noteworthy than a manager.'
SCALAR READING

[3]This use of the term PREJACENT was introduced by Horn (1996); however, Roberts (2006) observes that often with *only* the prejacent does not correspond to any overt syntactic constituent. von Fintel & Heim (2002) use the term for the constituent that is the sister of the operator at the level of logical syntax. Roberts (2006) addresses the debate on the status of the prejacent with *only* arguing that it is "closer to a presupposition than to an entailment or a conversational implicature". Crnič (2013) argues that the mixed behavior (prejacent presupposition vs. existential presupposition) results from two separate meaning components at LF with different scope.

Positive contribution:

c. 'Maria talked to (at least) the manager.' PREJACENT

The quantificational reading in (16a) can be seen as an instance of exclusion along a logical/entailment scale (Bonomi & Casalegno 1993, van Rooij 2002, Beaver 2004, Klinedinst 2005, Riester 2006, a.o.). The dimension of the scale is set relevant to the context, and where no such pragmatic ordering is readily available (such as the ranking of people in terms of importance), the logical scale is the default (e.g., all other people are excluded).

In the following sections I present a battery of tests to demonstrate that with all four Polish adverbial particles, *tylko*, *zaledwie*, *dopiero* and *aż*, exclusivity is asserted, the prejacent is presupposed, and scalarity, i.e., the 'low/high on the scale' reading, is presupposed. In addition to the exclusive assertion and the prejacent presupposition, it will be shown that scalarity comes from focus association, so that the focal presupposition supplies the set of alternatives for the ranking.

3 Focus association

Only belongs to so-called FOCUS-SENSITIVE adverbs because its surface position interacts with the placement of focus. The semantic effects of focus on the interpretation of *only* go hand in hand with syntactic/phonological effects, i.e., the way *only* modifies the meaning of the sentence depends on which constituent is focused, although the syntactic associate needs not always be marked by easily perceptible prosodic prominence. In Polish focus is marked prosodically, with pitch and intensity as the main acoustic correlates (Hamlaoui et al. 2019). The focused constituent is called the SYNTACTIC ASSOCIATOR. It can be a constituent that is immediately modified by *only*, as we have seen above and as shown below in (17): 'only for a loan'.

(17) Hanka poprosiła tylko [$_{PP}$ o [$_{DP}$ pożyczkę]], i o nic więcej / innego.
Hanka asked only for loan and for nothing more else
'Hanka only asked for a loan and nothing more/else.'

But the associator can also be more distant, as can be seen in (18). The different continuations in (18a)–(18d) result from different focus association configurations (associated with different prosodic patterns). Here *tylko* precedes the verb but it can associate with the object DP so that the sentence has the exact same interpretation as in (17). It can also associate with the verb itself, as shown by the compatibility with the continuation in (18b), and it can associate with the whole VP, as evidenced by the continuation in (18c). In contrast to English, *tylko* in the preverbal position can also associate with the subject DP, (18d). When *tylko* precedes the subject it can, unsurprisingly, associate with it, (19a), but it can also associate with the whole sentence, (19b), which, again, is not possible in English.

(18) [$_{\text{DP}}$ Hanka] tylko [$_{\text{VP}}$ [$_{\text{V}}$ poprosiła] o [$_{\text{DP}}$ pożyczkę]],
Hanka only asked for loan

a. … i o nic innego nie poprosiła. OBJECT DP FOCUS
and for nothing else not asked
'Hanka asked only for a loan and did not ask for anything else.'

b. … a nie zażądała. V FOCUS
and not demanded
'Hanka only asked for a loan and did not demand it.'

c. … i nic innego nie zrobiła. VP FOCUS
and nothing else not did
'Hanka only asked for a loan and did not do anything else.'

d. … i nikt inny nie prosił. SUBJECT DP FOCUS
and nobody else not asked
'Only Hanka asked for a loan and nobody else did.'

(19) (To) tylko [$_{\text{S}}$ [$_{\text{DP}}$ Hanka] poprosiła o pożyczkę],
TOP only Hanka asked for loan

a. … i nikt inny nie prosił. SUBJECT DP FOCUS
and nobody else not asked
'Hanka only asked for a loan and nobody else did.'

b. … i nic innego się nie stało. SENTENCE FOCUS
and nothing else self not happened
'It is only that Hanka asked for a loan and nothing else happened.'

The associate determines the implicit comparison with alternatives of the same type. So when *tylko* in the sentence-initial position associates with the whole clause (sentence focus in (19b)), what is excluded are other things that could have happened that are ranked higher on the scale of importance. A typical context for (19b) is a question 'Did anything happen?' and an answer such as 'No worries. It is just …'.

In (20) I provide an example from a Polish comedy TV series "Alternatywy 4" (episode 9), from a scene where what appears to be a woman but is in fact a robot, jumps down from a balcony, breaks her neck severing the head, then stands up and picks up the head. The people witnessing this are screaming in shock and a man is trying to calm them down saying that nothing out of the ordinary is happening.

(20) To nic takiego, to nic takiego! To tylko ten psychiczny dostał szału!
it nothing such it nothing such it only this lunatic got amok
'It is nothing, it is nothing! It is just that lunatic going crazy!'

I am pointing out the differences between Polish and English because the syntax (and prosody) of focus association may vary considerably between languages. The extra options available in Polish are not limited to Slavic languages but are also possible in German (Büring & Hartmann 2001, Steindl 2013, 2017). In (21) German *nur* 'only' associates with the subject DP that precedes it, while in (22) *nur* is in the sentence-initial position and can associate either with the subject or the whole clause. In English, on the other hand, the associates of *only* must be in its surface c-command domain (Taglicht 1984, Beaver & Clark 2008, Toosarvandani 2010, a.o.).

(21) $[_{\text{DP}}$ Juliane$]_{\text{F}}$ hat nur die Blumen gegossen.
Juliane has only the flowers watered
'It was only Juliane that watered the flowers.' (German; Steindl 2017)

(22) Nur $[_{\text{S}}$ $[_{\text{DP}}$ Juliane$]_{\text{F}_a}$ spielt Klavier$]_{\text{F}_b}$.
only Juliane plays piano
a. 'Only Juliane is playing the piano.' SUBJECT DP FOCUS
b. 'Nothing is going on except for Juliane playing the piano.' SENTENCE FOCUS
(German; Steindl 2013)

Of course, the different focus association patterns are associated with different prosodic contours of the sentence; specifically, the location of pitch accent and nuclear stress identify the constituents in focus. Without going into details of phonology, syntactic evidence for obligatory focus association comes from clitics. Clitics and reduced pronouns force a wider domain reading ((23b) vs. (23a)), which shows that *tylko* cannot associate with weak (unfocused) pronouns. In (23b) *tylko* has to associate with the adverbial or the verb, which can be independently focused (cf. Hoeksema & Zwarts 1991, Beaver & Clark 2008 on focus association of *only* in Dutch).

(23) a. [$_{\text{Adv}}$ Wieczorem] tylko widziałem [$_{\text{DP}}$ niego].
evening only saw him

i. 'In the evening I saw him and nobody else.'

ii. 'In the evening, and at no other time, I saw him.'

b. [$_{\text{Adv}}$ Wieczorem] tylko [$_{\text{VP}}$ go [$_{\text{V}}$ widziałem]].
evening only him saw

i. 'In the evening I saw him and did not e.g. hear him.'

ii. 'In the evening, and at no other time, I saw him.'

iii. *Unavailable:* 'In the evening I saw him and nobody else.'

The syntactic tests above, the interaction between the size of the focused constituent and the syntactic position of *tylko* (18)–(19), and the inability of *tylko* to associate with non-focused elements such as clitics (23), suggest that *tylko* obligatorily associates with focus. The same can be shown for other Polish exclusives, as demonstrated for *aż* in Tomaszewicz (2012), Tomaszewicz (2013a,b).

4 Different ways to exclude alternatives

4.1 *tylko*

Only is standardly assumed to contribute both to the assertion and the presupposition of the sentence in which it occurs (Horn 1969, Klinedinst 2005, Roberts 2006, 2011, van Rooij & Schulz 2007, Beaver & Clark 2008). The asserted meaning can be either scalar (the dedicated use of *merely*) or quantificational. In (11)–(12) we saw that *tylko* like *only* can have both scalar and quantificational readings; (12) is repeated below in (24), and the examples for the two types of scales are given in (25). In (24) *tylko* associates with the focus on the PP 'with the manager'.

(24) Maria rozmawiała tylko z menedżerem.
Maria talked only with manager
'Maria only talked to the/a manager.'
Scalar reading: 'Maria talked to nobody ranked higher than the/a manager.'
quantificational reading: 'Maria talked to nobody other than the/a manager.'

(25) Possible scales for (24):

a. Contextual	b. Logical
'Maria talked with the president'	'Maria talked with 10 people'
'Maria talked with vice president'	'Maria talked with 9 people'
…	…
'Maria talked with a manager'	'Maria talked with 2 people'
…	'Maria talked with 1 person'
'Maria talked with an employee'	'Maria talked with nobody'

Quantificational *only* asserts that no other alternative is true, e.g., Maria only talked to the manager and nobody else. As noted in the previous section, the alternatives are either seen as not ordered or as an instance of exclusion along a logical/entailment scale, such as a numeral scale, as shown in (25b) (van Rooij 2002, Klinedinst 2005, Riester 2006, Coppock & Beaver 2011, a.o.). The dimension of the scale is set pragmatically, and in contexts where no pragmatic ordering is available, the logical scale is the default.

Using tests for presupposition, such as embedding under a question operator or negation (Chierchia & McConnell-Ginet 1990), we can identify three meaning components of *only*, one asserted (exclusivity) and two presupposed components (the prejacent and the 'low on the scale' reading).[4] We will use these tests on the three meaning components of the sentence in (24) identified in (26a–c). The asserted component is the exclusion of all alternatives higher on the scale: the rank of the person that Maria talked to is no higher than a manager, (26a).[5] Additionally, *only/merely* contributes two presuppositions, (26b), (26c). First, it presupposes that the alternatives under consideration are at least as high on the scale as the prejacent (Klinedinst 2005, Beaver & Clark 2008). Other relevant ranks of the person that Maria talked to are at least as high as the manager, (26b) (lower ranks are simply not relevant). The second presupposition is that the prejacent is low on the scale of alternative propositions, which I paraphrase as 'is not significant', (26c) (Klinedinst 2005).

[4]Note that there is extensive debate on the nature of the non-truth conditional, i.e., projective content. Many different subtypes have been discussed, including pragmatic and semantic presuppositions (Stalnaker 1973), conversational and conventional implicature (Grice 1981, Bach 1994, Potts 2005, 2015), F-implicature (Horn 2013), proffered content (Roberts 1996) and several other types and their groupings (e.g., Classes A–D in Tonhauser et al. 2013).

[5]The alternative 'Maria talked to nobody' is also excluded, which, as suggested by Radek Šimík, could be seen as lower, and not higher, on the scale. However, the logical/entailment scale on the quantificational reading is not one-dimensional but it is a mereological structure (Riester 2006, Coppock & Beaver 2011).

(26) Maria rozmawiała tylko z menedżerem.
Maria talked only with manager
'Maria only talked to the manager.'

a. Maria talked to nobody ranked higher than a manager. ASSERTION
b. Maria talked to at least a manager. PRESUPPOSITION
c. Maria's talking to a manager is not significant. PRESUPPOSITION

Embedding under a question operator and under negation shows that the exclusive component in (26a) is an assertion: it is targeted both by the question operator, (27a), and by the negation operator, (28a). It is now readily apparent that the presupposition in (26b) projects in a question, (27b), and under negation, (28b). The low-on-the-scale inference in (26c) also clearly projects, (27c), (28c).

(27) Czy Maria rozmawiała tylko z menedżerem?
Q Maria talked only with manager
'Did Maria only/merely talk to a manager?'

a. Did Maria talk to anybody ranked higher than a manager? QUESTION
b. Maria talked to at least a manager. PRESUPPOSITION
c. Maria's talking to a manager is not significant. PRESUPPOSITION

(28) Maria nie rozmawiała tylko z menedżerem.
Maria not talked only with manager
'Maria did not merely talk to a manager.'

a. Maria talked to somebody ranked higher than a manager. NEGATIVE ASSERTION
b. Maria talked to at least a manager. PRESUPPOSITION
c. Maria's talking to a manager is not significant. PRESUPPOSITION

Beaver & Clark (2008) and Zeevat (2009) call the low-on-the-scale reading of *only* MIRATIVE. As a mirative marker *only* implies that the prejacent falls short of what is expected. This reading is easily detected with numerals, for instance, if you hear *The driver is only 1 stop away*, you know that 1 stop is very close. However, the sentence *The driver is only 15 stops away* also tells you that the driver is close – contrary to what you would expect given the usual duration of 15 stops (so for this statement to be true these 15 stops must be very short). Without *only*, the sentence *The driver is 15 stops away* does not create any expectation whether the driver is going to reach you soon or not.

Consider also that *tylko* can scope both below and above negation, as shown in (29). Both readings can be detected: quantificational, (29a), (29b), and scalar, (29c), (29d).

(29) Maria nie rozmawiała tylko z menedżerem.
Maria not talked only with manager

a. 'Maria talked to somebody else in addition to the manager.' NOT > ONLY

b. 'Maria talked to everyone else except the manager.' ONLY > NOT

c. 'Maria talked to somebody more noteworthy than the manager in addition to the manager.' NOT > ONLY

d. 'Maria talked to everyone more noteworthy than the manager except the manager.' ONLY > NOT

Another test to distinguish between asserted and presupposed content are EMOTIVE FACTIVE predicates, such as 'disappointed' in (30) (Chierchia & McConnell-Ginet 1990, Beaver & Clark 2008, Simons et al. 2011, Coppock & Beaver 2014). These predicates target the asserted negative component with *tylko*, showing, again, that exclusivity is asserted and scalarity presupposed, (30).

(30) Jestem zawiedziony, że Maria rozmawiała tylko z menedżerem.
AUX disappointed that Maria talked only with manager
'I am disappointed that Maria did not talk to anybody else/more noteworthy.'

Example (30) nicely illustrates the contribution of the prejacent presupposition 'Maria talked to at least a manager', because clearly the speaker is not disappointed that Maria did talk to the manager. A further test, utterance of assent/dissent (Roberts 2011) is shown in (31): The exclusive component is asserted (targeted by direct affirmation) and the prejacent projects.

(31) A: Maria rozmawiała tylko z menedżerem.
Maria talked only with manager
'Maria only talked to the manager.'

B: To prawda. {Nie rozmawiała z nikim innym. / #Rozmawiała z menedżerem.}
this true not talked with nobody else talked with manager
'That is true. {She did not talk to anybody else. / *[Incongruous:]* She talked to the manager.}'

B′: To prawda. {Nie rozmawiała z kimś ważniejszym. /
this true not talked with somebody more.important
#Rozmawiała z menedżerem.}
talked with manager
'That is true. {She did not talk to somebody more noteworthy. /
[Incongruous:] She talked to the manager.}'

In the diagnostics above, *tylko* patterns with *only*, in that it has both scalar and quantificational readings and its exclusive component is asserted, while the prejacent and the low-on-the-scale components are presupposed. In fact, *tylko* appears to be the most likely candidate for the counterpart of *only*, since it is the only exclusive that can have quantificational readings (in contrast to *zaledwie*, *dopiero* and *aż* discussed in the following sections). However, there is a significant difference between *tylko* and *only*: Although *tylko* has scalar readings, it is incompatible with a temporal scale. The sentence in (32a) can only mean that Marek woke up once (excluding all other 'times of waking up', i.e., the quantificational reading). The 'no later than'-reading is available with *(za)ledwie*, *dopiero* and *aż* but not with *tylko* (32b).

(32) a. Martwi mnie, że Marek obudził się tylko o godzinie 13.
worries me that Marek woke self only at hour 13
'I am worried that Marek woke up just once at 1 PM.'

b. Martwi mnie, że Marek obudził się {zaledwie / dopiero / aż /
worries me that Marek woke self merely only$_{\text{TEMP}}$ AŻ
#tylko} o godzinie 13.
only at hour 13
'I am worried that Marek woke up only at 1 PM and no sooner.'

The contrast between (33a) and (33b) shows that the presence of *tylko* results in an anomaly, which indicates how strong the restriction on the type of scale is – it is not possible to coerce a temporal reading.

(33) a. Jest {dopiero / zaledwie} luty.
is only$_{\text{TEMP}}$ merely February
'It is only February.' ('It is no later than February.')

b. # Jest tylko luty.
is only February
Incongruous: 'It is no other month now than February.'

If the dimension of the scale is underspecified in the semantics of an exclusive adverb, we expect it to be easily manipulated by the context (the semantics of the focused constituent the adverb associates with, e.g. 'February' above, and the pragmatics). The four Polish exclusives, however, appear to be more contextually restricted. Distance, temporal and numeral scales are all natural, inherent, dimensional scales, so the prediction should be that an exclusive that works with any kind of a scale is available in any language. This seems to be the case for *only* in English:

(34) We only reached Wilson Peak on that trip and were turned back [...] by an ice storm.

(35) I am usually up at 8:30am, but this morning for some reason I only woke up at 10am.

(36) Brussels was ranked only 62$^{\text{nd}}$ in Mercer's list of the world's most expensive cities for expats.

If Polish *tylko* can act scalar with individuals ordered on a contextual scale (importance, etc.) why is it incompatible with a temporal scale? In the next sections we will see that *zaledwie* is dedicated to pragmatic scales (just like *merely*) and *dopiero* to temporal scales. This means that *dopiero* might be blocking *tylko* (but not *zaledwie*) with a temporal scale, and that *zaledwie* does not block *tylko* with contextual scales (which perhaps suggests an interplay between the conventional nature of a scale and frequency in language use given that *zaledwie* is less frequent than *tylko*).

4.2 *(za)ledwie*

While we saw in the previous section that *tylko* has both quantificational and scalar readings, *zaledwie* is specified for scalar readings – it is a counterpart of English *merely*. What makes it particularly interesting though, is that it is the exact scalar opposite of the exclusive *aż*, which we will see in the next section. The example in (37) shows that *zaledwie* disprefers quantificational readings: In an impoverished context where no pragmatic ranking for Maria and others is available, the default is a scale of numbers of people who came (recall the two options for two scales in (25) above) and so the use of *tylko* is preferred.

(37) {Tylko / ?Zaledwie} Maria przyszła.
 only merely Maria came
 'Only/Merely Maria came.'

a. Quantificational reading: 'Nobody other than Maria came.'

b. Unavailable scalar reading: 'Nobody ranked higher than Maria came.'

Accordingly, (38) shows that the most accessible interpretation for *zaledwie* is that the excluded alternatives involve ranks higher than the manager (38B′) and not people other than the manager (38B).

(38) A: Maria rozmawiała zaledwie z menedżerem.
Maria talked merely with manager
'Maria only talked to the manager.'

B: To prawda. {#Nie rozmawiała z innymi. / #Rozmawiała z menedżerem.}
this true not talked with others talked with manager
'That is true. {*[Incongruous:]* She did not talk to other people. / *[Incongruous:]* She talked to the manager.}'

B′: To prawda. {Nie rozmawiała z kimś ważniejszym. / #Rozmawiała z menedżerem.}
this true not talked with somebody more.important talked with manager
'That is true. {She did not talk to somebody more noteworthy. / *[Incongruous:]* She talked to the manager.}'

Like *tylko*, *zaledwie* can scope both below and above negation, (39), which is expected given that *tylko* can do so on both scalar and quantificational readings.

(39) Maria nie rozmawiała zaledwie z menedżerem.
Maria not talked merely with manager

a. 'Maria talked to somebody more noteworthy than the manager in addition to the manager.' NOT > ONLY

b. 'Maria talked to everyone more noteworthy than the manager except the manager.' ONLY > NOT

The emotive factive diagnostic, just like with *tylko* in (28), reveals the positive presupposition of the prejacent, (40) (i.e., the speaker is not disappointed that Maria talked to the manager.)

(40) Jestem zawiedziony, że Maria rozmawiała zaledwie z menedżerem.
AUX disappointed that Maria talked merely with manager
'I am disappointed that Maria did not talk to anybody more noteworthy than the manager.'

What is interesting about *zaledwie* is that of the four exclusive particles, it is the only one that can work as the counterpart of English *barely* and receive APPROXIMATIVE (41) (Roberts 2011) and COUNTERFACTUAL (42) (Toosarvandani 2010) readings. On approximative readings, (41), the prefix *za-* can be dropped. On counterfactual readings, (42), *ledwie* occurs without *za-*.

(41) {(Za)ledwie / #Tylko / #Dopiero / #Aż} Marek otworzył drzwi, a
merely only $\text{only}_{\text{TEMP}}$ AŻ Marek opened door and
zadzwonił telefon.
rang telephone
'Marek barely opened the door, when the telephone rang.'

(42) Marek {(*za)ledwie / #tylko / #dopiero / #aż} otworzył drzwi.
Marek merely only $\text{only}_{\text{TEMP}}$ AŻ opened door
'Marek barely opened the door. (It was more likely for him to not open the door.)'

Ledwie is syntactically an adverbial (also *ledwo*), i.e. can only modify the VP, as shown by the contrast in (43): In (43a) *(za)ledwie* associates with Berlin, in (43b) *ledwie* associates with the VP 'reached Berlin' as can be seen from the translation, and here the prefix *za-* is not possible.

(43) a. Dotarłem (za)ledwie do Berlina.
reached merely to Berlin
'I only reached Berlin.'

b. (*Za)ledwie dotarłem do Berlina.
merely reached to Berlin
'I barely made it to Berlin.'

Summing up, the exclusive *zaledwie* prefers scalar readings (like *merely* does), but its variant *ledwie* has approximative and counterfactual readings.

4.3 *dopiero*

In (32a) above we saw that 'It is only February' requires *dopiero* in its translation, repeated below in (44), because *tylko* results in an anomaly ('It is no other month now than February'). This suggests that *dopiero* is a specialized TEMPORAL *only* and I gloss it as *only*$_{\text{TEMP}}$.

(44) a. It is only February.
b. Jest {dopiero / *tylko} luty.
is only$_{\text{TEMP}}$ only February
'It is only February. (It is no later than February.)'

The use of *dopiero*, in contrast to the other exclusives, coerces the scale associated with the predicate, e.g. quantity in (45)–(46), to an interpretation with respect to time.

(45) Maria rozmawiała dopiero z 2 osobami.
Maria talked only$_{\text{TEMP}}$ with 2 people
'Maria only talked to 2 people so far.'

While in the example (46) below temporal alternatives are obligatorily evoked, which results in the reading 'so far Jan bought 2 tomatoes and he will buy more', there is no such expectation with *tylko* in (47).

(46) Jan kupił dopiero 2 pomidory.
Jan bought only$_{\text{TEMP}}$ 2 tomatoes
'Jan bought only two tomatoes and we expect him to buy more.'

(47) Jan kupił tylko 2 pomidory.
Jan bought only 2 tomatoes
'Jan bought only two tomatoes.'

Logically, the temporal alternatives should be later in time: In (45) more people will come later, in (46) Jan will buy more tomatoes later, so in a context where such alternatives are not possible, as in (48) where Jan is not going to write any more, *dopiero* is infelicitous.

(48) Jan skończył pisać swój esej. Ma {tylko / #dopiero} 10 stron.
Jan finished writing his essay has only only$_{\text{TEMP}}$ 10 pages
'Jan finished writing his essay. It is only 10 pages long.'

The example above provides an interesting contrast to the sentence in (1) where the team ended up in 30th place (i.e., was not going to compete anymore) and *dopiero* is felicitous. Indeed, we find *dopiero* with scales that are not (necessarily) correlated with temporal progression, such as the example in (49) about a partial achievement, idiomatically referred to as halfway to success.

(49) Dobra drukarka to jednak dopiero połowa sukcesu. Drugą, równie
good printer is however only$_{\text{TEMP}}$ half success second equally
istotną częścią jest odpowiedni toner.
important part is suitable toner
'A good printer is only half of the success. Another equally important thing is a good toner.'

Like the other exclusives we considered, *dopiero* can take scope both above and below negation as we can see in (50)–(51).[6]

(50) Dopiero 2 osoby przyszły.
Only$_{\text{TEMP}}$ 2 people came
'So far only 2 people showed up.'

(51) Nie przyszły dopiero 2 osoby.
not came only$_{\text{TEMP}}$ 2 people
a. 'More than 2 people came.' NOT > ONLY
b. 'Everyone came except 2 people.' ONLY > NOT

The translations for the examples above indicate that *dopiero* is similar to the temporal adverbs *so far* (*jak dotąd*) / *yet* (*jeszcze*). However, *dopiero* also comes with an exclusive component which these temporal adverbs do not have by themselves. First, let us note that *dopiero*, like *yet*, and unlike *so far*, can occur with negation and in questions, (52). Note also that *so far* is not an NPI in contrast to *only* and *yet*, (53).

[6]Native speakers of Polish can reflect on how additional context facilitates the interpretation of the two scope configurations. Consider also the different prosodic contours with different focus structures. In (51a) *dopiero* associates with focus on '2 people', see (i.a) below. In (51b) *dopiero* takes wide scope and associates with sentential focus, which can be made overt with scrambling as in (i.b).

(i) a. Nie, nie przyszły dopiero [2 osoby]$_F$, są już [3 osoby]$_F$.
no not came only$_{\text{TEMP}}$ 2 people are already 3 people
'So far not just 2 people showed up, but 3.' NOT > ONLY

b. Wydaje mi się, że {nie przyszły dopiero 2 osoby / dopiero [2 osoby nie
seems me self that not came only$_{\text{TEMP}}$ 2 people only$_{\text{TEMP}}$ 2 people not
przyszły]$_F$}, ale może się pomyliłam w liczeniu.
came but maybe self mistook in counting
'I think that only 2 people didn't show up, but maybe I counted wrong' ONLY > NOT

(52) Kto wie, czy to nie jest dopiero początek problemów
who knows if this not is only$_{\text{TEMP}}$ beginning problems
premiera?
prime.minister.GEN
'Who knows if this is only the beginning of the prime minister's problems?'

(53) Who knows, maybe this is {only/yet/*so far} the beginning of the prime minister's problems?

Secondly, with *so far* in (54a) there is an implication that Maria talked to just one person, but this is not an assertion because it is possible to continue the sentence as in (54b) in contrast to (54c) containing an exclusive. The counterpart of (54c) in Polish, (55), is compatible with both *tylko* and *dopiero*.

(54) a. So far Maria has talked to the manager.
b. So far Maria has talked to the manager and a couple of other people.
c. So far Maria has only talked to the manager #and a couple of other people.

(55) Jak dotąd Maria rozmawiała {dopiero / tylko} z menedżerem.
how till.now Maria talked only$_{\text{TEMP}}$ only with manager
'So far Maria has only talked to the manager.'

That *so far* does not assert the exclusion of alternatives is also apparent in (56)–(58) below. In (56) we can expect that the person will call later, and accordingly in the Polish translation in (57) *dopiero* is not felicitous.

(56) So far he has not called.

(57) {Jak dotąd / *dopiero} nie zadzwonił.
how till.now only$_{\text{TEMP}}$ not called
'So far he has not called.' (Lit. 'Up till now he has not called.')

The presence of *so far* and *only* in (58) together results in an implication that documenting 373 species is a lot, because 'it is still only February', as said in 2012, indicates that this number was reached earlier than expected.[7]

[7]The source of (58) is https://james-champion.com/friday-24th-february-2012/ (accessed on 29 March 2012). In fact, John's initial goal for 2012 was 500 species and by the end of the year he documented 591 species.

(58) John Cahill is going for an all-out mega-bird list for Guatemala in 2012. […]. So far he has reached 373 species in Guatemala this year – and it is still only February!

In contrast to *so far* in (58), *dopiero* in the sentence in (59) contributes the reading that 373 species is not much, because, as the utterance of assent test shows, (60), the exclusive component 'no more than 373 yet' is asserted. The same test shows that *jak dotąd* (*so far*) lacks the exclusive component.

(59) A: Jan dotarł dopiero do 373 gatunków.
Jan reached only$_{\text{TEMP}}$ to 373 species
'Jan has only reached 373 species so far.'

B: To prawda. {Nie dotarł do więcej niż 373 gatunków. / #Dotarł do 373 gatunków.}
this true not reached to more than 373 species reached to 373 species
'That's true. {He hasn't reached more than 373 species. / *[Incongruous:]* He has reached 373 species.}'

(60) A: Jan dotarł jak dotąd do 373 gatunków.
Jan reached so far to 373 species.
'Jan has so far reached 373 species.'

B: To prawda. Dotarł (już) do 373 gatunków.
this true reached already to 373 species
'That's true. He has (already) reached 373 species.'

The data above demonstrates that *dopiero* specializes for temporal scales and now we will see that the diagnostics for assertion and presupposition replicate the results we found for the other exclusives. The utterance of assent test, (61), the emotive factive predicate test, (62) and negation test, (63), show that the exclusion of alternatives higher on the (temporal) scale is asserted, the prejacent is presupposed and the low-on-scale reading is presupposed.

(61) A: Maria rozmawiała dopiero z menedżerem.
Maria talked only$_{\text{TEMP}}$ with manager
'Maria only talked to the manager so far.'

B: To prawda. {Nie rozmawiała (jeszcze) z innymi. / #Rozmawiała z menedżerem.}
this true not talked yet with others talked with manager
'That is true. {She has not talked to other people yet. / *[Incongruous:]* She talked to the manager.}'

B′: To prawda. {Nie rozmawiała (jeszcze) z nikim ważniejszym. /
this true not she.talked yet with nobody more.important
#Rozmawiała z menedżerem.}
she.talked with manager
'That is true. {She has not talked to anybody more noteworthy yet. / *[Incongruous:]* She talked to the manager.}'

(62) Jestem zawiedziony, że Maria rozmawiała dopiero z menedżerem.
AUX disappointed that Maria talked only$_{\text{TEMP}}$ with manager
'I am disappointed that Maria hasn't talked to anybody else/more noteworthy yet.' (*Implied:* 'Maria talked to the manager.')

(63) Maria nie rozmawiała dopiero z menedżerem.
Maria not talked only$_{\text{TEMP}}$ with manager
a. 'Maria has already talked to somebody more noteworthy than the manager in addition to the manager.' NOT > ONLY
b. 'Maria has already talked to everyone more noteworthy than the manager except the manager.' ONLY > NOT

Finally, we must note that while *dopiero* excludes stronger alternatives on a temporal scale, they are not necessarily later, as is clear in (64). What is excluded in (64a) are earlier alternatives for finishing the writing. In this context, the exclusive that I am going to present in the next section, *aż*, is also possible. The fact that in (64b) *aż* is not felicitous, but instead it requires the preposition *do*, lit. 'to', (64c), which carries the meaning of 'until' (i.e., here *aż* is optional but *do* is not; see also (104) in Section 6) indicates that *aż* is not inherently specified for temporal scales and what allows it in (64a) is the presence of the numeral.

(64) a. Skończę pisać {dopiero / aż / #tylko / #zaledwie} za 2 miesiące.
will.finish writing only$_{\text{TEMP}}$ AŻ only merely in 2 months
'I will finish writing no sooner than in 2 months.'
b. Skończę pisać {dopiero / #aż / #tylko / #zaledwie} w sobotę.
will.finish writing only$_{\text{TEMP}}$ AŻ only merely on Saturday
'I will finish writing no sooner than on Saturday.'
c. Skończę pisać {aż / #dopiero / #tylko / #zaledwie} do soboty.
will.finish writing AŻ only$_{\text{TEMP}}$ only merely until Saturday
'I will finish writing until Saturday, which is late.'

We can conclude that *dopiero* has the same meaning components as *tylko*/*zaledwie*, but it is specified for numeral, especially temporal scales. In the next section, I will show that *aż* is a scalar opposite of *dopiero*.

4.4 *aż*

The exclusive *aż* is a scalar opposite of *zaledwie* (Tomaszewicz 2012, Tomaszewicz 2013a,b). The assent test in (65) shows us first that *aż* has no quantificational reading. Neither of the continuations in (65B) are possible – just like with *zaledwie* in (38B). Secondly, we see in (65B′) that the asserted component is 'She did not talk to somebody less noteworthy', which is the direct opposite of what we saw in (38B′) with *zaledwie*: 'She did not talk to somebody more noteworthy'.

(65) A: Maria rozmawiała aż z menedżerem.
Maria talked AŻ with manager
'Maria talked to somebody so important as the manager.'

B: To prawda. {#Nie rozmawiała z nikim innym. / #Rozmawiała
this true not talked with nobody other talked
z menedżerem}.
with manager
'That is true. {*[Incongruous:]* She did not talk to other people. /
[Incongruous:] She talked to the manager.}'

B′: To prawda. {Nie rozmawiała z kimś mniej ważnym. /
this true not talked with somebody less important
#Rozmawiała z menedżerem.}
talked with manager
'That is true. {She did not talk to somebody less noteworthy. /
[Incongruous:] She talked to the manager.}'

Native speaker intuitions are particularly robust for the juxtaposition of *aż* and *zaledwie* in (66)–(69) where the reason clause identifies the asserted content (Dretske 1972, Beaver & Clark 2008). In (66)–(67) the reason for Maria's (un)happiness is the position of the manager on the scale: high with *aż*, low with *zaledwie*. In (68)–(69) the reason for the party's success/failure is the fact that the expected number of guests was either exceeded, (68), or not achieved, (69). With *aż* in (66) Maria is happy because she is not talking to somebody less important than the manager and in (68) the party was a success because no less than 15 people came. The mere fact that Maria is talking to the manager is not the reason for her happiness – the fact that the manager is high on the scale of the relevant people is the reason. Similarly, the fact that 15 people came to the party is not the reason for the success – the fact that this number exceeded the expectations is the reason. Conversely, with *zaledwie* in (67) Maria is unhappy because her goal is to talk to someone more important than the manager and in (69) the party was a failure because more than 15 people were expected. The reason clause test demonstrates that exclusivity is asserted, which means that *aż* works like the scalar opposite of *zaledwie*, *tylko* and *dopiero*: It carries a high-on-the-scale presupposition.

(66) Maria jest zadowolona, bo rozmawia, aż z menedżerem.
Maria is happy because talks AŻ with manager
'Maria is happy because she is not talking to somebody less important than the manager.'

(67) Maria jest niezadowolona, bo rozmawia zaledwie z menedżerem.
Maria is unhappy because talks merely with manager
'Maria is unhappy because she is not talking to somebody more important than the manager.'

(68) Impreza się udała, bo przyszło aż 15 osób.
party self succeeded because came AŻ 15 people
'The party was a success because no less than 15 people came.'

(69) Impreza się nie udała, bo przyszło {tylko / zaledwie} 15 osób.
party self not succeeded because came only merely 15 people
'The party was not a success because no more than 15 people came.'

A parallel example with *dopiero* does not work, because it evokes a temporal alternative when more people come.

(70) Impreza się (nie) udała, #bo przyszło dopiero 15 osób.
party self not succeeded because came $\text{only}_{\text{TEMP}}$ 15 people
'The party was (not) a success *[Incongruous:]* because so far only 15 people came.'

The high-on-the-scale presupposition found with *aż* projects under negation and in questions, and the orientation of the scale stays the same, as shown in (71c), (72c), (73c). Embedding under question and negation operators clearly reveals the asserted exclusive component – the scalar opposite of the contribution of *only*. Although (71) could simply be taken to convey that Maria talked to a manager and that this is significant, its negative version, (72), crucially asserts not only that Maria did not talk to a manager, but also that she talked to somebody lower on the scale than the manager, (72a). This is evidence for an asserted exclusive component.

(71) Maria rozmawiała aż z menedżerem.
Maria talked aż with manager
'Maria talked to somebody so important as the manager.'

a. Maria talked to nobody ranked lower than a manager. Assertion
b. Maria talked to at most a manager. Presupposition
c. Maria's talking to a manager is significant. Presupposition

(72) Maria nie rozmawiała aż z menedżerem.
Maria not talked aż with manager
'Maria talked to somebody less important than the manager.'

a. Maria talked to somebody ranked lower than a manager. Assertion
b. Maria talked to at most a manager. Presupposition
c. Maria's talking to a manager is significant. Presupposition

Exclusivity also emerges in the question in (73) which asks whether the position of the person Maria talked to is lower than a manager, (73a). (This is exactly the opposite of the effect of the presence of scalar *tylko* 'only' in the question in (28)).

(73) Czy Maria rozmawiała aż z menedżerem?
Q Maria talked aż with manager
'Did Maria talk to somebody so important as the manager?'

a. Did Maria talk to somebody ranked higher than a manager? Question
b. Maria talked to at most a manager. Presupposition
c. Maria's talking to a manager is significant. Presupposition

The exclusive component in (71a), (72a) and (73a), i.e. the assertion that the alternatives lower on the scale are excluded, is the reverse of the contribution of *only*/*merely*, where higher alternatives are excluded, as we saw in (26a), (27a) and (28a). The two other meaning components of *aż* that project are scalar reversals of the two presuppositions of *only/merely* identified in (26b/c), (27b/c) and (28b/c).

In (71b), (72b), (73b), the presupposition is that the alternatives under consideration can be at most as high on the scale as the prejacent. The presupposition in (71c), (72c), (73c) is that the prejacent is significantly high on the scale. For (71) this gives us the interpretation that considering the alternative positions that the person Maria talked to could hold (71b), being a manager is the most significant among them (71b), (71c), and that that person is indeed no less than a manager (71a).

Beaver & Clark (2008) sum up the contribution of *only* as "contra expectation, nothing stronger holds" (p. 279). The scalar reversal of each of the meaning components of *only*, yields the interpretation for *aż* that can be described as 'contra expectation, nothing weaker holds'.

It is frequently noticed that *even* functions as an antonym of *only* at the level of presupposition (Horn 1969, König 1991, Beaver & Clark 2008, Greenberg 2022). The following contrasts between *aż*, *zaledwie* and *nawet* 'even' show that the scalar presupposition of *aż* and *zaledwie* picks up the contextually salient scale, namely, distance, while *even* requires the likelihood scale (contra the speculation about *aż* vs. *nawet* in Bajaj 2016). Negation in the presence of *aż* in (74) conveys that the distance that Jan traveled was shorter than to Warsaw; the sentence cannot mean that Jan did not travel at all, which is a possible interpretation in (12) containing negation and *even*. The sentence in (74) tells us both that (i) Jan went as far as Łódź, but not Warsaw (see Figure 1 below for the illustration of the distance), and that (ii) Jan's trip was not noteworthy, because it was not far. This meaning results from negation scoping above *aż* and is exactly parallel to the meaning of (76) where negation scopes above *zaledwie*. (76) says that (i) Jan went further than Łódź, Jan went as far as Warsaw, and that (ii) Jan's trip was noteworthy, because it was far. Both the use of *aż* and *zaledwie* induce a distance reading in contrast to *nawet* 'even' in (75). In (75) *nawet* needs to scope above negation, the scalar presupposition is that the prejacent 'Jan went to Łódź' is the most likely among the alternatives (i.e., the least significant).

(74) Jan nie pojechał aż do Warszawy, {pojechał zaledwie do Łodzi / #został
Jan not drove AŻ to Warsaw drove merely to Łódź stayed
w domu}.
at home
'Jan did not drive as far as Warsaw, {he drove to Łódź / *[Incongruous:]* he stayed at home}.'

(75) Jan nie pojechał nawet do Łodzi, {#pojechał zaledwie do Warszawy. /
Jan not drove even to Łódź drove merely to Warsaw
został w domu}.
stayed at home
'Jan did not even drive to Łódź, {*[Incongruous:]* he drove merely to Warsaw / he stayed at home}.'

(76) Jan nie pojechał zaledwie do Łodzi, {pojechał aż do Warszawy. / #został
Jan not drove merely to Łódź drove AŻ to Warsaw stayed
w domu}.
at home
‘Jan did not go merely to Łódź, {he went all the way to Warsaw / *[Incongruous:]* he stayed at home}.’

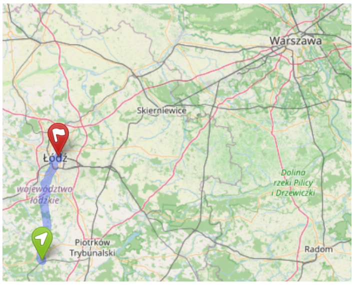

Figure 1: An illustration of the distances in the situation in example (74): ‘Jan did not go as far as Warsaw, he went merely to Łódź.’

In addition to its use as a scalar particle, *aż* functions in Polish as a complementizer, equivalent to *until* in English, (77a). As a complementizer, it is obligatorily sentence-initial, but as we saw in Section 3 focus associators can associate with foci of all sizes, including the whole sentence in Polish. I propose that the lexical semantics of *aż* is the same when it is interpreted as *until*, (77a)–(77b), and when it is a scalar adverb, (77c), but the syntax determines whether the ordering of alternatives is on a temporal scale or a pragmatic scale.[8] I observe that when the main clause explicitly mentions duration, *aż* can either work as *until*, (77b), or as the scalar exclusive in which case it is preceded by the complementizer *that*, (77c). In (77c) *aż* can either associate with the subject so that the alternatives are people who could tell Ann to stop singing, or with the whole clause so that alternatives are other less serious things that may happen, e.g., the whole auditioning committee laughing.

(77) a. Anna śpiewała, aż [$_{\text{IP}}$[$_{\text{DP}}$ Maria] kazała jej przestać].
Anna sang AŻ Maria ordered her stop
‘Anna sang until Maria told her to stop.’

[8]I thank Radek Šimík for pointing out that the other exclusive particles cannot switch in a parallel way with respect to the syntax, suggesting that the semantics of *aż* alone cannot be responsible for its function as a complementizer. This is an interesting topic for (a cross-linguistic) investigation.

b. Anna śpiewała tak długo, aż [$_{\text{IP}}$[$_{\text{DP}}$ Maria] kazała jej przestać].
Anna sang so long AŻ Maria ordered her stop
'Anna kept on singing until Maria told her to stop.'

c. Anna śpiewała tak {słabo / długo}, że aż [$_{\text{IP}}$[$_{\text{DP}}$ Maria] kazała jej
Anna sang so badly long that AŻ Maria ordered her
przestać].
stop

i. 'Anna sang so badly/long, that out of all things that could happen Maria told her to stop.'

ii. 'Anna sang so badly/long, that out of all people Maria told her to stop.'

In this section we established that *aż* is a scalar opposite of *zaledwie*, scalar *tylko* and *dopiero.* Similarly to *dopiero*, it receives a temporal interpretation when in the complementizer position. In the next section, I show how the lexical entries for the four exclusives can be related by a common core semantics.

5 Analysis

I propose that the four exclusives, *tylko*, *zaledwie*, *dopiero* and *aż* are a family related by having the same three semantic components: the exclusive assertion, the scalar presupposition and the prejacent/factive presupposition. The differences between them involve the direction of the scale and extra requirements on the sets of alternatives resulting in their idiosyncrasies. The facts are summarized in Table 1. Below I define the lexical entries for each member of the family in such a way that the common core of the three components is present and only the necessary parameters are tweaked.

The meaning of the scalar exclusive, *tylko/zaledwie* involves the assertion that no higher alternative is true, and the factive presupposition (78a), as well as the low-on-the-scale presupposition (78b). Scalarity is thus both asserted and presupposed.

(78) $[\![\textit{tylko/zaledwie}]\!]^{w}(C)(p)$ is defined iff

a. $\exists q \in C[q \geq p \wedge q(w) = 1]$
[Factivity: an alternative at least as strong as p is true]
and

b. $\forall q \in C[q \neq p \rightarrow q > p]$
[Scalarity: p is the weakest alternative]

Table 1: The meaning components of *tylko*, *zaledwie*, *dopiero* and *aż*

tylko	*(za)ledwie*	*dopiero*	*aż*
STATUS OF PREJACENT p			
p presupposed	p presupposed	p presupposed	p presupposed
SCALAR PRESUPPOSITION (p HIGH/LOW ON SOME SCALE S)			
p low on S	p low on S	p low on S	p high on S
EXCLUSIVE ASSERTION (ALTERNATIVES EXCLUDED, HIGHER/LOWER ON S THAN p)			
higher than p	higher than p	higher than p	lower than p
IDIOSYNCRASIES			
S is logical or pragmatic but not temporal	S is pragmatic; approximative readings	S is numeral/ temporal	S is pragmatic; temporal ‘until’ meaning

if defined then
$[\![\textit{tylko/zaledwie}]\!]^{w}(C)(p) = \forall q[q \in C \wedge q \neq p \wedge q > p) \rightarrow q(w) = 0]$

The FACTIVE presupposition is the positive meaning component that projects under negation, therefore it is not just the prejacent proposition, but an alternative proposition that is at least as strong on the relevant scale (I am following Klinedinst 2005 and Coppock & Beaver 2014, but see also Beaver & Clark 2008 and Zeevat 2009 for a mirative presupposition that the prejacent falls short of what is expected). This positive contribution is paraphrasable with ‘at least’ as we first saw in the example in (16): ‘Maria only talked to the manager’ implies that ‘Maria talked to (at least) the manager’, (16c). Coppock & Beaver (2014) point out that this component is easier to see on scalar readings, in examples such as (79), than on quantificational readings, as in (80).

(79) This is not {just/a mere} down payment.
Positive contribution: ‘This is at least a down payment.’
SCALAR READING Coppock & Beaver 2014: (8–9) p. 374

(80) Mary did not invite only John and Mike.
Positive contribution: ‘Mary invited at least John and Mike.’
QUANTIFICATIONAL READING Coppock & Beaver 2014: (18) p. 379

Since my analysis follows the accounts where the quantificational and scalar readings are derived from the same lexical entry (Bonomi & Casalegno 1993, van

Rooij 2002, Beaver 2004, Klinedinst 2005, Riester 2006, Beaver & Clark 2008, Coppock & Beaver 2013, 2014), the factive presupposition of *tylko/zaledwie* in (78a) allows for both readings. As mentioned in Section 2, the dimension of the scale is set by the context and quantificational readings arise with logical scales, whereas scalar readings result from pragmatic orderings of alternatives in *C*. In impoverished contexts the logical scale is the default.

Greenberg (2022), building on Guerzoni (2003), proposes that the scalar presupposition of *only* is "superlative" so that p is the weakest alternative in the context of focus alternatives, C. She also discusses how a general notion of stronger alternatives, symbolized as '>', suffices, given that C is similar to domain restriction on quantifiers and thus contains only contextually relevant alternatives, but at the same time the focus structure is constrained by the Question Under Discussion that is salient in the context (Rooth 1992, Roberts 1996, Beaver & Clark 2008, Coppock & Beaver 2013, 2014). Following Guerzoni (2003), Greenberg (2022) treats *only* and *even* as antonyms, with mirroring superlative presuppositions, but adds special constraints on C with respect to the QUD in order to account for the mirror image felicity patterns with *only* and *even*. She suggests that the superlative formulation of the scalar presupposition can be used to capture the differences in the types of scales with different operators, which is a question for future research. In this chapter, I focus on the common core of the exclusives, therefore I simply adopt the idea of stronger/weaker alternatives.

In (78) above, I proposed a single lexical entry for *tylko/zaledwie* because *zaledwie* does not block *tylko* from pragmatic scales and its use with numeral scales seems to only be dispreferred, as was shown in example (37) in Section 4.2. However, the approximative and counterfactual uses of *(za)ledwie* require an extra presupposition: the lexical entry for *(za)ledwie*$_{\text{APPROX}}$ makes it salient that *not p* could happen, (81c). I observed that those uses require obligatory wide focus (VP or sentence focus), which I take to correlate with the fact that $\neg p$ is a salient alternative. Given that C is determined also by the QUD, if $\neg p$ is salient, the context disambiguates between the lexical entries in (78) and (81).

(81) $[\![\textit{(za)ledwie}_{\text{APPROX}}]\!]^{w}(C)(p)$ is defined iff

a. $\exists q \in C[q \geq p \wedge q(w) = 1]$
[Factivity: an alternative at least as strong as p is true]
and

b. $\forall q \in C[q \neq p \rightarrow q > p]$
[Scalarity: p is the weakest alternative]
and

c. $\neg p > p$
[Scalarity: p is weaker than $\neg p$]

if defined then
$[\![\textit{(za)ledwie}_{\text{APPROX}}]\!]^w(C)(p) = \forall q[(q \in C \wedge q \neq p \wedge q > p) \rightarrow q(w) = 0]$

A separate entry for *dopiero* is needed. This is the only exclusive that is limited to temporal scales and blocks *tylko* and *zaledwie* from temporal uses (in contrast to *only* in English). In Section 4.3 we showed that in contrast to *tylko/zaledwie*, whose scalarity is pragmatically determined, *dopiero* requires a temporal scale. We can build this into its semantics by keeping exactly the same meaning components as for *tylko/zaledwie* in (78) but adding a presupposition specifically about a temporal scale such that the time of p is more noteworthy than the times of the alternatives, either because it was early or because it was late. The exclusive assertion stays the same, because as I noted in Section 4.3, *dopiero* excludes stronger alternatives on the contextual scale, not necessarily later ones (e.g., buying more than 2 tomatoes, (46), vs. finishing earlier than in 2 months, (64a)).[9]

(82) $[\![\textit{dopiero}]\!]^w(C)(p)$ is defined iff

a. $\exists q \in C[q \geq p \wedge q(w) = 1]$
[Factivity: an alternative at least as strong as p is true]
and

b. $\forall q \in C[q \neq p \rightarrow q > p]$
[Scalarity: p is the weakest alternative]
and

c. $\exists t_1 \in T[p = \lambda w[t_1 \text{ in } w]] \wedge \forall q \in C[\exists t_2 \in T[q = \lambda w[t_2 \text{ in } w]] \wedge q \neq p \wedge t_1 >_{\text{NOTEWORTHY}^w} t_2]$
[Temporality: Time t of p is the most noteworthy.]

if defined then
$[\![\textit{dopiero}]\!]^w(C)(p) = \forall q[(q \in C \wedge q \neq p \wedge q > p) \rightarrow q(w) = 0]$

As established in Section 4.4, *aż* is the scalar opposite of *tylko/zaledwie*, thus the scale is reverted in the asserted and the presupposed components: (83) vs. (78). The factive presupposition, (83a), states that the prejacent or an alternative at most as strong is true. The scalar presupposition, (83) (ii), is that the prejacent is the highest on the scale. Exclusivity but also scalarity, ‘no lower alternative’, is asserted in the last line of (83).

[9] My intuition is that the exclusion of weaker alternatives should account for why the time of the event in the prejacent is more noteworthy than that of a stronger alternative and then the extra presupposition in (82c) should be superfluous. This remains a topic for future work.

(83) $[\![a\dot{z}]\!]^w(C)(p)$ is defined iff

a. $\exists q \in C[q \leq p \wedge q(w) = 1]$
[Factivity: an alternative at most as strong as p is true]
and

b. $\forall q \in C[q \neq p \rightarrow p > q]$
[Scalarity: p is the strongest alternative]

if defined then
$[\![a\dot{z}]\!]^w(C)(p) = \forall q[(q \in C \wedge q \neq p \wedge p > q) \rightarrow q(w) = 0]$

Compare now how *aż* differs from the additive scalar adverb *even*. The prejacent presupposition in (84a) expresses additivity, and the scalar presupposition involves a scale of likelihood, (84b) (Guerzoni 2003). There is no contribution to the assertion of the sentence.

(84) $[\![even]\!]^w(C)(p)$ is defined iff

a. $\exists q \in C[q \neq p \wedge q(w) = 1]$
[Factivity and additivity: an alternative q is true]
and

b. $\forall q \in C[q \neq p \rightarrow q >_{\text{LIKELY}^w} p]$
[Scalarity: p is the least likely alternative]

if defined then
$[\![even]\!]^w(C)(p) = p(w)$

Greenberg (2022) observes that although *even* and *only* are scalar opposites, *even* must place the prejacent high on the scale, "above the norm" (formulated as a universal scalar presupposition in Guerzoni 2003: the least likely among all the alternatives, (84b)). However, *only* "does not inherently express 'a little', but just needs to indicate 'less' than its alternative(s) in C" (Greenberg 2022: 23). For this reason *only* is felicitous in the scenario in (85a), but *even* is not felicitous in (85b).

(85) a. John is tall. He is 1.92m. Bill is shorter – he is only 1.88m (but he is still tall). Greenberg 2022: (52a) p. 21

b. John is short. He is 1.52m. Bill is taller. He is (#even) 1.58m (but he is still short). Greenberg 2022: (52b) p. 21

In contrast to *even*/*nawet*, *aż* is indeed felicitous in the Polish counterpart of Greenberg's example in (86b), thanks to its contextually defined scale, as opposed to a likelihood scale. (As expected, (86a) shows that *tylko*/*zaledwie* behave in the exact same way as *only*.)

(86) a. Jan jest wysoki. Ma 192 cm wzrostu. Bill jest niższy. Ma {tylko / zaledwie} 188 cm wzrostu (ale i tak jest wysoki.)
Jan is tall has 192 cm height Bill is shorter has only merely 188 cm height but even so is tall
'Jan is tall. He is 192 cm. Bill is shorter – he is only 188 cm (but he is still tall).'

b. Jan jest niski. Ma 152 cm wzrostu. Bill jest wyższy. Ma {aż / #nawet} 158 cm wzrostu (ale i tak jest niski.)
Jan is short has 152 cm height Bill is taller has AŻ even 158 cm height but even so is short
'Jan is short. He is 152 cm. Bill is taller. He is *aż*/(#even) 158 cm (but he is still short).'

Moreover, it can be demonstrated that *aż* cannot use a likelihood scale that matches the wider discourse/world-knowledge. The scale with *aż* must be constructed with reference to the lexical content of the associate, e.g., a non-professional in (87). In (87) *even* adds the reading that hiring a non-professional is very unlikely (under normal circumstances). *Aż* is infelicitous in this sentence because it requires that a non-professional is ranked high on the scale of people we want to hire.

(87) Potrzebujemy sobowtóra głównego aktora. Zatrudnimy {nawet / #aż} nieprofesjonalistę.
we.need body.double lead actor will.hire even AŻ non.professional
'We need a body double of the lead actor. We will even hire a non-professional.'

The factive presupposition of *aż*, (83a), is not additive like the presupposition of *even*, (84a). This means that *aż* can occur in additive contexts, (88a)–(88b), and it can even co-occur with *nawet* 'even', (89).[10]

(88) a. Janek zjadł aż 5 jabłek.
Janek ate AŻ 5 apples
'Janek ate even 5 apples.'

[10] I thank the anynonymous reviewer for raising this issue.

b. Janek nie rozmawiał zaledwie z dziekanem, lecz aż z rektorem.
Janek not talked merely with dean but AŻ with chancellor
'Janek did not talk merely to the dean but to the chancellor himself.'
(Tomaszewicz 2012: 325, ex. (7b))

(89) Maria rozmawiała z szefem zakładu, rozmawiała z dziekanem, rozmawiała nawet aż z rektorem.
Maria talked with head institute talked with dean talked even AŻ with chancellor
'Maria talked to the institute head, she talked to the dean, she talked even to the chancellor himself.'

The emotive factive test in (90) shows that with *nawet* the assertion is the prejacent, but with *aż* it is the excluded component, namely, Maria's happiness in (90b) is about the fact that she got to talk to somebody high on the scale of important people, not about the fact that she talked to these particular people (the same effect was shown in (66)).

(90) Maria rozmawiała z szefem zakładu, rozmawiała z dziekanem, …
Maria talked with head institute talked with dean
'Maria talked to the institute head, she talked to the dean, …'

a. i jest zadowolona, bo rozmawiała nawet z rektorem.
and is happy because talked even with chancellor
'and she's happy because she talked even to the chancellor.'
Implied: 'Maria is happy because she talked to the chancellor.'

b. i jest zadowolona, bo rozmawiała aż z rektorem.
and is happy because talked AŻ with chancellor
'and she's happy because she talked to the chancellor himself.'
Implied: 'Maria is happy because she talked to somebody no less important than the chancellor.'

This also indicates that a shift in the QUD (from 'Who did Mary talk to?' to 'How important a person did Mary talk to?') changes the set of relevant alternatives in C, pruning some of those included in the prior discourse as shown in Greenberg (2022) for both *even* and *only*. In (88a) *aż* is felicitous although it is true that Janek ate 4/3/2/1 apples.

The formal mechanism of focus association derives the set of alternatives in C (so that the size and location of focus determine the contents of the set) and this set can be further restricted by the current QUD. Below, I illustrate this for *aż*; the mechanism works exactly the same for any focus-associating item taking sentential scope.

(91) Condition on focus association (Rooth 1992)
$C \subseteq [\![\alpha]\!]^f$, or $C \subseteq \bigcup [\![\alpha]\!]^f$
where C is the restrictor of a quantificational adverb and α the sister to $\sim$

In (92) I present the derivation for association with the focus 'Warsaw'. The meaning of the sentence in (92b) results from the specification of the lexical entry in (83). Note how it does not tell us what the weaker alternatives are. This information is derived from focus association: the focus operator $\sim$ that interprets the feature F on the syntactic constituent is attached at the sentential level, (92a), and thus its restrictor variable S is a subset of the focus alternative value of the TP. The focal presupposition is the set of alternative propositions {'that Mary drove to Łódź', 'that Mary drove to Grójec', 'that Mary drove to Berlin', …} in (92c). The restrictor of *aż*, C, is a subset of that set satisfying the condition on focus association in (91). Because $q \in C$, the excluded alternatives are a subset of the set in (92c). This subset does not include stronger alternatives such as 'that Mary drove to Berlin' because Berlin is further than Warsaw in the context where Mary is driving from Bełchatów (as seen in Figure 1). Obviously, on her way to Warsaw Mary drove through other places, but the QUD is 'How significantly far did Mary drive?' so the alternatives, 'Mary drove to Łódź', 'Mary drove to Grójec' are true, but do not answer the QUD because those places are not significantly far from Bełchatów.

(92) Mary pojechała aż do [Warszawy]$_F$.
Mary drove AŻ to Warsaw
'Mary drove all the way to Warsaw.'

a. [AŻ C] [$_{TP_2}$ [$\sim S$] [$_{TP_1}$ Mary drove to [Warsaw]$_F$]]
b. $[\![(92)]\!] = \lambda w.\forall q[(q \in C \wedge q \neq [\![\text{Mary drove to Warsaw}]\!] \wedge q <$ $[\![\text{Mary drove to Warsaw}]\!] \rightarrow \neg q(w)]$
c. $S \subseteq [\![\text{TP}_1]\!]^f = \{q : \exists x[q = \lambda w[\textsc{drive to}(\textsc{Mary}, x) \text{ in } w)]]\}$
d. $C \subseteq [\![\text{TP}_1]\!]^f$ [focus association; see (91)]

When *aż* associates with VP focus, the set C is a subset of the set of different things that Mary did, (93c).

(93) Mary aż [pojechała do Warszawy]$_F$.
Mary AŻ drove to Warsaw
'Mary even drove to Warsaw (of all things she could do).'

a. [AŻ C] [$_{TP_2}$ [$\sim S$] [$_{TP_1}$ Mary [drove to Warsaw]$_F$]]
b. $[\![(93)]\!] = \lambda w.\forall q[(q \in C \wedge q \neq [\![\text{Mary drove to Warsaw}]\!] \wedge q < [\![\text{Mary drove to Warsaw}]\!]) \rightarrow q(w)]$
c. $S \subseteq [\![\text{TP}_1]\!]^f = \{q : \exists f \exists x[q = \lambda w[f(\text{Mary}, x) \text{ in } w]]\}$
d. $C \subseteq [\![\text{TP}_1]\!]^f$ [focus association; see (91)]

When *aż* associates with sentence focus, the alternatives are all alternative things that could happen, (94c).

(94) Aż [Mary pojechała do Warszawy]$_F$.
AŻ Mary drove to Warsaw
'Of all things that could happen, Mary drove to Warsaw.'

a. [AŻ C] [$_{TP_2}$ [$\sim S$] [$_{TP_1}$ [Mary drove to Warsaw]$_F$]]
b. $[\![(94)]\!] = \lambda w.\forall q[(q \in C \wedge q \neq [\![\text{Mary drove to Warsaw}]\!] \wedge q < [\![\text{Mary drove to Warsaw}]\!]) \rightarrow \neg q(w)]$
c. $S \subseteq [\![\text{TP}_1]\!]^f = \{q \in D_{\langle s,t\rangle}\}$
d. $C \subseteq [\![\text{TP}_1]\!]^f$ [focus association; see (91)]

Since *aż* may but does not have to have a temporal reading like *dopiero*, I propose that the temporal interpretation is determined by pragmatics and syntax. Namely, TEMPORAL *aż* functions as a complementizer in Polish so it precedes a clause and the focus association mechanism derives a set of alternative propositions which are ordered with respect to time. As noted in footnote 8 the semantics and syntax alone cannot be the reason why *aż* can function as a complementizer but the other exclusive particles cannot. I propose that when *aż* associates with the whole sentence, as in (95b), the set of focus alternatives is the set of answers to the QUD, and pragmatics determines that the temporal ordering is the most relevant. What possibly contributes to this interpretation is the fact that without context a sort of default ordering for multiple events is a temporal ordering. In other words, I assume that, unlike for *dopiero*, for TEMPORAL *aż* no extra presupposition is necessary in the lexical entry. The set C in (95d) evokes multiple events and times for the set of propositions derived from wide focus. So what ends up being excluded are alternative propositions describing different events that happened at different times. The scalar presupposition says that those times are ordered on a scale. The truth conditions say that no weaker proposition, i.e. nothing that happened earlier, is true.[11]

(95) Zostaniemy tu aż zajdzie słońce.
will.stay here AŻ goes.down sun
'We will stay here until the sun goes down.'

a. Aż [sun goes down]$_F$.

b. [AŻ C] [$_{TP2}$ [$\sim S$] [$_{TP1}$ [sun goes down]$_F$]]

c. $[\![(95)]\!] = \lambda w.\forall q[(q \in C \wedge q \neq [\![\text{sun goes down}]\!] \wedge q < [\![\text{sun goes down}]\!]) \rightarrow \neg q(w))]$

d. $S \subseteq [\![\text{TP}_1]\!]^f = \{q : \exists t \in T[q = \lambda w[t \text{ in } w]]\}$

e. $C \subseteq [\![\text{TP}_1]\!]^f$ [focus association; see (91)]

Finally, let us see why the four exclusives are interchangeable in the sentence in (1), repeated below in (96). The logical syntax is given in (97a). The focus is on the numeral and the exclusive takes sentential scope. This derives the set of alternatives in (97c). With *tylko/zaledwie/dopiero* the truth conditions say that all of the stronger alternatives are excluded, e.g. 'that the team achieved 1st place, that the team achieved 15th place, …', (97b) and (98b). The scalar presupposition is that the prejacent 'that the team achieved 30th place' is the weakest alternative.

(96) Zespół był słaby i zajął {tylko / zaledwie / dopiero / aż} [30]$_F$
team was poor and achieved only merely only$_{TEMP}$ AŻ 30th
miejsce. (cf. (1))
place
'The team performed poorly and earned only 30th place.'

[11]There is an interesting contrast in Polish between two types of *until*-clauses: *aż* and *dopóki*. In contrast to *aż*, (95) repated as (i.a) below, *dopóki* requires negation for an identical interpretation, (i.b) and my intuition is that this is because no exclusivity is built into its semantics.

(i) a. Zostaniemy tu aż zajdzie słońce. = (95)
will.stay here AŻ goes.down sun
'We will stay here until the sun goes down.'

b. Zostaniemy tu dopóki nie zajdzie słońce.
will.stay here DOPÓKI not goes.down sun
'We will stay here until the sun goes down.'

Since *aż* excludes lower alternatives on the time scale, we obtain the time point where all alternatives to the sun going down are no longer true. But if *dopóki* works like 'as long as', negation is necessary to compositionally derive the endpoint. For extensive work on the Czech counterpart of *dopóki*, *dokud*, see Dočekal (2012, 2015), Iatridou & Zeijlstra (2021).

(97) a. [TYLKO/ZALEDWIE C] [$_{\text{TP}_2}$ [$\sim S$] [$_{\text{TP}_1}$ the team earned [30$^{\text{th}}$]$_\text{F}$ place]]

b. $[\![(97\text{a})]\!] = \lambda w.\forall q[(q \in C \wedge q \neq [\![\text{the team earned } 30^{\text{th}} \text{ place}]\!] \wedge q > [\![\text{the team earned } 30^{\text{th}} \text{ place}]\!]) \rightarrow \neg q(w)]$

c. $S \subseteq [\![\text{TP}_1]\!]^f = \{q : \exists n \in \mathbb{N}[q = \lambda w.\textsc{the team earned } n^{\text{th}} \textsc{ place} \text{ in } w]\}$

d. $C \subseteq [\![\text{TP}_1]\!]^f$ [focus association; see (91)]

With *dopiero* the excluded alternatives are further restricted by the presupposition about noteworthiness of alternative times, (98d), e.g., 'that the team achieved 1$^{\text{st}}$ place at time t_1, that the team achieved 15$^{\text{th}}$ place at time t_{15}, …'. So although the inherent ordering between numerals is such that 30 > 15, the context defines the scale where achieving 15$^{\text{th}}$ is a stronger alternative than achieving 30$^{\text{th}}$ place.

(98) a. [DOPIERO C] [$_{\text{TP}_2}$ [$\sim S$] [$_{\text{TP}_1}$ the team earned [30$^{\text{th}}$]$_\text{F}$ place]]

b. $[\![(98\text{a})]\!] = \lambda w.\forall q[(q \in C \wedge q \neq [\![\text{the team earned } 30^{\text{th}} \text{ place}]\!] \wedge q > [\![\text{the team earned } 30^{\text{th}} \text{ place}]\!]) \rightarrow \neg q(w)]$

c. $S \subseteq [\![\text{TP}_1]\!]^f = \{q : \exists n \in \mathbb{N}[q = \lambda w.\textsc{the team earned } n^{\text{th}} \textsc{ place} \text{ in } w]\}$

d. $C \subseteq [\![\text{TP}_1]\!]^f$ [focus association; see (91)]

e. $\exists t_1 \in T[p = \lambda w.\textsc{the team earned } 30^{\text{th}} \textsc{ place} \text{ at } t_1 \text{ in } w] \wedge \forall q \in C[\exists t_2 \in T\ \exists n \in \mathbb{N}[q = \lambda w.\ \textsc{the team earned } n^{\text{th}} \textsc{ place} \text{ at } t_2 \text{ in } w]] \wedge q \neq p \wedge t_1 >_{\textsc{noteworthy}^w} t_2]$

[presupposition in (82c) restricting the set in (98d)]

With *aż* the scalar presupposition is different: the prejacent is the strongest alternative and the ordering in C is 'that the team achieved 30$^{\text{th}}$ place' > 'that the team achieved 15$^{\text{th}}$ place'. The wider context makes it clear that achieving 30$^{\text{th}}$ place is 'less desirable', 'less than expected' than achieving a place lower on the numeral scale.

(99) a. [AŻ C] [$_{\text{TP}_2}$ [$\sim S$] [$_{\text{TP}_1}$ the team earned [30$^{\text{th}}$]$_\text{F}$ place]]

b. $[\![(99\text{a})]\!] = \lambda w.\forall q[(q \in C \wedge q \neq [\![\text{the team earned } 30^{\text{th}} \text{ place}]\!] \wedge q < [\![\text{the team earned } 30^{\text{th}} \text{ place}]\!]) \rightarrow \neg q(w)]$

c. $S \subseteq [\![\text{TP}_1]\!]^f = \{q : \exists n \in \mathbb{N}[q = \lambda w.\textsc{the team earned } n^{\text{th}} \textsc{ place} \text{ in } w]\}$

d. $C \subseteq [\![\text{TP}_1]\!]^f$ [focus association; see (91)]

Thus, the example in (1) is a special case where the truth conditions end up the same with opposite scalar orderings of alternatives, thanks to the context.

6 Family members in other Slavic languages

In Section 3, I discussed the different syntactic configurations for focus association and noted that not all of them may be available in every language; e.g., in English *only* in the clause-initial position cannot associate with the whole clause as it is possible in Polish or German. Now that I identified the minimal semantic differences between the four exclusives in Polish, we can expect that those same parameters will have different settings cross-linguistically and not all of the options may be available or, conversely, there could be more options for variation than in Polish.

For instance, the limits on scale dimension that are lacking with English *only*, are also found with German *erst* (König 1991), a temporal exclusive like *dopiero*. Several other Slavic languages have a counterpart of *aż*, *až* in Czech, Slovak and Russian, and *čak* in Bulgarian, Serbian and other South Slavic languages (Tomaszewicz 2013a,b), but they seem to have various requirements on the type of scale. While in Russian *až* is highly expressive and its use is considered colloquial, at first sight, the same range of contexts I have considered here for Polish seems to be available except for the two scope options with negation (Katja Jasinskaja, p.c.). In Czech and Slovak, on the other hand, there seems to be a strong requirement for temporal or distance scales, specifically with DP focus. Thus (100) in Czech does not have the reading that the manager is a noteworthy person for Maria to talk to, but instead a temporal scale needs to be accommodated, as indicated by the translation (Radek Šimík, p.c.)[12]. In Polish this reading is obtained with *dopiero* in this sentence.

(100) Maria mluvila až s manažerem.
Maria talked až with manager

a. 'Maria talked to the manager who appeared late and Maria talked to him but not the people who appeared before him.'

b. *Unavailable:* 'Maria talked to somebody so important as the manager.' (Czech; Radek Šimík, p.c.)

[12] There may be variation among the speakers. As reported in Tomaszewicz 2013b for the example in (i) some of my informants preferred the scalar interpretation in (a), rather than the temporal interpretation in (b). A dialectal difference is possible (Věra Dvořák, p.c.) – the speakers who preferred (i)a are from Moravia.

(i) Hanka poprosila o pomoc až prezidenta. (Czech; Tomaszewicz 2013b: (54))
Hanka asked for help až president

a. 'It was nobody less important than the president who Hanka asked for help.'

b. 'It was no sooner than when talking to the president that Hanka asked for help.'

With temporal scales *až* in Czech has the same exclusive reading as Polish *zaledwie*, *dopiero* and *aż* in the example we saw in (32), i.e., where it can be translated as *only* in English but is incompatible with Polish *tylko*. The sentence in (101) is the counterpart of the Polish (32). Just like in Polish, the reading with *až* is different from the reading with *even*. With *až* the sentence tells us that 6 o'clock is a noteworthy time to wake up, because it is late, i.e., 'no sooner than'. With *even* the meaning is 'Dan woke up several times in addition to the unlikely time at 6 AM'.

(101) Daneček se vzbudil až v 6 ráno.
Dan self woke.up AŽ at 6 morning
a. 'Little Dan woke up only/no sooner than at 6 AM.'
b. *Unavailable:* 'Little Dan woke up even at 6 AM.'
(Czech; Tomaszewicz 2013b: (6))

However, *až* in Czech can also associate with a VP and evoke a pragmatic scale (Tomaszewicz 2013a,b). In (102) the alternatives are different things that could result from the boy's surprise; there is no implication of a temporal ordering.

(102) Chlapec v zeleném overalu se překvapením až [$_{VP}$ začervenal].
boy in green overalls self surprise AŽ reddened
'The boy in a green overalls as much as blushed with surprise.'
(Czech; K. Čapek: *Od člověka k člověku III*)

In Slovak *až* can have temporal readings when modifying a DP with a temporal reading, such as 'Friday' in (103a)–(103b), without the high-on-the-scale implication that is found in such cases in Polish, i.e., in both (103a)–(103b) the meaning is 'until Friday' with the implication that Friday is included ('through Friday' in American English). Compare how the Polish counterpart of (103a) contains a preposition *do* '*to*', (104), and in the presence of *aż* ('*aż do Piątku*') the meaning would be that Friday is late, which would be very odd in this context (we saw this effect in (64c)). The temporal *až* in Slovak is reminiscent of the complementizer *aż* in Polish, in that a temporal endpoint is specified without evoking noteworthiness.

(103) a. V dňoch pondelok až piatok bude predajňa zatvorená.
in days Monday AŽ Friday AUX store closed
'In days from Monday to Friday, the store will be closed.'

b. Keby budeš budúci týždeň na službobke? Utorok až piatok.
when AUX next week on business.trip Tuesday AŽ Friday
'When will you be on a business trip next week? Tuesday through Friday.'
(Slovak; Andrea Kellö Žačoková, Jana Bašnáková, p.c.)

(104) W dniach od poniedziałku do piątku sklep będzie zamknięty.
in days from Monday to Friday store AUX closed
'In days from Monday to Friday, the store will be closed.'

The following example, (105), shows that in Slovak *až* can be used with intervals in general (not just with temporal endpoints as above) and again in Polish in such cases the preposition *do* must be used without *aż*, (106), or else the sentence would carry the implication that 400 is significantly more than 384 pages.

(105) Naša kniha bude mať 384 až 400 strán.
our book AUX have 384 AŽ 400 pages
'Our book will be 384 to 400 pages long.'
(Slovak; Andrea Kellö Žačoková, Jana Bašnáková, p.c.)

(106) Nasza książka będzie mieć od 384 do 400 stron.
our book AUX have from 384 to 400 pages
'Our book will be 384 to 400 pages long.'

The differences in the requirements on the scalar dimension with *aż*/*až* in Polish, Czech and Slovak predict that in different languages we may find exclusives specified for different scales. With temporal scales either an inherent ordering is lexicalized (Czech and Slovak *až*) or the context specifies the direction of the scale (so that the prejacent is either later or earlier than expected, as with Polish *dopiero*). And while for Polish *dopiero* I opted to assume an extra presupposition to account for temporality, for the Slovak *až* I would consider a meaning without a scalar presupposition but encoding scalarity in the assertion (see Charnavel 2015 for French scalar opposites of *even* and *only*).

In South Slavic languages, the adverbial particle *čak* has the same semantics as *aż* (Tomaszewicz 2013a,b), and in Bulgarian, which lacks a counterpart of *dopiero*, *čak* fulfills this role. Just like in Polish, in (107) *čak* contributes the reading that Mary is a noteworthy person for me to talk to. In (108) *čak* would be translated as *dopiero* rather than *aż* to yield the meaning that 'now' is a noteworthy time for you to tell me, because it is late, because as we saw in Section 4.3 it is more specific: It calls for a contextually relevant ranking of times.

(107) Govorix čak s Mary.
talked ČAK with Mary
'I talked to somebody so important as Mary.'
(Bulgarian; Tomaszewicz 2013a: (4), Tomaszewicz 2013b: (1))

(108) Zašto mi kazvaš čak sega?
why me tell ČAK now
a. 'Why are you telling me only now?'
b. *Unavailable:* 'Why are you telling me even now?'
(Bulgarian; Tomaszewicz 2013a: (10))

Notably, Bosnian-Croatian-Serbian (BCS) lacks a single lexical item with the meaning of *even* (recall the semantics of *even* in (84), contrasting it with *aż* in (83)). Instead, the two meaning components of *even*, scalarity and additivity, are lexicalized separately by *čak* and *i*, where *i* is the conjunction *and* (Jasinskaja & Tomaszewicz 2016). In example (109a) *čak* appears alone, while in (109b) it is accompanied by *i*, which contributes additivity absent in (109a), as evidenced by the continuation.

(109) a. Poslao sam žalbu čak dekanu, ali nisam nikome drugome.
sent AUX complaint ČAK dean but NEG.AUX nobody other
'I sent my complaint to somebody so important as the dean, but I did not send it to anyone else.'
b. Poslao sam žalbu čak i dekanu, #ali nisam nikome drugome.
sent AUX complaint ČAK and dean but NEG.AUX nobody other
'I sent my complaint even to the dean *[Incongruous:]* but I did not send it to anyone else.' (BCS; Jasinskaja & Tomaszewicz 2016: (27))

In (110a) *čak* together with negation yields the reading 'low on the scale of distance', while in (110b) the presence of the additive particle (its negated variant *ni*) yields the 'low on the scale of likelihood' reading, which is exactly the reading with *even* in its English translation.

(110) a. Ivan nije čak u Salzburgu, sad je u Zagrebu.
Ivan NEG.AUX ČAK in Salzburg now is in Zagreb
'Ivan is not as far as Salzburg, he is in Zagreb now.'

b. Ivan nije čak ni u {Zagrebu / #Salzburgu}.
Ivan NEG.AUX ČAK NEG.and in Zagreb Salzburg
'Ivan is not even in Zagreb / *[Incongruous:]* Salzburg.'
(BCS; Jasinskaja & Tomaszewicz 2016: (28–29))

The facts from BCS suggest that the different meaning components that I identified for the four Polish exclusives can be lexicalized separately, which adds to the other known cases of compositional combination of particles from the exclusive and additive family, e.g. *auch nur* in German (Guerzoni 2003) (*auch* means 'also', *nur* means 'only', *auch nur* means 'even'), *mégis* in Hungarian (Katja Jasinskaja, p.c.; Csirmaz & Slade 2020) (*még* means 'still', *is* is an additive particle and *mégis* is a scalar additive particle).

The case of BCS *čak i* shows that the repertoire of exclusive and additive scalar particles uses the same building blocks at the levels of assertion and presupposition.[13] The presupposed and asserted components may also end up being swapped, as argued in Guerzoni (2003) for subspecies of *even* and *only*, or the strength of particular components may differ, e.g. additivity may be weaker as proposed by Crnič (2011, 2013) for concessive additive particles. Moreover, the differences in presuppositions have grammatical effects, i.e., affect the distribution of different particles in different syntactic environments (Giannakidou 2007).

7 Conclusion

My analysis of the four Polish exclusives *tylko*, *(za)ledwie*, *dopiero* and *aż* adds to the body of work arguing that we can go beyond the identification of the different flavors of exclusive and additive meanings (as in König 1991) but track the origin of those flavors in their minimal building blocks (Rullmann 1997, Guerzoni 2003, Lahiri 2010, Crnič 2011, Coppock & Beaver 2011, 2013, 2014, a.o.). I identified which meaning components are core and thus reoccur in the different exclusive particles and which components require specific tweaking of parameters. Different settings for specific parameters may influence the flavor of exclusivity, e.g., force the temporal interpretation with *dopiero*, but may also extend their uses beyond that of variants of *only*: I showed that in Polish the restrictions on the alternative set with *(za)ledwie* yield approximative/counterfactual uses, and with *aż* they result in the meaning of *until*. This work could also be a starting point for the exploration of other expressions with exclusive meaning, such as gradable adjectives, non-focus associating adverbs, etc., for the identification of parameters

[13] The importance of the distinction between the asserted and presupposed content in the meaning of *even* and *only* goes back to Horn 1969.

other than the semantic ingredients that I isolated here. Coppock & Beaver (2011, 2013, 2014) argue that English adverbial *only* and *just*, and adjectival exclusives *mere*, *sole*, *only*, *single*, *exclusive* vary along two parameters: the ontological type of their arguments and constraints on the question under discussion. Greenberg (2016, 2022) suggests that the variability in the dimensions of the scale could be captured if the scalar presupposition is defined in terms of degrees, with respect to a contextually supplied gradable property. For Polish the next step would be to investigate the building blocks and find a common core with *even*-like particles (*nawet*, which we saw in this chapter, and *choć(by)*), additives (*i*, *też*, *także*, *również*, …), approximatives (the approximative *ledwie* that we saw here belongs to a rich repertoire: *prawie*, *niemal*, *niemalże*, *nieomal*, *omal*, *mało co*,…) and exceptives (*oprócz*, *poza*, *ponadto*, *zresztą*, …).

Abbreviations

AUX	auxiliary	TOP	topic particle
NEG	negation	Q	question particle

Acknowledgements

I would like to express my very special thanks to Berit Gehrke and Radek Šimík, without whose assistance and encouragement this chapter would not have been possible. I would also like to thank the anonymous reviewers for this volume for their insightful comments. The data from languages other than Polish comes from my prior work and its analysis benefited greatly from my exchanges with Katja Jasinskaja, Roumi Pancheva, Anastasija Smirnova, Andrea Kellö Žačoková, Jana Bašnáková, Věra Dvořák, Mojmír Dočekal, Petr Biskup, Denisa Lenertová, Dorothee Fehrmann, Hana Strachoňová, Natasha Fitzgibbons, Ivan Kapitonov, Natalia Korotkova, Tania Ionin, Agnieszka Łazorczyk, Agata Renans, Elena Guerzoni, Jim Higginbotham, Sabine Iatridou, Jessica Rett, Barry Schein and Yael Sharvit. A preliminary version of this work was presented at the *Association with focus* workshop at the GLOW 35 conference (University of Potsdam, 31 March 2012) and was greatly enriched by discussions with Roumi Pancheva and by valuable comments from the audience. Any mistakes in this work remain my own.

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Chapter 13

Numerals and their kin: The view from Slavic

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In this paper, I focus on the meaning of numerals, i.e., lexical items that express a relationship with numbers. I discuss different functions of numerals including their quantifying and arithmetical use and compare various attempts to capture the relationship between those functions. Furthermore, I examine several types of complex numerical expressions such as ordinals, multiplicatives, taxonomic and group numerals in Slavic and beyond. Finally, I investigate data concerning Slavic label numerals, i.e., expressions used to identify entities via association with a number and propose a morpho-semantic analysis of such expressions. The core of the proposal is that the arithmetical meaning is the numerals' underlying semantic core from which both the quantifying and the label meaning are derived.

1 Introduction

Numerals are funny words. On the one hand, it seems that they share some core component that justifies grouping them together as if they formed a single category, a practice common in both descriptive and theoretic approaches to language.[1] On the other hand, though, morphologically and syntactically they form a very heterogeneous class of expressions with different items often exhibiting

[1]This intuition dates back at least to the Neo-Latin grammarian tradition, which distinguished between three subclasses within the category *nomen*, i.e., *nomen substantivum*, *nomen adjectivum* and *nomen numerale*. At the same time, different linguistic traditions, including descriptive grammars of Slavic, have proposed conflicting classifications of what counts as a numeral; thus, *…and their kin* in the title of this paper.

Marcin Wągiel. 2026. Numerals and their kin: The view from Slavic. In Berit Gehrke & Radek Šimík (eds.), *Topics in the semantics of Slavic languages*, 703–746. Berlin: Language Science Press. DOI: 10.5281/zenodo.22078593

distinct properties. This is the case not only when one compares various subclasses of numerals, e.g., cardinals, ordinals and multiplicatives, but also often within the subclass of cardinals different items have adjectival, nominal, or both nominal and adjectival features while in some languages they seem to behave as verbs (Donohue 2005, Ionin & Matushansky 2018).

In generative linguistics, a lot of work has been focused on establishing the syntactic status of numerals, with different approaches differing in whether they are lexical or functional categories or whether they are heads or maximal projections. These questions have been studied thoroughly with respect to Slavic numerals, with their well-known idiosyncratic properties (e.g., Pesetsky 1982, Babby 1987, Franks 1994, Rutkowski 2002, Klockmann 2012).

On the other hand, since the early days of analytic philosophy and formal semantics much attention has been dedicated to capturing numerals' meaning. At least since Frege (1884), it has been recognized that this is far from trivial. Intuitively, it seems straightforward that numerals are linguistic expressions that somehow describe a numeric quantity. However, beyond this vague characterization it is much less obvious how to state exactly what they actually are.

In this paper, I will discuss the main formal approaches to the meaning of numerals. In doing so, I will focus on two aspects that only recently attracted due attention in the literature. Specifically, I will investigate different uses of basic numerals such as *five* and the behavior of various types of derivationally complex numerical expressions in Slavic. The two sets of data indicate that basic numerals and numerical expressions in general are semantically complex. I will argue that a proper approach to their meaning should describe a compositional mechanism deriving different functions from the underlying arithmetical meaning.

The paper is outlined as follows. In Section 2, I discuss various functions of basic numerals. Section 3 provides an overview of semantic approaches to their quantifying function, as in *five cats*. Section 4 explores the relationship between the quantifying meaning and the ability of numerals to refer to numbers as abstract arithmetical concepts. In Section 5, I review the literature on Slavic complex numerical expressions which presents evidence calling for a compositional treatment. Section 6 is a case study exploring the so-far neglected label use of numerals. Section 7 concludes the paper.

2 Basic numerals and their various flavors

Let us start with basic numerals like English *five*. One of the challenges in capturing their meaning is that they can be used in very different ways (Bultinck 2005, Geurts 2006). Consider (1). It turns out that a simple word such as *five* is in fact quite tricky. Of course, it can be used to count entities denoted by the modified NP when it appears prenominally (1a) but that is not the only function it can have. When it combines with a measure word (1b), it designates a quantity of a substance rather than a plurality of objects. It can also appear in a predicative context (1c) or as part of a measure phrase (1d). And that is not it since it can also be used to refer to an abstract arithmetical concept (1e) or to label an entity (1f).

(1) a. Five cats meowed.
    b. Five liters of milk got spilled.
    c. These are five cats.
    d. That cat is five years old.
    e. Five is a Fibonacci number.
    f. Player number five twisted her ankle.

Below, I will briefly discuss how the measure, predicative and arithmetical function of numerals relate to their most deeply studied quantifying use. A thorough investigation into properties of the label function will be undertaken in Section 6.

2.1 Quantifying vs. measure use

The mainstream research has been primarily focused on the quantifying use of numerals (1a). This seems justified since counting objects is perhaps the first thing that comes to mind when we think about expressions such as *five.* However, when a numeral appears in a measure phrase (1b), it seems to involve measuring within a certain dimension, e.g., volume, rather than counting objects.

Intuitively, the difference between counting and measuring is that the former is about specifying how many discrete objects of a certain kind there are, whereas the latter determines some quantity in relevant measure units. Admittedly, some proposals attempt to reduce measuring to a particular type of counting based on the individuation in terms of quantity (Lyons 1977, Matushansky & Zwarts 2017). Consequently, measuring would simply be counting units determined by measure words. An opposing approach treats counting as a form of measuring, i.e., measuring a quantity of a plural individual in terms of natural or object units (Krifka 1989, 1995). Yet, there is also the third account which views counting and measuring as two independent operations (Rothstein 2017).

The contrast between the two functions becomes clear when one considers classifier constructions involving container nouns (2). Such expressions are ambiguous between a measure (content) and counting (container) interpretation (e.g., Landman 2004, Grimshaw 2007, Partee & Borschev 2012). On the measure reading of (2a), Tomek flushed wine to the quantity of two bottles. On the counting reading, on the other hand, he did something quite spectacular since two actual bottles containing wine went down. Moreover, derivational morphology seems to be sensitive to the distinction since the suffix *-ful* selects only for the measure sense of a container noun (2b)–(2c) (Rothstein 2011).

(2) a. Tomek flushed two bottles of wine through the toilet.
b. Romek added two { glasses / glassfuls } of wine to the soup.
c. Marek brought two { glasses / #glassfuls } of wine for our guests.

Another argument for distinguishing counting and measuring as two distinct operations comes from distinct restrictions on the domain of quantification (Wągiel 2018). Though both measuring and counting quantify over entities that need to be disjoint (Landman 2016), only the latter requires objects individuated as coherent integrated wholes. To realize the contrast, imagine someone has spilled some wine on the table in such a way that there are two separate blobs *a* and *b* whose volume is exactly one and a half milliliters each. In such a scenario, (3a) is true despite the fact that one of the three milliliters must be split between a portion of *a* and a portion of *b*. On the other hand, (3b) is simply false. This contrast shows that measuring, unlike counting, ignores individuation in terms of integrity.

(3) Scenario: There are exactly two 1.5 ml blobs of wine on the table.
a. There are three milliliters of wine on the table. true
b. There are three objects on the table. false

Though monotonic systems of measurement track certain part-whole relations (Schwarzschild 2002), they are not sensitive to the spatial arrangement of parts making up a whole. This suggests that despite sharing a common core counting and measuring are two distinct things.

2.2 Quantifying vs. predicative use

It is well known that cardinals used as quantifiers have both lower and doubly bounded readings.[2] For instance, *five cats* in (4a) allows for an *at least* reading, as witnessed by the compatibility with the *in fact* clause. Similarly, (4b) is typically interpreted in a way that Tomek must take at least three cards.

[2] For an overview of the literature on numerals and scalar implicatures, see, e.g., Spector (2013).

(4) a. Five cats live in the barn; in fact, six cats live there.
b. Tomek must take three cards.

However, numeral NPs in predicate position systematically lack a lower-bounded interpretation (Partee 1987, Landman 2003, Geurts 2006). For instance, (5a) can only get the *exactly* reading. This is further corroborated by the infelicity of (5b).

(5) a. These are five cats.
b. # The guests are three filmmakers; in fact, they are four filmmakers.

Furthermore, it was observed that also bare cardinals can appear as predicates in predicate position (6) (Rothstein 2017). This resembles the behavior of adjectives rather than determiners.

(6) a. The apostles are twelve.
b. The planets are eight.

Similar to numeral NPs, bare cardinals in predicate position lack an *at least* reading. Thus, (6a) and (6b) are true if there are exactly twelve apostles and exactly eight planets, respectively. Moreover, sentences such as (7) are infelicitous.

(7) # My reasons for saying this are four; in fact they are five.

The contrasts discussed above indicate that the meaning of the predicative use of cardinals differs from their semantics in the quantifying function.

2.3 Quantifying vs. arithmetical use

It is not always the case that numerals designate a plurality or measure. In (1e), *five* refers to an abstract number concept rather than to a collection of entities. In this use, numerals behave differently than in their quantifying function (Bultinck 2005, Rothstein 2017, Wągiel & Caha 2020). Specifically, arithmetical concepts can have special properties such as being prime (8a) and they can occur in mathematical statements (8b) and dedicated grammatical constructions (8c). Finally, when compared, the dimension of comparison is based on their relative ordering rather than, e.g., on the physical size (8d).

(8) a. Five is prime.
b. Ten divided by five equals two.
c. Hanna can count up to five.
d. Five is bigger than four.

This behavior contrasts with quantifying numerals. (9a) is odd since being prime is not a property that can be attributed to a collection of things. Similarly, expressions denoting a plurality of entities are illicit in constructions calling for a numeric value such as (9b)–(9c) and (9d) has different truth conditions than (8d), e.g., it would not be true if one compared five cherries with four watermelons.

(9) a. # Five things are prime.
  b. # Ten things divided by five things equals two things.
  c. # Hanna can count up to five things.
  d. # Five things are bigger than four things.

Furthermore, numerals referring to number concepts are incompatible with modifiers such as *more than* and *at least* (10a). Finally, the arithmetical function is always associated with an *exactly* reading (10b).

(10) a. # More than five is prime.
  b. # Two multiplied by five equals ten, if not more.

As we saw, numerals can be used in various ways each of which has different semantic characteristics. In the next two sections, I will review major approaches to their quantifying meaning and compare how they account for at least some aspects of the flexibility discussed above.

3 Theories of cardinals

Given the variety of uses discussed in the previous section, any quest for *the* meaning of cardinal numerals is probably a misunderstanding. Rather, a theory of cardinals should be able to capture the relationship between meanings of cardinals in their various functions. And yet, the mainstream research has been mainly focused on the quantifying use often ignoring how it relates to other uses (with Bylinina & Nouwen's 2020 systematic inquiry being a notable exception).

3.1 Cardinals as determiners

Let us start with the earliest and most prominent account of the meaning of cardinal numerals, namely the Generalized Quantifiers (GQ) approach (Barwise & Cooper 1981; though it can be traced back via Montague to Frege). The intuition behind this analysis is that *five* is in fact a determiner similar to *some*, *most* or *all*, as suggested by its occurrence in prenominal position (11).

(11) { Some / Most / All / Five } cats meowed.

In the GQ theory, a determiner expresses a particular relation holding between two sets (type $\langle\langle e,t\rangle,\langle\langle e,t\rangle,t\rangle\rangle$), i.e., a set denoted by the NP and a set denoted by the VP. In (11), *some* yields the truth value True if the intersection between the set of cats and the set of entities that meowed is non-empty (12a). Similarly, *five* returns True if the cardinality of the set of cats that meowed equals 5 (12b).

(12) a. $[\![\text{some}]\!] = \lambda P \lambda Q[P \cap Q \neq \emptyset]$
b. $[\![\text{five}]\!] = \lambda P \lambda Q[|P \cap Q| = 5]$

The GQ approach aims to develop a unified analysis of all quantified DPs. Further, it can be enriched with a type-shifting mechanism which allows for systematic mapping between different semantic types (Partee 1987). Consequently, the theory can account for predicative uses of numeral NPs like (1c) by lowering the type of a general quantifier ($\langle\langle\langle e,t\rangle,t\rangle$) to the type of a predicate ($\langle e,t\rangle$).

However, for some time it has been realized that the GQ approach is most probably not a good way of capturing what cardinals are (e.g., Krifka 1999, Landman 2003). In the next sections, I will discuss the well-known problems with the GQ account and main alternative approaches.

3.2 Cardinals as predicates

There are several problems with the GQ approach to cardinals. First of all, there is a well-known asymmetry between DPs with numerals and DPs with regular determiners in predicate position. For instance, DPs headed by *every* cannot be used predicatively (13a). Similarly, (13b) is infelicitous on a non-partitive interpretation. Given the felicity of (1c), repeated here as (13c), and other sentences discussed in Section 2.2, this fact is puzzling (Landman 2003).

(13) a. # Hanna is every filmmaker at the party.
b. # The guests are { most / some } filmmakers.
c. These are five cats.

Furthermore, (14a) shows that bare determiners cannot be used predicatively. This contrasts with (6), in which the cardinal patterns with adjectives (14b). Since there is no standard type-shifting rule allowing for mapping the type of determiners ($\langle\langle e,t\rangle,\langle\langle e,t\rangle,t\rangle\rangle$) onto the type of predicates ($\langle e,t\rangle$), the predicative use of bare cardinals is unaccounted for. But even if such a rule were postulated, it would still fail to explain the contrast between cardinals and determiners.

(14) a. # My reasons for saying this are { most / all / every }.
b. My reasons for saying this are { serious / personal / five }.

The last problem concerns the fact that cardinals can appear along with bona fide determiners within a single DP (e.g., Rothstein 2017). From a GQ perspective, the compatibility of *five* with the definite article (15a) and *every* (15b) is unexpected and raises questions regarding the semantic contribution of each of the elements.

(15) a. The five cats that I saw meowed.
b. Every two students got to share a hotel room.

The evidence discussed above motivated an alternative approach, which treats cardinals as predicates (Landman 2003, 2004). Within this framework, cardinals get an interpretation very similar to that of intersective adjectives, specifically they express a cardinality property. For this approach to work, it is required to distinguish between two types of entities within the domain of individuals, namely atoms, i.e., singular entities, and pluralities (Link 1983). While atoms are minimal elements of a nominal denotation, pluralities are formed from atoms via a pluralization operation * which closes the predicate under sum, i.e., takes a set of atomic entities and returns that set extended with all the pluralities that can be formed by summing the atoms. In addition, the two types of individuals are associated with each other via a part-whole relation defined in terms of mereological parthood, e.g., an atomic entity a is part of a plurality $a \oplus b$.[3]

According to the semantics in (16a), *five* denotes a set of pluralities each of which contains 5 atomic entities. Similar to intersective adjectives, cardinals combine with NPs they modify via Predicate Modification (Heim & Kratzer 1998). Therefore, when *five* is composed with *cats* (16b), the resulting phrase denotes the intersection of a set of pluralities of cats and a set of plural individuals that consist of five atomic entities, i.e., a set of sums of five cats.

(16) a. $[\![\text{five}]\!] = \lambda x[\#(x) = 5]$
b. $[\![\text{five cats}]\!] = \lambda x[{}^{*}\textsc{cat}(x) \wedge \#(x) = 5]$

The approach discussed above gives a straightforward explanation for the contrasts between cardinals and determiners like *every* and *most* in predicate position as well as the felicity of constructions such as (15). However, it faces some challenges as well.

[3] For an introduction to mereological theories of plurality, see, e.g., Champollion & Krifka (2016).

3.3 Cardinals as predicate modifiers

A problem with treating cardinals as cardinal properties is that cross-linguistically the use of bare cardinals in predicate position is very restricted (Ionin & Matushansky 2018: 33–34). For instance, in Russian (17) is ungrammatical. Similarly, Dutch (18) can only mean that the children are two years old and not that they are two in number. And even in English the predicative use of bare cardinals is heavily restrained, as witnessed by the fact that examples such as (19) are highly degraded. If cardinals are assigned type $\langle e, t\rangle$, then this is rather startling.

(17) * Deti byli { dva / dvoe }.
children were two two.GNDR
Intended: 'The children were two in number.' (Russian)

(18) # De kinderen zijn twee.
the children are two
Intended: 'The children are two in number.' (Dutch)

(19) ?? The chairs in this room are twelve.

Another issue concerns the fact that what counts as 'one' seems to be highly dependent on the meaning of the modified NP. Arguably, atoms should be defined relative to a particular property rather than in absolute terms. To see why, let us consider partitive constructions (20) (see Chierchia 2010, Wągiel 2018). Here, (20a) is weird since the numeral NP seems to designate a pair of body parts that disappeared, i.e., it is not referring to two subsets of boys. Yet, in (20b) quantification over subgroups of boys is possible. Importantly, neither *parts of the boys* nor *parts of the group* denote entities that are atomic in any absolute sense since each can consist of parts that themselves fall into the extension of the corresponding phrase. This suggests that the cardinal determines what counts as 'one' relative to the denotation of the modified NP and it is not obvious how this fact could be captured if cardinals were interpreted simply as intersective predicates.

(20) Scenario: While hiking at night, a group of boys divided into five teams to explore a vast area but only three teams arrived at the meeting point.
a. # Two parts of the boys went missing.
b. Two parts of the group went missing.

An alternative to treating cardinals as cardinal properties is to analyze them as predicate modifiers (type $\langle\langle e, t\rangle, \langle e, t\rangle\rangle$) equipped with a pluralization operation * and a measure function $\#(P)$ (Krifka 1989, 1995). In (21a), * closes the predicate under sum, i.e., takes a set of atomic entities and returns that set extended with all the pluralities one can form by summing atoms. On the other hand, the operation # returns for a property P a measure function that yields pluralities of five individuals having that property. In other words, *five* maps pluralities of entities onto the natural number 5.[4] When the cardinal combines with an NP, we obtain a set of relevant plural individuals (type $\langle e,t\rangle$). For instance, *five cats* denotes a set of pluralities of cats each of which consists of five cats (21b). Thus, what counts as 'one' in this system is relativized to the denotation of the NP, and this together with some additional assumptions allows for capturing partitives like (20) where reference to absolute atoms is problematic (see also Chierchia 2010, Wągiel 2018).

(21) a. $[\![\text{five}]\!] = \lambda P_{\langle e,t\rangle}\lambda x_e[{}^*P(x) \wedge \#(P)(x) = 5]$
b. $[\![\text{five cats}]\!] = \lambda x_e[{}^*\textsc{cat}(x) \wedge \#(\textsc{cat})(x) = 5]$

Another option is to treat cardinals as predicate modifiers providing the cardinality of a partition of a plural entity (Ionin & Matushansky 2006, 2018). A partition of a plurality x is a cover of x, i.e., a set of entities such that the sum of all those entities forms x; in addition, it is a cover whose cells do not overlap, i.e., do not share any parts (see Gillon 1987, Schwarzschild 1996). In (22a), S is a partition Π of an individual x and the cardinality of S equals 5. The cardinal combines with the NP via Function Application and the result is of type $\langle e,t\rangle$. For instance, *five cats* gets the meaning in (22b), i.e., it denotes a set of pluralities divisible into 5 non-overlapping entities each of which has a property of being a cat.[5]

(22) a. $[\![\text{five}]\!] = \lambda P_{\langle e,t\rangle}\lambda x_e\exists S_{\langle e,t\rangle}[\Pi(S)(x) \wedge |S| = 5 \wedge \forall s \in S[P(s)]]$
b. $[\![\text{five cats}]\!] = \lambda x_e\exists S_{\langle e,t\rangle}[\Pi(S)(x) \wedge |S| = 5 \wedge \forall s \in S[[\![\text{cat}]\!](s)]]$

Notice, however, that both approaches discussed above require the denotation of NPs the cardinal combines with to be singular, i.e., to consist only of atoms. It is postulated that the source of plurality is the numeral and the plural marking on the noun is, e.g., due to agreement with no semantic interpretation. Supporting evidence comes from expressions such as (23) where the plural is not associated with a plurality (Krifka 1989, Bylinina & Nouwen 2018) and from languages like Finnish and Turkish which display the singular/plural distinction but in which unlike in, say, English cardinals systematically require singular NPs (24).

[4] This is a slight simplification since in fact Krifka (1989) postulates a special operation NU for measuring pluralities in terms of 'natural units' they consist of, whereas Krifka (1995) proposes OU for measuring the number of 'object units' realizing a particular kind.

[5] The system is devised this way to provide a unified analysis of both simple and complex numerals like *five hundred*.

(23) a. zero students
b. 1.0 students

(24) üç { çocuk / *çocuklar }
three child children
'three children' (Turkish)

Finally, analyzing cardinals as predicate modifiers makes it easier to account for restrictions on bare cardinals in predicate position (18)–(19), by appealing to the well-described peculiarities of the copula which would allow for a type-shift only under particular circumstances (Ionin & Matushansky 2018: 33–34).

3.4 Cardinals as degree quantifiers

The last approach to be discussed here builds on two observations regarding the behavior of cardinals. The first one is that sentences with existential modals (25) are ambiguous (Kennedy 2013, 2015). On the strong reading, (25a) means that Hanna is allowed to eat five cookies and she is not allowed to eat more. The strong reading is probably the most natural interpretation of (25a). There is, however, also a weak reading which merely states that eating five cookies by Hanna is allowed without saying anything about eating other quantities of cookies. This interpretation becomes more prominent in questions (25b).

(25) a. Hanna is allowed to eat five cookies.
b. Is Hanna allowed to eat five cookies?

The second observation is that numerals can take scope independently of the rest of the NP. The evidence comes from decimals (Kennedy & Stanley 2009). Crucially, (26a) does not entail the existence of families with 2.1 cats. This contrasts with (26b) which is infelicitous due to such an entailment.

(26) a. The average Polish family has 2.1 cats.
b. # A normal Polish family has 2.1 cats.

This suggests that after all cardinals are quantifiers. We already saw that the analysis of *five* as a determiner is most probably incorrect and clearly treating it as a quantifier over individuals (type $\langle\langle e,t\rangle,t\rangle$) would not make much sense. However, an interesting idea is to interpret cardinals as quantifiers over degrees, i.e., expressions of type $\langle\langle d,t\rangle,t\rangle$ (Kennedy 2013, 2015). Degrees (type d) are objects similar to individuals (type e) with the crucial difference that the domain of degrees D_d, unlike the domain of individuals D_e, is ordered.[6] Hence, on the degree

[6]For more discussion and an introduction to degree semantics, see, e.g., Morzycki (2016: Ch. 3).

quantifier analysis *five* denotes a set of degree properties whose maximal value equals 5 (27a). A sentence with a numeral NP is then interpreted as in (27b), e.g., the maximal number of entities that Hanna had and that are cats is 5.

(27) a. $[\![\text{five}]\!] = \lambda D_{\langle d,t\rangle}[\textsc{max}(D) = 5]$

b. $[\![\text{Hanna had five cats}]\!] =$
$= \textsc{max}\{n \mid \exists x[\textsc{had}(x)(\textsc{hanna}) \wedge {}^{*}\textsc{cat}(x) \wedge \#(x) = n)\} = 5$

The analysis of cardinals as degree quantifiers provides a promising perspective to explain interactions between modals and modified cardinals such as *more than five* and *at least five* (Nouwen 2010). In the next section, I will discuss the arithmetical function of numerals and how it relates to the theories described so far.

4 Relating cardinalities and number concepts

The theories of numerals discussed in the previous section focus on the quantifying function. However, we already saw in Section 2.3 that numerals can also be used to refer to abstract number concepts.

4.1 Numerals as names of numbers

In the arithmetical function, numerals seem to behave as proper names of number concepts (type n) or, alternatively, degrees (type d).[7] From this perspective, the arithmetical meaning of *five* is simply the number 5 (28a). Expressions such as *prime* would then denote properties of numbers (type $\langle n, t\rangle$), e.g., the extension of (28b) would be the set $\{2, 3, 5, 7, 11, 13, \dots\}$. On the other hand, expressions used to formulate mathematical statements such as *plus*, *times*, *divided by* and *equals* seem to describe relations between number concepts. For instance, (28c) is interpreted in terms of addition of two numeric values.

(28) a. $[\![\text{five}]\!] = 5$

b. $[\![\text{prime}]\!] = \lambda n_n[\textsc{prime}(n)]$

c. $[\![\text{plus}]\!] = \lambda n_n \lambda m_n[n + m]$

The key question concerns the relationship between names of numbers, i.e., the arithmetical use, and meanings of cardinals used in the quantifying function. In the next sections, I will discuss two possible approaches to this issue.

[7] I assume here that there is no difference between the two notions and following the widespread convention in the literature on the arithmetical use of numerals I will simply use the label *n*.

4.2 Deriving number concepts from cardinalities

The mainstream view is to take the quantifying meaning as basic, be it the predicate, predicate modifier or degree quantifier analysis, and to derive the arithmetical meaning from it. Let us discuss three variants of that view.

As we already saw in Section 3.2, on the predicate analysis cardinals denote a cardinal property (29a). Rothstein (2017) assumes this semantics to be basic.[8] The arithmetical meaning is then derived via a special type-shifting operation $^{\cap}$ which when applied to a cardinal property yields a proper name of a number (29b).

(29) a. $[\![\text{five}]\!]_{\langle e,t\rangle} = \lambda x_e[\#(x) = 5]$

b. $[\![\text{five}]\!]_n = {}^{\cap}[\![\text{five}]\!]_{\langle e,t\rangle} = 5$

The proposal is motivated by an analogy with adjectival and nominal uses of expressions describing properties such as *white* and *wise* ~ *wisdom*. It builds on Chierchia's (1985) theory of predication according to which every property has an entity correlate derived by $^{\cap}$. For example, the predicate *white* can be turned into the name of the property of being white. Similarly, $^{\cap}$ turns *five* into the name of the property of being five in number. The reverse shift is also possible via $^{\cup}$.

However, as argued by Ionin & Matushansky (2018: 31–33) the above analogy is problematic since an arithmetical statement such as those in Section 2.3 requires an argument to be the number 5 itself rather than the name of the property of being five in number. One can add that though from a philosophical perspective it is unclear what numbers ultimately are, from a linguistic point of view it is crucial that while there are many contexts in which, e.g., *wisdom* and *being wise* seem to be entirely synonymous (30), there are few (if any) contexts in which *five* and *being five in number* are equivalent (31).

(30) a. Being wise is the art of knowing what to overlook.

b. Wisdom is the art of knowing what to overlook.

(31) a. Five is the fourth prime number.

b. # Being five in number is the fourth prime number.

Consequently, in order to derive the arithmetical function Ionin & Matushansky (2018: 26–27) propose a null nominalizing suffix NOMNUM, which applied to a predicate modifier, turns it into a numeric singular term. More specifically, for any

[8] In fact, this is a slight simplification since the theory distinguishes typally between lower numerals and 'lexical powers', i.e., multiplicands such as *hundred* in *five hundred*.

cardinal numeral it yields the corresponding cardinality. An empirical argument for such an account comes from the fact that numerals as names of numbers typically display full morpho-syntactic uniformity suggesting they are derived expressions. On the other hand, cardinals in the quantifying function often show variation with respect to ϕ-features, e.g., grammatical gender.

Another approach is Kennedy's (2015) proposal to derive the arithmetical meaning from the quantifying one by the standard type-shifting operations BE and IOTA defined in (32) (Partee 1987). BE takes a quantifier and returns a property, whereas IOTA yields the unique entity of which the relevant property is true.

(32) a. $\textsc{be} = \lambda \mathcal{P}_{\langle\langle e,t\rangle,t\rangle} \lambda x_e [\mathcal{P}(\lambda y_e[y = x])]$
b. $\textsc{iota} = \lambda P_{\langle e,t\rangle} \iota x[P(x)]$

Generalizing the type-shifts defined above so that they can also apply to quantifiers over degrees and properties of degrees (or numbers), respectively, allows to derive the arithmetical meaning by applying BE and IOTA consecutively to the degree quantifier semantics. Specifically, when BE applies to a set of degree properties whose maximal value equals 5 (27a), the result in (33a) is the singleton set {5}. The subsequent application of IOTA yields the number 5 (33b).

(33) a. $\textsc{be}([\![\text{five}]\!]) = \lambda n_n[n = 5]$
b. $\textsc{iota}(\textsc{be}([\![\text{five}]\!])) = \textsc{iota}(\lambda n_n[n = 5]) = 5$

An advantageous feature of this approach is that it allows for relating the quantifying and the arithmetical meanings using standard and independently motivated type-shifting machinery. Importantly, however, by doing so it can only derive the arithmetical function from the quantifying one but not vice versa. We will see that this restriction, shared with Ionin & Matushansky's approach, makes a problematic prediction concerning morphological marking. In the next section, I will discuss accounts that derive cardinalities from the arithmetical meaning.

4.3 Deriving cardinalities from number concepts

In opposition to the view that the quantifying function of numerals is basic, a number of approaches build on a contrary intuition, namely that underlyingly numerals are simply names of number concepts (e.g., Scha 1981, Krifka 1989, Scontras 2014, Wągiel 2018). According to that alternative view, cardinals in prenominal position are in fact syntactically and semantically complex

An early theory of cardinals postulating that they are derived from number concepts was developed by Scha (1981). This system distinguishes between numbers, numerals and numerical determiners. The first simply denote number concepts. On the other hand, the NUMERAL head shifts an integer (type *n*) to a cardinal predicate while the numerical D head transforms such a cardinal predicate into a determiner with a GQ-style semantics (34).[9]

(34) $[\ \textsc{d}_{\langle\langle e,t\rangle,\langle\langle e,t\rangle,t\rangle\rangle}\ [\ \textsc{numeral}_{\langle e,t\rangle}\ [\ \textsc{number}_n\ \textit{five}\ ]\ ]\ ]$

A related idea is the core of the system developed by Hackl (2000) who makes use of a covert quantificational operator MANY to turn a number concept into a determiner (35a). On this approach, the quantifying cardinal is accompanied by an additional null element that ensures the shift (35b). The result combines with the denotation of the NP to form a generalized quantifier.

(35)
a. $[\![\textsc{many}]\!] = \lambda n_n \lambda P_{\langle e,t\rangle} \lambda Q_{\langle e,t\rangle} \exists x_e [\#(x) = n \wedge P(x) \wedge Q(x)]$
b. $[\![\text{five cats}]\!] = [\![[\ [\ 5\ \textsc{many}\ ]\ \text{cats}\ ]]\!]$

Another proposal on how to derive the quantifying meaning from the arithmetical one dates back to Krifka (1995). On this view, all numerals are assumed to underlyingly designate number concepts but they also involve an additional classifier element that depending on a language can be either overt or covert. There are several variants of this approach but what they all share is that the classifier element takes an integer and returns a counting device equipped with an appropriate measure function (36a) (Scontras 2014, Wągiel 2018). For instance, in the derivation of the phrase *five cats* the arithmetical entity is first turned into a quantifying predicate modifier, which then combines with the NP (36b).

(36)
a. $[\![\textsc{cl}]\!] = \lambda n_n \lambda P_{\langle e,t\rangle} \lambda x_e [{}^*P(x) \wedge \#(P)(x) = n]$
b. $[\![\text{five cats}]\!] = [\![[\ [\ 5\ \textsc{cl}\ ]\ \text{cats}\ ]]\!] = \lambda x_e [{}^*\textsc{cat}(x) \wedge \#(x) = 5]$

In yet another variant of the approach, the classifier semantics turns a numeric value into a predicate (Sudo 2016). The classifier takes a number and yields a cardinal property (type $\langle e, t\rangle$) rather than a predicate modifier. It can also introduce a special presupposition concerning the nature of the referents of the modified noun. The resulting expression then combines with the NP via Predicate Modification.

[9] The exact denotation of the numerical determiner depends on whether the whole sentence gets a collective, distributive or cover reading, i.e., there is a distinct D for each interpretation.

From a theoretical perspective both types of the relationship between the arithmetical and quantifying function, i.e., the derivation from the quantifying meaning to the arithmetical meaning and vice versa, seem plausible. Thus, the question which one is correct is an empirical issue. In the next section, I will discuss morphological evidence useful in determining the direction of the derivation.

4.4 Morphological evidence

Many languages distinguish formally between so-called counting, i.e., arithmetical, and attributive, i.e., quantifying, numerals (Hurford 1998, 2001). For instance, in Mandarin the form *èr* 'two' is used in arithmetical environments whereas *liǎng* 'two' is an attributive cardinal appearing in the quantifying use. Likewise, the Maltese *tnejn* 'two' indicates the arithmetical function while *żewġ* 'two' has the quantifying meaning. Though the distinction is typically restricted to a subset of numerals in a language and there are a number of patterns of how the two forms can differ, it is a relatively widespread phenomenon across languages.

Interestingly, the two families of theories discussed above make different predictions regarding the morphological make-up of numerals cross-linguistically. On the assumption that morphology expresses meaning, if the arithmetical function is derived from the quantifying function, see Section 4.2, then we would expect that across languages a pattern with arithmetical numerals being morphologically more complex than quantifying cardinals should be relatively widespread. This is because an additional affix is expected to introduce a shift from a counting device to a number concept. On the other hand, if the quantifying function is derived from the arithmetical one, see Section 4.3, then we would expect an inverse pattern to be common. Namely, quantifying cardinals should include an additional morpheme encoding a shift from number concepts to quantifying modifiers.

It turns out that the cross-linguistically widespread asymmetry attested, e.g., in many obligatory classifier languages, is of the latter type (Wągiel & Caha 2020, Wągiel 2023). For instance, bare numerals in Japanese cannot modify NPs and require an additional element, namely a classifier (37a).[10] However, such an element is illicit in an arithmetical environment like (37b) despite the fact that *ko* being a general classifier could be used to indicate any type of inanimate entity.

[10] Typically, classifiers also introduce certain restrictions on the referents of the NP, e.g., being a round or a flat object, a plant, a big animal etc. This aspect of their meaning can be captured by accommodating various presuppositions about the nature of the denoted entities (Sudo 2016).

(37) a. { *go-no / go-ko-no } ringo
five-GEN five-CLF-GEN apple
'five apples'

b. juu waru { go-wa / #go-ko-wa } ni-da.
ten divided five-TOP five-CLF-TOP two-COP
'Ten divided by five equals two.' (Japanese)

Therefore, it seems that the approaches that derive the arithmetical semantics from the quantifying one make incorrect predictions with respect to meaning/form correspondences regarding numerals cross-linguistically.[11]

To conclude, the patterns examined above suggest that the arithmetical meaning of numerals is the basic one and that other uses are derived from it, and thus supporting the approaches discussed in Section 4.3 and challenging those in Section 4.2. In the next section, I will discuss complex numerical expressions in Slavic and suggest how this type of evidence can provide further hints on which of the two ways of relating number concepts and cardinalities discussed so far is on the right track.

5 Complex numerical expressions

While the mainstream semantic research focuses on basic cardinal numerals, there is a growing body of formal literature acknowledging that they are in fact only a subset of various numerical expressions. This fact has been well known in Slavic descriptive traditions. Since Slavic languages have a rich morphology, it is not surprising that one can find abundant formal complexity also in numerals. In this section, I will briefly review recent developments in the study of different types of derivationally and semantically complex numerical expressions.

5.1 Ordinals

The cross-linguistically most widespread type of derived numerical expressions are ordinal numerals, i.e., forms like English *fifth* and Russian *pjatyj* 'fifth'. Intuitively, ordinals represent position or rank in a sequential order. Modulo suppletion, e.g., *one* ~ *first*, if a language distinguishes morphologically a class of ordinals, they are always derived by an additional affix (Stolz & Veselinova 2013).

Perhaps somewhat surprisingly, the semantics of ordinals has been typically linked to superlatives. Cross-linguistically, the two types of expressions often

[11]For an analysis of the rare German pattern *ein-s* ~ *ein* 'one', see Wągiel & Caha (2020).

share identical or related morphemes (Hurford 1987, Veselinova 1998)[12] and have the same syntax (Barbiers 2007). Furthermore, the acquistion of ordinals seems to be influenced by superlative morphology (Meyer et al. 2018). Even more importantly, ordinals, just like superlatives, give rise to an ambiguity between an absolute and a comparative interpretation (Bhatt 2006, Sharvit 2010). On the absolute reading of (38a), John gave Mary a telescope that is older than all other telescopes. On a comparative reading, however, he might have given her a relatively new telescope than happens to be older than the telescopes other people gave her, i.e., John's gift is compared to other gifts rather than to telescopes in general. Similarly, (38b) can mean either that Mary got a telescope that was older than other telescopes, or that John's telescope was the first Mary has received.

(38) a. John gave Mary the oldest telescope.
b. John gave Mary the first telescope.

However, ordinals, unlike superlatives, do not give rise to so-called upstairs *de dicto* readings in sentences with intensional operators such as *want* (Bylinina et al. 2015). For instance, (39a) can have a comparative interpretation which makes it true in the scenario below. Yet, that reading is unavailable for (39b).

(39) Scenario: There are many trains throughout the day. John wants to take a train. Any of the trains between 3 pm and 4 pm is fine for him. Similarly, Bill and Steve want to take a train, and they are fine as long as the departure time is between 5 pm and 6 pm and between 7 pm and 8 pm, respectively. These people do not know anything about one another.
a. John wants to take the earliest train. true
b. John wants to take the first train. false

As a result, Bylinina et al. (2015) argue that superlatives and ordinals differ with respect to where they are interpreted. According to that proposal, while superlatives involve movement, ordinals are always interpreted *in situ*.

5.2 Group numerals

One of the key topics in the literature on plurality regards distributive and collective interpretations of sentences including plural DPs (40) (Scha 1981, Link 1983,

[12]Notice that this is also the case in some Slavic languages. For instance, Polish *pierw-szy* and Ukrainian *per-šyj*, both 'first', contain the comparative markers *-szy* and *-šyj* which also appear in superlatives, e.g., *naj-star-szy* and *naj-star-šyj*, both 'the oldest', respectively.

Landman 1989, 2000, Schwarzschild 1996). For instance, (40b) is ambiguous between a reading on which a total of three letters were written, i.e., a distributive interpretation, and a reading on which only one letter was written, i.e., a collective interpretation. The source of the ambiguity has often been located inside the VP (Hoeksema 1983, Link 1984, Schwarzschild 1991; Lasersohn 1995: Ch. 7).

(40) a. The boys wrote a letter.
b. Three boys wrote a letter.

Interestingly, Slavic languages have special numerical expressions which force a collective interpretation of the whole sentence. For instance, consider (41) (Dočekal 2012, 2013).[13] While (41a) patterns with (40b) in that it is ambiguous between a distributive and a collective reading, (41b) rules out the distributive interpretation, and thus the sole possible reading is that only one letter was written.

(41) a. Tři chlapci napsali dopis. ✓COLL, ✓DISTR
three boys wrote letter.ACC
'Three boys wrote a letter.'
b. Trojice chlapců napsala dopis. ✓COLL, # DISTR
three.GRP boys.GEN wrote letter.ACC
'A group of three boys wrote a letter.' (Czech)

Since the difference between the two sentences boils down to the alternation between the basic cardinal numeral *tři* 'three' and the morphologically complex derived form *trojice* 'group of three', the suffix *-ice* was proposed to be a morphological reflex of a semantic operation forcing a collective interpretation, specifically the group-forming operator ↑ proposed by Landman (1989).[14]

Importantly, group numerals cannot be used to refer to number concepts (42). This fact is surprising if the arithmetic meaning were derived from the quantifying one, see Section 4.2. Despite the collective meaning component, NPs with group numerals denote pluralities and one would expect that the same shift that is posited to turn the quantificational meaning of a basic numeral into the name of a number concept should be applicable also in this case, contrary to facts.

[13]In the formal semantic literature, expressions such as *trojice* 'group of three' are also referred to as denumeral group nouns (due to their nominal-like behavior) and collective numerals. However, since the traditional Slavic linguistics uses the latter term for a different type of complex numerals (see Section 5.6), in order to avoid potential confusion I will not use it here.

[14]The affix *-oj-* appears only in complex numerical expressions derived from the roots for 2–3 and is usually assumed to have a purely structural function, i.e., it marks non-basic stems.

(42) a. # Trojice je prvočíslo.
three.GRP is prime.number
Intended: 'Three is prime.'

b. # Šestice děleno trojicí je dvojice.
six.GRP divided three.GRP.INS is two.GRP
Intended: 'Six divided by three is two.' (Czech)

Group numerals were identified also in other Slavic languages. In particular, Polish *trójka* 'group of three' patterns with (41b) (Wągiel 2015) and similar examples include Slovak *trojica*, Bosnian/Croatian/Montenegrin/Serbian (BCMS) *trojka* and Russian *trojka*, all 'group of three'.[15] The existence of the discussed phenomenon in Slavic is unexpected since it is typically postulated to hold cross-linguistically that if a language has a scopal quantifier, it is a quantifier forcing a distributive interpretation (Gil 1992).[16] Hence, the Slavic data discussed above compel to revise that generalization. In any case, a systematic theory-driven cross-linguistic research of group numerals has not been carried out so far.

5.3 *Both*

An interesting feature of many languages is that alongside a regular basic numeral corresponding to English *two* there is also another form corresponding to *both*. In an early GQ account, *both* was analyzed as a quantificational determiner with the same meaning as *the two* (Barwise & Cooper 1981). However, the two expressions are not equivalent since *both* is incompatible with collective predicates (43a) and cannot appear inside partitives (43b) (Ladusaw 1982). The same behavior is also attested in Slavic (44) (Łazorczyk & Pancheva 2009).

(43) a. { The two / #Both } students are a happy couple.

b. One of { the two / #both } children sneezed.

(44) a. # Obe ženščiny prišli vmeste.
both woman.GEN came together
Intended: 'Both women came together.'

[15] Russian shows a strong tendency to lexicalize group numerals, e.g., *trojka* is also a word for a carriage drawn by a team of three horses abreast and a special tribunal in the 30s.

[16] Hebrew has expressions that resemble Slavic group numerals, e.g., *šlišiya* 'threesome of'. However, they were reported to differ in that they do not force a collective reading (Gil 1992).

b. Kto iz ėtix {dvoix / *oboix} sdelal to, čto xotel
who of these.GEN two.GEN both.GEN did this what wanted
otec?
father
'Which of these two did what his father wanted?' (Russian)

The data presented above led to proposals postulating that *both*, unlike basic cardinals, involves an additional distributive component (e.g., Ladusaw 1982, Landman 1989).[17] Interestingly, however, there is yet another respect in which the two differ. Specifically, *both* and its counterparts in other languages cannot be used to refer to a number concept in an arithmetical environment (45).

(45) a. # Oba to liczba parzysta.
both that number even
Intended: 'Both is an even number.'

b. # Dziesięć dzielone przez oba równa się pięć.
ten divided by both.ACC equals REFL five
Intended: 'Ten divided by both equals five.' (Polish)

As group numerals, *both* and its Slavic equivalents pose a challenge for approaches deriving the arithmetical function from the allegedly basic quantifying meaning.

5.4 Taxonomic numerals

A unique property of some Slavic languages is that they have taxonomic numerals. Like group numerals, they are derived by a special suffix which triggers a nontrivial semantic effect. Unlike basic numerals, such expressions do not quantify over object-level entities, i.e., tokens of a type, but rather over subkinds of a particular kind corresponding to the meaning of the modified noun.[18] For instance, let us consider the taxonomic numeral *dvojí* 'two kinds of' in Czech (Dočekal 2012, 2013, Grimm & Dočekal 2021). (46a) does not refer to two musical instruments but rather to two types of violin, e.g., a set of classical violins and a set of electric violins. Likewise, (46b) denotes two kinds of the beverage, e.g., white and red wine, irrespective of the amount.[19]

[17] For recent research on distributive numerals, see Gil (2013), Cable (2014) and Wohlmuth (2019).
[18] For an introduction to the kind/object distinction, see, e.g., Krifka et al. (1995).
[19] Note that (46) would be more faithfully translated as 'violins of two kinds' and 'wine of two kinds', but for consistency I follow the convention adopted in Dočekal (2012, 2013).

(46) a. dvojí housle
two.TAX violin
'two kinds of violin'

b. dvojí víno
two.TAX wine
'two kinds of wine' (Czech)

Unlike basic numerals, taxonomic numerals specify the domain of quantification to be within the realm of kinds exclusively. Therefore, the suffix *-í* was proposed to introduce a quantificational operation dedicated to counting subkinds, specifically the KU (for 'kind unit') operation proposed by Krifka (1995).

Though taxonomic numerals seem to be best preserved in Czech, Polish *dwojaki* and *dwoisty*, Bulgarian *dvojak* and Russian *dvojakij*, all 'twofold', are arguably expressions of a similar type. However, there appear to be certain differences with respect to their distribution and behavior that have not been described sufficiently so far. Crucially, none of the taxonomic numerals can be used to refer to an abstract number concept.

5.5 Aggregate numerals

Another intriguing type of Slavic derivationally complex numerical expressions is sometimes referred to as aggregate numerals, e.g., forms such as Czech *dvoje* 'two collections' (Dočekal 2012, 2013) as well as BCMS *dvoji* (Lučić 2015). Like group numerals, they are derived by a special suffix which triggers a non-trivial semantic effect and lack the arithmetical meaning. While basic numerals simply count atomic individuals in the denotation of the modified NP, aggregate numerals are used to quantify over collections of entities that typically form a particular spatial configuration. For instance, (47a) refers to two aggregates of cards, e.g., two decks of cards and (47b) denotes two pairs of shoes.

(47) a. dvoje karty
two.AGGR cards
'two sets of cards'

b. dvoje boty
two.AGGR shoes
'two pairs of shoes' (Czech)

Aggregate numerals are not limited to Slavic. Similar expressions seem to have existed in Latin, e.g., *trēs* 'three' ~ *trīnī* 'three collections' (Ojeda 1997), and Estonian uses pluralized numerals for this purpose, e.g., *kaks* 'two' ~ *kahed* 'two collections' (Norris 2018). However, it is the Slavic data which was taken as evidence for the relevance of mereotopology in natural language, specifically the significance of topological relations holding between atomic members of a plurality within a part-whole structure.[20] In particular, the suffix *-e* in (47) is sensitive to how parts making up a whole are arranged, and thus it was proposed to introduce a semantic operation dedicated to counting clusters, i.e., topologically structured spatial groupings of entities (Grimm & Dočekal 2021).

5.6 Gender-sensitive numerals

All Slavic languages mark grammatical gender on low numerals whereas some of them, e.g., Bulgarian, Polish and Slovak, distinguish between virile (personal masculine) and non-virile forms also in higher numerals (Cinque & Krapova 2007, Pancheva 2018, Wągiel 2023). However, what is probably even more interesting is the existence of complex expressions which I will refer here to as gender-sensitive numerals.[21] Such expressions introduce non-trivial restrictions with respect to the natural gender of individuals making up a plurality.

Let us consider Polish gender-sensitive numerals such as *troje* 'three (at least one male and at least one female)' (Wągiel 2014, 2015). Unlike virile and non-virile cardinals in (48a)–(48b), they do not show grammatical gender agreement with modified NPs. Instead, they are neuter (48c). And yet they introduce a gender-related effect. While (48a) denotes a plurality of either males or students whose natural gender is not known to the speaker and (48b) refers to three females, (48c) is interpreted as designating a mixed-gender group.

(48) a. tych trzech studentów
these.V three.V students.GEN.V
'these three students (male or indeterminate)'

b. te trzy studentki
these.NV three.NV students.NOM.NV
'these three female students'

c. to troje studentów
this.N three.GNDR.N students.GEN.V
'these three students (at least one male and one female)' (Polish)

[20]For an introduction to mereotopology, see Grimm 2012: Ch. 4, Wągiel 2018: Ch. 6.

[21]In traditional Slavic linguistics such forms are typically termed "collective numerals". In order to avoid confusion with expressions discussed in Section 5.2, I will not use that term.

Based on the evidence above, numerals such as Polish *troje* were proposed to introduce a presupposition that a designated plurality is heterogeneous and consists of both male and female individuals. Though similar expressions seem to be only attested in BCMS, e.g., *dvoje*, ‘two (one male and one female)’, there are also other types of gender-sensitive numerals in Slavic. For instance, BCMS *dvojica* and Russian *dvoe*, both ‘two (male)’, presuppose a homogeneous group consisting of male individuals only (Kim 2009, Lučić 2015, Khrizman 2020). Interestingly, none of the gender-sensitive numerals can be used to refer to a number concept.

5.7 Indefinite numerals

Another intriguing class of numerical expressions concerns indefinite numerals such as English *several* (Kayne 2007). Similar quantifiers appear in all branches of Slavic, e.g., compare Czech *několik*, Russian *neskol’ko*, Bulgarian *nyakolko* and BCMS *nekoliko*, all ‘several’. Particularly interesting data come from Polish, which distinguishes between two different expressions of this type, specifically *kilka* ‘several, a few’ and *ileś (tam)* ‘some, some amount’. While the former is only compatible with count NPs and designates a cardinality between 3 and 9, the latter also combines with mass NPs and can indicate any quantity (Wągiel 2020b).

In terms of morphosyntax and many aspects of semantics, *several* and its Slavic equivalents pattern with cardinals. However, one crucial respect in which the two differ is that indefinite numerals do not fit contexts calling for number concepts, see (49a) and (50) (Schwarzschild 2002, Wągiel 2020b). Despite the fact that one can easily imagine the intended meaning and paraphrase it using the existential quantifier (49b), this is not what one gets in (49a) and (50).

(49) a. # Four plus several is less than ten.
b. There is a number *n* such that four plus *n* is less than ten.

(50) # Cztery plus { kilka / ileś (tam) } to mniej niż dziesięć.
four plus several some this less than ten
Intended: ‘Four plus several/some is less than ten.’ (Polish)

Polish indefinite numerals were analyzed as complex expressions with an incorporated measure function and choice function. The difference between *kilka* and *ileś (tam)* is due to a different type of set the choice function selects a number from and a different kind of built-in measure function (Wągiel 2020b).

5.8 Multipliers

In many languages there is an understudied class of numerical expressions sometimes referred to as multipliers, e.g., English *double*. Though Germanic and Romance languages have borrowed their multipliers from Latin, in Slavic they are morphologically transparent and one can easily notice that they are derived from numerical roots by various affixes, e.g., compare Polish *podwójny*, Russian *dvojnoj* and Czech *dvojitý*, all 'double'.

Interestingly, multipliers display non-trivial quantificational behavior. Specifically, they do not quantify over whole entities or events, as basic cardinals would, but rather over parts thereof (Wągiel 2018, 2020a). For instance, (51a) denotes a set of singular objects having a complex internal structure. Each of those objects is a crown but crucially it also has two parts that could be considered as having the property of being a crown themselves, e.g., entities such as the ancient Egyptian Pschent or the papal tiara. Similarly, (51b) refers to a murdering event with two victims. But this means that there were two parts of that event each of which could be described as a murder in its own right.

(51) a. dvostruka kruna
two.MULT.F crown.N
'double crown'

b. dvostruko ubistvo
two.MULT.N murder.N
'double murder' (BCMS)

The behavior of multipliers stems from the phenomenon of subatomic quantification (Wągiel 2018). According to the proposal, grammar is sensitive to the arrangement of parts within a singular entity. Certain expressions, e.g., proportional partitives and multipliers, can access subatomic part-whole structures and quantify over parts conceptualized as contiguous objects within a whole.

5.9 Multiplicatives

Yet another class of frequent though surprisingly understudied numerical expressions consists of multiplicatives (but see Landman 2006, Donazzan 2013, Kayne 2015). They include adverbial expressions such as English *twice* and *two times* which are primarily used to quantify over events but can also appear in comparatives and equatives. Multiplicatives are attested in all Slavic languages with BCMS *dvaput* and Czech *dvakrát* 'two times' being representative examples.

Intuitively, multiplicatives are part of a broader class of adverbs of quantification which includes frequency adverbs like English *often* (Abeillé et al. 2004, Doetjes 2007). However, close examination reveals interesting differences between them, e.g., in terms of degree modification (Dočekal & Wągiel 2018). For instance, (52a) is ambiguous between the quantified-event and the quantified-degree reading. On the former interpretation, there were two events of increasing the demand and the total value of the increase is unknown whereas on the latter there was a twofold increase in the demand irrespective of the number of events on which it increased. Crucially, (52b) lacks the quantified-degree reading.

(52) a. Poptávka po dotacích vzrostla dvakrát. ✓EVENT, ✓DEG
demand after subsidies.LOC increased.PFV twice
'The demand for subsidies increased (by) two times.'

b. Poptávka po dotacích rostla často. ✓EVENT, #DEG
demand after subsidies.LOC increased.IPFV often
'The demand for subsidies increased often.' (Czech)

In addition, while multiplicatives are perfectly fine in comparatives and equatives, frequency adjectives are infelicitous in such constructions (53).

(53) Petr je { dvakrát / #často } vyšší než Marie.
Petr is twice often taller than Marie
'Petr is two times taller than Marie.' (Czech)

Unlike all other types of numerical expressions discussed so far, multiplicatives can be used in arithmetical environments provided that they express multiplication. However, not all multiplicatives can do that. For instance, Polish has two sets of multiplicatives but only one is felicitous in (54) which suggests that the element *razy* 'times' is a homonymous form for a multiplicative morpheme used to quantify over events and an expression of an arithmetical relation.

(54) { Dwa razy / #Dwukrotnie } trzy równa się sześć.
two times twice three equals REFL six
'Two times three equals six.' (Polish)

Though multiplicatives are key to understanding event and degree quantification, so far they have received little attention compared to cardinals.

5.10 Frequency numerals

Finally, in many languages there are expressions I will refer to as frequency numerals, i.e., multiplicative adjectives equivalent to English *two-time*. Slavic is no exception here and Polish *dwukrotny*, Czech *dvojnásobný* and Russian *dvukratnyj*, all 'two-time', are representative examples of the class in question.

At first sight, it seems that frequency numerals such as *two-time* are expressions of the same type as frequency adjectives like *occasional* and *frequent* (Zimmermann 2003, Schäfer 2007, Gehrke & McNally 2015). However, while *occasional* can have the adverbial reading, *two-time* patterns with *frequent* in that it lacks such an interpretation. For instance, while (55a) means that occasionally a sailor strolled by, both (55b) and (55c) cannot be understood that way.

(55) a. An occasional sailor strolled by. ✓ADVERBIAL
b. A frequent sailor strolled by. #ADVERBIAL
c. A two-time senator strolled by. #ADVERBIAL

Further, frequency numerals fail to combine with sortal nouns and require role nouns denoting socially salient capacities that can be acquired repetitively, e.g., positions with a term and award recipients. That is because they quantify over conventionalized events of acquiring a property, e.g., via a ceremony (56).

(56) a. Jan to dwukrotny { mistrz / #człowiek }.
Jan this two-time champion man
'Jan is a two-time champion.'
b. Jan został { mistrzem / #człowiekiem } dwa razy.
Jan became champion.INS man.INS two times
'Jan became a champion twice.' (Polish)

Let us now summarize what the complex numerical expressions data show us.

5.11 Summary

Complex numerical expressions give rise to various non-trivial semantic effects. The dataset reviewed in this section is far larger and more diverse than typically analyzed in the semantic literature, see Section 3. And yet, there is an intuition that a general theory of numerals should cover all of the examined cases. Therefore, I believe that the relevance of the discussed Slavic data is twofold.

First, morphological transparency of the discussed forms clearly shows that (unless suppletion is involved) all complex numerical expressions share a common component, i.e., a numerical root designating a number. That number is somehow employed in counting but what exactly is counted may vary depending on the type of numeral, e.g., whole objects in the case of basic cardinals, parts in the case of multipliers and subkinds in the case of taxonomic numerals. Hence, a theory of numerals should aim at providing a general mechanism for capturing not only the behavior of basic numerals but also other numerical expressions.

Second, despite sharing an element designating a number, none of the complex numerical expressions can be used to refer to number concepts, i.e., the arithmetical function is never available except for basic numerals. This fact is problematic for the approaches deriving the arithmetical meaning from the quantifying one, see Section 4.2. The reason is that they shift a counting-device semantics to a number name. But if so, why can that shift not be also applied in other types of counting expressions, e.g., taxonomic or indefinite numerals, or *both*? Semantically, they do not seem to differ radically from basic cardinals (they all count something) to justify the unavailability of the shifting. If, on the other hand, the arithmetical meaning is basic, one expects to derive various quantificational flavors from it. Therefore, it would not be surprising that only basic numerals can have the arithmetical function while other types of numerical expressions lack it.

Based on the two conclusions reached above, in the next section I will propose an analysis of yet another type of numerical expressions, namely label numerals.

6 Case study: Numerals as labels

Let us now return to the label use of numerals introduced in (1f), repeated here as (57), and discuss how it relates to the quantifying and arithmetical function.

(57) Player number five twisted her ankle.

So far, the label meaning has received little attention in the literature (see Hurford 1987: 167–168; Wiese 2003: 37–42; Bultinck 2005: 119–129). I will demonstrate that not only is this function conceptually different from other functions discussed in Section 2 but also that the distinction is reflected in grammar. I will argue that the label function is derived from the arithmetical meaning and I will propose a morpho-semantic account that allows for deriving both the form and the meaning of dedicated label numerals in Slavic. The proposed analysis will combine standard compositional semantics with a Nanosyntax model of morphology.

6.1 Label uses are distinct

Label numerals provide means for identification of an object via its association with a unique number, which allows for the recognition of that object among other entities. This mechanism is sometimes referred to as NOMINAL NUMBER ASSIGNMENT (Wiese 2003: 37–38). A set of numbers are consistent labels as long as the numerals corresponding to distinct numbers in that set are uniquely used for each distinct object. This can be achieved by one-to-one mapping between a set of labeled entities and a given set of numbers. Importantly, the actual values of the numbers represented by label numerals are irrelevant since they do not indicate quantity, order or any other type of measurement. Furthermore, the set of labels can change over time, e.g., it can grow as new entities are labeled. The only thing that matters is that at a given time every labeled entity is associated with a unique number.

It is clear that from a conceptual point of view the label function differs from the quantifying and the arithmetical meaning. Importantly, however, the distinction is also encoded in grammar. First, numerals used as labels do not allow for modification by numeral modifiers and prequantifiers (58a). Moreover, label constructions do not give rise to scalar implicatures. For instance, in the context of (58b), taking tram number six would not guarantee getting to the university.

(58) a. # Tram { more than / at least / all } number five has left.
b. You must take tram number five to get to the university. #AT LEAST

The data in (58) differ from the quantifying function and pattern with the properties of the arithmetical use of numerals discussed in Section 2.3. However, these two functions can be distinguished linguistically as well. For instance, English constructions with numerals used as labels differ from other count NPs in that they are ungrammatical with the indefinite article (59a). On the other hand, names of number concepts cannot appear without the definite article (59b) (for a detailed discussion of English phrases such as *number five*, see Bultinck 2005: 124–129).

(59) a. Kim took (*a) tram number five.
b. * (The) number five is odd.

Above, I argued that the label meaning of numerals is both conceptually and linguistically different from the quantifying and the arithmetical function. In the following two sections, I will discuss further evidence for the relevance of the label function as well as its relationship with the other two discussed meanings.

6.2 Evidence from co-lexicalizations

In many languages, an additional expression can be optionally used alongside the numeral in order to indicate its quantifying, arithmetical or label use. Sometimes, the same lexical item can introduce all of the functions above. For instance, English employs a single noun, namely *number*, to indicate cardinality, to designate a mathematical object and to signal that an entity is identified via association with a relevant integer (60).

(60) a. The cats are five in number.
b. The number five is odd.
c. Tram number five left.

While English shows full co-lexicalization, i.e., the noun *number* can be used to indicate all three functions, more often lexicons develop a different word for at least one of the functions in question. For instance, Bulgarian exhibits a pattern with no co-lexicalization, i.e., each of the discussed functions is signalled by a morphologically distinct expression, specifically *broj* is for quantification, *čislo* is for arithmetic and *nomer* is for the label function, all ‘number’.

Though ways in which meanings and forms correspond to each other vary, it seems that they do not vary in an unrestricted manner. The co-lexicalization patterns found in Slavic are summarized in Table 1. Interestingly, the pattern that is not attested in the sample is a case of non-contiguous co-lexicalization where the quantifying and label function are expressed with the same form while the arithmetical function is expressed by a different lexical item. Perhaps surprisingly, this result resembles the well-known *ABA principle (Bobaljik 2012).

Table 1: Co-lexicalization of ‘number’

LANGUAGE	QUANTIFYING	ARITHMETICAL	LABEL
BCMS	A broj	A broj	A broj
Polish	A liczba	A liczba	B numer
Slovak	A počet	B číslo	B číslo
Bulgarian	A broj	B čislo	C nomer
unattested	A	B	A

Examining permissible patterns of co-lexicalization is rarely considered evidence in formal semantics. However, I believe that looking at how certain parts of the lexicon are structured can be instructive. Arguably, it can reveal that some meanings have more in common than others and in some cases that certain categories, though theoretically possible, are empirically unrealized.[22] Consequently, one can gain valuable insights regarding what those meanings are.

On the assumption that co-lexicalization does in fact reflect the relationship between concepts, the data in Table 1 suggest that while the quantifying and the label meaning are unrelated, they both seem to stem from the arithmetical function. The reason is that if both the quantifying and the label meaning are derived from the arithmetical meaning, the attested co-lexicalization patterns are unsurprising since the unrelated derived functions are not expected to be expressed by a single term that is different from the one expressing the underlying meaning (ABA). On the other hand, if the quantifying meaning were basic, then one would expect to find the ABA pattern instead of the Czech and Slovak ABB.

Whether a language which has a term for the quantifying and the label function to the exclusion of the arithmetical function can be found is an empirical issue. So far, the Slavic co-lexicalization data suggest a strong relationship between the label and the arithmetical meaning and no relationship between the label and the quantifying meaning. Let us now move to Slavic label numerals.

6.3 Derived label numerals

An interesting property of Slavic languages is that they have specialized label numerals. Such dedicated expressions are derived with the suffixes *-ka* and *-ica*. Table 2 gives the label forms for the numerals corresponding to 5 across Slavic.

Table 2: Cardinal and label 'five' in Slavic

LANGUAGE	NUMBER	CARDINAL	LABEL
Czech	5	pět	pětka
Polish	5	pięć	piątka
Russian	5	pjat'	pjaterka
Slovenian	5	pet	petka/petica
BCMS	5	pet	petica

Though there are a number of interesting differences in their productivity and distribution, e.g., Polish derives label forms with *-ka* up to 999 while Russian has

[22]This way of viewing data has been successful in lexicology (see Hage 1997 on kinship terms).

them only for 2–10, 20 and 30, all Slavic label numerals are morphosyntactically nominal expressions used to refer to objects that can be identified via nominal number assignment. Though very often they appear bare, the noun specifying what entity is labeled can optionally precede the label numeral (61). In the absence of the noun this information is typically provided by the context.

(61) (Tramvaj) pět-ka vyjede na novou trať v listopadu.
tram five-LBL will.go.out on new.ACC track.ACC in November.LOC
'Tram number five will head out on the new track in November.' (Czech)

Similar to classifier constructions and other complex numerical expressions, recall Section 4.4 and Section 5, Slavic label numerals exhibit uniform behavior in that they cannot be felicitously used in mathematical statements like (62). This, in turn, indicates that their semantics is more intricate than the arithmetical meaning.

(62) a. Dva-krát pět se rovná deset.
two-times five REFL equals ten
'Two times five equals ten.'

b. # Dvoj-ka krát pět-ka se rovná desít-ka.
five-LBL times two-LBL equals ten-LBL
Intended: 'Two times five equals ten.' (Czech)

Thus, Slavic label numerals are both morphologically and semantically complex. Let us now propose an account that will capture their morpho-semantics and the relationship between the arithmetical meaning, quantification and labeling.

6.4 Analysis

In Section 2 and 4, we saw the theoretical relevance of the fact that numerals are often ambiguous. In order to account for the relationship between the three uses of basic numerals this paper focuses on, i.e., their arithmetical, quantifying and label function, I develop a morpho-semantic system combining a standard inventory of formal semantics with the Nanosyntax model of morphology (see Wągiel & Caha 2020). The core of the proposal is that the quantifying and the label function are both derived from the arithmetical meaning which is taken to be basic.

I propose the ingredients in (63) as a set of morpho-semantic components that numeral expressions conveying the relevant functions are formed of. Specifically, I postulate the bottom-most NUMP$_n$ (for 'number') layer along with two additional heads, namely CL (for 'classifier') and LBL (for 'label').

(63) a. $[\![\textsc{NumP}_n]\!]_n = n$
b. $[\![\textsc{Cl}]\!]_{\langle n,\langle\langle e,t\rangle,\langle e,t\rangle\rangle\rangle} = \lambda n \lambda P \lambda x[{}^*P(x) \wedge \#(P)(x) = n]$
c. $[\![\textsc{Lbl}]\!]_{\langle n,\langle\langle e,t\rangle,e\rangle\rangle} = \lambda n \lambda P \iota x[P(x) \wedge \textsc{label}(n,x)]$

The meaning of $\textsc{NumP}_n$ is simply an abstract number concept (type *n*) (63a). The subscript indicates that $\textsc{NumP}_n$ is lexically encoded. Though for the purpose of this paper I assume that it is the basic structural component of each numeral, I do not claim that it is primitive. It might very well be further decomposed into a number of features.[23] On the other hand, the components Cl and Lbl are interpreted as shifts from *n* to a more complex type.

I will assume here that Cl is interpreted as a function from a number concept to a predicate modifier devised for counting individuals (type $\langle n,\langle\langle e,t\rangle,\langle e,t\rangle\rangle\rangle$) (63b).[24] Cl takes a number and yields an expression with the built-in pluralization operation * and measure function $\#(P)$, see Section 3.3. In cases such as English *five* or Czech *pět* 'five', Cl is already incorporated in the structure of a numeral but in some other languages it is encoded in an additional morpheme, e.g., a classifier.

On the other hand, Lbl is a labeling device (type $\langle n,\langle\langle e,t\rangle,e\rangle\rangle$) which selects a number and returns a function from properties to unique individuals such that they have the relevant property and are labeled with the corresponding number by the label operation (63c). Lbl is typically introduced by a free morpheme like *number* or a specialized affix like the Czech suffix *-ka*, as discussed in Section 6.2–6.3.

As a result of combining the components in (63), we obtain the structures in Figure 1 and 2. Figure 1 represents the meaning of the quantifying function of a numeral corresponding to 5 whereas Figure 2 depicts the parallel label function.

When a quantifying numeral, e.g., Czech *pět* 'five', and a label expression, e.g., Czech *číslo pět* 'number five', combine with an NP, we obtain denotations as in (64). In particular, (64a) denotes a set of pluralities of players such that each plurality in that set consists of five players. On the other hand, (64b) refers to the contextually unique player identified via their association with the number 5.

(64) a. $[\![\text{pět hráčů}]\!]_{\langle e,t\rangle} = \lambda x[{}^*\textsc{player}(x) \wedge \#(\textsc{player})(x) = 5]$
b. $[\![\text{hráč číslo pět}]\!]_e = \iota x[\textsc{player}(x) \wedge \textsc{label}(5,x)]$

[23] This is indicated by the triangle in Figure 1 and 2. For arguments why one would want to further decompose $\textsc{NumP}_n$, see Wągiel & Caha (2020).

[24] For the purpose of this paper, it is not important what the exact interpretation of Cl is. The proposed system is compatible with other theories of quantifying numerals discussed in Section 3 as long as it is possible to shift the primitive type *n* to a type for the quantifying function.

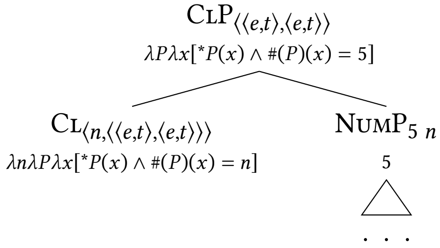

Figure 1: The quantifying function

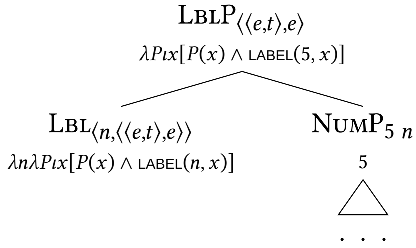

Figure 2: The label function

But how to account for the fact that *pět* 'five' can be used to express both the quantifying and the arithmetical meaning? This ambiguity can be captured as a consequence of a core property of late insertion, namely that lexical entries are not tailor-made for one specific use.[25] In Nanosyntax, a lexical entry pairs a well-formed syntactic structure with phonology. For instance, let us assign to the numeral *pět* the lexical entry in Figure 3. When syntax builds a structure interpreted as the quantifying meaning, as in Figure 1, the entry in Figure 3 can be used to lexicalize it. Consequently, the structure or, more precisely, its top-most phrasal node is pronounced by /*pjɛt*/, as indicated by the big circle in Figure 4.

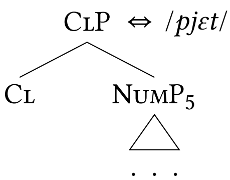

Figure 3: Lexical entry

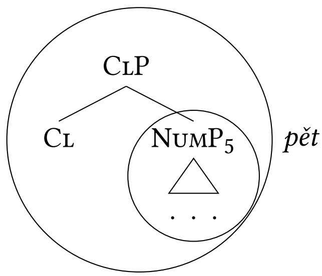

Figure 4: Lexicalizations

Crucially, however, when syntax builds a structure interpreted as the arithmetical meaning, i.e., the sole NumP$_{5}$, it can also be pronounced by Figure 3, as indicated by the small circle in Figure 4. This is due to the Superset Principle, defined in (65) (Starke 2009), which states that a lexical entry can be used to pronounce a particular structure iff it contains that structure as a sub-part (either proper or improper). As a result, the Czech numeral *pět* 'five' is ambiguous between the quantifying and the arithmetical meaning since both circled structures in Figure 4 are contained in the tree in Figure 3.

[25]For an introduction to late insertion models such as Distributed Morphology and Nanosyntax, see, e.g., Bobaljik (2017) and Baunaz & Lander (2018), respectively.

(65) The Superset Principle (Starke 2009)
A lexically stored tree matches a syntactic node iff the lexically stored tree contains the syntactic node.

Let us now consider how the label meaning arises. For this purpose, I will demonstrate how specialized label numerals such as Czech *pětka* 'number five' are derived. Since *pět* 'five' is stored as Figure 3, and thus cannot express the label meaning, the Lbl component needs to be spelled out by a separate entry, e.g., as the suffix *-ka* in Figure 5. Notice that the semantics of Lbl is not compatible with the quantifying meaning and requires type *n* as its input (63c). Thanks to the Superset Principle (65), *pět* can also pronounce the arithmetical meaning, i.e., the sole NumP$_5$, as indicated by the small circle in Figure 4. However, in order for this structure to apply at the LblP node, NumP$_5$ must move from its base-position. In Figure 6, the lower copy of the displaced element is shaded.[26] After the NumP$_5$ is moved, Figure 5 matches the lower LblP, as shown in Figure 7, where the label suffix *-ka* is inserted at the relevant node.

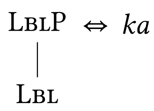

Figure 5: Czech label suffix

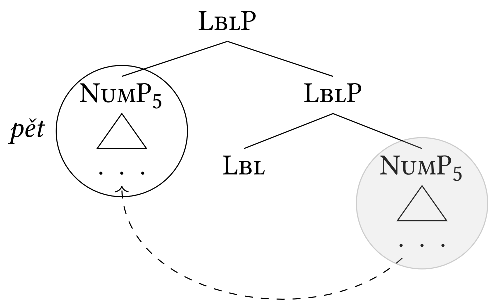

Figure 6: Movement

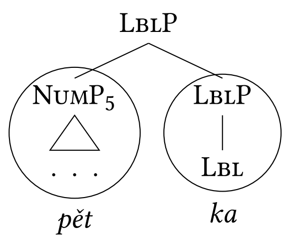

Figure 7: Spellout

The proposed analysis has several advantages. First of all, it provides a unified approach that captures the relationship between the arithmetical, the quantifying and the label meaning. Second, it explains when and how a numeral can be

[26]In Nanosyntax, the movement in Figure 6 is driven by the need to lexicalize the phrasal LblP node by the lexical entry in Figure 5. For a detailed explanation of how precisely the movement in Figure 6 is triggered, see Baunaz & Lander (2018) and Caha et al. (2019).

ambiguous between different meanings. Finally, it gives a detailed account for the meaning/form correspondences both in cardinal and in label numerals.

7 Conclusion

In this paper, I explored the semantics of numerals in Slavic and beyond. I discussed various functions that basic numerals can have as well as different approaches to capturing the relationships between these functions. Furthermore, I examined many types of complex numerical expressions that only recently attracted attention in the literature. I also provided novel evidence in favor of the view suggesting that the arithmetical meaning is basic while other functions are derived from it. Finally, I explored the so-far understudied label function and proposed a morpho-semantic analysis of Slavic derived label numerals that could be further extended to other types of complex numerical expressions.

Abbreviations

ACC	accusative
AGGR	aggregate numeral
CLF	classifier
COP	copula
F	feminine
GEN	genitive
GNDR	gender-sensitive numeral
GRP	group numeral
INS	instrumental
IPFV	imperfective
LBL	label numeral
LOC	locative
MULT	multiplier
N	neuter
NBR	number-denoting numeral
NOM	nominative
NV	non-virile
PFV	perfective
REFL	reflexive pronoun
TAX	taxonomic numeral
TOP	topic
V	virile

Acknowledgements

I would like to sincerely thank the anonymous reviewers as well as Lisa Bylinina, Pavel Caha, Katie Fraser, Jovana Gajić, Nina Haslinger, Ora Matushansky, Rick Nouwen, Marina Pantcheva, Dolf Rami, Viola Schmitt, Yasu Sudo, Peter Sutton, Lucia Vlášková and Kata Wohlmuth for their judgments, insightful comments and inspiring discussions. All errors are, of course, my own responsibility. The research was supported by a Czech Science Foundation (GAČR) grant to the Department of Czech Language at the Masaryk University in Brno (GA24-10073S).

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Name index

Topics in the semantics of Slavic languages

Topics in the semantics of Slavic languages contains twelve chapters, each devoted to a particular topic in formal semantics, as well as an introductory chapter. The topic chapters consist of two parts: an overview part and a new analytical contribution or case study. The volume fulfills a number of goals: (i) to summarize existing analyses and theories, (ii) to provide a representative bibliographic overview of the relevant literature, (iii) to demonstrate what Slavic data and theories built upon them have to offer to the general linguistic discourse, and (iv) to provide a novel theoretical and/or empirical contribution. The chapters in this collection discuss data from the following Slavic languages: Bosnian/Croatian/Montenegrin/Serbian (BCMS), and more specifically Croatian and Serbian, Bulgarian, Czech, Macedonian, Polish, Resian, Russian, Slovak, Slovenian, Upper Sorbian, and Ukrainian.

The collection is of interest to a broad readership – Slavic as well as general linguistic scholars, experienced researchers as well as advanced undergraduate or postgraduate students. Individual contributions can serve as background reading to topical seminars. The volume covers areas of semantics in which Slavic languages have traditionally or more recently played an important role in the general formal semantic discourse (e.g. aspect, event structure, genericity, tense, information structure, negation and NPIs, (in)definiteness and specificity, bare NP semantics), as well as topics that have received less attention in their application to Slavic languages, but are pertinent to the theory of semantics, including its interfaces (e.g. focus participles, questions, imperatives, numerals, passives).